DATASET TODO — master collection list (dataset-grain)

1114 dataset rows across 15 domain volumes (generated by build_todo_merge.py; source files `todo/NN-*.md`). Verification split — SURVEY-EVIDENCE: 384 · UNVERIFIED-CANDIDATE: 352 · VERIFIED-UPSTREAM-DOC: 378. Wave mix — W0: 319 · W1: 352 · W2: 282 · W3: 161. Actions — backfill: 113 · build-parser: 64 · connect: 749 · license-review: 40 · quote: 69 · verify: 79. License mix — : 1 · **AGPL-3: 1 ·** AGPL-3.0: 2 · **CC: 1 ·** CC-BY-NC: 1 · **NOASSERTION: 2 ·** WEO: 1 · **unofficial**: 1 · ABN:: 1 · BCCh: 1 · BIS: 5 · BSP: 1 · Bloomberg: 1 · BoC: 2 · BoI: 1 · BoJ: 3 · BoK: 1 · CC: 10 · CC-BY: 2 · CC-BY-SA?: 1 · CME: 1 · CN: 1 · CNB: 1 · Cboe: 3 · CertiK: 1 · Coinglass: 1 · Databento: 17 · Dune: 1 · GPL-3.0: 1 · HKMA: 1 · Hippocratic-2.1: 1 · IMF: 6 · MAS: 1 · MIT: 8 · NFIB: 1 · OGL-UK: 3 · OGL-UK?: 1 · OSS: 1 · OSS;: 1 · Pro: 1 · RBA: 1 · SARB: 1 · SIP: 1 · SNB: 1 · StockTwits: 1 · TronGrid: 1 · UK: 2 · UMich: 1 · US: 1 · US-gov: 19 · academic: 4 · add-on: 1 · archive: 1 · attribution: 1 · attribution+share-alike: 1 · attribution-license: 1 · attribution-needed: 1 · brokerage: 1 · chain: 13 · commercial: 39 · commercial-vendor: 23 · contractual;: 1 · copyright:: 1 · data: 3 · delayed: 1 · display-class: 1 · enterprise: 12 · enterprise,: 2 · enterprise-quote: 6 · exchange: 20 · exchange-fee: 9 · exchange-free: 18 · exchange-free?: 2 · fee-for-service: 1 · fees: 1 · flat-rate: 1 · free: 468 · free,: 4 · free-attribution: 24 · free-attribution;: 1 · free-for-display: 1 · free-ish: 11 · free-ish,: 1 · free-of-charge: 1 · free-read: 1 · free-tier: 1 · free-to-read;: 1 · free?: 1 · freemium: 22 · freemium→paid: 1 · gov: 1 · gov-published;: 1 · legacy: 1 · levels: 1 · licence: 1 · license: 8 · license-risk: 1 · license:: 1 · licensed: 32 · licensed-ish: 1 · mandated: 1 · mixed: 9 · mixed:: 4 · most: 1 · n: 1 · no: 4 · non-display: 1 · on-chain: 1 · open: 4 · paid: 20 · paid-API: 1 · paid-extract: 1 · paid-per-doc: 1 · paid-portal: 4 · partial: 1 · partially: 1 · per-repo: 2 · per-venue: 1 · pointer: 1 · portal-only: 7 · public: 34 · public,: 1 · public-domain: 21 · public;: 9 · published: 1 · re-use: 1 · regulatory-confidential: 2 · research: 1 · restricted: 1 · same: 3 · self: 1 · self-captured: 3 · site: 2 · state: 1 · strictly: 1 · subgraph: 2 · subscription: 11 · subscription-gated: 2 · undocumented: 1 · unlicensed: 1 · varies: 1 · vendor: 1 · vendor-license: 33 · vendor-tier: 33 · —: 4 · ⚠: 2 · ⚠**sanctioned: 1 · ⚠-blocked-never-commercial: 3. Duplicates dropped: 1 · schema problems: 0 (kept in build log, not in table).

domain volume	rows	schema problems
India equities & F&O (`todo/01-*`)	50	0
Asia-Pacific equities (`todo/02-*`)	90	0
US equities & microstructure (`todo/03-*`)	81	0
EU & global-rest equities (`todo/04-*`)	72	0
Fundamentals · filings · PIT (`todo/05-*`)	121	1
Derivatives (options/futures) (`todo/06-*`)	84	0
FX · rates · macro · credit · ESG (`todo/07-*`)	103	0
Commodities · energy · power · shipping (`todo/08-*`)	86	0
Alternative data (`todo/09-*`)	75	0
Registries · legal · IP · clinical · procurement (`todo/10-*`)	72	0
Crypto full-stack (`todo/11-*`)	90	0
Cross-domain historical archives (`todo/12-*`)	76	0
Prediction markets (`todo/13-*`)	7	0
India depth (alphalens gap-closure) (`todo/14-*`)	89	0
External resources (S43 intake) (`todo/15-*`)	18	0

Consolidated table

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	NSE equity bhavcopy (cash market OHLCV, all scrips)	price/trades	nseindia.com all-reports → daily CM bhavcopy zip (UDiFF-format since Jul-2024; legacy format before)	daily bar	1994→ (approx; pre-2010 sparse survey claim), daily zips, backfill-able now	EOD	file-drop	free (MVP-replace note: primary free base for INR equity closes)	PIT-complete post-close; dual parser needed across Jul-2024 format break	W0	backfill	SURVEY-EVIDENCE
2	NSE F&O bhav-copy, full chain (every contract OHLC + OI)	derivative price	nseindia.com deriva-tives bhavcopy → per-contract daily CSV, month-zips	daily bar per contract	F&O 2000→, month-zips, backfill-able now	EOD	file-drop	free (MVP-replace note)	per-expiry PIT chain; instrument-token churn across expiries needs roll book	W0	backfill	SURVEY-EVIDENCE
3	NSE F&O pre-2010 legacy archive (pre-UDiF-era files, 2000–2009)	derivative price	same month-zip family re 2000s	daily bar	2000→ per-row above; pre-2010 files in legacy columns	EOD	file-drop	free	legacy header/column dialect vs 2024+ UDiFF — build parser isolating eras	W0	build-parser	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
4	NSE index bhavcopy (all published indices daily)	index	nseindia.com all-reports → indices section	daily bar	~2007→ (approx survey claim), backfill-able	EOD	file-drop	free	index value as pub-lished; constituent-count changes create level breaks	W0	connect	SURVEY-EVIDENCE
5	NSE indices official back-calculated series (Nifty 50 1995→, Bank Nifty 2000→, per-index to incep-tion)	index	niftyindices.com historical-data down-loads (CSV per index)	daily bar	back-calc to inception (Nifty 50: 1995), one-shot bulk	EOD batch	REST bulk	free (reg-istration wall)	back-calculated ≠ as-trailed; PIT usable only from series-start of live method-ology	W0	backfill	UNVERIFIED-CANDIDATE
6	NSE delivery quantity & delivery %	market structure	all-reports → MTO delivery file + bhavcopy delivery column	daily	2017→ public archive (survey claim), backfill-able	EOD	file-drop	free	columns empty on T2T/other series — filter series PIT	W0	backfill	SURVEY-EVIDENCE
7	NSE MWPL (market-wide position limits, per F&O scrip)	risk	nseindia.com MWPL disclo-sure page	snapshot (daily-ish)	current values only on page; history requires own con-tinuous capture	delayed	own-capture	free (page) / ⚠-blocked-never-commercial if scraped at scale	take times-tamped snapshot; MWPL changes intraday possible	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
8	NSE F&O ban list (trading-ban scrips)	risk	nseindia.com derivatives AL page + daily file	daily	published continuously; archive depth uncertain → own-capture from today	delayed	own-capture	⚠-blocked-never-commercial (scrape family)	ban status is PIT per exchange-hour only	W0	connect	SURVEY-EVIDENCE
9	NSE bulk deals (≥0.5% turnover disclosures)	flow	nseindia.com live-analysis bulk-deals CSV	daily rows	~2010→ approx daily CSVs, backfillable	EOD	file-drop	free	as-disclosed PIT; counterparty name spelling drifts	W0	backfill	SURVEY-EVIDENCE
10	NSE block deals (SEBI block-window disclosures)	flow	same bulk/block deals disclosure plane	daily rows	disclosure-regime dated (pre/post 2024 rules differ); archive depth to verify	EOD	file-drop	free	two regimes (pre/post new block-window rules) need regime tags	W0	verify	SURVEY-EVIDENCE
11	NSE shareholding patterns (quarterly promoter + list, XLSX per quarter)	ownership	nseindia.com shareholding archive → quarter XLSX zip	quarterly	~2007→ (approx survey claim); XLSX backfiles manual crawl	batch	file-drop	free	revisions possible post-filing; capture-as-filed PIT + revision flag	W0	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
12	Shareholding backfile pre-2013 quarters (older-format filings)	ownership (archive)	NSE quarter-archive + inspection of earliest available quarters	quarterly	earliest-quarter intact depth must be proven by sampling	batch	file-drop	free	format drift across years; promoter-code taxonomy changes	W0	verify	UNVERIFIED-CANDIDATE
13	NSE corporate announcements API (regulation-30 feed JSON)	corporate event	nseindia.com corporate-announcements REST JSON	event rows	API exposes sliding window (~90d); web archive older	near-realtime	REST	⚠-blocked-never-commercial (scrape family; own-use research ok)	as-published timestamps; PDF attachments are the PIT truth	W0	connect	SURVEY-EVIDENCE
14	NSE corporate actions / split-bonus / calendar + new listings & IPO history	corporate event	all-reports corporate-actions + new-listing pages	event rows	~2000→ approx; issue-wise IPO facts patchier pre-2010	EOD	file-drop	free	record-date joins to bhavcopy need as-of logic (pre-close vs post-close sequencing)	W0	backfill	SURVEY-EVIDENCE
15	NSE ASM/GSM surveillance lists (stage/press releases)	regulatory	all-reports surveillance files	daily/batch	ASM regime ~2014→ survey-approx, backfillable	EOD	file-drop	free	stage-changes create overlapping eligibilities — model as events	W0	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
16	NSE pre-open session order snap-shots (09:00–09:08 window)	market structure	nseindia.com pre-open pages/pages CSV	intraday snap-shots	current sessions only; own-capture builds archive	delayed	own-capture	free / ⚠-blocked-never-commercial at scale	snapshot sequence must be stored raw; no vendor restate-ment	W0	connect	SURVEY-EVIDENCE
17	India VIX official history	vol index	NSE India-VIX historical CSV + Upstox V3 instru-ment key `NSE_INDEX|India VIX` (verified route ^9)	daily	2007→ (survey claim), backfill-able	EOD	REST bulk	free	methodology rebase across years (see NSE circulars before stitching)	W0	backfill	SURVEY-EVIDENCE
18	NSE live option chain (minute-poll own capture, all strikes)	derivative analytics	nseindia.com option-chain JSON polled	intraday	forward-only from capture date	realtime	own-capture	⚠-blocked-never-commercial (NSE scraping terms; internal research use only)	store raw poll dumps; NSE re-strikes/updates chain con-stantly	W0	build-parser	SURVEY-EVIDENCE
19	NSE stock-wise historical data product (paid per-symbol down-loads)	price/trades	NSE data-products portal, stock-wise historical down-loads	intraday	price: (survey corpus claim) ₹10,000/yr; depth per NSE tariff sheet to verify	EOD	REST bulk	vendor-license (NSE tariff; no redistri-bution)	verify tariff sheet + depth before buy; compare vs True-Data/GDFL overlap	W1	quote	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
20	NSE FII/DII cash+F&O activity & participant-wise OI	flow	all-reports partici-pant/activity files	daily	participant OI ~2002?→ / FII-DII 2001?→ to verify	EOD	file-drop	free	category defini-tions changed over years (FPI/FII, partici-pant buckets)	W0	backfill	SURVEY-EVIDENCE
21	BSE equity bhavcopy (EQ market, all scrips incl. BSE-only names)	price/trades	bseindia.com bhavcopy archive (per-day down-load)	daily bar	~1999→ approx (survey claim), backfill-able	EOD	file-drop	free (MVP-replace note; covers names absent from NSE)	BSE scrip churn differs from NSE; ISIN-key join first	W0	backfill	SURVEY-EVIDENCE
22	BSE an-nounce-ments & corporate-disclosure plane	corporate event	bseindia.com an-nounce-ments XML/API	event rows	long tail pre-2010 spotty; API window sliding	near-realtime	REST	free / ⚠-blocked-never-commercial at scale	BSE-announced items overlap NSE ~90%; dedupe on both	W0	connect	SURVEY-EVIDENCE
23	BSE indices official history (Sensex back-calculated 1979→, index bhavcopy set)	index	bseindia.com index pages / index historical download	daily bar	Sensex claimed back-calc 1979→ (strong free win to verify); other BSE indices later	EOD	REST bulk	free	older decades index-method metadata scarce — note level-break points	W0	verify	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
24	BSE pre-2010 archive (Wayback-era bseindia.com walk-through)	price/trades (archive)	web.archive.org CDX walk of old BSE bhavcopy/report URLs	daily bar	fragments pre-2005; completeness unproven	EOD	S3-archive walk	free (archive content)	fragile URLs; missing days documented as gaps	W3	backfill	UNVERIFIED-CANDIDATE
25	MCX bhavcopy (commodity derivatives OHLCV+OI)	derivative price	mcxindia.com bhavcopy download	daily bar	2003→ (launch), backfillable per survey	EOD	file-drop	free (MVP-replace note; commodity leg adjacency)	contract-naming conventions changed over the years	W0	backfill	SURVEY-EVIDENCE
26	NCDEX bhavcopy (agri derivatives)	derivative price	ncdex.com bhavcopy download	daily bar	2004→ (launch), backfillable per survey	EOD	file-drop	free	agri contract expiries/dates asymmetric vs financial derivatives	W0	backfill	SURVEY-EVIDENCE
27	GIFT Nifty (NSE International Exchange flagship NIFTY contracts)	derivative price	nseix.com EOD/bhavcopy (host timed out on fetch)	daily bar	Jul-2023→ (post-execution migration from SGX); SGX pre-2023 chain is a separate legacy series	EOD	file-drop	free (verify)	cross-venue continuity break at Jul-2023 — never stitch without flag	W0	verify	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
28	NSE IX other GIFT segments (currency, debt, IASP/commodity exports)	derivative price	nseix.com reports pages	daily bar	since segment launches (2022→)	EOD	file-drop	free (verify)	GIFT-session timezone (IST/UTC) mapping must be explicit	W0	verify	UNVERIFIED-CANDIDATE
29	AMFI NAV daily full-universe dump (NAVAll.txt)	fund NAV	amfiindia.com → NAVAll.txt (live fetch verified ^4)	daily snapshot	full scheme universe of latest NAVs (verified file emits ~18k rows/scheme-dated); full-history via own daily append + portal	EOD	REST bulk	free	NAV is PIT per scheme; discontinued schemes keep last NAV date — do not forward-fill silently	W0	connect	VERIFIED-UPSTREAM-DOC
30	AMFI NAV full-history backfile (per-scheme per-date series)	fund NAV (archive)	AMFI NAV-history portal (form POST) or longer archives	daily	oldest schemes ~1995→ (survey claim); oldest-intact depth to sample	batch	REST bulk	free	scheme-ID renumbering on AMCs and scheme mergers → ID-mapping table required	W0	backfill	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
31	MF monthly portfolio disclo-sures (scheme holdings)	ownership	AMFI/AMC monthly portfolio publica-tions	monthly	regime ~2011→ (survey claim), depth varies per AMC	batch	file-drop	free	heterogeneous formats across 40+ AMCs — normal-ization layer needed	W3	verify	UNVERIFIED-CANDIDATE
32	Kite Connect historical candles (Zerodha; NSE/BSE/NFO/MCX)	price/trades	api.kite.trade /instru-ments/historical/:token/:interval (verified contract ^1)	minute/3m/5m/10m/15m/30m/60m/day + OI flag	60 days (minute); “several years” (claim); commu-nity figure: minute ≈2016→; price: (survey claim) ₹2,000/m per app	batch (day-fresh)	REST bulk	vendor-license (Kite Connect; no redis-tribution)	expired F&O tokens not ad-dressable — cache instru-ments + continu-ous=1 only for day bars	W1	connect	VERIFIED-UPSTREAM-DOC
33	Dhan Historical API v2 (daily to inception; intraday 5y)	price/trades	api.dhan.co/v2 /charts/histor-ical + /charts/intraday (verified ^2)	1/5/15/25/60-min + day	daily to inception (claim ^2); intraday last 5 years; 90-day poll chunks	batch (day-fresh)	REST bulk	vendor-license (free with Data API access)	claims not field-tested for exotic tickers; F&O needs ex-piryCode book-keeping	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	Dhan Expired Options Data (rolling expired chains w/ IV, OI, spot)	derivative analytics	api.dhan.co/v2 `/charts/rollingoption` (verified ^3)	minute-level	last 5 years (claim ^3), ATM ±10 strikes (index) / ±3 (others), 30-day poll chunks	batch (day-fresh)	REST bulk	vendor-license (free)	strike-relative re-keying means strike identity moves day to day — PIT recon-struct per expiry	W1	connect	VERIFIED-UPSTREAM-DOC
35	Upstox V3 historical candles (+ India VIX + global indices incl. GIFT NIFTY keys)	price/trades	api.upstox.com/v3 `/historical-candle/...` (verified ^9)	min 1–300 / hours 1–5 / day/week/month	minutes from Jan-2022; days/weeks/months from Jan-2000 (claims ^9); chunk limits: min 1mo–1qtr, day 1 decade	batch (day-fresh)	REST bulk	vendor-license (free with Upstox account)	depth claims = vendor self-report; spot-check 2000s equities before trusting	W0	connect	VERIFIED-UPSTREAM-DOC
36	Upstox v2 historical candles (depre-cated path)	price/trades	api.upstox.com/v2 (verified ^8)	1m/30min/day/week/month	1m/30min 1mo; day/week/month 1y; 30min 1y; day 1y; week/month 10y (claims ^8)	batch	REST bulk	vendor-license	superseded by V3 — keep only for schema cross-check	W0	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
37	Upstox expired-instruments dataset	derivative analytics	upstox developer “Expired Instruments” API (nav seen ^8, endpoint page not fetched)	daily/contract	to verify	batch	REST bulk	vendor-license	expiry-chain continuity vs Kite `continuous` approach — compare before committing	W1	verify	UNVERIFIED-CANDIDATE
38	Angel One SmartAPI historical candle API	price/trades	smartapi.angelbroking.com historical-candleREST	minute-hour-day	(survey claim) minute from ~2011; docs portal non-rendered on fetch	batch	REST bulk	vendor-license (free)	rate-limit tight (survey claim); depth claim needs live test	W0	verify	SURVEY-EVIDENCE
39	Fyers v3 historical data (chart end-points)	price/trades	myapi.fyers.in v3 organic historical endpoint (claimed ^10: 29y EOD, up to 7y intraday)	tick?/min-hour-day	vendor claims: 29 years EOD; up to 7 years intraday	batch (day-fresh)	REST bulk	vendor-license (free)	29y claim likely composed EOD — verify pre-2000 daily series provenance (exchange or vendor-composed)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
40	Fyers expired F&O data (with historical Greeks)	derivative analytics	myapi.fyers.in v3 expired-fo end-points (claimed ^10: up to 7y weekly/monthly expiries + historical Greeks claim)	intraday/contract	up to 7 years of expired contracts + Greeks (claim ^10)	batch	REST bulk	vendor-license (free)	Greeks method-ology for expired contracts is opaque → ask Fyers before trust	W1	connect	VERIFIED-UPSTREAM-DOC
41	TrueData Velocity (autho-rized tick+minute feed NSE/BSE/MCX; 1-min & tick backfill)	price/trades	truedata.in Velocity plu-gin/API (verified ^6)	tick/2s/1-min ups	plans: (claim ^6 ^7) ₹1,699–₹3,299/m Lite→Ultima; historical backfill depth per tier to verify	realtime / EOD	library(CCXT-class) (plugin + custom API)	vendor-license (exchange-authorized tariff; commer-cial per NSE/BSE/MCX rules)	back-adjusted split/bonus series destroy PIT — request raw + ad-justment times-tamps	W1	quote	VERIFIED-UPSTREAM-DOC
42	TrueData API family (Corpo-rate An-nounce-ment + MF/NAV + Corpo-rate Data/results/fundamentals + News APIs)	corporate event/fundamental	truedata.in market-data-api section (verified listing ^6)	event/fundamental rows	depth per API tier to verify	near-realtime	REST	vendor-license	corporate-data PIT discipline: amend-ments flagged? — probe in trial	W1	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
43	GlobalDatafeeds historical intraday archives (GDFL)	price/trades	globaldatafeeds.in historical packages	tick/min/day	(survey claim) intraday from ~2008; premium global markets; price: see-site (site fetch 403)	realtime / EOD	library(CCXT-class)	vendor-license	redistribution by default not allowed — negotiate; tick-storage compression + retrieval cost; epoch/timezone alignment must be tested	W1	quote	SURVEY-EVIDENCE
44	RBI gilt auction results & SGL transaction data (equities-adjacent liquidity context)	regulatory/fixed income	rbi.org.in auction results + SGL trades listings	daily/batch	auction results decade-deep per survey; SGL depth to verify	batch	file-drop	free	only as context feed for equities-adjacent risk/spillover studies	W3	verify	SURVEY-EVIDENCE
45	SEBI data portal (FPI holdings monthly, MF AUM, SCORES/orders)	regulatory	sebi.gov.in dataset pages	monthly	monthly series decade-deep per survey	batch	file-drop	free	manually-published PDFs dominate — OCR layer needed	W3	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	Short-selling disclosure dataset (segregated short-sell orders)	flow	exchange disclosure pages (presence unconfirmed)	daily	existence and depth to verify — candidate only	EOD	file-drop	free (verify)	if absent public, note-and-skip rather than scrape	W3	verify	UNVERIFIED-CANDIDATE
47	Wayback-era NSE site CDX walk (pre-2007 archives; old weekly/daily report URLs)	price/trades (archive)	web.archive.org CDX index of nseindia.com old report paths	daily bar	fragments 2000ish→ era; completeness unproven	EOD	S3-archive walk	free (archive content)	missing days permanent gaps — document coverage matrix per year	W3	backfill	UNVERIFIED-CANDIDATE
48	NSE delisted / excluded securities archive (Z-series, delisted list, suspended list)	universe/archive	NSE delisted-companies listings + exclusion reports	snapshot/batch	depth partial; to verify	batch	file-drop	free	survivorship-bias killer: join historical bhavcopy only where delisted asset still has bars	W0	verify	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
49	Paid deep intraday backfile 2000s (TrueData/GDFL/NSE stock-wise cross-quote)	price intraday archive	procurement across rows 19/41/43 candidates (pick one vendor)	minute	2008–2016 NSE minute coverage guaranteed only via paid tier (survey claim)	EOD	licensed terminal	vendor-license	obtain one overlapping-window sample and cross-validate against an independent source before pick	W1	quote	SURVEY-EVIDENCE
50	Norgate Data India stock coverage — checklist rejection row	price/trades (archive)	norgatedata.com official site (verified ^5)	—	Norgate explicitly serves US/AU/CA stocks + world futures/forex only — NO India equity package	—	—	—	site checked; do not budget effort here (action-note: (closed: not offered))	W3	verify	VERIFIED-UPSTREAM-DOC
1	CK A-share daily quotes (`daily` full A-board OHLCV+amount)	EOD price	Tushare Pro REST (`daily`/`daily_basic`), token-keyed	daily	since 1990-12-19; full backfill yes	same-evening T+1	REST bulk	freemium (points tiers)	adj factor separate endpt; join pre-rel/delist_date for survivorship; unit=shares not hands (action-note: connect+)	W0	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
2	CK A-share minute history (`stk_mins` family)	intraday bar	Tushare Pro REST, high-points tier	1/5/15/30/60 min	recent years only, shallow backfill per tier	minutes after bar	REST bulk	vendor-license (points)	session/halt align-ment; tier gate; verify depth caps before budget-ing	W1	verify	VERIFIED-UPSTREAM-DOC
3	CK A-share realtime endpts (□□□□, □□□□-□□□)	realtime snapshot	Tushare Pro realtime REST	intraday snapshot	T0 only, no history	realtime polling	REST polling	vendor-license (points)	no PIT (live only); snapshot cadence unpub-lished	W2	verify	VERIFIED-UPSTREAM-DOC
4	CK financial state-ments (`income/balancesheet/cashflow/forecast/express`)	fundamentals	Tushare Pro REST, re-port_type param	quarterly	since 2000s (metric coverage varies), backfill yes	same-day after disclo-sure	REST bulk (paged)	freemium	PIT = an-nounce-ment date field, not report period; revenue-recognition calendar traps (action-note: con-nect+)	W0	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	CK reference/ownership events (top-10 holders, pledges, unlock, block trades, insider moves)	corporate events	Tushare Pro □□□□ family REST	per-event/daily	multi-year per family	T+1 publish	REST bulk	freemium	PIT by disclose_date; some families survey-only (no PIT label)	W1	connect	VERIFIED-UPSTREAM-DOC
6	CK margin + LHB boards (`margin`/`margin_detail`/`top_list` family)	flow/badge	Tushare Pro REST	daily	margin since 2010-03; LHB since 2005	T+1 publish	REST bulk	freemium	flag join via margin_targets; LHB badge semantics entrepreneurship	W1	connect	VERIFIED-UPSTREAM-DOC
7	CK ETF dataset family (info/shares/NAV/adjusted/minute)	fund	Tushare Pro ETF REST	daily	since 2004-12-30 (first ETF)	EOD+1	REST bulk	freemium	NAV vs close PIT; AUM changes as market data not fundamentals	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
8	CK index family (index_daily/index_weight/index_classify + SW industry)	index	Tushare Pro REST	daily	major indices since 1991; weights history since ~2005	same-day EOD	REST bulk	freemium	weights PIT at published rebal-ance rows; SW industry classifica-tion version churn (action-note: con-nect+)	W0	backfill	VERIFIED-UPSTREAM-DOC
9	CK public-fund family (fund list/NAV/dividends/holdings)	fund	Tushare Pro □□ REST	daily	open-fund era from 2001	EOD+1	REST bulk	freemium	NAV PIT at nav_date; holdings quarterly-lag only	W2	connect	VERIFIED-UPSTREAM-DOC
10	CK futures family (con-tracts/daily/adjusted/positions/settle/warehouse)	futures	Tushare Pro □□ REST	daily	SHFE/DCE/CZCE from mid-1990s; CFFEX from 2010	evening EOD	REST bulk	freemium	roll conti-nuity joins; □ lot units; settle vs close dis-tinction	W1	connect	VERIFIED-UPSTREAM-DOC
11	CK HK/US daily + funda-mentals (□/□□ family)	EOD price + fun	Tushare Pro REST	daily	HK from ~2004, US from 2000s (verify per market)	EOD	REST bulk	freemium	US sur-vivorship weak; HK board suffix rules; currency normal-ization	W2	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
12	CK macro/rates + text corpus (Shibor/LPR/CPI/PPI + LLM news/announce corpus)	macro/text	Tushare Pro REST	daily/monthly	Shibor 2006+; corpus decade+	episodic	REST bulk	freemium	text PIT by publish ts; requires text-qa pipeline	W2	connect	VERIFIED-UPSTREAM-DOC
13	AKShare `stock_zh_a_hist` (A-share daily, Eastmoney-underlying, qfq/hfq)	EOD price	pip akshare lib wrapper on eastmoney JSON	daily	since 1991/listing; full backfill	delayed same-day	library	free / ⚠ (EM ToS scraping plane)	adjust method param controls history rewrite; hands-vs-shares unit verify; EM can change endpoints (action-note: connect+)	W0	backfill	VERIFIED-UPSTREAM-DOC
14	AKShare minute (`stock_zh_a_hist_min_em`, `stock_zh_a_minute` sina)	intraday bar	akshare lib	1/5/15/30/60 min	short window per request; deep minute not archived	realtime/delayed	library	free / ⚠	per-request caps; bar timezone alignment (Beijing)	W1	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
15	AKShare A-sh funda-mentals (`stock_financial_abstract`, `stock_financial_report_sina`)	fundamentals	akshare lib sina tables	quarterly	since ~2000	T+1	library	free / ⚠	PIT via an-nounce-ment date; CN-GAAP mapping (action-note: con-nect+)	W1	build-parser	VERIFIED-UPSTREAM-DOC
16	AKShare in-dex+constituents (`stock_zh_index_daily_em`, `in-dex_stock_cons_csindex`, `in-dex_stock_cons_weight_csindex`)	index	akshare lib (EM + CSI offical files)	daily	1991+; CSI cons/weights official history	EOD	library	free	weights only at publish rows; CSI xlsx ingest	W0	connect	VERIFIED-UPSTREAM-DOC
17	AKShare futures (`futures_zh_daily_sina`, `fu-tures_main_sina`, `fu-tures_delivery`)	futures	akshare lib	daily	per-venue 1990s+	delayed EOD	library	free / ⚠	symbol canoni-caliza-tion; settle vs prev_close	W1	connect	VERIFIED-UPSTREAM-DOC
18	AKShare funds/ETF (`fund_etf_hist_em`, `fund_nav_em`)	fund	akshare lib	daily	ETF 2004+, funds 1998+	EOD	library	free / ⚠	NAV vs price; dividend adj	W2	connect	VERIFIED-UPSTREAM-DOC
19	AKShare HK plane (`stock_hk_daily`, `stock_hk_spot_em`)	EOD price	akshare lib	daily	from list-ing/IPO per name	EOD	library	free / ⚠	HKD units; HK half-day sessions; survivor-ship absent	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
20	AKShare US plane (`stock_us_daily`, `stock_us_spot_em`)	EOD price	akshare lib	daily	~decade deep (verify per name), no survivorship	EOD	library	free / ⚠	USD; uppercase tick swaps (BRK.A); delisting gaps	W2	verify	VERIFIED-UPSTREAM-DOC
21	AKShare flow/ownership blocks (`stock_zh_a_gdhs`, `stock_individual_fund_flow`, `stock_lhb_detail_em`)	flow/sentiment	akshare lib on EM datacenter	daily	per block 2000s+	EOD	library	free / ⚠	no per-record PIT; classification noise; block schema churn	W2	connect	VERIFIED-UPSTREAM-DOC
22	Baostock daily + 5-min K-line (A-share SDK)	EOD + 5-min bar	baostock free Python lib (socket protocol)	daily + 5-min	daily since 2006-01-01; 5-min shallow; backfill bulk yes	EOD evening	library	free	separate `query_adjust_factor`; join delist_date for survivorship (Baostock has market=sh/sz only) (action-note: connect+)	W0	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
23	Baostock dividends & adjust factors (`query_dividend_data`, `query_adjust_factor`)	corporate acts	baostock lib	per-event	since 2006	EOD	library	free	cross-check vs announcements; only A-share ex-div calendar (action-note: connect+)	W0	backfill	SURVEY-EVIDENCE
24	Baostock quarterly fundamentals (`query_profit_data/balance/growth/operation`)	fundamentals	baostock lib	quarterly	since 2006	post-disclosure cadence	library	free	PIT via PUBDATE field, not reportDate (action-note: connect+)	W1	backfill	SURVEY-EVIDENCE
25	RiceQuant RQData equities (price/fundamentals/calendar/trade dates)	multi	RQSDK (`rqdatac` Python, HTTP API)	daily→minute→tick tiered	deep daily (verify start year); minute from 2000s	realtime available	library	vendor-license (paid; trial)	PIT via `get_fundamentals` with query_date; adj='pre' semantics; contract delisting included	W1	quote	VERIFIED-UPSTREAM-DOC
26	RiceQuant RQData futures/options (contracts/settle/rolls)	futures	RQSDK	daily/tick	deep (2000s); options from 2015	realtime	library	vendor-license	roll conventions; settlement price vs close	W2	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
27	JoinQuant jqdata (`get_price`, `get_bars`, `get_fundamentals`, `normalize_code`)	multi	jqdata REST/SDK	daily+minute	daily 1990s; minute from 2005	EOD/realtime	library+REST	vendor-license (points; trial)	`get_fundamentals(date=)` PIT normal-ization; warmed history counts in quota	W2	evaluate	UNVERIFIED-CANDIDATE
28	Wind (□□) Terminal / WindPy / EDB	multi	Wind desktop + WindPy server-side	daily→tick	comprehensive multi-decade CN archive	realtime terminal	licensed terminal	vendor-license (quote)	entitlements per terminal; redistri-bution prohib-ited (action-note: quote+)	W3	license-review	UNVERIFIED-CANDIDATE
29	CSMAR □□□ CN academic archive (mar-ket/firms/events)	multi archive	CSMAR web bulk + SFTP crates	daily/annual	CN since 1990 full history	static exports	file-drop/SFTP	vendor-license (aca-demic quote)	PIT via list-ing/delist tables; survivor-ship handled; licensed redistri-bution N-OK (action-note: quote+)	W3	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
30	cninfo disclosure REST (P0ST `/new/hisAnnouncement/query`) SZ/SSE-listed co. announcements + PDFs	corporate disclosure	cninfo.com.cn public API	per-announcement	since ~1999 (digitized; verify earliest)	near-realtime	REST polling + file fetch	free / ⚠ (redistribution of doc PDFs per cninfo terms)	announce_time per-name per-announce batched query; PDF parse pipeline (action-note: connect+)	W0=pit;	build-parser	VERIFIED-UPSTREAM-DOC
31	cninfo □□□□ (□□□□ + □□□□□ □□□-data service)	disclosure-derived	cninfo portal (login-gated)	daily/quarterly	varies per table	T+1	web portal (contract unverified)	freemium (paid plans)	XBRL instances PIT; browser quota gates	W1	license-review	VERIFIED-UPSTREAM-DOC
32	SHFE (+INE) official daily EOD (settlement/trade/warehouse)	futures	shfe.com.cn market-data pages (JSON/HTML)	daily	since late-1990s (verify); INE since 2018	EOD evening	file-drop/HTTP	free	date published T+1; units (tonne/lot); verify archive endpoint depth (action-note: connect+)	W1	build-parser	VERIFIED-UPSTREAM-DOC
33	DCE official daily EOD	futures	dce.com.cn static pages	daily	since 1993-11; full archive	EOD	file-drop/HTTP	free	CN-side latency; publish T+1; HTML table parse	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	CZCE official daily EOD	futures	czce.com.cn static files	daily	since 1990s	EOD	file-drop/HTTP	free	**bot-block: GET / returned 412** from non-CN IP [f8] → needs CN egress or SFTP agreement (action-note: +verify)	W1	connect	SURVEY-EVIDENCE
35	GFEX official daily EOD (industrial-silicon/lithium-carbonate)	futures	gfex.com.cn data pages	daily	since 2022-04 founding	EOD	file-drop/HTTP	free	transport-blocked from this host [f9]; young archive only	W2	connect	UNVERIFIED-CANDIDATE
36	CFFEX official daily EOD (IF/IH/IC/IM equity-index fu-tures+options)	futures	cffex.com.cn static files	daily	since 2010-04-16	EOD	file-drop/HTTP	free	daily settle only; verify CSV archive availabil-ity	W1	connect	SURVEY-EVIDENCE
37	Eastmoney data-center planes (funds-flow/LHB/margin/announcements/stock-connect)	flow/sentiment	data.eastmoney.com + datacenter-web JSON REST	daily	per block 2000s-2010s	intraday-refresh some blocks	REST polling	free / ⚠ (scraping plane)	no per-record PIT; schema churn; EM rename field risk (action-note: +verify)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
38	Eastmoney realtime quote plane (push2.eastmoney.com `secid=` snapshot)	realtime snapshot	EM quote endpoints	tick-ish poll	live only, no history	realtime	REST polling	free / ⚠	no PIT; header/anti-bot measures; endpoint migration risk	W2	verify	UNVERIFIED-CANDIDATE
39	Sina hq quote plane (`hq.sinajs.cn list=` realtime quotes)	realtime snapshot	sina HTTP GET (needs `Referer: finance.sina.com.cn`)	tick-ish poll	live only	realtime	REST polling	free / ⚠ (unpublished)	mandatory Referer header (since ~2021, else 401); informal, breakage-prone	W2	verify	UNVERIFIED-CANDIDATE
40	Tencent gtimg plane (`qt.gtimg.cn/q=` live quotes; `web.ifzq.gtimg.cn/appstock/app/fqkline` CN/HK/US daily)	EOD price + snapshot	Tencent quote endpoints	daily	CN/HK/US daily since 1990s-2000s (verify)	delayed EOD (CN intraday live)	REST polling	free / ⚠ (unpublished)	fqkline param controls adj; text-array parse; informal endpoint (action-note: verify+)	W1	build-parser	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
41	CSI Index official index files (csindex.com.cn: constituents/weights/returns/factor downloads)	index	csindex official download center	daily/monthly	000300 since 2004-04; per-index archive	EOD	file-drop/HTTP	free (⚠ redistribution restricted per CSI terms)	methodology PIT via rebalance announcements; JS-heavy site; XLSX ingest (action-note: connect+)	W0	backfill	SURVEY-EVIDENCE
42	Stock-Connect northbound flows & holdings	flow	HKEX Northbound stats pages + Eastmoney datacenter blocks [f10]	daily	Shanghai-Connect since 2014-11-17; Shenzhen-Connect since 2016-12-05	intraday summary; holdings T+1 morning	REST polling + scrape	free / ⚠	HKD vs CNY fields; holdings PIT once per day only (action-note: +verify)	W0	connect	VERIFIED-UPSTREAM-DOC
43	A-share delisted/ST survivorship archive (delist_date + daily rows for delisted names)	archive	Tushare Pro ST lists + `daily` endpoints for retired tickers [f1]	daily	delistings since 1990s (first majors 2001); complete pre-delist backfill	n/a	REST bulk	freemium	survivorship-bias corrector; ST flag transition PIT (action-note: verify+)	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	HKEXnews disclosure archive (title-search + postings PDFs & XBRL)	filings	www1.hkexnews.hk/search/titlesearch.xhtml + ePortal	per-filing	~1999 (digital); complete coverage from ~2004	near-realtime	web POST + PDF/xbrl fetch	free / ⚠ (HKEX data-portal terms)	filing taxonomy (□□/□□/□□□□); PIT by release date; XBRL parse for structured (action-note: connect+)	W0	build-parser	VERIFIED-UPSTREAM-DOC
45	HKEX CCASS Shareholding Search (per-stock shareholding by CCASS participants incl. HKSCC Nominees)	shareholding	HKEX CCASS pages (shareholding search + participant list)	daily	archive from ~2010 s (current format); earlier via third-party only	next-morning T+1	own-capture (scrape)	free / ⚠	shareholding % = percent of shares held; float-based; daily changes need diff pipeline (action-note: +own-capture)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	HKEX realtime market data (Equip retail L1 / OMD licensed feed)	realtime L1/L2	hkex.com.hk Equip store + OMD feed	tick / EOD	live; EOD pur-chasable sepa-rately	realtime	licensed termi-nal/feed	vendor-license (quote)	exchange entitle-ments; redis-tribute prohib-ited; latency tiers (delayed L1 free vs realtime paid) (action-note: quote+)	W3	license-review	VERIFIED-UPSTREAM-DOC
47	HKEX historical data mar-ketplace (equities tick / EOD pur-chases)	tick/EOD archive	HKEX historical data mar-ketplace portal	tick→daily	paid archive (start year via license call; expected late-2000s)	batch	paid down-load/SFTP	vendor-license	attribution; no recurring distribu-tions without license (action-note: quote+)	W2	license-review	UNVERIFIED-CANDIDATE
48	Hang Seng Index series (HSI/HSCEI/HSTECH daily returns + con-stituents/weights files)	index	hsi.com.hk data down-loads	daily	HSI since 1969-11; HSCEI since 1994; HSTECH since 2020-07	EOD	file-drop/HTTP	free / ⚠ (HSIL terms)	PIT = index-methodology version; fund-family filing style; con-stituents CSV ingest	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
49	AASTOCKS-style HK realtime quotes/economic data	realtime/quote	aastocks.com portal (license quote)	tick-ish	live	realtime	licensed portal/terminal	vendor-license (quote)	full entitlement vendor; no historical via this plane	W3	quote	UNVERIFIED-CANDIDATE
50	Futu OpenAPI — HK/US/CN-A equity history + realtime (OpenD gateway; get_history_kline etc.)	EOD+quote	futunn OpenAPI SDK (Python/REST/WS via OpenD)	daily/minute + realtime	HK/CN/US daily since ~1990s-2000s (verify per market); HK realtime tick	realtime (with account)	library+WS	vendor-license (freemium w/ broker acct; quote row)	per-symbol-and-day quote quota; broker account eligibility; redistribution N-OK for feed (action-note: +quote)	W1	connect	SURVEY-EVIDENCE
51	TWSE OpenAPI daily trading data (TWT-family reports + exchangeReport/STOCK_DAY JSON)	EOD price	twse.com.tw OpenAPI/JSON endpoints	daily	modern series since 2004-01; deeper via monthly PDFs (1990s)	EOD evening	REST bulk (JSON)	free	units in NT$-thousand; halt/delisting; per-endpoint year caps (action-note: connect+)	W0	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
52	TWSE foreign/institutional investor daily flows (□□□□ per-security T86)	flow	TWSE TWT86U JSON endpoint	daily	since 2004	T+1 afternoon	REST bulk	free	investor-category semantics (□□/□□/□□□); cross-check against futures positions for sentiment	W0	connect	SURVEY-EVIDENCE
53	TWSE MOPS (□□□□□□□: monthly revenue / insider / major-shareholder filings)	filings/events	mops.twse.com.tw JSON endpoints + own scrapes	monthly/per-filing	digitized since ~2000s (verify earliest)	T+1	REST polling + file fetch	free / ⚠	insider PIT by □□□□; monthly-revenue release timing drives drift (action-note: connect+)	W1	build-parser	SURVEY-EVIDENCE
54	TPEx OpenAPI (□□ daily quotes + TPEx index)	EOD price	tpex.org.tw end-points	daily	similar depth to TWSE (~2004+); verify archive	EOD	REST bulk	free	units NT$-thousand; consistent with TWSE parse layer	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	FinMind datasets (TaiwanStockPrice etc.: daily/minute/fundamentals/flow)	EOD+intraday+flow	FinMind API (api.finmindtrade.com) + docs	daily/minute	daily since 2004; minute per dataset varies	EOD	REST polling	freemium (quota)	`TaiwanStockPriceAdj` for backtest; `TaiwanStockPrice` raw for consistency; per-dataset calendar (action-note: +verify)	W1	connect	UNVERIFIED-CANDIDATE
56	TEJ Taiwan Economic Journal (academic/enterprise archive: TW market, financials, ownership, analyst)	multi archive	TEJ proprietary database / file-push	daily/monthly	TW daily since 1980s; financials since 1990s	batch	proprietary terminal	vendor-license (quote)	PIT via announcement-date fields; strong TW extension to HK/CN (action-note: quote+)	W3	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
57	TW intraday own-capture plane (TWSE MIS realtime quote pages, 5s poll)	intraday	own-capture of twse.com.tw realtime JSON/HTML	5s snapshot / 1-min rollup	ingest-only from connec-tion date	realtime→rollup	own-capture	free / ⚠	per-session snapshot not tick; anti-bot throttling risk; needs JVM-compatible parse layer? (verify field formats)	W2	build-parser	UNVERIFIED-CANDIDATE
58	J-Quants API prices/daily_quotes	EOD price	J-Quants REST (api.jquants.com; tiered)	daily	tier-based depth; sample free tier	EOD	REST bulk	freemium	per-tier range caps; Ad-justment-Close flag for quotes (action-note: +verify)	W1	connect	SURVEY-EVIDENCE
59	J-Quants API financials (fins/statements + fins/announcement)	fundamentals	J-Quants REST	quarterly/annual	from EDINET-based state-ments (start year verify)	daily	REST bulk	freemium	PIT by Dis-closed-Date fields	W1	connect	SURVEY-EVIDENCE
60	J-Quants API in-dices/TOPIX (`indices/topix` + TOPIX families)	index	J-Quants REST (+ JPX official files for deep)	daily	TOPIX since 1968 (via JPX); JQ-API tier depth verify	EOD	REST bulk	freemium	tall-hist via JPX official files; index reconsti-tution events	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	EDINET API v2 (documents list + PDF/XBRL fetch)	filings	EDINET REST (disclosure2.edinet-fsa.go.jp)	per-filing	EDINET digitized since 2008; XBRL from 2012-08 (verify earliest)	near-realtime (within 24h)	REST bulk + file fetch	free / ⚠ (attribution/redistribution TOS)	doc-type map (□□□□ etc.); PIT by submitDateTime; XBRL instance parse (action-note: connect+)	W0	build-parser	VERIFIED-UPSTREAM-DOC
62	JPX Data Cloud (historical market statistics bulk releases)	EOD/intraday archive	marketdata.jpx portal	daily	archive start year unverified — cautious: **“expect 2000s archive; verify window by release calendar”**	batch	file-drop/download	freemium (⚠ redistribution terms)	PIT by release date; per-release coverage (action-note: verify+)	W1	connect	UNVERIFIED-CANDIDATE
63	JPX arrowhead (real-time MBO/L2 feed)	tick	JPX proprietary connect (arrowhead MBO)	tick	live; historical via licensed replay	realtime	licensed binary feed	vendor-license (quote via JPX member)	colocation/entitlement cost; redistribution prohibited (action-note: quote+)	W3	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
64	Nikkei NEEDS (price/fundamental/ownership databases)	multi archive	NEEDS licensed feed (CSV/file push or terminal)	daily	deep (JP daily since 1949)	batch	licensed terminal	vendor-license (quote; high cost)	PIT by release; JP-only vendor UX; per-table license scope (action-note: quote+)	W3	license-review	UNVERIFIED-CANDIDATE
65	kabutan □□ web data (daily quotes/fundamentals screens)	EOD+fund screens	kabutan.jp web scrape	daily	from ~2010 s (verify)	delayed EOD	own-capture	free / ⚠ (scraping ToS)	DOM churn; personal-use only per site policy; no PIT label on records	W3	build-parser	UNVERIFIED-CANDIDATE
66	kabusapi (au□□□□ KabuSta-tion API: JP/US realtime REST/WS)	realtime	kabusapi REST/WS (per-session token)	tick/1-min	live + short own-chart history	realtime	REST/WS	vendor-license (broker account free; redistri-bution N-OK)	rate limits; needs own bro-kerage acct; live-only (PIT only for own trades)	W2	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
67	DART filing search (corp Code master + per-filing list + PDF parse)	filings	DART portal + public search API (dart.fss.or.kr)	per-filing	DART digitized since 1999-09; complete from ~2003	near-realtime	REST polling + file fetch	free	corp_code refresh cadence; classification map (□□□□□/□□□□□/□□□□ flags); PIT by □□□ (action-note: connect+)	W0	build-parser	VERIFIED-UPSTREAM-DOC
68	OpenDART financial statements APIs (□□□□ □□□□□□ + □□□□□□□□□□ + XBRL □□)	fundamentals	OpenDART REST (opendart.fss.or.kr; free API key)	quarterly/annual	XBRL statements from 2012-12 (verify earliest)	daily	REST bulk	free	is□□□ vs is_consolidated fs_div; PIT by rm (report name); batch download quota (action-note: connect+)	W0	build-parser	VERIFIED-UPSTREAM-DOC
69	OpenDART □□□□ □□□ (major-shareholder/bulky filing aggregate)	shareholding	OpenDART REST (□□□□ □□ API)	per-filing	from 2016 (□□□□ robust coverage; verify earlier depth)	daily	REST bulk	free	shareholder PIT rounds; change_reason codes; XML parse	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
70	KRX Data Marketplace (□□·□ · daily quotes/statistics; + analyst/platform)	EOD price + stats	data.krx.co.kr queries + KRX Data Marketplace	daily	deep daily archive (per-dataset years vary; verify per table)	EOD (T+0 evening)	web query + REST-ish POST (verify contract)	freemium (base free; advanced needs subscription)	per-table headers/units; KRW; index vs price split; PIT at publish (action-note: +verify)	W1	connect	VERIFIED-UPSTREAM-DOC
71	KOSPI/KOSDAQ index history (via KRX/official index series)	index	KRX data portal + index official series	daily	KOSPI since 1980-01-04; KOSDAQ since 1996-07	EOD	REST bulk	free	methodology histories; currency KRW	W1	connect	SURVEY-EVIDENCE
72	NICE InfoService-type (KR credit/research data analog)	multi archive vendor	NICE info portal (license quote)	daily/monthly	varies by product	batch	proprietary terminal	vendor-license (quote)	per-product license scope; credit vs market split	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
73	SGX Securities market data (daily quotes/indices/STI + data-API)	EOD price + index	sgx.com market-data pages + SGX Data Centre	daily	since 1999 (SGX merger day) backfill per-series	EOD	REST/portal (verify per-dataset contract)	freemium (base free; full data licensed)	SGX data-license for redistribution; SGD; session (morning/luncheon break + afternoon); PIT at publish (action-note: +quote)	W1	connect	VERIFIED-UPSTREAM-DOC
74	SGX historical data marketplace (intraday/EOD purchases)	tick/EOD archive	SGX Market Data store	tick→daily	paid archive (start year via license call)	batch	paid download	vendor-license	redistribution restrictions; per-dataset license (action-note: quote+)	W2	license-review	UNVERIFIED-CANDIDATE
75	SET Thailand official dataset plane (daily quotes, indices, disclosures)	EOD price + corp	set.or.th / settrade portals	daily	since 1990 s? (per-venue backfill verify)	EOD	REST/portal (unverified contract)	free / ⚠	THB units; trading-session (morning/luncheon + afternoon); PIT by publication (action-note: +verify)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
76	SET intraday quote history files (settrade per-stock minute pages)	intraday bar	settrade quote-history static pages	1-min	per-stock minute archive (verify depth; expect rolling months-to-years window per stock)	delayed	file-drop/HTTP	free / ⚠	archive windows may rotate; per-request naming (action-note: verify+)	W2	build-parser	UNVERIFIED-CANDIDATE
77	IDX Indonesia dataset plane (daily trading summary, disclo-sures, ID-list)	EOD price + corp	idx.co.id Market Data/Announcement pages	daily	since 1990s? (official archive verify)	EOD	portal/REST-ish (unveri-fied contract)	free / ⚠	IDR units; IDX Micro-port contract; PIT by publish (action-note: +verify)	W1	connect	VERIFIED-UPSTREAM-DOC
78	Bursa Malaysia dataset plane (daily quotes + company an-nounce-ments)	EOD price + corp	bursamalaysia.com portals	daily	since 1980s (com-pany backfill verify)	EOD	portal scrape / PDFs (un-verified contract)	free / ⚠	MYR units; sessions; an-nounce-ments only in HTML/PDF (no single API) (action-note: +verify)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
79	PSE Edge portal (company disclosures / financial reports / index summary)	filings + EOD	edge.pse.com.ph (disclosure list + forms)	per-filing + daily	digital disclosures since ~2016? (verify); earlier via SEC	intraday (live)	web portal + PDF/JSON	free / ⚠	PH units (PHP); filing taxonomy (17-A/8); PIT by publish (action-note: connect+)	W1	build-parser	VERIFIED-UPSTREAM-DOC
80	Vietnam HOSE/HNX dataset plane (VND30/HNX-INDEX; SSI iBoard as unofficial plane)	EOD price	HOSE/HNX official + SSI iBoard screape	daily	since 2000 (HOSE from 2000; official archive verify)	EOD (delayed)	portal scrape + file-drop	free / ⚠	VND units; T+1 settlement norms; login-gated; no PIT label (action-note: +verify)	W2	connect	UNVERIFIED-CANDIDATE
81	NZX market-data plane (announcements + NZX daily data downloads)	EOD price + corp	nzx.com announcements + NZX data services	daily	since 1980 s (per market verify)	EOD	portal + paid data API	freemium (list-on-request; quote)	NZD; per-product license terms on redistribution (action-note: +quote)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
82	ASX official dataset plane (daily CSV archive + company announcements)	EOD price + corp	asx.com.au data pages + announcements	daily	since 1997 (ASX-path founding); deeper via annual/legacy archives	EOD	file-drop/HTTP	free / ⚠ (redistribution restricted)	AUD; codes include share-class suffixed; PIT by publish (action-note: connect+)	W1	backfill	UNVERIFIED-CANDIDATE
83	Tadawul API (SDMC/Saudi Exchange market data API v1 realtime + historical)	EOD price + realtime	api.tadawul.com.sa (v1 REST; **SDMC = Saudi Data & Metadata?**; verify scope: real-time/EOD)	daily + intraday	per-API tier (verify backfill depth)	realtime/delayed	REST	vendor-license (key; redistribution N)	SAR units; Sunday-Thursday calendar; entitlement needed (action-note: quote+)	W2	verify	UNVERIFIED-CANDIDATE
84	ADX market-data plane (Abu Dhabi Securities Exchange: daily summaries + announcements)	EOD price + corp	adx.ae portals	daily	since founding 2000-11	EOD	portal scrape / file	free / ⚠	AED; no single REST contract yet; holiday calendar (action-note: +verify)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
85	DFM market-data plane (Dubai Financial Market: daily quotes/indices)	EOD price	dfm.ae portals	daily	since 2000-03	EOD	portal scrape	free / ⚠	AED; verify per-endpoint	W3	connect	UNVERIFIED-CANDIDATE
86	EGX Egypt Open API (daily EOD + static files)	EOD price	egx.com.eg official Ope-nAPI/static	daily	since 2000s (verify)	EOD	REST (un-verified contract)	free	EGP units; official archive window unclear (action-note: +verify)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
87	MOEX ISS (securities/history/candles/trades REST reference; index constituents via listing endpoints)	EOD + tick-level	iss.moex.com REST (JSON/CSV/XML)	daily + intraday candle	EOD history deep (sample data back to 2010-11 in official examples; deeper via ISS dates endpoints) — write **“since ~2010 via ISS queries; deep via dates discovery”**	EOD/delayed	REST bulk	free (ISS public; redistribution check per MOEX TOS)	RUB/USD?; index boards; per-board histories (board-level dates listings) (action-note: connect+)	W1	backfill	VERIFIED-UPSTREAM-DOC
88	Stooq APAC long EOD archive (CN/HK/JP/IN/SG daily CSV)	EOD price	stooq.com CSV HTTP files	daily	since 1980-1990s per symbol (free bulk download)	delayed EOD	file-drop/HTTP	free	survivorship unknown; symbol reeling (index suffix); vendor-free but not PIT consistent (action-note: +verify)	W1	backfill	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
89	Yahoo Finance APAC backfill (HK/JP/TW/KR/SG daily via yahoo/yfinance)	EOD price	query1/quoteJSON (via yfinance)	daily	from late 1980s/1990s per venue; per-symbol from IPO	delayed EOD	library/REST	free / ⚠ (yahoo ToS scraping plane)	delisting gaps (yahoo only lists active); raw=uh parse; price-format changes (action-note: (cheap corpus for cross-check))	W1	backfill	UNVERIFIED-CANDIDATE
90	Norgate Data AU/NZ survivor-ship archive (ASX/NZX incl. delisted)	archive EOD	Norgate desktop + API/lift SDK	daily	AU back to 1980s; NZ 1990s; incl. delisted-firms full path	EOD	desktop-app+API	vendor-license (paid)	survivorship handled (Norgate curated); annual license	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	nasdaq-totalview-itch-5.0	exchange-official feed	Nasdaq TotalView-ITCH 5.0 via MoldUDP64 UDP multicast (+ Soup-BinTCP replay); spec PDF on Nasdaq-Trader	event-by-event full L3 order stream (add/exec/replace/trade/NOII/cross)	realtime feed; paid history via Data Link NTV (2014→ claim, re-verify)	realtime (exchange-native)	ITCH-5.0 feed handler; Databento MBO normal-ization is a proven template	exchange license — non-display family (display redistri-bution = separate tier; sales quote)	PIT: order-by-order only from ITCH (no overlay dup feeds); order on ts_event; NOII auction-phase gates (row 6) (action-note: → build-parser on free sample)	W1	license-review	VERIFIED-UPSTREAM-DOC
2	nasdaq-ntv-datalink-history	licensed historical archive	Nasdaq Data Link NTV database (prior session verified listing; site timed out this session)	per-day TotalView archives / leveled products	2014→ per prior-session claim (re-verify)	T+1 archive	NDL REST API or bulk download	exchange license via NDL; display/non-display billing split (vendor claim)	PIT: dataset-version pinning; re-download full days on spec bumps, never partial-patch (action-note: quote →)	W2	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	nasdaq-basic-NLS+	exchange-official (display L1)	Nasdaq Basic (BBO + NLS last, Nasdaq-listed); Databento mirror dataset “Nasdaq Basic with NLS Plus” listed	L1 quotes + last sale	realtime; archive depth via Nasdaq sales (re-verify)	realtime	MoldUDP64 handler, or Databento mirror	display-class license (cheapest Nasdaq tier; NLS redistribution rules)	PIT: Nasdaq-listed issue scope only; NOT SIP NBBO — merge with Tape C BBO for consolidated NBBO (action-note: (cheap live NBBO-half))	W2	quote	VERIFIED-UPSTREAM-DOC
4	nasdaq-itch-50-daily-archive	raw archive (low/free cost)	Nasdaq ITCH 5.0 daily message logs (sample library; e.g. emi sample routes — fetch failed this session) + free Databento TotalView samples	raw A/C/D/E/F/U/T/X/Y message files	sample from ~2013 (re-verify; paid depth via rows 1–2)	T+1 batch	bulk file walk + binary ITCH parser	free/low-cost sample entitlement; redistribution limits per Nasdaq terms (verify)	PIT: ground-truth lineage for every ITCH-derived dataset — pin spec version per day-file (action-note: + backfill)	W1	build-parser	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	nasdaq-bx-psx-ntx-totalview	exchange-official feed	Nasdaq satellite-venue order feeds; Databento dataset ids XBOS-ITCH (Nasdaq Texas) + XPSX-ITCH (PSX) listed TotalView	L3 order stream per satellite exchange	realtime; mirror archives per Databento ranges	realtime	reuse row-1 handler per feed, or Databento live/hist	exchange license, satellite-venue fee tier (small)	PIT: venue-scoped books; never merge XBOS/XPSX msgs into XNAS book state (action-note: (handler reuse))	W3	build-parser	VERIFIED-UPSTREAM-DOC
6	nasdaq-noii-auction-imbalance	administrative/auction	Nasdaq ITCH NOII mes-sages + Nasdaq cross statistics (open-ing/closing/EMC) — rides NTV enti-tlement, no separate product	imbalance snap-shots (paired/total-imbalance/near/far price/auction type)	realtime; cadence verified: 1s at 9:25–9:28 & 15:50–15:55, then 1s with near/far popu-lated	realtime	feed handler (row 1) + statistics schema for official cross prints	non-display (inside NTV)	PIT: far/near clearing prices unpopu-lated before 9:28/15:55 — explicit NULLs pre-threshold; official open/close prints land in statistics schema	W1	build-parser	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
7	nasdaq-halts-mcb-sscb	regulatory/administrative	Nasdaq Trader: Trading Halts / Circuit Breakers / Short Sale Circuit Breakers / Reg SHO Threshold (live nav links)	per-symbol halt/MCB/SSCB status events	intraday status; historical depth on pages (re-verify)	~minutes-class latency	web scrape + status watch	free public	PIT: keyed to halt-event timestamps only; never infer LULD state between published snapshots (action-note: connect +)	W0	build-parser	VERIFIED-UPSTREAM-DOC
8	nasdaq-daily-market-stats	free exchange stats	Nasdaq Trader: Daily Market Summary / Daily Market Files / Intraday Full Volume Summary (live nav links)	aggregate volume / market-share stats	daily files	refreshed daily	daily scrape	free public	PIT: file-date columns; snapshot-vs-revision flag discipline	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
9	nyse-integrated-feed-5venues	exchange-official feed	NYSE Integrated Feed via Pillar: NYSE / NYSE American / NYSE Arca / NYSE National / NYSE Texas; Pillar-equities common spec + Integrated-Feed spec v2.5 PDFs linked on page	order-by-order + trades + order imbalances + security status, in-sequence	realtime; sample data at ftp.nyse.com/Real Time Data Samples	realtime	Pillar binary feed handler	exchange license; NYSE pricing PDF (NYSE_Market_Data_Pricing.pdf) linked on page	PIT: in-sequence Pillar ordering is authoritative per venue; NBBO is a separate SIP product — never derive cross-venue NBBO from one venue (action-note: + build-parser)	W1	license-review	VERIFIED-UPSTREAM-DOC
10	nyse-openbook-ultra	exchange-official depth	NYSE Open-Book Ultra (real-time depth add-on) + historical mirror “TAQ NYSE Open-Book incl. Ultra & Aggregated files”	full venue LOB rebuild	realtime + T+1 flat files	realtime + T+1	feed handler + flat-file loader	exchange license (separate Open-Book fee product)	PIT: rebuild strictly from snapshot + Ultra events; never mix Aggregated and Ultra views in one book state (action-note: quote +)	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	nyse-taq-integrated-historical	licensed historical archive	TAQ per-venue Integrated flat files (NYSE / American / Arca / National / Texas): “depth of book, trades, order imbalances, stock summary, security status”	per-venue order streams	T+1 flat files	T+1	SFTP/flat-file walk	exchange license (NYSE TAQ subscription, venue-by-venue products)	PIT: schema-version per product; venue-scoped sequencing; batch freshness windows (action-note: + quote)	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
12	nyse-daily-taq-consolidated	licensed historical archive	NYSE Daily TAQ: “trades and quotes for all issues traded on NYSE, Nasdaq and regional exchanges” incl. NBBO, Master file, Quote/Trade Admin + CTA/UTP Admin Message files	consolidated day tape (trades + NBBO)	T+1 flat files	T+1	SFTP/flat-file walk	exchange license (NYSE TAQ)	PIT: SIP timestamps authoritative; NBBO as-of joins only from this tape’s coverage — not from merged venue feeds (action-note: (primary historical NBBO baseline))	W1	backfill	VERIFIED-UPSTREAM-DOC
13	nyse-taq-short-sales-short-volume	licensed historical archive	TAQ Group Short Sales & Short Volume (all five NYSE-side venues)	daily short-sale transactions + short volume summary	T+1	T+1	flat-file loader	exchange license	PIT: short flags are per-print venue attributes; complement (not substitute) for FINRA bi-weekly positions (row 36)	W2	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
14	nyse-taq-closing-prices-volume-summary	licensed historical archive	TAQ Closing Prices (O/H/L/Last + total volume at 16:00) + TAQ Volume Summary (share size, short sales, odd-lots, VWAP, by-market-center)	EOD aggregates	daily	daily	flat-file loader	exchange license	PIT: 16:00-close convention per venue; cross-check vs free EOD rows (rows 65–66); adjustment lineage explicit	W2	backfill	VERIFIED-UPSTREAM-DOC
15	nyse-market-data-pricing-tariff	license/metadata quote	NYSE Market Data Pricing PDF (public-docs/nyse/data/NYSE_Market_Data_Pricing.pdf) + “Pricing, Policies & Contracts” section	fee schedule	n/a	n/a	PDF fetch	free	PIT: fee schedules are version-dated — capture revision history per quote (action-note: quote +)	W0	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
16	nyse-arcabook	exchange-official (display, standalone)	ARCA standalone ArcaBook display product (venue feed covered in row 9; product page NOT fetched this session)	L2 display depth (ETF-heavy scope)	realtime	realtime (terminal-display class)	vendor/display terminal	exchange license	PIT: ARCX-only scope; do not use for non-ETF book studies	W3	verify	UNVERIFIED-CANDIDATE
17	nyse-auction-imbalance	administrative/TAQ	NYSE/American/Arca Order Imbalances — “buy and sell imbalances sent at specified intervals during auctions throughout the trading day” — + live imbalance msgs inside row 9	auction-window imbalance snapshots	realtime + T+1	realtime + T+1	feed handler + flat file	exchange license	PIT: only auction-window intervals are defined events; NULL pre-window (same discipline as row 6)	W2	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
18	cboe-us-equities-feeds	exchange-official	Cboe Data Services (“Exchange Data and Access” nav) + DataShop All-Access APIs; cboe.com equities product pages 404 this session — re-discover URL structure	BBO/depth + trades (BZX/BYX/EDGA/EDGX/C2)	realtime; historical via DataShop	realtime	binary feed handler/API — formats re-verify	exchange license (quote)	PIT: per-venue feed families distinct; cross-fusion only at Cboe-licensed level (action-note: quote +)	W2	verify	UNVERIFIED-CANDIDATE
19	cboe-datashop-us-equities	vendor archive (exchange-owned)	Cboe DataShop (datashop.cboe.com) — asset-class filter “Equities, options, ETFs, indices, futures, crypto”; data types Historical/tick/intraday/daily	per-product tick/intraday/daily sets	per-product (decades-dependent; verify)	EOD/intraday downloads	SFTP / my-account downloads / All-Access APIs	Cboe license (no wholesale redistribution)	PIT: per-product version tags; “Calcs” (IV/Greeks) are vendor-computed — recompute or label derived (action-note: verify +)	W2	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
20	cboe-open-close-volume-summary	vendor archive	Cboe Open-Close Volume Summary (DataShop trending product: EOD + 10-min intervals by capacity Customer/Pro-Customer/Broker/Firm/MM)	venue aggregates	T+1	T+1	DataShop	Cboe license	PIT: capacity buckets are exchange classifications — never re-map into own buckets	W3	verify	VERIFIED-UPSTREAM-DOC
21	cboe-livevol-pro	retail platform	Cboe LiveVol (DataShop Platforms > LiveVol) — near-real-time monitoring platform	UI/screens	recent tick windows (DataShop archive is the store)	delayed/NRT	web platform	Cboe subscription	PIT: platform claims not PIT-consistent; monitor-only, archive via row 19	W3	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
22	cme-mdp3-equity-index-futures	exchange-official feed	CME Globex MDP 3.0 (ES/NQ/RTY/YM + micros): multicast + Cloud MDP (Google Pub/Sub)	MBO/MBP-10 snapshots/trades/bbo/ohlcv	realtime; history via Databento or Databento GLBX mirror (Jun 2010→ claim)	realtime	MDP3 handler or Databento library	exchange license (MDP subscription; display/non-display tiers)	PIT: exchange-server timestamps; roll/settlement schedules PIT; index-futures vs cash microstructure adjacency explicit (action-note: quote +)	W2	license-review	SURVEY-EVIDENCE
23	cme-mdp3-other-classes	exchange-official feed	same MDP3 platform for Rates/FX/Energy/Ag/Crypto (Pub/Sub delivery verified prior session)	per-class depth/trades	realtime + historical via mirrors	realtime	same as row 22	exchange license	PIT: (a) each class lives in a distinct Globex product family; (b) message cadence/depth conventions differ by class — keep per-class pipelines	W3	quote	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
24	cme-datamine-history	licensed historical archive	CME DataMine self-service historical	tick/EOD for equity-index futures	range by dataset (quote)	T+1 batch	catalog flat-file delivery	CME license	PIT: CME dataset revision IDs vs “fix” annotations	W3	quote	SURVEY-EVIDENCE
25	databento-glbx-mdp3	mirror-of-exchange	Databento GLBX-MDP3 dataset (docs catalogue)	MBP-10/TBBO/Trades/OHLCV schemas	realtime + historical (2010→ claim)	realtime + batch	Databento library; boundary truth via get_dataset_range metadata API	Databento license (per-GB self-service)	PIT: range check via API metadata, not marketing claims; ranges vary by schema	W2	verify	VERIFIED-UPSTREAM-DOC
26	iex-tops	exchange-official	IEX Exchange Market Data — TOPS section (top-of-book + last) verified on live page	L1 snapshots	realtime; history via HIST	realtime	IEX-TP feed handler	exchange license (fee class re-verify)	PIT: IEX-only BBO (~2.5% tape share gate); never conflate with SIP	W1	connect	VERIFIED-UPSTREAM-DOC
27	iex-deep	exchange-official	IEX DEEP (book depth + trade break + auction info) verified on live page	book depth + auction indicators	realtime	realtime (exchange-native)	IEX-TP feed handler	exchange license (separate DEEP product)	PIT: DEEP auction channel reflects IEX-only auctions — not NYSE/Nasdaq cross state	W2	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
28	iex-hist-t1-pcap	free public archive	IEX HIST (T+1 pcap down-loads; IEX-TP ver-sioned; e.g. 20170515_IEXTP1_DEEP1.0.pcap.gz); Historical Data Terms of Use bound	pcap TOPS/DEEP/TRADES per day	T+1; archive coverage from 2017-05-15 example (map exact start)	T+1 batch	IEX-TP pcap decoder	free w/ terms-of-use	PIT: decoder patched to spec version named per-file; enter-date cata-loging before any backfill (action-note: backfill +)	W1	build-parser	VERIFIED-UPSTREAM-DOC
29	iex-cloud-retired	EXCLUDE (dead)	IEX Cloud shut 2024-08-31 (prior session)	n/a	n/a	n/a	n/a	n/a	migration gate: repoint con-sumers to rows 26/28 (IEX exchange direct) or aggrega-tor rows 72–74 (action-note: + unlink configs)	W0	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
30	cta-consolidated-quotes-tapeab	consolidated-tape	CTA SIP (Tapes A/B) — regime migrating to unified CT Plan	SIP consolidated quotes (NYSE-listed space)	realtime	realtime (SIP latency class)	SIP subscription handler	license via CTA Plan → CT Plan (row 34)	PIT: SIP timestamps; SIP-BBO vs direct-feed gap must be latency-bucketed, never merged silently (action-note: quote +)	W2	license-review	VERIFIED-UPSTREAM-DOC
31	utp-consolidated-quotes-tapec	consolidated-tape	UTP SIP (Tape C; Nasdaq-listed)	SIP consolidated quotes	realtime	realtime (SIP latency class)	SIP subscription handler	license via UTP → CT Plan	PIT: SIP quotes vs Nasdaq Basic BBO (row 3) divergence audit	W2	quote	VERIFIED-UPSTREAM-DOC
32	consolidated-last-sale-tapesabc	consolidated-tape	consolidated-last-sale/trades, Tapes A/B/C (CTA-side + UTP-side operators)	tape prints	realtime	realtime (SIP latency class)	SIP subscription handler	exchange/SIP license	PIT: consolidated stream includes off-exchange (S-segment) prints — flag segments, never strip S-prints silently	W2	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
33	finra-otc-trf-tas	consolidated tape	FINRA OTC trade reporting (TRF) + FINRA daily trade/quote file family (naming inconsis-tent across pages this session)	OTC prints + daily file deriva-tions	realtime (vendors) + daily files	realtime + T+1	licensed feed + daily-file ingest	exchange/vendor license (quote)	PIT: Form-T flags for late-reporting; OTC marker columns pre-served (action-note: quote +)	W2	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	us-ct-plan-transition	consolidation-ea admin	New unified CT Plan (DataCT LLC administrator) replacing CTA+UTP: migration portal Sep 1 2026, licensing from Oct 1 2026, agreements due Mar 1 2027, operative Apr 1 2027; “no SIP technology changes required”	regulatory regime	n/a	n/a	regime-transition tracker	mandated plan	PIT: contractual regime switch (feeds unchanged) — re-quote every SIP-era agreement before Mar 1 2027	W0	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
35	sip-playback-vendors	vendor class	historical consolidated tape playback via licensed SIP vendors (Nasdaq Trader “Market Data Vendors” list nav route)	tape replay	historical	vendor-delivered (latency n/b)	vendor playback API / flat feeds	SIP license redistribution trace	PIT: replay must preserve original tape sequence/thrust	W3	verify	UNVERIFIED-CANDIDATE
36	finra-equity-short-interest	free public file	FINRA Rule 4560 bi-weekly short interest: twice-monthly settlement (15th + EOM); published 7th business day after settlement; 5y rolling grid + historical files + Equity API	per-issue short positions/ratios	5y rolling grid + historical files	T+7 business days (publication lag)	interactive grid / CSV files / API	free (non-commercial basis stated)	PIT: as-of-join on publication date (T+7bd), settlement date must never be read as availability; Revision Flag tracking; no intra-cycle imputation	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
37	finra-ats-transparency-darkpool	free public file	FINRA OTC Transparency portal (fetched shell) + ATS Transparency catalog entries inside No-Fee catalog	weekly/monthly ATS pairs incl. off-exchange (dark-pool) aggregates	rolling archive depth per item (re-verify)	weekly/monthly publication cadence	portal files/APIs	free	PIT: weekly/monthly roll semantics; pairs-universe rebinding per period; revision re-download	W0	connect	VERIFIED-UPSTREAM-DOC
38	finra-reg-sho-threshold-lists	free public file	FINRA Reg SHO threshold-list page + Nasdaq Trader RegSHOThreshold / ShortSaleCircuitBreaker (nav-verified)	daily symbol threshold lists	site archive depth (re-verify)	daily	scrape (CSV/TXT)	free	PIT: union-with-dual-exchange mapping; gates-join with rows 7 + 36; threshold flags only for flag dates	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
39	finra-daily-short-sale-volume	free public file	FINRA “Daily Short Sale Volume” per-venue daily files (well-known file routes; catalog page not landed this session)	daily venue short-sale volume by symbol	daily (start year claims re-verify)	daily file cadence	file ingest	free	PIT: file cadence + revision practice re-verify; reconcile against ATS Trans-parency aggre-gates (row 37)	W1	connect	UNVERIFIED-CANDIDATE
40	finra-no-fee-catalog-sweep	free public file	FINRA Data Catalog (60 items; No Fee filter; Equity Data category)	sweep/registry	file-registry	catalog-refresh pace (re-verify)	catalog crawl/API	free	PIT: catalog items re-name/deprecate — enu-merate every Equity-Data slug and pin statuses	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
41	databento-xnas-itch	mirror-of-exchange	Databento dataset XNAS.ITCH (docs/datasets/xnas-itch fetched): normalized Nasdaq TotalView-ITCH with MoldUDP64 message layer documented + free sample downloads	full schema family (trades / mbp-10 / mbp-1 / tbbo / imbalance / stats / definition)	product-range claims cover 2010s-era onward; exact per-schema bounds via get_dataset_range API (verify)	realtime (Live) + historical batch	Databento lib → Delta/parquet landing, batch by explicit date-window	Databento license (per-GB self-serve + free credits)	PIT: ts_event vs ts_recv dual-timestamps kept; spec-version per day; raw Nasdaq ITCH files (row 4) stay ground truth alongside the mirror (action-note: backfill +)	W1	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
42	databento-nyse-pillar-family	mirror-of-exchange	Databento venue datasets `XNYS.ITCH` / `XASE.ITCH` / `ARCX.ITCH` (+ `XNYS.BASIC` display variant) listed in venue cata-logue	MBO / tbbo / trades per venue	per-dataset bounds via get_dataset_range (verify)	realtime + batch	same row-41 pipeline forked per venue	Databento license	PIT: Pillar in-sequence venue se-mantics (row 9); never merge NYSE-side venues into one book state (action-note: (phased by venue))	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
43	databento-definitions-schema	mirror-of-exchange (reference/security master)	Databento `definition` schema + DEFINITIONS datasets: security-level corporate actions (splits, dividends, name changes, listings) maintained venue-side	point-in-time definition snapshots per symbol	full-history per venue dataset (verify)	batch	definition snapshots keyed (symbol, eff_ts) → SCD2 spine	Databento license	PIT: effective-dated only — no falling-forward joins; predecessor/successor chains from venue-sourced rows; never mix lookups across venue datasets in one master row (action-note: (security-master spine))	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	databento-equs-mini	mirror-of-exchange (display)	Databento `EQUS-MINI` dataset (Nasdaq Basic w/ NLS Plus mirror; row-3 product) listed in catalogue	L1 quotes + last sale	dataset bounds via get_dataset_range (verify)	realtime + batch	Databento lib	Databento license	PIT: Nasdaq-listed scope only — consolidated NBBO still needs Tape C companion (row 31); display-class entitlement flow-through (action-note: range + quote)	W2	verify	VERIFIED-UPSTREAM-DOC
45	databento-equs-summary	mirror-of-exchange (stats)	Databento `EQUS-SUMMARY` dataset listed — per-session summary stats for the Nasdaq tape	OHLCV/VWAP per-class session bars	per-dataset bounds (verify)	realtime + batch	Databento lib → daily stats tables	Databento license	PIT: session-stat definitions per exchange calendar; derived-aggregation label pinned per run	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	databento-imbalance-schema	mirror-of-exchange	Databento `imbalance` schema documented — normalized auction-imbalance events (NOII family)	auction-window imbalance events	rides rows 41–42 entitlements	realtime + batch	mapper → auction-imbalance tables (row-6 schema)	Databento license	PIT: far/near fields empty before 9:28 / 15:55 ET must map to NULL, not 0 (row-6 rule); auction-type discrimination per event (action-note: build-mapper)	W1	connect	VERIFIED-UPSTREAM-DOC
47	databento-trades-schema	mirror-of-exchange	Databento `trades` schema documented — normalized prints with per-venue trade conditions	last-sale prints	per-venue bounds (verify)	realtime + batch	prints ingest + dedup on ts_recv	Databento license	PIT: venue-only prints ≠ consolidated SIP tape (row 32) — off-exchange S-prints absent by construction; label “not consolidated tape” in schema	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
48	databento-mbp-10-schema	mirror-of-exchange	Databento `mbp-10` schema documented — 10-level price-ladder snapshots + tick deltas	10-deep MBP	widely covered era (futures 2010s; equities ranges per dataset — verify)	realtime + batch	ladder rebuild (snapshot + delta merge)	Databento license	PIT: L3/order-book truth only where raw ITCH/Pillar rows (rows 1/4/9) are loaded; mbp-10 labeled as the coarser derived view	W1	backfill	VERIFIED-UPSTREAM-DOC
49	databento-mbp-1-schema	mirror-of-exchange	Databento `mbp-1` schema documented — L1 quote-change stream with derived best bid/offer	L1 ticks	per-source dataset	realtime + batch	quote-tick ingest	Databento license	PIT: derived view of MBO — label derived; audit against exchange-native L1 (rows 3/26/44) for drift	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
50	databento-tbbo-schema	mirror-of-exchange	Databento `tbbo` schema documented — trades with BBO-at-trade-time	BBO + prints	per-source dataset	realtime + batch	tbbo interval tables	Databento license	PIT: join key = (ts_event, price) pinned per run; treat as derived join — never present as exchange-defined BBO	W2	connect	VERIFIED-UPSTREAM-DOC
51	databento-bbo-1s-schema	mirror-of-exchange	Databento `bbo_1s` schema documented — 1-second interval best bid/offer	1s BBO intervals	coverage era claims (verify per venue)	realtime + batch	bbo interval tables	Databento license	PIT: 1s roll rule pinned per run; free-credit sample per dataset before any paid backfill	W2	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
52	databento-consolidated-nbbo	derived-consolidated	Databento consolidated-NBBO-class product (legacy connbbo-style range; current catalogue status re-verify — product pages not fetched this session)	consolidated-NBBO	legacy range era claims (2010s)	historical only (claims)	Databento lib	Databento license	PIT: labeled derived-vs-official-SIP; reconcile against NYSE Daily TAQ NBBO (row 12) before trusting any gap statistics (action-note: status)	W2	verify	UNVERIFIED-CANDIDATE
53	databento-stats-schema	mirror-of-exchange	Databento `stats` schema documented — session stats (open/close, high/low, volume, first/last)	per-symbol session stats	rides venue datasets	realtime + batch	daily-stat tables	Databento license	PIT: stats = exchange-defined session frames, not self-calculated; cross-check vs official daily files (rows 8/14)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
54	databento-live-streaming	vendor streaming	Databento Live API (realtime streaming; per-exchange latency-class claims in docs)	all schemas in realtime	stream (no archive)	realtime	live consumer → event tables; replay-vs-stream compare harness	Databento license (live per-venue subscription)	PIT: ts_recv-vs-ts_event gap monitoring from day 1; per-venue watermarks; capture raw messages before normalization (action-note: connect (parallel to row-41))	W1	backfill	VERIFIED-UPSTREAM-DOC
55	databento-ocea-blue-ocean	mirror-of-exchange	Databento OCEA dataset id listed (Blue Ocean overnight venue; product-naming per venue list — re-verify)	overnight-session orders/trades	venue-era start (2023 claims; re-verify)	realtime + batch	Databento lib	Databento license	PIT: overnight session ≠ regular-session continuity — never bridge across the open/close session gap silently	W3	verify	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
56	databento-cboe-memx-family	mirror-of-exchange	Databento venue dataset ids: MEMX, BATS (BZX), BATY (BYX), EDGA, EDGX, XCIS in venue cata-logue	per-venue MBO/trades/stats	per-venue via get_dataset_range	realtime + batch	Databento lib forked per venue	Databento license	PIT: Cboe book-fusion rules are not defined by Databento — stay per-venue until an exchange license (row 18) defines fusion (action-note: per venue)	W3	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
57	kibot-tick-bidask	vendor archive	Kibot Buy.aspx (fetched): tick data with B/A from 2009; tier bundles $990 / $2,400 / $4,800/yr (vendor claims); all-stocks tick+B/A $9,000/yr	scaled/raw tick	2009→ (tick+B/A)	T+1 batch	SFTP/HTTP flat files	subscription (vendor claims)	PIT: download-date + revision-date columns; “scaled” tick is vendor-adjusted — re-derive from raw where possible; last-quote/settlement defini-tions per Kibot specs (action-note: quote +)	W2	license-review	VERIFIED-UPSTREAM-DOC
58	kibot-1min-bars	vendor archive	Kibot 1-min bars since 1998; all-stocks 1-min $3,000/yr (vendor claim)	1-min bars	1998→	T+1 batch	flat-file loader	subscription	PIT: corporate-action adjust-ment method fixed per file — version it; validate vs free EOD rows (65–66) (action-note: min-bars)	W2	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
59	lobster-academic-lob	academic	LOBSTER academic limit-order-book datasets (per-day LOB slices + message file pairs; JS-blank site this session — re-verify domain & access route)	L1–L10 LOB slices + message file	2010s era (re-verify)	offline batch	flat files + loader via academic creden-tials	academic license (no redis-tribution)	PIT: not a continu-ous feed — venue/window filters per study design; message-file lineage must stay paired with its level-file	W3	verify	UNVERIFIED-CANDIDATE
60	tickdata-inc-archives	vendor archive	Tick Data Inc. (tick-data.com fetched): TickWrite 7 desktop, Web-based, API access, Deep History; parent OneMar-ket-Data/OneTick	tick (+ B/A) + delta/interval products	1980s-depth claims for equities (verify per product)	T+1 batch	TickWrite/API flat delivery	license (quote)	PIT: file-version + re-download policy per product; venue-vs-consolidated pair columns pre-served	W2	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	nanex-nx2-tick	vendor archive	Nanex .nx2 tick archives (order-book + trade frames)	frame files	2010s era claims (re-verify)	offline batch	.nx2 decoder	license	PIT: proprietary-format vendor-lock risk; frame-timing defini-tions re-verify before any latency work	W3	verify	UNVERIFIED-CANDIDATE
62	onetick-platform	vendor platform	OneTick / OneTick Cloud (OneMar-ketData) — tick-analytics + query platform; parent company of Tick Data (row 60)	platform (query/stream)	feeds via Tick Data + own	platform	OneTick query/stream API	license	PIT: as-of-time query primi-tives — keep compute-in-platform vs export lineage explicit in metadata	W3	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
63	wrds-taq	academic mirror	WRDS-hosted NYSE TAQ (Wharton): academic redistribution of the TAQ family incl. Daily TAQ	consolidated day tape	coverage per WRDS catalog (verify)	T+1 pace	WRDS table extract (SAS/CSV)	academic subscription (WRDS + exchange pass-through)	PIT: WRDS vintage tagging per file; revision flags tracked — never merge vintages silently (action-note: (institutional))	W2	license-review	SURVEY-EVIDENCE
64	crsp-us-equities-hist	academic/licensed	CRSP US Stock Databases (daily/monthly/stock master/delisted returns)	daily + monthly + security history	1926→ (vendor claim)	EOD	CRSP access / WRDS extract	license (CRSP subscription)	PIT: CRSP dataset-vintage editions; delisted-return series is the canonical survivorship fix; price-adjustment-factor chain preserved (action-note: + backfill)	W1	license-review	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	stooq-free-eod	free web CSV	Stooq public CSV end-points (daily EOD; site not fetched this session)	daily EOD	long history (verify span)	daily scrape	HTTP CSV fetch	free / personal-use terms (verify)	PIT: unlicensed-provenance flag — cross-check tier only, never system-of-record; corporate-action adjust-ment basis unknown (re-verify) (action-note: (cross-check tier))	W0	connect	UNVERIFIED-CANDIDATE
66	alphavantage-eod	retail API	Alpha Vantage TIME_SERIES_DAILY(_ADJUSTED) end-points (free-tier caps)	daily EOD	decade-scale (verify per symbol)	daily API	REST	free key + paid tiers (claim)	PIT: adjusted-vs-raw series split explicit; provider outage / rate-limit fallback needed; aggrega-tor prove-nance = licensed-display risk	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
67	polygon-massive-realtime	retail/vendor API	Polygon.io — footer rebrand verified: “© Massive.com, Inc.” (domain/API continuity migration in progress; endpoints still served)	realtime aggregates/WS/REST	per-product (Aggregates 2015→ era claims; verify per endpoint)	realtime + T+1	REST + WebSocket SDKs → event tables	subscription (free tier + paid)	PIT: rebrand = continuity risk — pin base-URL + API-version docs snapshot before building; data plane still Polygon (action-note: (after domain continuity check))	W1	connect	VERIFIED-UPSTREAM-DOC
68	polygon-trades-v2	retail/vendor API	Polygon Trades V2 endpoint (tick prints with exchange/condition codes)	prints	per-product claims (re-verify post-rebrand)	realtime + historical	paginated GET + WS	subscription	PIT: vendor-merged prints ≠ consolidated tape — exchange-id fields must stay; conditions mapping versioned (action-note: (cross-check vs row 47))	W2	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
69	polygon-quotes-nbbo	retail/vendor API	Polygon Quotes V2 / snapshot end-points (NBBO-class)	best bid/offer	per-product claims (re-verify)	realtime + historical	GET + WS	subscription	PIT: vendor NBBO ≠ SIP NBBO — latency-bucket compari-son vs row 12; never merge silently	W2	connect	SURVEY-EVIDENCE
70	polygon-aggregates-historical	retail/vendor API	Polygon Aggre-gates (minute/day bars, adjusted multipli-ers)	bars	2015→ era claims (verify per ticker)	batch	GET grouped/windowed	subscription (lower tier caps)	PIT: ad-justment multipli-ers ver-sioned; span boundary from pag-ination meta-data, not wall-clock assump-tions	W2	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
71	polygon-options-chains	retail/vendor API	Polygon Options (chains/snapshots; options aggregates claims)	chains + option aggregates	per-product claims	realtime + batch	REST	subscription	PIT: options depth lives in the options-domain census (row 80) — surface only chain-identity fields for equities work (action-note: defer)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
72	alpaca-marketdata-free	retail/vendor-API	Alpaca Market Data API (docs root fetched: Market Data API / Historical / WebSocket sections; /reference page 404 → use docs root + guides)	bars/quotes/trades + streams	free-tier = IEX plan claims; paid = full-SIP claims	realtime + historical	REST + WS SDK	free/paid plans (vendor)	PIT: free tier ≈ IEX-only (~2.5% tape share) — provenance flag mandatory before any tape calc; paid plan = SIP-sourced (verify at contract) (action-note: (free tier))	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
73	tiingo-flat-rate-eod	retail/vendor API	Tiingo (pricing/coverage pages fetched): $30/mo ind + $50/mo commercial flat-rate claims; “IEX Feed”; 30+ yr EOD claims; 10k req/hr	daily EOD + intraday	30+ years (vendor claim)	EOD/API	REST + daily CSV	flat-rate subscription	PIT: IEX-feed provenance disclosure (row-26 gate) — secondary feed only; 30-yr span is a vendor claim — verify per symbol before system-of-record use	W1	connect	VERIFIED-UPSTREAM-DOC
74	finnhub-free-api	retail vendor API	Finnhub REST (quotes/candles/etc.; not fetched this session — scope + rate caps re-verify)	quotes/candles per endpoint	claims (re-verify)	realtime-ish	REST	free-tier claims	PIT: aggregator provenance-gate class — unknown sourcing, unlicensed-display risk (same discipline as rows 72/73 flags)	W3	verify	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
75	twelvedata-free-api	retail vendor API	Twelve Data REST API (claims not fetched this session)	quotes/candles	per-endpoint claims	realtime-ish	REST	free tier + sub-scription	PIT: same aggrega-tor provenance-gate class as row 74	W3	verify	UNVERIFIED-CANDIDATE
76	barchart-on-demand	retail vendor API	Barchart OnDe-mand (vendor claims; not fetched)	quotes/history/f	per-endpoint claims	delayed/rt claims	REST/XML	subscription tiers	PIT: vendor-hedged data (delayed defaults) — verify feed class per endpoint before any PIT use	W3	verify	UNVERIFIED-CANDIDATE
77	ibkr-historical-tws	retail/broker API	Interactive Brokers TWS API historical windows (pacing limits; maturity-of-data rules)	bars/prints by window	per-resolution-capped lookback (maturity rules)	on-demand queries	TWS socket API	brokerage account license (display, non-redistribution)	PIT: pacing limits throttle backfills; depth is per-resolution-capped — log request-window metadata per call (action-note: + proto-type)	W2	verify	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
78	sec-13f-holdings	EXCLUDE (out-of-scope)	SEC EDGAR 13F institutional holdings filings	quarterly positions	long	filings lag	—	public	exclusion: filings/ownership domain, not market-data tape — handled in a separate holdings census; do NOT fold into this file's W0/W1 builds (action-note: move to holdings census)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
79	yfinance-unofficial-scrapers	EXCLUDE	yfinance + similar unofficial scrape routes (Yahoo-class end-points)	mixed	mixed	scrape	—	unlicensed/unstable	exclusion: ToS-violation + silent schema drift + non-PIT adjust-ment lineage — banned from any ingest allowlist or gold-layer feed (action-note: block in ingest allowlist)	W3	connect	UNVERIFIED-CANDIDATE
80	options-domain-pointer	POINTER	OPRA feeds (Databento 0PRA id among venue cata-logue) + Cboe DataShop options sets (rows 18–19 surface only identity here)	options domain	—	—	—	pointer	out-of-scope for equities mi-crostruc-ture — full options census main-tained sepa-rately (action-note: create options-domain census)	W3	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
81	nasdaq-nordic-intl-pointer	POINTER	Nasdaq Nordic + international equities venues (prior-census scope note)	intl eq	—	—	—	per-venue intl license	out-of-scope: US equities file; Nordic datasets enter via the international census (venue-local + delayed-data rules differ) (action-note: maintain in intl census)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	LSE Data Shop — End-of-Day Summary	historical-archive	londonstockexchange.com Data Shop → docs.londonstockexchange.com HDP_EOD_Summary spec [1]	EOD OHLCV per in-stock cash market, depth sold per window	LSE+Turquoise from Feb-2023	EOD next-session	file purchase + platform download (Data Shop self-service)	commercial license, internal use	exchange-issued OHLC; no PIT ad-justment — corporate-action join needed (action-note: spec already ingested — write COTAHIST-style loader for EOD Summary format)	W2	connect	VERIFIED-UPSTREAM-DOC
2	LSE Data Shop — tick (time & sales)	historical-archive	LSE Data Shop factsheet [2]	tick T&S	LSE from Jan-2004; Turquoise from Feb-2023	N/A (delivery file)	file purchase (CSV bundles, per-period)	commercial license	raw tape, as-traded codes; needs Turquoise currency-table join (action-note: price-sample, then decide tick de-pendency vs EOD-only GFI UK pillar)	W3	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	LSE Data Shop — best prices / full order book	historical-archive	Data Shop factsheet [2]	L1 best prices from Apr-2003; L2 full rebuild from Oct-1997	multi-decade	N/A	file purchase	commercial license	as-traded order book; symbol-consolidation join to LSPD events (action-note: only if GFI UK branch needs mi-crostruc-ture)	W3	connect	VERIFIED-UPSTREAM-DOC
4	LSE Data Shop — corporate actions + reference series	historical-archive	Data Shop cat-alogue (self-service filter per factsheet [2])	event-level	archive sold per window	N/A	file purchase	commercial license	vendor PIT events; reconcile against LSE notices (action-note: catalogue-sample; s corporate-action spine for row 1)	W2	backfill	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	LSE London Data Shop — news datasets (RNS archive product)	news-archive	Data Shop catalogue	notice-level	sold per window	N/A	file purchase	commercial license	RNS reconciliation against free RNS web UI (action-note: confirm exact RNS-archive SKU before pricing)	W2	connect	UNVERIFIED-CANDIDATE
6	LSE securities master + LSES reference files	reference-master	londonstockexchange.com market-data pages + LSE reference-data specs	instrument-level	current + full/delta	daily	registered download / vendor license	licensed (redistribution restricted)	ISIN/tradable-code joins; daily deltas must ack before replace (action-note: fetch LSE reference-data spec PDF; wire delta poller)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
7	LSE RNS feed (live)	news	londonstockexchange.com news / RNS	exchange real-time	web archive recent years	near-realtime site	web scrape (unofficial) or RNS Select subscription	site TOS grey; ⚠ decisions pending	dedupe by announcement id; timezone normalisation (action-note: if Data Shop news SKU exists (row 5) prefer it; scrape only as fallback)	W1	connect	UNVERIFIED-CANDIDATE
8	Turquoise MiFID II files	post-trade-equity	Turquoise Reference Repository “MiFID II Files” (verified in site nav)	per-venue transparency	rolling web	batch	anonymous web files	free (MiFIR Art.13/14 compliance data)	venue-issued; 15-min disclosure rules apply downstream (action-note: wget mirrors once Euronext EU-equivalent RTs land)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
9	LSPD — London Share Price Database	historical-archive	London Business School via WRDS	daily/monthly	1955→ (sample), daily series from Jan-1984 [4]	academic refresh irregular	institutional WRDS login	academic license, no redis-tribution	LBS research-grade with delist-ing/capital returns; best sur-vivorship spine for UK (action-note: partner affiliation or ; else mimic from rows 1+4)	W2	license-review	SURVEY-EVIDENCE
10	TRADEcho (LSE trade re-porting) data — Data Shop scope	post-trade-equity	Data Shop cat-alogue (fact-sheet [2] names TRADE-cho as included venue)	transaction-level	window purchase	N/A	file purchase	commercial license	SIs+venues pot; good PIT vs STS/ATS reports (action-note: only if ADR/MiFID transaction-pool refresh needed)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	Deutsche Börse MD+S — CEF realtime feeds (CEF-XTF Xetra / CEF-EFF Eurex)	realtime-tape	mds.deutsche-boerse.com CEF product sheet [5]	tick L1/L2	realtime config	realtime	licensed API connectivity	commercial license (redistribution contract-gated)	contract-gated; feed contract must pin bandwidth windows (action-note: only if realtime DE branch funded)	W3	connect	VERIFIED-UPSTREAM-DOC
12	Deutsche Börse MD+S — historical market data (Xetra/Eurex/futures tick+1-min)	historical-archive	MD+S historical catalogue + Host File Service [5]	tick–1min	Xetra/Eurex full session depth, multi-decade	batch	file server/host delivery	commercial license	seller-side data; PIT join to reference data products (action-note: for EUR-vintage Xetra EOD + 1-min: GFI DE pillar leapfrog)	W3	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
13	Deutsche Börse Data Shop / Statistics down-loads	EOD-stats	cashmarket.deutsche-boerse.com Market Data + mds popular topics ([5], [6])	EOD	Xetra/FSE since 1997, monthly series deep	EOD after session	anonymous web PDF/CSV or Prod-uct+Services cata-logue	free statistics (redistri-bution re-stricted)	official turnover/stats; no price PIT (action-note: mirror monthly Xetra statistics CSVs)	W0	connect	VERIFIED-UPSTREAM-DOC
14	Eurex historical time series (TSS bundles + settle-ment statistics)	historical-archive	eurex.com statis-tics/menu (reach-able via MD+*, [5])	EOD after session	multi-decade	EOD	licensed file	commercial license	per-contract fixes; roll chains must map to expiry calendar (action-note: iff GFI DE-futures row is built)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
15	SIX Swiss Exchange — trading data + MiFID II export drive	EOD-intraday	six-group.com market-data tree (MiFID II flag export facility)	EOD+tick licence	multi-decade	EOD+	license portal + SFTP	commercial license	SIX MiFID flag per instrument drives PIT classification (action-note: request SIX data-drive samples; MiFID-flag semantics vs FITRS)	W2	verify	UNVERIFIED-CANDIDATE
16	BME — trading data files + BME Data Hub	EOD	bme.es market-data (CSV,“BBDD” files)	EOD	1990s→	EOD	registered file delivery	commercial license (redistribution restricted)	Spanish-market EOHLC; separate BME corporate-events files for CA joins (action-note: archive-fetch converter for BME CSV layout)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
17	BME Latibex (IBEX→LatAm cross-listings)	EOD	bme.es Latibex pages	EOD	2001→	EOD	same channel as row 16	commercial license	duplicates BMV tickers — dual-listing dedupe rule (action-note: skip unless GFI cross-listing branch opens)	W2	connect	UNVERIFIED-CANDIDATE
18	Euronext — historical market data files (per country)	historical-archive	euronext.com data-offering (portal behind Connect login; per-Paris/AMS/Milan/Lisbon/Oslo)	EOD→tick	multi-market multi-year	batch	Connect2 portal or vendor	commercial license	sold per-market; solve PIT via Euronext CA files (action-note: obtain Paris vendor first; portal creden-tials check)	W2	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
19	Euronext — company regulated news archive	news-archive	euronext.com/en/products/equities/company-news-archive (live nav)	per-issuer level	to-inception web archive	near-daily	web scrape/API	free, TOS-restricted	announcement dedupe vs prior EU rows (action-note: scrape feed; map to GFI news schema)	W0	connect	VERIFIED-UPSTREAM-DOC
20	Euronext Dublin — Daily Official List (DOL)	EOD-reference	euronext.com/en/markets/dublin/daily-official-list (live nav)	instrument/market-level official list	historic series	EOD	web	free-ish, terms per page	official DOL is the PIT marker for Dublin-listed instruments (action-note: fetch DOL PDF; add to reference spine)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
21	live.euronext.com — jerboa quote/ENU API	real-time-quotes	live.euronext.com end-points (public keys)	15-min delayed spot	online only	15-min	REST JSON	free delayed (MiFIR 15-min rule)	quoting end-points and dis-closure rules differ per market (action-note: poller + cache; respect per-market disclo-sure)	W1	connect	UNVERIFIED-CANDIDATE
22	Nasdaq Nordic + Baltic — Histori-calView	historical-archive	nasdaq.com Nordic-Baltic Histori-calView product [7]	EOD daily files	eq/derivatives/ETF/index, multi-market multi-decade	EOD	paper order + file delivery	commercial license, **internal-use only clause** [7]	direct exchange-issued; PIT from Nasdaq CA files (action-note: signed agree-ment then)	W2	backfill	VERIFIED-UPSTREAM-DOC
23	Nasdaq Nordic — File Delivery Service (SFTP subscrip-tions)	EOD-intraday	nasdaq.com/product-levels/data/nordic-baltic/nordic-file-delivery-service [7]	file-based daily	trailing	EOD	SFTP sub-scription	commercial license	file-cadence SLA; missed-file alarm required (action-note: only if Nordic branch funded)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
24	Nasdaq Nordic free nominal-data files (globe-newfiles path)	EOD-free	globenewfiles.nasdaqomx.com public dirs (nominal data)	EOD blocks	recent files	EOD	anonymous web	free nominal (license posted per file)	re-verify robots + file-naming tem-plates (action-note: probe directory + build sched-uler)	W1	connect	UNVERIFIED-CANDIDATE
25	GPW — offered value-added data (CSV/XLS daily after close)	historical-archive	gpw.pl value-added-data-and-indicators page — CSV/XLS delivered daily after session via mail/SSL [8]	EOD full tape deriva-tives/instruments	archive sold per date window	EOD	e-mail/SSL file delivery	licensed (some free statistical rows, some paid)	daily session issue; PIT CA joins by GPW CA files (action-note: hook SSL delivery; reuse COTAHIST-style OHLC loader across venues)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
26	GPW — free daily trading-data files (session results archive)	historical-archive	gpw.pl/gpw-statistics + gpw.pl/market-data pages [9]	EOD blocks	multi-year	EOD	anonymous web files	free, redistri-bution restricted	official figures; limited per-instrument daily files to-date — confirm span (action-note: fetch daily session files; confirm earliest)	W0	backfill	SURVEY-EVIDENCE
27	GPW NewCon-nect + Catalyst files	EOD	gpw.pl market-data tree (NewCon-nect, Catalyst, TBSP)	EOD	mixed	EOD	same channel as row 25/26	licensed/free mix	separate listing-source registry (action-note: subscribe once cascade from row 25 is running)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
28	GPW Benchmark — WIBOR/WIBID historical CSV	reference-rates	gpwbenchmark.pl/en-dane-historyczne ([10]: SSL/mail CSV delivery 9:00 next business day)	daily fixing	full benchmark history	T+1	e-mail/SSL CSV	licensed benchmark (BMR-regulated)	rate PIT: use fixing-issued csv only, no re-cast (action-note: register supply channel; loader for fixing series)	W1	connect	VERIFIED-UPSTREAM-DOC
29	Prague SE (PSE) — trading data `dll` historical files	EOD-cz	pse.cz trading-data pages	EOD	multi-decade	EOD	registered/anonymous web	free-ish	Czech-language file specs; anchor legacy ticker codes to ISINs (action-note: file cadence + history window)	W1	verify	UNVERIFIED-CANDIDATE
30	Budapest SE (BSE) — data files	EOD-hu	bse.hu / data-bse.hu	EOD	multi-year	EOD	web files	free-ish	HU tax/foreigner codes (action-note: probing fetch)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
31	Wiener Börse — Data Services	EOD-at	wienerborse.at data-services	EOD	multi-market AT/CEE	EOD	account + file delivery	commercial license	separate index+prim-listed catalogs (action-note: info fetch; CEEplus via GPW cascade better?)	W2	license-review	UNVERIFIED-CANDIDATE
32	Luxembourg SE — market data	EOD-lu	bourse.lu data	EOD	shallow	EOD	web	free-ish	smallest EU node — buy-only-if-needed (action-note: skip until branch scope fixed)	W2	connect	UNVERIFIED-CANDIDATE
33	Athens SE — analytical data products	EOD-gr	athexgroup.gr data sections (linked from Euronext nav [3])	EOD	Greek crisis eras	EOD	web/API license	commercial license	separate Athens Group entity, Greek-char tradids (action-note: cata-logue fetch (GR/EN))	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	MOEX ISS — historical candles/history endpoints (per-endpoint census)	historical-archive	iss.moex.com per-endpoint tree /iss/history/engines/... (endpoint census from upstream docs + community API lists) [11]	EOD history endpoints + candles	official archive per-board incl. history_from per security	delayed free / realtime paid	REST GET, anonymous or auth'd	⚠ informational — commercial reuse needs written MOEX contract [11]	free tier + **familiarisation only**; no real-time/PIT entitlement without contract (action-note: **sanctions/jurisdiction gate first**; only then map each engine/market endpoint)	W0	connect	VERIFIED-UPSTREAM-DOC
35	MOEX ISS — securities/boards metadata (snapshot + per-item metadata)	reference-master	iss.moex.com securities/…, boards/… endpoints [11]	instrument-level	full master + history_from	realtime (delayed ok)	REST JSON rows	⚠ same informational gate	per-board mapping table required; MOEX metadata drifts daily (action-note: same jurisdiction gate as row 34)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
36	SPB Exchange (spbexchange.ru) — equities EOD/API	EOD	ISS-like endpoints / site downloads	EOD	multi-year	delayed	REST/web	⚠**sanctioned/EU-flagged** (2022+ restriction cycle) — informational flag only	**sanctions-screening before ANY compute** (action-note: **skip until sanctions review clears**; log flag in catalog)	W3	connect	UNVERIFIED-CANDIDATE
37	ESMA FIRDS — instrument reference data (full + delta files)	reference-master	registers.esma.europa.eu publication + official `esma_data` loader [12]	instrument level (full/delta)	since 3-Jan-2018 launch [13]	next-day (delta files)	anonymous machine-readable file downloads	free (regulator-published)	authoritative EU instrument spine; LEI joins from GLEIF (row 38); completeness caveats published by ESMA [13] (action-note: wire daily full+delta poll; join to GLEIF + row 8/40)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
38	GLEIF LEI — Golden Copy + Delta + Concate-nated + GLEIF API	reference-master	gleif.org Golden-Copy download page (3×/day + 4 delta windows) [14]	entity-level	pre-2017 issue dates → present, historic business-report archive	T+0 (3×/day)	anonymous file download + REST API	free, open data	LEI PIT — use delta file windows (8h/24h/7d/31d) for as-of-time joins (action-note: daily delta scheduler + as-of-time index)	W0	connect	VERIFIED-UPSTREAM-DOC
39	ESMA EMIT — MiFID II transaction-report data (data.esma.eu, UR2)	post-trade-equity	data.esma.eu EMIT portal (fetch blocked this session)	transaction-level	archive per UR	periodic publishes	anonymous file down-loads	free regulator data, heavy reuse TOS	**known PIT/completeness caveats** from ESMA [13]; big files (action-note: retry data.esma.eu from different egress; park UR citation)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
40	ESMA FITRS — MiFIR trans-parency data (pre/post-trade) publica-tion	post-trade-equity	registers.esma.europa.eu trans-parency files (`citation_date/latest.zip`, registers gate 502 this session)	period-level equity trans-parency data	historical files published multi-year;	periodic	anonymous file down-loads	free regulator data	venue/lagging time-window reconcili-ation against venue RTs (row 8, 68) (action-note: re-attempt portal fetch; snapshot old-est→newest files)	W1	connect	UNVERIFIED-CANDIDATE
41	STS — in-strument national-Competent-Authority venue lists (TRV)	reference-master	national CA published venue/MIC supersets (esma indexes)	venue-level	snapshot-per-update	ad hoc	web	free	cross-check against opera-tional venue/MIC reference (action-note: consoli-date NCA venue lists into `venues.csv`)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
42	CboE Europe — RTS13 public trade-data files (15-min delayed)	historical-archive	cboe.com/europe/equities/trade_data — free CSVs hourly wheel `rts13_public_trade_data_dxe_YYYY-MM-DD_HH.csv` [15]	transaction-level hourly buckets	online	hourly-after publica-tion	free wget automa-tion (sample com-mands on page)	free (MiFIR 15-min data-use clause), TOS-abide	per-venue ESMA RT-13 data; strong PIT vs SIs (action-note: set wget job (pattern cfg on page); nightly Δ aggrega-tor)	W0	connect	VERIFIED-UPSTREAM-DOC
43	B3 CO-TAHIST — historical quota-tions (daily/monthly/yearly)	historical-archive	b3.com.br historical-quote-data ([16]: prices since 1986, original-currency/format **without** adjust-ments)	EOD per instru-ment	since 1986	post-session (daily), retro batches monthly/yearly	anonymous download (fixed-width CO-TAHIST file)	public published data; redistri-bution restricted	unadjusted contexts — corporate-action chain must be designed over BVBG-086 + CA files (action-note: ingest daily CO-TAHIST; monthly+yearly to 1986)	W0	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	B3 BVBG-086 — market price report files	historical-archive	B3 historical-data tree (b3_bvbg-086 upstream template in rb3 ODM [17])	daily price-report	daily-to-retro file window	EOD	anonymous/public file	registered redistri-bution restricted	venue-issued CSV/XML; feeds CA spine for row 43 (action-note: loader + CA deriva-tion job)	W0	connect	VERIFIED-UPSTREAM-DOC
45	B3 — daily bulletins & settle-ment prices (de-rivs/fx/fi)	EOD-stats	b3.com.br historical data tree — Daily bulletins / Settle-ment prices / FX retro [16]	EOD	multi-decade	EOD	anonymous web	free/published	settlement-price PIT; separate fx-summary mixing rules (action-note: mirror per-section file trees)	W0	connect	VERIFIED-UPSTREAM-DOC
46	B3 — Market Data Platform (realtime, via dis-tributors)	realtime-tape	b3.com.br market-data-platform [18]	realtime L1/L2	realtime config	realtime	licensed distribu-tor contract	commercial license	per-product broad-cast; SLA off-wire (action-note: only if BR-branch funded)	W3	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
47	B3 — index portfolio/composition files (b3-indices templates)	index-components	B3 indices-files (rb3 template catalog row `b3-indexes-composition` + historical portfolio) [17]	composition per month	historical portfolio archive	monthly	anonymous web	published	index-membership as-of = file publication date (monthly PIT) (action-note: parse monthly portfolio + composition history)	W0	connect	VERIFIED-UPSTREAM-DOC
48	BMV — Grupo BMV daily CSV/DB market-data products (INTRA / SiBolsa / Bases de Datos)	EOD-intraday	bmv.com.mx/information-products/market-data (INTRA) + market-data contact [19]	EOD→intraday	deep (per-product)	EOD	licensed product delivery	commercial license	BMV tax codes `Serie`/`Servicio` list mapping (action-note: request INTRA product sheet + sample)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
49	Grupo BMV — weekly/annual statistical bulletins	historical-archive	bmv.com.mx press-room (annual+weekly statistical-bulletin downloads) [19]	period EOD	multi-year archive on site	weekly/annual	anonymous file download	public	primary for index/turnover reconstruction; equity CA joins still needed (action-note: ingest weekly bulletin files; keep annual snapshot)	W0	connect	SURVEY-EVIDENCE
50	BIVA Mexico (merged into BMV)	EOD-mx	historical BIVA files / BMV historical archive	EOD	BIVA listed 2018→	closure	web	license	**venue-merge deprecation** — join BIVA series into BMV master before 2025 cross-impl (action-note: confirm merge-date & file availability)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
51	Chile CMF — market-instruments securities/turnover datasets	EOD-numeric	cmfchile.cl statistics (institutional satellite files)	EOF	multi-year	T+1	anonymous/reg web	registration regulator data (non-market-data redistribution caveats)	turnover standard incl. institution dashboard rows (action-note: fetch stats page + sample)	W1	connect	UNVERIFIED-CANDIDATE
52	CSE Chile — premium market data (historical, licensed)	EOD-cl	cse.cl licensed files (CMF cross-checked)	EOD	announced products	T+1	license portal	commercial license	official license gate for redistribution (action-note: sample request)	W2	quote	UNVERIFIED-CANDIDATE
53	Colombia BVC — daily market files	EOD-co	bvc.com.co Files-Trading	EOD	multi-year	EOD	registered web	free-ish	limited vendors for processed history (action-note: probe files; MARAVATIA rows)	W1	connect	UNVERIFIED-CANDIDATE
54	Peru BVL — COM/CAP daily quotation files	EOD-pe	bvl.com.pe downloadable quotes	EOD	multi-year	EOD	anonymous web	free-ish	symbol-join rules (COM vs CAP) (action-note: probe file pattern)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	Argentina BCBA/CASLA — EOD down-loads + CAVAL	EOD-ar	bcba.com.ar / caslasite / lape-nas… file trees	EOD	1980s→ but gap-riddled	irregular	web + manual	free-ish ⚠	worst data-gap regimes of the set — **flag as out-of-scope unless hard-need** (action-note: skip; revisit only with LatAm-4-countries stage)	W3	connect	UNVERIFIED-CANDIDATE
56	JSE — market data portal + down-loadable files	EOD-za	clientportal.jse.co.za downloadable-files (portal files; login-gated) [20]	EOD	per product	EOD	account + file/API license	commercial license	ZAR-currency scripts; race dedupe (action-note: portal account + file list)	W2	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
57	JSE — licensed networks (ICE/ICE Data); JSE Market-Related via vendors	realtime/historical	ICE Developer Portal Entry (JSE) etc.	realtime+historical	GT vs orchestrator	near-realtime	commercial license (ICE contract)	commercial license (dual-class JSE product & vendor resale)	PIT redraw from ICE consolidated feed (action-note: skip incremental; revisit when ZA premium buyer exists)	W3	connect	UNVERIFIED-CANDIDATE
58	NSE Nigeria — daily quotes window	EOD-ng	nse.com.ng daily files (XLS)	EOD	rolling years	EOD	web scrape	free-ish	NSE file-quality rules; not NSE-India (action-note: probe cadence)	W1	connect	UNVERIFIED-CANDIDATE
59	EGX Egypt — daily market archives + disclosure	EOD-eg	egx.com.eg statistics + disclosure pages	EOD	moderate archive	EOD	anonymous web (unofficial) ⚠	site TOS; no SLA	USD/EGP and listing-restriction flags; wash dedupe (action-note: probe; use only if GFI needs MEA breadth)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
60	Nairobi NSE Kenya — daily data	EOD-ke	nse.co.ke data pages (listed-co stats)	EOD	portal-depth unknown	EOD	web	free-ish	same treat-ment as row 59 (action-note: probe; catalog-light)	W2	connect	UNVERIFIED-CANDIDATE
61	ASE Casablanca — official quota-tions/notes files	EOD-ma	casablanca-bourse.ma (or official version site)	EOD	archive on-site official images	EOD	PDF/CSV web	free-ish	image-release → OCR risk; keep flattened site variant (action-note: official file server (bourse-cma))	W2	verify	UNVERIFIED-CANDIDATE
62	Ghana GSE — daily lists	EOD-gh	gse.com.gh pages (list+changes)	EOD	shallow	EOD	web	free-ish	symbol drift; NII (action-note: probe)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
63	TSX/TSXV — Datalinx Trading Summary (TSX daily + V-file for TSXV)	historical-archive	tmxwebstore.com TSX-trading-summary-daily ([21]: EOD fields incl. O/H/L/C, volume, bid/ask, **history from Oct-1998 TSX / Mar-2009 TSXV**)	EOD per security	from Oct-1998 (TSX), Mar-2009 (TSXV)	daily file 18:00 ET	webstore subscription	commercial license, internal-use oriented	single daily file per venue; seasonal contract renewal (action-note: product-spec sample; fit into GFI CA ingest)	W1	connect	VERIFIED-UPSTREAM-DOC
64	TSX/TSXV issuer-side daily files (Daily Summary, Daily Historical Trading Activity, CDNX archive files)	historical-archive	tsx.com TSXV market-information (files description) [21]	EOD	CDNX composite to Dec-2001 (archived)	daily	Datalinx subscription	commercial license	same spine as row 63; CDNX archived set needed for pre-Sep-2001 GFI (action-note: order CDNX historical file for pre-Dec-2001)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	TSX/TSXV — NCIB corporate-action files (daily/monthly)	corporate-actions	tmxwebstore NCIB-listed-issuer product (daily for TSX+TSXV, monthly file released 3rd-last business day) [21]	CA-event	rolling + archive	monthly/daily	webstore subscrip-tion	commercial license	issuer-reported data delays; pivot-day load (action-note: parse into GFI CA spine)	W1	connect	VERIFIED-UPSTREAM-DOC
66	TMX — Daily Record (bulletins, dividend notices, most-active, index sum-mary)	corporate-actions/journal	tmxwebstore Daily Record ([21]: history back to 1998)	daily record	since 1998	08:30 ET daily	webstore subscrip-tion	commercial license	text-file OCR-style file; parse ro-bustness (action-note: order sample; convert to GFI event log)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
67	Montréal Exchange — daily derivative named constant data files	EOD-mx-derivs	m-x.ca/en/trading/data/historical (files **not official end-of-day source** per page note)	EOD	session history only	EOD	anonymous web	free (general-information basis only)	**explicitly non-official** — do not retain as compliant copy of record (action-note: use only as fallback; official copies via TMX Datalinx)	W2	connect	VERIFIED-UPSTREAM-DOC
68	SEDAR+ — Canadian issuer filings portal (disclosures)	filings-can	sedarplus.ca new portal	filing	1997→ archive	rolling	web/API pending bulk data-access	free for search; structured extraction manual ⚠	JS-only portal (action-note: migrate old-SEDAR files → bulk; filings-domain overlap — cross-ref the 02 fundamentals/filings census)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
69	TASE — Data Hub API (Israeli market data)	EOD-IL API	tase.co.il/en/content/products_services — API + daily Excel reports [23]	end-of-day + historical	full per-product archive	EOD/on-demand	API sub-scription	commercial license, TASE gated	TASE codes; ILS rounding; payment calendar (action-note: Data Hub subscrip-tion tier; build loader)	W1	connect	VERIFIED-UPSTREAM-DOC
70	TASE — Data Store (data files / reports / one-time historical data)	historical-archive	tase.co.il Data Store — one-time files, historical datasets, monthly plans [23]	file-level	per-product history archive	ad hoc/orders	purchase + download	commercial license	CSV/Excel layouts — verify per product from store samples (action-note: choose historical-data SKU; load)	W1	backfill	VERIFIED-UPSTREAM-DOC
71	NZX — market data portal + data services	EOD-nz	nzx.com data/market-data sections	EOD-licensed	mid-90s→	EOD	file + license	commercial license	NZ tick data format (CSV? channel-specific) (action-note: d sample)	W2	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
72	ATHENS-group pure data-portal republish (out of GFI-scope mirror check)	audit-only	cross-check row 33 vs Euronext nav [3]	—	—	—	—	—	conflicting licence paths — **do not auto-merge** with row 33 source (action-note: mark duplicate-entry check)	W3	connect	UNVERIFIED-CANDIDATE
1	SEC EDGAR company-submissions API	PIT-SPINE	data.sec.gov/submissions/ + nightly submis-sions.zip	per-CIK filing index (accession-keyed)	≈1993→present	T+0 dissemi-nation; zip nightly ~03:00 ET	REST + nightly bulk zip	free / US public domain	cap User-Agent string; 10 req/s; join ev-erything by acces-sion no. (action-note: nightly CIK-canvas pull → filings spine shared with rows 2–9)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
2	SEC EDGAR company-facts XBRL	PIT-SPINE	data.sec.gov/api/xbrl/companyfacts/CIK#.json	per-filer (tag, unit, end/fy/fp/**filed**)	XBRL facts 2009+ (iXBRL era)	<1 h from dissemination	REST + company-facts.zip nightly	free	filed-date per fact ⇒ restatement time-series reconstructable free; pin accession per fact (action-note: primary numeric fundamentals spine for US/backtest)	W0	connect	VERIFIED-UPSTREAM-DOC
3	SEC EDGAR company-concept	PIT-SPINE	data.sec.gov/api/xbrl/companyconcept/CIK#/us-gaap/TAG.json	per-concept full time-series per CIK	2009+	T+0/1 min	REST	free	same PIT gate as #2; taxonomy-name drift (us-gaap vs ifrs-full) (action-note: cheap targeted-tag pulls on hot signals)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
4	SEC EDGAR frames API	PIT-SPINE	data.sec.gov/api/xbrl/frames/us-gaap/TAG/USD/CY####Q#.json	calendar-period cross-section per metric	2009→present	T+0	REST	free	frames mix latest filings into a quarter — re-derive strict PIT by filed-dates from #2 (action-note: ready-made cross-sectional snap-shots for screens (live only))	W0	connect	VERIFIED-UPSTREAM-DOC
5	SEC EDGAR full-text search (EFTS)	PIT-SPINE	efts.sec.gov/LATEST/search-index?q=&forms=&dateRange=	full-text hits	2001→present	T+0	REST (rate-limited)	free	text w/o structure; keep hits keyed to acces-sion #s (action-note: event/keyword scanner corpus (8-K text, risk lan-guage))	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
6	EDGAR full-index nightly + Archives backfile	PIT-SPINE	www.sec.gov/Archives + indexes (master daily, CY/QTR full-index zips)	filing-level index universe	full machine-readable archive 1993→present; pre-1993 in Archives only (sparse/legacy)	T+1 (index file night after filing)	file-drop dirs + zips, no key	free	pre-1993 hole = survivorship-first backfile (pair with #117/#121) (action-note: enumer-ate everything-submitted back-ground job; popu-lates CSS dedupe)	W0	connect	VERIFIED-UPSTREAM-DOC
7	EDGAR 10-K per-item parse	SPINE-PARSER	primary docs → Items 1/1A/7/8 + segment data via iXBRL tags + NLP chunking	per-item per-filing	HTML-era ≈1996+; iXBRL 2009+	T+0/T+1	doc fetch + own parser (no API for “items”)	free	item structure not machine-stable pre-iXBRL; tag drift year-over-year (action-note: risk-factor/MD&A qualita-tive PIT extrac-tion)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
8	EDGAR 10-Q per-item parse	SPINE-PARSER	same route as #7, form=10-Q (incl. 10-Q/A)	per-item per-filing	≈1996+	T+0/T+1	doc fetch + parser	free	quarterly item granularity finer; amendments create vintages (action-note: intra-year fundamentals + going-concern language)	W1	connect	SURVEY-EVIDENCE
9	EDGAR 8-K per-item parse	CA-SPINE	same route, form=8-K; Item-code filter (1.01, 2.02, 2.03, 3.03, 4.02, 5.02, 8.01…)	per-item per-filing	1996+; item codes standardized 2004 (pre-2004 numeric 1–9)	T+0	doc fetch + item-map parser	free	item-code regime changes Aug-2004 → two-phase parser; 4-day filing window ≠ T+0 (action-note: event spine for CA/material-contract triggers)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
10	SEC Financial Statement Data Sets (Dera FS-DS)	SPINE-DERIVED	sec.gov/dera/data/financial-statement-data-sets (num/pre/sub/txt CSVs)	quarterly fact/period.csv	2009 (Q1)→present	quarterly batch, ~1 mo after quarter	file-download, no key	free	derived from as-filed XBRL = filing-date safe; but lag makes it a baseline not the live spine (#2 outruns it) (action-note: quick standard-ized ETL baseline; jumpstart cross-sectional labs)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	DEF 14A proxy state-ments	REGISTRANT	EDGAR form filter (DEF 14A/14C) + exhibit parse	per-item: exec comp, propos-als, votes	≈1997+	T+0/T+1	filings + parser	free	pay-date ≠ filing-date (vote BEFORE meeting but after record); PIT = filing date (action-note: gover-nance/comp factor layer)	W2	connect	SURVEY-EVIDENCE
12	S-1/F-1 prospec-tuses (incl. 424B4s)	REGISTRANT	EDGAR form filter (S-1/-A, F-1/-A, 424B4)	per-filing: proceeds, shares, lead under-writers, cliff table	1993+	T+0	filings + extrac-tion (parties, goals)	free	two-stage vintages (pre-pricing vs priced 424B4) both belong in corpus (action-note: IPO pipeline + issuance-event layer)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
13	20-F/40-F foreign-issuer XBRL	REGISTRANT	EDGAR form filter; IFRS tax-onomy facts (ifrs-full) + 6-K stream	per-fact/filing (IFRS)	iXBRL/XBRL ≈2011+; text filings 1990s+	T+0	filings + XBRL parse	free	IFRS tag coverage thinner than US-GAAP; currency-heavy (action-note: international-access schema (ADR/US listings))	W1	connect	SURVEY-EVIDENCE
14	13F-HR quarterly institu-tional positions	POSITION-REG	EDGAR 13F efilings (13F-HR/-A XML; no official SEC bulk API — WRDS/reseller bulk needed)	per-manager per-holding per-quarter	positions since 1980s (digi-tized); struc-tured clean ≈2013→ present	filings by T+45d after quarter-end	EDGAR parse + vendor bulk	free (public info)	45-day lag = step-event, by-design stale; 13F-HR/A restate-ments; holdings ≠ market value truth (action-note: quarterly institutional-layer frame (screen, crowd-ing))	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
15	13D/G beneficial ownership (>5%)	POSITION-REG	EDGAR form filter (SC 13D/13G/-A)	event-level ownership snapshots	1970s era covered; machine-readable bulk via accumulated filings	T+0–T+2	filings + parser	free	“calculation date” can precede filed date — use calc-date for position, filing for public knowledge (action-note: activism/intent signals)	W1	connect	SURVEY-EVIDENCE
16	Form 3/4 (SEC insider transactions)	POSITION-REG	EDGAR + data.sec.gov submissions → Form 4 XML exhibits	transaction-level (shares, price, role, relation)	electronic ≈1991→ (paper 1980s in Archives)	T+0/T+1 (2-BD deadline)	XML exhibits parse	free	amendments v4/A create vintages; dense footnote text; affiliate ≠ director-only (action-note: insider-flow factor + directors/officers lattice)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
17	SC 13E-3 (going private)	POSITION-REG	EDGAR form filter	deal-stage filings + exhibits	1985+	T+0	filings + stage parser	free	multi-stage (prelimi-nary→final) — stage-keyed (action-note: M&A/arb event set)	W2	connect	SURVEY-EVIDENCE
18	SC TO tender offers	POSITION-REG	EDGAR form filter (SC TO-I/T, TO-C) + 425 comms	per-offer per-day (shares sought)	1990s+	T+0	filings + parser	free	pairs with DST offer expiry clock (action-note: arb pack feeder)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
19	Form SHO short-sale reports	POSITION-REG (NON-PUBLIC)	submitted to exchanges/FINRA — **not publicly disseminated at firm level**	weekly aggregate regulatory	never public	n/a	NOT FETCHABLE	regulatory-confidential	public proxies only: FINRA daily short-sale vol (#25), 4560 (#26), Nasdaq-Trader threshold list (#49), SEC FTD bi-weekly files (action-note: investigate FTD/threshold composite as the public SHO layer)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
20	Form D / Reg D offerings	REGISTRANT	EDGAR form filter + SEC Reg D tables (sec.gov/reg-d offerings data)	per-offering (amount raised, investor count)	electronic 2008→ (paper pre-2008 not EDGAR)	T+0 filings; reg-data refreshes n-yearly	filings + CSV	free	2008 electronic-divide = backfile cliff; PERMIT = offering, not listing (action-note: private-market issuance tracker (not securities uni-verse))	W2	connect	SURVEY-EVIDENCE
21	Form ADV (IA filings)	REGISTRY	adviserinfo.sec.gov (IAPD portal) + re-ports.adviserinfo.sec.gov archives (bulk ADV files)	per-firm per-section (AUM, fees, custody)	bulk archives back to early 2000s; an-nual/amend cadence	T+0 portal; archives T+~annual	portal + export files (XLSX/XML)	free / public	firm-level (not security-level); CRD-keyed; 403 on reports host for raw bots → use export files (action-note: adviser entity-master + AUM signal)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
22	Form PF (private funds)	REGISTRY NON-PUBLIC	SEC-held; aggregate stats published only	quarterly/annual per-fund	2012+	n/a	NOT FETCHABLE (aggregate via SEC reports)	regulatory-confidential	primary research value is SEC's own aggregate releases (action-note: skip unless SEC aggregate feed sparks a thesis)	W3	connect	SURVEY-EVIDENCE
23	Form 13H (large trader)	REGISTRY	EDGAR form filter	registration/assignment level	2013+	T+0	filings	free	few IDs float publicly; identifies big-flow actors (action-note: optional big-trader label join)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
24	EMMA / MSRB municipal disclo-sures	FIDUCIARY-MUNI	emma.msrb.org (muni state-ments, continu-ing disclo-sures, official state-ments; MSRB Gateway for 15G)	per-issue (CUSIP6 govern-ment class)	2009 (EMMA launch)→; pre-2009 scattered via issuers	T+0 portal	portal/HTML search; gateway APIs limited	free / public data	⚠ host blocked auto-mated fetch (403 this session) — ToS-pacing; CUSIP-gated by design; muni facts ≠ equity-equivalent facts (action-note: muni-fiduciary layer if credit stream opens)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
26	FINRA Rule 4560 consolidated short interest	POSITION-REG	finra.org/finra-data/browse-catalog/equity-short-interest	per-symbol bi-monthly (15th & EOM settlement)	5 rolling years online; longer via historical files/WRDS	published 7th BD after settlement (T+2 bi-monthly)	web files + free-tier API (non-commercial)	free non-commercial / commercial entitlement	step-events not series; Revision Flag amends; OTC-covered; missed-venue bi-monthly from NYSE/ARCA files (action-note: bi-monthly short-interest panel)	W1	connect	VERIFIED-UPSTREAM-DOC
27	Motley Fool earnings-call transcripts (free)	TRANSCRIPTS	fool.com/earnings/call-transcripts (HTML, no auth)	per-call HTML	≈2003→present, continuous	posted within hours of call	scheduled HTML fetch + cache	free-to-read; publisher copyright — internal use only, no redelivery	⚠ no bulk API; robots/ToS respect; small-cap gaps (action-note: free-tier transcript engine; pair with CapIQ if budget lands)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
28	SeekingAlpha tran-scripts (paid)	TRANSCRIPTS	Seekingalpha.com (unoffi-cial end-points exist; session-auth scrape)	per-call HTML	deep archive (≈2005+)	posted within hours of call	unofficial REST + session cookies	paid frontend (~$239/yr claim)	⚠ ToS explicitly forbids extrac-tion — account-ban risk; NEVER redeliver text (action-note: do not build until priced/compliance decision)	W3	connect	UNVERIFIED-CANDIDATE
29	Enterprise tran-scripts (CapIQ Pro / FactSet StreetEvents / LSEG)	TRANSCRIPTS (LIC)	Terminal/API entitle-ment (S&P CapIQ Pro has audio natively)	global tran-scripts	≈2003→present	daily API/entitlement feeds	licensed	enterprise	strict no-redistribution of transcript text (action-note: buy-or-substitute decision (only fully licit struc-tured path))	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
30	Compustat Funda-mentals (XPF + an-nual/quarterly)	PIT-VENDOR	S&P Global Market Intelli-gence; flat files/fields; XPF = as-originally-filed history files	per-firm per-period	annual 1950s→; quarterly 1976→; XPF archived eras	daily batch; PIT not native	licensed flat files (SFTP)	enterprise, call-only	base Compus-tat ≠ PIT (latest-restated paradigm); XPF restores as-filed for those periods (action-note: enter-prise buy-gate; XPF is the as-filed rescue for vintage US)	W2	connect	SURVEY-EVIDENCE
31	Compustat Point-in-Time (PIT)	PIT-VENDOR	separate S&P product; published with vintages (released dates per data point)	per-firm per-period per-vintage	PIT vintages stacked for decades (annual archive era ≈1956 base)	published with vintages	licensed (flat + SFTP)	enterprise call-only (often $xx k/yr — UNVER-IFIED range)	the true as-of-date gold standard; budget-gated (action-note: enter-prise buy-gate; only path to global PIT beyond free #2 hierar-chy)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
32	S&P Capital IQ Pro (fundamentals suite)	PIT-VENDOR	CapIQ Pro terminal/API: fundamentals, estimates, calendar, ownership, deals, transcripts	global, deepest US	transcriptions 2003+; fundamentals deep	near-realtime→daily	licensed	enterprise, call-only	no free tier; pos: transcripts+estimates+fundamentals under one entitlement (action-note: station-level only if a seat materializes)	W3	connect	SURVEY-EVIDENCE
33	FactSet Fundamentals + PIT shared files	PIT-VENDOR	FactSet batch/API + separate PIT shared package	global fundamentals; PIT vintages largely monthly	global coverage; PIT shared set years-long	monthly PIT vintages; realtime port daily	licensed	enterprise	without PIT add-on, FactSet = restatement-blind (documented); no raw-redistribution rights (action-note: global-PIT breadth (US ≈ 2nd-best to Compustat-PIT))	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	LSEG (ex-Refinitiv) DataScope/TR fundamentals	PIT-VENDOR	DataScope Select / TR Fundamentals API	global	≈1980s+ (varies by field)	daily batch	licensed API/flat	enterprise	Workspace/EIKON fundamentals are as-reported only — no native PIT vintages there (action-note: studio-scope provider once global build begins)	W3	connect	SURVEY-EVIDENCE
35	Bloomberg (terminal FA fields + Data License PIT)	PIT-VENDOR	Terminal BPS/FA + Bloomberg Data License (PIT entitlement path)	global	long coverage (field-dependent)	realtime terminal; daily batch	licensed (terminal / enterprise feed)	enterprise	PIT only via Data License entitlement — terminal snapshots are NOT PIT (action-note: terminal seat already covers; PIT entitlement = buy decision)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
36	MSCI data (security master + index funda-mentals snap-shots)	PIT-VENDOR	MSCI data feed / idp/instrument files	security-across-geography master + index snap-shots	long	daily batch	licensed	enterprise	strengths in in-dex/ESG master data, not per-firm state-ments detail (action-note: signal if MSCI-entitlement leverage exists through repo contacts)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
37	Sharadar SF1 (Nasdaq Data Link)	PIT-VENDOR (VERIFY)	data.nasdaq.com SHARADAR/SF1	quarterly per-firm state-ments (incl/excl restate-ment views)	≈1998→present continu-ous	daily 17:30 & 23:30 ET	API/table pulls + bulk	paid tiers (pricing behind login)	publisher-page claims “Point-in-time. Incl/Excl Restate-ments” — must SPOT-VERIFY restate-ment deltas + known ticker-merge gaps (AM/AM1/AM2-type) before backtest role (action-note: 2-week verifica-tion sprint → adopt/reject per stream-02 plan)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
38	Sharadar SF2 (daily funda-mentals)	PIT-VENDOR (VERIFY)	data.nasdaq.com SHARADAR/SF2	daily-updated funda-mentals rows	follows SF1 vintages	daily	API/table	paid tier	same PIT claims/audit as #37; ensures intra-quarter revision visibility (action-note: fold into #37 verifica-tion sprint)	W0	connect	VERIFIED-UPSTREAM-DOC
39	Sharadar SF3 (insider transac-tions)	PIT-VENDOR	data.nasdaq.com SHARADAR/SF3	per-transaction	~2003→	daily	API/table	paid	mirrors Form 3/4 — check vs #16 primary (action-note: redundant-first, primary Form4 wins — keep as QC twin)	W0	connect	VERIFIED-UPSTREAM-DOC
40	Sharadar SF4 (insti-tutional holdings)	PIT-VENDOR	data.nasdaq.com SHARADAR/SF4	quarterly per-manager holdings	≈2000s→	quarterly	API/table	paid	45-day lag + revision windows — QC vs #14 (action-note: QC twin for 13F parse)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
41	Sharadar SEP (equity prices incl. delisted instru-ment flags)	PIT-VENDOR	data.nasdaq.com SHARADAR/SEP	per-symbol per-day	long (delisted instru-ments kept — label flags)	daily	API/table	paid	delisted-flag quality check vs Nor-gate/CRSP lists (action-note: universe+CA-carry glue (survivorship-smart))	W0	connect	VERIFIED-UPSTREAM-DOC
42	Nasdaq Sharadar Core US Equities bundle	PIT-VENDOR	data.nasdaq.com Core US Equities bundle packag-ing (SEP+SF1 com-bined)	bundle entitle-ment	same as #37/#41	daily	single en-titlement	paid	discipline: bundle ≠ new data; just pricing conve-nience (action-note: decision with #37)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
43	FMP fundamentals	⚠ NEVER-PIT	financialmodelingprep.com API (statements/ratios/filings/13F/transcripts/end-points)	per-symbol period **latest only (overwritten on restatement)**	30+ yrs claim (transcripts tiers); free=limited	daily (EOD-ish)	REST + bulk ZIP tiers	commercial subscription ($19–99/mo ladder); display/redistribution needs separate Display & Licensing Agreement	⚠⚠ NEVER-PIT by design: restated numbers rewrite history — **any as-of backtest built on this row reconstructs today-in-history fiction; EDGAR-as-filed/Compustat-PIT is the backtest spine** (action-note: screening/live + coverage-shim US/global lens only)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	EODHD fundamentals	⚠ NEVER-PIT	eodhd.com fundamentals API (also SEC-statements, calendar, insider)	per-symbol latest views	~30 yrs claim (fund-tilde tiers)	EOD daily	REST + bulk (burns 10–100 calls each)	paid $59.99/$99.99 mo	⚠ NEVER-PIT; All-In-One = cheapest everything-row for live protos (action-note: same gate discipline as #43)	W0	connect	VERIFIED-UPSTREAM-DOC
45	Tiingo fundamentals (add-on)	⚠ NEVER-PIT	tiingo.com/fundamentals API	per-symbol latest	~5 yrs individual / 15+ yrs business claim	EOD	REST	add-on priced-on-request (third-party provider — NOT inside $30/mo)	⚠ NEVER-PIT; add-on packaging fee surprises (action-note: prices-first row; fundamentals only if add-on priced sane)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	SimFin (standardized fundamentals)	⚠ NEVER-PIT	simfin.com API + B2B datasets	standardized statement dictionaries US/DE	~2005→ (varies)	periodic	REST + tables	freemium→paid (license pivot-history risk)	⚠ standardized ≠ PIT; restatement-versioning unverified (action-note: drop unless verification answers both PIT +)	W2	license-review	UNVERIFIED-CANDIDATE
47	Intrinio fundamentals	⚠ NEVER-PIT	intrinio.com fundamentals API	standardized views	multi-year claim	daily	REST — entitlement-tiers	commercial (tier quoting)	provider contract landscape shift-prone; pit claim unverified (action-note: raise to W1 only if structured US needs it)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
48	Finnhub fundamentals	⚠ NEVER-PIT	finnhub.io company profile/financials endpoints	per-symbol latest	varied (short)	daily	REST, free tier	paid tiers for history/bulk	⚠ NEVER-PIT; thin depth; screening-only (action-note: screening swap-in candidate)	W3	connect	SURVEY-EVIDENCE
49	NasdaqTrader free files (listing-status + CA alerts + Reg SHO threshold)	CA-ACTIONS (FREE)	nasdaqtrader.com (nasdaqlisted.txt, otherlisted.txt) + Corporate Action Alerts + threshold list	per-symbol per-day files	current-snapshot + rolling alerts	daily file-drop	file-download, no key	free / publication-of-SRO data	listing-status ≠ full CA history (alerts have shallow look-back); pair with #54/#53 (action-note: universe bootstrap + delist-class triggers)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
50	Nasdaq official corporate actions feed	CA-ACTIONS (LIC)	Nasdaq market-data Corp Actions product (file-based, licensed)	per-security per-event (splits, dividends, name-changes, mergers)	deep vendor archive claim	T+1 file cadence	licensed file-drop/SFTP	commercial entitlement	authoritative for Nasdaq-list only; pairing gate vs #49/#53 (action-note: price-gate + evaluate vs Norgate/CRSP substitute)	W2	connect	SURVEY-EVIDENCE
51	NYSE / ICE official corporate actions	CA-ACTIONS (LIC)	NYSE market-data (Corporate Actions/Extraction Tools, XSEL-CA files via ICE)	per-security per-event	long historical depth claim	T+1	licensed file delivery	commercial	same authoritative-vs-license logic as #50 for NYSE-listed (action-note: price-gate; substitute = CRSP+Norgate until single-exchange-only deltas matter)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
52	S&P DJI index corporate actions	CA-ACTIONS (LIC)	S&P DJI CGS/index events (additions/deletions/share changes)	per-index per-event	full-back history claim	daily/announcement-driven	licensed-API/files	commercial	index membership ≠ universe management; crossing into stream-02 (index data) (action-note: feed the survivorship-merge unless already in stream-02)	W2	connect	SURVEY-EVIDENCE
53	CRSP corporate actions + delisting returns	CA-ACTIONS GOLD	CRSP daily/Security level CA files (CRSP via WRDS or direct)	per-security per-event incl. delist-return computation	1926→present	daily/annual data-cycle refresh	licensed (WRDS academic or direct)	enterprise subscription (academic-friendly price tier)	survivorship gold standard; history includes US exchanges' closed eras (action-note: enterprise pricing decision; academic route likely cheapest exit)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
54	Norgate delistings + historical index con-stituents	CA-ACTIONS RETAIL	norgatedatapackages (US/ASX/CA)	com-symbol delist-event + con-stituent map	US delists 1990 (Plat-inum) / 1950 (Di-amond); ASX 1992→	daily EOD	desktop lib (nor-gatedata python) + API	subscription $270–$790/yr (tier ladder verified)	delist universe + historically-true con-stituents at retail price = cheapest disci-plined survivor-ship layer; ⚠ funda-mentals tier is current-only (action-note: plat-inum+ buy decision — anchors universe layer)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	EDGAR-borne delist/change signals (Form 25 + 8-K Items 3.03/5.03)	CA-ACTIONS (FREE)	EDGAR form-25 filings + 8-K parsed items	per-event	1990s+	T+0/T+1	event stream from #6/#9	free	no-delist-return (needs #53 for return calc); de-dupe against exchange feeds (action-note: cheap exchange-decision join (most matches CRSP/Norgate))	W0	connect	VERIFIED-UPSTREAM-DOC
56	FMP delisted-companies list	⚠ NEVER-PIT SHIM	FMP delisted-company end-points	per-symbol snapshot	recent-era empha-sis; NOT a historical universe	daily	REST	commercial tier	⚠ NOT a survivor-ship universe (mini-mum coverage; no returns) — never dress it up as one (stream-02 verdict) (action-note: screening shim only)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
57	OpenFIGI symbol-ogy API	IDENTIFIER	openfigi.com/api (POST map-pings)	/v3/ instrument mapping (FIGI↔tickers↔ISIN↔CUSIP↔exchange)	full coverage claim	realtime/batch	REST (free key, unlimited-ish)	free / CC-BY-4.0	tokens for bursts; keep private key off-log (action-note: symbol-ogy glue + join spine with LEI/MIC)	W0	connect	VERIFIED-UPSTREAM-DOC
58	GLEIF LEI records (+ daily golden copy)	IDENTIFIER	api.gleif.org/api/v1 (lei-records; golden copy files)	per-LEI registry entries	3.47M+ records live; rela-tionship chains shipped	daily golden copy	REST + daily file	free / CC-BY (open data)	relationship graph (fund par-ent→child); keep daily snapshot for CI ver-sioning (action-note: daily master snapshot into CI)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
59	ISO 10383 MIC codes	IDENTIFIER	iso20022.org/market-identifier-codes (RA-maintained list)	per-MIC (per-market)	full recording maintenance	maintained list release	file-download	free-of-charge per ISO RA	name/location data only — no venue metadata beyond the list (action-note: exchange-master join)	W0	connect	VERIFIED-UPSTREAM-DOC
60	ANNA/ISIN + CUSIP Global Services	IDENTIFIER (LIC)	ANNA member catalog + CGS (S&P-run)	per-NSIN	full	n/a — master sync	licensed entitlements	paid/restricted (NO bulk republication)	CUSIP/ISIN equivalent master tables cannot be redistributed; auto-identified via primary filings only; FIGI/LEI as open spine (action-note: zero-spend policy; -class firewall documented)	W0	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	FMP/EODHD corporate-activity calendars (dividend/split event feeds)	⚠ NEVER-PIT CA	FMP calendar endpoints + EODHD calendar endpoints	per-event per-symbol	short-to-mid depth claim	daily	REST	commercial tiers	⚠ events overwritten/late-fetched — reconcile vs #49/#53 before any historical backtest use (action-note: screening/live only)	W0	connect	VERIFIED-UPSTREAM-DOC
62	IBES consensus estimates (LSEG)	ESTIMATES	LSEG l/B/E/S (consensus + event-stamped estimates)	per-metric per-period (EPS/Rev/EPS-Award)	long consensus history claim	daily revision cadence	licensed API/flat (RSK feeds)	enterprise	event-stamped ⇒ intrinsically PIT-friendly — verify as-of-date discipline in feed semantics (action-note: consensus-moments layer; price-gate first)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
63	Zacks-class estimates	ESTIMATES	Zacks (consensus + detail datasets, typically via WRDS/licensed)	per-metric per-period	long-era claim	daily	licensed	commercial	competitive-class second source; anchor-avoidance (some pick-side bias) documented (action-note: secondary source for estimate-set triangulation)	W2	connect	SURVEY-EVIDENCE
64	Nasdaq earnings calendar API	ESTIMATES/CALENDAR	api.nasdaq.com/api/calendar/earnings (+ earnings-date endpoints)	per-symbol per-date (AMC/BMO fits)	short/window claim	daily	REST (unofficial-ish shape; no key)	free-for-display terms	shape not contractually documented → fragile client; bracket with #62/FMP (action-note: earnings-date master pre-earn (live tier))	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	FMP earnings calendar	⚠ NEVER-PIT CALEN-DAR	FMP calendar end-points	per-symbol per-date	short depth claim	daily	REST	commercial	⚠ date revisions over-write; cross-check #64 + 8-K Item 2.02 (action-note: live-schedule only)	W0	connect	VERIFIED-UPSTREAM-DOC
66	Company guidance datasets	ESTIMATES	CapIQ/FactSet guidance-format rows + 8-K Item 2.02/7.01 text mining	per-quarter guidance state-ment	vendor-history claim	daily	licensed + parser	commercial / free-field	holds company-said wording (no point-estimate strikethrough) unless keyed to 8-K (action-note: guidance-vs-actual delta as factor)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
67	UK Companies House filing-history API	REGISTRY	developer.company-information.service.gov.uk (filing-history endpoint)	per-company per-filing history	full registry history (decades)	live/realtime API	REST (free API key)	free / Crown-OGL	free real-time; end-points compos-ite richer with #68–#71 (action-note: UK registry spine)	W0	connect	VERIFIED-UPSTREAM-DOC
68	UK Companies House company-profile API	REGISTRY	same dev-portal (company profile endpoint)	per-company pro-file/status	full registry history	live	REST	free / OGL	status/formation/dissolution dates feed universe-lifecycle (action-note: entity-master UK)	W0	connect	VERIFIED-UPSTREAM-DOC
69	UK Companies House officers API	REGISTRY	same dev-portal (officers endpoint)	per-company per-officer-per-date (ap-point+resign)	registry history	live	REST	free / OGL	director-network panels (co-director graph across #58 iden-tities) (action-note: directors-lattice build)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
70	UK Companies House charges API	REGISTRY	same dev-portal (charges/SCC endpoint)	per-company charges	registry history	live	REST	free / OGL	secured-creditor + pledge signals (private-credit lens) (action-note: private-credit factor pull)	W0	connect	VERIFIED-UPSTREAM-DOC
71	UK Companies House bulk products + iXBRL accounts	REGISTRY (BULK)	CH data products (daily snapshot/changes, accounts data products & iXBRL bulk)	per-company bulk extract + full iXBRL accounts corpus	bulk snapshot history + iXBRL eras	daily/bi-annual products	file-drop + some paid product-tiers	free incl. re-publish (OGL); restricted items via bulk products	numeric-level financials require iXBRL-parse (not a facts API); lighter change-detection via daily diffs (action-note: daily-diff ingest + iXBRL build)	W0	build-parser	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
72	FCA NSM (National Storage Mecha-nism)	REGISTRY	data.fca.org.uk NSM portal (UK listed-company/ESEF-linked filings)	per-filing repository	≈2020→ (ESEF era); earlier listed-company filings via predeces-sor UK reposito-ries	T+0-ish portal	portal + man-ual/scripted fetch	free / public	weak link flagged in stream-02 (UI fetch, no numeric API) — verify bulk-capture pattern and retention for restate-ments (action-note: UK-listed filing relay; may merge into #71 route)	W1	connect	SURVEY-EVIDENCE
73	Ireland CRO (Compa-nies Registra-tion Office)	REGISTRY	cro.ie open services (WebExtract class extracts)	per-company registry + accounts extracts	decades (regula-tory pa-per→digital)	T+0-ish portal	portal/paid extracts	mixed: basics free; docs/extracts paid	no docu-mented bulk JSON API; per-doc fees (action-note: investi-gate; low priority vs UK)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
74	Esma/ESEF national reposito-ries map	REGISTRY-MAP	eurofiling.info / ESMA ESEF enabler list (per-country registry end-points)	per-country mapping	ESEF-fundament era ≈2020→	varies by NSR	per-country integra-tion	most NSRs free/open	national UI hetero-geneity; per-country excep-tions registry (action-note: per-country plug-in map (route table for EU registry rows))	W2	connect	SURVEY-EVIDENCE
75	Czech Registr obchod-ních (jus-tice.cz) + OVM open data	REGISTRY	justice.cz obchodní rejstřík + OVM/obchodní open datasets	per-company registry + selected docs	years-deep registry	T+0-ish portal	portal/HTML + partial open-data XML	free / public	Czech-lang financial docs; limited XBRL-era depth (action-note: probe per-country ingest; low priority)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
76	DE Unternehmensregister / Bundesanzeiger	REGISTRY	unternehmensregister.de (search incl. HRB/HRA docs) + bundesanzeiger.de (patrimony/announcements)	per-company Registry docs + annual accounts	years/decades (registrar paper→2007 e-Act)	T+0-ish portal	portal + paid fetch (~€5/doc-class; machine-access restricted)	mixed (per-doc fees)	⚠ machine/API access limited vs UK CH — DE = parsing-heavy; MaRIS-class resellers exist (action-note: probe → only if a German DL build opens)	W2	connect	SURVEY-EVIDENCE
77	FR INPI / data.inpi.fr (registry + annual accounts open data)	REGISTRY	data.inpi.fr portal + api.inpi.fr (OAuth; annual-accounts open data)	per-company registry + accounts PDF/structured	broad history claim	T+0-ish	OAuth REST + bulk	free open data (OAuth-keyed)	⚠ autom-fetch flags 403 this session; language + doc formats spur engineering (action-note: probe API-key signup + test endpoint reachability)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
78	NL KVK (Zoeken + Basis-bedrijfsgegevens + Jaar-rekenin-gen APIs)	REGISTRY	developers.kvk.nl official APIs (sub-scription) + ODAT open datasets	per-company + annual accounts API	registry history; accounts datasets	realtime-ish API; datasets cyclical	REST sub-scription (per-query fees; free test env)	subscription-paid (+ per-query)	⚠ Jaar-rekenin-gen (annual accounts) is an OFFICIAL API = rare NL find — the only NL route beyond scraping (action-note: subscription-vs-scrape cost probe; test env already usable free)	W1	connect	VERIFIED-UPSTREAM-DOC
79	ES Registro Mercantil (datos.gob.es + official certifica-dos)	REGISTRY	datos.gob.es datasets + Registro Mercantil official copies	per-company registry + accounts	decades	T+0, portal	open-data down-loads + paid certs	mixed (open bulk vs paid per-doc)	ES registry docs = per-doc costs; bulk open-data exists in parts (action-note: low-priority probe)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
80	IT Info-Camere / Registro Imprese open data	REGISTRY	infocamere.it + CCIAA registries open-data releases	per-company registry + filings overview	decades	cyclical releases	portal +_bulk	free open-data pieces; per-doc for official extracts	Italian doc formats; per-country variance high (action-note: low-priority probe)	W2	connect	UNVERIFIED-CANDIDATE
81	CH Zefix (central business name index) + register docu-ments	REGISTRY	zefix.ch (REST API exists) + commer-cial register extrac-tors	per-company registry	decades	T+0-ish portal	REST API + paid extracts	free basics / paid extracts	⚠ SPA-page fetches returned blank here — endpoint else flagged to verify via direct HTTP call (action-note: confirm API endpoint path by scripted HIT test)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
82	CH SIX price-sensitive (ad hoc) disclosures	DISCLOSURE	six-group.com disclosure platform (ad-hoc announcements + management transactions)	per-announcement	era-dependent	T+0 portal	portal/API-class (some data at SIX feeds)	mixed (some feeds paid)	pairs with SIX-K equivalent events; CH-listed filers (action-note: drop unless Swiss-listed stream matters)	W2	connect	SURVEY-EVIDENCE
83	DK CVR open data (data.virk.dk)	REGISTRY	data.virk.dk (CVR open data; Erhvervsstyrelsen)	per-company registry + selected accounts	~2005+ digital eras amid paper	T+0-ish; datasets cyclical	open-data files/API	free / public authority	danish-lang statements depth varies; transport-blocked here (F17) (action-note: probe; good candidate for free accounts extracts)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
84	NO Brønnøysund registers (incl. annual-accounts register)	REGISTRY	brreg.no (+ virksomhet.brreg.no lookup; annual-accounts submission pages; use-of-data policy)	per-company registry + accounts submissions	long eras (digital-forward from 2020s for accounts)	T+0-ish portal	open-data frameworks + APIs (API service exists per RA guidance)	free / public authority (use-of-data framework)	Norwegian-lang; accounts availability varies by filer class (action-note: probe accounts-register API + Norwegian account route)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
85	SE Bolagsverket registry + ESEF	REGISTRY	bolagsverket.se (+ per-registry extracts)	per-company registry	decades	T+0-ish portal	portal/paid extracts	mixed	full public accounts NOT universal (private filers pay via Ratsit/Upplysningarna-class vendors — cite in gates); ESEF era helps listed (action-note: low-priority probe; Ratsit-class paid vendor = later-decision)	W2	connect	UNVERIFIED-CANDIDATE
86	JP EDINET v2 filings-list API	REGISTRY-FILINGS	disclosure2.edinet-fsa.go.jp API v2 (documents.json list endpoint, date-window query)	per-filing (Japan last-window filings, incl. □□□□□□□□ “10-K-class”)	all digitized EDINET history (list-API deep; annual dates; XBRL eras newer)	T+0-publish → API T+1-class per FSA notes	REST (API key since 2023-08; v1 sunset 2024-03-29)	free / public sector	per-date polling + jitter; taxonomy = JP IFRS-compatible (action-note: nightly index + per-doc queue)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
87	JP EDINET v2 document download (HTML/XBRL/CSV)	REGISTRY-FILINGS	API v2 document-download endpoint (zip incl. Abstract + XBRL)	per-document bundle	vintages of attached formats	T+1	REST (rate-limited; doc-link series waits)	free	XBRL-by-EDINET-taxonomy parse = Japan numeric-fundamentals free path; 1-doc-per-req series waits (action-note: XBRL parse pipeline (doc-core per company))	W0	connect	VERIFIED-UPSTREAM-DOC
88	JP TDnet timely-disclosure (JPX)	DISCLOSURE	Tdnet — timely disclosure announcements (rinji hōkoku) via JPX/plug-feeds	per-announcement	era-dependent (not as deep as EDINET)	intraday/hourly (earnings + events real-time)	portal/equipment data-feed licensed (or MX feed)	mixed (feed licensed)	duplicates EDINET far faster for events; “same-day” earnings-announcement signal (action-note: price the d TDnet feed vs EDINET-delay analysis)	W1	license-review	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
89	KR DART OpenAPI (disclosure + key-info + financial statements with XBRL)	REGISTRY-FILINGS	opendart.fss.or.kr (list/company/fnlttSinglAcnt financial statements APIs incl. multi-year statements + XBRL raw download)	per-filing/period annual statement	deep historical (coverage from DART-era; XBRL bulk multi-year)	T+1 from DART publish	REST (free API key; tiered daily quotas)	free	K-GAAP restatements filing-time-versioned ⇒ pin `rcept_no` per period = PIT-safe; XBRL taxonomy drift year-dependent (action-note: statements+XBRL ingest; DART as KR filing spine)	W0	connect	VERIFIED-UPSTREAM-DOC
90	KR DART shareholding disclosures (□□□/□□)	POSITION-REG	opendart.fss.or.kr shareholding disclosure endpoints	per-disclosure per-holder	era-backed	T+1	REST	free	KR equivalent of Form 3/4-class ownership rings; holders unify chain (action-note: ownership panel build)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
91	KR DART bulk-download (company-docs ZIP per company/FY)	REGISTRY-FILINGS	opendart.fss.or.kr bulk-download endpoint (corp-download; per-FY/company zip)	per-kr company per-FY batch	deep	T+1	REST bulk (call quota heavy)	free	efficient backfill squeeze; onboarding quota vs plan (action-note: historical repeat-run tool)	W0	backfill	VERIFIED-UPSTREAM-DOC
92	CN cninfo (□□□□□) announcement / REST APIs	REGISTRY-FILINGS	cninfo.com.cn disclosure REST (hisAnnouncement/query-class; unofficial PyPI wrappers exist)	per-announcement per-listed-CN	deep archive claim (2000s+)	intraday publishing	REST portal + paced scrape	public disclosures (state-mandated); ToS-grey automation — pace; no formal open API documented	⚠ fetch probes SPA-blank (F17) — verify endpoint reachability (action-note: A-share/CN filings spine; CN financial-statement quality high)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
93	CN SSE announcements	REGISTRY-FILINGS	sse.com.cn disclosure endpoints	per-announcement (Shanghai)	deep chain	intraday	portal/JSON endpoints (undocumented)	public; ToS-grey paced	CN-language + announcement-parse burden; SSE vs cninfo overlap (action-note: secondary source to cninfo)	W1	connect	UNVERIFIED-CANDIDATE
94	CN SZSE announcements	REGISTRY-FILINGS	szse.cn disclosure endpoints	per-announcement (Shenzhen)	deep chain	intraday	portal/JSON (undocumented)	public; ToS-grey paced	same pattern as #93 (action-note: secondary source pair)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
95	CN corporate registry (Tianyancha/Qichacha-class)	REGISTRY (PAID)	third-party registry aggrega-tors (paid APIs; QCC/Tianyancha/□□□)	per-company reg-istry/shareholding/doc history	long claim	daily-ish refresh	vendor REST APIs	paid (quotes ¥3xx–year-plus tiers — UNVERI-FIED)	⚠ CN registry = NOT free (a red-line vs US/UK/JP); decide only when a CN build is real (action-note: vendor + ToS red-line review)	W3	quote	UNVERIFIED-CANDIDATE
96	HKEXnews (□□□) listed-company filings	REGISTRY-FILINGS	hkexnews.hk (title-search/search end-points; an-nounce-ment + financial reports)	per-announcement (HK)	varies by listing-class; years-deep generally	intraday publish-ing	portal search + polite per-company fetch agent (no docu-mented JSON API)	public filings; robots.txt likely gates bulk	⚠ no unified numeric XBRL (regime = document-parse); parsing burden triples vs US/JP/KR (stream-02 verdict) (action-note: HK filings relay agent, paced)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
97	India MCA (Master Data + RoC documents)	REGISTRY	mca.gov.in V3 (Master Data API, pro-state; RoC doc pay-per-view)	per-company registry + statutory docs	decades (RoC paper→digital mix)	T+0-ish portal	Master Data API (available per state) + doc fees	mixed: master free; docs ₹/doc-class	⚠ no clean new UK-CH-style numeric API; doc parsing (action-note: master-data spine for IND entities; doc packs later)	W1	connect	UNVERIFIED-CANDIDATE
98	NSE corporate announcements/CA	REGISTRY-FILINGS	nseindia.com corporate announcements + corp-filing endpoints (XBRL statements per SEBI)	per-announcement (results, board-date, M&A)	archive claim varies	intraday/T+1	unofficial-ish JSON endpoints, IP-aggressive	public; ToS-tight (nsearchives host blocks automation — [F17] prior evidence)	iXBRL per SEBI for statements; URL host has actively denied datacenter-style agents (action-note: paced relay agent; bypass = paid vendor (Trendlyne-class) decision)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
99	BSE announcements / corp actions	REGISTRY-FILINGS	bseindia.com announcements + CILAC endpoints	per-announcement (BSE)	archive claim	intraday	portal endpoints (undocumented)	public; ToS-grey	same-parse problem as #98 (action-note: pair with #98; two-host coverage)	W1	connect	SURVEY-EVIDENCE
100	Canada SEDAR+	REGISTRY-FILINGS	sedarplus.ca (search + document payment/extract; structured filings for ESEF-class iXBRL since ~2020)	per-filing (CA securities filers)	docs since 2000s; iXBRL era ≈2020+	T+0/T+1	portal + per-doc fees (official copies)	fees per document (statutory); search basic free	⚠ no free numeric API; the stream-02 matrix rows for CA flagged must complete; official copies costs add up at scale (action-note: iXBRL extraction + fee-model after probe)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
101	AU ASIC company register	REGISTRY	asic.gov.au register search + ASIC Connect extracts	per-company registry (+ historical extracts)	decades (registry digital-forward)	T+0-ish portal	portal + paid/bulk extracts (some bulk products)	mixed (basic free; extracts paid)	stream-02 matrix flagged SEDAR-stream GAAP-free numeric gap; ASIC extracts legiti-mate bulk but fee-classified (action-note: probe extract pricing; AU strength-ened if Norgate ASX pings improve)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
102	AU ASX Market An-nounce-ments Platform (MAP)	DISCLOSURE	market an-nounce-ments platform (asx.com.au MAP) — live listed-companies an-nounce-ments	per-announcement (ASX)	live platform era	intraday publish-ing	portal + paced fetch (no formal bulk API)	public; ToS-grey automa-tion	pairs with ASIC; an-nounce-ments = divi-dends/board/report dates relay (action-note: ASX an-nounce-ments relay agent)	W1	connect	SURVEY-EVIDENCE
103	NZ Com-panies Register (open data/API)	REGISTRY	companies-register.com (+ open-data sched-ules)	per-company registry (+ directors, share-holders)	long registry history	T+0-ish	open-data API sched-ules (free tier claim)	free (NZ Govern-ment open data)	⚠ 404 on busi-ness.govt.nz open-data path this session — locate the correct schedule page (action-note: probe; smaller stack weight)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
104	SA CIPC (ZAF company registry)	REGISTRY	cipc.co.za (register search/extract fees)	per-company registry	digital-forward	portal	fees/extracts	mixed (per-doc fees)	⚠ fetchability low — vendor-class work needed; global long-tail (action-note: skip unless SA thesis appears)	W3	connect	UNVERIFIED-CANDIDATE
105	Brazil CVM open data (dados.cvm.gov.br — incl. ITR/DFP standardized fundamentals)	REGISTRY-FILINGS	dados.cvm.gov.br bulk CSVs (ITR quarterly + DFP annual standardized fundamentals, shareholders, registry)	per-company per-period standardized CSVs	multi-year deep claim	monthly/yearly dataset releases	file-download (CSV bulk)	free / public authority	⚠ automated fetch currently 403 (bot-block this session) — use mirrors/Wayback or curl UA tuning; BR = one of the FEW free standardized non-US fundamentals (action-note: ITR/DFP + track monthly refresh cadence)	W1	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
106	MX BMV/BIVA issuer disclosures	REGISTRY-FILINGS	bmv.com.mx / biva.mx issuer pages (financial statements + relevant events)	per-issuer per-period	years-deep claim	T+0/T+1 via publish	portal + paced fetch	public filings; unfriendly bulk terms	PDF/Excel-class primary formats; Spanish (action-note: Mexico long-tail)	W3	connect	UNVERIFIED-CANDIDATE
107	IL TASE Maya disclosures	REGISTRY-FILINGS	maya.tase.co.il (mayas official disclosure portal; API reportedly usable)	per-disclosure (IL)	years-deep claim	intraday-ish	portal (+ unofficial API shape)	public disclosures; automation 403’d here (F17) — ToS-verify pace	⚠ bot-blocked this session; verify a tolerated fetch-agent path before building (action-note: probe; Hebrew/EN mixed docs)	W2	connect	SURVEY-EVIDENCE
108	SG ACRA BizFile registry	REGISTRY	acra.gov.sg BizFile+ (paid extracts for most; some open data)	per-entity registry	long	T+0-ish	paid extract fees + API-class services	paid-extract dominant	budget-gate; SGX-relay (below) covers listed side (action-note: probe cost)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
109	SGX Market An-nounce-ments	DISCLOSURE	sgx.com / SGX Net an-nounce-ments	per-announcement	years-deep	intraday	portal + paced fetch	public; ToS-grey	pairs with ACRA for the SG stack (action-note: SGX relay agent)	W2	connect	SURVEY-EVIDENCE
110	TWSE/TPEx MOPS filings	REGISTRY-FILINGS	mops.twse.com.tw / tamas (TWSE Market Observa-tion Post System)	per-filing (TW-listed)	years-deep claim	T+0/T+1	portal + paced fetch (docu-ment down-loads)	public; ToS probing needed	⚠ UI fetch blank here (JS-class; F17) — endpoint census needed; Chinese-language doc formats (action-note: MOPS relay agent; daily cadence)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
111	Turkey KAP (Public Disclosure Platform)	REGISTRY-FILINGS	kap.org.tr (EN/EN UI: notifications: Material/Financial/Regulatory/Other; downloadable financial reports 2009–2026 UI)	per-notification + per-financial-report (with detailed period/quarter filters)	2009→financial reports institutional (≈2005+ CA)	intraday-ish	portal + detailed search + report download	free / public (CMB-mandated disclosures)	⚠ VERIFIED live this session: categories incl. **Corporate Actions** + financial-statement download interface (2009–2026 quarterly/annual) (action-note: KAP ingest (Bist reports + CA events); TR = feasible free market outside majors)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
112	PL ESPI/EBI reports + KRS registry	REGISTRY-FILINGS	GPW ESPI/EBI notifica-tions + e-KRS / CEIDG registries	per-announcement + registry entries	years-deep	T+0/T+1	portal + paced fetch	public; KRS extracts fee-class	Polish-lang docs; KRS monthly extracts exist (action-note: probe; CEE slot)	W3	connect	UNVERIFIED-CANDIDATE
113	Malaysia Bursa an-nounce-ments	DISCLOSURE	bursamalaysia.com an-nounce-ments pages	per-announcement	years-deep	intraday	portal + paced	public; ToS-grey	Malaysia long-tail (action-note: probe slot only)	W3	connect	UNVERIFIED-CANDIDATE
114	Indonesia IDX an-nounce-ments / corp actions	DISCLOSURE	idx.co.id an-nounce-ments + IDX events pages	per-announcement	years-deep	intraday	portal + paced	public	Indonesian-long-tail; KSEI feeds paid (action-note: probe slot only)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
115	Compustat PIT vintages archive (NA annual 1956 base, quarterly 1976+)	ARCHIVE	S&P PIT product — historical vintage stacks (institutional procurement)	per-vintage per-firm per-period	academic archive eras back to ≈1956 (annual; quarterly 1976+)	archived vintages (static)	licensed datasets	enterprise procurement (WRDS-class access possible)	the historical PIT depth row — distinct from current-PIT subscription (#31); vintage semantics documented in WRDS notes (action-note: deep-archive purchase decision (W3 if budget-frozen))	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
116	IBES detail/vintage historical (estimate-detail history files)	ARCHIVE	LSEG IBES Detail History (daily estimate files, vintage-consistent)	per-analyst/consensus per-period	detail history back to ≈1984 (consensus ≈1976 era)	archived daily files	licensed (WRDS-class or direct)	enterprise	event-stamped estimates = vintage-native; the estimate-side archive gold (action-note: buy/WRDS decision if estimate factors enter the build)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
117	EDGAR full-history backfile (incl. pre-1993 gap handling)	ARCHIVE	www.sec.gov/Archives (1993→present machine-readable complete; full text 2001); pre-1993 = paper/film archive (vendors digitized; SEC reading room)	per-filing	full-machine 1993+; full-text 2001+; pre-1993 vendor-class	static archive	bulk archive fetch + third-party digitization	free (SEC); vendors paid for pre-1993	⚠ survivorship hole pre-1993 — pair with #121 + Compustat-PIT for as-filed older eras (action-note: build historical job; evaluate digitized-vendor pre-1993 need)	W0	backfill	VERIFIED-UPSTREAM-DOC
118	CRSP survivorship-equity database (incl. delisting returns 1926+)	ARCHIVE	CRSP Daily Stock file / CRSP-Compustat Merged (via WRDS or direct sub)	per-security per-day + CA/delist-return constructs	1926→present	daily/annual data-cycle	licensed	enterprise/academic	survivorship-gold standard incl. closed-bourse eras — complements Norgate at retail (#54) (action-note: WRDS/academic route pricing decision)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
119	Moody’s Manuals / Poor’s Register (pre-1956)	ARCHIVE	archive.org + HathiTrust scans (Moody’s Industrial/Utility/Manuals, Poor’s yearbooks)	per-company per-year page scans	pre-1956 fundamentals/officer data eras	static archive	OCR/curation pipeline	copyright: pre-1928 public domain; Depression-era mostly PD or controlled once cleaned	careful copyright-class; OCR-accuracy gates; heavy curation labor (action-note: long-tail deep-history only if a thesis demands)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
120	WRDS-hosted mirrors (Compustat/IBES/CRSP/Ent class)	ARCHIVE ACCESS	wharton WRDS subscription (per-dataset entitlements incl. PIT archives)	per-dataset institution-level	dataset-native depths (parent rows above)	per-dataset refresh	WRDS web/SAS/STATA/API/query	academic/subscription (internal use)	cheapest bona-fide route to enterprise PIT archives if an academic or institutional relationship exists; streaming-redelivery NOT permitted (action-note: negotiate/decide route (likely unlocks #115/#116/#118 cheaply))	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
121	SEC pre-1993 historical (digitized filings vendors + SEC reading room)	ARCHIVE	vendors (e.g., Historical-Filings-digitizers) + SEC reading-room microfilm	per-filing scans/text	1934→1993 (SEC era machine-originated 1993+)	static	vendor bulk (paid) + on-site	paid vendor / library	survivorship hole pre-1993; cost/deliverability to verify vendor pricing (action-note: vendor scope if deep pre-1993 backtests enter proof-of-concept)	W3	quote	UNVERIFIED-CANDIDATE
122	ESEF historical iXBRL corpus across EU registries	ARCHIVE	national ESEF reposito-ries + ESMA ESEF filings map (per-country fetch layer on #74 route table)	per-filing iXBRL	ESEF era ≈2020→	T+0–T+1 by registry	per-registry fetch + iXBRL parse	mixed (most NSRs free)	iXBRL tag hetero-geneity across national tax-onomies (hyper-local exten-sions) (action-note: one + per-country adapters)	W2	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	OPRA real-time feed (Cboe Pillar/NYSE etc. consolidated)	options-feed	OPRA plan via member marketplace [29]	tick (MBO)	n/a (live)	realtime	feed-handler	exchange-fee (display + usage license quote)	vendor/non-display gating; audit consumption (action-note: quote only; do not capture live until scale justified)	W3	license-review	SURVEY-EVIDENCE
2	Databento OPRA.PILLAR options trades/quotes/TBBO/OHLCV/definitions	options-feed	Databento OPRA.PILLAR (Cboe PCAP relay) [6]	tick + daily bars	per dataset-range (confirm earliest date)	near-realtime + batch	websocket + S3 batch	vendor-tier (metered)	confirm historical instrument-definition continuity & range start (action-note: get range metadata (`get_dataset_range`) + unit pricing)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	Databento DEFINITIONS schema incl. options numeric fields	symbology	Databento definitions/shemas [6]	daily	full vendor history	batch	S3 batch	vendor-tier	verify option-specific fields (strike/bid low limits etc.) (action-note: pull definitions snapshot for options universe)	W1	connect	SURVEY-EVIDENCE
4	Polygon.io options aggregates/trades/quotes	chains	polygon.io REST [19]	tick + daily (+intraday tier)	vendor-limited window (confirm)	tier-dependent; delayed on lower tiers	REST SDK	vendor-tier (tier gates history+realtime)	PIT on splits/adjustments; rate limits (action-note: tier pricing vs depth; sample SPX chain)	W1	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	Cboe DataShop Option EOD Summary (quotes/OHLC/vol + 3:45pm snapshot, optional Calcs IV/Greeks)	chains	datashop.cboe.com/option-eod-summary [9]	daily + 3:45pm ET	shopping-cart dependent (deep multi-year possible EOD)	EOD	web download/SFTP	vendor-tier per-file	optional Calcs only when purchased; snapshot vs close semantics (action-note: ingest all-equity + index EOD history)	W1	connect	VERIFIED-UPSTREAM-DOC
6	Cboe DataShop Option Quote Intervals (1-min / n-min NBBO w/ size + optional OI, IV/Greeks)	chains	datashop.cboe.com/option-quote-intervals [9]	1-min / n-min intervals	multi-year per symbol	EOD	download/SFTP	vendor-tier per-file	NBBO from ops experts; OI timing (action-note: purchase sample month across SPX/SPY/AAPL)	W1	connect	VERIFIED-UPSTREAM-DOC
7	Cboe DataShop Option Trades (daily prints + contextual market info, optional IV/Delta)	chains	datashop.cboe.com/option-trades [9]	tick / daily file	multi-year	EOD	download/SFTP	vendor-tier per-file	trade+quote context alignment (action-note: sample file, parse to parquet)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
8	Cboe DataShop Enhanced US Options Trade-by-Trade (full OPRA-level execution detail, all exchanges)	OPRA-lite	datashop.cboe.com/enhanced-us-options-trade-by-trade-execution-detail [9]	tick	vendor-stated history depth (confirm)	EOD	download/SFTP	enterprise-quote	consolidation delays vs OPRA; NBBO reconstruction (action-note: request for (a) 1 y + (b) deep history)	W2	quote	VERIFIED-UPSTREAM-DOC
9	Cboe DataShop Open-Close Volume Summary (Customer/Pro/Firm/MM capacity, 10-min buckets)	flows	datashop.cboe.com/cboe-options-open-close-volume-summary [9]	10-min / EOD	multi-year	EOD	download	vendor-tier per-file	Cboe-only venue scope (BZX,C1,C2,EDGX) not All-OPRA (action-note: ingest for flow studies)	W1	connect	VERIFIED-UPSTREAM-DOC
10	ORATS Options Data API (EOD surface, IV30, SG, vol-forecast)	vol-surface	orats.com/data-api [8]	daily	2007 →	EOD	REST/library	vendor-tier	SV backfill consistency; earnings events (action-note: tier list & v2 endpoints)	W1	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	ORATS Intraday Data API (1-min options, 5,000+ symbols)	vol-surface	orats.com/intraday-data-api [7]	1-min	2020-08 →	intraday + delayed + historical	REST/cloud delivery	vendor-tier	$496/mo tier price not con-firmed (survey shows other tiers) (action-note: $496 tier & coverage scope)	W2	verify	VERIFIED-UPSTREAM-DOC
12	IvyDB US (option EOD prices, IV surfaces, greeks)	vol-surface	OptionMetrics [10]	daily	1996 →	EOD	database/bulk-export	enterprise-quote	standardized rate/dividend assump-tions; corporate-action handling (action-note: quote + sample quarter)	W2	license-review	VERIFIED-UPSTREAM-DOC
13	IvyDB US Intraday (on Snowflake)	vol-surface	OptionMetrics / Snow Market-place [10]	intraday	recent window (confirm)	intraday	Snowflake data share	enterprise-quote	snapshot cadence & field list (action-note: confirm coverage start & pricing)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
14	IvyDB Futures (US futures options), IvyDB Europe/Asia/Canada	vol-surface	OptionMetrics [10]	daily	US futures + regional ~2010s (confirm per-region)	EOD	database	enterprise-quote	per-venue availability dates vary (action-note: region-by-region availability matrix)	W3	connect	VERIFIED-UPSTREAM-DOC
15	LiveVol (Cboe) Pro & volume-premium datasets	chains	livevol.com (Cboe)	tick/1-min	~2006 →	delayed/real-time tiers	platform + API	vendor-tier	product line post-Cboe acq; endpoint reachability failed in survey (action-note: price quote; SKU list)	W2	verify	SURVEY-EVIDENCE
16	CBOE VIX index history + VX futures daily settlement term structure files	vol-index	cboe.com index & VX settlement data	daily	VIX 1990 →; VX futures 2005/2007 →	EOD	free web files	exchange-free (attribution)	settlement-vs-close skew (action-note: ingest monthly archives)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
17	CBOE market-place daily statistics (vol-ume/OI by exchange & product)	vol-stat	cboe.com daily market statistics	daily	multi-decade	EOD	free download	exchange-free	cross-check vs OCC totals (action-note: ingest)	W0	connect	SURVEY-EVIDENCE
18	OCC volume & open-interest reports (daily/weekly/monthly/historical statistics)	reference	theocc.com market-data-reports [15]	daily + monthly + annual	multi-decade	EOD / weekly	web CSV	exchange-free	OI snap-shots not PIT restated (action-note: ingest)	W0	connect	VERIFIED-UPSTREAM-DOC
19	Chain/vendor fallback option chains (Tradier / theta-data free tiers)	chains	vendor REST	tick/daily	vendor windows	varies	REST SDK	freemium	not primary source; PIT gaps (action-note: keep logged as fallback only)	W1	connect	SURVEY-EVIDENCE
20	CBOE foreign-exchange-style deriva-tive securities (FX chains on listed se-curities)	fx-options	cboe listings	daily	?	EOD	web	exchange-free	naming changes (action-note: confirm whether actually needed before ingest)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
21	CME MDP3 Globex realtime feed (raw L3)	futures-feed	CMEMDP MDP3	tick MBO	n/a (live)	realtime	feed-handler (SBE)	exchange-fee (usage-based)	colo/non-colo sizing; entitle-ment audit (action-note: only if vendor feed inad-equate)	W3	connect	SURVEY-EVIDENCE
22	CME MDP3 via Databento GLBX.MDP3 — equity-index (ES/NQ/YM + spreads)	futures-feed	Databento GLBX.MDP3 [6]	tick + daily	per dataset-range (confirm earliest)	near-realtime + S3	websocket + S3 batch	vendor-tier	symbol-map & roll events; OI/settle stats schedule (action-note: pricing + full range test)	W1	connect	VERIFIED-UPSTREAM-DOC
23	CME MDP3 via GLBX — rates (SOFR 1m/3m, ZT/ZN, incl. SONIA-type SR3)	futures-feed	GLBX.MDP3 [6]	tick + daily	as row 22	near-realtime + S3	websocket + S3	vendor-tier	term-structure continu-ity (action-note: same plan probe as row 22)	W1	connect	VERIFIED-UPSTREAM-DOC
24	CME MDP3 via GLBX — energy (CL/NG/HO/RB)	futures-feed	GLBX.MDP3 [6]	tick + daily	as row 22	same	websocket + S3	vendor-tier		W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
25	CME MDP3 via GLBX — metals (GC/SI/HG)	futures-feed	GLBX.MDP3 [6]	tick + daily	as row 22	same	websocket + S3	vendor-tier		W1	connect	VERIFIED-UPSTREAM-DOC
26	CME MDP3 via GLBX — agricul-ture (ZC/ZS/ZW/CT/OJ)	futures-feed	GLBX.MDP3 [6]	tick + daily	as row 22	same	websocket + S3	vendor-tier		W1	connect	VERIFIED-UPSTREAM-DOC
27	CME MDP3 via GLBX — FX (6E/6B)	futures-feed	GLBX.MDP3 [6]	tick + daily	as row 22	same	websocket + S3	vendor-tier		W2	connect	VERIFIED-UPSTREAM-DOC
28	CME MDP3 via GLBX — crypto futures (MBT/MET/SBT etc.)	futures-feed	GLBX.MDP3 [6]	tick + daily	contract-by-contract since listing	same	websocket + S3	vendor-tier		W2	connect	VERIFIED-UPSTREAM-DOC
29	CME options-on-futures chains (incl. de-ferred contract detec-tion)	options-on-futures	GLBX.MDP3 + defini-tions [6]	chain daily	as rows 22-28	near-realtime	S3	vendor-tier	per-contract expiry rollover (action-note: chain render for vol-through-futures)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
30	CME volume/OI/settlement statistics (statistics schema, publication schedule)	OI-settlement	GLBX statistics via Databento [6]	daily	years	EOD per schedule	S3/REST	vendor-tier	verify publication timestamps (action-note: ingest daily OI/settlement)	W0	connect	VERIFIED-UPSTREAM-DOC
31	CME DataMine bulk historical (per-product subsets)	archive	cmegroup.com CME DataMine	tick/daily varies	varies by product	batch	vendor-bulk	enterprise-quote	redistribution restrictions (action-note: request for equities + energy deep history)	W2	quote	SURVEY-EVIDENCE
32	CME curve/term-structure via vendor (continuous contracts, back-adjustment modes)	curve	Databento symbology/continuous [6]	daily	decades	EOD	S3	vendor-tier	roll-convention pitfall (volume/OI switching) (action-note: build term structure + roll feature)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
33	CME SOFR futures transac-tions public dataset (free subset)	rates	CMES / cme free data portal	daily/tick	?	?	?	exchange-free?	? (action-note: confirm existence & route)	W3	connect	UNVERIFIED-CANDIDATE
34	ICE Europe Com-modities (IFEU) via Databento	futures-feed	Databento ifeu-impact [6]	tick + daily	vendor range (confirm start)	near-realtime	websocket + S3	vendor-tier	verify symbol continu-ity (action-note: pricing & range test)	W2	connect	VERIFIED-UPSTREAM-DOC
35	ICE Futures US (IFUS: cot-ton/coffee/SB) via Databento	futures-feed	Databento ifus-impact [6]	tick + daily	vendor range	near-realtime	websocket + S3	vendor-tier		W2	connect	VERIFIED-UPSTREAM-DOC
36	ICE Europe Fi-nancials (IFLL) via Databento	futures-feed	Databento ifll-impact [6]	tick + daily	vendor range	near-realtime	websocket + S3	vendor-tier		W2	connect	VERIFIED-UPSTREAM-DOC
37	ICE Endex (NDEX gas/power) via Databento	futures-feed	Databento ndex-impact [6]	tick + daily	vendor range	near-realtime	websocket + S3	vendor-tier		W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
38	ICE Pet-Connect (real-time suite feed)	feed	ice.com [27]	tick	n/a	realtime	feed	exchange-fee	naming/access to confirm (action-note: confirm current product name & access path)	W3	connect	UNVERIFIED-CANDIDATE
39	Eurex T7 market data via Databento XEUR.EOBI	futures-feed	Databento xeur-eobi [6]	tick + daily	vendor range (EOBI start confirm)	near-realtime	websocket + S3	vendor-tier	verify DAX/Stoxx chains (action-note: pricing & range test)	W2	connect	VERIFIED-UPSTREAM-DOC
40	Eurex historical daily data (official archive)	archive	eurex.com data services [26]	daily seg-ments	1998 → (Eurex in-ception)	EOD	download	exchange-fee	per-market-segment files; splits vs consoli-dated (action-note: request & earliest-segment samples)	W2	quote	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
41	DCE daily futures stats (OHLCV + OI per contract)	futures-feed	dce.com.cn daily files	daily	years (per product inception)	T+1 (published next morning)	web download	exchange-free (attribution)	OI is session-final; grain conventions differ (action-note: free-file ingest)	W0	connect	SURVEY-EVIDENCE
42	CZCE daily futures/options stats	futures-feed	czce.com.cn daily files	daily	per-product inception window	T+1	web	exchange-free	per-contract queries (action-note: free-file ingest)	W0	connect	SURVEY-EVIDENCE
43	GFEX daily stats (lithium carbonate / industrial silicon futures/options)	futures-feed	gfex.com.cn daily files	daily	since contracts started (2022-23)	T+1	web	exchange-free	new exchange; format changes (action-note: free-file ingest)	W0	connect	SURVEY-EVIDENCE
44	SHFE + INE daily stats (copper/crude/rubber etc.)	futures-feed	shfe.com.cn daily files	daily	years	T+1	web	exchange-free	includes INE under same open-fit pattern (action-note: free-file ingest)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
45	CFFEX daily futures/options stats (bond/index derivatives)	futures-feed	cffex.com.cn daily files	daily	since bond/index product launches (2010s)	T+1	web	exchange-free	dividend-adjusted on main contracts (action-note: free-file ingest)	W1	connect	SURVEY-EVIDENCE
46	SGX iron-ore / freight derivatives actual & EOD data	futures-feed	sgx.com derivatives iron-ore page [18]	daily	multi-year (per contract)	EOD	download	exchange-fee (SDE)	page reached only as browser shell in survey — extract file format (action-note: request sample file & fees)	W1	connect	SURVEY-EVIDENCE
47	LME inventories + daily prices / SDE statistical data	metals	lme.com market-data + licensing pages [12,13]	daily	decades of official data (1980s+)	EOD	SFTP/LME portal	exchange-fee (license quote)	redistribution re-stricted; 15-month averaging rare (action-note: obtain quote via market-data-licensing page)	W2	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
48	KRX KOSPI200 futures/options (data.krx)	index-derivs	data.krx.co.kr [21]	daily	KOSPI200 options 1997-ish onwards	EOD/T+1	web/API (KRD product export)	exchange-free? (some paid products)	Korean encoding; file-format changes (action-note: download route + product names)	W1	verify	SURVEY-EVIDENCE
49	HKEX derivatives statistics (consolidated daily reports incl. options OI by strike)	index-derivs	hkex.com.hk market data [20]	daily	~2000s →	T+1	web	exchange-free	stock options vs index-option separation (action-note: ingest)	W1	connect	SURVEY-EVIDENCE
50	TAIFEX Taiwan futures/options daily data	index-derivs	taifex.com.tw [22]	daily	1998 →	T+1	web (TAIFEX files)	exchange-free	naming locale quirks (action-note: ingest)	W1	connect	SURVEY-EVIDENCE
51	B3 (Brasil) derivatives data	futures	b3.com.br	daily	long archive	EOD	web	exchange-free	B3 portal layout changes (action-note: defer)	W3	connect	SURVEY-EVIDENCE
52	DGCX (Dubai) futures data	futures	dgcx.ae	daily	multi-year	EOD	web	exchange-free	(action-note: defer)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
53	Euronext derivatives (EURIBOR/Stoxx legacy LIFFE) data	rates-index	euronext.com	daily	decades	EOD	web	exchange-fee	(action-note: defer)	W3	connect	SURVEY-EVIDENCE
54	DTCC GTR Public Reports (US OTC derivatives: IRS/CDS/FX/Equity classes)	regulatory	dtcc.com GTR public reports [16,17]	transaction/aggregated	2015 →	T+0/T+1 publication	web + API endpoints	public (free subsets)	only public subset; trade-level filing is restricted (action-note: ingest public report catalog)	W1	connect	VERIFIED-UPSTREAM-DOC
55	DTCC GTR other jurisdictions (EU/HK/Japan/Australia public subsets)	regulatory	GTR publications [17]	aggregated	varies	T+1	web	public minimum bars	per-jurisdiction scope varies (action-note: enumerate per-jurisdiction availability)	W2	connect	SURVEY-EVIDENCE
56	ESTR futures (Eurex)	rates	eurex.com products	daily	2019 →	EOD	web/download	exchange-fee	(action-note: confirm product & file access)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
57	EURIBOR futures (Euronext/LIFFE legacy) + STIR futures touchstones	rates	euronext.com market data	daily	decades	EOD	web	exchange-fee	(action-note: defer)	W3	connect	SURVEY-EVIDENCE
58	CFTC COT legacy Futures-only archive	regulatory	CFTC Historical-Compressed [11]	weekly (per commodity)	1986 →	T+3 (week after Tue)	zip/txt download	public/free	grain-in-bushels until 1998 (action-note: ingest full archive)	W0	connect	VERIFIED-UPSTREAM-DOC
59	CFTC COT legacy futures-and-options combined archive	regulatory	CFTC Historical-Compressed [11]	weekly	combined availability start is earlier than futures-only in places (confirm exact per-commodity boundary)	T+3	zip	public/free	combined vs futures-only OI difference (action-note: ingest; combined start year)	W0	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
60	CFTC Disaggregated COT (futures-only + combined; per-commodity fences)	regulatory	CFTC Historical-Compressed [11]	weekly	2006-06-30 → (files from 2009+ report years; 2006-2016 backfill zip)	T+3	zip	public/free	only reporting-funded universe (action-note: ingest)	W0	connect	VERIFIED-UPSTREAM-DOC
61	CFTC Traders-in-Financial-Futures (TFF) futures-only & combined	regulatory	CFTC Historical-Compressed [11]	weekly	2006 → (2010-07 file boundary in some; confirm)	T+3	zip	public/free	dealer vs LSP vs AR vs OR classification (action-note: ingest)	W0	connect	VERIFIED-UPSTREAM-DOC
62	COT Public Reporting Environment (Socrata API)	regulatory	publicreporting.cftc.gov (Socrata)	ingestable	see rows 58-61	T+3	REST (Socrata/Socrata open-data)	public	API pagination (action-note: bulk pulls via Socrata endpoints)	W0	connect	VERIFIED-UPSTREAM-DOC
63	BIS FX derivatives pivot (FX swaps, forwards, NDF, CCS notional outstanding)	macro-derivs	bis.org statistics	quarterly (+ triennial detail)	1998 →	T+months	web CSV/Excel	public	reporting-dealer basis; geog breakouts (action-note: ingest quarterly tables)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
64	BIS OIS / interest-rate deriva-tives pivot (IR swaps & options outstand-ing)	macro-derivs	bis.org statistics	quarterly	1998 →	T+months	web	public	(action-note: ingest)	W1	connect	SURVEY-EVIDENCE
65	NDF & FX-forward quotes (vendor)	fx-derivs	vendor-bulk	daily quotes	vendor window	EOD	vendor-bulk	vendor-tier	corresponds with 06-vendor diligence stream (action-note: pick vendor after stream 06)	W3	connect	SURVEY-EVIDENCE
66	BIS global-liquidity / FX-swaps outstand-ing series	fx-derivs	bis.org statistics	quarterly	decade	T+months	web	public	(action-note: ingest)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
67	Tardis.dev per-exchange crypto tick history (incl. Deribit options, delisted exchanges: BitMEX/FTX)	crypto-tick	tardis.dev HTTP/replay [1,2]	tick (100ns ts)	per-exchange (many since ~2017; Deribit-collection start later than exchange inception — confirm per exchange)	T-6min	HTTP + python/node libs	vendor-tier (subscription)	incidents published; resubscription gaps (action-note: org pricing + per-exchange earliest-date pull)	W1	connect	VERIFIED-UPSTREAM-DOC
68	Tardis.dev downloadable CSVs (S3)	crypto-tick	tardis.dev CSV service [2]	tick	as row 67	batch	S3	subscription	normalization re-derivable (action-note: cost compare vs API)	W1	connect	VERIFIED-UPSTREAM-DOC
69	Deribit realtime API (WS/REST/FIX + Starbase SBE multicast for colo)	crypto-derivs	docs.deribit.com [14]	tick	n/a (live)	realtime	WS/FIX/SBE	exchange-free (rate limits)	inception 2016 (action-note: capture later; docs verified)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
70	Deribit DVOL volatility index + historical (exchange inception ~2016)	vol-index	Deribit API / iv-index [14]	daily	confirm DVOL history depth (inception claim)	EOD + realtime	REST	exchange-free	DVOL chain composition (30d) (action-note: DVOL retrieval endpoint + history start)	W0	verify	SURVEY-EVIDENCE
71	Bybit derivatives historical (funding/OI/mark/liquidations)	crypto-derivs	api-docs.bybit.com [25]	endpoint-dependent	per product inception	EOD+realtime	REST	exchange-free	per-venue O42/liq-rules differences (action-note: REST ingest)	W0	connect	SURVEY-EVIDENCE
72	OKX derivatives historical (funding/OI/mark/liquidations)	crypto-derivs	okx.com docs-v5 [24]	endpoint-dependent	per product	EOD+realtime	REST	exchange-free	funding interval changes (action-note: REST ingest)	W0	connect	SURVEY-EVIDENCE
73	Binance derivatives historical (USDT-M/COIN-M futures + European options; funding/OI/mark/liq)	crypto-derivs	developers.binance.com derivatives [23]	endpoint-dependent	long histories via API sub-archive	realtime+bars	REST	exchange-free	DELISTED-series handling (action-note: REST ingest)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
74	Coinglass OI dataset (per-pair & aggregated OHLC history incl. stable/coin-margin splits)	aggregator	coinglass API [3]	OHLC hist per pair	multi-year via API	intraday	REST	vendor-tier	aggregation across venues (action-note: ingest)	W1	connect	VERIFIED-UPSTREAM-DOC
75	Coinglass funding-rate dataset ((hist + OI-weight + vol-weight + exchange lists)	aggregator	coinglass API [3]	OHLC hist	multi-year	intraday	REST	vendor-tier	cross-exchange coverage (action-note: ingest)	W1	connect	VERIFIED-UPSTREAM-DOC
76	Coinglass liquidation dataset (pair/coin history, heatmaps, 7d orders, max-pain)	aggregator	coinglass API [3]	event+aggregate	historical liquidations	near-realtime	REST	vendor-tier	heatmap models are derived (not actual fills) — PIT not guaranteed (action-note: ingest actual liq events only for PIT)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
77	Coinglass L/S ratios, taker flow, CVD, futures netflow	flows	coinglass API [3]	multi-frequency	multi-year	near-realtime	REST	vendor-tier	Top-Trader subsets (action-note: ingest)	W1	connect	VERIFIED-UPSTREAM-DOC
78	Coinglass Hyperliquid whale/position distribution endpoints	flows	coinglass API [3]	wallet-level	recent	realtime	REST	vendor-tier	(action-note: ingest)	W2	connect	VERIFIED-UPSTREAM-DOC
79	Amberdata derivatives (options surface / DVOL / OData API)	crypto-opts	amberdata.io	tick/daily	vendor archive	intraday	API	enterprise-quote	(action-note: ; de-prioritize vs Tardis)	W3	quote	SURVEY-EVIDENCE
80	Laevitas derivatives analytics	crypto-opts	laevitas.ch	daily	vendor archive	intraday	API	vendor-tier		W3	quote	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
81	Pyth Pro price feeds (incl. commodity futures symbols like WTI, plus equities/crypto — subscription product, prev. Lazer)	oracle	docs.pyth.network [4]	tick	via History API	sub-second (configure per subscription)	WS + REST + compass	subscription-gated	verify per-feed latency & symbol set vs VTL? (action-note: confirm feed IDs + history API depth)	W1	connect	VERIFIED-UPSTREAM-DOC
82	Chainlink Data Streams (crypto/forex out-of-band oracle feeds)	oracle	docs.chain.link/data-streams [5]	tick	n/a (live)	~sub-second (verify page spec)	WS/REST	subscription-gated	latency claims not read from page (fetch returned sparse) (action-note: pull Streams latency spec + pricing)	W2	connect	VERIFIED-UPSTREAM-DOC
84	WFE derivatives volume statistics (regional annual)	reference	focus.world-exchanges.org [28]	annual/monthly	decades	T+months	web	public/free	scope includes futures+options (action-note: ingest as context metric)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
85	Exchange market-data auctions (e.g., OCC data-sale? venues auction OPRA-class rights)	meta	various (OCC/venedie?)						(action-note: research actual existence & mecha-nism)	W3	connect	UNVERIFIED-CANDIDATE
1	ecb-fx-ref	fx-ref	ECB Data Portal EXR family (SDMX 2.1/3.0 CSV+JSON REST, keyless)	daily 16:00 CET fix	1999-01-04→; ≈30-currency set (USD JPY GBP CHF CNY AUD BRL CAD HKD IDR KRW MXN MYR NZD PHP SGD THB ZAR + EU non-euro)	D0 fix	SDMX REST; Frank-furter fail-over	free-attribution	official ECB fix; No free intraday leg any-where (gap — intraday = licensed rows 20/21) (action-note: build fetcher; 1999→)	W0	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
2	frankfurter	fx-ref	Frankfurter v2 REST `api.frankfurter.dev` — 223 curren-cies / 105 official providers; blended + per-provider routes; CSV/NDJSON; Docker self-host	daily (some provider-series weekly/monthly)	1948→ (provider-dependent)	daily	REST keyless; no quotas; self-host	open (blended = Frank-furter calc, free incl. com-mercial); per-provider route = provider terms	compliance → always filter `providers=ecb` etc.; blended recalcu-lated as data arrives → pin date+provider key for PIT (action-note: integrate as unified FX spine + provider-attribution log)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	fed-h10	fx-fix	Fed H.10/G.5 Foreign Exchange Rates (`releases/h10/current` + DDP; Nominal/Real Indexes + Currency Weights sub-pages)	daily (G.5 weekly)	1971→ (per-currency start varies)	T+1 ~16:00 ET	DDP CSV/JSON/XML	US-gov public	NY noon buying rates ≠ market close/4pm fix; single-price statistic (no bid/ask); PIT = release calendar (action-note: build fetcher (H.10 + G.5))	W0	connect	VERIFIED-UPSTREAM-DOC
4	boe-fx	fx-fix	BoE Statistical Interactive Database (`/statistics/exchangerates`; boeapps CSV endpoint)	daily + monthly averages	GBP/$ daily 1975→ (verify per-series starts)	T+1	boeapps CSV URL builder	UK OGL attribution	long monthly averages pre-1975 into historical runs — verify coverage at build (action-note: daily + monthly)	W1	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	boj-ttss	fx+rates	BoJ Time-Series Statistics Search (TTSS, SDMX 2.1)	daily/monthly	FX + rate panels decades (1974 FX leg — verify)	T+1	TTSS SDMX/CSV catalog	BoJ stat-use terms (no open redistribution)	statutory re-use agreement required before redistribution (action-note: parse TTSS catalog; JPY set)	W1	connect	UNVERIFIED-CANDIDATE
6	rba-f11	fx-fix	RBA Statistical Tables F11/F11.1 (`/statistics/tables/csv/*.csv` direct endpoints + xls)	daily (F11.1) / monthly (F11)	AUD/USD since Dec-1983 float; monthly deeper (verify per-table)	D+1 16:10 AEDT	direct CSV/xlsx GET, keyless	RBA fair-use attribution	RBA publishes series-break + discontinued pages officially; revisions in release notes (action-note: ingest csv endpoints +)	W0	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
7	boc-valet	fx-fix	Bank of Canada Valet API (`/valet/observations/FXUSDCAD`, JSON/CSV)	daily	1971-01→ (verify)	T+1	Valet REST, keyless	BoC terms attribution	Valet page returned empty twice on 2026-10-24 fetch — confirm keyless route live before reliance; CORRA piggybacks same rail (action-note: probe API; CAD sets)	W1	backfill	UNVERIFIED-CANDIDATE
8	snb-fx	fx-fix	SNB data portal (data.snb.ch CSV/ZIP + REST candidates)	daily + monthly avg	CHF reference daily; monthly averages long-history (verify)	T+1	CSV/XLSX download + API candidate	SNB terms (attribution)	verify depth claim before storing long-run monthly averages (action-note: map SNB topic codes)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
9	hkma-fx	fx-fix	HKMA statistics (USD/HKD spot fixings, HIBOR fixings)	daily	2005?→ verify	D+0	web CSV/XLSX	HKMA terms	fixing publica-tion conven-tion to verify; HIBOR tenor panel check (action-note: loop CSV; HKD/HIBOR legs)	W2	connect	UNVERIFIED-CANDIDATE
10	bok-ecos	fx+rates	BoK ECOS database (ecos.bok.or.kr JSON API + CSV)	daily/monthly	KRW FX + rates decades (verify)	D+1	ECOS JSON API	BoK terms	unofficial English docs; series-code mapping community-driven (action-note: probe ECOS series)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	tsy-rates-of-exchange	fx-monthly	Treasury Fiscal Data v1/accounting/od/rates_of_exchange (docs example verified)	quarterly in docs example (periodicity verify)	accounting-rates span (verify)	periodic	REST JSON/CSV/XML keyless	US-gov public — no restrictions	USG accounting rates, NOT market quotes — exclude from benchmark panels; cross-check rail only (action-note: ingest for cross-check panel)	W0	connect	VERIFIED-UPSTREAM-DOC
12	imf-er	fx-monthly	IMF IFS/ER exchange-rate family (data.imf.org SDMX + DataMapper)	monthly	1957→ (IFS print era)	monthly/quarterly	SDMX + DataMapper API	IMF free attribution	representational rates regime-dependent (fix vs float encoding) — country use-tags needed (action-note: ingest ER family)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
13	bis-eer	fx-index	BIS Effective Exchange Rates (data.bis.org + `/bulk-download`; SDMX API v2)	monthly + daily (publishes daily EER per docs)	narrow 26/27-ccy 1964→; broad 64-ccy 1994→; 2020=100	latest period live	SDMX + bulk-download xlsx/csv	BIS permitted-use attribution	3-yr rolling trade weights documented (pre-1990 narrow back-ext uses 1990-92 weights); base switch = break to log (action-note: ingest daily+monthly EER set)	W0	connect	VERIFIED-UPSTREAM-DOC
14	bis-triennial-fx	fx-vol·archive	BIS Triennial FX Survey datasets (rpfx editions; coordinated with 52 jurisdictions)	every 3 yrs (April)	FX leg since 1986 (IR-derivatives leg since 1995); Apr-2022 global $7.5tn/day	triennial	edition XLSX/CSV + PDF release	BIS permitted use	gauge/coverage changes between editions; not time-series-comparable bin-to-bin (methodology notes per edition) (action-note: archive all editions ≤ latest)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
15	dukascopy-bi5	fx-tick	Dukascopy historical-data feed (public per-instrument per-hour `.bi5` LZMA files)	tick ms bid/ask (+vol)	~2003→ majors	archived intraday	raw-HTTP URL template + LZMA decode → parquet	research/no-resale (licence text to confirm)	page fetch 2026-10-24 rendered nav-only — confirm bi5 URL template + DST/timezone conven-tion at build (action-note: crawler + LZMA→parquet)	W1	connect	UNVERIFIED-CANDIDATE
16	truefx-ticks	fx-tick	TrueFX historical down-loads (monthly zips; Integral Develop-ment Corp)	tick ms top-of-book bid/ask	16+ majors, 2009→	monthly zips	registration → ZIP fetch	free/research; redistri-bution prohib-ited (Integral ©2009–2026)	indicative interbank aggre-gate — not executed tape; gap-audit each month-zip (action-note: bulk month-zips + gap audit)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
17	histdata-fx	fx-tick	HistData.com per-pair M1 + tick (1s resolution) ASCII CSV zips	M1 / tick-1s	2000s→ (per-pair varies)	monthly (last update 2026-10-18 per site)	web zips; FTP/SFTP $7/mo via PayPal; GoogleDrive auto-update $7/mo	free-attribution; paid rails voluntary	T1/T2 bid-ask split conventions; volume fields unreliable; treated as approximation, not official tape (action-note: majors + monthly diff job)	W0	backfill	VERIFIED-UPSTREAM-DOC
18	stooq-fx	fx-eod	Stooq plain CSV endpoints (`stooq.com/q/d/l/`)	daily	decades per pair	D+0 EOD	plain CSV GET, keyless	undocumented — no explicit licence found	unverified upstream; do not use before licence path confirmed (action-note: licence check → then)	W2	backfill	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
19	oanda-rates	fx-eod	OANDA v20 Rates API (practice account) + FXDS paid service (32y/38k-pairs vendor claim)	daily/hourly aggre-gates	free-tier recent; paid deep (32y claim)	near-realtime (live endpoint)	REST v20	contractual; paid fixings licence-gated	indicative quotes ≠ official fix; quote-age = PIT marker (action-note: practice rail now; FXDS RFQ later)	W1	connect	SURVEY-EVIDENCE
20	wm-4pm-fix	fx-fix	LSEG WM/Reuters 4pm London fixings (+ intraday hourly markers)	daily + intraday	1990s?→ verify	D0 16:00 London	LSEG feed/API	licensed quote	THE insti-tutional buy-side bench-mark; BMR referen-cability rules — audit trail required (action-note: RFQ scope G10+ majors)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
21	bloomberg-bfix	fx-fix	Bloomberg BFIX bench-mark (New York 4pm et al.)	daily fix	2016?→ verify	D0	Bloomberg feed/library	licensed quote	BFIX method-ology docs; NDF coverage check at RFQ (action-note: RFQ parallel w/ [20])	W3	connect	SURVEY-EVIDENCE
22	cls-data	fx-vol	CLS settle-ment/volume data (ag-gregate CSV/PDF; granular via CLS)	monthly	2002?→ verify	monthly	web CSV aggre-gates	free ag-gregates; granular licensed quote	CLS-covered turnover only — NOT com-parable to BIS triennial totals (action-note: aggre-gates first; granular later RFQ)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
23	sofr-nyfed	rates-fix	NY Fed SOFR + SOFR Averages & Index + BGCR + TGCR	daily	2018-04-02→	T+1 ~08:45 ET	CSV/XLSX release pages	US-gov public	T+1 publication calendar; corrections/re-publications logged; first-print = PIT value (action-note: ingest all 4 rails)	W0	connect	VERIFIED-UPSTREAM-DOC
24	effr-obfr	rates-fix	NY Fed EFFR/OBFR (+ FRED mirrors)	daily	EFFR 1954→	T+1	NY Fed CSV + FRED API	US-gov public	floor-regime shifts (2008+, 2021) as break-tags (action-note: ingest)	W0	connect	SURVEY-EVIDENCE
25	estr-ecb	rates-fix	ECB €STR (data portal, keyless)	daily	2019-10-01→	T+1 08:00 CET	SDMX REST	free-attribution	€STR = blended unsecured euro O/N; volume metric co-published (action-note: ingest)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
26	sonia-boe	rates-fix	BoE SONIA (IUDSOIA) + SONIA Compounded Index (boeapps CSV, free)	daily	Mar-1997→; BoE admin since 2016; reformed 2018; Compounded Index from 2020-08-03	T+1 10:00	boeapps CSV, free	free (administered benchmark)	pre-2016 RFR = legacy methodology (break tag); pre-2018 reform breaks tagged (action-note: 1997→ + compounded index)	W0	backfill	VERIFIED-UPSTREAM-DOC
27	saron-snb	rates-fix	SNB/SIX SARON	daily	2009?→ verify	T+1 10:00	SNB web (free display)	free reference value; SIX DB licensed for depth	free route may lag SIX history — SWISS arms-length check (action-note: free daily route now; SIX later)	W1	connect	UNVERIFIED-CANDIDATE
28	tona-boj	rates-fix	BoJ TONA (uncollateralised O/N call)	daily	2008?→ verify	T+1	BoJ TTSS/fin-market CSV	BoJ terms	history/availability verify at build (action-note: map TONA code)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
29	corra-boc	rates-fix	BoC CORRA + CORRA index (Valet rail)	daily	2018-07→	T+1 ~12:00 ET	Valet REST	BoC terms	same Valet de-pendency as [7] (action-note: piggy-back Valet probe)	W1	connect	UNVERIFIED-CANDIDATE
30	sora-mas	rates-fix	MAS SORA (SG)	daily	SORA comp. index 2016?→ verify	T+1	MAS web CSV	MAS terms	route + depth verify at build (action-note: add SG RFR leg)	W2	connect	UNVERIFIED-CANDIDATE
31	bbsw-asx	rates-fix	ASX BBSW / bid rate (AUD bench-marks)	daily	1986?→ verify	D0	ASX licensed feed	licensed quote	AUD float-rate bench-mark — licence scope check (non-AUS use) (action-note: RFQ)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
32	libor-legacy	rates-fix·archive	Legacy LIBOR panel rates — cessation archive (ICE; FCA mandated windows; USD ended 2023-06-30)	daily	1980s?→ cessation	static archive	ICE/published archive files	Legacy licence — reuse restricted; **new benchmarking prohibited**	usable for historical/fallback-transition matrices only — never model-forward (action-note: archive free portions only)	W2	connect	UNVERIFIED-CANDIDATE
33	fred-h15-daily	rates-curve	FRED H.15 daily family (DGS1MO→DGS30; CP rates; discount; ratios)	daily	DGS 1962→; long-term averages 1953→	D0/D+1	FRED API v2 (free key)	US-gov public	CMT-based constant-maturity — not spot; no forward curve implied (action-note: panel to 1953/1962)	W0	backfill	SURVEY-EVIDENCE
34	fred-tips	rates-curve	FRED TIPS real yields (DFII1–DFII30) + breakevens (T5YIE, T10YIE)	daily	real 1997→; BE 2003→	D0	FRED API	US-gov public	CB-computed indirect breakeven ≠ swap-market BE (action-note: ingest)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
35	fred-monthly-weekly	rates	FRED weekly/monthly rate set (FF, MORT-GAGE30US 1971→, BANKPRIME, WPesos)	weekly/monthly	decades	per release	FRED API	US-gov public	MORTGAGE30US = Freddie Mac survey, not CMT (action-note: ingest)	W0	connect	SURVEY-EVIDENCE
36	alfred	rates·archive	ALFRED vintage-dated FRED series (`fred/alfred` params)	vintage per revision	back to each series' first release	per-vintage	FRED API (vintage params)	US-gov public	**the** PIT-safe macro rail; store observa-tion_date + vin-tage_date (action-note: PIC factory + PIT merge engine)	W0	connect	SURVEY-EVIDENCE
37	tsy-cmt-par	rates-curve	Treasury par curve (daily CSV/XML; long-term averages + TIPS real curve)	daily	1990→ (LT averages deeper; verify)	D0	treasury.gov/interest-rates CSV	US-gov public	par vs CMT method (boot-strap-pable); fetch of page timed out 2026-10-24 — retry at build (action-note: + align vs FRED DGS)	W0	backfill	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
38	tsy-auctions	rates-auction	Treasury Direct Auction Search (CSV/JSON/TSV/XML end-points)	per auction	TIPS 1997→; all others 1998-07-27→; fields extended Apr-2008 + Oct-2026	per auction	auctionQuery endpoint, free	US-gov public	settlement/note/coupon cross-check vs NFP auction PR; some string-typed fields (action-note: ingest per security type)	W0	connect	VERIFIED-UPSTREAM-DOC
39	tsy-fiscaldata	rates-fiscal	Treasury Fiscal Data API families (MTS tables 1–9, MSPD, Debt-to-Penny, avg interest rates, DTS)	daily→monthly	per table (DTS daily)	daily	REST JSON/CSV/XML keyless (182 tabular end-points)	US-gov public no-restrictions	record_date = as-published → PIT; re-reporting normal (keep history) (action-note: bulk-ingest MTS + DTS)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
40	frb-h41-h8	macro-bank	Fed H.4.1 (weekly factors) + H.8 (weekly bank aggregates)	weekly	H.4.1 2002?→; H.8 1973?→ verify	weekly	DDP CSV/JSON	US-gov public	H.8 = commercial banks only (not full Fed framework); reserve breaks documented (action-note: ingest weekly rails)	W0	connect	SURVEY-EVIDENCE
41	frb-z1	macro·archive	Fed Z.1 flow-of-funds (DDP bulk CSV)	quarterly	1945→	quarterly	DDP bulk CSV/JSON	US-gov public	annual revision cycle (Sep) → recompute derived sector flows (action-note: ingest DDP bulk)	W0	connect	SURVEY-EVIDENCE
42	frb-g19	macro	Fed G.19 consumer credit	monthly	1943 annual origin / monthly ~1945 (verify)	monthly	DDP CSV	US-gov public	seasonal revisions reset bucket shares (action-note: ingest)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
43	infl-swaps	rates-deriv	Bloomberg/ICE 10y zC CPI inflation swaps implied breakevens	daily	market-driven	D0	Bloomberg/ICE licensed	licensed quote	swap BE vs CMT BE — validate basis; zC conven-tions differ (action-note: RFQ only if basis-critical)	W3	connect	UNVERIFIED-CANDIDATE
44	bis-otc-deriv	rates-vol	BIS OTC deriva-tives statistics (notion-als; triennial since 1995 + semian-nual/quarterly detail)	3-yr/semiannual	1995→ IR lag	periodic	SDMX + bulk-download	BIS per-mitted use	notional-type ≠ risk type (convert to paired gross/market-value); legacy Ten-view conven-tions (action-note: ingest OTC suites)	W1	connect	SURVEY-EVIDENCE
45	ecb-policy-rates	rates	ECB key rates (DFR/MRO/MLF; key-rates page)	event-based	1999-01→	event	SDMX keyless	free-attribution	operational-facility changes (e.g. clear route since <1981) — tag (action-note: ingest)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	ecb-govcurves	rates-curve	ECB €-area gov bond curves (AAA + non-AAA spot; EMU convergence yields)	daily	2004?→ verify	D0	SDMX keyless	free-attribution	read ECB curve-construction docs once, pin version (action-note: ingest)	W0	connect	SURVEY-EVIDENCE
47	eurostat-sdmx	macro	Eurostat SDMX REST (HICP, LFS unemployment, trade)	monthly	1960s→ many series	monthly	SDMX keyless	free-attribution	country-specific HICP re-basing chains (action-note: ingest core families)	W0	connect	SURVEY-EVIDENCE
48	bis-credit-gaps	macro	BIS credit-to-GDP + credit-gap stats (bis.org/statistics/crediggy)	quarterly	credit series: verify 1948 claim; gap computation 1999→	quarterly	SDMX + bulk-download	BIS permitted use	brief's "since 1948?" → validate at build; not all countries start same (action-note: ingest + span audit)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
49	bis-long-run-hist	macro·archive	BIS long-run historical monetary/credit statistics (bulk-download)	monthly/annual	claimed 1948→ (verify)	static	bulkdownload TXT/CSV	BIS permitted use	pre-modern-convention mix risk; verify spans before merging into modern frame (action-note: fetch bulk; validate spans)	W2	connect	UNVERIFIED-CANDIDATE
50	imf-ifs	macro-archive	IMF IFS monthly database (data.imf.org)	monthly	1957→ (print began Jan-1957)	monthly	SDMX (data.imf.org)	IMF free attribution	country coverage/breaks historically uneven (action-note: ingest IFS core)	W1	connect	SURVEY-EVIDENCE
51	imf-ifs-vintages	macro·archive	IFS vintage archive (eLibrary print scans + archived editions)	monthly editions	1957→ scans	static	eLibrary PDF → OCR	IMF use-terms	OCR pipeline cost; PIT = printed vintage (action-note: targeted OCR probes)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
52	imf-weo-archives	macro·archive	IMF WEO vintage archives (April + October edition datasets)	semiannual	prior editions back-separated (earliest edition verify)	biannual	data.imf.org edition files (xLS/SDMX)	IMF free	edition date-stamp = forecast PIT; use not only latest (action-note: archive all editions ≤ latest)	W1	connect	SURVEY-EVIDENCE
53	imf-cofer	macro-archive	IMF COFER (official FX reserve composi-tion; SDR rates)	quarterly	1999Q1→ pool currency split window	quarterly	SDMX/CSV	IMF free-attribution	currency-assignment revisions possible; shares may ex-ceed/under tend due to unallo-cated (action-note: ingest quar-terly)	W0	connect	SURVEY-EVIDENCE
54	imf-fsi	macro	IMF Financial Sound-ness Indica-tors	quarterly	2000s→ (verify per-country)	quarterly	SDMX	IMF free-attribution	NPL defi-nitions diverge; curation grid needed (action-note: ingest FSI core)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	wb-wdi	macro	World Bank WDI API (api.worldbank.org/v2; verified URL/arg queries, XML/JSON; bulk ZIPs + BigQuery mirror)	annual (some sub-annual)	1960→	annual	REST v2 + bulk ZIP	CC BY 4.0	no vintages — append revision LOG of published correc-tions (action-note: ingest (+ BigQuery mirror))	W0	connect	VERIFIED-UPSTREAM-DOC
56	wb-gpi	macro	WDI gender-parity (GPI) indicator family (SE.* / SG.*)	annual	per indicator	annual	same WDI API codes	CC BY 4.0	pin exact indicator codes + defini-tions at build (action-note: map GPI codes)	W2	connect	UNVERIFIED-CANDIDATE
57	wb-long-run	macro-archive	World Bank legacy historical long-run (older World Tables volumes + assisted series)	annual	mid-20th-C → 1960	static	bulk CSV + archive contain-ers	attribution	pre-1960 per-country coverage patchy; treat as supple-ment to WDI spine (action-note: deep-history fills)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
58	oecd-mei	macro	OECD SDMX (MEI/CPI monthly; composite leading indicators; quarterly national accounts)	monthly	1960s→	monthly	SDMX REST (JSON/XML)	CC BY 4.0	CLI composition revisions = factor breaks (action-note: ingest MEI + CLI)	W0	connect	SURVEY-EVIDENCE
59	boe-rates-curve	rates-curve	BoE rates + gilt curve (Bank Rate 1694→ documented; sterling 5/10/20y daily curve)	daily	Bank Rate 1694→	T+1	boeapps CSV/XLSX	UK OGL	read curve-construction docs once, pin version (HM Treasury linkage) (action-note: ingest Bank Rate + curve)	W0	connect	SURVEY-EVIDENCE
60	boj-jgb	rates-curve	BoJ JGB daily yield list (TTSS route)	daily	1970s?→ verify	D0	TTSS/BoJ CSV	BoJ terms	maturity panel depth varies by era (action-note: JGB curve ingest)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	pboc-chinamoney	rates	PBoC / Chinamoney (CFETS) — Shibor (2006→), LPR (2019→)	daily	per fixings	D+1	Chinamoney EN web CSV	CN public info — non-commercial	no API for org redistribution; scraping resilience needed (action-note: parse daily fixings)	W1	connect	SURVEY-EVIDENCE
62	rbi-dbie	rates	RBI DBie (SDMX; repo, G-sec curve, CPI/WPI)	daily/monthly	1991→ mostly	daily	DBie SDMX/CSV	public gov-data attribution	some legacy series PDF-only (action-note: SDMX ingest)	W0	connect	SURVEY-EVIDENCE
63	safe-cn	macro	SAFE reserves/settlement data (PDF/HTML releases)	monthly	2010→ verify	monthly	web scrape	public CN info (no API)	manual QA; CN capital-flow proxy (action-note: monthly scrape)	W2	connect	SURVEY-EVIDENCE
64	cnb-arad	rates	Czech National Bank ARAD time-series system (confirmed URL `cnb.cz/arad`)	daily/monthly	deep CZK curve + macro (1990s?→ verify)	D+1	ARAD app/API	CNB terms	confirm REST/CSV API spec at build (action-note: probe + ingest)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	boi-rates	rates	Bank of Israel data portal (rates/FX/macro)	daily/monthly	verify	daily	BoI web CSV?	BoI terms	bilingual (HE/EN) metadata QA; API existence uncon-firmed (action-note: probe)	W1	connect	UNVERIFIED-CANDIDATE
66	cbs-israel	macro	Central Bureau of Statistics — Israel (CPI)	monthly	deep (verify)	monthly	CBS web CSV	public gov	CPI rebase method-ology notes per year (action-note: CPI leg)	W2	connect	UNVERIFIED-CANDIDATE
67	sarb-stats	rates	South African Reserve Bank statdata (repo curve, S.A. macro)	daily/monthly	verify	daily	SARB statdata JSON?/web	SARB terms	API shape undocu-mented — probe first (action-note: probe)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
68	bcb-sgs	rates	BCB Brazil Série-Históricas API (`api.bcb.gov.br/dados/serie/bcdata.sgs.*`)	daily/monthly	Selic 1986?→; CPI 1980s?→ verify	D+1	REST JSON, keyless	open BCB data (CC-BY? confirm)	fetch 2026-10-24 got HTTP 406 (UA-block) — retry w/ browser UA; map SGS codes (e.g. 432 Selic) (action-note: probe w/ UA; code map)	W0	connect	UNVERIFIED-CANDIDATE
69	bcch-sdmx	rates	Banco Central de Chile SDMX REST (si3.bcentral.cl; TPM, UF, CPI)	daily/monthly	verify depth	daily	SDMX REST	BCCh terms	coverage nuances; personal data regs (action-note: probe)	W1	connect	UNVERIFIED-CANDIDATE
70	bsp-ph	rates	Bangko Sentral ng Pilipinas statistics (RRP, CPI, PHP fx)	daily/monthly	verify	daily	BSP web XLS	BSP terms	XLS/PDF mix; extraction fragility (action-note: probe)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
71	bls-cpi	macro-archive	BLS CPI (CU/CW/SU) — series database + Public Data API v2 + down-load.bls.gov/pub/time.series/cu text files	monthly	1913→ national	monthly	API v2 (free key) + text files	US-gov public	annual baseline revisions; chained-CPI (SU) separate lineage — never mix SA/NSA (action-note: ingest + seasonal-ity flags)	W0	connect	VERIFIED-UPSTREAM-DOC
72	bls-ppi	macro	BLS PPI (pc / wp commod-ity + FD-ID)	monthly	1980s?→ deep (verify)	monthly	API + text files	US-gov public	2011 FD-ID reform breaks (action-note: ingest)	W0	connect	VERIFIED-UPSTREAM-DOC
73	bls-jolts	macro	BLS JOLTS (job open-ings/hires/quits)	monthly	2000-12→	~5-6 wk lag	API + /jlt/ files	US-gov public	3-release revision ladder; flag final only (action-note: ingest)	W0	connect	VERIFIED-UPSTREAM-DOC
74	bls-eci-ecec	macro	BLS ECI + ECEC (compen-sation)	quarterly	ECI 1975?→ verify	quarterly	API	US-gov public	seasonally-adjusted subset only (action-note: ingest)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
75	census-adv-trade	macro	Census advance interna-tional trade (FT900 monthly)	monthly	1992?→ verify	~6-weeks lag	Census EP API (key free) + bulk	public	advance→final revision chain; joint Cen-sus/BP ITA con-solidation (action-note: ingest)	W0	connect	UNVERIFIED-CANDIDATE
76	census-construction	macro	Census construc-tion spending (VIP)	monthly	1964?→ verify	~1mo	Census API	public	seasonal + 1-month revision ladder (action-note: ingest)	W0	connect	UNVERIFIED-CANDIDATE
77	census-retail	macro	Census retail (MRTS / advance retail)	monthly	1992?→ verify	~2wk	Census API	public	NAICS re-basing cycles (action-note: ingest)	W0	connect	UNVERIFIED-CANDIDATE
78	bea-nipa	macro	BEA NIPA (GDP/investment/personal income; annual + quarterly tables)	quarterly + annual	annual 1929→; quarterly 1947→	monthly esti-mates/revision calendar	BEA API (free key)	US-gov public	summer annual-revision resets; PIT only via ALFRED [36] (action-note: ingest + PIT walk)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
79	conference-board	macro	Conference Board LEI/CLI + Consumer Confidence	monthly	LEI 1959?→ verify	monthly	subscription/API	licensed quote	proprietary index construction; component weights change (action-note: RFQ)	W3	connect	UNVERIFIED-CANDIDATE
80	spf-philly	macro-archive	Philadelphia Fed Survey of Professional Forecasters	quarterly	1968Q4→	quarterly	free XLS/CSV data files	US public (Fed bank)	free vintage archive of forecasts (consensus) — rare PIT grader (action-note: ingest + forecaster breakdowns)	W0	connect	SURVEY-EVIDENCE
81	michigan-sca	macro-archive	University of Michigan Surveys of Consumers (ICS/MCS)	monthly	1978 monthly (earlier reconstruction verify)	monthly	free site CSV/XLS archive	UMich terms (research)	keep prelim vs final distinction; revisions history (action-note: archive + register if needed)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
82	nfib-sbet	macro-archive	NFIB Small Business Economic Trends	monthly	1974?/1986?→ verify	monthly	NFIB web PDF/CSV	NFIB terms (some restricted)	archive availability + cost unconfirmed (action-note: archive index + subcomponents)	W1	connect	UNVERIFIED-CANDIDATE
83	ism	macro	ISM Manufacturing/Services Indices	monthly	mfg 1948?→ (verify)	monthly	ISM web (paywalled core)	licensed quote	free townhall PDFs lack historical depth (action-note: assess ISM web data routes)	W2	connect	UNVERIFIED-CANDIDATE
84	spg-pmi	macro	S&P Global (ex-Markit) PMIs (manufacturing/services/composite, 30+ countries)	monthly	1998→ majors (some 1980s-era legacy)	monthly	licensed (data/API)	licensed quote	PMI revision ladder — pin first print for PIT (action-note: RFQ with historical-archive scope)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
85	caixin-pmi	macro	Caixin China PMIs (manufacturing + services, CN)	monthly	2011→	monthly	licensed	licensed quote	CN business-cycle factor alternative (pairs w/ [61]) (action-note: RFQ)	W3	connect	UNVERIFIED-CANDIDATE
86	fred-credit-spreads	credit	FRED credit spreads (Moody’s Baa-Aaa 1983→; ICE BofA HY/IG OAS 1996→)	daily	per-row	D0/D+1	FRED API	public (underlying index values mirror ICE/Moody’s)	index-owner licence references documented; HY OAS = option-adjusted, not raw spread (action-note: ingest + licence notes)	W0	connect	SURVEY-EVIDENCE
87	moody-yield-arch	credit·archive	Moody’s long corporate yield averages (Aaa/Baa via published archives; FRED daily/pre-1983 monthly)	monthly→daily	1919→	historic	FRED (1983→) + archive PDFs	archive use (verify)	splice 1919→1983 monthly with 1983→ daily — break-tag the join (action-note: assemble archive panel)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
88	finra-trace	credit-trades	FINRA TRACE (corporate bond trade reports + aggregates)	intraday trades / EOD	1990s?→ verify era bounds	realtime/EOD	FINRA data subscriptions	licensed quote	fair-use rules; no redistribution; enhanced tier = extra (action-note: RFQ breadth + fair-use review)	W3	connect	UNVERIFIED-CANDIDATE
89	bbg-barclays-indices	credit-index	Bloomberg Barclays Aggregates (US/EU/Global + sub-indices)	daily	US Agg 1986?→ (verify per index)	D0	Bloomberg library	licensed quote	redistribution prohibited; TR vs price-index gates (action-note: RFQ only parent+travel series needed)	W3	connect	UNVERIFIED-CANDIDATE
90	jpm-embi	credit-index	JPMorgan EMBI (EM sovereign spread index family)	daily	1993?→ verify	D0	JPM subscription	licensed quote	index membership changes; hard vs local-currency split (action-note: RFQ)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
91	cma-cds	credit	CMA (S&P Global MI) sovereign CDS quotes	daily	2010s?→ verify	D0	enterprise API	licensed quote (row pre-verified in [S1])	daily close markers only (not intraday); currency/tenor panel spreads (action-note: RFQ)	W3	connect	SURVEY-EVIDENCE
92	sp-default-studies	credit-archive	S&P Ratings annual default/upgrade studies (public PDFs / XLS tables)	annual	1981→ cohort studies	annual	spglobal research publishes PDFs	public publication (data tables vary)	cohort methodology changes by year — pin per-edition definitions (action-note: scrape + parse yearly studies)	W2	connect	UNVERIFIED-CANDIDATE
93	cfed-nfci	credit	Chicago Fed National Financial Conditions Index	weekly	1971→	weekly	FRED/cfed CSV	public	weekly updates; sub-indices available (action-note: ingest)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
94	stlfsi4	credit	St. Louis Fed Financial Stress Index (STLFSI4)	weekly	1994→	weekly	FRED	public	v4 construction differs from prior versions — break-tag (action-note: ingest)	W1	connect	UNVERIFIED-CANDIDATE
95	msci-esg	esg-ratings	MSCI ESG ratings (AAA–CCC; feed/API)	monthly refresh (verify)	rating-change history needed	monthly	MSCI data licensed	licensed quote	PIT = rating-change events (need change log, not snapshot); methodology revisions frequent (action-note: RFQ — demand event history)	W3	connect	SURVEY-EVIDENCE
96	sustainalytics-esg	esg	Sustainalytics (Morningstar) ESG Risk Ratings	company snapshot	coverage list + rating history	monthly	data/API	licensed quote	risk-score scale ≠ MSCI ordinal; standardization layer required (action-note: RFQ)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
97	iss-esg	esg	ISS ESG ratings	monthly	verify	monthly	licensed	licensed quote	methodology version-ing check (action-note: RFQ)	W3	connect	UNVERIFIED-CANDIDATE
98	lseg-esg	esg	LSEG/Refinitiv ESG (pillar and com-ponent scores)	monthly	2002?→ verify	D+1	workspace/API	licensed quote	heavy backfill available but licence-limited (row pre-verified in [S1]) (action-note: RFQ)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
99	eex-eua-auctions	esg-carbon	EEX EUA primary-market auction results (free CSV down-loads)	per-auction	Germany 2013?→; UK 2021→; per-platform	post-auction	EEX CSV down-loads	free (EU auction data open)	distribution pattern differs per member state (EEX/other platform) — my fetch of EEX path 404’d 2026-10-24; re-discover path at build (action-note: discover path + parse EUA results)	W0	connect	UNVERIFIED-CANDIDATE
100	icap-status	esg-carbon	ICAP Status Report + ETS Map data (annual Excel)	annual	2005?→ verify	annual	site Excel	free (verify licence)	coverage/inclusion rules differ per ETS — map schema first (action-note: annual Excel ingest)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
101	cdp	esg	CDP corporate climate disclo-sures	annually	2002?→ verify	annually	CDP data/API	licensed quote (some open)	disclosure vs score linkage; coverage gaps (action-note: RFQ)	W3	connect	UNVERIFIED-CANDIDATE
102	wb-esg	esg	World Bank sovereign ESG dataset (`data.worldbank.org` ESG col-lection)	annual	aligns to WDI rail 1960→ subset	annual	WB API/CSV (same rail [55])	CC BY 4.0 (assumed same as WDI — verify)	curated ESG-framework subset ≠ WDI superset — keep separate ID names-pace (action-note: piggy-back on WDI ingest)	W0	connect	UNVERIFIED-CANDIDATE
103	sdg-index	esg	UN/SDSN Global SDG Index (SDR Ex-cel/CSV)	annual	2015→ (vs baseline 2000/15) verify	annually	free Ex-cel/CSV	CC BY-NC? verify licence	methodology changes between editions — break-tag (action-note: annual ingest)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	wb-pink-sheet	archive	World Bank CMO / Pink Sheet monthly + annual price XLSX (the-docs.worldbank.org fixed doc URLs)	monthly + annual	annual 1960→; monthly long-run 1990s-era	monthly (fixed ~1st of month)	file-drop XLSX → parquet; per-release vintage dir	free (WB dataset terms, CC BY catalog [^1])	rebasing changes histori-cals ⇒ snapshot per release date (action-note: ingest monthly+annual each release; keep vintage store)	W0	connect	VERIFIED-UPSTREAM-DOC
2	imf-pcp	archive	IMF Primary Commod-ity Price System (monthly XLSX + pre-rebase supple-ment)	monthly	current base: 1992→; supple-ment covers pre-1992 decades	monthly	XLSX file-drop	free w/ at-tribution	base-year rebases rewrite history ⇒ vintage; fetch blocked 429×3 today (action-note: resolve w/ retry/wget from another IP; then bulk)	W0	backfill	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	lme-official-prices	benchmark	LME Data licensing portal — Official daily metal prices	daily	1990s→	daily (official close)	licence portal / SFTP drop	exchange licence; **next-day distribution one-off ~$4k** vs $25k/yr realtime [^3]	delayed-only ⇒ no PIT perfect; cross-check CME/SHFE same metal (action-note: purchase next-day tier; map Cu/Al/Ni/Zn/Pb/Sn to CME twins)	W1	connect	VERIFIED-UPSTREAM-DOC
4	lme-stocks	benchmark	LME ware-house stocks (per metal, per region)	daily	2000s→	daily	same licence portal [^3]	same licence	warehouse-cornering months distort; join with CME de-stock proxy (action-note: buy with prices; build stocks-change signal)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	cme-comex-metalshist	benchmark	CME DataMine / COMEX per-metal per-contract (GC, SI, HG, PL) settle-ments+volume	daily (tick-class W2)	multi-decade	settlements EOD; ticks W2	DataMine API/request (paid)	enterprise licence; no redis-tribution	contract-roll math manda-tory; expired-contract historics (action-note: pilot GC+HG sample; wire to roll calendar (#18))	W2	connect	UNVERIFIED-CANDIDATE
6	databento-glbx	benchmark	Databento GLBX.MDP3 historical (CME Globex futures: metals, WTI, NG, agri)	1-min→tick	~15y Rt-era; per-GB	<1s live	REST batch → DBN files	paid per-GB ~$0.40–0.50, credits $125→$199 plan [^4]	roll math; schema stable (owned files) (action-note: free-credit prototype CL/GC/ZC bars)	W2	connect	VERIFIED-UPSTREAM-DOC
7	ice-eu-futures	benchmark	ICE Futures Europe Data Services (Brent, Gasoil, Cocoa, Coffee, Sugar, WTI-Europe?)	daily/tick	decades	realtime/delayed tiers	ICE Data Portal (paid line-items)	exchange licence	roll math; Euronext-ICE split pre-2012 caveats (action-note: first; only if energy-contango sim needed)	W2	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
8	platts-assessments	benchmark	S&P Global Commodity Insights (Platts) — crude/products/LNG assessments	daily	1980s→	realtime	enterprise feed	no redistribution	MOG window conventions; subscriber-only basis (action-note: first procurement; cross-check EIA spot free)	W3	quote	SURVEY-EVIDENCE
9	argus-media	benchmark	Argus Media (Argus-DC) non-US energy/chemicals assessments	daily	1990s→	daily	enterprise feed	no redistribution	same class as Platts (action-note: ; pick ONE of Platts/Argus first)	W3	quote	SURVEY-EVIDENCE
10	icis-assessments	benchmark	ICIS petrochemical / energy / LNG assessments	daily	1990s→	daily	enterprise feed	no redistribution	overlapping w/ Platts/Argus (action-note: only if chemical leg opens)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	fastmarkets-metallic’s	benchmark	Fastmarkets (battery-metals, Li/Co/Ni) assess-ments	daily	2010s→	daily	enterprise feed	no redis-tribution	new-index basis flux (action-note: only if battery-metals leg)	W3	quote	UNVERIFIED-CANDIDATE
12	sp-gsci	benchmark	S&P GSCI (S&P DJI) index levels + method-ol-ogy/constituents	daily	1970→	daily	S&P DJI portal; fact-sheet free; con-stituents paid	levels free-ish; method-ology licence	roll-window defini-tions in method-ology PDF (action-note: free index levels now; con-stituents via)	W1	quote	UNVERIFIED-CANDIDATE
13	bloomberg-bcom	benchmark	Bloomberg Commod-ity Index levels + annual con-stituents files	daily	1991→	daily	BCOM portal download / own terminal	Bloomberg licence (con-stituents)	index-vs-constituent Lump bindings (action-note: levels scrape; con-stituents one-off contract)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
14	yf-futures-proxy	benchmark	yfinance commodity front-months (CL=F, GC=F, BZ=F, NG=F, ZC=F…)	intraday/daily	full history of each prompt contract	delayed ~15 min	yfinance REST (no key)	delayed TOS; non-commercial	roll contamination; venue mix (CME/ICE); use proxy only (action-note: cross-check Pink/EIA vs futures spread; compute fu-adj)	W0	connect	SURVEY-EVIDENCE
15	shfe-ine-daily	benchmark	SHFE + INE (Shanghai Intl Energy crude) settlement/quotes via official site CSV/HTML	daily	SHFE 2000s→; INE crude 2018→	EOD +1d	web CSV/HTML (semi-official)	gov-published; TOS grey	unofficial endpoint drift (action-note: scrape with throttle; CN commodity anchor)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
16	cftc-disagg-cot	positioning	CFTC Commitments of Traders — Disaggregated (physical metals/energy) via Socrata API	weekly	disagg 2006→; legacy 1995→	weekly Fri EOD	REST publicreporting.cftc.gov (Socrata, keyless)	free US-gov	open-interest basis; contract codes rename (action-note: pull disagg + legacy; build positioning z-scores)	W0	connect	UNVERIFIED-CANDIDATE
17	eia-spot-crudes	benchmark	EIA PMM/APIv2 Cushing WTI spot + landed/refiner acquisition costs	daily product spot (monthly series)	Cushing WTI 1986→	monthly+weekly releases	APIv2	free US-gov	monthly revisions; PMM presets (action-note: fold into EIA spine row family (#19))	W0	connect	VERIFIED-UPSTREAM-DOC
18	roll-calendar-cme	methodology	contract-roll mapping for WTI/Brent/metal/agri front months	daily	spans full history	n/a	internal derivation from #5/#6 conventions	—	wrong roll ⇒ phantom PnL (action-note: build once, test vs Databento continuous)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
19	eia-api-v2-spine	energy-balance	EIA APIv2 — full family flagship (petroleum/petroleum, nat-gas/NG, electric-ity/EPB, coal, STEO)	→ annual	60+ years (per family varies)	15–60 min (rto)	REST free key per family	free US-gov	revisions disclosed per report (action-note: central spine already selected in stream-05 □)	W0	connect	VERIFIED-UPSTREAM-DOC
20	eia-grid-hourly	power-grid	EIA Hourly Electric Grid Monitor (rto family, BA-level demand/net-gen/interchange)	hourly	2019→ current; pre-2019 bulk EBA zip [^2]	≤1h	REST + EBA.zip bulk backfill [^2]	free [^2]	BA-code mapping; timezone normal-ization (action-note: bulk both zips + daily delta)	W0	backfill	VERIFIED-UPSTREAM-DOC
21	eia-wpsr-weekly	energy-balance	EIA Weekly Petroleum Status Report (crude/product stocks, refinery utiliza-tion, imports)	weekly	1980s→	weekly Thu 10:30 ET (market-moving)	APIv2 + PDF/CSV	free US-gov	TBD revisions; monthly bench-mark reset vs weekly (action-note: one-shot ETL + revision tracker)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
22	eia-ng-storage-weekly	energy-balance	EIA Weekly Natural Gas Storage Report (working gas, L48/regional)	weekly	1994→	weekly Thu 10:30 ET	APIv2 (nat-gas/stor) [^2]	free US-gov	Sept SGC audit dis-locations (action-note: deliver W0; volatility trigger row)	W0	connect	VERIFIED-UPSTREAM-DOC
23	eia-steo-current	energy-balance	STEO — monthly projec-tions (US sup-ply/demand/prices + global liquids)	monthly (1–25-month horizon)	forecasts ~2y	monthly Tue	APIv2 steo browser + tables [^2]	free US-gov	vintage store needed (STEO re-estimates) (action-note: pull each edition)	W0	connect	VERIFIED-UPSTREAM-DOC
24	eia-steo-archive	archive	STEO editions archive (vintage stack)	monthly	1990s→ vintage	monthly	ZIP/PDF archive	free US-gov	THE PIT spine for oil/US energy forecasts (action-note: snapshot every edition on release)	W0	connect	VERIFIED-UPSTREAM-DOC
25	eia-international	energy-balance	EIA Inter-national Energy Statistics (produc-tion, consump-tion, prices, trade)	monthly/annual	1980→ish	monthly	APIv2 interna-tional browser [^2]	free US-gov	country-renames; Iran-style gaps (action-note: ingest selected series)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
26	jodi-oil	energy-balance	JODI-Oil World DB (production, demand, refinery runs, stocks, trade)	monthly	Jan 2002→ (CSV annual files 2002-2026 confirmed) [^6]	monthly (+1-month lag)	annual CSV bulk + REST API v1.2 free account [^6]	free	participation quality colour-codes 1/2/3 in-file [^6]; country coverage gaps (action-note: bulk CSVs + REST deltas; keep colour flag)	W0	connect	VERIFIED-UPSTREAM-DOC
27	jodi-gas	energy-balance	JODI-Gas World DB (production, demand, LNG trade, storage)	monthly	2009→	monthly	CSV bulk down-loads [^6]	free	same quality codes class (action-note: bulk)	W0	backfill	VERIFIED-UPSTREAM-DOC
28	opec-momr-archive	archive	OPEC MOMR historical editions (global supply/demand/shipping tables)	monthly	2000s→ editions	monthly	PDF (slot 2/wk)	free w/ attribution	PDF parse brittleness; vintage store (action-note: parse library + release snapshotter)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
29	eurostat-nrg	archive	Eurostat nrg_* SDMX datasets (supply, stocks, gas/electricity prices by con-sumer band)	monthly/annual	1990s→ (prices 2007→)	monthly/annual	SDMX CSV REST	free w/ at-tribution	shortcode churn (nrg changes names) (action-note: select nrg_cb_gal/nrg_pc core series)	W0	connect	UNVERIFIED-CANDIDATE
30	entsoe-gen	power-grid	ENTSO-E Trans-parency — actual genera-tion per unit / per produc-tion type	15–60 min	2015→	near-RT (release ~1h)	REST API (token, per-dataset params)	free w/ ToS token	per-country gaps; unit-code churn; probe reach □ [^7] (action-note: bulk + daily)	W0	backfill	VERIFIED-UPSTREAM-DOC
31	entsoe-load	power-grid	ENTSO-E actual total load + day/week/season-ahead forecasts	15–60 min	2015→	near-RT	REST API [^7]	free w/ token	forecast-vintage retention (action-note: ingest + store forecast vintages)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
32	entsoe-transfers	power-grid	ENTSO-E cross-border physical flows + scheduled commercial exchanges	15–60 min	2015→	near-RT	REST API [^7]	free	border direction sign conventions (action-note: ingest)	W0	connect	VERIFIED-UPSTREAM-DOC
33	entsoe-capacity	power-grid	ENTSO-E installed generation capacity per unit/type	annual	2015→	annual	REST API [^7]	free	decommission date lag (action-note: annual refresh)	W0	connect	VERIFIED-UPSTREAM-DOC
34	entsoe-outages	power-grid	ENTSO-E unavailability of generation units + transmission (outage RDF)	event	2015→	up to RT	REST API	free	flag mergers/withdrawals (planned vs unplanned) (action-note: ETL + event store)	W0	connect	VERIFIED-UPSTREAM-DOC
35	entsoe-daa-prices	power-grid	ENTSO-E day-ahead auction (EUPHEMIA) + intraday/continuous prices (legacy free-download platform)	hourly	2015→	next-day	REST API + legacy CSV.gov	free	re-run/corrections of published prices (action-note: ingest for EU zonal price spine)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
36	ember-electricity	power-grid	Ember electricity dataset (generation mix per country, monthly/yearly CSVs on S3)	monthly/annual	2000→	monthly	CSV bulk (S3-published)	**CC BY 4.0**	methodology changes between editions (action-note: full download + hash pin)	W0	connect	VERIFIED-UPSTREAM-DOC
37	ember-wholesale-prices	power-grid	Ember wholesale day-ahead price dataset (EU zonal, daily)	daily	2015→	monthly	CSV	CC BY 4.0	market quirks per BZ (action-note: download)	W0	connect	VERIFIED-UPSTREAM-DOC
38	ei-statistical-review	archive	Energy Institute **Statistical Review of World Energy** incl. pre-fusion BP archive (75th edition, since 1952; XLSX tables 1965→) [^9]	annual	1965→ (tables); 1952 inception	annual June	email-gated XLSX download	free (“full, first, free, fair”)	2026 methodology updates; Ember partnership columns (action-note: download all-archive XLSX; harmonize ex-BP splice)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
39	iea-wreo-datasets	energy-balance	IEA WEO + IEA Data Browser free tiers (world energy balances)	annual	decades	annual	IEA portal down-loads	**WEO data = CC BY-NC-SA ⚠ NOT commercial/derivable** [^10]	year-base changes; NC-SA gating (action-note: only for dis-play/consensus; negotiate IEA reuse if core)	W1	connect	VERIFIED-UPSTREAM-DOC
40	iea-tech-datasets	energy-balance	IEA per-technology free datasets (solar PV, wind, EV fleet, heat pumps; CC BY subset)	annual	2000s→	annual	IEA CSV/portal	CC BY 4.0 for tech data folder	WEO subset ≠ tech subset — per-file licence check (action-note: tech panels)	W0	backfill	UNVERIFIED-CANDIDATE
41	owid-energy	archive	Our World in Data energy (legacy repo + new catalog CSV)	annual	1900→ country panel	annual	GitHub + cata-log.ourworldindata.org CSV/XLSX + codebook **[legacy frozen; switched 2026 method-ology]** [^11]	CC BY	2026 primary-energy re-measurement ⇒ old≠new (action-note: pin col-umn_mapping.csv; use new catalog)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
42	caiso-oasis-art-prices	power-grid	CAISO **OASIS** — PRICES publication group (Locational Marginal Prices DAM/RTM/interval, HASP, FMM, congestion shadow prices) [^12]	5-min / hourly / daily	DAM LMP 1998→	daily post-market + interval RT	OASIS `oasisapi` REST (keyless, reportName+querystring, ZIP download)	free ISO TOS	constant price-corrections log; ACR flags [^12] (action-note: by SPP reportName month-by-month)	W0	backfill	VERIFIED-UPSTREAM-DOC
43	caiso-demand-netload	power-grid	CAISO OASIS demand forecast / net-demand / fuel mix dashboards	5–60 min	2010s→	realtime-after	OASIS + public dashboards CSV	free ISO	jettisoned? no (action-note: attach to OASIS loop)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	ercot-rt-lmp	power-grid	ERCOT EMS Real-Time Market LMPs (15-min) + DAM hourly points / price caps	15-min / hourly	2010s→	realtime-after (2-min)	MIS public REST + CSV filedrop	free TOS	**doc blocked 403** today; endpoint naming unverified (action-note: curl-probe /mis paths; then bulk)	W0	connect	UNVERIFIED-CANDIDATE
45	ercot-fuelmix-outages	power-grid	ERCOT fuel mix (gen by source incl. weather-sensitive), outages (SCED telemetry summaries)	5/15-min	2010s→	realtime-after	MIS CSV + dashboards [^13]	free	mirror EMS naming (action-note: probe + ingest)	W0	connect	UNVERIFIED-CANDIDATE
46	pjm-dataminer2-lmp	power-grid	PJM **DataMiner 2** — DA/RT LMP per pnode, hourly/5-min	hourly / 5-min	2000s→	realtime-after	DataMiner2 REST (keyless; feed-class, month windows)	free ISO TOS	node mapping churn; price-corrections file (action-note: bulk monthly via page-fence)	W0	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
47	pjm-gen-fuel	power-grid	PJM generation by fuel (eDART/eDART Archive CSV) + interchange, load	hourly	2011→	daily	DataMiner2 / eDART CSV	free ISO	unit-type roll-ups (action-note: ingest alongside LMP)	W0	connect	UNVERIFIED-CANDIDATE
48	miso-openapi	power-grid	MISO Market Data APIs (DA/RT LMPs, gen by fuel, load, outages; REST + CSVs)	5–60 min	2005-ish→	realtime-after	MISO REST	free TOS	MISO south/CBC swap (action-note: probe; ingest)	W0	connect	UNVERIFIED-CANDIDATE
49	iso-ne-ais-express	power-grid	ISO-NE Express API (hourly gen by fuel, LMP by zone, demand+forecast JSON)	hourly	2000s→	1–5 min	Express REST JSON	free TOS	fuzzy time-epoch (posix vs local) (action-note: probe + ingest)	W0	connect	UNVERIFIED-CANDIDATE
50	nyiso-portal	power-grid	NYISO portal (DAM LMP, RT 5-min, gen by fuel, capacity, forecast-vs-actual)	5–60 min	2000s→	realtime-after	portal CSV/ZIP + custom MIS	free TOS	11-zone → generator-level translation (action-note: monthly CSV bulk loop)	W0	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
51	spp-mis	power-grid	SPP Market Data Portal (DA/RT LMP MP valuation slicing, gen by fuel, operating reserve)	5–60 min	2007-ish→	realtime-after	MDP CSV/report download	free TOS	MDP UI JS-app; API path fuzziness (action-note: probe; schedule)	W0	connect	UNVERIFIED-CANDIDATE
52	aemo-mms	power-grid	AEMO NEM MMS data model (dispatch 5-min, price tables, PLAP archives)	5-min	2000s→	realtime-after (settle-ment 30-min, **5-min pricing since Oct-2021**)	MMS CSV/TRI archives (month-fold)	free (AEMO TOS)	DST/Adelaide offset traps; Apr 2021 5-min rule transition (action-note: monthly MMS bulk + daily delta)	W0	connect	UNVERIFIED-CANDIDATE
53	aemo-aggregate-demand	power-grid	AEMO ag-gregator price & demand datasets (regional 30-min/now 5-min)	30→5-min	1998→ (price/demand run back)	daily/hourly	aggregate CSV per region	free	30-min→5-min stitch step (action-note: 1998→)	W0	backfill	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
54	elexon-bmrs	power-grid	Elexon **BMRS per-dataset** (System Prices, Market Index, DA/WD prices, FPN)	30–60 min + events	2000s→	realtime-after	BMRS site + Insights REST API [^14]	free (GFD attribu-tion)	app JS-only — probe Swagger via curl (action-note: probe Insights end-points)	W0	connect	UNVERIFIED-CANDIDATE
55	elexon-bods	power-grid	Elexon BODS — physical notifica-tions (FPN) / BM unit solar-wind forecast	30-min	2000s→	realtime	BODS REST API	free	FPN update fre-quency noise (action-note: probe + ingest for GB commodity-renewables z-score)	W0	connect	UNVERIFIED-CANDIDATE
56	gie-agsi	power-grid	GIE **AGSI+** — gas storage (working gas, injec-tion/withdrawal, per coun-try/operator; REMIT-aligned) [^15]	daily (prior day)	2010s→	daily	AGSI REST API (account token) [^15]	free w/ at-tribution	operator double-counting guards (action-note: token + REST pull)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
57	gie-alsi	power-grid	GIE ALSI — LNG terminal inventories, send-out, cargoes arrived/expected [^15]	daily	2010s→	daily	ALSI REST API	free w/ attribution	terminal granularity (action-note: pull with AGSI)	W0	connect	VERIFIED-UPSTREAM-DOC
58	gie-umm-iip	power-grid	GIE UMM / IIP Inside Information platform (TSO/operator urgent notices) [^15]	event	2010s→	realtime	iip.gie.eu	free w/ attribution	non-structured text (action-note: event feed → alert bus)	W1	connect	VERIFIED-UPSTREAM-DOC
59	jao-tcf	power-grid	JAO/ENTSO-E **TCF** (Transparency Capacity) cross-border capacity [^16]	daily	2010s→	daily	JAO CSV/REST feed	free	TCF versioning (action-note: probe; attach to ENTSO-E ingest)	W0	connect	UNVERIFIED-CANDIDATE
60	energinet-datashop	power-grid	DK Energienet Datashop API (elpriser, gen 15-min, gaslib)	5–60 min	2000s→	realtime-after	REST + CSV bulk	free	DK DK1/DK2 split (action-note: probe; Danish quality anchor)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	tennet-transparency	power-grid	NL TenneT Trans-parency portal (15-min gen, trade, system reserve, balance)	15-min	2010s→	realtime-after	portal REST/XML/CSV	free	just-sourced naming align-ment (action-note: probe; ingest)	W0	connect	UNVERIFIED-CANDIDATE
62	elia-opendata	power-grid	BE Elia Open Data portal (gen/load/balancing, per-CSV)	15-min	2014-ish→	realtime-after	REST open-data API	free	BE-A midjör split (action-note: probe; ingest)	W0	connect	UNVERIFIED-CANDIDATE
63	de-gen-by-source	power-grid	DE SMARD (Bun-desnetza-gentur) gen per source (incl. now-cast), EUR prices	15-min	2015→	up to 15-min	SMARD REST + CSV	free	national→TSO reconcili-ation (action-note: probe; ingest)	W0	connect	UNVERIFIED-CANDIDATE
64	baltic-exchange-indices	shipping	Baltic Ex-change BDI/BCI/BDTI/BDSI per-index historical	daily	1985→ (BDI base 1000)	daily	Baltic Exchange data licence	strictly licensed	composition re-bases (action-note: first procure-ment)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	bdi-open-mirror	shipping	BDI via public secondary mirrors/sites (non-licensed)	daily	1990s→	daily	web scrape (mirror risk)	**unofficial** = no basis to re-quote	lag/accuracy — proxy only, never quote-generating (action-note: only as compute-less proxy until W3)	W2	license-review	UNVERIFIED-CANDIDATE
66	xeneta-xsi	shipping	Xeneta XSI ocean/air benchmark indices	weekly	2010s→	weekly	enterprise feed	licensed	tender-pool basis	W3	quote	UNVERIFIED-CANDIDATE
67	freightos-fbx	shipping	Freightos **FBX** (Freightos Baltic Index ocean) + **FAX** (air); Terminal API portal (api-dos.freightos.com) [^17]	weekly (daily intraweek variants)	2010s-era→ weekly tracks	weekly	Terminal API / product pages [^17]	vendor TOS; header-registered data downloads	route-average basis (22 lanes) (action-note: register + API probe)	W1	connect	VERIFIED-UPSTREAM-DOC
68	drewry-wci	shipping	Drewry World Container Index weekly spot assessment + long-run CSVs	weekly	2011→	weekly	CSV download (email/register)	free-ish research-with-attribution	route-set changes (action-note: register downloader)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
69	tac-air-cargo	shipping	TAC Index (TAC Shanghai/TAC HK air cargo rates)	weekly	2010s→	weekly	enterprise feed	licensed	US-ban Asia-basis flux (action-note: only if air leg)	W3	quote	UNVERIFIED-CANDIDATE
70	aisstream-live-rail	shipping	aisstream.io free WebSocket AIS rail (PositionReport, ShipProperty, bounding-box filters) [^18]	realtime event (<5s)	live only (no history)	<5s	WS JSON (APIKey+v1 stream) [^18]	free live; **3 conn/IP**; browser-banned [^18]	event-driven drop risk — fast consumer needed [^18] (action-note: bounding-box subscribe + persist)	W0	connect	VERIFIED-UPSTREAM-DOC
71	kpler-ais-history	shipping	Kpler (incl. MarineTraffic+FleetMon since 2023) AIS events + history, vessel class flows	event / aggregates	2008→	realtime	enterprise	licensed — **one contract covers sector** [^19]	history pricing tiers (action-note: ; single contract)	W3	quote	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
72	kpler-lng-cargoes	shipping	Kpler LNG cargo-level flows (load/discharge) + GIE ALSI cross-check	daily	2008→	realtime	enterprise	licensed	AIS-inference basis (action-note: ; pre-post gas shocks)	W3	quote	UNVERIFIED-CANDIDATE
73	emodnet-port-calls	shipping	EMODnet Human Activities — port calls, vessel density rasters	monthly/annual	2017→	monthly/annual	CSV download + WMS/WFS	CC BY 4.0	raster pixel approxi-mations (action-note: download density + port-call tables)	W1	connect	UNVERIFIED-CANDIDATE
74	unctad-maritime	shipping	UNCTAD maritime statistics (freight-rate index, port calls, tonnage, canal transits)	annual/quarterly	2000s→	annual	web CSV/API	free w/ at-tribution	low fre-quency — context only (action-note: bulk static down-load)	W0	connect	UNVERIFIED-CANDIDATE
75	eia-psm-movements	shipping	EIA Petroleum Supply Monthly (regional product move-ments, pipelines, PADD transfers)	monthly	1981→	monthly	APIv2 + PDF	free US-gov	inter-PADD move-ments revisions (action-note: fold into APIv2 ETL)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
76	usda-nass-quickstats	agri	USDA NASS QuickStats (surveys + census: crops area/yield, livestock, prices; county/national)	annual/weekly	**1850/1860s→** (year selector) [^20]	daily/weekly (NASS schedule)	QuickStats REST API (free key)	free US-gov	census vs survey gap (action-note: keyset-iterate commodities×years)	W0	connect	VERIFIED-UPSTREAM-DOC
77	usda-fas-psd	archive	USDA FAS PSD Online — world crop supply/use long-run (grains, oilseeds, cotton, sugar)	monthly	1960-ish→	monthly (WASDE cadence)	PSD Online query/API + CSV	free US-gov	mid-month vintage ⇒ snapshot each PSD update (action-note: bulk tables + vintage store)	W0	connect	UNVERIFIED-CANDIDATE
78	wasde-archive	archive	USDA WASDE monthly report archive	monthly	1970s→	monthly 12:00 ET	usda.gov archive PDF+XLSX	free US-gov	**12:00 release shock**; fetch blocked 403 today (action-note: schedule pull at 12:05 ET)	W0	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
79	faostat-fppi	agri	FAO FAOSTAT (production, trade, prices) + FAO Food Price Index (FPPI nom/real)	monthly/annual	FPPI 1990→; FAOSTAT 1961→	monthly	CSV bulk + FAO REST	CC BY 4.0	base-period resets (action-note: bulk series)	W0	connect	UNVERIFIED-CANDIDATE
80	cme-agri-hist	agri	CME Group agri contracts (CBOT ZC/ZS/ZW/ZM/ZO, ICE-11 sugar) via CME DataMine / Databento	daily/tick	decades	EOD→tick	API paid	licence	roll math (#18) (action-note: share pipeline with #5/#6)	W2	connect	UNVERIFIED-CANDIDATE
81	icco-cocoa-prices	archive & agri	ICCO daily cocoa prices (London £, New York $, ICCO avg) + monthly averages to 1960 in Statistics portal [^21]	daily (2-lag on site) + monthly	monthly 1960→ [^21]	daily	site table scrape + Statistics section	free w/ attribution	NY/LDN fix cadence; holiday gaps (action-note: pull daily + monthly series)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
82	igc-grains	agri	International Grains Council — grains/rice/oilseeds market report + supply-demand	monthly	2000s-era→	monthly	web/PDF + paid data portal	partial licence (archives paid)	annual projec-tion vintages (action-note: ; free monthly summary parse)	W1	quote	UNVERIFIED-CANDIDATE
83	mpob-palm	archive/agri	MPOB BEPI — Malaysian palm stats (produc-tion, stocks, exports, FFB; monthly release 10th 12:30 + annual Overview 2016→ PDFs) [^22]	monthly	deep (portal stats); Overviews 2016→ [^22]	monthly 12:30	portal login + monthly tables	free regis-tration (login-gated) [^22]	login-gate; MY-only official, world palm via PSD (action-note: register; sched-uled pull + snapshot [#22])	W0	connect	VERIFIED-UPSTREAM-DOC
84	iso-sugar-daily	agri	International Sugar Organiza-tion daily world sugar price	daily	1985-ish→	daily	website web/PDF + data subscrip-tion	free headline / archives paid	NY11-White premium context (action-note: pull daily series)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
85	eia-seds-longrun	archive	EIA State Energy Data System (SEDS) — 1960→ state energy by source/sector, annual [^2]	annual	1960→	annual	APIv2 + bulk [^2]	free US-gov	benchmark updates each fall (action-note: bulk ingest for US-energy panel)	W0	connect	VERIFIED-UPSTREAM-DOC
86	copper-warehouse-verify	benchmark	copper LME-COMEx-SHFE tri-market spread validation panel (per #3/#5/#15)	daily	full overlap window	daily	derived from 3/4/5/15	mixed	inventory-correction impossibility — must reconcile (action-note: build arbitrage panel test)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	Similarweb	web/app	Similarweb Digital Intelligence (site traffic, web segments)	site/per-month visits & channels	panel-derived backfill (publisher claim; years)	weekly-ish refresh	licensed REST API + file-export entitlement	commercial-vendor ⚠no-redistribute	estimates are panel-modelled, not census — record panel mix + revision policy per release (action-note: request + API T&Cs; pilot one competitor set before committing)	W2	quote	UNVERIFIED-CANDIDATE
2	Semrush/Ahrefs-class web intel	web/app	Semrush Traffic Analytics (second opinion vs Similarweb)	site/per-month	years backfill (vendor claim)	near-realtime to weekly	licensed API (per-plan limits)	commercial-vendor ⚠no-redistribute	clickstream/panel estimates with thin methodology disclosure — use as triangulation only, never spine (action-note: only; optional second-source for web-coverage)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	Sensor Tower (absorbed data.ai 2024)	web/app	Sensor Tower Digital Market Intelligence (app downloads/DAU/MAU, IAP estimates)	app/cohort/global-weekly-monthly	global-year backfill (publisher claim)	weekly-ish refresh	licensed Query API / BI export	commercial-vendor ⚠no-redistribute	panel extrapolation; store-policy shocks create breakpoints — record panel version; **one invoice: data.ai/Sensor Tower consolidated** (action-note: request ; scope = 100–500 app universe only)	W2	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
4	App-store charts (App Store / Google Play)	web/app	public top-charts pages	app/daily rank by category	rank snapshot only (no history) → self-archive from day 0	realtime (page)	self-capture (⚠ both stores restrict auto-mated access — ToS-gray)	self-captured ⚠ToS-risk	charts re-rank intraday — pin fetch time; keep raw HTML/PNG evidence (action-note: build daily rank harvester (top ~200, core geos, price/type flags))	W2	connect	UNVERIFIED-CANDIDATE
5	Amazon market-place monitor-ing	web/app	Amazon best-seller ranks, offer counts, buybox price per ASIN	ASIN/daily (hourly pilots)	self-archive only	near-realtime	self-capture (⚠ Amazon actively bans scraping; consider partner vendors)	self-captured ⚠ToS-risk	rank is category-local and re-scores hourly; snapshot full PDP; dedupe offer set (action-note: pilot 1k-ASIN tracker; evaluate d alterna-tive before scale-up)	W2	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
6	Keepa (Amazon price/rank history)	web/app	Keepa API tokens (pre-collected price & rank series by ASIN)	ASIN/daily-hourly	multi-year price history (popular ASINs back to ~2015)	daily token quota (~monthly token packs; pricing UNVERI-FIED)	licensed API	commercial-vendor ⚠no-redistribute	rank units differ per category — normal-ize; gaps for cold ASINs; price-reads vs rounded market prices mixed (action-note: price 500-ASIN pilot coverage before commit-ting)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
7	Glassdoor (employer sentiment)	web/app	Glassdoor data partnerships (reviews, ratings, salary bands)	employer/month	vendor-claim years	event-level on page	licensed partnership feed	commercial-vendor ⚠PII (self-selected reviewer panel)	extreme selection bias (disgruntled/leavers); moderation edits republish — store first-seen timestamps (action-note: via partnerships desk; low priority vs LinkUp/Lightcast)	W3	quote	UNVERIFIED-CANDIDATE
8	G2 (software peer sentiment)	web/app	G2 research program (grid scores, review trends)	vendor/quarter/product-scorecards	launch forward only	quarterly	licensed research-data program	commercial-vendor ⚠no-redistribute	tiny review volume per vendor pre-2022 — mid-quality confirmation only (action-note: ; bundle with Glassdoor decision)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
9	Google Trends	web/app	Google Trends UI + unofficial `pytrends` library	term/geo daily (hourly last 90d)	back to 2004	hours	library self-capture (⚠ unofficial endpoint, no SLA; rate-limit hard)	free (⚠ ToS-gray on bulk)	sampling noise — re-runs differ; snapshot per run and never join across reruns (action-note: `pytrends` pipeline for ~200-term watchlist; zero-vendor depen-dency)	W1	connect	UNVERIFIED-CANDIDATE
10	Wikimedia pageviews	web/app	`wikimedia.org/api/rest_v1/metrics/pageviews` + dumps	per-project-article daily-hourly	2015-08 →	near-realtime	REST API batch; per-month HTML dump files	free CC0-attribution (WMF)	page re-names/redirect chains break series — resolve canonical titles first (action-note: company wiki pages + ticker redirects as interest proxy)	W0	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	Indeed Hiring Lab posting aggregates	employment	hiringlab.org sector/state/geo postings trackers (vs Feb-2020 base)	country/state/sector weekly	2020→ (indices, rebased — not raw counts)	weekly	free PDF/CSV attachments (data page WAF-403 to bots — browser headers)	free-attribution (publisher credit terms)	index rebasing + methodology footnotes per release — snapshot every release; revisions not versioned (action-note: ingest weekly US/EU trackers; store attachments immutably)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
12	LinkUp (raw postings)	employment	LinkUp STS postings API (posting counts, skills)	posting/geo/pub daily	publisher claims 2008+ archive	daily	licensed API / flat-file entitle-ment	commercial-vendor ⚠no-redistribute	posting dedupe & spam-post inflation are known issues — demand method-ology changelog (action-note: ; bench-mark 1 country vs Indeed free + JOLTS)	W2	quote	UNVERIFIED-CANDIDATE
13	Lightcast (ex-Emsi Burning Glass)	employment	Lightcast post-ings/skills subscrip-tion	posting/firm/skill daily	~2010+ (claim)	daily	licensed API + flat files	commercial-vendor ⚠no-redistribute	multi-source scrape consoli-dation — request post-dedupe count method-ology doc (action-note: (firms + skills bundle))	W2	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
14	layoffs.fyi	employment	layoffs.fyi crowd-tracked tracker (firm/region/sector, staged per date)	firm/event daily	2020-03 →	near-realtime (community-edited)	self-capture weekly CSV/sheet snapshot	free-attribution (site terms)	crowd-edit races; severance numbers arrive delayed — store last-modified timestamps; cross-check WARN notices + 8-Ks (action-note: CSV ingest; free panel; add cross-check gate in pipeline)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
15	Sentinel-1 SAR	satellite/geo	Copernicus Data Space S1 GRD+SLC	swath/12d revisit	2014-10 →	hours	STAC product catalogue / openEO / S3 + OData; `sentinel-sat`/openeo SDKs ([^1])	free (Copernicus open-data licence)	processing-baseline versions change back-compat — archive by baseline ID (action-note: STAC + openEO extract for watch-list ports/fields)	W1	connect	VERIFIED-UPSTREAM-DOC
16	Sentinel-2 L1C/L2A	satellite/geo	Copernicus Data Space S2 MSI L1C/L2A (cloud-masked)	tile/10m	L2A from 2016-06 (L1C 2015-06)	hours	STAC / openEO / S3 ([^1])	free (Copernicus)	Sen2Cor version shifts L2A values — pin baseline; cloud-mask QA layer mandatory (action-note: same pipeline as row 15; NDVI/urban-sprawl indices)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
17	Landsat 5-9 archive	satellite/geo	CDSE Complementary Data (Landsat-5/7/8/9 + Bi-monthly Mosaics) + USGS EarthExplorer / Machine-to-Machine API	scene/pass ≈16d	**1972 →**	hours	USGS M2M API (bulk) — head of site is bot-challenged (202); CDSE STAC mirrors Landsat products ([^1])	free (USGS public-domain; Copernicus licence for CDSE path)	Collection 2 standard — pin collection version; CDSE coverage caveats — re-confirm geos (action-note: primary = USGS M2M bulk; CDSE as convenient mirror)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
18	Planet Insights Platform (incl. NICFI)	satellite/geo	Planet ~3 m daily + monthly mosaics; NICFI tropics basemaps	tile/daily	NICFI from 2015/2016 (tropics); commercial scope varies	daily	licensed APIs (tasking/orders)	commercial-vendor ⚠no-redistribute	cloud artifacts; NICFI program phases change mosaic availability — record mosaic release ID (action-note: for 3 m mosaics over watch-atlas zones (pilot))	W3	quote	UNVERIFIED-CANDIDATE
19	ICEYE	satellite/geo	ICEYE X-band SAR constellation + analytics	tile/task daily	archive on request	day (tasking)	licensed download & API	commercial-vendor ⚠no-redistribute	tasking vs archive pricing split; SLC/L1 access tiers; flood/freeze analytics separate SKUs (action-note: (flood/train-disruption case study))	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
20	Umbra	satellite/geo	Umbra SAR tasking + open-data distribu-tion	spot-task tiles	archive on request; some free samples	day	licensed API + free dis-tributor tracks	commercial-vendor (limited free tier)	task price varies by area/angle — record task plan (action-note: ; low priority vs IC-EYE/Planet)	W3	quote	UNVERIFIED-CANDIDATE
21	NASA FIRMS active fire	satellite/geo	FIRMS area API (MODIS/VIIRS/Landsat fire de-tections)	point/daily	2000+ MODIS; 2012+ VIIRS; 2024+ Landsat (US/CA NRT)	≤3h NRT; URT <1 min US	REST `/api/area/csv/…` with free MAP_KEY; 5000 tx/10-min cap ([^2])	free (US gov)	VIIRS vs MODIS pixel-size differ; confidence/low-confidence filtering manda-tory (action-note: ingest for crop/forest watch-geos; register MAP_KEY)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
22	VIIRS night lights (S-NPP/NOAA-20/21)	satellite/geo	NOAA/Colorado Mines EOG data center (daily + monthly composites)	grid/daily-monthly	2012-01 →	hours–days	file-drop (free w/ account) — **/products/viirs/ path returned 404 this census; domain live but portal restructured** ([^6])	free-attribution (EOG)	stray-light correction versions; radiance units nW/cm²/sr (action-note: re-locate current EOG catalogue page + bulk repo before ingest)	W2	connect	SURVEY-EVIDENCE
23	DMSP-OLS 1992–2013 night lights	satellite/geo	EOG/NOAA DMSP-OLS intercalibrated annual composites	grid/annual	1992 → 2013 (frozen)	n/a (never updated)	one-time bulk download	free-attribution	inter-calibration across F10–F18 satellites debated — use distributed intercalibrated set, pin version (action-note: pull full archive once; historical light index)	W0	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
24	GPM IMERG rainfall	satellite/geo	NASA GPM IMERG V07 early/final	grid/0.1°/30-min	2000-06 →	4h (early) / weeks (final)	GES DISC file-drop + Giovanni extracts	free (NASA)	Early vs Final differ — always FINAL for back-tests; griddle bias vs station rain (action-note: scripted GES DISC pull for pilot basins)	W1	connect	SURVEY-EVIDENCE
25	ERA5	weather	Copernicus Climate Data Store re-analysis	grid/0.25°/hourly	**1940-01** →	~5-day tail	`cdsapi` file-drop (NetCDF/GRIB), monthly release runs	free-attribution (licence per-dataset; re-check CDS licence banner)	ERA5T prelimi-nary values revised next cycle — snapshot release month (action-note: CDS bulk pulls for key watch-grids)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
26	WorldPop population grids	satellite/geo	WorldPop hub (counts, density, age-sex, migration, settlement)	grid/100m country-year	2000 → 2030 projections	annual	file-drop + WOPR REST API ([^3])	free CC-BY-4.0 (confirmed on hub footer)	UN-adjusted vs unadjusted vintages per country; census covariate lag by country (action-note: ingest v2020/2021 rasters for GFI country set)	W1	connect	VERIFIED-UPSTREAM-DOC
27	Google Open Buildings v3 (+Temporal)	satellite/geo	GCS `gs://open-buildings-data/v3/` polygons (S2-sharded CSV/WKT); v3-Temporal = per-epoch presence	building footprint	v3 inference May-2023 on 58M km² (Africa/S/S-E Asia/LatAm/Caribbean); Temporal = presence series	batch (rare updates)	gsutil download / per-country Colab; Earth Engine asset; HDX packs ([^4])	free dual-licence CC-BY 4.0 / ODbL 1.0	publisher recommends ≥90%-precision confidence filter; Global-South coverage only (no G7) (action-note: download pilot footprints + Temporal presence)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
28	OSM-derived tables	satellite/geo	OpenStreetMap planet dump + full-history diffs + Overpass	Maple/daily	continuous 2007 → (history via diffs)	minutes–daily	weekly `.osm.pbf` file-drop + `pyosmium` diff application; Overpass self-capture	free ODbL 1.0 (attribution + share-alike on derived DB)	bulk-import waves create fake growth jumps — count real additions from full-history `.osh`; tag-usage drift (action-note: nightly POI-count time-series DB for watch city-set)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
29	Copernicus Marine (SST/chlorophyll)	satellite/geo	Copernicus Marine Service (model + satellite products)	grid/daily	SST 1982+; reanalysis 1992+	hours–days	`copernicusmarine` python file-drop	marine (Copernicus)	product version migrations break series — archive toolchain logs per version (action-note: optional for crop/fish-adjacent commodity overlay)	W2	connect	SURVEY-EVIDENCE
30	PortWatch (IMF + World Bank)	shipping/logistics	portwatch.imf.org — AIS-derived port calls, congestion, chokepoints	port/daily; chokepoint/daily	historical archive from ~2019 baseline (publisher claim — verify on first load)	daily (1–2d source lag)	public dashboard + curated Python package (package name/access to re-verify on portal) ([^7])	free-attribution (IMF open data — confirm licence on portal)	congestion-window definitions disclosed in pipeline notes; AIS panel composition not disclosed (action-note: ingest daily flow for listed ports; historical depth immediately)	W0	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
31	UNCTAD ports & liner connectivity	shipping/log	UNCTADstat datacentre (TEU throughput, PLSCI)	port/country monthly-yearly	throughput from ~1970s; PLSCI 2004+	yearly-quarterly	datacentre CSV/SDMX/Excel exports (JS-rendered UI — direct file URLs needed) ([^8])	free-attribution (UN)	port renames / operator change break continuity — use UNCTAD continuity notes (action-note: locate direct CSV/SDMX endpoints; ingest long TEU series)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
32	Kpler (post-2023 hub, absorbed MarineTraffic)	shipping/log	Kpler One (tankers/dry/LNG AIS flows, port calls, flow analytics)	vessel/port daily	vendor-claim multi-year backfill varies by product	hours	licensed platform/API entitlement	commercial-vendor ⚠no-redistribute	dark-fleet AIS gaps compress into estimates — flag; MarineTraffic integrations one-vendor since 2023 — record vendor version (action-note: (commodity-flow tier only))	W3	quote	UNVERIFIED-CANDIDATE
33	MarineTraffic (Kpler-owned)	shipping/log	MarineTraffic vessel/port AIS	vessel/port realtime-daily	platform holds historic AIS per plan	near-realtime	licensed API / platform	commercial-vendor ⚠no-redistribute	same AIS pool as Kpler — treat as one vendor from 2023; avoid double-purchase (action-note: fold into Kpler quotation (one invoice))	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	FleetMon	shipping/log	FleetMon vessel tracking & port ETD/ETA API (independent, DE)	vessel/port realtime-daily	long AIS history (claim)	near-realtime	licensed API/Explorer	commercial-vendor ⚠no-redistribute	independent vendor = hedge vs Kpler consolidation; benchmark AIS match rate on common sample (action-note: ; keep as vendor-risk hedge)	W3	quote	UNVERIFIED-CANDIDATE
35	aisstream.io free AIS feed	shipping/log	aisstream.io WebSocket relay (free key)	vessel/realtime	stream only (we self-archive)	realtime	WebSocket + key; spool MMSI-level into DuckDB	free (per site ToS)	community receiver coverage uneven (good Americas/EU, thin elsewhere) (action-note: 30-day pilot spooler to quantify realistic coverage before paying a vendor)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
36	Global Fishing Watch open data	shipping/log	GFW API + open datasets (fishing presence, transshipment, vessel identity)	vessel/grid daily	2017 →	days	open REST API + weekly GeoJSON tiles	free CC-BY-4.0	fishing-gear class probabilities — keep gear-class caveats (action-note: vessel-flow overlay on commodity corridors)	W2	connect	SURVEY-EVIDENCE
37	EMODnet Human Activities	shipping/log	EMODnet Human Activities WMS/WPS/CSV (ports, shipping density, EU maritime zones)	port/grid monthly-yearly	~2005+ by layer	monthly batch	open datasets + WPS services	free-attribution (EU)	EU-scope only — incomplete outside EU waters (action-note: ingest only if EU corridors on GFI list)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
38	Container freight-rate indices	shipping/log	Drewry WCI / Freightos FBX / Xeneta	lane/weekly index	WCI 2011+; FBX 2017+	weekly	licensed dash-board export + free weekly press-release attribu-tion share	commercial-vendor (chart at-tribution lenient)	spot vs contract mix per index — stay index-consistent; publica-tion gaps docu-mented (action-note: subscribe weekly CSV for ~8 long-haul lanes)	W2	connect	UNVERIFIED-CANDIDATE
39	BofA Institute card ag-gregates	consumer/credit	institute.bankofamerica.com	household aggregate weekly-monthly by cohort; US	2019 →	weekly release cadence	free PDF/CSV attach-ments ([^9])	free-attribution (issuer's own ag-gregate publica-tion)	issuer-sample (BofA cardhold-ers skew NE US/wealthy); index rebases at period start — track release footnotes (action-note: weekly scrape → CSV normal-ization)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
40	Facteus (de-identified card panel)	consumer/credit	Facteus Mimic panel (US card spend trends)	spend/category/merchant daily-weekly	vendor-claim from ~2014	near-realtime	licensed feed/API	commercial-vendor ⚠no-redistribute	merchant-category allocation (NAICS cross-walk) is the main quality risk — verify before use; panel composition doc required (action-note: ; only if BofA/Placer gap confirmed)	W2	quote	UNVERIFIED-CANDIDATE
41	Consumer Edge	consumer/credit	Consumer Edge card-spend data (US/global)	spend/brand daily-weekly	vendor-claim multi-year	daily	licensed feed/API	commercial-vendor ⚠no-redistribute	bank-sample panel churn — request panel-cohort doc; spend-share ≠ transaction-count (action-note: ; big-cap consumer coverage pitch)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
42	Placer.ai (Placer Labs)	consumer/credit	Placer.ai retail foot traffic	POI/daily visits	vendor-claim years	daily	licensed portal reports only (no bulk API on standard tier)	commercial-vendor ⚠no-redistribute	GPS-device panel mix churn; store chain-level definitions; POI re-geocoding mid-series (action-note: ; scope to ~50 tickers' POI list)	W2	quote	UNVERIFIED-CANDIDATE
43	Unacast	consumer/credit	Unacast site analytics deliveries (visits, cross-shop, commute)	POI/daily	vendor-claim years	daily	licensed S3 deliveries	commercial-vendor ⚠no-redistribute	device-panel sampling shift; record panel-vintage per snapshot (action-note: secondary as Placer fallback)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	Google COVID-19 Commu-nity Mobility Reports	consumer/credit	Google health-research frozen archive (categori-cal mobility %)	country-region-category daily	2020-02 → 2022 (frozen)	n/a (frozen)	per-region PDFs/CSV via health.google research page (guessed help URL 404’d this census — locate via site nav)	free-attribution	frozen product — historical only; Jan/Feb-2020 baseline fixed (action-note: one-time pull of all target-region files)	W0	connect	SURVEY-EVIDENCE
45	Apple Mobility archive	consumer/credit	covid19.apple.com/mobility frozen CSVs per coun-try/region (driv-ing/walking/transit)	country-region daily	2020-01-13 → end-2022 (frozen; verify end date on pull)	n/a (frozen)	direct CSV/JSON download ([^12])	free-attribution	relative-vs-Jan2020 index only; transit series weak outside big cities (action-note: grab full historical JSON once)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	GTFS-RT transit feeds	consumer/crowd	Mobility Database catalog + per-agency GTFS real-time feeds	agency/realtime	self-archive from ingest	realtime	protobuf + JSON per feed; daily schedule roll-up	free (per-agency licences — most open, some restrict auto-mated use)	feed down-time churn; per-agency licence variance (action-note: collect feeds for ~10 major GFI metros)	W2	connect	SURVEY-EVIDENCE
47	Panjiva (S&P Global)	trade/customs	Panjiva supply-chain intelli-gence (cus-toms/shipment records)	shipment/entity; daily-monthly	US BoL baseline ~2007+; other countries frag-mented	daily–weekly	licensed plat-form/API	commercial-vendor ⚠no-redistribute	raw customs data quality issues (typos, delays) — S&P cleaning rules; CN data restricted (action-note: ; start with supplier-node graph for 1–2 in-dustries)	W2	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
48	ImportGenius	trade/customs	ImportGenius customs shipment data	shipment/daily	US BoL from ~2006 (claim)	daily–weekly	licensed platform/API	commercial-vendor ⚠no-redistribute	cleansing similar to Panjiva — buy only one of the two customs shops (action-note: ; prefer Census raw + Panjiva if budget limited)	W2	quote	SURVEY-EVIDENCE
49	US Census international trade (USA Trade Online + bulk API)	trade/customs	api.census.gov/data/timeseries/intltrade — imports/exports HS-2/4/10 by country + port + district	monthly	1989/1992/1996/2002 (varies by series; decades)	monthly (~T+6 wks after period close)	REST bulk API + USA Trade Online UI ([^15])	free (US gov public-domain)	HS/global classification revisions — store first-release vs latest-revision; cross-walks rebuilt yearly (action-note: full monthly imports/exports bulk pull to SQLite/S3; revision snapshotting)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
50	UN Comtrade Plus	trade/customs	comtradeplus.un.org + v2 API (v1 deprecated — migrate)	country × HS flow monthly-annual	**free back to 1962** (annual aggregates); monthly depth varies by reporter	T+4 weeks to months	REST API v2 + bulk files + BigQuery mirror ([^13])	free-attribution (UN)	reporter methodology changes (FOB/CIF, revisions); store reporter metadata; v1→v2 migration guard (action-note: monthly key flows (US/CN/EU27/JP/KR/IN) + historical annual full pull)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
51	Eurostat Comext	trade/customs	Eurostat international trade-in-goods (Comext) UI + bulk/API routes	reporter-partner-HS monthly	**1988 →**	monthly T+6 wks	REST bulk (`/api/comext`) + TSV bulkfiles ([^14])	free-attribution (EU)	intra-EU flows re-bound on enlargement (EU-27) — store break-down to avoid false series-shifts (action-note: bulk API for extra-EU27 focus flows)	W0	connect	VERIFIED-UPSTREAM-DOC
52	India DGFT / MoSPI trade statistics	trade/customs	DGFT tradestatistics + MoSPI open data (HS export-import)	HS/country monthly-yearly	~2010+ online; earlier via publications	monthly batch	web CSV/PDF self-capture + data.gov.in APIs	free-attribution (India gov)	formatting not versioned; HS alignment crosswalk maintained (action-note: bulk monthly pull; secondary to Comtrade for IN)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
53	China GACC customs HS statistics	trade/customs	GACC online statistics query (HS-8 monthly)	commodity/country-monthly	~2014+ reliably online; earlier in year-books	monthly (mid-month lag)	self-capture query UI (⚠ anti-bot) + yearbook PDFs	free-attribution (China gov; site discour-ages bulk scrape)	primary CN trade source; no public bulk API — record query cadence (mid-month) (action-note: build query-scraper capturing monthly CN im-port/export by com-modity)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
54	FEC API v2 (campaign finance)	politics/policy	api.open.fec.gov v2 — filings, contributions, eFiling	committee-contribution transaction	filings archives from 1980s; efilings stream 2017+	efilings near-realtime; rest T+0–7	REST API (free key; ~1000 calls/hr) + bulk flat files per cycle ([^17])	free (US gov)	form-type mapping (F3X/F3P…); committee re-registrations — dedupe `filer_ids`; itemized vs unitemized amounts (action-note: bulk 20-cycle historical pull + live efilings stream)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	OpenSecrets (contributors + lobbying)	politics/policy	opensecrets.org API + bulk data files (CRP); lobbying LDA remap	politician/committee/company cycle	1989/1990-present (claim)	yearly/half-yearly	free key-by-request API (redistribution restricted) — **docs page WAF-403 to scripts this census** ([^18])	free-attribution ⚠no-redistribute	LDA→standard industry mapping re-versioned — snapshot mapping; prefer FEC raw as spine, OpenSecrets for aggregates only (action-note: confirm API access/terms hand-on before use)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
56	GovInfo bulk (Congressional Record / bills)	politics/policy	govinfo.gov bulk XML/PDF + api.govinfo.gov	document/daily	CR electronic 1994+; pre-1994 print	next-day	bulk per-year packages + free-key API	free (US gov)	CR daily vs bound revisions — store with revision flag; bill text versions pinned by version ID (action-note: new-docs API feed + CR 1994→ bulk archive)	W0	connect	SURVEY-EVIDENCE
57	Federal Register API v1	politics/policy	federalregister.gov/api/v1 (rules/notices/PRs/comment windows)	agency/daily	1994→	next-day	keyless REST JSON (fields param) + bulk XML mirrors	free (US gov)	docket re-corrections — snapshot docs, don’t re-pull (action-note: new-doc feed as event calendar)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
58	Regulations.gov API v4	politics/policy	regulations.gov API — dockets, documents, comments	docket/comment daily	electronic comments ~2003+; pre-2003 scans	hours–day	REST v4 (free key; rate-limited) + bulk dataset files	free (US gov)	v4 schema reshaped — pin `/datasets` schema; form-letter campaign dedupe (action-note: dockets for financials/energy/crypto/AI watch sectors)	W1	connect	SURVEY-EVIDENCE
59	Congress.gov API v3	politics/policy	congress.gov API v3 (bills, amendments, members, votes, committees)	bill/member daily	full release history as published	next-day	REST (free key; chunked JSON)	free (US gov)	use `latestAction.date` not index; roll-call depth varies by era (action-note: key + incremental bill tracking + committee rosters)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
60	Open States	politics/policy	v3.openstates.org API (50-state bills, legislators, committees)	state-bill daily	backfill varies by state (up to ~2007+)	near-realtime bill actions	free API key + bulk data downloads ([^19])	free-attribution (per-state source licences vary)	per-state cadence/quality differs; snapshot rather than re-pull; API key confirmed in docs ([^19]) (action-note: ingest bills feed for ~5 focus legislatures)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	US LDA lobbying disclo-sure (primary)	politics/policy	official LDA search / quarterly report feeds (House/Senate disclo-sure portals)	registrant/filing quarterly	electronic LDA era 2008+; paper filings remain	quarterly	downloadable XML/bulk + portal queries	free (US gov)	issue-code standard-ization; client-name entity res-olution needed; portal URL churn — locate bulk end-points (action-note: locate bulk XML export end-points — TBD this census)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
62	FiveThirtyEight data archive	politics/policy	github.com/fivethirtyeight/data (polls, pollster ratings, generic-ballot)	poll-level + aggregate	**1948**→ historical-polling archives + modern scrape archives	frozen (repo static)	GitHub clone ([^20])	free CC-BY-4.0 (per-folder licences — read READMEs)	repo frozen post-2023-ish — historical only; pollster-weight defini-tions per project; new ag-gregator (ABC538) separate (action-note: clone archive; generic-ballot 1948→ series)	W0	connect	VERIFIED-UPSTREAM-DOC
63	RealClearPolitics aggre-gates	politics/policy	realclearpolitics.com daily average tables (embed-ded, no API)	race/day avg	race-level rolling — self-archive	daily	self-capture HTML (⚠ ToS check before scale)	self-captured ⚠ToS-risk	methodology opaque; race links break across cycles — snapshot per race (action-note: harvest final-weeks averages per election cycle)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
64	White House visitor logs archive	politics/policy	NARA / white-house.gov disclo-sure portals per adminis-tration	visitor/day-batch	Obama-era bulk 2009–2017; later admins par-tial/policy swings	n/a (archive)	file-drop PDF/XLSX per release	free (US gov)	redaction tiers vary by admin-istration; record release policy per admin (action-note: one-time archive pull per admin; low priority)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	SEC EDGAR full-text search + n-gram build	corp-own-signal	`efts.sec.gov/LATEST/search-index` (200 OK this census) + FTS UI; filings corpus cross-ref sheet 00	filing-level	**2001 →** (FTS coverage)	realtime-ish (index within ~min)	REST JSON; respect 10 req/s + declared User-Agent ([^21])	free (US gov)	FTS covers published text incl. exhibits (graphics excluded); filings resubmissions — dedupe versions in n-gram panel (action-note: corpus queries + n-gram extract for key themes (supply chain, layoffs, “generative AI”))	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
66	Earnings-call tran-scripts	corp-own-signal	→ cross-ref only to tran-scripts stream (sheet 05 overlap): Motley Fool free tier, Al-phaSense/CapIQ licensed, SA paywall	transcript/event	per source	hours after call	cross-ref only — licensing & scraper-risk live in sheet 05	mixed: free tier ⚠ToS; licensed ⚠no-redistribute	never splice n-grams across transcrip-tion providers — vendor-token differ-ences (action-note: depen-dency note only; owner-ship lives in sheet 05)	W1	connect	SURVEY-EVIDENCE
67	Annual-report text mining	corp-own-signal	→ cross-ref only to filings sheet (00): 10-K/20-F/annual reports + proxy state-ments PDFs	filing/doc-level	per filing	n/a	cross-ref only (text-extraction pipeline belongs to filings sheet)	free (US gov; foreign filings per juris-diction)	— (action-note: depen-dency note only)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
68	TSA passenger volumes	travel	tsa.gov/travel/passenger-volumes — daily checkpoint throughput CSV (with same-day prior-year compare)	checkpoint-level daily	2019 → (older years via archived releases)	daily	CSV file-drop per year ([^22])	free (US gov)	holiday-ops peaks; counts = travelers, not flights — proxy only (action-note: full CSV archive pull + monthly updater)	W0	connect	VERIFIED-UPSTREAM-DOC
69	BTS Airline On-Time Performance	travel	TranStats BTS “Reporting Carrier On-Time Performance”	flight/route-daily	**1987 →**	next-month batch	TranStats monthly CSV/bz2 downloads (site WAF-hard to scripts — browser headers)	free (US gov)	carrier code changes across years; cancelled/diverted flags stable; route-truncation pre-2003 (action-note: monthly historical pull to 1987)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
70	UNWTO (UN Tourism) statistics	travel	UN Tourism data dash-board + World Tourism Barome-ter	country monthly-yearly arrivals, receipts	decades (reporter-based)	monthly-quarterly	dashboard + licensed analytics channel for depth	partially free (head-line), depth-tier licensed	arrivals defini-tions differ (tourists vs day visitors) per country — store defining metadata (action-note: barome-ter subscrip-tion + free headline tables)	W2	connect	UNVERIFIED-CANDIDATE
71	STR / CoStar hotel data	travel	STR via CoStar (occu-pancy, ADR, RevPAR)	country/CITY monthly-weekly	multi-decade (claim)	weekly/monthly	licensed plat-form/API	commercial-vendor ⚠no-redistribute	market-code sets; pre-1996 chain-census differ-ences — pin method-ology year (action-note: ; only if hospital-ity sectors in GFI list)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
72	Open Ownership (BODS)	ownership	openownership.org beneficial-ownership dataset (BODS standard)	entity-beneficial-owner snapshots	per implementation 2015+; country coverage uneven	periodic refresh	bulk downloads + API	free-attribution (BODS; per-source national licences vary)	per-jurisdiction quality highly uneven — build scorecard; US CTA BOI status volatile (policy) (action-note: bulk ingest + country-coverage scorecard)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
74	Bureau van Dijk Orbis	ownership	Moody’s BvD Orbis (global firm ownership structure + financials)	company/annual (monthly embed)	deep, varies by country (claim)	monthly embed	licensed platform / flat-file delivery	commercial-vendor ⚠no-redistribute	large-firm bias in some EMs; ownership linkages update as filings change — record snapshot date; cross-check open registries (action-note: quote; scope to watch-list firms)	W3	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
75	PACER (US federal court filings)	ownership/legal	pacer.uscourts.gov (documents, dockets; ~$0.10/page, waived under $30/quarter)	docket/document daily	online records ~1990s+ (varies by court)	realtime-ish	pay-per-page portal/API (judgement/fraud)	fee-for-service (nominal; not free redistribution license — check)	redaction quality varies by court; fees prevent bulk pulls — strategic query lists only (action-note: scope: targeted GFI-litigation watchlist; check current fee waiver rules)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
76	CourtListener / RECAP	ownership/legal	courtlistener.com (free PACER mirror + REST API + bulk archives)	docket/document-daily	RECAP archive coverage grows via user pulls	near-realtime updates	REST API + bulk tarballs	free-attribution (RECAP; per-term judicial data subset)	coverage is subset of PACER (grows via contri-butions) — never treat as full federal corpus (action-note: ingest API for watchlist dockets; check bulk archive terms)	W2	connect	SURVEY-EVIDENCE
1	GLEIF LEI base pool (Level 1 LEI-CDF)	corporate-registry	GLEIF Golden Copy + LEI2 concate-nated files (lei-data.gleif.org/api/v1); daily full files	record-per-LEI (~3.3M; 2026-05 validated ~3,304,415)	full since 2012 LEI launch; **previous Golden Copies retained for down-load**; daily deltas (intra-day/day/week/month)	daily	file-download (ZIP XML CDF) + GLEIF JSON API	open/attribution (CC-class free; re-confirm term page)	ContentDate vintage per file; registra-tion.status alive-set for PIT (action-note: pin schema 2.1; ingest snap-shot+delta cadence)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
2	GLEIF relation-ships (Level 2 RR-CDF)	corporate-registry-graph	RR-CDF Golden Copy/concate-nated files; XML schema 2.1 (fund-relationship types since 2021)	edge-per-parent (direct/ultimate consoli-dating parent, fund-managed-by, sub-fund-of, feeder-to; ~647k records)	archived Golden Copies back to RR publi-cation (Nov 2016 format v1.0; May 2017 v1.1; May 2021 v2.1); valid-From/validTo on edges	daily (8h Golden Copy cycle)	file-download + JSON API (`direct-parent-relationship`)	same as LEI base	relationship.periods[] + validity dates = PIT spine for own-ership graphs (action-note: separate ingest stream from base pool; version history ≠ edge history (dedupe delta se-mantics))	W0	connect	VERIFIED-UPSTREAM-DOC
3	GLEIF Reporting Excep-tions (RepEx)	corporate-registry	RepEx CDF concate-nated files (~5.9M records)	record-per-non-reported-parent (state-owned, natural person, legal re-striction)	full archive via previous Golden Copies	daily	file-download	same	use as confi-dence labels, not edges (action-note: attach as edge-quality attribute)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
4	OpenCorporates	corporate-registry-aggregator	REST API JSON/XML (documented v0.4.8); free for open-data projects w/ attribution+share-alike; paid accounts remove share-alike restrictions	company record + officers + PSC/beneficial data + incorporation events (200d jurisdictions)	long-tail: some records back to 1600s-1800s; per-source depth varies	near-realtime API; snapshot via paid	API (key required)	attribution+share-alike (free) / paid-API	source provenance field; watch share-alike handling of indices (action-note: quote for commercial use BEFORE ingest)	W1	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	UK Companies House (registry)	corporate-registry	REST API (free, key) + bulk product downloads (free; company profile, filing history, persons-of-significant-control)	company/record; directors; PSC	bulk; filing history back to 1991 for electronic records; paper pre-1991 scanned; PSC since 2016	API near-realtime; bulk weekly	API + file-download	OGL-UK govt	cross-ref 05 skip duplicates: 05 handles accounts/filings consumed for fundamentals — this row = identity + directors + PSC only (action-note: dedupe vs class 05; PSC graph join to GLEIF)	W0	connect	VERIFIED-UPSTREAM-DOC
6	EU BRIS (business register interconnection)	corporate-registry	e-Justice BR portal (ejustice-justo gov) — search UI per national BR; **no official bulk/API**	company stub per member state	current only	web	scraping only (ToS risk)	portal-only	patchwork formats (action-note: low priority; prefer national registries (INPI/BRIS members map))	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
7	US state corporate filings	corporate-registry	per-state SOS: DE SOS search portal (no API), NY Open-Data CSV, WA SOS bulk DL, CA bizsearch…	filing-level per state	mixed; some states keep 100y+ (DE pre-1900), most online since 1990s	daily-ish per portal	scraping (no unified API)	state govt public records	entity-number keys; name-change events unreli-able (action-note: jurisdiction-by-jurisdiction; start DE+NY+WA)	W2	connect	UNVERIFIED-CANDIDATE
8	Cayman Islands registry (CIMA/General Registry)	corporate-registry	cayman registry site — pay-per-search; no public API; occa-sional blocks on offshore-scraper clouds	company record	current; filing index history on request	pay-per-query	manual/pay portal	paid-portal	— (action-note: flag: unreach-able for automa-tion; use OpenCor-porates as proxy)	W3	connect	UNVERIFIED-CANDIDATE
9	BVI FSRA (VIR-RGIN)	corporate-registry	bviassa fsra viewer — pay-per-search, no bulk	company record	current	pay-per-query	manual/pay portal	paid-portal	— (action-note: flag: unreach-able; OpenCor-porates proxy)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
10	India MCA Master Data	corporate-registry	mca.gov.in view CIN/DIN master (free); API LOGIN-gated?; private proxies (SignCare ₹5/lookup, Deepvue) exist → direct API channel unclear 2026	CIN record: company status, directors, DINs	filing history depth ~2006+ (CIN); master = current snapshot	near-self (gov'ed) day up	web scraping or private API (paid)	restricted (gov) — scraping not clearly permit-ted	director identity PIT via DIN; no master-PIT (action-note: locate official API/registration; else out-source via proxy)	W2	connect	UNVERIFIED-CANDIDATE
11	India GSTN (GSTIN registry)	corporate-registry/tax-id	gst.gov.in GSTIN search (free portal); bulk API requires PAN-registration + ToS; periodic "auto dealers list"	GSTIN→PAN entity map	current only	near-realtime portal	portal + restricted API	portal-only (ToS-check needed)	PAN=XOR for Holly Henry dedupe (action-note: read GSTN API ToS; test search-page rate limits)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
12	Singapore ACRA (bizfile)	corporate-registry	ACRA data products + Free Corpo-rate Search; API via ACRA API Gateway (subscrip-tion tiers; free “shell” master-data no con-firmed)	company full par-ticulars / UBO extracts (paid per doc HK$5.50-SGD free-tier? unclear 2026)	filings since 2014 digitized? — e-docs; pre-2014 microfilm	extract per-order	API gateway (paid) / manual extracts	paid-API	UBO/registered-status via BizFile extracts (action-note: current gateway pricing incl. free tier)	W2	verify	UNVERIFIED-CANDIDATE
13	Hong Kong Compa-nies Registry (CR e-Search)	corporate-registry	e-services.cr.gov.hk — free: name search, doc index, charges index, disqualifi-cation index; paid: par-ticulars HK$22/co, director index HK$11, images HK$9-35/doc	doc-image + field-report level	docs index back to incorpo-ration year 1845eso? — images from 2014+; pre-2014 docs archived (order)	T+0 to T+3 images	pay-per-doc / account HK$500/yr principal + HK$100 subse-quent	paid-portal	**shareholder data with-held** (NID/PI) — form route only (action-note: build doc-index free; budget §8(h)-(j) disclo-sure HK$10 for FI access)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
14	Japan □□□□ (Houjin Bangou, National Tax Agency CNPS)	corporate-registry	houjin-bangou.nta.go.jp — free CSV per prefecture (regular rolling)	legal-entity number record: corporate number, name, address, HOCR w/o historical field	published monthly per prefecture (full replacement); no history at source — archive-of-vintages needed	monthly (free ML)	file-download	public-domain (Japanese gov open data)	corporate-number PKeys; no prior names (action-note: store vintage-stamped archives (free → trivial))	W0	connect	VERIFIED-UPSTREAM-DOC
15	Korea DART (corp registry subset)	corporate-registry	dart.fss.or.kr — free w/ login-gated corp code list download; API via open-dart.fss.or.kr free (registered corp corp-publishers only)	corp code + entity metadata	since 1998 electronic filings (05-assets); corp-code file monthly full	API near-realtime; corp-code list monthly	OpenDART API (free key, gov open-data-portal license)	gov open data (KR open-API license)	cross-ref: mercados Stocks (03) skip duplicates — keep identity only (action-note: OpenDART API key; align corp-code vintage with filings)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
16	KVK (Nether-lands)	corporate-registry	kvk.nl — extracts (paid per doc), Cor-porate History product, “KVK APIs” bundle; **KVK Open Data** free since 2025? (open data file incl. ad-dresses/registration)	company + office + history product	extracts historical since 1983?; digital open data 2025	extracts on-order	manual extracts + data-API (licensed)	paid-portal + some open-data	SBI codes; multiples per KvK ID (action-note: Open-Data API scope + extract pricing)	W2	verify	SURVEY-EVIDENCE
17	Australia ASIC + ABR	corporate-registry	ASIC Connect extracts (search-free-info product, paid per doc); ABR (ABN Lookup) — free web services (abr.business.gov.au json/xml SOAP; data.gov.au free ABN bulk extracts)	ABN→entity; ASIC per-company	ABN since 2000; ASIC extracts current	extracts T+0	ABN web services free; ASIC paid extracts	ABN: free/attribution; ASIC: paid	ABN current-only; entity-status snap-shots (action-note: ABR first (free); ASIC extracts only for col issues)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
18	Germany Handel-sregister	corporate-registry	handelsregister.de (free web search + pdf extracts since 2022 free?); no bulk	company record, pdf extract	since 2007 elec-tronic register (pre-2007 paper)	extracts per-order	manual/scrape (base-rate offered by 3rd parties legally?)	portal-only	unique court+HRB key (action-note: no-bulk flag; use OpenCor-porates for multi-entity lift)	W2	connect	UNVERIFIED-CANDIDATE
19	France INPI Open Data	corporate-registry	data.inpi.fr free down-loads (com-pany identities, filings PSC-like since 2018, no bulk restric-tions), API-modifiable	SIREN record + filing index	DB (2001+ filings) 15y+ SIRENE registry	daily-ish	bulk CSV/API	free/attribution (CC-BY-class); declaration-based RSS	SIREN↔SIRET keys; Etalab identity join (action-note: bulk CSV archive loop + SIRENE cross-join)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
20	Brazil Receita Federal CNPJ	corporate-registry	arquivos.receitafederal.gov.br/cnpj/dados/ (free, monthly full CSV rotation) + gov.br page	establishment-per-CNPJ (~35M)	**monthly full-file archive avail-able** (2015+ online; older in file-set); historical rollovers	monthly	file-download (multi-table PB-classic fixed/CSV)	public-domain (BR gov open data)	CNPJ root+branch PIT via monthly vintages (action-note: download full historical stack; build CNPJ vintage DB)	W0	connect	SURVEY-EVIDENCE
21	IRS TEOS / EO BMF (chari-ties)	charitable-registry	irs.gov EO BMF extract (CSV per US state/region, updated monthly) + TEOS bulk (BMF + e-file + pub78 PDFs)	org record (EIN, sub-section code, ruling year)	**incl. “His-torical Con-tent” switch on TEOS page**; BMF monthly vintages; ruling-year on records	monthly	file-download (CSV)	public-domain (US Fed)	EIN unique-ness; subsec-tion codded path-fix for SOX501-type changes (action-note: pull historical BMF stack + pub78 letter versions)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
22	SEC EDGAR filer identities	corporate-registry-filing	sec.gov company tickers JSON + EDGAR full-text/company-search	filer identity (CIK, tickers, SIC)	CIK identities since 1993? filings; identities mostly historical-stable	EOD	API/file	public-domain	cross-ref 05 skip duplicates: 05 owns filings themselves (action-note: join CIK↔LEI↔CNPJ via GLEIF registeredAs)	W0	connect	SURVEY-EVIDENCE
23	UK Charity Commission register	charitable-registry	register-of-charities extracted CSV/documents (free; charity data incl. income, contact); historical register snapshots on archive	org record	**2007+ full register historic on archive**; annual returns per charity	weekly	file-download	OGL-UK	charity-number PIT (action-note: pair with IRS TEOS global charity coverage)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
24	OFAC SDN	sanctions	treasury sanction-slist + down-loads sdn.csv/sdn_xml.csv (2001+) + **OFAC SDN changes (delta) files**	entry-level; SDN changes delta (2003+)	SDN changes file incl. his-torical addi-tions/removals since 2016?; full list snap-shots — external archive	updated T+0-ish (list updates not sched-uled)	file-download, checker	public-domain	alias res-olution; update-date stamp; building-materials alias-match (action-note: download snapshot + changes-delta ingest)	W0	connect	VERIFIED-UPSTREAM-DOC
25	OFAC non-SDN lists (NS-MBS, NS-IC / CAPTA, NS-CMIC)	sanctions	same OFAC download tree (separate files)	entry-level	CMIC list since 2020 (EO 13959/14032 routes); separate delta	event-driven	file	public-domain	program-code routing (action-note: pair with SDN; program-tag)	W1	connect	VERIFIED-UPSTREAM-DOC
26	EU con-solidated sanctions list	sanctions	finance.ec.europa.eu consoli-dated list — CSV/XML/PDF per regime + entity-level unified feed	entry (per-son/organisation)	regime-by-regime archives; consoli-dated XML updated rolling (history: archived by month on fi-nance.ec.europa.eu)	event-driven	file-download	free (EU)	subject-type+regime-code mapping; OJ amend-ments (action-note: unified feed + regime codemap)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
27	UN 1267/Security Council consolidated list	sanctions	un.org security-council/sanctions consolidated list (free downloads; XML/HTML) + UNODC?	entry-level (individuals, entities)	since 1999 targeted sanctions; UN free	event-driven	file-download	free (UN)	narrative-report cross-check (action-note: ingest into canonical sanction DB)	W1	connect	SURVEY-EVIDENCE
28	UK OFSI consolidated list	sanctions	gov.uk OFSI consolidated list (CSV/XML) + OFSI case links	entry-level	since OFSI created 2016; consolidated includes 2002+ acts?	event-driven	file-download	OGL-UK? (gov.uk open)	sanctions-court-appeals track (action-note: ingest)	W1	connect	SURVEY-EVIDENCE
29	OpenSanctions	sanctions-PEP-aggregate	opensanctions.org — **data under AGPL/attribution license structure; hosted API paid**; datasets: Default (sanctions+PEP+others (maritime, securities, cross-KYus?)	entity + alias/network edges	own PEPs data since 2022; sources mirrored (OFSI/EU/UN/US regulatory),	daily	bulk pt.tar archives + API (licence-based)	**AGPL-3 + CC-BY?** verify licensing page (path moved)	entity dedupe/statement-level (datasetschema) — best PIT spine if attributed (action-note: free tier ingest + pricing for screening API)	W1	quote	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
30	World-Check-class (Refinitiv/LSEG World-Check One)	sanctions-PEP-commercial	LSEG — API + SFTP feeds; institutional subscription	entity + risk-level + adverse-media	curated global coverage years-deep	realtime-ish API; EOD file	SFTP/API	license quote RE-QUIRED	proprietary mapping (no source transparency) (action-note: vs OpenSanctions hosted; PIT vs OpenSanctions value debated)	W2	quote	UNVERIFIED-CANDIDATE
31	EveryPolitician / PEP-registry classes	sanctions-PEP	everypolitician.org (free CSV; per-country politicians data; background via github repo)	person-level (role, dates)	archived per country (static snapshot model; repo-based)	repo-based	file-download CSV	CC-BY-SA? (mySociety open)	merge into OpenSanctions-candidate PEPs (action-note: supplement to OpenSanctions PEPs; not primary)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
32	Interpol notices	sanctions/notices	interpol.int notices (free public search for Red/UN-UNRF notices); API-эп? limited	notice-level (person/entity)	search UI only; no bulk archive at source — archive via own scrape	web	scraping-safe? (rate-limits unknown)	portal-only (rights restrictions on reuse — check per-notice)	notice status withdrawal lag (action-note: scrape + respect robots; use OpenSanctions mirror)	W1	connect	VERIFIED-UPSTREAM-DOC
33	CourtListener / RECAP Archive	litigation	courtlistener.com — bulk data service (daily dumps; case-law, dockets, opinions, oral args); **free incl. full historical dockets archive**; RECAP extension pulls PACER free copy	docket/entry; document; opinion	dockets from RECAP (2009+); opinions back to 1750s (via court packs); documents backfill continuing	daily dumps	bulk file / API	public-domain (SC/US court docs; some courts?) — mostly CC/research	best docket PST; document judge-coverage gaps (action-note: mirror daily bulk; build PACER-coverage gauge)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	PACER (via RECAP exten-sion)	litigation	pacer.uscourts.gov — pay-per-page $(0.10/page per $3 cap/doc, oval) + RECAP free-mirror encour-ages sharing	docket docu-ment	1988+ ECF era elec-tronic; legacy court PDFs	T+0 (pay)	extension + per-case payment	paid-per-doc	dedupe vs CourtLis-tener mirror before paying (action-note: RECAP-first; pay only for uncov-ered districts)	W1	connect	SURVEY-EVIDENCE
35	SEC litigation releases + admin proceed-ings	litigation/financial enforcement	sec.gov/litigation (free since 1995 archives; older via sec.gov/rules?) + admin-istrative proceed-ings archive	release-level	releases back to 1995 (2000s com-plete); earlier manual	EOD	scraping/RSS	public-domain	release-number ordering (action-note: mirror releases → entity-to-publisher case match)	W1	connect	SURVEY-EVIDENCE
36	DOJ press + action pages	litigation	justice.gov/news (press releases), case pages	release-level	~2010+ online search-able; older via DOJ archive	event	web	public-domain	entity-tag extrac-tion needed (action-note: flow into sanc-tions/ enforce-ment graph)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
37	FINRA AWC + BrokerCheck/CRD disciplinary	litigation/regulator	finra.org BrokerCheck (free API/detail pages; AWC/CRD disciplinary via BrokerCheck PDF + finra arbitration awards)	rep/firm + action	CRD disciplinary history deep (decades); AWC/BR disclosure	event	API/scraper	free (non-commercial ToS)	CRD is firm-of-record for broker-identity PIT (action-note: BrokerCheck endpoint probe → CRD cross-join)	W1	connect	SURVEY-EVIDENCE
38	State AG enforcement actions	litigation	naag? per-office press pages (no unified dataset)	action-level	patchwork 2010+	web	scraping	varies	dedupe vs federal (action-note: low priority until CFPB/state-regulatorParsing unified)	W2	connect	UNVERIFIED-CANDIDATE
39	ICSID case database	arbitration	icsid.worldbank.org/cases-and-proceedings — free case records + awards PDF	case-level	since ICSID founding 1966; **full awards archive**	event	web/db download links	public + free (World Bank)	case status; award date ordering (action-note: archive PDFs + case metadata JSON)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
40	WTO dispute-settlement archive	arbitration	wto.org disputes (free case pages + docs since in-stitution 1995)	case-level	since 1995 full	event	web	free	panel/AB ruling dates (action-note: mirror)	W1	connect	SURVEY-EVIDENCE
41	PCA case database	arbitration	pca-cpa.org/cases — free	case-level	PCA cases since 1900s archives	event	web	free	— (action-note: mirror)	W1	connect	SURVEY-EVIDENCE
42	EU CELEX / EUR-Lex + Curia	litigation/regs	eur-lex.europa.eu (OJ, regu-lations) + cu-ria.europa.eu (judg-ments) — free, full historical	document-level	EU legal corpus complete since 1952; judg-ments all	EOD	basket scraping + cell (CELEX IDs)	re-use allowed (EU decision 2011/833)	procedure-type classifica-tion (action-note: CELEX ID graph join to sanctions regs)	W1	connect	SURVEY-EVIDENCE
43	UK BAILII (courts of UK/IE)	litigation	bailii.org — free full judg-ments; also “Find Case Law” (caselaw.nationalarchives.gov.uk, OGL)	judgment	deep (case law 1800s+, UKSC 2009+, courts archives)	updates irregular	scraping	free (BAILII condi-tions); TNA OGL for Find-Case-Law	— (action-note: prefer TNA UK Find-Case-Law (struc-tured))	W1	connect	SURVEY-EVIDENCE
44	Global Ar-bitration Review / Kluwer Arb	litigation-commercial	globalarbitrationreview.com; kluwerar-bitra-tionblog → license quote only	article/case commen-tary	subscription archive	paywall	license	paid license	— (action-note: if arbitrator-network analysis justifies)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
45	USPTO PatentsView bulk	IP/patents	patentsview.org → **migrated to data.uspto.gov (transition guide live)**; bulk downloads (grants 1976+ — full text), applications (2001+)	patent/assignee/inventor/cpc	**grants 1976+ full text historical; entity disambiguation versions**	weekly-ish (grant lag increased 2023+)	file-download (TSV/parquet per entity tables)	public-domain (US Fed)	assignee disambig versioning = PIT trap (action-note: adopt disambiguation version pinning + grant-lag watch)	W0	connect	VERIFIED-UPSTREAM-DOC
46	USPTO Patent Examination Data (PDES)	IP/patents	developer.uspto.gov PDES API/public — prosecution-history events	application-transaction	since 1981 applications?	monthly	API/bulk free	public-domain	office-action ↔ claim-text timing (action-note: pair with PatentsView grants)	W1	connect	SURVEY-EVIDENCE
47	EPO OPS	IP/patents	developers.epo.org — REST XML bibliographic (worldwide), free limited tier w/ key	doc-level worldwide	EPO worldwide DB deep (1900s worldwide claims?)	near-realtime-ish (biblio)	RESTful XML; weekly bulk (pay?)	free limited tier; bulk on fee?	family/INPADOC linking is the hard part (action-note: OPS free-tier coverage mapping)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
48	WIPO PATENTSCOPE	IP/patents	patentscope.wipo.int — free search; bulk via subscription/CCC	2w-level (PCT national phase)	PCT since 1978 full	live-ish search; scheduled bulk paid	search free; bulk license	free search / paid bulk	national-phase entry tracking (action-note: only if non-US/EPO filing weight justified)	W2	connect	SURVEY-EVIDENCE
49	Lens.org	IP/patents	lens.org — free API tier (attribution); Scholar+Patent linked graphs	doc/person/org graphs	deep patent collections (100y?); scholar layer growing	daily	REST API (key)	free tier attribution; commercial tiers	open-linkage graph (well-documented quality) (action-note: free-tier scoped pilot (citation graphs))	W1	connect	SURVEY-EVIDENCE
50	USPTO TSDR (trademark status)	IP/trademark	tsdr.uspto.gov — free per-mark docs/status; TSDR API	mark-level	marks since 1884? (records via TSDR since 1984 digital); pre-1984 paper	per-mark	free API, no bulk	public-domain	mark-status event log (action-note: status-list bulk alternate: USPTO trademark bulk XML)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
51	USPTO trade-mark bulk XML	IP/trademark	developer.uspto.gov trade-mark bulk XML (current applica-tion + registra-tion XML monthly)	nat'l	tradmark registry monthly snap-shots; historic dumps	monthly	file-download	public-domain	method: pair TSDR with bulk dumps (action-note: alternate to TSDR scraping)	W1	connect	SURVEY-EVIDENCE
52	EUIPO TMview / Design-View	IP/trademark	tmview.euipo.europa.eu/copo — free; EUIPO Open Data bulk (free CSV)	mark/design per OHIM	EU-wide harmo-nize (member-state feeds per national office)	daily-ish	search + open-data bulk	free (EU)	status date fields; national-office lag asymme-try (action-note: gener-ated CSV archive loop)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
53	ClinicalTrials.gov API v2	clinical-trials	clinicaltrials.gov/api/api — REST JSON; full-field incl. results + full study record; NLM **archive snapshots exist** (compressed monthly public data files)	study/data-record (NCT)	since registry 2000 (FDAAA); protocol revisions via record-history (API field)	daily (record revisions differential)	REST API; monthly full-file alternative	public-domain / CC0-class (US gov)	record revision timelines as PIT (action-note: API-v2 full snapshot crawl + archive file mirror)	W0	connect	VERIFIED-UPSTREAM-DOC
54	EU CTIS (EMA clinical data portal)	clinical-trials	ema.europa.eu clinical data portal (CTIS) — public data; API unclear	study/event	since CTIS launch Jan 2022 (EU Reg 536/2014); pre-2022 in EudraCT	event	web (API unconfirmed)	free (EU)	transparency-rule rollouts phased; commercial-info redactions (action-note: probe CTIS API/bulk availability)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	openFDA (drug events/devices/recalls/enforcement)	clinical/regulatory	api.fda.gov — free bulk JSON; end-points drug/event, drug/label, de-vice/event, de-vice/recall, food/enforcement, etc.; **reports back to 2004 (FAERS), enforce-ment histori-cal archives**	/data/downloads level	drug event 2004+; device adve 1990s? enforce-ment historical	monthly-ish updates	bulk JSON per endpoint + free API	public-domain (US Fed)	FAERS duplicate/lack-of-causality gates (action-note: full-archive mirror + endpoint row-limits plan)	W0	connect	VERIFIED-UPSTREAM-DOC
56	EMA company registry / XEVMPD	clinical/regulatory	ema.europa.eu medicines/compan data (free web); “compa-nies” list	company/product level	1995+ central-ized; MRP/DCP 2000s	EOD-ish	web/scraping + some data files	free (EU)	MA-holder↔manufacturer mapping quality (action-note: pair with openFDA global)	W1	connect	SURVEY-EVIDENCE
57	EudraGMDP	clinical/regulatory	eudragmdp.ema.europa — EU GMP cer-tificates + GDP; public web	certificate level	certificates EMA pub-lished; outages known	web	scraping	free (EU)	certificate with-drawal dates (action-note: low-priority scrape)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
58	WHO Prequalification (med products)	clinical/regulatory	who.int PQ database — free lists + PQ status docs (long-archived)	product/manufacturer	since 2001 WHO PQ; docs public	event	web + downloads	free (WHO)	product status “prequalified/suspended” (action-note: mirror lists + status PDFs)	W1	connect	SURVEY-EVIDENCE
59	SAM.gov (entity + procurement notices)	procurement/firmo	sam.gov/data-services — free API; bulk data downloads (`api.sam.gov` entity API, exclusions, HMDA? no) + forecast + notices	entity/notice/award record	entity-registrations historical?; **notices historical since ~2010 bulk**	near-realtime / daily extracts	REST API (key) + bulk download	public-domain (US Fed)	UEI replaces DUNS (2022) — ID migration gate (action-note: entity extract + notices bulk archive)	W0	connect	VERIFIED-UPSTREAM-DOC
60	EU TED (Tenders Electronic Daily)	procurement	ted.europa.eu — free API v3 + data extracts (**historical CSV dumps since ~2006/2007**; extended to 1980s-90s via national tegular sources?)	notice/cotract-award	historical extracts since 2006-ish online; older raw? (e-notices pre-2006 partial)	daily	API + bulk extracts	free (EU re-use)	award-decision date vs publication lag (action-note: extracts archive loop + API delta)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	UK Contracts Finder	procurement	contracts-finder.service.gov.uk — free API + CSV; OGL	notice/contract	since 2015 platform	daily	API/CSV	OGL-UK	award-value nulls common (action-note: CSV archive)	W1	connect	SURVEY-EVIDENCE
62	Australia AusTender	procurement	tenders.gov.au — free CSV exports; (403-bot-blocked this session)	contract notices	since 2007-ish consolidated	daily	portal CSV export	free (AU)	contract-value missing rates (action-note: retry with browser UA / schedule off-peak)	W1	connect	SURVEY-EVIDENCE
63	India GeM (Government e-Marketplace)	procurement	gem.gov.in — portal; public API limited; reports	bid/contract	since 2016	portal	scraping (limited)	portal-only	— (action-note: probe report downloads)	W2	connect	UNVERIFIED-CANDIDATE
64	Canada tender notices (Buyandsell.gc.ca; “Oppy?” token unresolved)	procurement	buyandsell.gc.ca (TBO/OJEC-class) — free; OPPA? token unparseable	notice	~2005+	daily-ish	web/CSV	free (CA)	token ambiguity to resolve with owner (action-note: owner-confirm “Oppy?” referent first)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	Korea Nara Market (□□□□) — PPS procurement	procurement	nara.go? (PPS portal) — free portal; API unclear	notice/bid	since ~2000	portal	unknown API	portal-only	— (action-note: probe emarket? availability)	W2	connect	UNVERIFIED-CANDIDATE
66	Japan government procurement (□□□□□□)	procurement	geppo? (□□□□□□/API?) — free portal; no unified API known	tender notices	~2010s	portal	scraping	portal-only	— (action-note: probe available CSV/feeds)	W2	connect	UNVERIFIED-CANDIDATE
67	World Bank procurement notices	procurement	/dev — **DNS NXDOMAIN on 2026-10-24 (host moved?)**; data.worldbank?	worldbank.org	vintage portal ~2001+	—	unknown post-move	free?	migrated to “World Bank Group procurement portal”? (action-note: re-locate current host; skip until)	W3	connect	UNVERIFIED-CANDIDATE
68	OpenOpps	procurement-commercial	openopps.com — commercial platform + paid API/data extracts	tender graph (cross-source EU+)	multi-year archive compiled	daily	API (paid)	paid license	aggregation-quality vs TED (action-note: only if TED-extract gaps empirically bite)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
69	USAspending (archive)	procurement_spend	/spending_transaction_download — free bulk CSV; **monthly archives back to FY 2001**; API	transaction-level	FY2001+ (mandatory); award-level	monthly archive + daily API	file-download + API	public-domain	agency/org PK migration; CFDA→assistance-listing re-key (action-note: PhD ingestation of full archive — highest-value free fiscal history)	W0	connect	VERIFIED-UPSTREAM-DOC
70	FPDS-NG (federal procurementdata system)	procurement_spend	/ezsearch / fpds.gov / army equires? free contract-reports export (search→CSV); daily reports published	contract-transaction	linked into USAspending (same transactions pre-2007 via legacy FPDS)	daily	search-export free	public-domain	NAICS/PSC code codemap versioning (action-note: feed USAspending overlap only; not standalone primary)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
71	FCC ULS database downloads	telecom-filings	fcc.gov/wireless/data/publ access-files + data.fcc.gov/download/pubs — free ZIPs (per radio service; SQL schema docs published)	application/license records per /complete,daily,weekly/	**full complete DB snapshots + daily/weekly transaction files; historical retention: rotating (own archive needed)**	daily	file-download ZIP (data + SQL defs)	public-domain (US Fed)	call-sign keyed; schema versioned (defs updated 2026-03-19) (action-note: standing snapshot+daily ingest w/ own archive)	W0	connect	VERIFIED-UPSTREAM-DOC
72	FCC spectrum auction results	telecom/spectrum	fcc.gov auction reports/results (public auction results since 1994; bidding-system docs)	auction-round/license	since Auction 1 (1994) PDF/reports; modern auctions CSV	per-auction	file-download	public-domain	bidder↔ULS entity cross-join (action-note: auction-history power users (charta PR foam))	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	binance-vision-spot-klines	CEX-archive	data.binance.vision → data/spot/{daily,monthly}/klines (URL-pattern walk)	OHLCV 1s,1m…1mo (12)	2017-08→ full (□C2)	daily T+1; monthly 1st-Monday	S3-archive walk → Bronze	free (site ToS)	spot timestamps become µs from 2025-01-01 (□C2) — schema gate; delist = files stop, kept (action-note: full-history walker; union daily+monthly dedup)	W1	connect	VERIFIED-UPSTREAM-DOC
2	binance-vision-spot-trades	CEX-archive	same tree /trades (historicalTrades fields per README)	tick	2017-08→ (□C2)	daily T+1	S3-archive walk	free	tradeId gap check (first/last id); µs regime 2025+ (action-note: bulk-walk all d pairs → Parquet Silver)	W1	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	binance-vision-spot-aggtrades	CEX-archive	/aggTrades files (/api/v3/aggTrades fields)	aggregated tick	2017-08→ (□C2)	daily T+1	S3-archive walk	free	aggId continuity; µs regime (action-note: cross-check agg vs raw trade counts)	W1	connect	VERIFIED-UPSTREAM-DOC
4	binance-vision-umfut-tape	CEX-archive (derivs)	data/futures/um/{daily,monthly}/{klines+tradesCHaGVTrades} (/fapi/v1 fields)	tick + 3H aggTrades	2019-09→ per listing (□C2)	daily/monthly	S3-archive walk	free	funding-hour PIT (00:00/08:00 UTC); ms timestamps retained spot≠fut (action-note: um tape load; funding alignment)	W1	connect	VERIFIED-UPSTREAM-DOC
5	binance-vision-dapi (COIN-M)	CEX-archive (derivs)	data/futures/cm/{daily,monthly}/{klines+tradesCHaGVTrades} (/dapi/v1 fields)	tick + 3H aggTrades	since dapi listing (older than um → verify start)	daily/monthly	S3-archive walk	free	base-asset volume semantics differ vs um (action-note: second-tier after um)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
6	binance-vision-fut-bookticker-diffdepth	CEX-archive L2	data/futures tree bookTicker / diffDepth daily files	book-ticker tick; depth-diff	later than klines family (verify start)	daily	S3-archive walk	free	partial-depth only; L2 rebuild via diff+snapshot replay, keep incident log (action-note: L2 reconstruction stage for futures book)	W2	connect	UNVERIFIED-CANDIDATE
7	binance-vision-fundingrate	CEX-archive (derivs)	data/futures/um fundingRate daily/monthly files	8h prints	since listing (verify start)	daily	S3-archive walk	free	funding print PIT vs premiumIndexKlines; mag-rate-change flags (action-note: cross-venue funding panel input)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
8	binance-vision-liquidationSnapshot	CEX-archive (derivs)	data/futures/um liquidationSnapshot dumps (legacy product)	event	sweep window-limited (verify start)	daily	S3-archive walk	free	legacy dump vs live forceOrder stream semantics changed → PIT flag (action-note: archive-only; live liq via WS fapi)	W2	connect	UNVERIFIED-CANDIDATE
9	binance-spot-REST	CEX-API	api.binance.com /api/v3 klines, aggTrades, historicalTrades, depth	realtime	full via paging	realtime	REST keyless	free (weight caps)	geo-fencing caveat; weight-limit backoff; µs 2025+ (action-note: live tail fills W1 dump gaps)	W0	connect	SURVEY-EVIDENCE
10	binance-spot-WS	CEX-API	wss @trade @kline_1s @bookTicker @depth20 @diffDepth	tick/event	live tail	realtime	WS keyless	free	diff-depth lastUpdateId sync PIT; geo-fencing flag (row 12) (action-note: canonical realtime Bronze tail)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	binance-fapi-derivs-REST	CEX-API (derivs)	fapi/v1 fundingRate (full history), premiumIndexKlines; futures/data/openInterestHist	8h/5m	funding full since listing; OI history rolling-cap (verify window)	realtime	REST keyless	free	OI-history cap → use dumps for deep OI; funding PIT (action-note: full funding loader + OI cap workaround)	W1	connect	UNVERIFIED-CANDIDATE
12	cex-geo-fencing-metadata	compliance	Binance prohibited-jurisdictions page (bot-walled per stream-04); OKX DOJ DPA + 2025 plea ($505M) + US relaunch under ICE JV (□C11□C12); venue entity charts	ToS events	ToS history	batch	webfetch+Ujo care → GFI compliance table	public ToS text	ban-list drift detector; per-jurisdiction flag on every CEX row; Binance page bot-wall = curated manual copy (action-note: curate per-jurisdiction matrix; auto-diff releases)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
13	okx-v5-rest-ws	CEX-API	OKX V5 public REST + WS (candles w/ hist param since listing, funding-rate-history, OI history; WS books ≤400 levels)	tick + OHLCV 60s+	per-instrument since listing	realtime	REST+WS key-less/key	free tier	entity/jurisdiction pin (US policy line DOJ-verified □C11; re-launched US 2025 □C12); KYC-walled users; docs page timed out this session (re-verify params) (action-note: realtime tail; funding & OI loaders)	W0	connect	SURVEY-EVIDENCE
14	bybit-v5-rest-ws	CEX-API	Bybit V5 (spot, linear/inverse perps, funding, mark, insurance)	tick + 8h funding	candles since listing	realtime	REST+WS keyless	free	region-entity split; funding PIT; server-time skew (action-note: per-venue funding/liq live leg)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
15	bybit-derivatives-data-dumps	CEX-archive	bybit.com/derivatives-data downloadable exports	event/8h	Tardis anchors: inverse 2019-11-07; linear 2020-05-28; liquidations 2020-12-18 (□C5)	daily	batch-CSV walk	free	portal JS-rendered (listing □C21) — enumerate file index; Tardis mirror as cross-check (action-note: verify file list; ingest bybit)	W1	backfill	VERIFIED-UPSTREAM-DOC
16	coinbase-exchange-candles	CEX-API	api.exchange.coinbase.com /products/{id}/candles granularities 60/3600/86400…	1m–1d	~2014-12→ GDAX era (stream-04)	realtime	REST keyless	free	candle-gap detector; US-compliant reference venue; product rename continuity (BTC-USD etc.) (action-note: live leg; full candle W1)	W0	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
17	kraken-rest-ohlc	CEX-API	Kraken public REST /0/public/{OHLC,Trades,Depth}	1m+ OHLC, tick, L2 top	OHLC REST capped 720 candles back (stream-04)	realtime	REST keyless	free	720-candle cap → dumps (rows 18–19) for depth; Trades since market start via API (action-note: live tail only)	W0	connect	SURVEY-EVIDENCE
18	kraken-ohlcvt-dumps	CEX-archive	support.kraken.com/articles/360047124832 downloadable OHLCVT zips (per-market CSV ×8 intervals)	1/5/15/30/60/240/720/1440 min	from inception of each market (→2013+ era); quarterly incremental ZIPs	batch	batch-CSV + sha256 manifest verify	free	gaps NOT zero-filled (no-trade intervals absent); MANIFEST.json + checksum gate (action-note: full archive ingest + quarterly incrementals)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
19	kraken-trades-dumps	CEX-archive	support.kraken.com “time and sales” CSVs (one per pair)	tick	from beginning of each market	batch	batch-CSV	free	header schema (ts,price,vol,b,s,trade_id); manifest+sha; reassemble part-files in order (action-note: tick-grade free (US venue))	W1	backfill	VERIFIED-UPSTREAM-DOC
20	kraken-futures-paid-tick	CEX-derivs-paid	Kraken Futures tick+full-depth via CryptoStruct self-service	tick + L2	per-day instrument-days (vendor coverage July 2023→, verify)	batch purchase	CSV/Parquet per-day	paid (€1/day-class vendor)	vendor capture claims; low liquidity venue flag (action-note: slot only if Kraken perp features needed)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
21	bitstamp-dumps	CEX-archive	bitstamp.net/market/ data down-loadable trade files (page ×2 timeout this session) + CDD mirror (C20)	t/k/CHLC CSV	Amberdata claim: spot from 2011-08-18 (C8); Tardis mirror only 2019-03-30→ (C4)	batch	batch-CSV direct-dump walk	free	start date is vendor-claim (fetch to confirm); oldest continu-ous BTC tape (action-note: walk dump tree; fill 2011–2019 via dumps)	W1	connect	SURVEY-EVIDENCE
22	bitfinex-v2-rest-ws	CEX-API	Bitfinex v2 REST /candles (hist since listing), /trades; WS v2	1m+	2015→ (stream-04, verify)	realtime	REST+WS keyless	free	maintenance candle oddities (stream-04) → sec-ondary venue cap (action-note: sec-ondary his-tory/book)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
23	midtier-ccxt-venues-kucoin-gate-mexc	CEX-API	KuCoin v4 + Gate.io v4 + MEXC v3 REST/WS (via CCXT harness)	tick + 1m+	since listing each	realtime	REST+WS (CCXT)	free	≥1 filler venue read-only; new-listing signal from MEXC/Gate; volume-credibility discount (action-note: single adapter, three venue ids)	W2	connect	UNVERIFIED-CANDIDATE
24	upbit-bithumb-korea	CEX-API	Upbit V1 REST (KRW candles/trades; KCE/US companions) + Bithumb candle JSON	1m/1d	since listing (Upbit daily 2017-10→, verify)	realtime	REST keyless	free	KR fiat quote; regional walling; archive-depth verify (action-note: KRW parity strip)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
25	india-cex-wazirx-coindcx-zebpay	CEX-API	WazirX + CoinDCX (+ZebPay check-points) public OHLC/depth REST	1m/1d	since listing (verify)	realtime	REST	free	WazirX suspended-flow incidents → PIT gate; INR reference only (action-note: optional INR strip)	W3	connect	UNVERIFIED-CANDIDATE
26	coingecko-chart-history	aggregator-REST	CoinGecko /coins/{id}/market_chart/global (has /coins/{id}/history for exact ts)	5m/daily	~2013→ global series com-puted via coingecko history + mar-ket_chart range (paid depth)	near-realtime	REST + demo key	free tier tight keyless limits; paid tiers	free depth cap (~365d keyless; verify); volume credibility flags; delist/market PIT at global level (action-note: daily global snapshot job + 2013 history loader)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
27	coingecko-onchain-geckoterminal	DEX-agg	CoinGecko onchain DEX-candle endpoint + GeckoTerminal pools API	pool OHLCV	per-pool listing	realtime-ish	REST	free tier	pool rebasing; router identity; new-pool spam (action-note: DEX parity vs sub-graphs)	W3	connect	UNVERIFIED-CANDIDATE
28	coinmarketcap-freemium	aggregator-REST	CMC Pro API open/labels/global + v2 historical	hourly/daily	freemium history windows limited	near-realtime	REST keyed	free tier	CMC method-ology; rate caps (action-note: only for parity strip)	W2	connect	SURVEY-EVIDENCE
29	coindesk-data-cryptocompare	aggregator-REST	CoinDesk Data ex-CryptoCompare legacy min-api + CSV historical archives	1m+	2011-era reference windows	near-realtime	REST keyed	freemium + paid	support-life-cycle risk (stream-04 □D3): migrate or budget replace-ment; endpoint deprecа-tion watch (action-note: 2011-era fallback history only)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
30	coinapi-flat-files	archive-flatfiles	CoinAPI Flat Files (S3/Snowflake/CSV.gz, gzip partitions)	tick + agg	400+ exchanges / 599K symbols / 632TB (vendor-claim; verify partition map)	batch	S3 bulk + license	paid (negotiated)	per-REQUEST vs bulk budgeting; venue-symbol map drift (action-note: cost-quantify vs Tardis/Massive)	W2	connect	VERIFIED-UPSTREAM-DOC
31	coinapi-unified-api	aggregator-REST	CoinAPI REST/WS/FIX (multi-venue trade+quote unified)	tick	since 2013+ per plan	realtime	REST+WS keyed	paid (free ~100 req/d order-of-magnitude; re-verify)	per-request cost discipline; normalized schema drift (action-note: multi-venue fallback when CCXT too costly)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
32	tardis-dev-archive	archive-flatfiles+replay	docs.tardis.dev Market Data API + downloadable CSVs (50+ venues)	tick L2/trades/liq	per-venue anchors: Bitstamp 2019-03-30 (□C4); Bybit inverse 2019-11-07, linear 2020-05-28, liquidations 2020-12-18 (□C5)	batch	HTTP tick-list API + CSV; Python/Node libs	free recent-N-days (vendor-claim); full history paid	per-venue channel availability vary; incident pages per venue; no redistribution (action-note: full-history buy eval for L2)	W2	backfill	VERIFIED-UPSTREAM-DOC
33	coinpaprika-api	aggregator-REST	api.coinpaprika.com/v1 tickers, OHLCV, markets, contracts	daily OHLCV (+hourly Business-up)	BTC first-data 2010-07-17; full OHLCV→2010 on Ultimate tier (vendor docs; verify tier price)	realtime	REST keyless 20k calls/mo	free/personal; enterprise paid	SEC-hosted A-grade data-quality study Oct-2025 (□C14); trust-scores per market (action-note: reference/market cross-check source)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	massive-crypto-flat-files	archive-flatfiles	massive.com (ex-Polygon) flat files: Coin-base/Bitstamp/Kraken/Bitfinex trades	hourly-partitioned trades/aggs	per-vendor coverage; verify starts per ticker	batch	S3 CSV bulk	paid	UTC-standardized; hourly zstd; first-party doc □C19 lists subset venues (action-note: evaluate as Coin-base/Kraken tick mirror)	W2	connect	VERIFIED-UPSTREAM-DOC
35	cryptodatadownload-csv	archive-flatfiles	cryptodatadownload.com CSV archive (Bit-stamp, Binance, Gemini, FTX-pre-collapse, OKX, Bybit…)	1m/hr/d	2017→ refreshed daily	batch	direct CSV download	free (no login, no paywall)	cleaned/derived ware-house — NOT raw ticks; dead-venue archive lane (Bit-trex/FTX for back-testing) (action-note: dead-venue archive lane only)	W3	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
36	dormant-aggregators-nomics-coinlore	dormant-flag	Nomics (defunct-risk) + CoinLore free	—	era-varying	—	—	—	confirm operational status before integration; Nomics API key docs long stale (action-note: kill-confirm then exclude)	W3	connect	UNVERIFIED-CANDIDATE
37	ccxt-library-route	library	ccxt unified connectors (multi-venue REST/WS adapters)	n/a	n/a	realtime	library integration	OSS; venue ToS inherit	adapter quirks patched unevenly → pin CCXT version per venue; arms races on REST shapes (action-note: W0 tail adapter harness)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
38	deribit-options-chains-realtime	derivs-API	Deribit v2 WS/REST (book.{instrument}, ticker, get_volatility_index)	tick	2016→ (venue start)	realtime	WS keyless	free	BTC/ETH options ground truth (stream-04 □D1); testnet mirrors; quote-quality legs (action-note: L2 + IV joiner (non-colo path))	W0	connect	VERIFIED-UPSTREAM-DOC
39	deribit-dvol-history	derivs-index-history	DVOL index history via Deribit get_volatility_index_data + monthly bulk	1m/1d series	DVOL since 2016 launch	realtime	REST	free	DVOL method-ology updates; cross-ref vendor-diligence census (06) (action-note: index-series loader + index refresh)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
40	okx-funding-oi-history	derivs-archive	OKX /funding-rate-history, /open-interest-history	8h + intervals	since listing (verify)	realtime	REST	free	OKX entity split context (rows 12–13); funding PIT (action-note: third funding leg after Bi-nance/Bybit)	W1	connect	SURVEY-EVIDENCE
41	bybit-funding-liq-archive	derivs-archive	Bybit funding records + liquida-tions via API + Tardis coverage anchors (□C5)	event/8h	funding since listing; liq mir-rorable from 2020-12-18	realtime	REST+WS	free	liq vs forced-order seman-tics; PIT on instru-ment rename (action-note: build fund-ing+liq panels)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
42	derivs-aggregators-coinglass-coinalyze	derivs-agg	Coinglass v4 (paid liq-heatmap/history; MCP beta exposes skill) + Coinalyze free OI/funding/liq cross-section	1s–1d	multi-month by tier	near-realtime	REST keyed	Coinglass paid; Coinalyze free-attribution	free-tier history caps; vendor-claim pricing; Coinalyze attribu-tion req (action-note: free Coinalyze bench-mark first; Coinglass only for heatmap feature)	W2	connect	VERIFIED-UPSTREAM-DOC
43	bitmex-dumps	derivs-archive	bitmex.com/app/tradedQuotedLicensed zips	tick/trade/quote/liquidation	~2014-11→ (BitMEX perp era; verify)	batch	batch-CSV	free	legacy dump staleness check; earliest perp liqui-dation tape (action-note: pre-Binance perp-era cover-age)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	btc-fullnode-archive	onchain-node	bitcoind (full archive with txindex, or pruned tail node)	block/tx + mempool	since 2009-01-03 genesis	realtime (node RPC/ZMQ)	own-node RPC+ZMQ	chain data public; OSS client	reorg-window gate; disk budget; ZMQ rawtx vs hash-block subscrip-tion choice (action-note: node deploy plan; genesis→ present archive)	W1	connect	UNVERIFIED-CANDIDATE
45	mempool-space-rest	onchain-API	mempool.space REST explorer API (blocks, tx, addr, mempool, fees, mining-pool series incl. block audit scores, /api/v1/historical-price)	block/addr-level history	blocks from genesis queryable; mining series spans history	near-realtime	REST keyless	free	shared public instance rate-limit → throttle + self-host mempool for prod; index rebuilds PIT (action-note: BTC blocks/fees/ via API)	W0	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	mempool-space-ws	onchain-API	mempool.space Web-Socket API (blocks, mempool additions, stats)	tick	live	realtime	WS keyless	free	reconnect/backoff; sequence resync (action-note: BTC realtime tail (blocks+mem))	W0	connect	VERIFIED-UPSTREAM-DOC
47	eth-execution-archive-node	onchain-node	Geth/Reth/Erigon archive (eth_getLogs, ws subs; Reth snap-shots shorten sync)	block/log/tx	since 2015-07-30 genesis (archive mode)	realtime	own-node RPC/WS	chain data public; OSS clients	reorg handling; getLogs range caps vs Erigon full-index; disk ~2TB+ order-of-mag (stream-04) (action-note: ERC-20 transfer account-ing Gold layer)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
48	eth-mempool-subscribe	onchain-stream	self-host Geth/Reth ws eth_subscribe newPend-ingTrans-actions	tick	live	realtime	own-node WS	self	sampling NON-exhaustive (relay/private-orderflow blind); privacy handling for EMD (action-note: only for DEX execution-slippage research)	W2	connect	UNVERIFIED-CANDIDATE
49	beaconchain-api-eth	onchain-API	beaconcha.in API v2 (valida-tors, balances, effective balances, attesta-tion, Beacon-Score; endpoint v1→v2 migra-tion)	epoch/validator	beacon genesis 2020-12-01→	epoch cadence (near-realtime)	REST keyed	free w/ required attribu-tion (docs C15)	attribution require-ments; premium selectors pay-walled; rate-limit tiers (action-note: ETH staking yield panels)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
50	beacon-self-attestation	onchain-node	own consensus client (Lighthouse/Prysm) + Beacon API (duties, attestation history)	slot/epoch	since 2020-12-01	12s slots	own-node Beacon API	chain data public; OSS	attestation duties are not persisted by spec for old epochs → rebuild window; huge state sync cost (action-note: only if validator-level features beyond beacon-cha.in)	W2	connect	UNVERIFIED-CANDIDATE
51	base-node	onchain-node	op-stack (op-geth + op-node) self-host	block/log	since 2023 (target archive)	realtime	own-node	chain data public; OSS	rollup stage/fault-proofs (L2Beat); reorg semantics; L3 extras post-scope (action-note: L2 Gold feeds before buying indexers)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
52	arbitrum-nitro-node	onchain-node	Arbitrum nitro full/archive node	block/log	since 2021 (target archive)	realtime	own-node	chain data public; OSS	L1-posted data dependency for full replay; snapshot availability varies (action-note: L2 strip)	W1	connect	UNVERIFIED-CANDIDATE
53	optimism-node	onchain-node	op-stack (op-geth) self-host	block/log	since 2021 superchain era migration (verify pre-2023 depth)	realtime	own-node	chain data public; OSS	bedrock migration block semantics → PIT on pre/post while replaying (action-note: L2 strip)	W1	connect	UNVERIFIED-CANDIDATE
54	solana-rpc-geyser	onchain-node	validator-lite RPC + Yellowstone gRPC (geyser plugin streams)	slot/txn	RPC lookback bounded by snapshot window; archival beyond scope	realtime (sub-second slots)	own-node gRPC	chain data public; OSS	public RPC rate-caps; memory-heavy node; snapshot resurrection offsets (action-note: only if Solana-DEX venue play)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	avalanche-node	onchain-node	avalanchego node (C-Chain EVM, X/P chains)	block/log	since 2020 (C-Chain 2020-09)	realtime	own-node	chain data public; OSS	multi-chain index burden; subnet filtering (action-note: alt-L1 strip)	W3	connect	UNVERIFIED-CANDIDATE
56	polygon-pos-node	onchain-node	heimdall + bor (or erigon/LaGeth-class forks)	block/log	since 2020-05 mainnet PoS chain	realtime	own-node	chain data public; OSS	checkpoint vs block cadence split; snap-shot/pruning trade-offs (action-note: Polygon strip)	W2	connect	UNVERIFIED-CANDIDATE
57	bsc-node	onchain-node	geth-fork client (erigon-style for BSC)	block/log	since 2020-09 mainnet	realtime	own-node	chain data public; OSS	TB-class snapshot sync; parallel-exec reorg schema (2024+ eras) (action-note: BSC strip)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
58	tron-trongrid	onchain-API+node	TronGrid REST/HTTP (compat Java-tron RPC) or self-host java-tron + event indexer	block/contract-event	since 2018-05 mainnet	near-realtime	REST/node	TronGrid free tier keyed	energy/resource model events; indexer filter per contract (action-note: TRON strip via API first)	W3	connect	UNVERIFIED-CANDIDATE
59	ton-fullnode	onchain-node	ton-node (validator-lite) + tx explorer ingestion	block/tx mes-sages	since 2020-08 mainnet	realtime	own-node	chain data public; OSS	shard message splits; indexer immatu-rity — verify stability (action-note: TON strip)	W3	connect	UNVERIFIED-CANDIDATE
60	cardano-dbsync	onchain-DB	cardano-node + cardano-db-sync → Postgres SQL	block/UTxO/ledger	since 2017-09 mainnet	near-realtime (sync lag grows without snap-shot)	own-node SQL	chain data public; OSS	multi-day replay cycle; era-schema changes (Shel-ley/Vasil…) (action-note: Cardano strip (SQL-native))	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	aptos-sui-move-chains	onchain-node	Aptos fullnode + Indexer gRPC/GraphQL; Sui fullnode + GraphQL	block/tx events	Aptos 2022-10 / Sui 2023-05 →	realtime	own-node API	chain data public; OSS	Move event-model differences; indexer lag + version-manager maturity flags (action-note: Move-chain strip (both L1s, one worker))	W3	connect	UNVERIFIED-CANDIDATE
62	dune-sql-layer	onchain-warehouse	Dune SQL (Spellbook dbt, Trino connector, API credits, SDK-execute)	per-request	multi-chain from genesis	batch	SQL/rest + credits	free credits; paid tiers	query-credit cost management; Spellbook pinned model versions; output attribution policy (stream-04 □D5) (action-note: Gold-layer prototyping + s)	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
63	mev-eigenphi-api	mev-DB	EigenPhi API (arb/sandwich/liq MEV stats)	event + aggre-gated	~2021→ (verify)	near-realtime	REST keyed	free com-munity tier / paid	docs end-points unreach-able ×2 this session (trans-port error) — verify before paid commit (action-note: DEX exec-cost slippage features)	W2	connect	UNVERIFIED-CANDIDATE
64	flashbots-mevboost-relay-datasets	mev-DB	Flashbots MEV-Boost relay public datasets + ZeroMEV / mev-boost.pics Dune lineage	slot-level	since Merge 2022-09	batch (block-lag)	file + SQL (Dune-backed)	open dataset lineage	relay par-ticipation share shifts; SSZ→decoded schema version-ing (action-note: ETH execution-tax features; cross-check EigenPhi)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	uniswap-v2-v3-subgraphs	dex-subgraph	The Graph hosted sub-graphs (v2 factory 2018; v3 pools w/ sqrt-PriceX96 + tick liquidity)	per-block/event	since deploy block	near-realtime (minutes)	GraphQL	subgraph data open; Graph 100k free queries/mo per subgraph (stream-04 □D4)	index lag = minutes — NOT tick-exact; pin de-ployment ids (depreca-tions) (action-note: pool-day flows + liquidity-tick snap-shots)	W1	connect	VERIFIED-UPSTREAM-DOC
66	uniswap-v4-subgraph	dex-subgraph	V4 hooks/state indexing (per-chain subgraph deploy-ments)	per-block	2024→	near-realtime	GraphQL	data open	hooks make pools heteroge-neous → schema-drift vigilance; verify per-chain deploy-ment coverage (action-note: v3 indexer fork with hook-state columns)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
67	curve-balancer-sushi-subgraphs	dex-subgraph	Curve pools/rates + Balancer weighted pools + SushiSwap pairs (hosted sub-graphs)	per-block	since 2020-era deploys	near-realtime	GraphQL	subgraph data open	veCRV gauge ac-counting; Balancer weight drift; Sushi fork-of-v2 seman-tics (action-note: stable-swap + weighted-pool strips)	W1	connect	UNVERIFIED-CANDIDATE
68	aave-compound-lending-subgraphs	lending-subgraph	Aave v2/v3 + Com-pound v2/v3 (sup-ply/borrow/utilization/liquidations)	per-block	Aave 2020→; Com-pound 2020/2021→	near-realtime	GraphQL	data open	utilization PIT from index snap-shots; reserve-factor config history (action-note: DeFi lend/borrow rate panels)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
69	lido-eigenlayer-staking-subgraphs	staking-subgraph	Lido stETH subgraph + EigenLayer operator/AVS datasets	per-block	Lido 2020-12→; EigenLayer 2023→	near-realtime	GraphQL	data open	yield attribution split vs beacon-chain validators (row 49); restaking risk (action-note: staking-yield joins for ETH yield models)	W2	connect	UNVERIFIED-CANDIDATE
70	defillama-tvl	defi-aggregate	DefiLlama API /protocols + /charts (endpoint families per D7)	chain/protocol/token	2019→ per protocol	realtime-approx (m-refresh)	REST keyless	free + attribution	per-adapter TVL methodology quirks; double-count flags; stale-adapter audit (action-note: core DeFi aggregate feed)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
71	defillama-stablecoins	defi-aggregate	DefiLlama stable-coins mcap + pegSta-bility end-points	chain/token	issuers tracked since 2018-era	realtime	REST	free	mint/redemption double-count; mcap lag hours; issuance-bridge splits (action-note: stable-coin module pair w/ attesta-tions (77–79))	W1	connect	VERIFIED-UPSTREAM-DOC
72	defillama-yields	defi-aggregate	DefiLlama yields/pools history	pool-level	2021→ estab-lished pools	realtime-approx	REST	free	APY com-position (base vs reward) seman-tics; TVL-shift noise (action-note: yield bench-mark set (Stak-ingRe-wards cross-check))	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
73	defillama-bridges	defi-aggregate	DefiLlama bridges vol-ume/day	chain	2020→	near-realtime	REST	free	bridge-vendor volume overlap; bridge incident PIT (action-note: cross-chain flow features)	W2	connect	VERIFIED-UPSTREAM-DOC
74	defillama-volume-fees-raises	defi-aggregate	DefiLlama DEX spot volume + protocol fees/revenue + raises DB	chain/protocol	volume 2020+ era; raises listed	realtime	REST	free	fees vs revenue seman-tics differ per protocol; raises PIT (action-note: money-flow + cap-formation panels)	W1	connect	VERIFIED-UPSTREAM-DOC
75	l2beat-tvs-risk	L2-aggregate	L2Beat API (TVS, Stage 0/1/2, risk dic-tionary)	chain	since rollup launches	hours	REST unauth/json	free (at-tribution)	risk-dictionary method-ology changes → version pin; TVS ≠ TVL (action-note: L2 trust-scoring enrich-ment)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
76	dexscreener-pools	DEX-agg	DexScreener API (new-token pool pairs, liquid-ity/vol)	per-pool	since listing	realtime-ish	REST	free tier limited	rebase/exploit filter; canonical subgraph fallback (action-note: alt-DEX parity/scam-filter lane)	W3	connect	UNVERIFIED-CANDIDATE
77	chainlist-metadata	chain-metadata	Chainlist (De-fiLlama EVM registry: chain-ids, RPC/WSS end-points)	static registry	n/a	batch	REST/static	OSS (MIT)	RPC hygiene (public endpoint quality varies); registry drift (action-note: feed GFI or config)	W1	connect	UNVERIFIED-CANDIDATE
78	circle-usdc-attestations	stablecoin-attestation	Circle trans-parency (attesta-tion service + reserve disclo-sures)	monthly attesta-tion	from 2018	batch	PDF + public API fetch	public disclo-sure	issuer-commissioned attesta-tion; malformed-archive paths → check-sum gating (action-note: USDC collateral-risk feature)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
79	tether-attestations	stablecoin-attestation	Tether quarterly (BDO Italy assurance + CRR disclosure)	quarterly	from ~2014	batch	PDF	public disclosure	assurance breadth selectively scoped → directional only (action-note: USDT counterpart feature)	W1	connect	UNVERIFIED-CANDIDATE
80	paxos-pyusd-attestations	stablecoin-attestation	Paxos attestation service (USDP, PYUSD, USDG monthly)	monthly	from 2018	batch	PDF, URL-param per issuer	public disclosures	issuer coverage timeline; PYUSD conversion-PIT (verify) (action-note: lesser-issuer stable lane)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
81	pyth-price-feeds	oracle	Pyth Pro (ex-Lazer) WS/REST + history endpoint; classic on-chain pull feeds (stream-04 □D8)	tick	history via Pro REST endpoint	ms-class (vendor claim)	keyed WS/REST	Pro paid; on-chain pull free	confidence-interval attached per tick; per-asset feed ids drift; enti-tlement scope (action-note: multi-asset realtime overlay (incl. non-crypto))	W0	connect	VERIFIED-UPSTREAM-DOC
82	chainlink-feed-history	oracle	on-chain aggrega-tor reads (Round-Data/EVENT logs of classic feeds; Data Streams out of initial scope, stream-04 □D9)	round-based	since pair-feed deploy-ment (~2020+)	heartbeat/deviation-bound (NOT tick)	own-node eth_getLogs	on-chain output public	round PIT = update-block times-tamp (not heart-beat); devia-tion+heartbeat config per pair; L2/L1 chain splits (action-note: consensus-fallback leg of GFI price ag-gregator)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
83	redstone-api3-oracles	oracle	Redstone (WS/REST per feed model) + API3 OEV/dAPI claims	tick/round	since deploy per chain (verify)	near-realtime	REST/WS	free read	per-chain feed coverage verify first; coverage decay after Pyth/CL domi-nance (action-note: sec-ondary oracle strips only)	W3	connect	UNVERIFIED-CANDIDATE
84	tally-snapshot-governance	governance-votes	Tally API (EVM on-chain propos-als/votes historical) + Snapshot Hub API (propos-als/votes via IPFS archive)	proposal-level	Snapshot 2020→; Tally per DAO deploy	batch	REST	free (rate-limited)	Snapshot IPFS per-sistence gap risk; Tally = EVM-only; quorum-params PIT (action-note: governance-vote panel cross-vendor check)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
85	certik-immunefi-security	security-DB	CertiK Skynet scores (paid API) + Immunefi boun-ties/loss data (free REST)	asset-level	since audit listings	near-realtime	REST	CertiK paid; Immunefi free	score-methodology churn; bounty-TVL mapping staleness (action-note: risk-score enrich-ment features)	W3	connect	UNVERIFIED-CANDIDATE
86	rekt-chainabuse-incident-db	security-DB	rekt.news (histori-cal hacks w/ loss series) + Chainabuse reports + De-fiLlama raises (□D7)	incident-level	rekt archive 2020→	batch	web/REST scrape	attribution-license	incident attribu-tion error risk; loss-figure spread across sources; scam-report quality noise (action-note: loss-episode feature set for risk models)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
87	nft-dune-blur-os	nft-market	Dune-lineage NFT trade datasets (OpenSea/Blur/X2Y2; royalty-free era break-points)	per-tx	2021→ core marketplaces	batch	SQL credits (Dune □D5 lineage)	Dune output attribution	floor/trait quality; indexer lags; royalty-era PIT break-points (action-note: optional NFT feature lane)	W3	connect	UNVERIFIED-CANDIDATE
88	crypto-social-stocktwits-twitter	social-sentiment	StockTwits crypto streams (free API) + paid crypto-Twitter/social vendors (RavenPack/Meltwater-class)	msg-level	varies by vendor	near-realtime	REST	StockTwits free; paid-class for X/Telegram coverage	free Twitter API limits (2023-era pricing regime) — paid wedge for X; spam-filter PIT (action-note: optional sentiment lane)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
89	fear-greed-index-archive	sentiment-index	alternative.me F&G historical API (daily)	daily	from 2018-02-01	daily T+	REST keyless	free	index methodology opaque/vol-weighted; census anchor claims vol-11 verification — archive URL NOT re-fetched this session → re-verify endpoint live-ness (action-note: daily sentiment series (cheap W1))	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
90	paid-onchain-metrics-cryptoquant-glassnode-santiment	paid-metrics	CryptoQuant (exchange-flow), Glassnode Studio (limited free series), Santiment (API)	multi-resolution	multi-year per metric	daily/RT-class	REST keyed	paid tiers; limited free tiers	licenses forbid redistribution into vaults; cross-vendor triangulation needed (no single source of truth) (action-note: paid wedge only when exchange-flow deltas required)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	Jordà–Schularick–Taylor Macrohistory Database (JST R6)	market-history/macro	macrohistory.net direct file download (Stata/XLSX) [1]	Annual; 48 vars; 18 advanced economies	1870–present (~155y)	batch (release-based)	archive-HTTP bulk file; pin R.6 snapshot	free, CC BY-NC-SA 4.0 (non-commercial; no vendor redistribution)	fixed release tag; crisis-dummy alignment vs JST crisis chronology PDF; document splices per 100pp documentation (action-note: ingest R6 XLSX → country×variable long table; freeze source snapshot hash)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
2	JST crisis chronol-ogy + RORE returns docs	financial-stress/history	macrohistory.net/ data doc PDFs (same release) [1]	Annual event/series	1870–2015	batch	archive-HTTP PDF/XLSX parse	free CC BY-NC-SA	cross-check crisis flags in main table vs chronol-ogy PDF; PIT: cite release year (action-note: parse crisis chronol-ogy into event table (cross-ref row 1))	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	Shiller US equity/CAPE/US home prices (ie_data.xls, Fig3-1)	market-history/equity/	shillerdata.com direct XLSX [2]	Monthly; US	1871+ stock/CAPE; 1890+ home prices	batch (monthly)	archive-HTTP XLSX; scripted re-pull + diff	free (attribution; no warranty)	spliced Cowles/1926 sources documented; CPI splice at 1913; recompute CAPE from constituents to validate (action-note: ingest ie_data.xls all columns; version-by-download-date snapshot)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
4	BoE “A Millennium of Macroeconomic Data” (UK v3.1)	market-history/macro	bankofengland.co.uk/research/datasets page, 28MB XLSX direct [3]	Annual (some pre-1830 benchmarks); UK	C13th–2016 (1086 Domesday benchmarks)	batch (rare updates)	archive-HTTP XLSX	free (Crown/attributed via BoE)	v3.1 published; email-contact annotation; pre-1830 treated as benchmark-point, not continuous (action-note: ingest UK mega-sheet into long format; flag pre-1700 series provenance)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	MeasuringWorth US/UK (GDP, CPI, wages, $/£ER, interest rates, S&P/DJIA)	macro-history/deflator	measuringworth.com, datasets pages [12]	Annual, US+UK (+AU, ES)	US GDP 1790+; CPI 1774+; ER 1791+	batch	scrape/download per-dataset pages (no bulk API)	free for non-profit educational use w/ credit; permission needed otherwise	method-essay awareness (7 measures of worth); cite staff/Officer papers (action-note: ingest US CPI + $/£ ER as deflator backbone; keep per-dataset source notes)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
6	Global Financial Data (GFD)	market-history (longest runs)	globalfinancialdata.com (licensed subscrip-tion; custom data-file delivery)	Annual/monthly, 200+ countries	1265+ (header series), equities 1600s+	batch (subscrip-tion dumps)	licensed bulk file (SFTP/email zip)	licensed (paid, per-user/redistribution limits)	license quote req: seats, redistri-bution, annual update fee; cross-validate overlap vs Shiller/JST (action-note: request quote; scope: JST gap-fill + pre-1870 bench-marks only)	W2	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
7	Ibbotson SBBI (Morningstar)	market-history/equity	Morningstar Direct/IDX; licensed	Annual/monthly; US asset classes	1926–present	batch	licensed feed; CSV export	licensed (paid; chart-licensing separate)	quote cost; replicate-access only via WRDS unless licensed; PIT: annual ed. vintage (action-note: ; only if GFD fails for 1926–1970 returns)	W2	quote	UNVERIFIED-CANDIDATE
8	CRSP US equities (cross-ref 05)	securities/history	WRDS/Crsp; see vol 05	Daily; US listed universe	1925–present	batch	WRDS (sas xport ftplib)	licensed (academic quote)	coverage-gaps for delisted shares — survivorship-free requirement (action-note: cross-ref 05; here only for pre-1962 of 05’s route)	W2	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
9	Compustat North America (cross-ref 05)	securities/fund-history	WRDS; see vol 05	Annual/quarterly	1950s–	batch	WRDS	licensed	restatement vintages via comp_history tables (action-note: cross-ref 05 only)	W2	connect	SURVEY-EVIDENCE
10	WFE (World Federation of Exchanges) annual market stats archive	exchange-history	focus.world-exchanges.org; down-loaded WFE workbook archive	Annual by exchange	1970s+ (common members backfill)	batch	archive-HTTP XLSX per year	licensed-ish (terms: no redistribution, non-commercial survey use)	member-change breaks continuity; document coverage by exchange-year (action-note: pull archive year sheets; build exchange-years matrix)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	FRED bulk CSV (full history, by series id)	macro-archive bulk	fred.stlouisfed.org/fred/series/{id}/ + fred-graph + fredapi/FRED API + full-file dumps via external mirrors	Varies: daily-annual); global	Series-specific (many 1900s+)	batch	REST + bulk-CSV	free	cite St. Louis Fed as redistrib-utor of underly-ing owners; per-series owner/license check (action-note: nightly delta via FRED API; store vin-tage_id=term)	W0	connect	SURVEY-EVIDENCE
12	ALFRED (archival FRED vintage)	PIT/vintage	alfred.stlouisfed.org [10]; API base fred.stlouisfed.org/docs/api/fred/	Varies	Vintages since 2006 (data itself older)	batch (already archived)	REST API (`vintage_dates`) + CSV	free	the desig-nated PIT spine for US macro: ingest real-time-vintage series first (action-note: fetch `observa-tion_start`+vintage paths for top-200 US macro series)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
13	Maddison Project Database (GDP per capita, popula-tion)	academic/macro-history	ggdc.net/maddison (Project 2020 release, direct XLSX)	Admin1; 169+ coun-tries/regions	AD 1–2018 core; 1500+ denser	batch	archive-HTTP XLSX	free (CC BY 4.0-type; cite Bolt & van Zanden)	year-0 nan pattern; region-vs-country bound-aries; use 2012-boundary CLIO harmo-nization if mixing (action-note: ingest maddi-son2020; derive growth/level panels)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
14	CLIO-INFRA (86 datasets: GDP/capita 1500+, bond yields 1727+, FX 1500+, central debt 1692+, metals production 1681+, demography)	academic/multi-domain	clio-infra.eu direct Indicator XLSX downloads [4]	Annual-decadal; 224 countries	1500–2013 (varies 1543–2018)	batch (archived)	archive-HTTP XLSX per indicator	free (attribution / varied per dataset — check per-file terms)	boundaries=V2012-reference; mixed annual/decadal already harmonized; download per-indicator as-of-date provenance (action-note: bulk-pull finance block first (yields, FX, debt) then production)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
15	Our World in Data (owlid bulk + per-topic grapher CSVs)	academic/topic-archive	ourworldindata.org/grapher/{slug}.csv per chart + owid dataset repo (quiltify)	Annual; global topic-mapped	varies; energy 1800s+, poverty 1820+	batch	bulk-CSV (grapher endpoint) + GitHub repo mirroring	free (CC BY 4.0)	chart-slug volatility → pin slugs; underlying-source attribution chain per column (action-note: enumerate target slugs (energy, emissions, growth); subscribe repo diffs)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
16	Barro–Lee educational attainment 1950–2010	academic/human-capital	barrolee.com — direct XLSX	5-year interval; 146 countries	1950–2010	batch (static)	archive-HTTP XLSX	free (cite Barro & Lee 2013)	2010 home-age cutoff; no forward updates — pair with post-2010 Lutz witt (CLIO/OWED) (action-note: one-time ingest; register as frozen reference)	W1	connect	SURVEY-EVIDENCE
17	WID.world (DINA top wealth/income shares)	inequality-history	wid.world data portal full-dataset CSV export + `wid` Stata package [7]	Annual; 100+ countries	1820–present (col. vary)	batch (infrequent quarterly)	portal bulk CSV + REST-ish WID API	free (CC BY-NC? verify per-dataset; attribution required)	DINA-ratio method editions; collection-specific selection (match JST era) (action-note: bulk download WID-DINA; build top-1%/top-10% panels)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
18	World Bank WDI (bundled long-run archive + API archive)	macro/long-run	databank.worldbank.org/source/world-development-indicators + api.worldbank.org (M119.MWI/M…); plus legacy “WDI archives” page	Annual; 217 countries	1960–current (archival pdf back to 1980s)	batch	REST (JSON/CSV) `source=2`	free (CC BY 4.0)	country-code-history (e.g., remnant fran-chises), missing-x pattern; fallback to World Bank CSV dumps (action-note: API script by indicator-list; keep wrangler yearly in-stance_id)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
19	IMF IFS historical (classic IFS lines incl. exchange-era FX)	macro-archive	IMF Data (data.imf.org) "International Financial Statistics"; legacy line-set cut off series-wise (classic IFS retired, successor tables absorb)	Monthly; ~200 countries	1948–2022 (some pre-1948 via IFS PDFs-cans)	batch (monthly, archived)	SDMX bulk CSV (IFS dataset id: IFS_6.0-style key family)	free (IMF CC BY-NC condition; IFS carry copyright of underlying countries)	IFS line-number stability; skip IFS data-reference codes with patch notes (action-note: bulk SDMX (IFS) monthly; keep pre-1980 PDF scan backlog as W3 manual)	W0	connect	UNVERIFIED-CANDIDATE
20	UN Statistics Monthly Bulletin of Statistics (MBS) archive	macro-archive	unstats.un.org MBS archive (PDF/CD-era)	Monthly; ~200 countries	1950s–2010s (online archive)	batch (static)	archive-HTTP PDF	free (public UN)	table-code churn; only as gap-fill vs IFS/WDI (action-note: select ~20 tables (minting/M2/FX) if IMF gaps remain)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
21	OECD Historical Statistics (book-era series + OECD.Stat archive)	macro-archive	OECD Data Explorer “Historical Statistics 1960–1996” bound dataset + scanned Historical Statistics compilations (1947–2010 era) as OCR fallback	Annual	1960–1996 (book compilations)	batch (static)	SDMX bulk or manual scan	license: OECD reproduction terms (non-commercial ok w/ attribution for stats)	scanned-table OCR QC; series-merge notes vs current OECD.Stat (action-note: only where pre-1960 OECD gaps)	W2	connect	UNVERIFIED-CANDIDATE
22	Historical Statistics of the United States (Bicentennial Edition)	.US-history	1975 Bicentennial-edition scans via FRASER (fraser.stlouisfed.org); pay-walled on-line ed. at hsus.cambridge.org	Mixed (annual proxy-specific)	1610–1975 (Bicentennial edition)	batch (static)	archive-HTTP PDF scan + manual table extraction	public-domain scans (FRASER); hsus.cambridge.org licensed	OCR QC gates for scanned tables; series-code dropout doc (action-note: FRASER-selected tables (finance, prices, FX): manual extract)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
23	PWT (Penn World Table v10.x)	academic/macro	pwt.ruhby? — rug.nl/ggdc/productivity/pwt; University of Groningen direct download	annual; 183 countries	1950–2022 (v10.0)	batch (release-based)	archive-HTTP XLSX	free (CC BY 4.0)	version pinning (use v10.x); rgdpna vs rgdpe marker (action-note: ingest v10.x — one-time + release watch)	W0	connect	SURVEY-EVIDENCE
24	GGDC productivity/structural-change databases (incl. 10-sector historical backfills pre-1947 for ~10 countries)	academic/industry	rug.nl/ggdc/productivity + ggdc historical database	annual; NACE-grouped	pre-1947 backfills via GGDC historical	batch	archive-HTTP XLSX	free (attribution)	pre-1947 output-concept v1 inconsistencies; cite-by-database-specific papers (action-note: ingest 10-sector historical block only (economies coverage thin))	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
25	UNIDO INDSTAT 2 (2022+ public interna-tional standard)	industry/history	stat.unido.org (public data down-loads since 2022)	Annual; 4-digit ISIC	1963–current (cover-age varies)	batch	portal bulk CSV	free w/ citation (UNIDO public)	classification revisions (ISIC rev 2→3→4 bridges) (action-note: download country-wide manufac-turing value-added & output)	W2	connect	UNVERIFIED-CANDIDATE
26	Freeman–Oostendorp wages database (IISG)	academic/wages	socialhistory.org project archive (CLIO per-petuation via CLIO-INFRA real-wages)	Annual; many countries	1200–2000 (frag-mented)	batch (static)	archive	free w/ citation	coverage sparse — prefer CLIO real-wage (row 14) for conti-nuity (action-note: skip unless target country wage-history needed)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
27	BoE daily GBP spot rates (exchange-rate database)	FX archive	bankofengland.co.uk/boeapps/database (Series “A” spot FX)	Daily; ~50 currencies vs GBP	1919–present (some w/ gaps)	batch	HTTP CSV export per series ID	free (attribution)	spliced pre-1971 string with dollar regimes; BoE-only series (not market-close) — note source; splice-flags required (action-note: export A-series FX block; document splice points)	W0	connect	VERIFIED-UPSTREAM-DOC
28	BIS Nominal/Real Effective Exchange Rate series (BIS/NBHR)	FX archive	bis.org/statistics/eer; historical long-run EERs	Monthly; 60–100 economies	Jan 1964–present (nominal; broader since 1994)	batch (monthly)	SDMX bulk CSV	free (BIS)	1964–93 thin trading caveat; rebase-pair vs 2010/2015 weights revisions (action-note: ingest NOMER/REALER; join vs row 27 splices)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
29	IMF IFS exchange-rate era series (FX cross of row 19)	FX archive	data.imf.org IFS RF-lines (end-of-period FX)	Monthly	1948 onward (cover-age historic via IFS)	batch	SDMX as row 19	free w/ condition	era breaks: gold/, SDR unifi-cation periods (action-note: extracted from row 19 dataset (RF% line subset))	W0	connect	UNVERIFIED-CANDIDATE
30	FRB FRASER H.10 archive (US $ daily/weekly FX 1971–)	FX archive	fraser.stlouisfed.org H.10 collection (scanned+API-era) + FRED series EX…	Daily/weekly	1971–present	batch	FRED API + FRASER PDF for pre-API classics	free	1971 start (Bretton-Woods collapse); pre-1971 via row 27 (action-note: FRED DEXUSUK… id-block pull)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
31	ICE/USDXY (USDX) index history	FX archive	ICE DataServices (licensed); free partial via market-media mirrors	Daily	1973–present	batch	licensed feed or WSJ-Yahoo free snapshot	licensed if ICE official; media-mirror licensed-free (non-commercial)	PIT: index-methodology changes (1999 euro); rebase weights history (action-note: methods note first; media-keyed snapshot acceptable for daily-close research)	W2	connect	UNVERIFIED-CANDIDATE
32	HISTDATA (free historical FX / MT5 tick)	FX tick archive	histdata.com per-pair per-month free zip	tick (2000s era) & MT5	1990s?–present	manual-batch	archive-HTTP zip per pair	free (non-commercial terms on site)	session-basis server-side; checksum; volume dn flags (action-note: only for pairs missing from Dukas/Yahoo)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
33	Dukascopy tick reposi-tory (cross-ref 07)	FX tick	datafeed.dukascopy.com historical tick HTTP (free)	Tick, 30+ pairs	2003–present	batch	free-release HTTP (cross-ref vol 07 tech)	free (terms: no redis-tribution of raw tick)	gap-patching + tick-clean; overlap vs row 32 (action-note: cross-ref 07; here for 2003–2010 span)	W2	backfill	SURVEY-EVIDENCE
34	RMB official/NDF-history (CNY fixed, managed float eras)	FX archive	PBoC + SAFE published daily USD/CNY from 1994-01; NDF-era via news/research captures	Daily	1994–2005 fixed; 2005–2015 crawling +NDF era	batch	portal CSV/web page parse (PBoC has limited API)	free (govt data)	regime-change labels manda-tory (peg/crawl/floating periods); use row 30 DEX-CHUS as US$-side cross-check (action-note: build CNY era-tagged series; NDF s from FRB/press archives only if gap)	W2	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
35	IMF Primary Commodity Prices (PCPS/PCP amenable long-run)	commodity archive	imf.org PCPS monthly bulk (index-level + USD/unit)	Monthly; 50+ commodities, 9 aggregates	1962–present (varies by commodity)	batch (monthly)	SDMX/API bulk CSV	free (IMF, attribution)	base-era rebases; unit-USD history; check long-spin cadence variance for softs (action-note: bulk pull PCPS; index s vs Pink Sheet (row 36))	W0	backfill	SURVEY-EVIDENCE
36	World Bank Pink Sheet Commodity Price Data archive	commodity archive	worldbank.org commodity-markets monthly XLSX archive (back decades)	Monthly; 45+ commodities	1960–present (archive extent varies)	batch (monthly)	direct XLSX archive-HTTP	free (attribution: World Bank Global Price Monitoring System)	pink-sheet monthly tiny-format churn; annual “commodity price data archive” join to herded archive (action-note: ingest full archive incl. annual-history sheets)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
37	EIA WTI/Brent/NYM spot & inventory long history	commodity archive	api.eia.gov / eia.gov series browser (WTI from Jan 1946, Brent from Jan 1987)	Monthly (daily post-1986)	1946–present	batch (monthly)	REST API v2 (api key)	free (US govt public domain)	use Cushing WTI era-notes; join vs FRED DCOILWTICO (action-note: API script for oil-gas-coal-nuclear spot set)	W0	connect	SURVEY-EVIDENCE
38	USDA PSD (Production, Supply & Distribution) historical archive	agriculture archive	apps.fas.usda.gov/psdonline/ historical bulk downloads (CSV/XLSX)	Annual marketing-year; commodities by country	1960+ (fixed backfill) to current	batch (monthly)	bulk zipped CSV + dataset-API (individual commodities CSV)	free (USDA public domain)	marketing-year (not cal-yr) alignment; rolling revisions — store snapshot code/year (action-note: bulk download PSD all-commodities; year-grained panel)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
39	FAOSTAT production/trade/supply archives	agriculture archive	fao.org/faostat bulk downloads (All-Data zip + FAOSTAT API)	Annual yields; 1961+ ; land use 1961+	1961–present	batch (annual)	bulk-HTTP zip + ETL coming FAOSTAT-API	free (CC BY 4.0)	production-country boundary corrections (post-USSR split handled natively) (action-note: All-Data pull for production + trade matrix)	W0	connect	SURVEY-EVIDENCE
40	Energy Institute (ex-BP) Statistical Review of World Energy	energy archive	energyinst.org/statistical-review (2023+); BP archive 1950s+ PDFs per decade	Annual; country×fuel	global 1950s+; per-country 1965+ with v2.2 OWID reframe	batch (annual)	XLSX download per year-reviewed; PDF archives pre-2001	free (attribution)	decade-boundary backfills; fuel-category reclass (nuclear appear later; wind cap IAWS) (action-note: pieced: XLSX 2001–now + PDF table-scan for pre-2001)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
41	USGS Historical Mineral Commodity Summaries + commodities consumption statistics	metals archive	usgs.gov MCA (annual PDFs/XLSX) + minerals consumption historical	Annual; ~90 minerals; countries	commodity consumption back to 19th century for many (HSUS-linked); summaries 1995+	batch (annual Jan/Feb)	XLSX/PDF bulk per year	free (public domain, US govt)	year-of-printed MCA version pinning; expect country-of-record break patterns (action-note: pull MCA last 10y + historical consumption stats heavy-mineral set)	W0	connect	SURVEY-EVIDENCE
42	NOAA ISD (Integrated Surface Data) global station archive	weather-archive	ncei.noaa.gov ISD: GHCN-D? separate — ISD via NCEI bulk (ftp-era + WebAPI now)	Hourly/sub-hourly stations	1901–present (Mexico? US 1901+; global thin pre-1940)	batch (annual)	bulk HTTPS zip per year	free (public domain)	station-id churn & relocations; pre-1940 QC flags quality-code (action-note: bulk pipeline for 15 pilot exchange-region stations 1901–2026)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
43	NASA GISTEMP v4 global surface temp 1880+	climate archive	data.giss.nasa.gov/gistemp/ GLB tsp tables + netCDF [9]	Monthly & annual	1880–present	batch (monthly ~10th)	plain-text table + netCDF archive-HTTP	free (public-domain-ish; cite GISTEMP team)	1951–1980 base-period anchor-ing; re-base to 1991-2020 when combin-ing (action-note: pull GLB TSV monthly; join vs HadCRUT row 44)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	HadCRUT5 (CRU/Met Office) 1850+	climate archive	crudata.uea.ac.uk/cru/data/temperature/ [8]	Monthly & annual; 5°×5° grid	1850–present	batch (every 1–2 mos)	CRU text format (global/nh/sh) + netCDF grid	free (CRU licence; attribution + scientific citation)	non-infilled vs Analysis choice: pick Analysis for wider coverage; late-arrival revisions near-real-time (action-note: CRU text GL/NH/SH series; netCDF only if grid modeling later)	W1	connect	VERIFIED-UPSTREAM-DOC
45	ICE softs (cof-fee/cocoa/sugar) official historical	commodity-market (gar-den)	ICE DataSer-vices (RTH + daily historical per-mar.ket)	Daily	market birth (NYBOT-era continua-tions back to 1890s via stocks&commodity splices)	batch	licensed data feed (ICE Con-nect/API)	licensed (paid; per-market onboard-ing)	continuous contracts splice policy; volume caps by license tier (action-note: only if cof-fee/cocoa MVP ships (pilot 2 roster))	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	MPOB palm-oil long-run (ITC-pioneered histori-cal)	agriculture archive	MPOB (Malaysian Palm Oil Board) apsis mpob.gov.my an-nual/monthly statistical tables	Monthly/annual	1970s+ (statisti-cal bulletins)	batch	portal down-loads (XLSX/PDF)	free w/ at-tribution (MPOB Malaysian govt)	revise-panels; freight & FFA hedging refs only as index (action-note: monthly OPR/CPO price history manual collect)	W2	connect	UNVERIFIED-CANDIDATE
47	FRED Moody’s Aaa/Baa yields (long-run US credit)	rates archive	FRED AAA/BAA (origi-nally Moody’s; now ICE BofA carried), monthly series from Dec 1919	Monthly	Dec 1919–present	batch (monthly)	FRED REST/CSV (row 11 route)	free (require Moody’s→Moody’s/ICE 2016 source at-tribution)	series-switch (Moody’s exiting; FED’s-other-publisher churn); verify with release notes (action-note: FRED AAA/BAA ingest; yield-spread derived)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
48	H.15 vintages (Treasury/motional rates release archive)	rates archive	FRB H.15 via FRASER (scanned pre-API) + FRED (DGS-alle series)	Daily/weekly/monthly	H.15 earliest scanned 1950s/1960s; DGS backfill: 1962+, DFF 1954+	batch	FRED API + FRASER PDF scans	free (public domain)	vintage-aware for PIT: dailyTN concept changes (Wednesday-cycle pre-1997) (action-note: FRED daily-treasury block + FRASER scan backlog only if pre-1962 debt-model needs)	W1	connect	SURVEY-EVIDENCE
49	CRSP US Treasury Database (bond database incl. pre-1941 roots)	rates securities	WRDS CRSP fixed-income legacy + CRSP TreasuryF-BOND	Monthly	1926–present (bond db 1926+ via CRSP legacy)	batch	WRDS	licensed	missing-calls on pre-1926; PIT uses CRSP access dates (action-note: as complement to FRED H.15 (row 48))	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
50	JGB long-run yields (Japan MoF)	rates archive	mof.go.jp interest-rate historical (long-term/short-term rates, monthly since 1974, 10yJGB matrix)	Daily/monthly, JPY curve tenors	1974–present (long 10y rate from 1874 via aca-demic?” long-run DAIWAR)	batch (monthly/daily)	HTTP CSV/XLSX per page	free (govt data)	market-era discon-tinuity pre-1971 (fixed-rate issues); partici-pating vs settle-ment conven-tions (action-note: pull MoF rate history pages; mark era labels)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
51	ECB long-run sovereign yields (EMU aggrega-tion)	rates archive	ecb.europa.eu SDW series (e.g., monthly long-term yields for EMU-continuation & per-country)	Monthly	1960s–present via con-structed “continua-tion” member series, 1999+ direct	batch (monthly)	ECB SDW API/CSV	free (ECB data terms)	COMBI-of-country series construc-tion (rotate members joins); pre-1999 synthetic labeling (action-note: SDW API: long rates (LT) for EA countries; tag pre-euro coyness)	W0	connect	SURVEY-EVIDENCE
52	St. Louis Fed STLFSI2/STLFSI4 (financial-stress index)	stress archive	research.stlouisfed.org FRED STLFSI/STLFSI2/STLFSI4	Weekly/daily	1993+ (v1), 1994+ (v2/3/4)	batch (weekly)	FRED API	free	component-set key differ-ences across versions — keep all 3 con-tiguous (action-note: pull all three stlfsi series)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
53	Chicago Fed CFNAI (+CFNAI-MA3)	stress/archive	chicagofed.org + FRED CFNAI	Monthly	1967–present	batch	FRED/HTTP	free	month-of-release publication lag (release t+1) — PIT: use release-vintage mode of ALFRED if available (action-note: FRED pull CFNAI/CFNAIMA3)	W0	connect	SURVEY-EVIDENCE
54	OECD CLI (composite leading indicator) long run	archive indicator	oecd.org CLI SDMX	Monthly	1960s+ (varies country)	batch (monthly)	SDMX CSV (as row 21 route family)	free (OECD terms)	CLI historical construction revisions; per-station base-month labeling (action-note: OECD API pull per-country CLI)	W0	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	Conference Board LEI/US + world LEI historical	archive indicator	conferenboard.org (official, licensed); free partial series via FRED (USLEI)	Monthly	1959+ (US); later interna-tional	batch	FRED-part + licensed direct	mixed: FRED US-LEI free; full-international licensed	FRED partial cross-check only; PIT from CB release dates (action-note: FRED USLEI + USSLIND; CB if non-US LEI needed)	W1	quote	UNVERIFIED-CANDIDATE
56	NBER US business-cycle dates (peak/trough chronol-ogy)	archive reference	nber.org bcdc-page JSON/xlsx [6]	Monthly reference points	Dec 1854–present (32 cycles)	batch (committee-driven)	data.nber.org/cycles/ JSON API	free (fact-tribution)	announcement-date (not turning-point-date) as PIT for “when was it known” vs “when it was” (action-note: JSON pull; freeze an-nounce-ment dates)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
57	Reinhart–Rogoff crisis chronology (historical currency/banking/debt crises)	stress archive	Harvard/PIIE hosted dataset page (Reinhart's crisis dataset page; canonical "external-banking-debt-crises" XLSX + IMF WP annex files)	Annual event table; 70 countries	1800–2016 (varies)	batch (static)	XLSX download from creator-affiliated post	free (academic; cite Reinhart & Rogoff)	cannot claim current-governance; update path = Laeven-Valencia (row 58) (action-note: locate canonical XLSX; parse events 1800–2020)	W1	connect	SURVEY-EVIDENCE
58	Laeven–Valencia Systemic Banking Crises Database	stress archive	IMF WP/20/112 annex dataset (4th ed 2020)	Annual events; 132 countries	1970–2018 (4th ed)	batch (static)	XLSX via imf.org article page	free (IMF)	event-code discipline; 5th-update watch (action-note: parse annex XLSX into crisis table)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
59	IMF Financial Soundness Indicators (FSI)	stress archive	data.imf.org FSI dataset	Quarterly/annual	1990s+ (country varies), guidance-2000-era	batch	SDMX (row 19 route)	free w/ attribution	coverage-depth varies; compliance with IMF FSI guide quality flags (action-note: bulk FSI core set (NPL ratio, capital adequacy))	W2	connect	UNVERIFIED-CANDIDATE
60	GDELT (events, GKG, translingual, 1979+ full archive)	news/text archive	gdeltproject.org; BigQuery public dataset + raw file HTTP (data.gdeltproject.org) [5]	Event/article, 15-min update	Jan 1, 1979–present	near-realtime (this volume treats as batch historical sweeps)	BigQuery SQL + raw HTTP firehose	free (open data; attribution requested)	dedupe vs vol 08 real-time consumer; CAMEO coverage density pre-1990s thinner (action-note: BigQuery historical sweeps for W1 sample (cross-ref 08))	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	Common Crawl (quarterly web snapshots)	news/web text archive	commoncrawl.org; s3://commoncrawl + data.commoncrawl.org per-crawl indexes [11]	WARC/WET/WAT snapshot	2008 Q3–present (~50+ crawls)	batch (quarterly)	S3/HTTP + CDX index; parallel compute via AWS Open Data Sponsorship	free (site-content rights vary — crawl-level free)	per-domain rights only apply downstream; pin crawl ID per research corpus (action-note: choose sample crawl for pilot text-measure (narrative-history proxy))	W2	connect	VERIFIED-UPSTREAM-DOC
62	Chronicling America (LoC historical US newspapers)	news/text archive	chroniclingamerica.loc.gov JSON/text API (bot-403 on root: use api path with UA)	issue/page OCR	1776–1963 (per digitization waves)	batch (static)	HTTP JSON (search API) + plain-text OCR	free (public domain)	OCR quality varies by era; accept per-page with OCR-confidence flag (action-note: targeted fetch for event-anchored narrative windows (~12 events))	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
63	Wayback Machine (CDX API snap-shots)	web/archive	web.archive.org CDX API (`/cdx?url=…&output=json`)	Snapshot times-tamps	1996–present	batch	REST CDX + WARC download	free (Internet Archive terms)	snapshot-density skew post-2015; capture-status (4xx/redirect) flags (action-note: CDX for pilot-firm domains (e.g., annual-report URLs))	W1	backfill	SURVEY-EVIDENCE
64	Internet Archive books/texts scholar collec-tions	text archive	archive.org ad-vanced-search API + item-download	Page-text (books/journals)	Pre-20th-century–present	batch	REST search API + HTTPS files	free (most items public-domain or IA lending)	inbound license bound-aries per item; OCR con-fidence noted (action-note: only for GFI-narrative feature search (pre-internet events))	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	Nikkei 225 long-run index archive	dormant/dead-venue	indexes.nikkei.co.jp historical “long-term data” XLSX (base 1986 adjusted for returns to 1949)	Daily (pre-1960s recon-structed monthly in official down-load)	1949–present	batch	portal XLSX download	free for research w/ attri-bution (Nikkei Inc. dis-claimer)	splice: index-methodology changes 1970 (50→225) & 2010s changeovers; distribute via download-snapshots (action-note: pull official long-history XLSX; docu-ment method-change dates)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
66	Berlin Stock Exchange historical (dead venue)	dormant	pre-1945 Berlin Börse indices/turnover via academic digitization (FU Berlin / digitized course-of-the-day bulletins) — no official archive	Annual/monthly	1870s–1944	batch	digitized journal/PDF scans	free (scan archives) w/ citation	high manual-friction (bulletin scans where no digital index); defer if GFD (row 6) covered (action-note: scoped test: 1870s–1913 aggregate share-price index only)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
67	DNB/De Neder-landsche Bank historical money & debt stats	dormant/central-bank-archive	dnb.nl historical statistical publica-tions (Dutch guilder era scans)	Monthly/annual	1860s–2014 guilder-era	batch	portal XLSX/PDF	free (govt)	OCR + Dutch-language headers; prefer JST/CLIO where overlap-ping (action-note: pull guilder-era money-credit set if Nether-lands pilot)	W2	connect	UNVERIFIED-CANDIDATE
68	RBA 150-year statistics archive? Bank of Australia? RBA 2000-series historical	central-bank-archive	rba.gov.au “Measur-ing the size of the econ-omy” legacy — concrete: RBA statistical historical-150-year series XLSX (2021 release)	Annual/monthly	1901–2021	batch	XLSX direct	free (govt)	splice notes native in publica-tion; pre-1940 thin series flagged (action-note: ingest 150-year historical XLSX (Aus-tralia))	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
69	Danmarks Nationalbank historical statistics (pre-1950 stat-archive/“data on central interest 1824”)	central-bank-archive	nationalbanken.dk historical statistics page (long-run rates & credit series incl. Abildgren backfills to 1818)	Monthly/annual	1824–present (key rates series)	batch	XLSX download + published research-doc series	free (govt)	Abildgren-construction notes binding; document overlaps vs JST Denmark rows (action-note: pull Danmarks historical long-rate file (Abildgren 2020 update))	W2	connect	UNVERIFIED-CANDIDATE
70	Norges Bank historical monetary statistics	central-bank-archive	norges-bank.no/en/about/statistics/historical-monetary-statistics	Annual/monthly	1819–present (nominal series); FX & capital flows back to 1938	batch	XLSX direct	free (govt)	series-rebasing listed in historical-statistics com-pendium (action-note: pull kronie money+FX block)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
71	BoJ long-run historical statistics (BOJ “100-year” historical stats)	central-bank-archive	boj.or.jp/en/statistics/stat_history/classics archive (Historical Statistics, hundred-year series PDF/CGHTML)	Annual/step	1885/1886–2000 (hundred-year vol), post-2000 in main DB	batch	PDF/HTML transcribe + BOJ time-series lookup (BOJ main stats DB continues)	free (govt)	typo-reviewed transcription; weigh vs JST Japan rows (action-note: extract M2/long-rates columns from hundred-year table)	W2	connect	UNVERIFIED-CANDIDATE
72	Dead/delisted asset archive: CoinMetrics Community (free CSV incl. delisted assets incl. Mt.-Gox-era data)	dead-asset record	github.com/coinmetrics/data free CSV incl. delisted/dead markets	Daily	2009/2010–present (per asset)	batch (daily)	GitHub-HTTPS raw CSV	free (CC BY 4.0)	delisted assets freeze at last-print date — must not forward-fill; exchange-coverage flags per asset (action-note: pull CSV; split live vs delisted universe (cross-ref 07 for dex-live-live))	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
73	Dead-coin/delisted marker: CoinGecko dead status + delisted-event history	dead-asset record	api.coingecko.com (public end-points; includes delisted coins history where rate-limited)	Daily + status flag	2013–now (per asset varies)	near-realtime (treat batch curate weekly)	REST (public, no key)	free (at-tribution, rate-limited)	delist-date cannot guaran-tee last-price quality (thin tape) — pair with row 72 oracle (action-note: weekly delist-flag delta; archive removed assets)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
74	CoinMarketCap historical archive dumps (early CSV snapshots incl. dead coins)	dead-asset record	community-crafted dumps (CMC 2013–2015 era official snapshot CSVs; CMC discontinued free full backfill long ago)	Daily	2013–2021 (community-file dependent)	batch (static archive)	download archive CSVs	license-risk (CMC data not redistributable freely) → use for PIT research only	snapshot completeness verified by row-count-per-date sanity; do not publish externally (action-note: acquire community dumps; internal-only use; no publication)	W2	connect	UNVERIFIED-CANDIDATE
75	DeadCoins.com / delisted-project registry	dead-asset record	deadcoins.com catalog (scrapeable) + cross-ref modest GitHub “deadcoin-dataset” mirrors	Event/record	2013–2022 (site state varies)	batch (static scrape)	HTTP scrape → CSV	attribution-needed (site unclear OTRL) — treat reference-only	site abandoned risk = archive.org mirror WARC as fallback (row 63 route) (action-note: archive.org WARC scrape into registry table)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
76	Delisted ex-change/failure record archive (exchange-collapse dataset: Mt.Gox 2010–2014, Cryptsy 2014–16, Fcoin 2019)	dead-venue record	open-source recon: Mt.Gox recovered-trades commu-nity releases (github MtGox-recovered datasets) + published post-mortem/FMA-recovery reports	Tick/daily	2010–2019 era-specific	batch (static)	GitHub/archive release	free (forensic-community releases)	recovered-trade complete-ness labels; volume excluded from live-market aggre-gates (action-note: pull Mt.Gox recovery block; docu-ment coverage fraction)	W2	connect	UNVERIFIED-CANDIDATE
1	Polymarket market metadata + prices	prediction	Gamma API `gamma-api.polymarket.com/markets` + `/events` (S43 fetch 200 OK; alphalens spike-proven)	market/outcome snapshot	launch 2020→	minutes (poll)	REST polling	free (per site ToS; display-derived only)	outcome-token splits re-denominate markets — key on condi-tionId not slug; mar-ket→event 1:N	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
2	Polymarket CLOB order-book + trades	prediction	CLOB API `clob.polymarket.com` book + trades (alphalens spike-proven: 200)	book/tick	2020→ prices/markets/trades	poll (no free WS guarantee)	REST polling + own-spool	free (ToS)	thin long-tail books — store best-N levels only; resolution disputes move prices post-hoc	W1	connect	VERIFIED-UPSTREAM-DOC
3	Kalshi event-contract market data	prediction	Kalshi Trade API v2 `api.elections.kalshi.com/trade-api/v2` markets/trades/tickers (S43 rate-limited this census; alphalens probed)	contract/tick	2021→	poll	REST (key optional for reads)	free-read (trading needs account)	CFTC-regulated settlement = clean resolution truth; our IP got 429 — polite-poll cadence mandatory	W1	verify	SURVEY-EVIDENCE
4	Manifold Markets play-money forecasts	prediction	Open API `api.manifold.markets` + bets + positions	market/user forecast	2021→	near-realtime	REST	free (open API)	play-money ≠ real odds — calibration signal only; Brier scoring needs resolution labels	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	Metaculus quantitative predictions	prediction	API2 `api2.metaculus.com` questions + community forecasts	question/forecast time-series	2015?→now (verify earliest)	daily-ish	REST (API key pending)	free w/ key	community vs author probabilities differ — store both; closed questions = resolution set	W2	verify	UNVERIFIED-CANDIDATE
6	PredictIt archive (retired)	prediction (archive)	predictit.org market files + web.archive.org CDX of market pages	market/day	2014?→retired (verify exact end)	n/a (frozen)	S3-archive walk	free (archive content)	retired venue — historical-only; fork (Manifold/Polymarket) continuity is NOT stitchable	W3	backfill	UNVERIFIED-CANDIDATE
7	India-relevant prediction-market odds	prediction	cross-venue roll-up from rows 1–5 filtered to India events (elections, RBI/rate, cricket World Cup, monsoon)	event/day	venue-limited	poll	derived view over rows 1–5	free (derived)	venue event-ID mapping is manual — keep venue-native IDs; odds ≠ realized probability (calibrate vs outcomes)	W2	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	NSE 52-week highs/lows + price bands	market structure	nseindia.com all-reports band files (alphalens in_004)	daily per scrip	regime-dated; own-capture forward	EOD	file-drop	free	band-change events PIT per session; T2T vs normal series tags	W1	build-parser	SURVEY-EVIDENCE
2	NSE SLBS securities-lending volumes	flow	NSE SLBS daily reports (alphalens in_012)	daily aggregate + per scrip	depth to sample	EOD	file-drop	free	borrower/lender category buckets; haircut changes break series	W1	verify	SURVEY-EVIDENCE
3	NSE SME (Emerge) + BSE SME platform stats	universe	exchange SME report pages (alphalens in_014)	daily/monthly	SME-regime 2012→ approx	EOD	file-drop	free	SME ≠ main-board liquidity — never mix in main-board breadth stats	W2	verify	SURVEY-EVIDENCE
4	NSE index reconsti-tution press releases	corporate event	nseindia.com index press page (alphalens in_016)	event rows	index-history-long	event-driven	file-drop	free	announcement→effective lag = tradeable window; store both dates	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	InvIT/REIT unit disclosures + metrics	corporate event	NSE/BSE InvIT-REIT sections (alphalens in_017)	per-filing + daily unit data	regime 2014→ approx (InvIT) / 2007→ (REIT)	EOD/batch	file-drop	free	distribution-yield definitions differ per trust — store trust methodology	W1	verify	SURVEY-EVIDENCE
6	NSE currency-derivative (CD) bhavcopy	derivative price	nseindia.com CD reports (alphalens in_018)	daily bar per contract	CD-regime 2008?→ to verify	EOD	file-drop	free	USD/INR dominance; expiry-calendar changes across regimes	W1	backfill	SURVEY-EVIDENCE
7	NSE commodity segment (NSE-COM) reports	derivative price	nseindia.com commodity reports (alphalens in_019)	daily bar	segment launch→	EOD	file-drop	free	thin segment — zero-volume days are real, not gaps	W2	verify	SURVEY-EVIDENCE
8	SGB secondary-market premium/discount	flow/valuation	exchange SGB series + RBI issue data (alphalens in_021)	daily per series	SGB era 2015→	EOD	file-drop	free	premium decays to gold-spot near maturity — model per series; series-ID mapping required	W2	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
9	Sovereign green-bond auctions	rates/policy	RBI auction results green tranche (alphalens in_022)	auction	green-bond era 2023→	auction-day	file-drop	free	twin-auction pricing vs conventional G-sec = spread signal	W2	verify	SURVEY-EVIDENCE
10	Combined-exchange OI report (NSE+BSE)	flow	exchange combined OI disclosure (alphalens in_031)	daily	recent regime	EOD	file-drop	free	combined vs single-exchange OI must not be stitched without regime flag	W1	verify	SURVEY-EVIDENCE
11	NSDL fortnightly FPI sector-wise holdings	flow	nsdl.com FPI reports (alphalens in_032)	fortnight × sector	FPI-regime 2019?→ (post-FEMA revamp)	fortnightly	file-drop	free	FPI→FPI-category renumbering 2019 — regime tag mandatory; sector taxonomy vintages	W1	backfill	SURVEY-EVIDENCE
12	Per-stock FII headroom disclosure	risk	exchange FII-headroom files (alphalens in_033)	daily per scrip	regime-dated	EOD	file-drop	free	headroom hits 0 = binding constraint event; MWPL cross-check	W1	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
13	Demat account additions (NSDL+CDSL)	flow/retail	nsdl + cdsl monthly stats (alphalens in_034)	monthly per depository	demat-boom 2015?→	monthly	file-drop	free	monthly net adds ≠ active accounts; KYC-rule changes shift series (Jan-2023 SHCIL halt documented)	W0	backfill	SURVEY-EVIDENCE
14	Broker-wise active clients (monthly mandate)	flow/retail	SEBI monthly broker-client disclosure (alphalens in_035)	monthly × broker	mandate-era 2018?→	monthly	file-drop	free	per-broker dedupe across NSE+BSE registrations; round-trip caps distort	W1	backfill	SURVEY-EVIDENCE
15	MF folio additions (equity vs debt split)	flow	AMFI monthly fact-sheet+statistical report (alphalens in_036)	monthly × category	depth to sample	monthly	file-drop	free	folio ≠ AUM; category reclassifications mid-series	W1	backfill	SURVEY-EVIDENCE
16	ETF AUM/flows + iNAV premium	flow	exchange ETF pages + AMFI (alphalens in_039)	daily iNAV; monthly AUM	ETF-era 2008?→	daily/monthly	file-drop	free	iNAV = indicative, not executable; premium spikes at close are artifacts	W2	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
17	PFRDA/NPS pension AUM + sub-scribers	flow	prda.gov.in + nsdl cra reports (al-phalens in_041)	monthly	NPS era 2004?→	monthly	file-drop	free	subscriber counts include exited accounts; scheme-tier splits	W1	backfill	SURVEY-EVIDENCE
18	IRDAI insurer premium + persis-tence	flow	irdai.gov.in annual + monthly state-ments (al-phalens in_042)	monthly/yearly × insurer	irdai digital era	monthly	file-drop	free	premium ≠ profit; renewal-persistence defini-tions per regulator vintage	W1	backfill	SURVEY-EVIDENCE
19	SEBI PIT insider-trading disclo-sures (SAST)	ownership	sebi.gov.in PIT filings (al-phalens in_046)	per-filing	SAST-regime 2002?→	T+2 publish	file-drop	free	trading-window rule changes; demat-pledge encum-bered sales misread as exits	W1	backfill	SURVEY-EVIDENCE
20	Credit-rating actions + rationale PDFs	corporate event	CARE/CRISIL/ICRA/SBIL + BRWR pages (al-phalens in_047)	per-action	rating-agency digital era	T+0-ish	file-drop + PDF	free	rating floor changes (SEBI scale revamps 2010s) — never stitch across scales; outlook vs action	W1	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
21	Auditor changes / resignations / qualified opinions	governance	MCA + exchange filings (alphalens in_048)	per-event	depth to sample	T+1	file-drop	free	qualified-opinion flag lives in financial-assertion PDFs — requires text parse; churn = distress signal	W1	build-parser	SURVEY-EVIDENCE
22	KMP exits (CEO/CFO/CS changes)	governance	exchange announcements + MCA DIN filings (alphalens in_049)	per-event	digital era	T+0	file-drop	free	name→DIN entity resolution; interim vs permanent appointments	W1	build-parser	SURVEY-EVIDENCE
23	ESOP grant/vesting filings	ownership	exchange filings + annual-report annexures (alphalens in_050)	per-event	ESOP-regime	T+1	file-drop	free	grant ≠ vest ≠ exercise — model 3 event types; dilution math per series	W2	build-parser	SURVEY-EVIDENCE
24	Buyback tenders + results	corporate event	exchange buyback pages (alphalens in_051)	per-buyback + tender window	buyback-regime 2000s→	event	file-drop	free	tender-vs-market route; acceptance-ratio math; buyback-entitlement date joins	W1	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
25	Reg-23 related-party transaction disclosures	governance	annual reports + quarterly RPT filings (alphalens in_052)	per-filing	LODR-regime 2015?→	quarterly	file-drop + PDF	free	RPT materiality thresholds revised 2021+ — threshold regime tags; arm's-length flag text	W2	build-parser	SURVEY-EVIDENCE
26	SEBI enforcement orders database	regulatory	sebi.gov.in orders page (alphalens in_056)	per-order	SEBI orders digital era 2000s→	weekly-ish	file-drop + PDF	free	order→entity resolution needed; appeal outcomes post-order	W1	build-parser	SURVEY-EVIDENCE
27	NCLT/CIRP insolvency process states	legal/corporate	nclt.gov.in orders + ibbi.org.in (alphalens in_058)	per-case state	IBC era 2016→	weekly	file-drop + PDF	free	CIRP stage machine (admission→CoC→plan→liquidation) — store state transitions as events; order PDFs are truth	W1	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
28	eCourts judgments entity-matched	legal	ecourts.gov.in + High Court portals (alphalens in_059)	per-case	eCourts digital era	weekly-ish	portal self-capture	free (portal ToS check)	case-number reuse across courts; party-name entity resolution is the hard part; scrape-politeness	W2	build-parser	SURVEY-EVIDENCE
29	RBI weekly credit/deposit growth	macro	rbi.org.in weekly statistical supplement (alphalens in_065)	weekly	deep weekly archive	weekly	file-drop	free	fortnightly-vs-weekly rhythm; festival-shifted reporting weeks	W1	backfill	SURVEY-EVIDENCE
30	CCIL yields/volumes + FBIL benchmarks	rates	ccil.in + fbil.org.in (alphalens in_068)	daily benchmarks	CCIL digital era	EOD	file-drop	free	FBIL fix methodology revisions (0900/1430/1600 cuts); GCA vs settlement volumes	W1	connect	SURVEY-EVIDENCE
31	G-sec auction results + curve	rates	rbi.org.in auctions (alphalens in_069)	per-auction	auction archive deep	auction-day	file-drop	free	cut-off vs notified amount; non-competitive bids; auction calendar shifts	W1	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
32	CGA central accounts (capex/receipts)	macro/fiscal	cgaindia.nic.in monthly state-ments (al-phalens in_070)	monthly	FY-digital era	monthly	file-drop + PDF	free	FY vs calendar tagging; revised-vs-provisional flags per release	W1	build-parser	SURVEY-EVIDENCE
33	CBDT direct-tax collec-tions	macro/fiscal	incometax.gov.in + CBDT releases (al-phalens in_071)	monthly × tax head	digital era	monthly	file-drop	free	gross vs net; refunds-netted series; FY pacing noise	W1	backfill	SURVEY-EVIDENCE
34	GST col-lections (state × month)	macro/fiscal	gstcouncil.gov.in + cbic releases (al-phalens in_072)	monthly × state	GST era 2017-07→	monthly	file-drop	free	gross-domestic vs IGST-import splits; rate-rationalization break-points (Sep-2019)	W1	backfill	SURVEY-EVIDENCE
35	Economic Survey corpus (annual PDF)	macro/doc	indiabudget.gov.in (al-phalens in_073)	annual docu-ment	1950s→ (digital copies ~2010→)	annual	file-drop	free (Govt of India)	one-shot per year; chapter text revisions between in-terim/final prints	W1	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
36	Union Budget document corpus	macro/doc	indiabudget.gov.in full budget + annexures (alphalens in_074)	annual document set	digital era 2010s→ (older via archive)	annual	file-drop	free (Govt of India)	interim vs full budget per FY; amendment finance bills; speech-vs-text deltas	W1	backfill	SURVEY-EVIDENCE
37	State budget documents	macro/doc	state finance portals (alphalens in_075)	annual × state	2010s→ varies per state	annual	portal self-capture	free (state gov)	FRBM classification changes per state; state fiscal-year tagging	W2	build-parser	SURVEY-EVIDENCE
38	RBI FSR + Trend & Progress of Banking	macro/doc	rbi.org.in publications (alphalens in_076)	half-yearly / annual	FSR digital era	half-yearly	file-drop	free	financial-stability narrative = regime label source; bank-sample changes (NPA cycles)	W2	backfill	SURVEY-EVIDENCE
39	CAG audit reports (union + state)	governance	cag.gov.in (alphalens in_077)	per-report	CAG digital era	periodic	file-drop + PDF	free (CAG)	paragraph-level extraction needed; audit→allegation lag	W2	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
40	NITI Aayog indices (SDG, MPI)	macro	niti.gov.in (al-phalens in_078)	annual × state/district	index-vintage only	annual	file-drop	free	index method-ology rebasing across editions — store per-edition, never stitch	W2	verify	SURVEY-EVIDENCE
41	PRS Leg-islative Research bill tracker	policy	prsindia.org + sansad.in (al-phalens in_079)	per-bill stage	PRS digital era	near-realtime	REST-ish + file-drop	free (PRS attribu-tion)	bill num-bering resets per session; standing-committee stage dates	W1	connect	SURVEY-EVIDENCE
42	SEBI con-sultation papers + final circulars	regulatory	sebi.gov.in consult pages (al-phalens in_080)	per-document	digital era	weekly	file-drop	free	paper→circular pairing = policy-reaction event pair; comment-window close date	W2	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
43	DPIIT startup recogni-tions + FDI inflows (sector × country)	macro/policy	dpiit.gov.in + rbi FDI reports (al-phalens in_081, in_082)	monthly FDI × sector-country; recogni-tion stock	FDI digital era	monthly	file-drop + PDF	free	press-note policy changes (2020 CASO carve-outs) = regime tags; route vs guess counts	W1	build-parser	SURVEY-EVIDENCE
44	EPFO payroll additions	macro/labor	epfindia.gov.in monthly press releases (al-phalens in_083)	monthly	EPFO digital era 2018?→	monthly	file-drop	free	establishment-count growth ≠ new jobs; re-subscription vs first-time split (2022+ split pub-lished)	W1	backfill	SURVEY-EVIDENCE
45	Naukri JobSpeak hiring index	macro/labor	naukri.com JobSpeak monthly PDF (al-phalens in_084)	monthly × sec-tor/city	JobSpeak era 2008?→	monthly	file-drop	free (pub-lisher terms)	panel-based index from one job-board — non-census; rebasing events; cross-check EPFO/PLFS	W1	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	PLFS employment-unemployment survey	macro/labor	mospi.gov.in PLFS annual + usual-status CWS (alphalens in_085)	annual × state/sector	PLFS 2017-18→ (NSS 68th superseded)	annual	file-drop	free (MoSPI)	PLFS ≠ NSS-68th basis — never splice; CWS vs UPS definitions	W1	backfill	SURVEY-EVIDENCE
47	NPCI UPI volumes + bank-wise share	macro/payments	npci.org.in transactions page (alphalens in_086)	monthly (+bank share)	UPI era 2016→	monthly (live counter volatile)	portal self-capture	free (NPCI terms)	counter values revised; bank-share restatements; value vs volume	W1	build-parser	UNVERIFIED-CANDIDATE
48	Grid-India 15-min all-India demand CSV	power/nowcast	grid-india.in reports (alphalens in_087; V1-verified 15-min CSV exists)	15-min × region	digital era, depth to sample	15-min	file-drop	free	regional-balance vs all-India; holiday dips real; despatch vs consumption	W0	backfill	SURVEY-EVIDENCE
49	PFC discom finances annual dataset	power	pfc.org.in DISCOM reports (alphalens in_090)	annual × discom	DISCOM dataset era	annual	file-drop	free (PFC)	AT&C loss definitions revised; discom-ID churn; state-vs-utility mapping	W2	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
50	IMD gridded rainfall 1901+ + AWS stations	weather	mausam.imd.gov.in + metgrid (al-phalens in_093)	grid/daily; sta-tion/hourly	gridded 1901→	days	file-drop	free (IMD)	gridded-product version shifts; AWS station-network churn; QA-flag columns	W1	backfill	SURVEY-EVIDENCE
51	CPCB AQI per city + GRAP stages	environment	cpcb.nic.in AQI bulletins (al-phalens in_095, in_096)	city/hourly + stage events	AQI era 2014→	hours	portal self-capture + file-drop	free	station additions mid-series; GRAP stage = event calendar for Delhi pollution trades	W1	build-parser	SURVEY-EVIDENCE
52	CWC reservoir storage (150+ reser-voirs)	water/power	cwc.gov.in weekly reservoir bulletins (al-phalens in_097)	reservoir/weekly	bulletin era	weekly	file-drop	free	live vs avg storage; monsoon-season tagging; reservoir set churn	W1	build-parser	SURVEY-EVIDENCE
53	Flood forecasts + cyclone warnings	weather/risk	CWC flood forecasts + IMD cyclone bulletins (al-phalens in_098, in_099)	event/alert	bulletin era	event-driven	portal self-capture	free	alert-grade escala-tion events; forecast-track revisions store first-seen	W2	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
54	Soil-moisture satellites + CGWB groundwater	water	SMAP/ESA CCI + cgwb.gov.in (alphalens in_101, in_102, in_160)	grid/3-day (sat); district/year (GW)	SMAP 2015→; GW rounds	days–annual	file-drop	free (NASA/ESA/GOVT)	SMOS vs SMAP product IDs; groundwater round cadence irregular; irrigation-season tagging	W2	build-parser	SURVEY-EVIDENCE
55	AGMARKNET mandi prices + arrivals	agri/nowcast	agmarknet.gov.in daily 3000+ mandis (alphalens in_103; parser-sprint target #1)	commodity × mandi/day	digital era, depth to sample	daily	file-drop + scrape	free (Agmarknet)	arrival = volume not price; quality-variant grades; mandi-ID churn; duplicate rows across reports	W1	build-parser	SURVEY-EVIDENCE
56	e-NAM integrated market trades	agri	enam.gov.in (alphalens in_104)	trade × mandi/day	e-NAM era 2016→	daily	portal self-capture	free	e-NAM subset of mandis — never treat as national; traded-vs-offered	W2	verify	SURVEY-EVIDENCE
57	Weekly sowing progress (state × crop)	agri/nowcast	agri.nic.in + ministry dashboards (alphalens in_105)	state × crop/weekly	sowing season era	weekly (kharif/rabi)	file-drop	free	sowing ≠ harvest; area-revision mid-season; kharif/rabi calendars	W1	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
58	MSP notifications (kharif/rabi)	agri/policy	agri.gov.in MSP pages (alphalens in_106)	crop × season	deep archive	annual (event)	file-drop	free	announcement-date PIT; effective-vs-announced pricing; A2+50% formula shifts	W1	backfill	SURVEY-EVIDENCE
59	Fertilizer sales (DoF monthly)	agri	fertilizer.gov.in (alphalens in_107)	month × state × product	digital era	monthly	file-drop	free	pant-season year (Oct–Sep) ≠ calendar; subsidy-rate regime changes	W2	build-parser	SURVEY-EVIDENCE
60	FCI food-grain stocks (central pool)	agri	fci.gov.in stock reports (alphalens in_108)	weekly/monthly	digital era	weekly	file-drop	free	buffer norms revised 2013+; season-tagged (Oct onset); state-wise split	W1	backfill	SURVEY-EVIDENCE
61	Daily retail prices of 22 essentials □	agri/nowcast	consumer affairs price monitoring (alphalens in_114; parser-proof committed)	commodity × city/day	digital era	daily	file-drop	free	parser already proven in alphalens sprint — port their extractor; city-set churn	W1	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
62	CEA daily coal stock per thermal plant □	energy/nowcast	cea.nic.in daily coal-stock reports (alphalens in_115; PDF spike proven)	plant/day	spike-era 2010s→	daily	file-drop (PDF/xlsx)	free (CEA)	norm-days math (15/20/25 by plant class); PDF table extractor needed; plant-ID mapping	W1	build-parser	SURVEY-EVIDENCE
63	PPAC fuel consumption + crude imports by source	energy/trade	ppac.gov.in monthly (alphalens in_118, in_131)	month × product × source-country	PPAC digital era	monthly	file-drop	free	source-country switch events (Russia flows 2022+) are the signal; SKU definitions	W1	build-parser	SURVEY-EVIDENCE
64	DGH crude/gas field production	energy	dgh.gov.in monthly (alphalens in_120)	field/month	digital era	monthly	file-drop	free	field-ID mapping; ODL vs gross; block transfers break continuity	W2	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	PNGRB city-gas network + CNG/PNG connections	energy	pngrb.gov.in (alphalens in_121)	network/month	PNGRB digital era	monthly	file-drop	free	network-extension phase tagging (2018–2030 phases); connection vs consumption	W2	verify	SURVEY-EVIDENCE
66	Toll revenue plaza × concessionaire □	logistics/nowcast	ihmcl.gov.in NHAI toll disclosures (alphalens in_122; format-proven)	plaza × month	concession-era 2010s?→	monthly	file-drop (PDF/xlsx)	free	HAM vs BOT concession mix; COVID-2020 toll waiver = documented gap; plaza additions mid-series	W1	build-parser	SURVEY-EVIDENCE
67	Major-port cargo traffic (IPA daily)	logistics/nowcast	indianports.gov.in IPA daily (alphalens in_123; table-scrape proven)	port/day	digital era	daily	scrape + file-drop	free	cargo ≠ container TEU — keep separate series; monsoon seasonality	W1	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
68	DGCA airline traffic by carrier	logistics/nowcast	dgca.gov.in monthly PDFs (alphalens in_125)	carrier × month	digital era	monthly	file-drop (PDF)	free (DGCA)	PDF extractor templates; carrier-code churn (Jet 2019 exit = structural break); ASM vs RPK vs passengers	W1	build-parser	SURVEY-EVIDENCE
69	AAI airport passengers + cargo	logistics/nowcast	aai.aero monthly (alphalens in_126)	airport × month	digital era	monthly	file-drop	free	airport reclassification (international status changes); COVID gap real	W2	build-parser	SURVEY-EVIDENCE
70	Railway freight loading	logistics/nowcast	indianrailways monthly stats (alphalens in_127)	commodity/zone × month	digital era	monthly	file-drop	free	freight-rate revamp 2015+; zone reorganizations; container-vs-break-bulk splits	W1	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
71	VAHAN vehicle registrations + EV dashboard	logistics/now-cast	parivahan.gov.in VAHAN (alphalens in_128)	category × state × month/week	VAHAN digital era	weekly/monthly	portal self-capture	free (VAHAN ToS check)	registration ≠ sales (dealer vs RTO lag); BS-IV→BS-VI switch Apr-2020 = break; EV split era 2019→	W1	build-parser	SURVEY-EVIDENCE
72	Metro ridership (7 metros)	logistics/now-cast	metro rail corporation stats (alphalens in_129)	metro × month	line-openings era	monthly	portal self-capture	free	new-lines create fake growth — per-line tagging; COVID gap real	W2	build-parser	SURVEY-EVIDENCE
73	US FDA import refusals + 483s (Indian plants)	pharma-regulatory	FDA DUINS/RII + import-alert queries (alphalens in_132; V1 query-builder proven)	per-event	FDA digital era	daily-ish	REST (openFDA-class) + file-drop	free (US gov)	firm-name entity resolution (Indian plant names in romanization/transliteration); 483 ≠ refusal — separate series	W1	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
74	DGTR anti-dumping / safe-guard cases	trade-regulatory	dgtr.gov.in case search (al-phalens in_133)	per-case	DGTR digital era	weekly-ish	portal self-capture	free	case→product HS mapping; duty-levy vs case-filing events; sunset reviews	W2	build-parser	UNVERIFIED-CANDIDATE
75	RoDTEP/RoSCTL incentive rates	trade/policy	dgft.gov.in RoDTEP sched-ules (al-phalens in_134)	HS × rate schedule	RoDTEP era 2021→	periodic	file-drop	free	rate sched-ules revised per review cycle — store schedule-vintage; scrip vs cash modes	W2	verify	SURVEY-EVIDENCE
76	BIS QCO import-restriction tracker	trade/regulatory	bis.gov.in + dgft QCO noti-fications (al-phalens in_135)	per-QCO × HS	QCO wave 2020s→	event	file-drop	free	notification→effective-date lag; scope creep per sector	W2	build-parser	SURVEY-EVIDENCE
77	Export councils monthly (GJEPC/Pharmexcil/EEPC/APEDA…)	trade/nowcast	council monthly releases (al-phalens in_136)	council × month	digital era	monthly	portal self-capture	free	per-council HTML/PDF hetero-geneity — 6 parser families; gem/jewellery volatility	W2	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
78	RBI services exports (quarterly)	macro/trade	rbi.org.in BoP services (alphalens in_137)	quarterly	BoP digital era	quarterly	file-drop	free	BoP revisions (2013 India-usage vs US-usage basis note); software vs other-services	W2	verify	SURVEY-EVIDENCE
79	India earnings-call transcripts (LODR-mandated)	news/corpus	exchange audio + LODR transcripts (alphalens in_141)	per-call	LODR era 2018?→	hours	file-drop + self-transcribe	free / self-derived	LODR mandates calls — audio archived by exchanges; ASR quality per language; Q&A segment splits	W1	build-parser	SURVEY-EVIDENCE
80	Investor presentations + annual-report library	corpus	exchange corp-interview pages + company IR (alphalens in_142, in_143)	per-doc	digital era	T+1	file-drop + PDF	free (company filings)	presentation decks are unaudited — flag vs financials; version churn post-IR edits	W1	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
81	Board-meeting / results calendar	corporate event	exchange board-meeting pages (alphalens in_145)	event rows	digital era	near-realtime	file-drop	free	outcome (approved/postponed) arrives after date — store both; extended-date events	W0	connect	SURVEY-EVIDENCE
82	Reddit investing-sub senti-ment	attention	PRAW pull r/IndianStreetBets + r/IndiaInvestments (alphalens in_149)	post/thread hourly	sub-era 2015?→	hourly	REST (PRAW; API terms)	free (Reddit API terms; rate-capped)	bot/spam share; API pricing shock 2023 — terms can change; PII retention policy needed	W2	connect	SURVEY-EVIDENCE
83	YouTube earnings-call video attention	attention	YouTube Data API v3 (alphalens in_154)	video/day	API-era	daily	REST (key; quota-capped)	free (API terms; quota)	view counts re-vised/removed; comment API terms; quota 10k units/key/day	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
84	TROPOMI NO2 / methane plumes (refineries, ports)	satellite/energy	Sentinel-5P via CDSE/Tropomi (alphalens in_158)	grid/3-day	2018→ (S5P launch)	days	STAC / file-drop	free (Copernicus)	overpass-filter (cloud); reprocessing-version pins; refinery-activity proxy only	W2	build-parser	SURVEY-EVIDENCE
85	Reservoir / water-body extent change	satellite/water	JRC Global Surface Water + Sentinel-1 (alphalens in_161)	grid/monthly	JRC 1984→	monthly	file-drop	free (EC JRC)	JRC extent vs CWC gauge = two different truths — keep separate; monsoon-seasonal	W2	build-parser	SURVEY-EVIDENCE
86	Seismic events, India region	macro/risk	USGS + NCS (alphalens in_164)	event	deep archive	near-realtime	REST	free (USGS public-domain)	magnitude/depth quality flags; aftershock clustering; NCS-vs-USGS dedupe	W2	connect	SURVEY-EVIDENCE
87	NOTAMs + GNSS-jamming advisories (aviation risk)	risk/aviation	notams.aim.gov.in + NAV CANADA-class mirrors (alphalens in_166)	advisory	digital era	hours	portal self-capture	free (ICAO terms)	NOTAM decode formats; jamming-event dedupe across issuers	W3	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
88	India news corpus + ticker sentiment (6-year HF set) □	news/corpus	HuggingFace `mveen3/Sixty_Ticker_Indian_Stock_Market_News_and_Ticker` — 4.2 GB, MIT ([HF-s43])	article × ticker	~6y window (verify exact span on pull)	daily-batch	HF datasets clone (parquet)	MIT (dataset card; source-article ToS sits up-stream; dedupe wire copies; ticker-tagging accuracy unknown — sample-audit before training; PIT = article publish times-tamp (first-seen)	source-article copyright still third-party — internal use, no redistri-bution)	W0	backfill	VERIFIED-UPSTREAM-DOC
89	GDELT news tone 15-min (cross-check rail)	news	GDELT DOC API (al-phalens in_152; global rows exist in TODO)	15-min tone	GDELT 2.0 2015→	15-min	REST	free (GDELT terms)	tone is model-derived; language-filter errors; cross-ref to existing GDELT row in sheet 09 — dedupe on ingest	W1	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	Jhanvi NSE/BSE 1-minute bars	price/trades	HuggingFace `Jhanvi-29112000/indian-stock-market-minute-data` — 10.5 GB snappy parquet (125 GB raw), MIT ([HF-jhanvi])	1-min bar × symbol	2022 → 2026-10 (rolling; verify exact start on pull)	daily-batch (mirror)	HF datasets clone → DuckDB/Parquet store	MIT (dataset card; underlying exchange data ToS still applies to redistribution — internal use)	1-min aggregation convention (bar close vs tick) unknown — sample vs TrueData/Kite on 1 overlapping day; symbol-set churn; delisted coverage unknown	W0	backfill	VERIFIED-UPSTREAM-DOC
2	Jhanvi NSE/BSE daily OHLCV	price/trades	HuggingFace same-repo daily partition — 2000 → 2026 ([HF-jhanvi])	daily bar × symbol	~2000 → 2026	daily-batch	HF clone → store	MIT	survivorship unknown (delisted names?); adjustment convention unknown (raw vs split-adjusted) — verify vs NSE bhavcopy row 01 on 5 sample days before use as spine	W0	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	Vishnu Indian market daily OHLCV	price/trades	HuggingFace `vishnun0027/indian-market-historical-ohlcv` — 290.1 MB, MIT ([HF-vishnu])	daily bar × symbol	multi-year (verify exact span on pull)	daily-batch	HF clone → store	MIT	smaller daily universe — cross-check coverage vs row 2 (union both, prefer official bhavcopy where overlapping)	W0	backfill	VERIFIED-UPSTREAM-DOC
4	India index-options 1-minute	derivative price	HuggingFace `thetrademarkk/india-index-options-1m` — 4 GB, CC-BY-NC 4.0 ([HF-options])	1-min bar × contract	rolling window (verify start on pull)	daily-batch	HF clone → store	**CC-BY-NC — internal-use only, no redistribution/commercial derivative display**	NC license conflicts with any future commercial product — quarantine rows by license-tier from day 1; IV/OI presence unknown; contract-ID mapping to exchange tokens needed	W0	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	Hyperliquid perpetuals full history	crypto derivative price	HuggingFace `Chainticks/hyperliquid-data` — 89.5 GB, CC-BY 4.0 ([HF-hyperliquid])	tick/1-minute bars; per-perp	Hyperliquid era 2023→ (verify exact start)	daily-batch	HF clone → object store	CC-BY 4.0 (attribution; redistribution OK with credit)	funding/OI columns presence to verify; cross-check vs in-house Hyperliquid node (latency UNVERIFIED per S38) — dataset is the backfill, node is the live rail	W1	backfill	VERIFIED-UPSTREAM-DOC
6	XAUUSD tick data	fx/commodity ticks	HuggingFace `CarlosSilvaJr/xauusd-ticks` — 2.65 GB, CC-BY 4.0 ([HF-xauusd])	tick × 21 pairs	multi-year (verify span)	daily-batch	HF clone → store	CC-BY 4.0	broker-feed provenance (not exchange tape) — spread/bid-ask conventions per feed; DST/timezone alignment; cross-check vs Dukascopy on 1 day	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
7	Shadowbroker OSINT feed spine	reference/pattern donor	GitHub BigBody-Cobain/Shadowbroker — AGPL-3.0, 11,399★, pushed 2026-10-27 ([GH-shadowbroker])	repo (FastAPI + Next.js, 60+ live OSINT feeds cat-alogued)	n/a (code + feed registry)	n/a	**pattern donor**: polling spine (60s fast / 5min slow cycles, gzip/ETag/dedup, feed registry = candi-date source list for sheet 09/10/14	**AGPL-3.0 — internal-use code only; any network deploy-ment of derived code must stay internal (no SaaS)**	feed registry freshness must be re-probed per feed (catalog ≠ live); AGPL derivative-use boundary = O11 checklist item	W2	license-review	VERIFIED-UPSTREAM-DOC
8	worldmonitor country-metrics + freshness tracker	reference/pattern donor	GitHub koala73/worldmonitor — AGPL-3.0, 89.7k★ ([GH-worldmonitor])	repo (feed-freshness tracker, Welford anomaly detector, MCP server)	n/a	n/a	**pattern donor**: source-health dash-board design (fresh/stale/error), anomaly detection for ingest QA, macro-regime context factor	**AGPL-3.0 — internal-use only**	same AGPL boundary as row 7; metric defini-tions per country need own valida-tion	W2	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
9	OpenBB integration patterns	reference/pattern	GitHub `openbq-org/OpenBB` — NOASSERTION, 74,821★, org renamed, pushed 2026-10-24 ([GH-openbb])	repo (Open Data Platform; provider-adapter architecture)	n/a	n/a	**pattern donor**: adapter-interface design (Vol-1 §3 one-adapter philosophy), provider fallback chains	**NOASSERTION license — treat as proprietary until legal review; no code copy**	License unresolved = no code reuse; design-pattern study only	W2	license-review	VERIFIED-UPSTREAM-DOC
10	FinanceDatabase asset-class taxonomy	reference/data	GitHub `JerBouma/FinanceDatabase` — MIT, 9.5k★ ([GH-financedb])	repo (structured ticker/asset-class/sector metadata, JSON)	continuous (repo current)	weekly-ish	clone → reference table (asset classification spine)	MIT (attribution)	taxonomy is editorial (not exchange-official) — use for classification joins, never as exchange universe truth	W1	backfill	VERIFIED-UPSTREAM-DOC
11	gods-eye-view county geoJSON	reference/data	GitHub `bilawalsidhu/gods-eye-view` — 60.8k★ ([GH-godseye])	repo (US county boundary geoJSON set)	static snapshot	n/a	clone → geo reference table	per-repo (verify license file on clone)	US-only geo; vintage must be pinned; overlaps sheet 09 WorldPop/OSM geo stack — keep OSM as spine	W2	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
12	OpenTerminal	reference/pattern	GitHub `ErTas-selli/OpenTerminal` — MIT, 2.1k★ ([GH-openterminal])	repo (terminal-style data UI)	n/a	n/a	**pattern donor**: terminal UX + data-fetch patterns	MIT	UI reference only — no data claims	W3	verify	VERIFIED-UPSTREAM-DOC
13	OracleX	reference/pattern	GitHub `Yigtwxx/OracleX` — MIT, 2★ ([GH-oraclex])	repo	n/a	n/a	low-star repo — evaluate for ideas only before adopting anything	MIT	tiny repo — credibility check before any reliance	W3	verify	VERIFIED-UPSTREAM-DOC
14	evans-0/finance	reference/pattern	GitHub `evans-0/finance` — MIT, 2★ ([GH-evans])	repo	n/a	n/a	low-star repo — same low-priority screen as row 13	MIT	tiny repo — credibility check	W3	verify	VERIFIED-UPSTREAM-DOC
15	osint-worldview	reference/pattern	GitHub `amanimran786/osint-worldview` — 69★ ([GH-osintwv])	repo (geocoded places + OSINT map layers)	n/a	n/a	candidate geo-OSINT layer for sheet 09/14 — probe layer sources before adopting	per-repo (verify on clone)	low-star; layer prove-nance unknown — audit each layer	W3	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
16	OpenSpaceData (Copernicus EO access)	reference/pattern	GitHub org `OpenSpaceData` — Hippocratic-2.1, dormant since 2021 ([GH-openspace])	repo (ESA Copernicus EO access tooling)	n/a	n/a	**dormant 2021** — read-only reference; sheet 09 CDSE/sentinel rails already cover this need	Hippocratic-2.1 (ethical-use license — restrictive; internal reference only)	dormant = unmaintained against CDSE API changes; do not build on it	W3	verify	VERIFIED-UPSTREAM-DOC
17	QGIS Dashboard plugin	reference/tool	GitHub `isaacenage/QGIS-Plugins` — GPL-3.0, v0.2 ([GH-qgis])	repo (QGIS visualization plugin)	n/a	n/a	**visualizer tool, NOT a data source** — optional local map-QA tool for AOI sanity checks	GPL-3.0 (desktop tool use fine; no redistribution of modified plugin)	v0.2 = immature; tool-only row — zero data claims	W3	verify	VERIFIED-UPSTREAM-DOC
18	FinceptTerminal	reference/pattern	GitHub `Fincept-Corporation/FinceptTerminal` — NOASSERTION, 33.2k★ ([GH-fincept])	repo (multi-asset terminal, broad data rails)	n/a	n/a	**pattern donor**: multi-asset rail inventory for coverage-gap cross-check	**NOASSERTION — proprietary-risk; design reference only, no code copy**	License unresolved; audit its rail list vs our catalog for missed sources (not for direct reuse)	W2	license-review	VERIFIED-UPSTREAM-DOC

By domain volume

01 — India equities & F&O (50 rows)

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	NSE equity bhavcopy (cash market OHLCV, all scrips)	price/trades	nseindia.com all-reports → daily CM bhavcopy zip (UDiFF-format since Jul-2024; legacy format before)	daily bar	1994→ (approx; pre-2010 sparse survey claim), daily zips, backfill-able now	EOD	file-drop	free (MVP-replace note: primary free base for INR equity closes)	PIT-complete post-close; dual parser needed across Jul-2024 format break	W0	backfill	SURVEY-EVIDENCE
2	NSE F&O bhav-copy, full chain (every contract OHLC + OI)	derivative price	nseindia.com deriva-tives bhavcopy → per-contract daily CSV, month-zips	daily bar per contract	F&O 2000→, month-zips, backfill-able now	EOD	file-drop	free (MVP-replace note)	per-expiry PIT chain; instrument-token churn across expiries needs roll book	W0	backfill	SURVEY-EVIDENCE
3	NSE F&O pre-2010 legacy archive (pre-UDiF-era files, 2000–2009)	derivative price	same month-zip family re 2000s	daily bar	2000→ per-row above; pre-2010 files in legacy columns	EOD	file-drop	free	legacy header/column dialect vs 2024+ UDiFF — build parser isolating eras	W0	build-parser	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
4	NSE index bhavcopy (all published indices daily)	index	nseindia.com all-reports → indices section	daily bar	~2007→ (approx survey claim), backfill-able	EOD	file-drop	free	index value as pub-lished; constituent-count changes create level breaks	W0	connect	SURVEY-EVIDENCE
5	NSE indices official back-calculated series (Nifty 50 1995→, Bank Nifty 2000→, per-index to incep-tion)	index	niftyindices.com historical-data down-loads (CSV per index)	daily bar	back-calc to inception (Nifty 50: 1995), one-shot bulk	EOD batch	REST bulk	free (reg-istration wall)	back-calculated ≠ as-trailed; PIT usable only from series-start of live method-ology	W0	backfill	UNVERIFIED-CANDIDATE
6	NSE delivery quantity & delivery %	market structure	all-reports → MTO delivery file + bhavcopy delivery column	daily	2017→ public archive (survey claim), backfill-able	EOD	file-drop	free	columns empty on T2T/other series — filter series PIT	W0	backfill	SURVEY-EVIDENCE
7	NSE MWPL (market-wide position limits, per F&O scrip)	risk	nseindia.com MWPL disclo-sure page	snapshot (daily-ish)	current values only on page; history requires own con-tinuous capture	delayed	own-capture	free (page) / ⚠-blocked-never-commercial if scraped at scale	take times-tamped snapshot; MWPL changes intraday possible	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
8	NSE F&O ban list (trading-ban scrips)	risk	nseindia.com derivatives AL page + daily file	daily	published continuously; archive depth uncertain → own-capture from today	delayed	own-capture	⚠-blocked-never-commercial (scrape family)	ban status is PIT per exchange-hour only	W0	connect	SURVEY-EVIDENCE
9	NSE bulk deals (≥0.5% turnover disclosures)	flow	nseindia.com live-analysis bulk-deals CSV	daily rows	~2010→ approx daily CSVs, backfillable	EOD	file-drop	free	as-disclosed PIT; counterparty name spelling drifts	W0	backfill	SURVEY-EVIDENCE
10	NSE block deals (SEBI block-window disclosures)	flow	same bulk/block deals disclosure plane	daily rows	disclosure-regime dated (pre/post 2024 rules differ); archive depth to verify	EOD	file-drop	free	two regimes (pre/post new block-window rules) need regime tags	W0	verify	SURVEY-EVIDENCE
11	NSE shareholding patterns (quarterly promoter + list, XLSX per quarter)	ownership	nseindia.com shareholding archive → quarter XLSX zip	quarterly	~2007→ (approx survey claim); XLSX backfiles manual crawl	batch	file-drop	free	revisions possible post-filing; capture-as-filed PIT + revision flag	W0	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
12	Shareholding backfile pre-2013 quarters (older-format filings)	ownership (archive)	NSE quarter-archive + inspection of earliest available quarters	quarterly	earliest-quarter intact depth must be proven by sampling	batch	file-drop	free	format drift across years; promoter-code taxonomy changes	W0	verify	UNVERIFIED-CANDIDATE
13	NSE corporate announcements API (regulation-30 feed JSON)	corporate event	nseindia.com corporate-announcements REST JSON	event rows	API exposes sliding window (~90d); web archive older	near-realtime	REST	⚠-blocked-never-commercial (scrape family; own-use research ok)	as-published timestamps; PDF attachments are the PIT truth	W0	connect	SURVEY-EVIDENCE
14	NSE corporate actions / split-bonus / calendar + new listings & IPO history	corporate event	all-reports corporate-actions + new-listing pages	event rows	~2000→ approx; issue-wise IPO facts patchier pre-2010	EOD	file-drop	free	record-date joins to bhavcopy need as-of logic (pre-close vs post-close sequencing)	W0	backfill	SURVEY-EVIDENCE
15	NSE ASM/GSM surveillance lists (stage/press releases)	regulatory	all-reports surveillance files	daily/batch	ASM regime ~2014→ survey-approx, backfillable	EOD	file-drop	free	stage-changes create overlapping eligibilities — model as events	W0	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
16	NSE pre-open session order snap-shots (09:00–09:08 window)	market structure	nseindia.com pre-open pages/pages CSV	intraday snap-shots	current sessions only; own-capture builds archive	delayed	own-capture	free / ⚠-blocked-never-commercial at scale	snapshot sequence must be stored raw; no vendor restate-ment	W0	connect	SURVEY-EVIDENCE
17	India VIX official history	vol index	NSE India-VIX historical CSV + Upstox V3 instru-ment key `NSE_INDEX|India VIX` (verified route ^9)	daily	2007→ (survey claim), backfill-able	EOD	REST bulk	free	methodology rebase across years (see NSE circulars before stitching)	W0	backfill	SURVEY-EVIDENCE
18	NSE live option chain (minute-poll own capture, all strikes)	derivative analytics	nseindia.com option-chain JSON polled	intraday	forward-only from capture date	realtime	own-capture	⚠-blocked-never-commercial (NSE scraping terms; internal research use only)	store raw poll dumps; NSE re-strikes/updates chain con-stantly	W0	build-parser	SURVEY-EVIDENCE
19	NSE stock-wise historical data product (paid per-symbol down-loads)	price/trades	NSE data-products portal, stock-wise historical down-loads	intraday	price: (survey corpus claim) ₹10,000/yr; depth per NSE tariff sheet to verify	EOD	REST bulk	vendor-license (NSE tariff; no redistri-bution)	verify tariff sheet + depth before buy; compare vs True-Data/GDFL overlap	W1	quote	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
20	NSE FII/DII cash+F&O activity & participant-wise OI	flow	all-reports partici-pant/activity files	daily	participant OI ~2002?→ / FII-DII 2001?→ to verify	EOD	file-drop	free	category defini-tions changed over years (FPI/FII, partici-pant buckets)	W0	backfill	SURVEY-EVIDENCE
21	BSE equity bhavcopy (EQ market, all scrips incl. BSE-only names)	price/trades	bseindia.com bhavcopy archive (per-day down-load)	daily bar	~1999→ approx (survey claim), backfill-able	EOD	file-drop	free (MVP-replace note; covers names absent from NSE)	BSE scrip churn differs from NSE; ISIN-key join first	W0	backfill	SURVEY-EVIDENCE
22	BSE an-nounce-ments & corporate-disclosure plane	corporate event	bseindia.com an-nounce-ments XML/API	event rows	long tail pre-2010 spotty; API window sliding	near-realtime	REST	free / ⚠-blocked-never-commercial at scale	BSE-announced items overlap NSE ~90%; dedupe on both	W0	connect	SURVEY-EVIDENCE
23	BSE indices official history (Sensex back-calculated 1979→, index bhavcopy set)	index	bseindia.com index pages / index historical download	daily bar	Sensex claimed back-calc 1979→ (strong free win to verify); other BSE indices later	EOD	REST bulk	free	older decades index-method metadata scarce — note level-break points	W0	verify	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
24	BSE pre-2010 archive (Wayback-era bseindia.com walk-through)	price/trades (archive)	web.archive.org CDX walk of old BSE bhavcopy/report URLs	daily bar	fragments pre-2005; completeness unproven	EOD	S3-archive walk	free (archive content)	fragile URLs; missing days documented as gaps	W3	backfill	UNVERIFIED-CANDIDATE
25	MCX bhavcopy (commodity derivatives OHLCV+OI)	derivative price	mcxindia.com bhavcopy download	daily bar	2003→ (launch), backfillable per survey	EOD	file-drop	free (MVP-replace note; commodity leg adjacency)	contract-naming conventions changed over the years	W0	backfill	SURVEY-EVIDENCE
26	NCDEX bhavcopy (agri derivatives)	derivative price	ncdex.com bhavcopy download	daily bar	2004→ (launch), backfillable per survey	EOD	file-drop	free	agri contract expiries/dates asymmetric vs financial derivatives	W0	backfill	SURVEY-EVIDENCE
27	GIFT Nifty (NSE International Exchange flagship NIFTY contracts)	derivative price	nseix.com EOD/bhavcopy (host timed out on fetch)	daily bar	Jul-2023→ (post-execution migration from SGX); SGX pre-2023 chain is a separate legacy series	EOD	file-drop	free (verify)	cross-venue continuity break at Jul-2023 — never stitch without flag	W0	verify	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
28	NSE IX other GIFT segments (currency, debt, IASP/commodity exports)	derivative price	nseix.com reports pages	daily bar	since segment launches (2022→)	EOD	file-drop	free (verify)	GIFT-session timezone (IST/UTC) mapping must be explicit	W0	verify	UNVERIFIED-CANDIDATE
29	AMFI NAV daily full-universe dump (NAVAll.txt)	fund NAV	amfiindia.com → NAVAll.txt (live fetch verified ^4)	daily snapshot	full scheme universe of latest NAVs (verified file emits ~18k rows/scheme-dated); full-history via own daily append + portal	EOD	REST bulk	free	NAV is PIT per scheme; discontinued schemes keep last NAV date — do not forward-fill silently	W0	connect	VERIFIED-UPSTREAM-DOC
30	AMFI NAV full-history backfile (per-scheme per-date series)	fund NAV (archive)	AMFI NAV-history portal (form POST) or longer archives	daily	oldest schemes ~1995→ (survey claim); oldest-intact depth to sample	batch	REST bulk	free	scheme-ID renumbering on AMCs and scheme mergers → ID-mapping table required	W0	backfill	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
31	MF monthly portfolio disclo-sures (scheme holdings)	ownership	AMFI/AMC monthly portfolio publica-tions	monthly	regime ~2011→ (survey claim), depth varies per AMC	batch	file-drop	free	heterogeneous formats across 40+ AMCs — normal-ization layer needed	W3	verify	UNVERIFIED-CANDIDATE
32	Kite Connect historical candles (Zerodha; NSE/BSE/NFO/MCX)	price/trades	api.kite.trade /instru-ments/historical/:token/:interval (verified contract ^1)	minute/3m/5m/10m/15m/30m/60m/day + OI flag	60 days (minute) / 200 days (3m/5m/10m/15m/30m/60m) / “several years” (day); community figure: minute ≈2016→; price: (survey claim) ₹2,000/m per app	batch (day-fresh)	REST bulk	vendor-license (Kite Connect; no redis-tribution)	expired F&O tokens not ad-dressable — cache instru-ments + continu-ous=1 only for day bars	W1	connect	VERIFIED-UPSTREAM-DOC
33	Dhan Historical API v2 (daily to inception; intraday 5y)	price/trades	api.dhan.co/v2 /charts/historical + /charts/intraday (verified ^2)	1/5/15/25/60-minute + day	daily to inception (claim ^2); intraday last 5 years; 90-day poll chunks	batch (day-fresh)	REST bulk	vendor-license (free with Data API access)	claims not field-tested for exotic tickers; F&O needs ex-piryCode book-keeping	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	Dhan Expired Options Data (rolling expired chains w/ IV, OI, spot)	derivative analytics	api.dhan.co/v2 `/charts/rollingoption` (verified ^3)	minute-level	last 5 years (claim ^3), ATM ±10 strikes (index) / ±3 (others), 30-day poll chunks	batch (day-fresh)	REST bulk	vendor-license (free)	strike-relative re-keying means strike identity moves day to day — PIT recon-struct per expiry	W1	connect	VERIFIED-UPSTREAM-DOC
35	Upstox V3 historical candles (+ India VIX + global indices incl. GIFT NIFTY keys)	price/trades	api.upstox.com/v3 `/historical-candle/...` (verified ^9)	minutes 1–300 / hours 1–5 / day/week/month	minutes from Jan-2022; days/weeks/months from Jan-2000 (claims ^9); chunk limits: min 1mo–1qtr, day 1 decade	batch (day-fresh)	REST bulk	vendor-license (free with Upstox account)	depth claims = vendor self-report; spot-check 2000s equities before trusting	W0	connect	VERIFIED-UPSTREAM-DOC
36	Upstox v2 historical candles (depre-cated path)	price/trades	api.upstox.com/v2 (verified ^8)	1m/30min/day/week/month	1m/30min 1y; day 1y; week/month 10y (claims ^8)	batch	REST bulk	vendor-license	superseded by V3 — keep only for schema cross-check	W0	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
37	Upstox expired-instruments dataset	derivative analytics	upstox developer “Expired Instruments” API (nav seen ^8, endpoint page not fetched)	daily/contract	to verify	batch	REST bulk	vendor-license	expiry-chain continuity vs Kite `continuous` approach — compare before committing	W1	verify	UNVERIFIED-CANDIDATE
38	Angel One SmartAPI historical candle API	price/trades	smartapi.angelbroking.com historical-candleREST	minute-hour-day	(survey claim) minute from ~2011; docs portal non-rendered on fetch	batch	REST bulk	vendor-license (free)	rate-limit tight (survey claim); depth claim needs live test	W0	verify	SURVEY-EVIDENCE
39	Fyers v3 historical data (chart end-points)	price/trades	myapi.fyers.in v3 organic historical endpoint (claimed ^10: 29y EOD, up to 7y intraday)	tick?/min-hour-day	vendor claims: 29 years EOD; up to 7 years intraday	batch (day-fresh)	REST bulk	vendor-license (free)	29y claim likely composed EOD — verify pre-2000 daily series provenance (exchange or vendor-composed)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
40	Fyers expired F&O data (with historical Greeks)	derivative analytics	myapi.fyers.in v3 expired-fo end-points (claimed ^10: up to 7y weekly/monthly expiries + historical Greeks claim)	intraday/contract	up to 7 years of expired contracts + Greeks (claim ^10)	batch	REST bulk	vendor-license (free)	Greeks method-ology for expired contracts is opaque → ask Fyers before trust	W1	connect	VERIFIED-UPSTREAM-DOC
41	TrueData Velocity (autho-rized tick+minute feed NSE/BSE/MCX; 1-min & tick backfill)	price/trades	truedata.in Velocity plu-gin/API (verified ^6)	tick/2s/1-min ups	plans: (claim ^6 ^7) ₹1,699–₹3,299/m Lite→Ultima; historical backfill depth per tier to verify	realtime / EOD	library(CCXT-class) (plugin + custom API)	vendor-license (exchange-authorized tariff; commer-cial per NSE/BSE/MCX rules)	back-adjusted split/bonus series destroy PIT — request raw + ad-justment times-tamps	W1	quote	VERIFIED-UPSTREAM-DOC
42	TrueData API family (Corpo-rate An-nounce-ment + MF/NAV + Corpo-rate Data/results/fundamentals + News APIs)	corporate event/fundamental	truedata.in market-data-api section (verified listing ^6)	event/fundamental rows	depth per API tier to verify	near-realtime	REST	vendor-license	corporate-data PIT discipline: amend-ments flagged? — probe in trial	W1	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
43	GlobalDatafeeds historical intraday archives (GDFL)	price/trades	globaldatafeeds.in historical packages	tick/min/day	(survey claim) intraday from ~2008; premium global markets; price: see-site (site fetch 403)	realtime / EOD	library(CCXT-class)	vendor-license	redistribution by default not allowed — negotiate; tick-storage compression + retrieval cost; epoch/timezone alignment must be tested	W1	quote	SURVEY-EVIDENCE
44	RBI gilt auction results & SGL transaction data (equities-adjacent liquidity context)	regulatory/fixed income	rbi.org.in auction results + SGL trades listings	daily/batch	auction results decade-deep per survey; SGL depth to verify	batch	file-drop	free	only as context feed for equities-adjacent risk/spillover studies	W3	verify	SURVEY-EVIDENCE
45	SEBI data portal (FPI holdings monthly, MF AUM, SCORES/orders)	regulatory	sebi.gov.in dataset pages	monthly	monthly series decade-deep per survey	batch	file-drop	free	manually-published PDFs dominate — OCR layer needed	W3	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	Short-selling disclosure dataset (segregated short-sell orders)	flow	exchange disclosure pages (presence unconfirmed)	daily	existence and depth to verify — candidate only	EOD	file-drop	free (verify)	if absent public, note-and-skip rather than scrape	W3	verify	UNVERIFIED-CANDIDATE
47	Wayback-era NSE site CDX walk (pre-2007 archives; old weekly/daily report URLs)	price/trades (archive)	web.archive.org CDX index of nseindia.com old report paths	daily bar	fragments 2000ish→ era; completeness unproven	EOD	S3-archive walk	free (archive content)	missing days permanent gaps — document coverage matrix per year	W3	backfill	UNVERIFIED-CANDIDATE
48	NSE delisted / excluded securities archive (Z-series, delisted list, suspended list)	universe/archive	NSE delisted-companies listings + exclusion reports	snapshot/batch	depth partial; to verify	batch	file-drop	free	survivorship-bias killer: join historical bhavcopy only where delisted asset still has bars	W0	verify	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
49	Paid deep intraday backfile 2000s (TrueData/GDFL/NSE stock-wise cross-quote)	price intraday archive	procurement across rows 19/41/43 candidates (pick one vendor)	minute	2008–2016 NSE minute coverage guaranteed only via paid tier (survey claim)	EOD	licensed terminal	vendor-license	obtain one overlapping-window sample and cross-validate against an independent source before pick	W1	quote	SURVEY-EVIDENCE
50	Norgate Data India stock coverage — checklist rejection row	price/trades (archive)	norgatedata.com official site (verified ^5)	—	Norgate explicitly serves US/AU/CA stocks + world futures/forex only — NO India equity package	—	—	—	site checked; do not budget effort here (action-note: (closed: not offered))	W3	verify	VERIFIED-UPSTREAM-DOC

02 — Asia-Pacific equities (90 rows)

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	CK A-share daily quotes (daily full A-board OHLCV+amount)	EOD price	Tushare Pro REST (daily/daily_basic), token-keyed	daily	since 1990-12-19; full backfill yes	same-evening T+1	REST bulk	freemium (points tiers)	adj factor separate endpt; join pre-rel/delist_date for sur-vivorship; unit=shares not hands (action-note: con-nect+)	W0	backfill	VERIFIED-UPSTREAM-DOC
2	CK A-share minute history (stk_mins family)	intraday bar	Tushare Pro REST, high-points tier	1/5/15/30/60 min	recent years only, shallow backfill per tier	minutes after bar	REST bulk	vendor-license (points)	session/halt align-ment; tier gate; verify depth caps before budget-ing	W1	verify	VERIFIED-UPSTREAM-DOC
3	CK A-share realtime endpts (□□□□, □□□□-□□□)	realtime snapshot	Tushare Pro realtime REST	intraday snapshot	T0 only, no history	realtime polling	REST polling	vendor-license (points)	no PIT (live only); snapshot cadence unpub-lished	W2	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
4	CK financial statements (income/balancesheet/cashflow/forecast/express)	fundamentals	Tushare Pro REST, report_type param	quarterly	since 2000s (metric coverage varies), backfill yes	same-day after disclosure	REST bulk (paged)	freemium	PIT = announcement date field, not report period; revenue-recognition calendar traps (action-note: connect+)	W0	backfill	VERIFIED-UPSTREAM-DOC
5	CK reference/ownership events (top-10 holders, pledges, unlock, block trades, insider moves)	corporate events	Tushare Pro □□□□ family REST	per-event/daily	multi-year per family	T+1 publish	REST bulk	freemium	PIT by disclose_date; some families survey-only (no PIT label)	W1	connect	VERIFIED-UPSTREAM-DOC
6	CK margin + LHB boards (margin/margin_detail/top_list family)	flow/badge	Tushare Pro REST	daily	margin since 2010-03; LHB since 2005	T+1 publish	REST bulk	freemium	flag join via margin_targets; LHB badge semantics entrepreneurship	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
7	CK ETF dataset family (info/shares/NAV/adjusted/minute)	fund	Tushare Pro ETF REST	daily	since 2004-12-30 (first ETF)	EOD+1	REST bulk	freemium	NAV vs close PIT; AUM changes as market data not funda-mentals	W2	connect	VERIFIED-UPSTREAM-DOC
8	CK index family (index_daily/index_weight/index_classify + SW industry)	index	Tushare Pro REST	daily	major indices since 1991; weights history since ~2005	same-day EOD	REST bulk	freemium	weights PIT at published rebal-ance rows; SW industry classifica-tion version churn (action-note: con-nect+)	W0	backfill	VERIFIED-UPSTREAM-DOC
9	CK public-fund family (fund list/NAV/dividends/holdings)	fund	Tushare Pro □□ REST	daily	open-fund era from 2001	EOD+1	REST bulk	freemium	NAV PIT at nav_date; holdings quarterly-lag only	W2	connect	VERIFIED-UPSTREAM-DOC
10	CK futures family (con-tracts/daily/adjusted/positions/settle/warehouse)	futures	Tushare Pro □□ REST	daily	SHFE/DCE/CZCE from mid-1990s; CFFEX from 2010	evening EOD	REST bulk	freemium	roll conti-nuity joins; □ lot units; settle vs close dis-tinction	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	CK HK/US daily + fundamentals (□/□□ family)	EOD price + fun	Tushare Pro REST	daily	HK from ~2004, US from 2000s (verify per market)	EOD	REST bulk	freemium	US survivorship weak; HK board suffix rules; currency normalization	W2	verify	VERIFIED-UPSTREAM-DOC
12	CK macro/rates + text corpus (Shibor/LPR/CPI/PPI + LLM news/announce corpus)	macro/text	Tushare Pro REST	daily/monthly	Shibor 2006+; corpus decade+	episodic	REST bulk	freemium	text PIT by publish ts; requires text-qa pipeline	W2	connect	VERIFIED-UPSTREAM-DOC
13	AKShare `stock_zh_a_hist` (A-share daily, Eastmoney-underlying, qfq/hfq)	EOD price	pip akshare lib wrapper on eastmoney JSON	daily	since 1991/listing; full backfill	delayed same-day	library	free / ⚠ (EM ToS scraping plane)	adjust method param controls history rewrite; hands-vs-shares unit verify; EM can change endpoints (action-note: connect+)	W0	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
14	AKShare minute (`stock_zh_a_hist_min_em`, `stock_zh_a_minute` sina)	intraday bar	akshare lib	1/5/15/30/60 min	short window per request; deep minute not archived	realtime/delayed	library	free / ⚠	per-request caps; bar timezone align-ment (Beijing)	W1	verify	VERIFIED-UPSTREAM-DOC
15	AKShare A-sh funda-mentals (`stock_financial_abstract`, `stock_financial_report_sina`)	fundamentals	akshare lib sina tables	quarterly	since ~2000	T+1	library	free / ⚠	PIT via an-nounce-ment date; CN-GAAP mapping (action-note: con-nect+)	W1	build-parser	VERIFIED-UPSTREAM-DOC
16	AKShare in-dex+constituents (`stock_zh_index_daily_em`, `in-dex_stock_cons_csindex`, `in-dex_stock_cons_weight_csindex`)	index	akshare lib (EM + CSI official files)	daily	1991+; CSI cons/weights official history	EOD	library	free	weights only at publish rows; CSI xlsx ingest	W0	connect	VERIFIED-UPSTREAM-DOC
17	AKShare futures (`futures_zh_daily_sina`, `fu-tures_main_sina`, `fu-tures_delivery`)	futures	akshare lib	daily	per-venue 1990s+	delayed EOD	library	free / ⚠	symbol canoni-caliza-tion; settle vs prev_close	W1	connect	VERIFIED-UPSTREAM-DOC
18	AKShare funds/ETF (`fund_etf_hist_em`, `fund_nav_em`)	fund	akshare lib	daily	ETF 2004+, funds 1998+	EOD	library	free / ⚠	NAV vs price; dividend adj	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
19	AKShare HK plane (`stock_hk_daily`, `stock_hk_spot_em`)	EOD price	akshare lib	daily	from listing/IPO per name	EOD	library	free / ⚠	HKD units; HK half-day sessions; survivorship absent	W1	connect	VERIFIED-UPSTREAM-DOC
20	AKShare US plane (`stock_us_daily`, `stock_us_spot_em`)	EOD price	akshare lib	daily	~decade deep (verify per name), no survivorship	EOD	library	free / ⚠	USD; uppercase tick swaps (BRK.A); delisting gaps	W2	verify	VERIFIED-UPSTREAM-DOC
21	AKShare flow/ownership blocks (`stock_zh_a_gdhs`, `stock_individual_fund_flow`, `stock_lhb_detail_em`)	flow/sentiment	akshare lib on EM datacenter	daily	per block 2000s+	EOD	library	free / ⚠	no per-record PIT; classification noise; block schema churn	W2	connect	VERIFIED-UPSTREAM-DOC
22	Baostock daily + 5-min K-line (A-share SDK)	EOD + 5-min bar	baostock free Python lib (socket protocol)	daily + 5-min	daily since 2006-01-01; 5-min shallow; backfill bulk yes	EOD evening	library	free	separate `query_adjust_factor`; join delist_date for survivorship (Baostock has market=sh/sz only) (action-note: connect+)	W0	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
23	Baostock dividends & adjust factors (`query_dividend_data`, `query_adjust_factor`)	corporate acts	baostock lib	per-event	since 2006	EOD	library	free	cross-check vs an-nounce-ments; only A-share ex-div calendar (action-note: con-nect+)	W0	backfill	SURVEY-EVIDENCE
24	Baostock quarterly funda-mentals (`query_profit_data/balance/growth/operation`)	fundamentals	baostock lib	quarterly	since 2006	post-disclosure cadence	library	free	PIT via PUBDATE field, not report-Date (action-note: con-nect+)	W1	backfill	SURVEY-EVIDENCE
25	RiceQuant RQData equities (price/fundamentals/calendar/trade dates)	multi	RQSDK (`rqdatac` Python, HTTP API)	daily→minute→tick tiered	deep daily (verify start year); minute from 2000s	realtime available	library	vendor-license (paid; trial)	PIT via `get_fundamentals` with query_date; adj='pre' seman-tics; contract delisting included	W1	quote	VERIFIED-UPSTREAM-DOC
26	RiceQuant RQData fu-tures/options (con-tracts/settle/rolls)	futures	RQSDK	daily/tick	deep (2000s); options from 2015	realtime	library	vendor-license	roll con-ventions; settle-ment price vs close	W2	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
27	JoinQuant jqdata (`get_price`, `get_bars`, `get_fundamentals`, `normalize_code`)	multi	jqdata REST/SDK	daily+minute	daily 1990s; minute from 2005	EOD/realtime	library+REST	vendor-license (points; trial)	`get_fundamentals(date=)` PIT normal-ization; warmed history counts in quota	W2	evaluate	UNVERIFIED-CANDIDATE
28	Wind (□□) Terminal / WindPy / EDB	multi	Wind desktop + WindPy server-side	daily→tick	comprehensive multi-decade CN archive	realtime terminal	licensed terminal	vendor-license (quote)	entitlements per terminal; redistri-bution prohib-ited (action-note: quote+)	W3	license-review	UNVERIFIED-CANDIDATE
29	CSMAR □□□ CN academic archive (mar-ket/firms/events)	multi archive	CSMAR web bulk + SFTP crates	daily/annual	CN since 1990 full history	static exports	file-drop/SFTP	vendor-license (aca-demic quote)	PIT via list-ing/delist tables; survivor-ship handled; licensed redistri-bution N-OK (action-note: quote+)	W3	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
30	cninfo disclosure REST (P0ST `/new/hisAnnouncement/query`) SZ/SSE-listed co. announcements + PDFs	corporate disclosure	cninfo.com.cn public API	per-announcement	since ~1999 (digitized; verify earliest)	near-realtime	REST polling + file fetch	free / ⚠ (redistribution of doc PDFs per cninfo terms)	announce_time per-name per-announce batched query; PDF parse pipeline (action-note: connect+)	W0=pit;	build-parser	VERIFIED-UPSTREAM-DOC
31	cninfo □□□□ (□□□□ + □□□□□ □□□-data service)	disclosure-derived	cninfo portal (login-gated)	daily/quarterly	varies per table	T+1	web portal (contract unverified)	freemium (paid plans)	XBRL instances PIT; browser quota gates	W1	license-review	VERIFIED-UPSTREAM-DOC
32	SHFE (+INE) official daily EOD (settlement/trade/warehouse)	futures	shfe.com.cn market-data pages (JSON/HTML)	daily	since late-1990s (verify); INE since 2018	EOD evening	file-drop/HTTP	free	date published T+1; units (tonne/lot); verify archive endpoint depth (action-note: connect+)	W1	build-parser	VERIFIED-UPSTREAM-DOC
33	DCE official daily EOD	futures	dce.com.cn static pages	daily	since 1993-11; full archive	EOD	file-drop/HTTP	free	CN-side latency; publish T+1; HTML table parse	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	CZCE official daily EOD	futures	czce.com.cn static files	daily	since 1990s	EOD	file-drop/HTTP	free	**bot-block: GET / returned 412** from non-CN IP [f8] → needs CN egress or SFTP agreement (action-note: +verify)	W1	connect	SURVEY-EVIDENCE
35	GFEX official daily EOD (industrial-silicon/lithium-carbonate)	futures	gfex.com.cn data pages	daily	since 2022-04 founding	EOD	file-drop/HTTP	free	transport-blocked from this host [f9]; young archive only	W2	connect	UNVERIFIED-CANDIDATE
36	CFFEX official daily EOD (IF/IH/IC/IM equity-index fu-tures+options)	futures	cffex.com.cn static files	daily	since 2010-04-16	EOD	file-drop/HTTP	free	daily settle only; verify CSV archive availabil-ity	W1	connect	SURVEY-EVIDENCE
37	Eastmoney data-center planes (funds-flow/LHB/margin/announcements/stock-connect)	flow/sentiment	data.eastmoney.com + datacenter-web JSON REST	daily	per block 2000s-2010s	intraday-refresh some blocks	REST polling	free / ⚠ (scraping plane)	no per-record PIT; schema churn; EM rename field risk (action-note: +verify)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
38	Eastmoney realtime quote plane (push2.eastmoney.com `secid=` snapshot)	realtime snapshot	EM quote endpoints	tick-ish poll	live only, no history	realtime	REST polling	free / ⚠	no PIT; header/anti-bot measures; endpoint migration risk	W2	verify	UNVERIFIED-CANDIDATE
39	Sina hq quote plane (`hq.sinajs.cn list=` realtime quotes)	realtime snapshot	sina HTTP GET (needs `Referer: finance.sina.com.cn`)	tick-ish poll	live only	realtime	REST polling	free / ⚠ (unpublished)	mandatory Referer header (since ~2021, else 401); informal, breakage-prone	W2	verify	UNVERIFIED-CANDIDATE
40	Tencent gtimg plane (`qt.gtimg.cn/q=` live quotes; `web.ifzq.gtimg.cn/appstock/app/fqkline` CN/HK/US daily)	EOD price + snapshot	Tencent quote endpoints	daily	CN/HK/US daily since 1990s-2000s (verify)	delayed EOD (CN intraday live)	REST polling	free / ⚠ (unpublished)	fqkline param controls adj; text-array parse; informal endpoint (action-note: verify+)	W1	build-parser	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
41	CSI Index official index files (csindex.com.cn: constituents/weights/returns/factor downloads)	index	csindex official download center	daily/monthly	000300 since 2004-04; per-index archive	EOD	file-drop/HTTP	free (⚠ redistribution restricted per CSI terms)	methodology PIT via rebalance announcements; JS-heavy site; XLSX ingest (action-note: connect+)	W0	backfill	SURVEY-EVIDENCE
42	Stock-Connect northbound flows & holdings	flow	HKEX Northbound stats pages + Eastmoney datacenter blocks [f10]	daily	Shanghai-Connect since 2014-11-17; Shenzhen-Connect since 2016-12-05	intraday summary; holdings T+1 morning	REST polling + scrape	free / ⚠	HKD vs CNY fields; holdings PIT once per day only (action-note: +verify)	W0	connect	VERIFIED-UPSTREAM-DOC
43	A-share delisted/ST survivorship archive (delist_date + daily rows for delisted names)	archive	Tushare Pro ST lists + `daily` endpoints for retired tickers [f1]	daily	delistings since 1990s (first majors 2001); complete pre-delist backfill	n/a	REST bulk	freemium	survivorship-bias corrector; ST flag transition PIT (action-note: verify+)	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	HKEXnews disclo-sure archive (title-search + postings PDFs & XBRL)	filings	www1.hkexnews.hk/search/titlesearch.xhtml + ePortal	per-filing	since ~1999 (digital); complete coverage from ~2004	near-realtime	web POST + PDF/xbrl fetch	free / ⚠ (HKEX data-portal terms)	filing tax-onomy (□□/□□/□□□□); PIT by release date; XBRL parse for struc-tured (action-note: con-nect+)	W0	build-parser	VERIFIED-UPSTREAM-DOC
45	HKEX CCASS Share-holding Search (per-stock share-holding by CCASS partici-pants incl. HKSCC Nomi-nees)	shareholding	HKEX CCASS pages (share-holding search + partici-pant list)	daily	archive from ~2010 s (current format); earlier via third-party only	next-morning T+1	own-capture (scrape)	free / ⚠	shareholding % = percent of shares held; float-based; daily changes need diff pipeline (action-note: +own-capture)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	HKEX realtime market data (Equip retail L1 / OMD licensed feed)	realtime L1/L2	hkex.com.hk Equip store + OMD feed	tick / EOD	live; EOD purchasable separately	realtime	licensed terminal/feed	vendor-license (quote)	exchange entitlements; redistribute prohibited; latency tiers (delayed L1 free vs realtime paid) (action-note: quote+)	W3	license-review	VERIFIED-UPSTREAM-DOC
47	HKEX historical data marketplace (equities tick / EOD purchases)	tick/EOD archive	HKEX historical data marketplace portal	tick→daily	paid archive (start year via license call; expected late-2000s)	batch	paid download/SFTP	vendor-license	attribution; no recurring distributions without license (action-note: quote+)	W2	license-review	UNVERIFIED-CANDIDATE
48	Hang Seng Index series (HSI/HSCEI/HSTECH daily returns + constituents/weights files)	index	hsi.com.hk data downloads	daily	HSI since 1969-11; HSCEI since 1994; HSTECH since 2020-07	EOD	file-drop/HTTP	free / ⚠ (HSIL terms)	PIT = index-methodology version; fund-family filing style; constituents CSV ingest	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
49	AASTOCKS-style HK realtime quotes/economic data	realtime/quote	aastocks.com portal (license quote)	tick-ish	live	realtime	licensed portal/terminal	vendor-license (quote)	full entitlement vendor; no historical via this plane	W3	quote	UNVERIFIED-CANDIDATE
50	Futu OpenAPI — HK/US/CN-A equity history + realtime (OpenD gateway; get_history_kline etc.)	EOD+quote	futunn OpenAPI SDK (Python/REST/WS via OpenD)	daily/minute + realtime	HK/CN/US daily since ~1990s-2000s (verify per market); HK realtime tick	realtime (with account)	library+WS	vendor-license (freemium w/ broker acct; quote row)	per-symbol-and-day quote quota; broker account eligibility; redistribution N-OK for feed (action-note: +quote)	W1	connect	SURVEY-EVIDENCE
51	TWSE OpenAPI daily trading data (TWT-family reports + exchangeReport/STOCK_DAY JSON)	EOD price	twse.com.tw OpenAPI/JSON endpoints	daily	modern series since 2004-01; deeper via monthly PDFs (1990s)	EOD evening	REST bulk (JSON)	free	units in NT$-thousand; halt/delisting; per-endpoint year caps (action-note: connect+)	W0	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
52	TWSE foreign/institutional investor daily flows (□□□□ per-security T86)	flow	TWSE TWT86U JSON endpoint	daily	since 2004	T+1 afternoon	REST bulk	free	investor-category semantics (□□/□□/□□□); cross-check against futures positions for sentiment	W0	connect	SURVEY-EVIDENCE
53	TWSE MOPS (□□□□□□□: monthly revenue / insider / major-shareholder filings)	filings/events	mops.twse.com.tw JSON endpoints + own scrapes	monthly/per-filing	digitized since ~2000s (verify earliest)	T+1	REST polling + file fetch	free / ⚠	insider PIT by □□□□; monthly-revenue release timing drives drift (action-note: connect+)	W1	build-parser	SURVEY-EVIDENCE
54	TPEx OpenAPI (□□ daily quotes + TPEx index)	EOD price	tpex.org.tw end-points	daily	similar depth to TWSE (~2004+); verify archive	EOD	REST bulk	free	units NT$-thousand; consistent with TWSE parse layer	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	FinMind datasets (TaiwanStockPrice etc.: daily/minute/fundamentals/flow)	EOD+intraday+flow	FinMind API (api.finmindtrade.com) + docs	daily/minute	daily since 2004; minute per dataset varies	EOD	REST polling	freemium (quota)	TaiwanStockPriceAdj for backtest; TaiwanStockPrice raw for consistency; per-dataset calendar (action-note: +verify)	W1	connect	UNVERIFIED-CANDIDATE
56	TEJ Taiwan Economic Journal (academic/enterprise archive: TW market, financials, ownership, analyst)	multi archive	TEJ proprietary database / file-push	daily/monthly	TW daily since 1980s; financials since 1990s	batch	proprietary terminal	vendor-license (quote)	PIT via announcement-date fields; strong TW extension to HK/CN (action-note: quote+)	W3	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
57	TW intraday own-capture plane (TWSE MIS realtime quote pages, 5s poll)	intraday	own-capture of twse.com.tw realtime JSON/HTML	5s snapshot / 1-min rollup	ingest-only from connec-tion date	realtime→rollup	own-capture	free / ⚠	per-session snapshot not tick; anti-bot throttling risk; needs JVM-compatible parse layer? (verify field formats)	W2	build-parser	UNVERIFIED-CANDIDATE
58	J-Quants API prices/daily_quotes	EOD price	J-Quants REST (api.jquants.com; tiered)	daily	tier-based depth; sample free tier	EOD	REST bulk	freemium	per-tier range caps; Ad-justment-Close flag for quotes (action-note: +verify)	W1	connect	SURVEY-EVIDENCE
59	J-Quants API financials (fins/statements + fins/announcement)	fundamentals	J-Quants REST	quarterly/annual	from EDINET-based state-ments (start year verify)	daily	REST bulk	freemium	PIT by Dis-closed-Date fields	W1	connect	SURVEY-EVIDENCE
60	J-Quants API in-dices/TOPIX (`indices/topix` + TOPIX families)	index	J-Quants REST (+ JPX official files for deep)	daily	TOPIX since 1968 (via JPX); JQ-API tier depth verify	EOD	REST bulk	freemium	tall-hist via JPX official files; index reconsti-tution events	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	EDINET API v2 (documents list + PDF/XBRL fetch)	filings	EDINET REST (disclosure2.edinet-fsa.go.jp)	per-filing	EDINET digitized since 2008; XBRL from 2012-08 (verify earliest)	near-realtime (within 24h)	REST bulk + file fetch	free / ⚠ (attribution/redistribution TOS)	doc-type map (□□□□ etc.); PIT by submitDateTime; XBRL instance parse (action-note: connect+)	W0	build-parser	VERIFIED-UPSTREAM-DOC
62	JPX Data Cloud (historical market statistics bulk releases)	EOD/intraday archive	marketdata.jpx portal	daily	archive start year unverified — cautious: **“expect 2000s archive; verify window by release calendar”**	batch	file-drop/download	freemium (⚠ redistribution terms)	PIT by release date; per-release coverage (action-note: verify+)	W1	connect	UNVERIFIED-CANDIDATE
63	JPX arrowhead (real-time MBO/L2 feed)	tick	JPX proprietary connect (arrowhead MBO)	tick	live; historical via licensed replay	realtime	licensed binary feed	vendor-license (quote via JPX member)	colocation/entitlement cost; redistribution prohibited (action-note: quote+)	W3	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
64	Nikkei NEEDS (price/fundamental/ownership databases)	multi archive	NEEDS licensed feed (CSV/file push or terminal)	daily	deep (JP daily since 1949)	batch	licensed terminal	vendor-license (quote; high cost)	PIT by release; JP-only vendor UX; per-table license scope (action-note: quote+)	W3	license-review	UNVERIFIED-CANDIDATE
65	kabutan □□ web data (daily quotes/fundamentals screens)	EOD+fund screens	kabutan.jp web scrape	daily	from ~2010 s (verify)	delayed EOD	own-capture	free / ⚠ (scraping ToS)	DOM churn; personal-use only per site policy; no PIT label on records	W3	build-parser	UNVERIFIED-CANDIDATE
66	kabusapi (au□□□□ KabuSta-tion API: JP/US realtime REST/WS)	realtime	kabusapi REST/WS (per-session token)	tick/1-min	live + short own-chart history	realtime	REST/WS	vendor-license (broker account free; redistri-bution N-OK)	rate limits; needs own bro-kerage acct; live-only (PIT only for own trades)	W2	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
67	DART filing search (corp Code master + per-filing list + PDF parse)	filings	DART portal + public search API (dart.fss.or.kr)	per-filing	DART digitized since 1999-09; complete from ~2003	near-realtime	REST polling + file fetch	free	corp_code refresh cadence; classification map (□□□□□/□□□□□/□□□□ flags); PIT by □□□ (action-note: connect+)	W0	build-parser	VERIFIED-UPSTREAM-DOC
68	OpenDART financial statements APIs (□□□□ □□□□□□ + □□□□□□□□□□ + XBRL □□)	fundamentals	OpenDART REST (opendart.fss.or.kr; free API key)	quarterly/annual	XBRL statements from 2012-12 (verify earliest)	daily	REST bulk	free	is□□□ vs is_consolidated fs_div; PIT by rm (report name); batch download quota (action-note: connect+)	W0	build-parser	VERIFIED-UPSTREAM-DOC
69	OpenDART □□□□ □□□ (major-shareholder/bulky filing aggregate)	shareholding	OpenDART REST (□□□□ □□ API)	per-filing	from 2016 (□□□□ robust coverage; verify earlier depth)	daily	REST bulk	free	shareholder PIT rounds; change_reason codes; XML parse	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
70	KRX Data Marketplace (□□·□ · daily quotes/statistics; + analyst/platform)	EOD price + stats	data.krx.co.kr queries + KRX Data Marketplace	daily	deep daily archive (per-dataset years vary; verify per table)	EOD (T+0 evening)	web query + REST-ish POST (verify contract)	freemium (base free; advanced needs subscription)	per-table headers/units; KRW; index vs price split; PIT at publish (action-note: +verify)	W1	connect	VERIFIED-UPSTREAM-DOC
71	KOSPI/KOSDAQ index history (via KRX/official index series)	index	KRX data portal + index official series	daily	KOSPI since 1980-01-04; KOSDAQ since 1996-07	EOD	REST bulk	free	methodology histories; currency KRW	W1	connect	SURVEY-EVIDENCE
72	NICE InfoService-type (KR credit/research data analog)	multi archive vendor	NICE info portal (license quote)	daily/monthly	varies by product	batch	proprietary terminal	vendor-license (quote)	per-product license scope; credit vs market split	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
73	SGX Securities market data (daily quotes/indices/STI + data-API)	EOD price + index	sgx.com market-data pages + SGX Data Centre	daily	since 1999 (SGX merger day) backfill per-series	EOD	REST/portal (verify per-dataset contract)	freemium (base free; full data licensed)	SGX data-license for redistribution; SGD; session (morning/luncheon break + afternoon); PIT at publish (action-note: +quote)	W1	connect	VERIFIED-UPSTREAM-DOC
74	SGX historical data marketplace (intraday/EOD purchases)	tick/EOD archive	SGX Market Data store	tick→daily	paid archive (start year via license call)	batch	paid download	vendor-license	redistribution restrictions; per-dataset license (action-note: quote+)	W2	license-review	UNVERIFIED-CANDIDATE
75	SET Thailand official dataset plane (daily quotes, indices, disclosures)	EOD price + corp	set.or.th / settrade portals	daily	since 1990 s? (per-venue backfill verify)	EOD	REST/portal (unverified contract)	free / ⚠	THB units; trading-session (morning/luncheon + afternoon); PIT by publication (action-note: +verify)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
76	SET intraday quote history files (settrade per-stock minute pages)	intraday bar	settrade quote-history static pages	1-min	per-stock minute archive (verify depth; expect rolling months-to-years window per stock)	delayed	file-drop/HTTP	free / ⚠	archive windows may rotate; per-request naming (action-note: verify+)	W2	build-parser	UNVERIFIED-CANDIDATE
77	IDX Indonesia dataset plane (daily trading summary, disclo-sures, ID-list)	EOD price + corp	idx.co.id Market Data/Announcement pages	daily	since 1990s? (official archive verify)	EOD	portal/REST-ish (unveri-fied contract)	free / ⚠	IDR units; IDX Micro-port contract; PIT by publish (action-note: +verify)	W1	connect	VERIFIED-UPSTREAM-DOC
78	Bursa Malaysia dataset plane (daily quotes + company an-nounce-ments)	EOD price + corp	bursamalaysia.com portals	daily	since 1980s (com-pany backfill verify)	EOD	portal scrape / PDFs (un-verified contract)	free / ⚠	MYR units; sessions; an-nounce-ments only in HTML/PDF (no single API) (action-note: +verify)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
79	PSE Edge portal (company disclosures / financial reports / index summary)	filings + EOD	edge.pse.com.ph (disclosure list + forms)	per-filing + daily	digital disclosures since ~2016? (verify); earlier via SEC	intraday (live)	web portal + PDF/JSON	free / ⚠	PH units (PHP); filing taxonomy (17-A/8); PIT by publish (action-note: connect+)	W1	build-parser	VERIFIED-UPSTREAM-DOC
80	Vietnam HOSE/HNX dataset plane (VND30/HNX-INDEX; SSI iBoard as unofficial plane)	EOD price	HOSE/HNX official + SSI iBoard screape	daily	since 2000 (HOSE from 2000; official archive verify)	EOD (delayed)	portal scrape + file-drop	free / ⚠	VND units; T+1 settlement norms; login-gated; no PIT label (action-note: +verify)	W2	connect	UNVERIFIED-CANDIDATE
81	NZX market-data plane (announcements + NZX daily data downloads)	EOD price + corp	nzx.com announcements + NZX data services	daily	since 1980 s (per market verify)	EOD	portal + paid data API	freemium (list-on-request; quote)	NZD; per-product license terms on redistribution (action-note: +quote)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
82	ASX official dataset plane (daily CSV archive + company announcements)	EOD price + corp	asx.com.au data pages + announcements	daily	since 1997 (ASX-path founding); deeper via annual/legacy archives	EOD	file-drop/HTTP	free / ⚠ (redistribution restricted)	AUD; codes include share-class suffixed; PIT by publish (action-note: connect+)	W1	backfill	UNVERIFIED-CANDIDATE
83	Tadawul API (SDMC/Saudi Exchange market data API v1 realtime + historical)	EOD price + realtime	api.tadawul.com.sa (v1 REST; **SDMC = Saudi Data & Metadata?**; verify scope: real-time/EOD)	daily + intraday	per-API tier (verify backfill depth)	realtime/delayed	REST	vendor-license (key; redistribution N)	SAR units; Sunday-Thursday calendar; entitlement needed (action-note: quote+)	W2	verify	UNVERIFIED-CANDIDATE
84	ADX market-data plane (Abu Dhabi Securities Exchange: daily summaries + announcements)	EOD price + corp	adx.ae portals	daily	since founding 2000-11	EOD	portal scrape / file	free / ⚠	AED; no single REST contract yet; holiday calendar (action-note: +verify)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
85	DFM market-data plane (Dubai Financial Market: daily quotes/indices)	EOD price	dfm.ae portals	daily	since 2000-03	EOD	portal scrape	free / ⚠	AED; verify per-endpoint	W3	connect	UNVERIFIED-CANDIDATE
86	EGX Egypt Open API (daily EOD + static files)	EOD price	egx.com.eg official OpenAPI/static	daily	since 2000s (verify)	EOD	REST (unverified contract)	free	EGP units; official archive window unclear (action-note: +verify)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
87	MOEX ISS (securities/history/candles/trades REST reference; index constituents via listing endpoints)	EOD + tick-level	iss.moex.com REST (JSON/CSV/XML)	daily + intraday candle	EOD history deep (sample data back to 2010-11 in official examples; deeper via ISS dates endpoints) — write **“since ~2010 via ISS queries; deep via dates discovery”**	EOD/delayed	REST bulk	free (ISS public; redistribution check per MOEX TOS)	RUB/USD?; index boards; per-board histories (board-level dates listings) (action-note: connect+)	W1	backfill	VERIFIED-UPSTREAM-DOC
88	Stooq APAC long EOD archive (CN/HK/JP/IN/SG daily CSV)	EOD price	stooq.com CSV HTTP files	daily	since 1980-1990s per symbol (free bulk download)	delayed EOD	file-drop/HTTP	free	survivorship unknown; symbol reeling (index suffix); vendor-free but not PIT consistent (action-note: +verify)	W1	backfill	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
89	Yahoo Finance APAC backfill (HK/JP/TW/KR/SG daily via yahoo/yfinance)	EOD price	query1/quoteSummary JSON (via yfinance)	daily	yfinance.yaml: late 1980s/1990s per venue; per-symbol from IPO	delayed EOD	library/REST	free / ⚠ (yahoo ToS scraping plane)	delisting gaps (yahoo only lists active); raw=uh parse; price-format changes (action-note: (cheap corpus for cross-check))	W1	backfill	UNVERIFIED-CANDIDATE
90	Norgate Data AU/NZ survivorship archive (ASX/NZX incl. delisted)	archive EOD	Norgate desktop + API/lift SDK	daily	AU back to 1980s; NZ 1990s; incl. delisted-firms full path	EOD	desktop-app+API	vendor-license (paid)	survivorship handled (Norgate curated); annual license	W3	quote	UNVERIFIED-CANDIDATE

03 — US equities & microstructure (81 rows)

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	nasdaq-totalview-itch-5.0	exchange-official feed	Nasdaq TotalView-ITCH 5.0 via MoldUDP64 UDP multicast (+ Soup-BinTCP replay); spec PDF on Nasdaq-Trader	event-by-event full L3 order stream (add/exec/replace/trade/NOII/cross)	realtime feed; paid history via Data Link NTV (2014→ claim, re-verify)	realtime (exchange-native)	ITCH-5.0 feed handler; Databento MBO normal-ization is a proven template	exchange license — non-display family (display redistri-bution = separate tier; sales quote)	PIT: order-by-order only from ITCH (no overlay dup feeds); order on ts_event; NOII auction-phase gates (row 6) (action-note: → build-parser on free sample)	W1	license-review	VERIFIED-UPSTREAM-DOC
2	nasdaq-ntv-datalink-history	licensed historical archive	Nasdaq Data Link NTV database (prior session verified listing; site timed out this session)	per-day TotalView archives / leveled products	2014→ per prior-session claim (re-verify)	T+1 archive	NDL REST API or bulk download	exchange license via NDL; display/non-display billing split (vendor claim)	PIT: dataset-version pinning; re-download full days on spec bumps, never partial-patch (action-note: quote →)	W2	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	nasdaq-basic-NLS+	exchange-official (display L1)	Nasdaq Basic (BBO + NLS last, Nasdaq-listed); Databento mirror dataset “Nasdaq Basic with NLS Plus” listed	L1 quotes + last sale	realtime; archive depth via Nasdaq sales (re-verify)	realtime	MoldUDP64 handler, or Databento mirror	display-class license (cheapest Nasdaq tier; NLS redistribution rules)	PIT: Nasdaq-listed issue scope only; NOT SIP NBBO — merge with Tape C BBO for consolidated NBBO (action-note: (cheap live NBBO-half))	W2	quote	VERIFIED-UPSTREAM-DOC
4	nasdaq-itch-50-daily-archive	raw archive (low/free cost)	Nasdaq ITCH 5.0 daily message logs (sample library; e.g. emi sample routes — fetch failed this session) + free Databento TotalView samples	raw A/C/D/E/F/U/T/X/Y message files	sample history since ~2013 (re-verify; paid depth via rows 1–2)	T+1 batch	bulk file walk + binary ITCH parser	free/low-cost sample entitlement; redistribution limits per Nasdaq terms (verify)	PIT: ground-truth lineage for every ITCH-derived dataset — pin spec version per day-file (action-note: + backfill)	W1	build-parser	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	nasdaq-bx-psx-ntx-totalview	exchange-official feed	Nasdaq satellite-venue order feeds; Databento dataset ids XBOS-ITCH (Nasdaq Texas) + XPSX-ITCH (PSX) listed TotalView	L3 order stream per satellite exchange	realtime; mirror archives per Databento ranges	realtime	reuse row-1 handler per feed, or Databento live/hist	exchange license, satellite-venue fee tier (small)	PIT: venue-scoped books; never merge XBOS/XPSX msgs into XNAS book state (action-note: (handler reuse))	W3	build-parser	VERIFIED-UPSTREAM-DOC
6	nasdaq-noii-auction-imbalance	administrative/auction	Databento ITCH NOII mes-sages + Nasdaq cross statistics (open-ing/closing/EMC) — rides NTV enti-tlement, no separate product	imbalance snap-shots (paired/total-imbalance/near/far price/auction type)	realtime; cadence verified: 1s at 9:25–9:28 & 15:50–15:55, then 1s with near/far popu-lated	realtime	feed handler (row 1) + statistics schema for official cross prints	non-display (inside NTV)	PIT: far/near clearing prices unpopu-lated before 9:28/15:55 — explicit NULLs pre-threshold; official open/close prints land in statistics schema	W1	build-parser	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
7	nasdaq-halts-mcb-sscb	regulatory/administrative	Nasdaq Trader: Trading Halts / Circuit Breakers / Short Sale Circuit Breakers / Reg SHO Threshold (live nav links)	per-symbol halt/MCB/SSCB status events	intraday status; historical depth on pages (re-verify)	~minutes-class latency	web scrape + status watch	free public	PIT: keyed to halt-event timestamps only; never infer LULD state between published snapshots (action-note: connect +)	W0	build-parser	VERIFIED-UPSTREAM-DOC
8	nasdaq-daily-market-stats	free exchange stats	Nasdaq Trader: Daily Market Summary / Daily Market Files / Intraday Full Volume Summary (live nav links)	aggregate volume / market-share stats	daily files	refreshed daily	daily scrape	free public	PIT: file-date columns; snapshot-vs-revision flag discipline	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
9	nyse-integrated-feed-5venues	exchange-official feed	NYSE Integrated Feed via Pillar: NYSE / NYSE American / NYSE Arca / NYSE National / NYSE Texas; Pillar-equities common spec + Integrated-Feed spec v2.5 PDFs linked on page	order-by-order + trades + order imbalances + security status, in-sequence	realtime; sample data at ftp.nyse.com/Real Time Data Samples	realtime	Pillar binary feed handler	exchange license; NYSE pricing PDF (NYSE_Market_Data_Pricing.pdf) linked on page	PIT: in-sequence Pillar ordering is authoritative per venue; NBBO is a separate SIP product — never derive cross-venue NBBO from one venue (action-note: + build-parser)	W1	license-review	VERIFIED-UPSTREAM-DOC
10	nyse-openbook-ultra	exchange-official depth	NYSE Open-Book Ultra (real-time depth add-on) + historical mirror “TAQ NYSE Open-Book incl. Ultra & Aggregated files”	full venue LOB rebuild	realtime + T+1 flat files	realtime + T+1	feed handler + flat-file loader	exchange license (separate Open-Book fee product)	PIT: rebuild strictly from snapshot + Ultra events; never mix Aggregated and Ultra views in one book state (action-note: quote +)	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	nyse-taq-integrated-historical	licensed historical archive	TAQ per-venue Integrated flat files (NYSE / American / Arca / National / Texas): “depth of book, trades, order imbalances, stock summary, security status”	per-venue order streams	T+1 flat files	T+1	SFTP/flat-file walk	exchange license (NYSE TAQ subscription, venue-by-venue products)	PIT: schema-version per product; venue-scoped sequencing; batch freshness windows (action-note: + quote)	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
12	nyse-daily-taq-consolidated	licensed historical archive	NYSE Daily TAQ: “trades and quotes for all issues traded on NYSE, Nasdaq and regional exchanges” incl. NBBO, Master file, Quote/Trade Admin + CTA/UTP Admin Message files	consolidated day tape (trades + NBBO)	T+1 flat files	T+1	SFTP/flat-file walk	exchange license (NYSE TAQ)	PIT: SIP timestamps authoritative; NBBO as-of joins only from this tape’s coverage — not from merged venue feeds (action-note: (primary historical NBBO baseline))	W1	backfill	VERIFIED-UPSTREAM-DOC
13	nyse-taq-short-sales-short-volume	licensed historical archive	TAQ Group Short Sales & Short Volume (all five NYSE-side venues)	daily short-sale transactions + short volume summary	T+1	T+1	flat-file loader	exchange license	PIT: short flags are per-print venue attributes; complement (not substitute) for FINRA bi-weekly positions (row 36)	W2	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
14	nyse-taq-closing-prices-volume-summary	licensed historical archive	TAQ Closing Prices (O/H/L/Last + total volume at 16:00) + TAQ Volume Summary (share size, short sales, odd-lots, VWAP, by-market-center)	EOD aggregates	daily	daily	flat-file loader	exchange license	PIT: 16:00-close convention per venue; cross-check vs free EOD rows (rows 65–66); adjustment lineage explicit	W2	backfill	VERIFIED-UPSTREAM-DOC
15	nyse-market-data-pricing-tariff	license/metadata quote	NYSE Market Data Pricing PDF (public-docs/nyse/data/NYSE_Market_Data_Pricing.pdf) + “Pricing, Policies & Contracts” section	fee schedule	n/a	n/a	PDF fetch	free	PIT: fee schedules are version-dated — capture revision history per quote (action-note: quote +)	W0	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
16	nyse-arcabook	exchange-official (display, standalone)	ARCA standalone ArcaBook display product (venue feed covered in row 9; product page NOT fetched this session)	L2 display depth (ETF-heavy scope)	realtime	realtime (terminal-display class)	vendor/display terminal	exchange license	PIT: ARCX-only scope; do not use for non-ETF book studies	W3	verify	UNVERIFIED-CANDIDATE
17	nyse-auction-imbalance	administrative/	TAQ auction NYSE/American/Arca Order Imbalances — “buy and sell imbalances sent at specified intervals during auctions throughout the trading day” — + live imbalance msgs inside row 9	auction-window imbalance snapshots	realtime + T+1	realtime + T+1	feed handler + flat file	exchange license	PIT: only auction-window intervals are defined events; NULL pre-window (same discipline as row 6)	W2	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
18	cboe-us-equities-feeds	exchange-official	Cboe Data Services (“Exchange Data and Access” nav) + DataShop All-Access APIs; cboe.com equities product pages 404 this session — re-discover URL structure	BBO/depth + trades (BZX/BYX/EDGA/EDGX/C2)	realtime; historical via DataShop	realtime	binary feed handler/API — formats re-verify	exchange license (quote)	PIT: per-venue feed families distinct; cross-fusion only at Cboe-licensed level (action-note: quote +)	W2	verify	UNVERIFIED-CANDIDATE
19	cboe-datashop-us-equities	vendor archive (exchange-owned)	Cboe DataShop (datashop.cboe.com) — asset-class filter “Equities, options, ETFs, indices, futures, crypto”; data types Historical/tick/intraday/daily	per-product tick/intraday/daily sets	per-product (decades-dependent; verify)	EOD/intraday downloads	SFTP / my-account downloads / All-Access APIs	Cboe license (no wholesale redistribution)	PIT: per-product version tags; “Calcs” (IV/Greeks) are vendor-computed — recompute or label derived (action-note: verify +)	W2	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
20	cboe-open-close-volume-summary	vendor archive	Cboe Open-Close Volume Summary (DataShop trending product: EOD + 10-min intervals by capacity Customer/Pro-Customer/Broker/Firm/MM)	venue aggregates	T+1	T+1	DataShop	Cboe license	PIT: capacity buckets are exchange classifications — never re-map into own buckets	W3	verify	VERIFIED-UPSTREAM-DOC
21	cboe-livevol-pro	retail platform	Cboe LiveVol (DataShop Platforms > LiveVol) — near-real-time monitoring platform	UI/screens	recent tick windows (DataShop archive is the store)	delayed/NRT	web platform	Cboe subscription	PIT: platform claims not PIT-consistent; monitor-only, archive via row 19	W3	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
22	cme-mdp3-equity-index-futures	exchange-official feed	CME Globex MDP 3.0 (ES/NQ/RTY/YM + micros): multicast + Cloud MDP (Google Pub/Sub)	MBO/MBP-10 snapshots/trades/bbo/1m	realtime; history via Databento GLBX mirror (Jun 2010→ claim)	realtime	MDP3 handler or Databento library	exchange license (MDP subscription; display/non-display tiers)	PIT: exchange-server timestamps; roll/settlement schedules PIT; index-futures vs cash microstructure adjacency explicit (action-note: quote +)	W2	license-review	SURVEY-EVIDENCE
23	cme-mdp3-other-classes	exchange-official feed	same MDP3 platform for Rates/FX/Energy/Ag/Crypto (Pub/Sub delivery verified prior session)	per-class depth/trades	realtime + historical via mirrors	realtime	same as row 22	exchange license	PIT: (a) each class lives in a distinct Globex product family; (b) message cadence/depth conventions differ by class — keep per-class pipelines	W3	quote	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
24	cme-datamine-history	licensed historical archive	CME DataMine self-service historical	tick/EOD for equity-index futures	range by dataset (quote)	T+1 batch	catalog flat-file delivery	CME license	PIT: CME dataset revision IDs vs “fix” annotations	W3	quote	SURVEY-EVIDENCE
25	databento-glbx-mdp3	mirror-of-exchange	Databento GLBX-MDP3 dataset (docs catalogue)	MBP-10/TBBO/Trades/OHLCV schemas	realtime + historical (2010→ claim)	realtime + batch	Databento library; boundary truth via get_dataset_range metadata API	Databento license (per-GB self-service)	PIT: range check via API metadata, not marketing claims; ranges vary by schema	W2	verify	VERIFIED-UPSTREAM-DOC
26	iex-tops	exchange-official	IEX Exchange Market Data — TOPS section (top-of-book + last) verified on live page	L1 snapshots	realtime; history via HIST	realtime	IEX-TP feed handler	exchange license (fee class re-verify)	PIT: IEX-only BBO (~2.5% tape share gate); never conflate with SIP	W1	connect	VERIFIED-UPSTREAM-DOC
27	iex-deep	exchange-official	IEX DEEP (book depth + trade break + auction info) verified on live page	book depth + auction indicators	realtime	realtime (exchange-native)	IEX-TP feed handler	exchange license (separate DEEP product)	PIT: DEEP auction channel reflects IEX-only auctions — not NYSE/Nasdaq cross state	W2	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
28	iex-hist-t1-pcap	free public archive	IEX HIST (T+1 pcap down-loads; IEX-TP ver-sioned; e.g. 20170515_IEXTP1_DEEP1.0.pcap.gz); Historical Data Terms of Use bound	pcap TOPS/DEEP/TRADES per day	T+1; archive coverage from 2017-05-15 example (map exact start)	T+1 batch	IEX-TP pcap decoder	free w/ terms-of-use	PIT: decoder patched to spec version named per-file; enter-date cata-loging before any backfill (action-note: backfill +)	W1	build-parser	VERIFIED-UPSTREAM-DOC
29	iex-cloud-retired	EXCLUDE (dead)	IEX Cloud shut 2024-08-31 (prior session)	n/a	n/a	n/a	n/a	n/a	migration gate: repoint con-sumers to rows 26/28 (IEX exchange direct) or aggrega-tor rows 72–74 (action-note: + unlink configs)	W0	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
30	cta-consolidated-quotes-tapeab	consolidated-tape	CTA SIP (Tapes A/B) — regime migrating to unified CT Plan	SIP consolidated quotes (NYSE-listed space)	realtime	realtime (SIP latency class)	SIP subscription handler	license via CTA Plan → CT Plan (row 34)	PIT: SIP timestamps; SIP-BBO vs direct-feed gap must be latency-bucketed, never merged silently (action-note: quote +)	W2	license-review	VERIFIED-UPSTREAM-DOC
31	utp-consolidated-quotes-tapec	consolidated-tape	UTP SIP (Tape C; Nasdaq-listed)	SIP consolidated quotes	realtime	realtime (SIP latency class)	SIP subscription handler	license via UTP → CT Plan	PIT: SIP quotes vs Nasdaq Basic BBO (row 3) divergence audit	W2	quote	VERIFIED-UPSTREAM-DOC
32	consolidated-last-sale-tapesabc	consolidated-tape	consolidated-last-sale/trades, Tapes A/B/C (CTA-side + UTP-side operators)	tape prints	realtime	realtime (SIP latency class)	SIP subscription handler	exchange/SIP license	PIT: consolidated stream includes off-exchange (S-segment) prints — flag segments, never strip S-prints silently	W2	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
33	finra-otc-trf-tas	consolidated tape	FINRA OTC trade reporting (TRF) + FINRA daily trade/quote file family (naming inconsis-tent across pages this session)	OTC prints + daily file deriva-tions	realtime (vendors) + daily files	realtime + T+1	licensed feed + daily-file ingest	exchange/vendor license (quote)	PIT-or Form-T flags for late-reporting; OTC marker columns pre-served (action-note: quote +)	W2	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	us-ct-plan-transition	consolidation-admin	New unified CT Plan (DataCT LLC administrator) replacing CTA+UTP: migration portal Sep 1 2026, licensing from Oct 1 2026, agreements due Mar 1 2027, operative Apr 1 2027; “no SIP technology changes required”	regulatory regime	n/a	n/a	regime-transition tracker	mandated plan	PIT: contractual regime switch (feeds unchanged) — re-quote every SIP-era agreement before Mar 1 2027	W0	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
35	sip-playback-vendors	vendor class	historical consolidated tape playback via licensed SIP vendors (Nasdaq Trader “Market Data Vendors” list nav route)	tape replay	historical	vendor-delivered (latency n/b)	vendor playback API / flat feeds	SIP license redistribution trace	PIT: replay must preserve original tape sequence/thrust	W3	verify	UNVERIFIED-CANDIDATE
36	finra-equity-short-interest	free public file	FINRA Rule 4560 bi-weekly short interest: twice-monthly settlement (15th + EOM); published 7th business day after settlement; 5y rolling grid + historical files + Equity API	per-issue short positions/ratios	5y rolling grid + historical files	T+7 business days (publication lag)	interactive grid / CSV files / API	free (non-commercial basis stated)	PIT: as-of-join on publication date (T+7bd), settlement date must never be read as availability; Revision Flag tracking; no intra-cycle imputation	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
37	finra-ats-transparency-darkpool	free public file	FINRA OTC Transparency portal (fetched shell) + ATS Transparency catalog entries inside No-Fee catalog	weekly/monthly ATS pairs incl. off-exchange (dark-pool) aggregates	rolling archive depth per item (re-verify)	weekly/monthly publication cadence	portal files/APIs	free	PIT: weekly/monthly roll semantics; pairs-universe rebinding per period; revision re-download	W0	connect	VERIFIED-UPSTREAM-DOC
38	finra-reg-sho-threshold-lists	free public file	FINRA Reg SHO threshold-list page + Nasdaq Trader RegSHOThreshold / ShortSaleCircuitBreaker (nav-verified)	daily symbol threshold lists	site archive depth (re-verify)	daily	scrape (CSV/TXT)	free	PIT: union-with-dual-exchange mapping; gates-join with rows 7 + 36; threshold flags only for flag dates	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
39	finra-daily-short-sale-volume	free public file	FINRA “Daily Short Sale Volume” per-venue daily files (well-known file routes; catalog page not landed this session)	daily venue short-sale volume by symbol	daily (start year claims re-verify)	daily file cadence	file ingest	free	PIT: file cadence + revision practice re-verify; reconcile against ATS Trans-parency aggre-gates (row 37)	W1	connect	UNVERIFIED-CANDIDATE
40	finra-no-fee-catalog-sweep	free public file	FINRA Data Catalog (60 items; No Fee filter; Equity Data category)	sweep/registry	file-registry	catalog-refresh pace (re-verify)	catalog crawl/API	free	PIT: catalog items re-name/deprecate — enu-merate every Equity-Data slug and pin statuses	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
41	databento-xnas-itch	mirror-of-exchange	Databento dataset `XNAS.ITCH` (docs/datasets/xnas-itch fetched): normalized Nasdaq TotalView-ITCH with MoldUDP64 message layer documented + free sample downloads	full schema family (trades / mbp-10 / mbp-1 / tbbo / imbalance / stats / definition)	product-range claims cover 2010s-era onward; exact per-schema bounds via get_dataset_range API (verify)	realtime (Live) + historical batch	Databento lib → Delta/parquet landing, batch by explicit date-window	Databento license (per-GB self-serve + free credits)	PIT: ts_event vs ts_recv dual-timestamps kept; spec-version per day; raw Nasdaq ITCH files (row 4) stay ground truth alongside the mirror (action-note: backfill +)	W1	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
42	databento-nyse-pillar-family	mirror-of-exchange	Databento venue datasets XNYS.ITCH / XASE.ITCH / ARCX.ITCH (+ XNYS.BASIC display variant) listed in venue cata-logue	MBO / tbbo / trades per venue	per-dataset bounds via get_dataset_range (verify)	realtime + batch	same row-41 pipeline forked per venue	Databento license	PIT: Pillar in-sequence venue se-mantics (row 9); never merge NYSE-side venues into one book state (action-note: (phased by venue))	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
43	databento-definitions-schema	mirror-of-exchange (reference/security master)	Databento `definition` schema + DEFINITIONS datasets: security-level corporate actions (splits, dividends, name changes, listings) maintained venue-side	point-in-time definition snapshots per symbol	full-history per venue dataset (verify)	batch	definition snapshots keyed (symbol, eff_ts) → SCD2 spine	Databento license	PIT: effective-dated only — no falling-forward joins; predecessor/successor chains from venue-sourced rows; never mix lookups across venue datasets in one master row (action-note: (security-master spine))	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	databento-equs-mini	mirror-of-exchange (display)	Databento `EQUS-MINI` dataset (Nasdaq Basic w/ NLS Plus mirror; row-3 product) listed in catalogue	L1 quotes + last sale	dataset bounds via get_dataset_range (verify)	realtime + batch	Databento lib	Databento license	PIT: Nasdaq-listed scope only — consolidated NBBO still needs Tape C companion (row 31); display-class entitlement flow-through (action-note: range + quote)	W2	verify	VERIFIED-UPSTREAM-DOC
45	databento-equs-summary	mirror-of-exchange (stats)	Databento `EQUS-SUMMARY` dataset listed — per-session summary stats for the Nasdaq tape	OHLCV/VWAP per-class session bars	per-dataset bounds (verify)	realtime + batch	Databento lib → daily stats tables	Databento license	PIT: session-stat definitions per exchange calendar; derived-aggregation label pinned per run	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	databento-imbalance-schema	mirror-of-exchange	Databento `imbalance` schema documented — normalized auction-imbalance events (NOII family)	auction-window imbalance events	rides rows 41–42 entitlements	realtime + batch	mapper → auction-imbalance tables (row-6 schema)	Databento license	PIT: far/near fields empty before 9:28 / 15:55 ET must map to NULL, not 0 (row-6 rule); auction-type discrimination per event (action-note: build-mapper)	W1	connect	VERIFIED-UPSTREAM-DOC
47	databento-trades-schema	mirror-of-exchange	Databento `trades` schema documented — normalized prints with per-venue trade conditions	last-sale prints	per-venue bounds (verify)	realtime + batch	prints ingest + dedup on ts_recv	Databento license	PIT: venue-only prints ≠ consolidated SIP tape (row 32) — off-exchange S-prints absent by construction; label “not consolidated tape” in schema	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
48	databento-mbp-10-schema	mirror-of-exchange	Databento `mbp-10` schema documented — 10-level price-ladder snapshots + tick deltas	10-deep MBP	widely covered era (futures 2010s; equities ranges per dataset — verify)	realtime + batch	ladder rebuild (snapshot + delta merge)	Databento license	PIT: L3/order-book truth only where raw ITCH/Pillar rows (rows 1/4/9) are loaded; mbp-10 labeled as the coarser derived view	W1	backfill	VERIFIED-UPSTREAM-DOC
49	databento-mbp-1-schema	mirror-of-exchange	Databento `mbp-1` schema documented — L1 quote-change stream with derived best bid/offer	L1 ticks	per-source dataset	realtime + batch	quote-tick ingest	Databento license	PIT: derived view of MBO — label derived; audit against exchange-native L1 (rows 3/26/44) for drift	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
50	databento-tbbo-schema	mirror-of-exchange	Databento `tbbo` schema documented — trades with BBO-at-trade-time	BBO + prints	per-source dataset	realtime + batch	tbbo interval tables	Databento license	PIT: join key = (ts_event, price) pinned per run; treat as derived join — never present as exchange-defined BBO	W2	connect	VERIFIED-UPSTREAM-DOC
51	databento-bbo-1s-schema	mirror-of-exchange	Databento `bbo_1s` schema documented — 1-second interval best bid/offer	1s BBO intervals	coverage era claims (verify per venue)	realtime + batch	bbo interval tables	Databento license	PIT: 1s roll rule pinned per run; free-credit sample per dataset before any paid backfill	W2	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
52	databento-consolidated-nbbo	derived-consolidated	Databento consolidated-NBBO-class product (legacy connbbo-style range; current catalogue status re-verify — product pages not fetched this session)	consolidated NBBO	legacy range era claims (2010s)	historical only (claims)	Databento lib	Databento license	PIT: labeled derived-vs-official-SIP; reconcile against NYSE Daily TAQ NBBO (row 12) before trusting any gap statistics (action-note: status)	W2	verify	UNVERIFIED-CANDIDATE
53	databento-stats-schema	mirror-of-exchange	Databento `stats` schema documented — session stats (open/close, high/low, volume, first/last)	per-symbol session stats	rides venue datasets	realtime + batch	daily-stat tables	Databento license	PIT: stats = exchange-defined session frames, not self-calculated; cross-check vs official daily files (rows 8/14)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
54	databento-live-streaming	vendor streaming	Databento Live API (realtime streaming; per-exchange latency-class claims in docs)	all schemas in realtime	stream (no archive)	realtime	live consumer → event tables; replay-vs-stream compare harness	Databento license (live per-venue subscription)	PIT: ts_recv-vs-ts_event gap monitoring from day 1; per-venue watermarks; capture raw messages before normalization (action-note: connect (parallel to row-41))	W1	backfill	VERIFIED-UPSTREAM-DOC
55	databento-ocea-blue-ocean	mirror-of-exchange	Databento OCEA dataset id listed (Blue Ocean overnight venue; product-naming per venue list — re-verify)	overnight-session orders/trades	venue-era start (2023 claims; re-verify)	realtime + batch	Databento lib	Databento license	PIT: overnight session ≠ regular-session continuity — never bridge across the open/close session gap silently	W3	verify	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
56	databento-cboe-memx-family	mirror-of-exchange	Databento venue dataset ids: MEMX, BATS (BZX), BATY (BYX), EDGA, EDGX, XCIS in venue cata-logue	per-venue MBO/trades/stats	per-venue via get_dataset_range	realtime + batch	Databento lib forked per venue	Databento license	PIT: Cboe book-fusion rules are not defined by Databento — stay per-venue until an exchange license (row 18) defines fusion (action-note: per venue)	W3	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
57	kibot-tick-bidask	vendor archive	Kibot Buy.aspx (fetched): tick data with B/A from 2009; tier bundles $990 / $2,400 / $4,800/yr (vendor claims); all-stocks tick+B/A $9,000/yr	scaled/raw tick	2009→ (tick+B/A)	T+1 batch	SFTP/HTTP flat files	subscription (vendor claims)	PIT: download-date + revision-date columns; “scaled” tick is vendor-adjusted — re-derive from raw where possible; last-quote/settlement defini-tions per Kibot specs (action-note: quote +)	W2	license-review	VERIFIED-UPSTREAM-DOC
58	kibot-1min-bars	vendor archive	Kibot 1-min bars since 1998; all-stocks 1-min $3,000/yr (vendor claim)	1-min bars	1998→	T+1 batch	flat-file loader	subscription	PIT: corporate-action adjust-ment method fixed per file — version it; validate vs free EOD rows (65–66) (action-note: min-bars)	W2	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
59	lobster-academic-lob	academic	LOBSTER academic limit-order-book datasets (per-day LOB slices + message file pairs; JS-blank site this session — re-verify domain & access route)	L1–L10 LOB slices + message file	2010s era (re-verify)	offline batch	flat files + loader via academic creden-tials	academic license (no redis-tribution)	PIT: not a continu-ous feed — venue/window filters per study design; message-file lineage must stay paired with its level-file	W3	verify	UNVERIFIED-CANDIDATE
60	tickdata-inc-archives	vendor archive	Tick Data Inc. (tick-data.com fetched): TickWrite 7 desktop, Web-based, API access, Deep History; parent OneMar-ket-Data/OneTick	tick (+ B/A) + delta/interval products	1980s-depth claims for equities (verify per product)	T+1 batch	TickWrite/API flat delivery	license (quote)	PIT: file-version + re-download policy per product; venue-vs-consolidated pair columns pre-served	W2	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	nanex-nx2-tick	vendor archive	Nanex .nx2 tick archives (order-book + trade frames)	frame files	2010s era claims (re-verify)	offline batch	.nx2 decoder	license	PIT: proprietary-format vendor-lock risk; frame-timing defini-tions re-verify before any latency work	W3	verify	UNVERIFIED-CANDIDATE
62	onetick-platform	vendor platform	OneTick / OneTick Cloud (OneMar-ketData) — tick-analytics + query platform; parent company of Tick Data (row 60)	platform (query/stream)	feeds via Tick Data + own	platform	OneTick query/stream API	license	PIT: as-of-time query primi-tives — keep compute-in-platform vs export lineage explicit in metadata	W3	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
63	wrds-taq	academic mirror	WRDS-hosted NYSE TAQ (Wharton): academic redistribution of the TAQ family incl. Daily TAQ	consolidated day tape	coverage per WRDS catalog (verify)	T+1 pace	WRDS table extract (SAS/CSV)	academic subscription (WRDS + exchange pass-through)	PIT: WRDS vintage tagging per file; revision flags tracked — never merge vintages silently (action-note: (institutional))	W2	license-review	SURVEY-EVIDENCE
64	crsp-us-equities-hist	academic/licensed	CRSP US Stock Databases (daily/monthly/security master/delisted returns)	daily + monthly + security hyster	1926→ (vendor claim)	EOD	CRSP access / WRDS extract	license (CRSP subscription)	PIT: CRSP dataset-vintage editions; delisted-return series is the canonical survivorship fix; price-adjustment-factor chain preserved (action-note: + backfill)	W1	license-review	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	stooq-free-eod	free web CSV	Stooq public CSV end-points (daily EOD; site not fetched this session)	daily EOD	long history (verify span)	daily scrape	HTTP CSV fetch	free / personal-use terms (verify)	PIT: unlicensed-provenance flag — cross-check tier only, never system-of-record; corporate-action adjust-ment basis unknown (re-verify) (action-note: (cross-check tier))	W0	connect	UNVERIFIED-CANDIDATE
66	alphavantage-eod	retail API	Alpha Vantage TIME_SERIES_DAILY(_ADJUSTED) end-points (free-tier caps)	daily EOD	decade-scale (verify per symbol)	daily API	REST	free key + paid tiers (claim)	PIT: adjusted-vs-raw series split explicit; provider outage / rate-limit fallback needed; aggrega-tor prove-nance = licensed-display risk	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
67	polygon-massive-realtime	retail/vendor API	Polygon.io — footer rebrand verified: “© Massive.com, Inc.” (domain/API continuity migration in progress; endpoints still served)	realtime aggregates/WS/REST	per-product (Aggregates 2015→ era claims; verify per endpoint)	realtime + T+1	REST + WebSocket SDKs → event tables	subscription (free tier + paid)	PIT: rebrand = continuity risk — pin base-URL + API-version docs snapshot before building; data plane still Polygon (action-note: (after domain continuity check))	W1	connect	VERIFIED-UPSTREAM-DOC
68	polygon-trades-v2	retail/vendor API	Polygon Trades V2 endpoint (tick prints with exchange/condition codes)	prints	per-product claims (re-verify post-rebrand)	realtime + historical	paginated GET + WS	subscription	PIT: vendor-merged prints ≠ consolidated tape — exchange-id fields must stay; conditions mapping versioned (action-note: (cross-check vs row 47))	W2	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
69	polygon-quotes-nbbo	retail/vendor API	Polygon Quotes V2 / snapshot end-points (NBBO-class)	best bid/offer	per-product claims (re-verify)	realtime + historical	GET + WS	subscription	PIT: vendor NBBO ≠ SIP NBBO — latency-bucket compari-son vs row 12; never merge silently	W2	connect	SURVEY-EVIDENCE
70	polygon-aggregates-historical	retail/vendor API	Polygon Aggre-gates (minute/day bars, adjusted multipli-ers)	bars	2015→ era claims (verify per ticker)	batch	GET grouped/window	subscription (lower tier caps)	PIT: ad-justment multipli-ers ver-sioned; span boundary from pag-ination meta-data, not wall-clock assump-tions	W2	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
71	polygon-options-chains	retail/vendor API	Polygon Options (chains/snapshots, options aggregates claims)	chains + option aggregates	per-product claims	realtime + batch	REST	subscription	PIT: options depth lives in the options-domain census (row 80) — surface only chain-identity fields for equities work (action-note: defer)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
72	alpaca-marketdata-free	retail/vendor-API	Alpaca Market Data API (docs root fetched: Market Data API / Historical / WebSocket sections; /reference page 404 → use docs root + guides)	bars/quotes/trades + streams	free-tier = IEX plan claims; paid = full-SIP claims	realtime + historical	REST + WS SDK	free/paid plans (vendor)	PIT: free tier ≈ IEX-only (~2.5% tape share) — provenance flag mandatory before any tape calc; paid plan = SIP-sourced (verify at contract) (action-note: (free tier))	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
73	tiingo-flat-rate-eod	retail/vendor API	Tiingo (pricing/coverage pages fetched): $30/mo ind + $50/mo commercial flat-rate claims; “IEX Feed”; 30+ yr EOD claims; 10k req/hr	daily EOD + intraday	30+ years (vendor claim)	EOD/API	REST + daily CSV	flat-rate subscription	PIT: IEX-feed provenance disclosure (row-26 gate) — secondary feed only; 30-yr span is a vendor claim — verify per symbol before system-of-record use	W1	connect	VERIFIED-UPSTREAM-DOC
74	finnhub-free-api	retail vendor API	Finnhub REST (quotes/candles/etc.; not fetched this session — scope + rate caps re-verify)	quotes/candles per endpoint	claims (re-verify)	realtime-ish	REST	free-tier claims	PIT: aggregator provenance-gate class — unknown sourcing, unlicensed-display risk (same discipline as rows 72/73 flags)	W3	verify	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
75	twelvedata-free-api	retail vendor API	Twelve Data REST API (claims not fetched this session)	quotes/candles	per-endpoint claims	realtime-ish	REST	free tier + sub-scription	PIT: same aggrega-tor provenance-gate class as row 74	W3	verify	UNVERIFIED-CANDIDATE
76	barchart-on-demand	retail vendor API	Barchart OnDe-mand (vendor claims; not fetched)	quotes/history/f	per-endpoint claims	delayed/rt claims	REST/XML	subscription tiers	PIT: vendor-hedged data (delayed defaults) — verify feed class per endpoint before any PIT use	W3	verify	UNVERIFIED-CANDIDATE
77	ibkr-historical-tws	retail/broker API	Interactive Brokers TWS API historical windows (pacing limits; maturity-of-data rules)	bars/prints by window	per-resolution-capped lookback (maturity rules)	on-demand queries	TWS socket API	brokerage account license (display, non-redistribution)	PIT: pacing limits throttle backfills; depth is per-resolution-capped — log request-window metadata per call (action-note: + proto-type)	W2	verify	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
78	sec-13f-holdings	EXCLUDE (out-of-scope)	SEC EDGAR 13F institutional holdings filings	quarterly positions	long	filings lag	—	public	exclusion: filings/ownership domain, not market-data tape — handled in a separate holdings census; do NOT fold into this file's W0/W1 builds (action-note: move to holdings census)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
79	yfinance-unofficial-scrapers	EXCLUDE	yfinance + similar unofficial scrape routes (Yahoo-class end-points)	mixed	mixed	scrape	—	unlicensed/unstable	exclusion: ToS-violation + silent schema drift + non-PIT adjust-ment lineage — banned from any ingest allowlist or gold-layer feed (action-note: block in ingest allowlist)	W3	connect	UNVERIFIED-CANDIDATE
80	options-domain-pointer	POINTER	OPRA feeds (Databento 0PRA id among venue cata-logue) + Cboe DataShop options sets (rows 18–19 surface only identity here)	options domain	—	—	—	pointer	out-of-scope for equities mi-crostruc-ture — full options census main-tained sepa-rately (action-note: create options-domain census)	W3	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
81	nasdaq-nordic-intl-pointer	POINTER	Nasdaq Nordic + international equities venues (prior-census scope note)	intl eq	—	—	—	per-venue intl license	out-of-scope: US equities file; Nordic datasets enter via the international census (venue-local + delayed-data rules differ) (action-note: maintain in intl census)	W3	connect	SURVEY-EVIDENCE

04 — EU & global-rest equities (72 rows)

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	LSE Data Shop — End-of-Day Summary	historical-archive	londonstockexchange.com Data Shop → docs.londonstockexchange.com HDP_EOD_Summary spec [1]	EOD OHLCV per in-stock cash market, depth sold per window	LSE+Turquoise from Jan-2004	EOD next-session	file purchase + platform download (Data Shop self-service)	commercial license, internal use	exchange-issued OHLC; no PIT ad-justment — corporate-action join needed (action-note: spec already ingested — write COTAHIST-style loader for EOD Summary format)	W2	connect	VERIFIED-UPSTREAM-DOC
2	LSE Data Shop — tick (time & sales)	historical-archive	LSE Data Shop factsheet [2]	tick T&S	LSE from Jan-2004; Turquoise from Feb-2023	N/A (delivery file)	file purchase (CSV bundles, per-period)	commercial license	raw tape, as-traded codes; needs Turquoise currency-table join (action-note: price-sample, then decide tick de-pendency vs EOD-only GFI UK pillar)	W3	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	LSE Data Shop — best prices / full order book	historical-archive	Data Shop factsheet [2]	L1 best prices from Apr-2003; L2 full rebuild from Oct-1997	multi-decade	N/A	file purchase	commercial license	as-traded order book; symbol-consolidation join to LSPD events (action-note: only if GFI UK branch needs mi-crostruc-ture)	W3	connect	VERIFIED-UPSTREAM-DOC
4	LSE Data Shop — corporate actions + reference series	historical-archive	Data Shop cat-alogue (self-service filter per factsheet [2])	event-level	archive sold per window	N/A	file purchase	commercial license	vendor PIT events; reconcile against LSE notices (action-note: catalogue-sample; s corporate-action spine for row 1)	W2	backfill	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	LSE London Data Shop — news datasets (RNS archive product)	news-archive	Data Shop cat-alogue	notice-level	sold per window	N/A	file purchase	commercial license	RNS reconcili-ation against free RNS web UI (action-note: confirm exact RNS-archive SKU before pricing)	W2	connect	UNVERIFIED-CANDIDATE
6	LSE securities master + LSES reference files	reference-master	londonstockexchange.com market-data pages + LSE reference-data specs	instrument-level	current + full/delta	daily	registered download / vendor license	licensed (redistri-bution re-stricted)	ISIN/tradable-code joins; daily deltas must ack before replace (action-note: fetch LSE reference-data spec PDF; wire delta poller)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
7	LSE RNS feed (live)	news	londonstockexchange.com news / RNS	exchange real-time	web archive recent years	near-realtime site	web scrape (unofficial) or RNS Select subscription	site TOS grey; ⚠ decisions pending	dedupe by announcement id; timezone normalisation (action-note: if Data Shop news SKU exists (row 5) prefer it; scrape only as fallback)	W1	connect	UNVERIFIED-CANDIDATE
8	Turquoise MiFID II files	post-trade-equity	Turquoise Reference Repository “MiFID II Files” (verified in site nav)	per-venue transparency	rolling web	batch	anonymous web files	free (MiFIR Art.13/14 compliance data)	venue-issued; 15-min disclosure rules apply downstream (action-note: wget mirrors once Euronext EU-equivalent RTs land)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
9	LSPD — London Share Price Database	historical-archive	London Business School via WRDS	daily/monthly	1955→ (sample), daily series from Jan-1984 [4]	academic refresh irregular	institutional WRDS login	academic license, no redis-tribution	LBS research-grade with delist-ing/capital returns; best sur-vivorship spine for UK (action-note: partner affiliation or ; else mimic from rows 1+4)	W2	license-review	SURVEY-EVIDENCE
10	TRADEcho (LSE trade re-porting) data — Data Shop scope	post-trade-equity	Data Shop cat-alogue (fact-sheet [2] names TRADE-cho as included venue)	transaction-level	window purchase	N/A	file purchase	commercial license	SIs+venues pot; good PIT vs STS/ATS reports (action-note: only if ADR/MiFID transaction-pool refresh needed)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	Deutsche Börse MD+S — CEF realtime feeds (CEF-XTF Xetra / CEF-EFF Eurex)	realtime-tape	mds.deutsche-boerse.com CEF product sheet [5]	tick L1/L2	realtime config	realtime	licensed API con-nectivity	commercial license (redistri-bution contract-gated)	contract-gated; feed contract must pin band-width windows (action-note: only if realtime DE branch funded)	W3	connect	VERIFIED-UPSTREAM-DOC
12	Deutsche Börse MD+S — historical market data (Xe-tra/Eurex/futures tick+1-min)	historical-archive	MD+S historical cata-logue + Host File Service [5]	tick–1min	Xetra/Eurex full session depth, multi-decade	batch	file server/host delivery	commercial license	seller-side data; PIT join to reference data products (action-note: for EUR-vintage Xetra EOD + 1-min: GFI DE pillar leapfrog)	W3	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
13	Deutsche Börse Data Shop / Statistics down-loads	EOD-stats	cashmarket.deutsche-boerse.com Market Data + mds popular topics ([5], [6])	EOD	Xetra/FSE since 1997, monthly series deep	EOD after session	anonymous web PDF/CSV or Prod-uct+Services cata-logue	free statistics (redistri-bution re-stricted)	official turnover/stats; no price PIT (action-note: mirror monthly Xetra statistics CSVs)	W0	connect	VERIFIED-UPSTREAM-DOC
14	Eurex historical time series (TSS bundles + settle-ment statistics)	historical-archive	eurex.com statis-tics/menu (reach-able via MD+*, [5])	EOD after session	multi-decade	EOD	licensed file	commercial license	per-contract fixes; roll chains must map to expiry calendar (action-note: iff GFI DE-futures row is built)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
15	SIX Swiss Exchange — trading data + MiFID II export drive	EOD-intraday	six-group.com market-data tree (MiFID II flag export facility)	EOD+tick licence	multi-decade	EOD+	license portal + SFTP	commercial license	SIX MiFID flag per instrument drives PIT classification (action-note: request SIX data-drive samples; MiFID-flag semantics vs FITRS)	W2	verify	UNVERIFIED-CANDIDATE
16	BME — trading data files + BME Data Hub	EOD	bme.es market-data (CSV,“BBDD” files)	EOD	1990s→	EOD	registered file delivery	commercial license (redistribution restricted)	Spanish-market EOHLC; separate BME corporate-events files for CA joins (action-note: archive-fetch converter for BME CSV layout)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
17	BME Latibex (IBEX→LatAm cross-listings)	EOD	bme.es Latibex pages	EOD	2001→	EOD	same channel as row 16	commercial license	duplicates BMV tickers — dual-listing dedupe rule (action-note: skip unless GFI cross-listing branch opens)	W2	connect	UNVERIFIED-CANDIDATE
18	Euronext — historical market data files (per country)	historical-archive	euronext.com data-offering (portal behind Connect login; per-Paris/AMS/Milan/Lisbon/Oslo)	EOD→tick	multi-market multi-year	batch	Connect2 portal or vendor	commercial license	sold per-market; solve PIT via Euronext CA files (action-note: obtain Paris vendor first; portal creden-tials check)	W2	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
19	Euronext — company regulated news archive	news-archive	euronext.com/en/products/equities/company-news-archive (live nav)	per-issuer level	rolling web archive	near-daily	web scrape/API	free, TOS-restricted	announcement dedupe vs prior EU rows (action-note: scrape feed; map to GFI news schema)	W0	connect	VERIFIED-UPSTREAM-DOC
20	Euronext Dublin — Daily Official List (DOL)	EOD-reference	euronext.com/en/markets/dublin/daily-official-list (live nav)	instrument-level official list	historic series	EOD	web	free-ish, terms per page	official DOL is the PIT marker for Dublin-listed instruments (action-note: fetch DOL PDF; add to reference spine)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
21	live.euronext.com — jerboa quote/ENU API	real-time-quotes	live.euronext.com end-points (public keys)	15-min delayed spot	online only	15-min	REST JSON	free delayed (MiFIR 15-min rule)	quoting end-points and dis-closure rules differ per market (action-note: poller + cache; respect per-market disclo-sure)	W1	connect	UNVERIFIED-CANDIDATE
22	Nasdaq Nordic + Baltic — Histori-calView	historical-archive	nasdaq.com Nordic-Baltic Histori-calView product [7]	EOD daily files	eq/derivatives/ETF/index, multi-market multi-decade	EOD	paper order + file delivery	commercial license, **internal-use only clause** [7]	direct exchange-issued; PIT from Nasdaq CA files (action-note: signed agree-ment then)	W2	backfill	VERIFIED-UPSTREAM-DOC
23	Nasdaq Nordic — File Delivery Service (SFTP subscrip-tions)	EOD-intraday	nasdaq.com/product-levels/data/nordic-baltic/nordic-file-delivery-service [7]	file drops, daily	trading-day	EOD	SFTP sub-scription	commercial license	file-cadence SLA; missed-file alarm required (action-note: only if Nordic branch funded)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
24	Nasdaq Nordic free nominal-data files (globe-newfiles path)	EOD-free	globenewfiles.nasdaqomx.com public dirs (nominal data)	EOD blocks	recent files	EOD	anonymous web	free nominal (license posted per file)	re-verify robots + file-naming tem-plates (action-note: probe directory + build sched-uler)	W1	connect	UNVERIFIED-CANDIDATE
25	GPW — offered value-added data (CSV/XLS daily after close)	historical-archive	gpw.pl value-added-data-and-indicators page — CSV/XLS delivered daily after session via mail/SSL [8]	EOD full tape deriva-tives/instruments	archive sold per date window	EOD	e-mail/SSL file delivery	licensed (some free statistical rows, some paid)	daily session issue; PIT CA joins by GPW CA files (action-note: hook SSL delivery; reuse COTAHIST-style OHLC loader across venues)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
26	GPW — free daily trading-data files (session results archive)	historical-archive	gpw.pl/gpw-statistics + gpw.pl/market-data pages [9]	EOD blocks	multi-year	EOD	anonymous web files	free, redistri-bution restricted	official figures; limited per-instrument daily files to-date — confirm span (action-note: fetch daily session files; confirm earliest)	W0	backfill	SURVEY-EVIDENCE
27	GPW NewCon-nect + Catalyst files	EOD	gpw.pl market-data tree (NewCon-nect, Catalyst, TBSP)	EOD	mixed	EOD	same channel as row 25/26	licensed/free mix	separate listing-source registry (action-note: subscribe once cascade from row 25 is running)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
28	GPW Benchmark — WIBOR/WIBID historical CSV	reference-rates	gpwbenchmark.pl/en-dane-historyczne ([10]: SSL/mail CSV delivery 9:00 next business day)	daily fixing	full benchmark history	T+1	e-mail/SSL CSV	licensed benchmark (BMR-regulated)	rate PIT: use fixing-issued csv only, no re-cast (action-note: register supply channel; loader for fixing series)	W1	connect	VERIFIED-UPSTREAM-DOC
29	Prague SE (PSE) — trading data `dll` historical files	EOD-cz	pse.cz trading-data pages	EOD	multi-decade	EOD	registered/anonymous web	free-ish	Czech-language file specs; anchor legacy ticker codes to ISINs (action-note: file cadence + history window)	W1	verify	UNVERIFIED-CANDIDATE
30	Budapest SE (BSE) — data files	EOD-hu	bse.hu / data-bse.hu	EOD	multi-year	EOD	web files	free-ish	HU tax/foreigner codes (action-note: probing fetch)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
31	Wiener Börse — Data Services	EOD-at	wienerborse.at data-services	EOD	multi-market AT/CEE	EOD	account + file delivery	commercial license	separate index+prim-listed catalogs (action-note: info fetch; CEEplus via GPW cascade better?)	W2	license-review	UNVERIFIED-CANDIDATE
32	Luxembourg SE — market data	EOD-lu	bourse.lu data	EOD	shallow	EOD	web	free-ish	smallest EU node — buy-only-if-needed (action-note: skip until branch scope fixed)	W2	connect	UNVERIFIED-CANDIDATE
33	Athens SE — analytical data products	EOD-gr	athexgroup.gr data sections (linked from Euronext nav [3])	EOD	Greek crisis eras	EOD	web/API license	commercial license	separate Athens Group entity, Greek-char tradids (action-note: cata-logue fetch (GR/EN))	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	MOEX ISS — historical candles/history endpoints (per-endpoint census)	historical-archive	iss.moex.com per-endpoint tree /iss/history/engines/… (endpoint census from upstream docs + community API lists) [11]	EOD history endpoints + candles	official archive per-board incl. history_from per security	delayed free / realtime paid	REST GET, anonymous or auth'd	⚠ informational — commercial reuse needs written MOEX contract [11]	free tier = **familiarisation only**; no real-time/PIT entitlement without contract (action-note: **sanctions/jurisdiction gate first**; only then map each engine/market endpoint)	W0	connect	VERIFIED-UPSTREAM-DOC
35	MOEX ISS — securities/boards metadata (snapshot + per-item metadata)	reference-master	iss.moex.com securities/…, boards/… endpoints [11]	instrument-level	full master + history_from	realtime (delayed ok)	REST JSON rows	⚠ same informational gate	per-board mapping table required; MOEX metadata drifts daily (action-note: same jurisdiction gate as row 34)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
36	SPB Exchange (spbexchange.ru) — equities EOD/API	EOD	ISS-like endpoints / site downloads	EOD	multi-year	delayed	REST/web	⚠**sanctioned/EU-flagged** (2022+ restriction cycle) — informational flag only	**sanctions-screening before ANY compute** (action-note: **skip until sanctions review clears**; log flag in catalog)	W3	connect	UNVERIFIED-CANDIDATE
37	ESMA FIRDS — instrument reference data (full + delta files)	reference-master	registers.esma.europa.eu publication + official `esma_data` loader [12]	instrument level (full/delta)	since 3-Jan-2018 launch [13]	next-day (delta files)	anonymous machine-readable file downloads	free (regulator-published)	authoritative EU instrument spine; LEI joins from GLEIF (row 38); completeness caveats published by ESMA [13] (action-note: wire daily full+delta poll; join to GLEIF + row 8/40)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
38	GLEIF LEI — Golden Copy + Delta + Concate-nated + GLEIF API	reference-master	gleif.org Golden-Copy download page (3×/day + 4 delta windows) [14]	entity-level	pre-2017 issue dates → present, historic business-report archive	T+0 (3×/day)	anonymous file download + REST API	free, open data	LEI PIT — use delta file windows (8h/24h/7d/31d) for as-of-time joins (action-note: daily delta scheduler + as-of-time index)	W0	connect	VERIFIED-UPSTREAM-DOC
39	ESMA EMIT — MiFID II transaction-report data (data.esma.eu, UR2)	post-trade-equity	data.esma.eu EMIT portal (fetch blocked this session)	transaction-level	archive per UR	periodic publishes	anonymous file down-loads	free regulator data, heavy reuse TOS	**known PIT/completeness caveats** from ESMA [13]; big files (action-note: retry data.esma.eu from different egress; park UR citation)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
40	ESMA FITRS — MiFIR trans-parency data (pre/post-trade) publica-tion	post-trade-equity	registers.esma.europa.eu trans-parency files (`citation_date/latest.zip`, registers gate 502 this session)	period-level equity trans-parency data	historical files published multi-year;	periodic	anonymous file down-loads	free regulator data	venue/lagging time-window reconcili-ation against venue RTs (row 8, 68) (action-note: re-attempt portal fetch; snapshot old-est→newest files)	W1	connect	UNVERIFIED-CANDIDATE
41	STS — in-strument national-Competent-Authority venue lists (TRV)	reference-master	national CA published venue/MIC supersets (esma indexes)	venue-level	snapshot-per-update	ad hoc	web	free	cross-check against opera-tional venue/MIC reference (action-note: consoli-date NCA venue lists into `venues.csv`)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
42	CboE Europe — RTS13 public trade-data files (15-min delayed)	historical-archive	cboe.com/europe/equities/trade_data — free CSVs hourly wheel `rts13_public_trade_data_dxe_YYYY-MM-DD_HH.csv` [15]	transaction-level hourly buckets	online	hourly-after publica-tion	free wget automa-tion (sample com-mands on page)	free (MiFIR 15-min data-use clause), TOS-abide	per-venue ESMA RT-13 data; strong PIT vs SIs (action-note: set wget job (pattern cfg on page); nightly Δ aggrega-tor)	W0	connect	VERIFIED-UPSTREAM-DOC
43	B3 CO-TAHIST — historical quota-tions (daily/monthly/yearly)	historical-archive	b3.com.br historical-quote-data ([16]: prices since 1986, original-currency/format **without** adjust-ments)	EOD per instru-ment	since 1986	post-session (daily), retro batches monthly/yearly	anonymous download (fixed-width CO-TAHIST file)	public published data; redistri-bution restricted	unadjusted contexts — corporate-action chain must be designed over BVBG-086 + CA files (action-note: ingest daily CO-TAHIST; monthly+yearly to 1986)	W0	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	B3 BVBG-086 — market price report files	historical-archive	B3 historical-data tree (b3_bvbg-086 upstream template in rb3 ODM [17])	daily price-report	daily-to-retro file window	EOD	anonymous/public file	registered redistri-bution restricted	venue-issued CSV/XML; feeds CA spine for row 43 (action-note: loader + CA deriva-tion job)	W0	connect	VERIFIED-UPSTREAM-DOC
45	B3 — daily bulletins & settle-ment prices (de-rivs/fx/fi)	EOD-stats	b3.com.br historical data tree — Daily bulletins / Settle-ment prices / FX retro [16]	EOD	multi-decade	EOD	anonymous web	free/published	settlement-price PIT; separate fx-summary mixing rules (action-note: mirror per-section file trees)	W0	connect	VERIFIED-UPSTREAM-DOC
46	B3 — Market Data Platform (realtime, via dis-tributors)	realtime-tape	b3.com.br market-data-platform [18]	realtime L1/L2	realtime config	realtime	licensed distribu-tor contract	commercial license	per-product broad-cast; SLA off-wire (action-note: only if BR-branch funded)	W3	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
47	B3 — index portfolio/composition files (b3-indices templates)	index-components	B3 indices-files (rb3 template catalog row b3-indexes-composition + historical portfolio) [17]	composition per month	historical portfolio archive	monthly	anonymous web	published	index-membership as-of = file publication date (monthly PIT) (action-note: parse monthly portfolio + composition history)	W0	connect	VERIFIED-UPSTREAM-DOC
48	BMV — Grupo BMV daily CSV/DB market-data products (INTRA / SiBolsa / Bases de Datos)	EOD-intraday	bmv.com.mx/information-products/market-data (INTRA) + market-data contact [19]	EOD→intraday	deep (per-product)	EOD	licensed product delivery	commercial license	BMV tax codes Serie/Servicio list mapping (action-note: request INTRA product sheet + sample)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
49	Grupo BMV — weekly/annual statistical bulletins	historical-archive	bmv.com.mx press-room (annual+weekly statistical-bulletin downloads) [19]	period EOD	multi-year archive on site	weekly/annual	anonymous file download	public	primary for index/turnover reconstruction; equity CA joins still needed (action-note: ingest weekly bulletin files; keep annual snapshot)	W0	connect	SURVEY-EVIDENCE
50	BIVA Mexico (merged into BMV)	EOD-mx	historical BIVA files / BMV historical archive	EOD	BIVA listed 2018→	closure	web	license	**venue-merge deprecation** — join BIVA series into BMV master before 2025 cross-impl (action-note: confirm merge-date & file availability)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
51	Chile CMF — market-instruments securities/turnover datasets	EOD-numeric	cmfchile.cl statistics (institutional satellite files)	EOF	multi-year	T+1	anonymous/reg web	registration regulator data (non-market-data redistribution caveats)	turnover standard incl. institution dashboard rows (action-note: fetch stats page + sample)	W1	connect	UNVERIFIED-CANDIDATE
52	CSE Chile — premium market data (historical, licensed)	EOD-cl	cse.cl licensed files (CMF cross-checked)	EOD	announced products	T+1	license portal	commercial license	official license gate for redistribution (action-note: sample request)	W2	quote	UNVERIFIED-CANDIDATE
53	Colombia BVC — daily market files	EOD-co	bvc.com.co Files-Trading	EOD	multi-year	EOD	registered web	free-ish	limited vendors for processed history (action-note: probe files; MARAVATIA rows)	W1	connect	UNVERIFIED-CANDIDATE
54	Peru BVL — COM/CAP daily quotation files	EOD-pe	bvl.com.pe downloadable quotes	EOD	multi-year	EOD	anonymous web	free-ish	symbol-join rules (COM vs CAP) (action-note: probe file pattern)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	Argentina BCBA/CASLA — EOD down-loads + CAVAL	EOD-ar	bcba.com.ar / caslasite / lape-nas… file trees	EOD	1980s→ but gap-riddled	irregular	web + manual	free-ish ⚠	worst data-gap regimes of the set — **flag as out-of-scope unless hard-need** (action-note: skip; revisit only with LatAm-4-countries stage)	W3	connect	UNVERIFIED-CANDIDATE
56	JSE — market data portal + down-loadable files	EOD-za	clientportal.jse.co.za downloadable-files (portal files; login-gated) [20]	EOD	per product	EOD	account + file/API license	commercial license	ZAR-currency scripts; race dedupe (action-note: portal account + file list)	W2	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
57	JSE — licensed networks (ICE/ICE Data); JSE Market-Related via vendors	realtime/historical	ICE Developer Portal Entry (JSE) etc.	realtime+historical	~10 yr	near-realtime	commercial license (ICE contract)	commercial license (dual-class JSE product & vendor resale)	PIT redraw from ICE consolidated feed (action-note: skip incremental; revisit when ZA premium buyer exists)	W3	connect	UNVERIFIED-CANDIDATE
58	NSE Nigeria — daily quotes window	EOD-ng	nse.com.ng daily files (XLS)	EOD	rolling years	EOD	web scrape	free-ish	NSE file-quality rules; not NSE-India (action-note: probe cadence)	W1	connect	UNVERIFIED-CANDIDATE
59	EGX Egypt — daily market archives + disclosure	EOD-eg	egx.com.eg statistics + disclosure pages	EOD	moderate archive	EOD	anonymous web (unofficial) ⚠	site TOS; no SLA	USD/EGP and listing-restriction flags; wash dedupe (action-note: probe; use only if GFI needs MEA breadth)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
60	Nairobi NSE Kenya — daily data	EOD-ke	nse.co.ke data pages (listed-co stats)	EOD	portal-depth unknown	EOD	web	free-ish	same treat-ment as row 59 (action-note: probe; catalog-light)	W2	connect	UNVERIFIED-CANDIDATE
61	ASE Casablanca — official quota-tions/notes files	EOD-ma	casablanca-bourse.ma (or official version site)	EOD	archive on-site official images	EOD	PDF/CSV web	free-ish	image-release → OCR risk; keep flattened site variant (action-note: official file server (bourse-cma))	W2	verify	UNVERIFIED-CANDIDATE
62	Ghana GSE — daily lists	EOD-gh	gse.com.gh pages (list+changes)	EOD	shallow	EOD	web	free-ish	symbol drift; NII (action-note: probe)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
63	TSX/TSXV — Datalinx Trading Summary (TSX daily + V-file for TSXV)	historical-archive	tmxwebstore.com TSX-trading-summary-daily ([21]: EOD fields incl. O/H/L/C, volume, bid/ask, **history from Oct-1998 TSX / Mar-2009 TSXV**)	EOD per security	from Oct-1998 (TSX), Mar-2009 (TSXV)	daily file 18:00 ET	webstore subscription	commercial license, internal-use oriented	single daily file per venue; seasonal contract renewal (action-note: product-spec sample; fit into GFI CA ingest)	W1	connect	VERIFIED-UPSTREAM-DOC
64	TSX/TSXV issuer-side daily files (Daily Summary, Daily Historical Trading Activity, CDNX archive files)	historical-archive	tsx.com TSXV market-information (files description) [21]	EOD	CDNX composite to Dec-2001 (archived)	daily	Datalinx subscription	commercial license	same spine as row 63; CDNX archived set needed for pre-Sep-2001 GFI (action-note: order CDNX historical file for pre-Dec-2001)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	TSX/TSXV — NCIB corporate-action files (daily/monthly)	corporate-actions	tmxwebstore NCIB-listed-issuer product (daily for TSX+TSXV, monthly file released 3rd-last business day) [21]	CA-event	rolling + archive	monthly/daily	webstore subscrip-tion	commercial license	issuer-reported data delays; pivot-day load (action-note: parse into GFI CA spine)	W1	connect	VERIFIED-UPSTREAM-DOC
66	TMX — Daily Record (bulletins, dividend notices, most-active, index sum-mary)	corporate-actions/journal	tmxwebstore Daily Record ([21]: history back to 1998)	daily record	since 1998	08:30 ET daily	webstore subscrip-tion	commercial license	text-file OCR-style file; parse ro-bustness (action-note: order sample; convert to GFI event log)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
67	Montréal Exchange — daily derivative named constant data files	EOD-mx-derivs	m-x.ca/en/trading/data/historical (files **not official end-of-day source** per page note)	EOD	session history only	EOD	anonymous web	free (general-information basis only)	**explicitly non-official** — do not retain as compliant copy of record (action-note: use only as fallback; official copies via TMX Datalinx)	W2	connect	VERIFIED-UPSTREAM-DOC
68	SEDAR+ — Canadian issuer filings portal (disclosures)	filings-can	sedarplus.ca new portal	filing	1997→ archive	rolling	web/API pending bulk data-access	free for search; structured extraction manual ⚠	JS-only portal (action-note: migrate old-SEDAR files → bulk; filings-domain overlap — cross-ref the 02 fundamentals/filings census)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
69	TASE — Data Hub API (Israeli market data)	EOD-IL API	tase.co.il/en/content/products_services — API + daily Excel reports [23]	end-of-day + historical	full per-product archive	EOD/on-demand	API sub-scription	commercial license, TASE gated	TASE codes; ILS rounding; payment calendar (action-note: Data Hub subscrip-tion tier; build loader)	W1	connect	VERIFIED-UPSTREAM-DOC
70	TASE — Data Store (data files / reports / one-time historical data)	historical-archive	tase.co.il Data Store — one-time files, historical datasets, monthly plans [23]	file-level	per-product history archive	ad hoc/orders	purchase + download	commercial license	CSV/Excel layouts — verify per product from store samples (action-note: choose historical-data SKU; load)	W1	backfill	VERIFIED-UPSTREAM-DOC
71	NZX — market data portal + data services	EOD-nz	nzx.com data/market-data sections	EOD-licensed	mid-90s→	EOD	file + license	commercial license	NZ tick data format (CSV? channel-specific) (action-note: d sample)	W2	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
72	ATHENS-group pure data-portal republish (out of GFI-scope mirror check)	audit-only	cross-check row 33 vs Euronext nav [3]	—	—	—	—	—	conflicting licence paths — **do not auto-merge** with row 33 source (action-note: mark duplicate-entry check)	W3	connect	UNVERIFIED-CANDIDATE

05 — Fundamentals · filings · PIT (121 rows)

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	SEC EDGAR company-submissions API	PIT-SPINE	data.sec.gov per-filer JSON + nightly submis-sions.zip	per-filing index (accession-keyed)	all EDGAR filings ≈1993→present	T+0 dissemi-nation; zip nightly ~03:00 ET	REST + nightly bulk zip	free / US public domain	cap User-Agent string; 10 req/s; join ev-erything by acces-sion no. (action-note: nightly CIK-canvas pull → filings spine shared with rows 2–9)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
2	SEC EDGAR company-facts XBRL	PIT-SPINE	data.sec.gov/api/xbrl/companyfacts/CIK#.json	per-filer (tag, unit, end/fy/fp/**filed**)	XBRL facts 2009+ (iXBRL era)	<1 min from dissemination	REST + company-facts.zip nightly	free	filed-date per fact ⇒ restatement time-series reconstructable free; pin accession per fact (action-note: primary numeric fundamentals spine for US/backtest)	W0	connect	VERIFIED-UPSTREAM-DOC
3	SEC EDGAR company-concept	PIT-SPINE	data.sec.gov/api/xbrl/companyconcept/CIK#/us-gaap/TAG.json	per-concept full time-series per CIK	2009+	<1 min	REST	free	same PIT gate as #2; taxonomy-name drift (us-gaap vs ifrs-full) (action-note: cheap targeted-tag pulls on hot signals)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
4	SEC EDGAR frames API	PIT-SPINE	data.sec.gov/api/xbrl/frames/us-gaap/TAG/USD/CY####Q#.json	calendar-period cross-section per metric	2009→present	T+0	REST	free	frames mix latest filings into a quarter — re-derive strict PIT by filed-dates from #2 (action-note: ready-made cross-sectional snapshots for screens (live only))	W0	connect	VERIFIED-UPSTREAM-DOC
5	SEC EDGAR full-text search (EFTS)	PIT-SPINE	efts.sec.gov/LATEST/search-index?q=&forms=&dateRange=	full-text hits	2001→present	T+0	REST (rate-limited)	free	text w/o structure; keep hits keyed to accession #s (action-note: event/keyword scanner corpus (8-K text, risk language))	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
6	EDGAR full-index nightly + Archives backfile	PIT-SPINE	www.sec.gov/Archives + indexes (master daily, CY/QTR full-index zips)	filing-level index universe	full machine-readable archive 1993→present; pre-1993 in Archives only (sparse/legacy)	T+1 (index file night after filing)	file-drop dirs + zips, no key	free	pre-1993 hole = survivorship-first backfile (pair with #117/#121) (action-note: enumer-ate everything-submitted back-ground job; popu-lates CSS dedupe)	W0	connect	VERIFIED-UPSTREAM-DOC
7	EDGAR 10-K per-item parse	SPINE-PARSER	primary docs → Items 1/1A/7/8 + segment data via iXBRL tags + NLP chunking	per-item per-filing	HTML-era ≈1996+; iXBRL 2009+	T+0/T+1	doc fetch + own parser (no API for “items”)	free	item structure not machine-stable pre-iXBRL; tag drift year-over-year (action-note: risk-factor/MD&A qualita-tive PIT extrac-tion)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
8	EDGAR 10-Q per-item parse	SPINE-PARSER	same route as #7, form=10-Q (incl. 10-Q/A)	per-item per-filing	≈1996+	T+0/T+1	doc fetch + parser	free	quarterly item granularity finer; amendments create vintages (action-note: intra-year fundamentals + going-concern language)	W1	connect	SURVEY-EVIDENCE
9	EDGAR 8-K per-item parse	CA-SPINE	same route, form=8-K; Item-code filter (1.01, 2.02, 2.03, 3.03, 4.02, 5.02, 8.01…)	per-item per-filing	1996+; item codes standardized 2004 (pre-2004 numeric 1–9)	T+0	doc fetch + item-map parser	free	item-code regime changes Aug-2004 → two-phase parser; 4-day filing window ≠ T+0 (action-note: event spine for CA/material-contract triggers)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
10	SEC Financial Statement Data Sets (Dera FS-DS)	SPINE-DERIVED	sec.gov/dera/data/financial-statement-data-sets (num/pre/sub/txt CSVs)	quarterly fact/period.csv	2009 (Q1)→present	quarterly batch, ~1 mo after quarter	file-download, no key	free	derived from as-filed XBRL = filing-date safe; but lag makes it a baseline not the live spine (#2 outruns it) (action-note: quick standard-ized ETL baseline; jumpstart cross-sectional labs)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	DEF 14A proxy state-ments	REGISTRANT	EDGAR form filter (DEF 14A/14C) + exhibit parse	per-item: exec comp, propos-als, votes	≈1997+	T+0/T+1	filings + parser	free	pay-date ≠ filing-date (vote BEFORE meeting but after record); PIT = filing date (action-note: gover-nance/comp factor layer)	W2	connect	SURVEY-EVIDENCE
12	S-1/F-1 prospec-tuses (incl. 424B4s)	REGISTRANT	EDGAR form filter (S-1/-A, F-1/-A, 424B4)	per-filing: proceeds, shares, lead under-writers, cliff table	1993+	T+0	filings + extrac-tion (parties, goals)	free	two-stage vintages (pre-pricing vs priced 424B4) both belong in corpus (action-note: IPO pipeline + issuance-event layer)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
13	20-F/40-F foreign-issuer XBRL	REGISTRANT	EDGAR form filter; IFRS tax-onomy facts (ifrs-full) + 6-K stream	per-fact/filing (IFRS)	iXBRL/XBRL ≈2011+; text filings 1990s+	T+0	filings + XBRL parse	free	IFRS tag coverage thinner than US-GAAP; currency-heavy (action-note: international-access schema (ADR/US listings))	W1	connect	SURVEY-EVIDENCE
14	13F-HR quarterly institu-tional positions	POSITION-REG	EDGAR 13F efilings (13F-HR/-A XML; no official SEC bulk API — WRDS/reseller bulk needed)	per-manager per-holding per-quarter	positions since 1980s (digi-tized); struc-tured clean ≈2013→ present	filings by T+45d after quarter-end	EDGAR parse + vendor bulk	free (public info)	45-day lag = step-event, by-design stale; 13F-HR/A restate-ments; holdings ≠ market value truth (action-note: quarterly institutional-layer frame (screen, crowd-ing))	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
15	13D/G beneficial ownership (>5%)	POSITION-REG	EDGAR form filter (SC 13D/13G/-A)	event-level ownership snapshots	1970s era covered; machine-readable bulk via accumulated filings	T+0–T+2	filings + parser	free	“calculation date” can precede filed date — use calc-date for position, filing for public knowledge (action-note: activism/intent signals)	W1	connect	SURVEY-EVIDENCE
16	Form 3/4 (SEC insider transactions)	POSITION-REG	EDGAR + data.sec.gov submissions → Form 4 XML exhibits	transaction-level (shares, price, role, relation)	electronic ≈1991→ (paper 1980s in Archives)	T+0/T+1 (2-BD deadline)	XML exhibits parse	free	amendments v4/A create vintages; dense footnote text; affiliate ≠ director-only (action-note: insider-flow factor + directors/officers lattice)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
17	SC 13E-3 (going private)	POSITION-REG	EDGAR form filter	deal-stage filings + exhibits	1985+	T+0	filings + stage parser	free	multi-stage (prelimi-nary→final) — stage-keyed (action-note: M&A/arb event set)	W2	connect	SURVEY-EVIDENCE
18	SC TO tender offers	POSITION-REG	EDGAR form filter (SC TO-I/T, TO-C) + 425 comms	per-offer per-day (shares sought)	1990s+	T+0	filings + parser	free	pairs with DST offer expiry clock (action-note: arb pack feeder)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
19	Form SHO short-sale reports	POSITION-REG (NON-PUBLIC)	submitted to ex-changes/FINRA — **not publicly dissemi-nated at firm level**	weekly aggre-gate regula-tory	never public	n/a	NOT FETCH-ABLE	regulatory-confidential	public proxies only: FINRA daily short-sale vol (#25), 4560 (#26), Nasdaq-Trader threshold list (#49), SEC FTD bi-weekly files (action-note: investi-gate FTD/threshold compos-ite as the public SHO layer)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
20	Form D / Reg D offerings	REGISTRANT	EDGAR form filter + SEC Reg D tables (sec.gov/regd offerings data)	per-offering (amount raised, investor count)	electronic 2008→ (paper pre-2008 not EDGAR)	T+0 filings; reg-data refreshes n-yearly	filings + CSV	free	2008 electronic-divide = backfile cliff; PERMIT = offering, not listing (action-note: private-market issuance tracker (not securities uni-verse))	W2	connect	SURVEY-EVIDENCE
21	Form ADV (IA filings)	REGISTRY	adviserinfo.sec.gov (IAPD portal) + reports.adviserinfo.sec.gov archives (bulk ADV files)	per-firm per-section (AUM, fees, custody)	bulk archives back to early 2000s; annual/amend cadence	T+0 portal; archives T+~annual	portal + export files (XLSX/XML)	free / public	firm-level (not security-level); CRD-keyed; 403 on reports host for raw bots → use export files (action-note: adviser entity-master + AUM signal)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
22	Form PF (private funds)	REGISTRY NON-PUBLIC	SEC-held; aggregate stats published only	quarterly/annual per-fund	2012+	n/a	NOT FETCHABLE (aggregate via SEC reports)	regulatory-confidential	primary research value is SEC's own aggregate releases (action-note: skip unless SEC aggregate feed sparks a thesis)	W3	connect	SURVEY-EVIDENCE
23	Form 13H (large trader)	REGISTRY	EDGAR form filter	registration/assignment level	2013+	T+0	filings	free	few IDs float publicly; identifies big-flow actors (action-note: optional big-trader label join)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
24	EMMA / MSRB municipal disclo-sures	FIDUCIARY-MUNI	emma.msrb.org (muni state-ments, continu-ing disclo-sures, official state-ments; MSRB Gateway for 15G)	per-issue (CUSIP6 govern-ment class)	2009 (EMMA launch)→; pre-2009 scattered via issuers	T+0 portal	portal/HTML search; gateway APIs limited	free / public data	⚠ host blocked auto-mated fetch (403 this session) — ToS-pacing; CUSIP-gated by design; muni facts ≠ equity-equivalent facts (action-note: muni-fiduciary layer if credit stream opens)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
26	FINRA Rule 4560 consolidated short interest	POSITION-REG	finra.org/finra-data/browse-catalog/equity-short-interest	per-symbol bi-monthly (15th & EOM settlement)	5 rolling years online; longer via historical files/WRDS	published 7th BD after settlement (T+2 bi-monthly)	web files + free-tier API (non-commercial)	free non-commercial / commercial entitlement	step-events not series; Revision Flag amends; OTC-covered; missed-venue bi-monthly from NYSE/ARCA files (action-note: bi-monthly short-interest panel)	W1	connect	VERIFIED-UPSTREAM-DOC
27	Motley Fool earnings-call transcripts (free)	TRANSCRIPTS	fool.com/earnings/call-transcripts (HTML, no auth)	per-call HTML	≈2003→present for large caps; continuous	posted within hours of call	scheduled HTML fetch + cache	free-to-read; publisher copyright — internal use only, no redelivery	⚠ no bulk API; robots/ToS respect; small-cap gaps (action-note: free-tier transcript engine; pair with CapIQ if budget lands)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
28	SeekingAlpha tran-scripts (paid)	TRANSCRIPTS	Seekingalpha.com (unoffi-cial end-points exist; session-auth scrape)	per-call HTML	deep archive (≈2005+)	posted within hours of call	unofficial REST + session cookies	paid frontend (~$239/yr claim)	⚠ ToS explicitly forbids extrac-tion — account-ban risk; NEVER redeliver text (action-note: do not build until priced/compliance decision)	W3	connect	UNVERIFIED-CANDIDATE
29	Enterprise tran-scripts (CapIQ Pro / FactSet StreetEvents / LSEG)	TRANSCRIPTS (LIC)	Terminal/API entitle-ment (S&P CapIQ Pro has audio natively)	global tran-scripts	≈2003→present	daily API/entitlement feeds	licensed	enterprise	strict no-redistribution of transcript text (action-note: buy-or-substitute decision (only fully licit struc-tured path))	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
30	Compustat Funda-mentals (XPF + an-nual/quarterly)	PIT-VENDOR	S&P Global Market Intelli-gence; flat files/fields; XPF = as-originally-filed history files	per-firm per-period	annual 1950s→; quarterly 1976→; XPF archived eras	daily batch; PIT not native	licensed flat files (SFTP)	enterprise, call-only	base Compus-tat ≠ PIT (latest-restated paradigm); XPF restores as-filed for those periods (action-note: enter-prise buy-gate; XPF is the as-filed rescue for vintage US)	W2	connect	SURVEY-EVIDENCE
31	Compustat Point-in-Time (PIT)	PIT-VENDOR	separate S&P product; published with vintages (released dates per data point)	per-firm per-period per-vintage	PIT vintages stacked for decades (annual archive era ≈1956 base)	published with vintages	licensed (flat + SFTP)	enterprise call-only (often $xx k/yr — UNVER-IFIED range)	the true as-of-date gold standard; budget-gated (action-note: enter-prise buy-gate; only path to global PIT beyond free #2 hierar-chy)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
32	S&P Capital IQ Pro (fundamentals suite)	PIT-VENDOR	CapIQ Pro terminal/API: fundamentals, estimates, calendar, ownership, deals, transcripts	global, deepest US	transcriptions 2003+; fundamentals deep	near-realtime→daily	licensed	enterprise, call-only	no free tier; pos: transcripts+estimates+fundamentals under one entitlement (action-note: station-level only if a seat materializes)	W3	connect	SURVEY-EVIDENCE
33	FactSet Fundamentals + PIT shared files	PIT-VENDOR	FactSet batch/API + separate PIT shared package	global fundamentals; PIT vintages largely monthly	global coverage; PIT shared set years-long	monthly PIT vintages; realtime port daily	licensed	enterprise	without PIT add-on, FactSet = restatement-blind (documented); no raw-redistribution rights (action-note: global-PIT breadth (US ≈ 2nd-best to Compustat-PIT))	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	LSEG (ex-Refinitiv) DataScope/TR fundamentals	PIT-VENDOR	DataScope Select / TR Fundamentals API	global	≈1980s+ (varies by field)	daily batch	licensed API/flat	enterprise	Workspace/EIKON fundamentals are as-reported only — no native PIT vintages there (action-note: studio-scope provider once global build begins)	W3	connect	SURVEY-EVIDENCE
35	Bloomberg (terminal FA fields + Data License PIT)	PIT-VENDOR	Terminal BPS/FA + Bloomberg Data License (PIT entitlement path)	global	long coverage (field-dependent)	realtime terminal; daily batch	licensed (terminal / enterprise feed)	enterprise	PIT only via Data License entitlement — terminal snapshots are NOT PIT (action-note: terminal seat already covers; PIT entitlement = buy decision)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
36	MSCI data (security master + index funda-mentals snap-shots)	PIT-VENDOR	MSCI data feed / idp/instrument files	security-across-geography master + index snap-shots	long	daily batch	licensed	enterprise	strengths in in-dex/ESG master data, not per-firm state-ments detail (action-note: signal if MSCI-entitlement leverage exists through repo contacts)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
37	Sharadar SF1 (Nasdaq Data Link)	PIT-VENDOR (VERIFY)	data.nasdaq.com SHARADAR/SF1	quarterly per-firm state-ments (incl/excl restate-ment views)	≈1998→present continu-ous	daily 17:30 & 23:30 ET	API/table pulls + bulk	paid tiers (pricing behind login)	publisher-page claims “Point-in-time. Incl/Excl Restate-ments” — must SPOT-VERIFY restate-ment deltas + known ticker-merge gaps (AM/AM1/AM2-type) before backtest role (action-note: 2-week verifica-tion sprint → adopt/reject per stream-02 plan)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
38	Sharadar SF2 (daily funda-mentals)	PIT-VENDOR (VERIFY)	data.nasdaq.com SHARADAR/SF2	daily-updated funda-mentals rows	follows SF1 vintages	daily	API/table	paid tier	same PIT claims/audit as #37; ensures intra-quarter revision visibility (action-note: fold into #37 verifica-tion sprint)	W0	connect	VERIFIED-UPSTREAM-DOC
39	Sharadar SF3 (insider transac-tions)	PIT-VENDOR	data.nasdaq.com SHARADAR/SF3	per-transaction	~2003→	daily	API/table	paid	mirrors Form 3/4 — check vs #16 primary (action-note: redundant-first, primary Form4 wins — keep as QC twin)	W0	connect	VERIFIED-UPSTREAM-DOC
40	Sharadar SF4 (insti-tutional holdings)	PIT-VENDOR	data.nasdaq.com SHARADAR/SF4	quarterly per-manager holdings	≈2000s→	quarterly	API/table	paid	45-day lag + revision windows — QC vs #14 (action-note: QC twin for 13F parse)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
41	Sharadar SEP (equity prices incl. delisted instru-ment flags)	PIT-VENDOR	data.nasdaq.com SHARADAR/SEP	permno-day	long (delisted instru-ments kept — label flags)	daily	API/table	paid	delisted-flag quality check vs Nor-gate/CRSP lists (action-note: universe+CA-carry glue (survivorship-smart))	W0	connect	VERIFIED-UPSTREAM-DOC
42	Nasdaq Sharadar Core US Equities bundle	PIT-VENDOR	data.nasdaq.com Core US Equities bundle packag-ing (SEP+SF1 com-bined)	bundle entitle-ment	same as #37/#41	daily	single en-titlement	paid	discipline: bundle ≠ new data; just pricing conve-nience (action-note: decision with #37)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
43	FMP funda-mentals	⚠ NEVER-PIT	financialmodelingprep.com API (state-ments/ratios/fil-end-points)	per-symbol filing-period **latest only (over-written on restate-ment)**	30+ yrs claim (transcript tiers); free=limited	daily (EOD-ish)	REST + bulk ZIP tiers	commercial subscrip-tion ($19–99/mo ladder); dis-play/redistribution needs separate Display & Licensing Agree-ment	⚠⚠ NEVER-PIT by design: restated numbers rewrite history — **any as-of backtest built on this row recon-structs today-in-history fiction; EDGAR-as-filed/Compustat-PIT is the backtest spine** (action-note: screen-ing/live + coverage-shim US/global lens only)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	EODHD fundamentals	⚠ NEVER-PIT	eodhd.com fundamentals API (also SEC-statements, calendar, insider)	per-symbol latest views	~30 yrs claim (fund-tilde tiers)	EOD daily	REST + bulk (burns 10–100 calls each)	paid $59.99/$99.99 mo	⚠ NEVER-PIT; All-In-One = cheapest everything-row for live protos (action-note: same gate discipline as #43)	W0	connect	VERIFIED-UPSTREAM-DOC
45	Tiingo fundamentals (add-on)	⚠ NEVER-PIT	tiingo.com/fundamentals API	per-symbol latest	~5 yrs individual / 15+ yrs business claim	EOD	REST	add-on priced-on-request (third-party provider — NOT inside $30/mo)	⚠ NEVER-PIT; add-on packaging fee surprises (action-note: prices-first row; fundamentals only if add-on priced sane)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	SimFin (standardized fundamentals)	⚠ NEVER-PIT	simfin.com API + B2B datasets	standardized statement dictionaries US/DE	~2005→ (varies)	periodic	REST + tables	freemium→paid (license pivot-history risk)	⚠ standardized ≠ PIT; restatement-versioning unverified (action-note: drop unless verification answers both PIT +)	W2	license-review	UNVERIFIED-CANDIDATE
47	Intrinio fundamentals	⚠ NEVER-PIT	intrinio.com fundamentals API	standardized views	multi-year claim	daily	REST — entitlement-tiers	commercial (tier quoting)	provider contract landscape shift-prone; pit claim unverified (action-note: raise to W1 only if structured US needs it)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
48	Finnhub fundamentals	⚠ NEVER-PIT	finnhub.io company profile/financials endpoints	per-symbol latest	varied (short)	daily	REST, free tier	paid tiers for history/bulk	⚠ NEVER-PIT; thin depth; screening-only (action-note: screening swap-in candidate)	W3	connect	SURVEY-EVIDENCE
49	NasdaqTrader free files (listing-status + CA alerts + Reg SHO threshold)	CA-ACTIONS (FREE)	nasdaqtrader.com (nasdaqlisted.txt, otherlisted.txt) + Corporate Action Alerts + threshold list	per-symbol per-day files	current-snapshot + rolling alerts	daily file-drop	file-download, no key	free / publication-of-SRO data	listing-status ≠ full CA history (alerts have shallow look-back); pair with #54/#53 (action-note: universe bootstrap + delist-class triggers)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
50	Nasdaq official corporate actions feed	CA-ACTIONS (LIC)	Nasdaq market-data Corp Actions product (file-based, licensed)	per-security per-event (splits, dividends, name-changes, mergers)	deep vendor archive claim	T+1 file cadence	licensed file-drop/SFTP	commercial entitlement	authoritative for Nasdaq-list only; pairing gate vs #49/#53 (action-note: price-gate + evaluate vs Norgate/CRSP substitute)	W2	connect	SURVEY-EVIDENCE
51	NYSE / ICE official corporate actions	CA-ACTIONS (LIC)	NYSE market-data (Corporate Actions/Extraction Tools, XSEL-CA files via ICE)	per-security per-event	long historical depth claim	T+1	licensed file delivery	commercial	same authoritative-vs-license logic as #50 for NYSE-listed (action-note: price-gate; substitute = CRSP+Norgate until single-exchange-only deltas matter)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
52	S&P DJI index corporate actions	CA-ACTIONS (LIC)	S&P DJI CGS/index events (additions/deletions/share changes)	per-index per-event	full-back history claim	daily/announcement-driven	licensed-API/files	commercial	index membership ≠ universe management; crossing into stream-02 (index data) (action-note: feed the survivorship-merge unless already in stream-02)	W2	connect	SURVEY-EVIDENCE
53	CRSP corporate actions + delisting returns	CA-ACTIONS GOLD	CRSP daily/Security level CA files (CRSP via WRDS or direct)	per-security per-event incl. delist-return computation	1926→present	daily/annual data-cycle refresh	licensed (WRDS academic or direct)	enterprise subscription (academic-friendly price tier)	survivorship gold standard; history includes US exchanges' closed eras (action-note: enterprise pricing decision; academic route likely cheapest exit)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
54	Norgate delistings + historical index con-stituents	CA-ACTIONS RETAIL	norgatedatapackages (US/ASX/CA)	com-symbol delist-event + con-stituent map	US delists 1990 (Plat-inum) / 1950 (Di-amond); ASX 1992→	daily EOD	desktop lib (nor-gatedata python) + API	subscription $270–$790/yr (tier ladder verified)	delist universe + historically-true con-stituents at retail price = cheapest disci-plined survivor-ship layer; ⚠ funda-mentals tier is current-only (action-note: plat-inum+ buy decision — anchors universe layer)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	EDGAR-borne delist/change signals (Form 25 + 8-K Items 3.03/5.03)	CA-ACTIONS (FREE)	EDGAR form-25 filings + 8-K parsed items	per-event	1990s+	T+0/T+1	event stream from #6/#9	free	no-delist-return (needs #53 for return calc); de-dupe against exchange feeds (action-note: cheap exchange-decision join (most matches CRSP/Norgate))	W0	connect	VERIFIED-UPSTREAM-DOC
56	FMP delisted-companies list	⚠ NEVER-PIT SHIM	FMP delisted-company end-points	per-symbol snapshot	recent-era empha-sis; NOT a historical universe	daily	REST	commercial tier	⚠ NOT a survivor-ship universe (mini-mum coverage; no returns) — never dress it up as one (stream-02 verdict) (action-note: screening shim only)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
57	OpenFIGI symbol-ogy API	IDENTIFIER	openfigi.com/api (POST map-pings)	/v3/ instrument mapping (FIGI↔tickers↔ISIN↔CUSIP↔exchange)	full coverage claim	realtime/batch	REST (free key, unlimited-ish)	free / CC-BY-4.0	tokens for bursts; keep private key off-log (action-note: symbol-ogy glue + join spine with LEI/MIC)	W0	connect	VERIFIED-UPSTREAM-DOC
58	GLEIF LEI records (+ daily golden copy)	IDENTIFIER	api.gleif.org/api/v1 (lei-records; golden copy files)	per-LEI registry entries	3.47M+ records live; rela-tionship chains shipped	daily golden copy	REST + daily file	free / CC-BY (open data)	relationship graph (fund par-ent→child); keep daily snapshot for CI ver-sioning (action-note: daily master snapshot into CI)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
59	ISO 10383 MIC codes	IDENTIFIER	iso20022.org/market-identifier-codes (RA-maintained list)	per-MIC (per-market)	full recording maintenance	maintained list release	file-download	free-of-charge per ISO RA	name/location data only — no venue metadata beyond the list (action-note: exchange-master join)	W0	connect	VERIFIED-UPSTREAM-DOC
60	ANNA/ISIN + CUSIP Global Services	IDENTIFIER (LIC)	ANNA member catalog + CGS (S&P-run)	per-NSIN	full	n/a — master sync	licensed entitlements	paid/restricted (NO bulk republication)	CUSIP/ISIN equivalent master tables cannot be redistributed; auto-identified via primary filings only; FIGI/LEI as open spine (action-note: zero-spend policy; -class firewall documented)	W0	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	FMP/EODHD corporate-activity calendars (dividend/split event feeds)	⚠ NEVER-PIT CA	FMP calendar endpoints + EODHD calendar endpoints	per-event per-symbol	short-to-mid depth claim	daily	REST	commercial tiers	⚠ events overwritten/late-fetched — reconcile vs #49/#53 before any historical backtest use (action-note: screening/live only)	W0	connect	VERIFIED-UPSTREAM-DOC
62	IBES consensus estimates (LSEG)	ESTIMATES	LSEG l/B/E/S (consensus + event-stamped estimates)	per-metric per-period (EPS/Rev/EPS-Award)	long consensus history claim	daily revision cadence	licensed API/flat (RSK feeds)	enterprise	event-stamped ⇒ intrinsically PIT-friendly — verify as-of-date discipline in feed semantics (action-note: consensus-moments layer; price-gate first)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
63	Zacks-class estimates	ESTIMATES	Zacks (consensus + detail datasets, typically via WRDS/licensed)	per-metric per-period	long-era claim	daily	licensed	commercial	competitive-class second source; anchor-avoidance (some pick-side bias) documented (action-note: secondary source for estimate-set triangulation)	W2	connect	SURVEY-EVIDENCE
64	Nasdaq earnings calendar API	ESTIMATES/CALENDAR	api.nasdaq.com/api/calendar/earnings (+ earnings-date endpoints)	per-symbol per-date (AMC/BMO fits)	short/window claim	daily	REST (unofficial-ish shape; no key)	free-for-display terms	shape not contractually documented → fragile client; bracket with #62/FMP (action-note: earnings-date master pre-earn (live tier))	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	FMP earnings calendar	⚠ NEVER-PIT CALEN-DAR	FMP calendar end-points	per-symbol per-date	short depth claim	daily	REST	commercial	⚠ date revisions over-write; cross-check #64 + 8-K Item 2.02 (action-note: live-schedule only)	W0	connect	VERIFIED-UPSTREAM-DOC
66	Company guidance datasets	ESTIMATES	CapIQ/FactSet guidance-format rows + 8-K Item 2.02/7.01 text mining	per-quarter guidance state-ment	vendor-history claim	daily	licensed + parser	commercial / free-field	holds company-said wording (no point-estimate strikethrough) unless keyed to 8-K (action-note: guidance-vs-actual delta as factor)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
67	UK Companies House filing-history API	REGISTRY	developer.company-information.service.gov.uk (filing-history endpoint)	per-company per-filing history	full registry history (decades)	live/realtime API	REST (free API key)	free / Crown-OGL	free real-time; end-points compos-ite richer with #68–#71 (action-note: UK registry spine)	W0	connect	VERIFIED-UPSTREAM-DOC
68	UK Companies House company-profile API	REGISTRY	same dev-portal (company profile endpoint)	per-company pro-file/status	full registry history	live	REST	free / OGL	status/formation/dissolution dates feed universe-lifecycle (action-note: entity-master UK)	W0	connect	VERIFIED-UPSTREAM-DOC
69	UK Companies House officers API	REGISTRY	same dev-portal (officers endpoint)	per-company per-officer-per-date (ap-point+resign)	registry history	live	REST	free / OGL	director-network panels (co-director graph across #58 iden-tities) (action-note: directors-lattice build)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
70	UK Companies House charges API	REGISTRY	same dev-portal (charges/SCC endpoint)	per-company charges	registry history	live	REST	free / OGL	secured-creditor + pledge signals (private-credit lens) (action-note: private-credit factor pull)	W0	connect	VERIFIED-UPSTREAM-DOC
71	UK Companies House bulk products + iXBRL accounts	REGISTRY (BULK)	CH data products (daily snapshot/changes, accounts data products & iXBRL bulk)	per-company bulk extract + full iXBRL accounts corpus	bulk snapshot history + iXBRL eras	daily/bi-annual products	file-drop + some paid product-tiers	free incl. re-publish (OGL); restricted items via bulk products	numeric-level financials require iXBRL-parse (not a facts API); lighter change-detection via daily diffs (action-note: daily-diff ingest + iXBRL build)	W0	build-parser	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
72	FCA NSM (National Storage Mecha-nism)	REGISTRY	data.fca.org.uk NSM portal (UK listed-company/ESEF-linked filings)	per-filing repository	≈2020→ (ESEF era); earlier listed-company filings via predecessor UK repositories	T+0-ish portal	portal + manual/scripted fetch	free / public	weak link flagged in stream-02 (UI fetch, no numeric API) — verify bulk-capture pattern and retention for restatements (action-note: UK-listed filing relay; may merge into #71 route)	W1	connect	SURVEY-EVIDENCE
73	Ireland CRO (Companies Registration Office)	REGISTRY	cro.ie open services (WebExtract class extracts)	per-company registry + accounts extracts	decades (regulatory paper→digital)	T+0-ish portal	portal/paid extracts	mixed: basics free; docs/extracts paid	no documented bulk JSON API; per-doc fees (action-note: investigate; low priority vs UK)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
74	Esma/ESEF national reposito-ries map	REGISTRY-MAP	eurofiling.info / ESMA ESEF enabler list (per-country registry end-points)	per-country mapping	ESEF-fundament era ≈2020→	varies by NSR	per-country integra-tion	most NSRs free/open	national UI hetero-geneity; per-country excep-tions registry (action-note: per-country plug-in map (route table for EU registry rows))	W2	connect	SURVEY-EVIDENCE
75	Czech Registr obchod-ních (jus-tice.cz) + OVM open data	REGISTRY	justice.cz obchodní rejstřík + OVM/obchodní open datasets	per-company registry + selected docs	years-deep registry	T+0-ish portal	portal/HTML + partial open-data XML	free / public	Czech-lang financial docs; limited XBRL-era depth (action-note: probe per-country ingest; low priority)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
76	DE Unternehmensregister / Bundesanzeiger	REGISTRY	unternehmensregister.de (search incl. HRB/HRA docs) + bundesanzeiger.de (patrimony/announcements)	per-company Registry docs + annual accounts	years/decades (registrar paper→2007 e-Act)	T+0-ish portal	portal + paid fetch (~€5/doc-class; machine-access restricted)	mixed (per-doc fees)	⚠ machine/API access limited vs UK CH — DE = parsing-heavy; MaRIS-class resellers exist (action-note: probe → only if a German DL build opens)	W2	connect	SURVEY-EVIDENCE
77	FR INPI / data.inpi.fr (registry + annual accounts open data)	REGISTRY	data.inpi.fr portal + api.inpi.fr (OAuth; annual-accounts open data)	per-company registry + accounts PDF/structured	broad history claim	T+0-ish	OAuth REST + bulk	free open data (OAuth-keyed)	⚠ autom-fetch flags 403 this session; language + doc formats spur engineering (action-note: probe API-key signup + test endpoint reachability)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
78	NL KVK (Zoeken + Basis-bedrijfsgegevens + Jaar-rekenin-gen APIs)	REGISTRY	developers.kvk.nl official APIs (sub-scription) + ODAT open datasets	per-company + annual accounts API	registry history; accounts datasets	realtime-ish API; datasets cyclical	REST sub-scription (per-query fees; free test env)	subscription-paid (+ per-query)	⚠ Jaar-rekenin-gen (annual accounts) is an OFFICIAL API = rare NL find — the only NL route beyond scraping (action-note: subscription-vs-scrape cost probe; test env already usable free)	W1	connect	VERIFIED-UPSTREAM-DOC
79	ES Registro Mercantil (datos.gob.es + official certifica-dos)	REGISTRY	datos.gob.es datasets + Registro Mercantil official copies	per-company registry + accounts	decades	T+0, portal	open-data down-loads + paid certs	mixed (open bulk vs paid per-doc)	ES registry docs = per-doc costs; bulk open-data exists in parts (action-note: low-priority probe)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
80	IT Info-Camere / Registro Imprese open data	REGISTRY	infocamere.it + CCIAA registries open-data releases	per-company registry + filings overview	decades	cyclical releases	portal +_bulk	free open-data pieces; per-doc for official extracts	Italian doc formats; per-country variance high (action-note: low-priority probe)	W2	connect	UNVERIFIED-CANDIDATE
81	CH Zefix (central business name index) + register docu-ments	REGISTRY	zefix.ch (REST API exists) + commer-cial register extrac-tors	per-company registry	decades	T+0-ish portal	REST API + paid extracts	free basics / paid extracts	⚠ SPA-page fetches returned blank here — endpoint else flagged to verify via direct HTTP call (action-note: confirm API endpoint path by scripted HIT test)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
82	CH SIX price-sensitive (ad hoc) disclosures	DISCLOSURE	six-group.com disclosure platform (ad-hoc announcements + management transactions)	per-announcement	era-dependent	T+0 portal	portal/API-class (some data at SIX feeds)	mixed (some feeds paid)	pairs with SIX-K equivalent events; CH-listed filers (action-note: drop unless Swiss-listed stream matters)	W2	connect	SURVEY-EVIDENCE
83	DK CVR open data (data.virk.dk)	REGISTRY	data.virk.dk (CVR open data; Erhvervsstyrelsen)	per-company registry + selected accounts	~2005+ digital eras amid paper	T+0-ish; datasets cyclical	open-data files/API	free / public authority	danish-lang statements depth varies; transport-blocked here (F17) (action-note: probe; good candidate for free accounts extracts)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
84	NO Brønnøysund registers (incl. annual-accounts register)	REGISTRY	brreg.no (+ virksomhet.brreg.no lookup; annual-accounts submission pages; use-of-data policy)	per-company registry + accounts submissions	long eras (digital-forward from 2020s for accounts)	T+0-ish portal	open-data frameworks + APIs (API service exists per RA guidance)	free / public authority (use-of-data framework)	Norwegian-lang; accounts availability varies by filer class (action-note: probe accounts-register API + Norwegian account route)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
85	SE Bolagsverket registry + ESEF	REGISTRY	bolagsverket.se (+ per-registry extracts)	per-company registry	decades	T+0-ish portal	portal/paid extracts	mixed	full public accounts NOT universal (private filers pay via Ratsit/Upplysningarna-class vendors — cite in gates); ESEF era helps listed (action-note: low-priority probe; Ratsit-class paid vendor = later-decision)	W2	connect	UNVERIFIED-CANDIDATE
86	JP EDINET v2 filings-list API	REGISTRY-FILINGS	disclosure2.edinet-fsa.go.jp API v2 (documents.json list endpoint, date-window query)	per-filing (Japan last-window filings, incl. □□□□□□□□ “10-K-class”)	all digitized EDINET history (list-API deep; annual dates; XBRL eras newer)	T+0-publish → API T+1-class per FSA notes	REST (API key since 2023-08; v1 sunset 2024-03-29)	free / public sector	per-date polling + jitter; taxonomy = JP IFRS-compatible (action-note: nightly index + per-doc queue)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
87	JP EDINET v2 document download (HTML/XBRL/CSV)	REGISTRY-FILINGS	API v2 document-download endpoint (zip incl. Abstract + XBRL)	per-document bundle	vintages of attached formats	T+1	REST (rate-limited; doc-link series waits)	free	XBRL-by-EDINET-taxonomy parse = Japan numeric-fundamentals free path; 1-doc-per-req series waits (action-note: XBRL parse pipeline (doc-core per company))	W0	connect	VERIFIED-UPSTREAM-DOC
88	JP TDnet timely-disclosure (JPX)	DISCLOSURE	Tdnet — timely disclosure announcements (rinji hōkoku) via JPX/plug-feeds	per-announcement	era-dependent (not as deep as EDINET)	intraday/hourly (earnings + events real-time)	portal/equipment data-feed licensed (or MX feed)	mixed (feed licensed)	duplicates EDINET far faster for events; “same-day” earnings-announcement signal (action-note: price the d TDnet feed vs EDINET-delay analysis)	W1	license-review	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
89	KR DART OpenAPI (disclosure + key-info + financial statements with XBRL)	REGISTRY-FILINGS	opendart.fss.or.kr (list/company/fnlttSinglAcnt financial statements APIs incl. multi-year statements + XBRL raw download)	per-filing/period annual statement	deep historical (coverage from DART-era; XBRL bulk multi-year)	T+1 from DART publish	REST (free API key; tiered daily quotas)	free	K-GAAP restatements filing-time-versioned ⇒ pin `rcept_no` per period = PIT-safe; XBRL taxonomy drift year-dependent (action-note: statements+XBRL ingest; DART as KR filing spine)	W0	connect	VERIFIED-UPSTREAM-DOC
90	KR DART shareholding disclosures (□□□/□□)	POSITION-REG	opendart.fss.or.kr shareholding disclosure endpoints	per-disclosure per-holder	era-backed	T+1	REST	free	KR equivalent of Form 3/4-class ownership rings; holders unify chain (action-note: ownership panel build)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
91	KR DART bulk-download (company-docs ZIP per company/FY)	REGISTRY-FILINGS	opendart.fss.or.kr bulk-download endpoint (corp-download; per-FY/company zip)	per-kr company per-FY batch	deep	T+1	REST bulk (call quota heavy)	free	efficient backfill squeeze; onboarding quota vs plan (action-note: historical repeat-run tool)	W0	backfill	VERIFIED-UPSTREAM-DOC
92	CN cninfo (□□□□□) announcement / REST APIs	REGISTRY-FILINGS	cninfo.com.cn disclosure REST (hisAnnouncement/query-class; unofficial PyPI wrappers exist)	per-announcement per-listed-CN	deep archive claim (2000s+)	intraday publishing	REST portal + paced scrape	public disclosures (state-mandated); ToS-grey automation — pace; no formal open API documented	⚠ fetch probes SPA-blank (F17) — verify endpoint reachability (action-note: A-share/CN filings spine; CN financial-statement quality high)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
93	CN SSE announcements	REGISTRY-FILINGS	sse.com.cn disclosure endpoints	per-announcement (Shanghai)	deep chain	intraday	portal/JSON endpoints (undocumented)	public; ToS-grey paced	CN-language + announcement-parse burden; SSE vs cninfo overlap (action-note: secondary source to cninfo)	W1	connect	UNVERIFIED-CANDIDATE
94	CN SZSE announcements	REGISTRY-FILINGS	szse.cn disclosure endpoints	per-announcement (Shenzhen)	deep chain	intraday	portal/JSON (undocumented)	public; ToS-grey paced	same pattern as #93 (action-note: secondary source pair)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
95	CN corporate registry (Tianyancha/Qichacha-class)	REGISTRY (PAID)	third-party registry aggrega-tors (paid APIs; QCC/Tianyancha/□□□)	per-company reg-istry/shareholding/doc history	long claim	daily-ish refresh	vendor REST APIs	paid (quotes ¥3xx–year-plus tiers — UNVERI-FIED)	⚠ CN registry = NOT free (a red-line vs US/UK/JP); decide only when a CN build is real (action-note: vendor + ToS red-line review)	W3	quote	UNVERIFIED-CANDIDATE
96	HKEXnews (□□□) listed-company filings	REGISTRY-FILINGS	hkexnews.hk (title-search/search end-points; an-nounce-ment + financial reports)	per-announcement (HK)	varies by listing-class; years-deep generally	intraday publish-ing	portal search + polite per-company fetch agent (no docu-mented JSON API)	public filings; robots.txt likely gates bulk	⚠ no unified numeric XBRL (regime = document-parse); parsing burden triples vs US/JP/KR (stream-02 verdict) (action-note: HK filings relay agent, paced)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
97	India MCA (Master Data + RoC documents)	REGISTRY	mca.gov.in V3 (Master Data API, pro-state; RoC doc pay-per-view)	per-company registry + statutory docs	decades (RoC paper→digital mix)	T+0-ish portal	Master Data API (available per state) + doc fees	mixed: master free; docs ₹/doc-class	⚠ no clean new UK-CH-style numeric API; doc parsing (action-note: master-data spine for IND entities; doc packs later)	W1	connect	UNVERIFIED-CANDIDATE
98	NSE corporate announcements/CA	REGISTRY-FILINGS	nseindia.com corporate announcements + corp-filing endpoints (XBRL statements per SEBI)	per-announcement (results, board-date, M&A)	archive claim varies	intraday/T+1	unofficial-ish JSON endpoints, IP-aggressive	public; ToS-tight (nsearchives host blocks automation — [F17] prior evidence)	iXBRL per SEBI for statements; URL host has actively denied datacenter-style agents (action-note: paced relay agent; bypass = paid vendor (Trendlyne-class) decision)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
99	BSE announcements / corp actions	REGISTRY-FILINGS	bseindia.com announcements + CILAC endpoints	per-announcement (BSE)	archive claim	intraday	portal endpoints (undocumented)	public; ToS-grey	same-parse problem as #98 (action-note: pair with #98; two-host coverage)	W1	connect	SURVEY-EVIDENCE
100	Canada SEDAR+	REGISTRY-FILINGS	sedarplus.ca (search + document payment/extract; structured filings for ESEF-class iXBRL since ~2020)	per-filing (CA securities filers)	docs since 2000s; iXBRL era ≈2020+	T+0/T+1	portal + per-doc fees (official copies)	fees per document (statutory); search basic free	⚠ no free numeric API; the stream-02 matrix rows for CA flagged must complete; official copies costs add up at scale (action-note: iXBRL extraction + fee-model after probe)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
101	AU ASIC company register	REGISTRY	asic.gov.au register search + ASIC Connect extracts	per-company registry (+ historical extracts)	decades (registry digital-forward)	T+0-ish portal	portal + paid/bulk extracts (some bulk products)	mixed (basic free; extracts paid)	stream-02 matrix flagged SEDAR-stream GAAP-free numeric gap; ASIC extracts legiti-mate bulk but fee-classified (action-note: probe extract pricing; AU strength-ened if Norgate ASX pings improve)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
102	AU ASX Market An-nounce-ments Platform (MAP)	DISCLOSURE	market an-nounce-ments platform (asx.com.au MAP) — live listed-companies an-nounce-ments	per-announcement (ASX)	live platform era	intraday publish-ing	portal + paced fetch (no formal bulk API)	public; ToS-grey automa-tion	pairs with ASIC; an-nounce-ments = divi-dends/board/report dates relay (action-note: ASX an-nounce-ments relay agent)	W1	connect	SURVEY-EVIDENCE
103	NZ Com-panies Register (open data/API)	REGISTRY	companies-register.com (+ open-data sched-ules)	per-company registry (+ directors, share-holders)	long registry history	T+0-ish	open-data API sched-ules (free tier claim)	free (NZ Govern-ment open data)	⚠ 404 on busi-ness.govt.nz open-data path this session — locate the correct schedule page (action-note: probe; smaller stack weight)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
104	SA CIPC (ZAF company registry)	REGISTRY	cipc.co.za (register search/extract fees)	per-company registry	digital-forward	portal	fees/extracts	mixed (per-doc fees)	⚠ fetchability low — vendor-class work needed; global long-tail (action-note: skip unless SA thesis appears)	W3	connect	UNVERIFIED-CANDIDATE
105	Brazil CVM open data (dados.cvm.gov.br — incl. ITR/DFP standardized fundamentals)	REGISTRY-FILINGS	dados.cvm.gov.br bulk CSVs (ITR quarterly + DFP annual standardized fundamentals, shareholders, registry)	per-company per-period standardized CSVs	multi-year deep claim	monthly/yearly dataset releases	file-download (CSV bulk)	free / public authority	⚠ automated fetch currently 403 (bot-block this session) — use mirrors/Wayback or curl UA tuning; BR = one of the FEW free standardized non-US fundamentals (action-note: ITR/DFP + track monthly refresh cadence)	W1	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
106	MX BMV/BIVA issuer disclosures	REGISTRY-FILINGS	bmv.com.mx / biva.mx issuer pages (financial statements + relevant events)	per-issuer per-period	years-deep claim	T+0/T+1 via publish	portal + paced fetch	public filings; unfriendly bulk terms	PDF/Excel-class primary formats; Spanish (action-note: Mexico long-tail)	W3	connect	UNVERIFIED-CANDIDATE
107	IL TASE Maya disclosures	REGISTRY-FILINGS	maya.tase.co.il (mayas official disclosure portal; API reportedly usable)	per-disclosure (IL)	years-deep claim	intraday-ish	portal (+ unofficial API shape)	public disclosures; automation 403'd here (F17) — ToS-verify pace	⚠ bot-blocked this session; verify a tolerated fetch-agent path before building (action-note: probe; Hebrew/EN mixed docs)	W2	connect	SURVEY-EVIDENCE
108	SG ACRA BizFile registry	REGISTRY	acra.gov.sg BizFile+ (paid extracts for most; some open data)	per-entity registry	long	T+0-ish	paid extract fees + API-class services	paid-extract dominant	budget-gate; SGX-relay (below) covers listed side (action-note: probe cost)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
109	SGX Market An-nounce-ments	DISCLOSURE	sgx.com / SGX Net an-nounce-ments	per-announcement	years-deep	intraday	portal + paced fetch	public; ToS-grey	pairs with ACRA for the SG stack (action-note: SGX relay agent)	W2	connect	SURVEY-EVIDENCE
110	TWSE/TPEx MOPS filings	REGISTRY-FILINGS	mops.twse.com.tw / tamas (TWSE Market Observa-tion Post System)	per-filing (TW-listed)	years-deep claim	T+0/T+1	portal + paced fetch (docu-ment down-loads)	public; ToS probing needed	⚠ UI fetch blank here (JS-class; F17) — endpoint census needed; Chinese-language doc formats (action-note: MOPS relay agent; daily cadence)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
111	Turkey KAP (Public Disclosure Platform)	REGISTRY-FILINGS	kap.org.tr (EN/EN UI: notifications: Material/Financial/Regulatory/Other; downloadable financial reports 2009–2026 UI)	per-notification + per-financial-report (with detailed period/quarter filters)	2009→financial reports institutional (≈2005+ CA)	intraday-ish	portal + detailed search + report download	free / public (CMB-mandated disclosures)	⚠ VERIFIED live this session: categories incl. **Corporate Actions** + financial-statement download interface (2009–2026 quarterly/annual) (action-note: KAP ingest (Bist reports + CA events); TR = feasible free market outside majors)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
112	PL ESPI/EBI reports + KRS registry	REGISTRY-FILINGS	GPW ESPI/EBI notifica-tions + e-KRS / CEIDG registries	per-announcement + registry entries	years-deep	T+0/T+1	portal + paced fetch	public; KRS extracts fee-class	Polish-lang docs; KRS monthly extracts exist (action-note: probe; CEE slot)	W3	connect	UNVERIFIED-CANDIDATE
113	Malaysia Bursa an-nounce-ments	DISCLOSURE	bursamalaysia.com an-nounce-ments pages	per-announcement	years-deep	intraday	portal + paced	public; ToS-grey	Malaysia long-tail (action-note: probe slot only)	W3	connect	UNVERIFIED-CANDIDATE
114	Indonesia IDX an-nounce-ments / corp actions	DISCLOSURE	idx.co.id an-nounce-ments + IDX events pages	per-announcement	years-deep	intraday	portal + paced	public	Indonesian-long-tail; KSEI feeds paid (action-note: probe slot only)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
115	Compustat PIT vintages archive (NA annual 1956 base, quarterly 1976+)	ARCHIVE	S&P PIT product — historical vintage stacks (institutional procurement)	per-vintage per-firm per-period	academic archive eras back to ≈1956 (annual; quarterly 1976+)	archived vintages (static)	licensed datasets	enterprise procurement (WRDS-class access possible)	the historical PIT depth row — distinct from current-PIT subscription (#31); vintage semantics documented in WRDS notes (action-note: deep-archive purchase decision (W3 if budget-frozen))	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
116	IBES detail/vintage historical (estimate-detail history files)	ARCHIVE	LSEG IBES Detail History (daily estimate files, vintage-consistent)	per-analyst/consensus per-period	detail history back to ≈1984 (consensus ≈1976 era)	archived daily files	licensed (WRDS-class or direct)	enterprise	event-stamped estimates = vintage-native; the estimate-side archive gold (action-note: buy/WRDS decision if estimate factors enter the build)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
117	EDGAR full-history backfile (incl. pre-1993 gap handling)	ARCHIVE	www.sec.gov/Archives (1993→present machine-readable complete; full text 2001); pre-1993 = paper/film archive (vendors digitized; SEC reading room)	per-filing	full-machine 1993+; full-text 2001+; pre-1993 vendor-class	static archive	bulk archive fetch + third-party digitization	free (SEC); vendors paid for pre-1993	⚠ survivorship hole pre-1993 — pair with #121 + Compustat-PIT for as-filed older eras (action-note: build historical job; evaluate digitized-vendor pre-1993 need)	W0	backfill	VERIFIED-UPSTREAM-DOC
118	CRSP survivorship-equity database (incl. delisting returns 1926+)	ARCHIVE	CRSP Daily Stock file / CRSP-Compustat Merged (via WRDS or direct sub)	per-security per-day + CA/delist-return constructs	1926→present	daily/annual data-cycle	licensed	enterprise/academic	survivorship-gold standard incl. closed-bourse eras — complements Norgate at retail (#54) (action-note: WRDS/academic route pricing decision)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
119	Moody’s Manuals / Poor’s Register (pre-1956)	ARCHIVE	archive.org + HathiTrust scans (Moody’s Industrial/Utility/Manuals, Poor’s yearbooks)	per-company per-year page scans	pre-1956 fundamentals/officer data eras	static archive	OCR/curation pipeline	copyright: pre-1928 public domain; Depression-era mostly PD or controlled once cleaned	careful copyright-class; OCR-accuracy gates; heavy curation labor (action-note: long-tail deep-history only if a thesis demands)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
120	WRDS-hosted mirrors (Compustat/IBES/CRSP/Ent class)	ARCHIVE ACCESS	wharton WRDS subscrip-tion (per-dataset entitle-ments incl. PIT archives)	per-dataset institution-level	dataset-native depths (parent rows above)	per-dataset refresh	WRDS web/SAS/STATA/API/query	academic/subscription (internal use)	cheapest bona-fide route to enter-prise PIT archives if an academic or institu-tional relation-ship exists; streaming-redelivery NOT per-mitted (action-note: negoti-ate/decide route (likely unlocks #115/#116/#118 cheaply))	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
121	SEC pre-1993 historical (digitized filings vendors + SEC reading room)	ARCHIVE	vendors (e.g., Historical-Filings-digitizers) + SEC reading-room microfilm	per-filing scans/text	1934→1993 (SEC era machine-originated 1993+)	static	vendor bulk (paid) + on-site	paid vendor / library	survivorship hole pre-1993; cost/deliverability to verify vendor pricing (action-note: vendor scope if deep pre-1993 backtests enter proof-of-concept)	W3	quote	UNVERIFIED-CANDIDATE
122	ESEF historical iXBRL corpus across EU registries	ARCHIVE	national ESEF reposito-ries + ESMA ESEF filings map (per-country fetch layer on #74 route table)	per-filing iXBRL	ESEF era ≈2020→	T+0–T+1 by registry	per-registry fetch + iXBRL parse	mixed (most NSRs free)	iXBRL tag hetero-geneity across national tax-onomies (hyper-local exten-sions) (action-note: one + per-country adapters)	W2	build-parser	SURVEY-EVIDENCE

06 — Derivatives (options/futures) (84 rows)

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	OPRA real-time feed (Cboe Pillar/NYSE etc. consolidated)	options-feed	OPRA plan via member marketplace [29]	tick (MBO)	n/a (live)	realtime	feed-handler	exchange-fee (display + usage license quote)	vendor/non-display gating; audit consumption (action-note: quote only; do not capture live until scale justified)	W3	license-review	SURVEY-EVIDENCE
2	Databento OPRA.PILLAR options trades/quotes/TBBO/OHLCV/definitions	options-feed	Databento OPRA.PILLAR (Cboe PCAP relay) [6]	tick + daily bars	per dataset-range (confirm earliest date)	near-realtime + batch	websocket + S3 batch	vendor-tier (metered)	confirm historical instrument-definition continuity & range start (action-note: get range metadata (`get_dataset_range`) + unit pricing)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	Databento DEFINITIONS schema incl. options numeric fields	symbology	Databento definitions/shemas [6]	daily	full vendor history	batch	S3 batch	vendor-tier	verify option-specific fields (strike/bid low limits etc.) (action-note: pull definitions snapshot for options universe)	W1	connect	SURVEY-EVIDENCE
4	Polygon.io options aggregates/trades/quotes	chains	polygon.io REST [19]	tick + daily (+intraday tier)	vendor-limited window (confirm)	tier-dependent; delayed on lower tiers	REST SDK	vendor-tier (tier gates history+realtime)	PIT on splits/adjustments; rate limits (action-note: tier pricing vs depth; sample SPX chain)	W1	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	Cboe DataShop Option EOD Summary (quotes/OHLC/vol + 3:45pm snapshot, optional Calcs IV/Greeks)	chains	datashop.cboe.com/option-eod-summary [9]	daily + 3:45pm ET	shopping-cart dependent (deep multi-year possible EOD)	EOD	web download/SFTP	vendor-tier per-file	optional Calcs only when purchased; snapshot vs close semantics (action-note: ingest all-equity + index EOD history)	W1	connect	VERIFIED-UPSTREAM-DOC
6	Cboe DataShop Option Quote Intervals (1-min / n-min NBBO w/ size + optional OI, IV/Greeks)	chains	datashop.cboe.com/option-quote-intervals [9]	1-min / n-min intervals	multi-year per symbol	EOD	download/SFTP	vendor-tier per-file	NBBO from ops experts; OI timing (action-note: purchase sample month across SPX/SPY/AAPL)	W1	connect	VERIFIED-UPSTREAM-DOC
7	Cboe DataShop Option Trades (daily prints + contextual market info, optional IV/Delta)	chains	datashop.cboe.com/option-trades [9]	tick / daily file	multi-year	EOD	download/SFTP	vendor-tier per-file	trade+quote context alignment (action-note: sample file, parse to parquet)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
8	Cboe DataShop Enhanced US Options Trade-by-Trade (full OPRA-level execution detail, all exchanges)	OPRA-lite	datashop.cboe.com/enhanced-us-options-trade-by-trade-execution-detail [9]	tick	vendor-stated history depth (confirm)	EOD	download/SFTP	enterprise-quote	consolidation delays vs OPRA; NBBO reconstruction (action-note: request for (a) 1 y + (b) deep history)	W2	quote	VERIFIED-UPSTREAM-DOC
9	Cboe DataShop Open-Close Volume Summary (Customer/Pro/Firm/MM capacity, 10-min buckets)	flows	datashop.cboe.com/cboe-options-open-close-volume-summary [9]	10-min / EOD	multi-year	EOD	download	vendor-tier per-file	Cboe-only venue scope (BZX,C1,C2,EDGX) not All-OPRA (action-note: ingest for flow studies)	W1	connect	VERIFIED-UPSTREAM-DOC
10	ORATS Options Data API (EOD surface, IV30, SG, vol-forecast)	vol-surface	orats.com/data-api [8]	daily	2007 →	EOD	REST/library	vendor-tier	SV backfill consistency; earnings events (action-note: tier list & v2 endpoints)	W1	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	ORATS Intraday Data API (1-min options, 5,000+ symbols)	vol-surface	orats.com/intraday-data-api [7]	1-min	2020-08 →	intraday + delayed + historical	REST/cloud delivery	vendor-tier	$496/mo tier price not con-firmed (survey shows other tiers) (action-note: $496 tier & coverage scope)	W2	verify	VERIFIED-UPSTREAM-DOC
12	IvyDB US (option EOD prices, IV surfaces, greeks)	vol-surface	OptionMetrics [10]	daily	1996 →	EOD	database/bulk-export	enterprise-quote	standardized rate/dividend assump-tions; corporate-action handling (action-note: quote + sample quarter)	W2	license-review	VERIFIED-UPSTREAM-DOC
13	IvyDB US Intraday (on Snowflake)	vol-surface	OptionMetrics / Snow Market-place [10]	intraday	recent window (confirm)	intraday	Snowflake data share	enterprise-quote	snapshot cadence & field list (action-note: confirm coverage start & pricing)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
14	IvyDB Futures (US futures options), IvyDB Europe/Asia/Canada	vol-surface	OptionMetrics [10]	daily	US futures + regional ~2010s (confirm per-region)	EOD	database	enterprise-quote	per-venue availability dates vary (action-note: region-by-region availability matrix)	W3	connect	VERIFIED-UPSTREAM-DOC
15	LiveVol (Cboe) Pro & volume-premium datasets	chains	livevol.com (Cboe)	tick/1-min	~2006 →	delayed/real-time tiers	platform + API	vendor-tier	product line post-Cboe acq; endpoint reachability failed in survey (action-note: price quote; SKU list)	W2	verify	SURVEY-EVIDENCE
16	CBOE VIX index history + VX futures daily settlement term structure files	vol-index	cboe.com index & VX settlement data	daily	VIX 1990 →; VX futures 2005/2007 →	EOD	free web files	exchange-free (attribution)	settlement-vs-close skew (action-note: ingest monthly archives)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
17	CBOE market-place daily statistics (vol-ume/OI by exchange & product)	vol-stat	cboe.com daily market statistics	daily	multi-decade	EOD	free download	exchange-free	cross-check vs OCC totals (action-note: ingest)	W0	connect	SURVEY-EVIDENCE
18	OCC volume & open-interest reports (daily/weekly/monthly/historical statistics)	reference	theocc.com market-data-reports [15]	daily + monthly + annual	multi-decade	EOD / weekly	web CSV	exchange-free	OI snap-shots not PIT restated (action-note: ingest)	W0	connect	VERIFIED-UPSTREAM-DOC
19	Chain/vendor fallback option chains (Tradier / theta-data free tiers)	chains	vendor REST	tick/daily	vendor windows	varies	REST SDK	freemium	not primary source; PIT gaps (action-note: keep logged as fallback only)	W1	connect	SURVEY-EVIDENCE
20	CBOE foreign-exchange-style deriva-tive securities (FX chains on listed se-curities)	fx-options	cboe listings	daily	?	EOD	web	exchange-free	naming changes (action-note: confirm whether actually needed before ingest)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
21	CME MDP3 Globex realtime feed (raw L3)	futures-feed	CMEMDP MDP3	tick MBO	n/a (live)	realtime	feed-handler (SBE)	exchange-fee (usage-based)	colo/non-colo sizing; entitle-ment audit (action-note: only if vendor feed inad-equate)	W3	connect	SURVEY-EVIDENCE
22	CME MDP3 via Databento GLBX.MDP3 — equity-index (ES/NQ/YM + spreads)	futures-feed	Databento GLBX.MDP3 [6]	tick + daily	per dataset-range (confirm earliest)	near-realtime + S3	websocket + S3 batch	vendor-tier	symbol-map & roll events; OI/settle stats schedule (action-note: pricing + full range test)	W1	connect	VERIFIED-UPSTREAM-DOC
23	CME MDP3 via GLBX — rates (SOFR 1m/3m, ZT/ZN, incl. SONIA-type SR3)	futures-feed	GLBX.MDP3 [6]	tick + daily	as row 22	near-realtime + S3	websocket + S3	vendor-tier	term-structure continu-ity (action-note: same plan probe as row 22)	W1	connect	VERIFIED-UPSTREAM-DOC
24	CME MDP3 via GLBX — energy (CL/NG/HO/RB)	futures-feed	GLBX.MDP3 [6]	tick + daily	as row 22	same	websocket + S3	vendor-tier		W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
25	CME MDP3 via GLBX — metals (GC/SI/HG)	futures-feed	GLBX.MDP3 [6]	tick + daily	as row 22	same	websocket + S3	vendor-tier		W1	connect	VERIFIED-UPSTREAM-DOC
26	CME MDP3 via GLBX — agricul-ture (ZC/ZS/ZW/CT/OJ)	futures-feed	GLBX.MDP3 [6]	tick + daily	as row 22	same	websocket + S3	vendor-tier		W1	connect	VERIFIED-UPSTREAM-DOC
27	CME MDP3 via GLBX — FX (6E/6B)	futures-feed	GLBX.MDP3 [6]	tick + daily	as row 22	same	websocket + S3	vendor-tier		W2	connect	VERIFIED-UPSTREAM-DOC
28	CME MDP3 via GLBX — crypto futures (MBT/MET/SBT etc.)	futures-feed	GLBX.MDP3 [6]	tick + daily	contract-by-contract since listing	same	websocket + S3	vendor-tier		W2	connect	VERIFIED-UPSTREAM-DOC
29	CME options-on-futures chains (incl. de-ferred contract detec-tion)	options-on-futures	GLBX.MDP3 + defini-tions [6]	chain daily	as rows 22-28	near-realtime	S3	vendor-tier	per-contract expiry rollover (action-note: chain render for vol-through-futures)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
30	CME volume/OI/settlement statistics (statistics schema, publication schedule)	OI-settlement	GLBX statistics via Databento [6]	daily	years	EOD per schedule	S3/REST	vendor-tier	verify publication timestamps (action-note: ingest daily OI/settlement)	W0	connect	VERIFIED-UPSTREAM-DOC
31	CME DataMine bulk historical (per-product subsets)	archive	cmegroup.com CME DataMine	tick/daily varies	varies by product	batch	vendor-bulk	enterprise-quote	redistribution restrictions (action-note: request for equities + energy deep history)	W2	quote	SURVEY-EVIDENCE
32	CME curve/term-structure via vendor (continuous contracts, back-adjustment modes)	curve	Databento symbology/continuous [6]	daily	decades	EOD	S3	vendor-tier	roll-convention pitfall (volume/OI switching) (action-note: build term structure + roll feature)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
33	CME SOFR futures transac-tions public dataset (free subset)	rates	CMES / cme free data portal	daily/tick	?	?	?	exchange-free?	? (action-note: confirm existence & route)	W3	connect	UNVERIFIED-CANDIDATE
34	ICE Europe Com-modities (IFEU) via Databento	futures-feed	Databento ifeu-impact [6]	tick + daily	vendor range (confirm start)	near-realtime	websocket + S3	vendor-tier	verify symbol continu-ity (action-note: pricing & range test)	W2	connect	VERIFIED-UPSTREAM-DOC
35	ICE Futures US (IFUS: cot-ton/coffee/SB) via Databento	futures-feed	Databento ifus-impact [6]	tick + daily	vendor range	near-realtime	websocket + S3	vendor-tier		W2	connect	VERIFIED-UPSTREAM-DOC
36	ICE Europe Fi-nancials (IFLL) via Databento	futures-feed	Databento ifll-impact [6]	tick + daily	vendor range	near-realtime	websocket + S3	vendor-tier		W2	connect	VERIFIED-UPSTREAM-DOC
37	ICE Endex (NDEX gas/power) via Databento	futures-feed	Databento ndex-impact [6]	tick + daily	vendor range	near-realtime	websocket + S3	vendor-tier		W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
38	ICE Pet-Connect (real-time suite feed)	feed	ice.com [27]	tick	n/a	realtime	feed	exchange-fee	naming/access to confirm (action-note: confirm current product name & access path)	W3	connect	UNVERIFIED-CANDIDATE
39	Eurex T7 market data via Databento XEUR.EOBI	futures-feed	Databento xeur-eobi [6]	tick + daily	vendor range (EOBI start confirm)	near-realtime	websocket + S3	vendor-tier	verify DAX/Stoxx chains (action-note: pricing & range test)	W2	connect	VERIFIED-UPSTREAM-DOC
40	Eurex historical daily data (official archive)	archive	eurex.com data services [26]	daily seg-ments	1998 → (Eurex in-ception)	EOD	download	exchange-fee	per-market-segment files; splits vs consoli-dated (action-note: request & earliest-segment samples)	W2	quote	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
41	DCE daily futures stats (OHLCV + OI per contract)	futures-feed	dce.com.cn daily files	daily	years (per product inception)	T+1 (published next morning)	web download	exchange-free (attribution)	OI is session-final; grain conventions differ (action-note: free-file ingest)	W0	connect	SURVEY-EVIDENCE
42	CZCE daily futures/options stats	futures-feed	czce.com.cn daily files	daily	per-product inception window	T+1	web	exchange-free	per-contract queries (action-note: free-file ingest)	W0	connect	SURVEY-EVIDENCE
43	GFEX daily stats (lithium carbonate / industrial silicon futures/options)	futures-feed	gfex.com.cn daily files	daily	since contracts started (2022-23)	T+1	web	exchange-free	new exchange; format changes (action-note: free-file ingest)	W0	connect	SURVEY-EVIDENCE
44	SHFE + INE daily stats (copper/crude/rubber etc.)	futures-feed	shfe.com.cn daily files	daily	years	T+1	web	exchange-free	includes INE under same open-fit pattern (action-note: free-file ingest)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
45	CFFEX daily futures/options stats (bond/index derivatives)	futures-feed	cffex.com.cn daily files	daily	since bond/index product launches (2010s)	T+1	web	exchange-free	dividend-adjusted on main contracts (action-note: free-file ingest)	W1	connect	SURVEY-EVIDENCE
46	SGX iron-ore / freight derivatives actual & EOD data	futures-feed	sgx.com derivatives iron-ore page [18]	daily	multi-year (per contract)	EOD	download	exchange-fee (SDE)	page reached only as browser shell in survey — extract file format (action-note: request sample file & fees)	W1	connect	SURVEY-EVIDENCE
47	LME inventories + daily prices / SDE statistical data	metals	lme.com market-data + licensing pages [12,13]	daily	decades of official data (1980s+)	EOD	SFTP/LME portal	exchange-fee (license quote)	redistribution re-stricted; 15-month averaging rare (action-note: obtain quote via market-data-licensing page)	W2	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
48	KRX KOSPI200 futures/options (data.krx)	index-derivs	data.krx.co.kr [21]	daily	KOSPI200 options 1997-ish onwards	EOD/T+1	web/API (KRD product export)	exchange-free? (some paid products)	Korean encoding; file-format changes (action-note: download route + product names)	W1	verify	SURVEY-EVIDENCE
49	HKEX derivatives statistics (consolidated daily reports incl. options OI by strike)	index-derivs	hkex.com.hk market data [20]	daily	~2000s →	T+1	web	exchange-free	stock options vs index-option separation (action-note: ingest)	W1	connect	SURVEY-EVIDENCE
50	TAIFEX Taiwan futures/options daily data	index-derivs	taifex.com.tw [22]	daily	1998 →	T+1	web (TAIFEX files)	exchange-free	naming locale quirks (action-note: ingest)	W1	connect	SURVEY-EVIDENCE
51	B3 (Brasil) derivatives data	futures	b3.com.br	daily	long archive	EOD	web	exchange-free	B3 portal layout changes (action-note: defer)	W3	connect	SURVEY-EVIDENCE
52	DGCX (Dubai) futures data	futures	dgcx.ae	daily	multi-year	EOD	web	exchange-free	(action-note: defer)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
53	Euronext derivatives (EURIBOR/Stoxx legacy LIFFE) data	rates-index	euronext.com	daily	decades	EOD	web	exchange-fee	(action-note: defer)	W3	connect	SURVEY-EVIDENCE
54	DTCC GTR Public Reports (US OTC derivatives: IRS/CDS/FX/Equity classes)	regulatory	dtcc.com GTR public reports [16,17]	transaction/aggregated	2015 →	T+0/T+1 publication	web + API endpoints	public (free subsets)	only public subset; trade-level filing is restricted (action-note: ingest public report catalog)	W1	connect	VERIFIED-UPSTREAM-DOC
55	DTCC GTR other jurisdictions (EU/HK/Japan/Australia public subsets)	regulatory	GTR publications [17]	aggregated	varies	T+1	web	public minimum bars	per-jurisdiction scope varies (action-note: enumerate per-jurisdiction availability)	W2	connect	SURVEY-EVIDENCE
56	ESTR futures (Eurex)	rates	eurex.com products	daily	2019 →	EOD	web/download	exchange-fee	(action-note: confirm product & file access)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
57	EURIBOR futures (Euronext/LIFFE legacy) + STIR futures touchstones	rates	euronext.com market data	daily	decades	EOD	web	exchange-fee	(action-note: defer)	W3	connect	SURVEY-EVIDENCE
58	CFTC COT legacy Futures-only archive	regulatory	CFTC Historical-Compressed [11]	weekly (per commodity)	1986 →	T+3 (week after Tue)	zip/txt download	public/free	grain-in-bushels until 1998 (action-note: ingest full archive)	W0	connect	VERIFIED-UPSTREAM-DOC
59	CFTC COT legacy futures-and-options combined archive	regulatory	CFTC Historical-Compressed [11]	weekly	combined availability start is earlier than futures-only in places (confirm exact per-commodity boundary)	T+3	zip	public/free	combined vs futures-only OI difference (action-note: ingest; combined start year)	W0	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
60	CFTC Disaggregated COT (futures-only + combined; per-commodity fences)	regulatory	CFTC Historical-Compressed [11]	weekly	2006-06-30 → (files from 2009+ report years; 2006-2016 backfill zip)	T+3	zip	public/free	only reporting-funded universe (action-note: ingest)	W0	connect	VERIFIED-UPSTREAM-DOC
61	CFTC Traders-in-Financial-Futures (TFF) futures-only & combined	regulatory	CFTC Historical-Compressed [11]	weekly	2006 → (2010-07 file boundary in some; confirm)	T+3	zip	public/free	dealer vs LSP vs AR vs OR classification (action-note: ingest)	W0	connect	VERIFIED-UPSTREAM-DOC
62	COT Public Reporting Environment (Socrata API)	regulatory	publicreporting.cftc.gov (Socrata)	weekly	see rows 58-61	T+3	REST (Socrata/Socrata open-data)	public	API pagination (action-note: bulk pulls via Socrata endpoints)	W0	connect	VERIFIED-UPSTREAM-DOC
63	BIS FX derivatives pivot (FX swaps, forwards, NDF, CCS notional outstanding)	macro-derivs	bis.org statistics	quarterly (+ triennial detail)	1998 →	T+months	web CSV/Excel	public	reporting-dealer basis; geog breakouts (action-note: ingest quarterly tables)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
64	BIS OIS / interest-rate deriva-tives pivot (IR swaps & options outstand-ing)	macro-derivs	bis.org statistics	quarterly	1998 →	T+months	web	public	(action-note: ingest)	W1	connect	SURVEY-EVIDENCE
65	NDF & FX-forward quotes (vendor)	fx-derivs	vendor-bulk	daily quotes	vendor window	EOD	vendor-bulk	vendor-tier	corresponds with 06-vendor diligence stream (action-note: pick vendor after stream 06)	W3	connect	SURVEY-EVIDENCE
66	BIS global-liquidity / FX-swaps outstand-ing series	fx-derivs	bis.org statistics	quarterly	decade	T+months	web	public	(action-note: ingest)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
67	Tardis.dev per-exchange crypto tick history (incl. Deribit options, delisted exchanges: BitMEX/FTX)	crypto-tick	tardis.dev HTTP/replay [1,2]	tick (100ns ts)	per-exchange (many since ~2017; Deribit-collection start later than exchange inception — confirm per exchange)	T-6min	HTTP + python/node libs	vendor-tier (subscription)	incidents published; resubscription gaps (action-note: org pricing + per-exchange earliest-date pull)	W1	connect	VERIFIED-UPSTREAM-DOC
68	Tardis.dev downloadable CSVs (S3)	crypto-tick	tardis.dev CSV service [2]	tick	as row 67	batch	S3	subscription	normalization re-derivable (action-note: cost compare vs API)	W1	connect	VERIFIED-UPSTREAM-DOC
69	Deribit realtime API (WS/REST/FIX + Starbase SBE multicast for colo)	crypto-derivs	docs.deribit.com [14]	tick	n/a (live)	realtime	WS/FIX/SBE	exchange-free (rate limits)	inception 2016 (action-note: capture later; docs verified)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
70	Deribit DVOL volatility index + historical (exchange inception ~2016)	vol-index	Deribit API / iv-index [14]	daily	confirm DVOL history depth (inception claim)	EOD + realtime	REST	exchange-free	DVOL chain composition (30d) (action-note: DVOL retrieval endpoint + history start)	W0	verify	SURVEY-EVIDENCE
71	Bybit derivatives historical (funding/OI/mark/liquidations)	crypto-derivs	api-docs.bybit.com [25]	endpoint-dependent	per product inception	EOD+realtime	REST	exchange-free	per-venue O42/liq-rules differences (action-note: REST ingest)	W0	connect	SURVEY-EVIDENCE
72	OKX derivatives historical (funding/OI/mark/liquidations)	crypto-derivs	okx.com docs-v5 [24]	endpoint-dependent	per product	EOD+realtime	REST	exchange-free	funding interval changes (action-note: REST ingest)	W0	connect	SURVEY-EVIDENCE
73	Binance derivatives historical (USDT-M/COIN-M futures + European options; funding/OI/mark/liq)	crypto-derivs	developers.binance.com derivatives [23]	endpoint-dependent	long histories via API sub-archive	realtime+bars	REST	exchange-free	DELISTED-series handling (action-note: REST ingest)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
74	Coinglass OI dataset (per-pair & aggregated OHLC history incl. stable/coin-margin splits)	aggregator	coinglass API [3]	OHLC hist per pair	multi-year via API	intraday	REST	vendor-tier	aggregation across venues (action-note: ingest)	W1	connect	VERIFIED-UPSTREAM-DOC
75	Coinglass funding-rate dataset ((hist + OI-weight + vol-weight + exchange lists)	aggregator	coinglass API [3]	OHLC hist	multi-year	intraday	REST	vendor-tier	cross-exchange coverage (action-note: ingest)	W1	connect	VERIFIED-UPSTREAM-DOC
76	Coinglass liquidation dataset (pair/coin history, heatmaps, 7d orders, max-pain)	aggregator	coinglass API [3]	event+aggregate	historical liquidations	near-realtime	REST	vendor-tier	heatmap models are derived (not actual fills) — PIT not guaranteed (action-note: ingest actual liq events only for PIT)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
77	Coinglass L/S ratios, taker flow, CVD, futures netflow	flows	coinglass API [3]	multi-frequency	multi-year	near-realtime	REST	vendor-tier	Top-Trader subsets (action-note: ingest)	W1	connect	VERIFIED-UPSTREAM-DOC
78	Coinglass Hyperliquid whale/position distribution end-points	flows	coinglass API [3]	wallet-level	recent	realtime	REST	vendor-tier	(action-note: ingest)	W2	connect	VERIFIED-UPSTREAM-DOC
79	Amberdata derivatives (options surface / DVOL / OData API)	crypto-opts	amberdata.io	tick/daily	vendor archive	intraday	API	enterprise-quote	(action-note: ; de-prioritize vs Tardis)	W3	quote	SURVEY-EVIDENCE
80	Laevitas derivatives analytics	crypto-opts	laevitas.ch	daily	vendor archive	intraday	API	vendor-tier		W3	quote	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
81	Pyth Pro price feeds (incl. commodity futures symbols like WTI, plus equities/crypto — subscription product, prev. Lazer)	oracle	docs.pyth.network [4]	tick	via History API	sub-second (configure per subscription)	WS + REST + compass	subscription-gated	verify per-feed latency & symbol set vs VTL? (action-note: confirm feed IDs + history API depth)	W1	connect	VERIFIED-UPSTREAM-DOC
82	Chainlink Data Streams (crypto/forex out-of-band oracle feeds)	oracle	docs.chain.link/data-streams [5]	tick	n/a (live)	~sub-second (verify page spec)	WS/REST	subscription-gated	latency claims not read from page (fetch returned sparse) (action-note: pull Streams latency spec + pricing)	W2	connect	VERIFIED-UPSTREAM-DOC
84	WFE derivatives volume statistics (regional annual)	reference	focus.world-exchanges.org [28]	annual/monthly	decades	T+months	web	public/free	scope includes futures+options (action-note: ingest as context metric)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
85	Exchange market-data auctions (e.g., OCC data-sale? venues auction OPRA-class rights)	meta	various (OCC/venedie?)						(action-note: research actual existence & mecha-nism)	W3	connect	UNVERIFIED-CANDIDATE

07 — FX · rates · macro · credit · ESG (103 rows)

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	ecb-fx-ref	fx-ref	ECB Data Portal EXR family (SDMX 2.1/3.0 CSV+JSON REST, keyless)	daily 16:00 CET fix	1999-01-04→; ≈30-currency set (USD JPY GBP CHF CNY AUD BRL CAD HKD IDR KRW MXN MYR NZD PHP SGD THB ZAR + EU non-euro)	D0 fix	SDMX REST; Frank-furter fail-over	free-attribution	official ECB fix; No free intraday leg any-where (gap — intraday = licensed rows 20/21) (action-note: build fetcher; 1999→)	W0	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
2	frankfurter	fx-ref	Frankfurter v2 REST `api.frankfurter.dev` — 223 curren-cies / 105 official providers; blended + per-provider routes; CSV/NDJSON; Docker self-host	daily (some providers intraday series weekly/monthly)	1948→ (provider-dependent)	daily	REST keyless; no quotas; self-host	open (blended = Frank-furter calc, free incl. com-mercial); per-provider route = provider terms	compliance → always filter `providers=ecb` etc.; blended recalcu-lated as data arrives → pin date+provider key for PIT (action-note: integrate as unified FX spine + provider-attribution log)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	fed-h10	fx-fix	Fed H.10/G.5 Foreign Ex-change Rates (`releases/h10/current` + DDP; Nomi-nal/Real Indexes + Currency Weights sub-pages)	daily (G.5 weekly)	1971→ (per-currency start varies)	T+1 ~16:00 ET	DDP CSV/JSON/XML	US-gov public	NY noon buying rates ≠ market close/4pm fix; single-price statistic (no bid/ask); PIT = release calendar (action-note: build fetcher (H.10 + G.5))	W0	connect	VERIFIED-UPSTREAM-DOC
4	boe-fx	fx-fix	BoE Sta-tistical Interac-tive Database (`/statistics/exchange-rates`; boeapps CSV endpoint)	daily + monthly averages	GBP/$ daily 1975→ (verify per-series starts)	T+1	boeapps CSV URL builder	UK OGL attribu-tion	long monthly averages pre-1975 into historical runs — verify coverage at build (action-note: daily + monthly)	W1	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	boj-ttss	fx+rates	BoJ Time-Series Statistics Search (TTSS, SDMX 2.1)	daily/monthly	FX + rate panels decades (1974 FX leg — verify)	T+1	TTSS SDMX/CSV catalog	BoJ stat-use terms (no open redistribution)	statutory re-use agreement required before redistribution (action-note: parse TTSS catalog; JPY set)	W1	connect	UNVERIFIED-CANDIDATE
6	rba-f11	fx-fix	RBA Statistical Tables F11/F11.1 (`/statistics/tables/csv/*.csv`, direct endpoints + xls)	daily (F11.1) / monthly (F11)	AUD/USD since Dec-1983 float; monthly deeper (verify per-table)	D+1 16:10 AEDT	direct CSV/xlsx GET, keyless	RBA fair-use attribution	RBA publishes series-break + discontinued pages officially; revisions in release notes (action-note: ingest csv endpoints +)	W0	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
7	boc-valet	fx-fix	Bank of Canada Valet API (`/valet/observations/FXUSDCAD`, JSON/CSV)	daily	1971-01→ (verify)	T+1	Valet REST, keyless	BoC terms attribution	Valet page returned empty twice on 2026-10-24 fetch — confirm keyless route live before reliance; CORRA piggybacks same rail (action-note: probe API; CAD sets)	W1	backfill	UNVERIFIED-CANDIDATE
8	snb-fx	fx-fix	SNB data portal (data.snb.ch CSV/ZIP + REST candidates)	daily + monthly avg	CHF reference daily; monthly averages long-history (verify)	T+1	CSV/XLSX download + API candidate	SNB terms (attribution)	verify depth claim before storing long-run monthly averages (action-note: map SNB topic codes)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
9	hkma-fx	fx-fix	HKMA statistics (USD/HKD spot fixings, HIBOR fixings)	daily	2005?→ verify	D+0	web CSV/XLSX	HKMA terms	fixing publica-tion conven-tion to verify; HIBOR tenor panel check (action-note: loop CSV; HKD/HIBOR legs)	W2	connect	UNVERIFIED-CANDIDATE
10	bok-ecos	fx+rates	BoK ECOS database (ecos.bok.or.kr JSON API + CSV)	daily/monthly	KRW FX + rates decades (verify)	D+1	ECOS JSON API	BoK terms	unofficial English docs; series-code mapping community-driven (action-note: probe ECOS series)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	tsy-rates-of-exchange	fx-monthly	Treasury Fiscal Data v1/accounting/od/rates_of_exchange (docs example verified)	quarterly in docs example (periodicity verify)	accounting-rates span (verify)	periodic	REST JSON/CSV/XML keyless	US-gov public — no restrictions	USG accounting rates, NOT market quotes — exclude from benchmark panels; cross-check rail only (action-note: ingest for cross-check panel)	W0	connect	VERIFIED-UPSTREAM-DOC
12	imf-er	fx-monthly	IMF IFS/ER exchange-rate family (data.imf.org SDMX + DataMapper)	monthly	1957→ (IFS print era)	monthly/quarterly	SDMX + DataMapper API	IMF free attribution	representational rates regime-dependent (fix vs float encoding) — country use-tags needed (action-note: ingest ER family)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
13	bis-eer	fx-index	BIS Effective Exchange Rates (data.bis.org + `/bulk-download`; SDMX API v2)	monthly + daily (publishes daily EER per docs)	narrow 26/27-ccy 1964→; broad 64-ccy 1994→; 2020=100	latest period live	SDMX + bulk-download xlsx/csv	BIS permitted-use attribution	3-yr rolling trade weights documented (pre-1990 narrow back-ext uses 1990-92 weights); base switch = break to log (action-note: ingest daily+monthly EER set)	W0	connect	VERIFIED-UPSTREAM-DOC
14	bis-triennial-fx	fx-vol·archive	BIS Triennial FX Survey datasets (rpfx editions; coordinated with 52 jurisdictions)	every 3 yrs (April)	FX leg since 1986 (IR-derivatives leg since 1995); Apr-2022 global $7.5tn/day	triennial	edition XLSX/CSV + PDF release	BIS permitted use	gauge/coverage changes between editions; not time-series-comparable bin-to-bin (methodology notes per edition) (action-note: archive all editions ≤ latest)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
15	dukascopy-bi5	fx-tick	Dukascopy historical-data feed (public per-instrument per-hour `.bi5` LZMA files)	tick ms bid/ask (+vol)	~2003→ majors	archived intraday	raw-HTTP URL template + LZMA decode → parquet	research/no-resale (licence text to confirm)	page fetch 2026-10-24 rendered nav-only — confirm bi5 URL template + DST/timezone conven-tion at build (action-note: crawler + LZMA→parquet)	W1	connect	UNVERIFIED-CANDIDATE
16	truefx-ticks	fx-tick	TrueFX historical down-loads (monthly zips; Integral Develop-ment Corp)	tick ms top-of-book bid/ask	16+ majors, 2009→	monthly zips	registration → ZIP fetch	free/research; redistri-bution prohib-ited (Integral ©2009–2026)	indicative interbank aggre-gate — not executed tape; gap-audit each month-zip (action-note: bulk month-zips + gap audit)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
17	histdata-fx	fx-tick	HistData.com per-pair M1 + tick (1s resolution) ASCII CSV zips	M1 / tick-1s	2000s→ (per-pair varies)	monthly (last update 2026-10-18 per site)	web zips; FTP/SFTP $7/mo via PayPal; GoogleDrive auto-update $7/mo	free-attribution; paid rails voluntary	T1/T2 bid-ask split conventions; volume fields unreliable; treated as approximation, not official tape (action-note: majors + monthly diff job)	W0	backfill	VERIFIED-UPSTREAM-DOC
18	stooq-fx	fx-eod	Stooq plain CSV endpoints (`stooq.com/q/d/l/`)	daily	decades per pair	D+0 EOD	plain CSV GET, keyless	undocumented — no explicit licence found	unverified upstream; do not use before licence path confirmed (action-note: licence check → then)	W2	backfill	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
19	oanda-rates	fx-eod	OANDA v20 Rates API (practice account) + FXDS paid service (32y/38k-pairs vendor claim)	daily/hourly aggre-gates	free-tier recent; paid deep (32y claim)	near-realtime (live endpoint)	REST v20	contractual; paid fixings licence-gated	indicative quotes ≠ official fix; quote-age = PIT marker (action-note: practice rail now; FXDS RFQ later)	W1	connect	SURVEY-EVIDENCE
20	wm-4pm-fix	fx-fix	LSEG WM/Reuters 4pm London fixings (+ intraday hourly markers)	daily + intraday	1990s?→ verify	D0 16:00 London	LSEG feed/API	licensed quote	THE insti-tutional buy-side bench-mark; BMR referen-cability rules — audit trail required (action-note: RFQ scope G10+ majors)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
21	bloomberg-bfix	fx-fix	Bloomberg BFIX bench-mark (New York 4pm et al.)	daily fix	2016?→ verify	D0	Bloomberg feed/library	licensed quote	BFIX method-ology docs; NDF coverage check at RFQ (action-note: RFQ parallel w/ [20])	W3	connect	SURVEY-EVIDENCE
22	cls-data	fx-vol	CLS settle-ment/volume data (ag-gregate CSV/PDF; granular via CLS)	monthly	2002?→ verify	monthly	web CSV aggre-gates	free ag-gregates; granular licensed quote	CLS-covered turnover only — NOT com-parable to BIS triennial totals (action-note: aggre-gates first; granular later RFQ)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
23	sofr-nyfed	rates-fix	NY Fed SOFR + SOFR Averages & Index + BGCR + TGCR	daily	2018-04-02→	T+1 ~08:45 ET	CSV/XLSX release pages	US-gov public	T+1 publication calendar; corrections/republications logged; first-print = PIT value (action-note: ingest all 4 rails)	W0	connect	VERIFIED-UPSTREAM-DOC
24	effr-obfr	rates-fix	NY Fed EFFR/OBFR (+ FRED mirrors)	daily	EFFR 1954→	T+1	NY Fed CSV + FRED API	US-gov public	floor-regime shifts (2008+, 2021) as break-tags (action-note: ingest)	W0	connect	SURVEY-EVIDENCE
25	estr-ecb	rates-fix	ECB €STR (data portal, keyless)	daily	2019-10-01→	T+1 08:00 CET	SDMX REST	free-attribution	€STR = blended unsecured euro O/N; volume metric co-published (action-note: ingest)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
26	sonia-boe	rates-fix	BoE SONIA (IUDSOIA) + SONIA Compounded Index (boeapps CSV, free)	daily	Mar-1997→; BoE admin since 2016; reformed 2018; Compounded Index from 2020-08-03	T+1 10:00	boeapps CSV, free	free (administered benchmark)	pre-2016 RFR = legacy methodology (break tag); pre-2018 reform breaks tagged (action-note: 1997→ + compounded index)	W0	backfill	VERIFIED-UPSTREAM-DOC
27	saron-snb	rates-fix	SNB/SIX SARON	daily	2009?→ verify	T+1 10:00	SNB web (free display)	free reference value; SIX DB licensed for depth	free route may lag SIX history — SWISS arms-length check (action-note: free daily route now; SIX later)	W1	connect	UNVERIFIED-CANDIDATE
28	tona-boj	rates-fix	BoJ TONA (uncollateralised O/N call)	daily	2008?→ verify	T+1	BoJ TTSS/fin-market CSV	BoJ terms	history/availability verify at build (action-note: map TONA code)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
29	corra-boc	rates-fix	BoC CORRA + CORRA index (Valet rail)	daily	2018-07→	T+1 ~12:00 ET	Valet REST	BoC terms	same Valet de-pendency as [7] (action-note: piggy-back Valet probe)	W1	connect	UNVERIFIED-CANDIDATE
30	sora-mas	rates-fix	MAS SORA (SG)	daily	SORA comp. index 2016?→ verify	T+1	MAS web CSV	MAS terms	route + depth verify at build (action-note: add SG RFR leg)	W2	connect	UNVERIFIED-CANDIDATE
31	bbsw-asx	rates-fix	ASX BBSW / bid rate (AUD bench-marks)	daily	1986?→ verify	D0	ASX licensed feed	licensed quote	AUD float-rate bench-mark — licence scope check (non-AUS use) (action-note: RFQ)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
32	libor-legacy	rates-fix·archive	Legacy LIBOR panel rates — cessation archive (ICE; FCA mandated windows; USD ended 2023-06-30)	daily	1980s?→ cessation	static archive	ICE/published archive files	legacy licence — reuse restricted; **new benchmarking prohibited**	usable for historical/fallback-transition matrices only — never model-forward (action-note: archive free portions only)	W2	connect	UNVERIFIED-CANDIDATE
33	fred-h15-daily	rates-curve	FRED H.15 daily family (DGS1MO→DGS30; CP rates; discount; ratios)	daily	DGS 1962→; long-term averages 1953→	D0/D+1	FRED API v2 (free key)	US-gov public	CMT-based constant-maturity — not spot; no forward curve implied (action-note: panel to 1953/1962)	W0	backfill	SURVEY-EVIDENCE
34	fred-tips	rates-curve	FRED TIPS real yields (DFII1–DFII30) + breakevens (T5YIE, T10YIE)	daily	real 1997→; BE 2003→	D0	FRED API	US-gov public	CB-computed indirect breakeven ≠ swap-market BE (action-note: ingest)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
35	fred-monthly-weekly	rates	FRED weekly/monthly rate set (FF, MORT-GAGE30US 1971→, BANKPRIME, WPesos)	weekly/monthly	decades	per release	FRED API	US-gov public	MORTGAGE30US = Freddie Mac survey, not CMT (action-note: ingest)	W0	connect	SURVEY-EVIDENCE
36	alfred	rates·archive	ALFRED vintage-dated FRED series (`fred/alfred` params)	vintage per revision	back to each series' first release	per-vintage	FRED API (vintage params)	US-gov public	**the** PIT-safe macro rail; store observa-tion_date + vin-tage_date (action-note: PIC factory + PIT merge engine)	W0	connect	SURVEY-EVIDENCE
37	tsy-cmt-par	rates-curve	Treasury par curve (daily CSV/XML; long-term averages + TIPS real curve)	daily	1990→ (LT averages deeper; verify)	D0	treasury.gov/interest-rates CSV	US-gov public	par vs CMT method (boot-strap-pable); fetch of page timed out 2026-10-24 — retry at build (action-note: + align vs FRED DGS)	W0	backfill	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
38	tsy-auctions	rates-auction	TreasuryDirect Auction Search (CSV/JSON/TSV/XML end-points)	per auction	TIPS 1997→; all others 1998-07-27→; fields extended Apr-2008 + Oct-2026	per auction	auctionQuery endpoint, free	US-gov public	settlement/note/coupon cross-check vs NFP auction PR; some string-typed fields (action-note: ingest per security type)	W0	connect	VERIFIED-UPSTREAM-DOC
39	tsy-fiscaldata	rates-fiscal	Treasury Fiscal Data API families (MTS tables 1–9, MSPD, Debt-to-Penny, avg interest rates, DTS)	daily→monthly	per table (DTS daily)	daily	REST JSON/CSV/XML keyless (182 tabular end-points)	US-gov public no-restrictions	record_date = as-published → PIT; re-reporting normal (keep history) (action-note: bulk-ingest MTS + DTS)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
40	frb-h41-h8	macro-bank	Fed H.4.1 (weekly factors) + H.8 (weekly bank aggregates)	weekly	H.4.1 2002?→; H.8 1973?→ verify	weekly	DDP CSV/JSON	US-gov public	H.8 = commercial banks only (not full Fed framework); reserve breaks documented (action-note: ingest weekly rails)	W0	connect	SURVEY-EVIDENCE
41	frb-z1	macro·archive	Fed Z.1 flow-of-funds (DDP bulk CSV)	quarterly	1945→	quarterly	DDP bulk CSV/JSON	US-gov public	annual revision cycle (Sep) → recompute derived sector flows (action-note: ingest DDP bulk)	W0	connect	SURVEY-EVIDENCE
42	frb-g19	macro	Fed G.19 consumer credit	monthly	1943 annual origin / monthly ~1945 (verify)	monthly	DDP CSV	US-gov public	seasonal revisions reset bucket shares (action-note: ingest)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
43	infl-swaps	rates-deriv	Bloomberg/ICE 10y zC CPI inflation swaps implied breakevens	daily	market-driven	D0	Bloomberg/ICE licensed	licensed quote	swap BE vs CMT BE — validate basis; zC conven-tions differ (action-note: RFQ only if basis-critical)	W3	connect	UNVERIFIED-CANDIDATE
44	bis-otc-deriv	rates-vol	BIS OTC deriva-tives statistics (notion-als; triennial since 1995 + semian-nual/quarterly detail)	3-yr/semiannual	1995→ IR lag	periodic	SDMX + bulk-download	BIS per-mitted use	notional-type ≠ risk type (convert to paired gross/market-value); legacy Ten-view conven-tions (action-note: ingest OTC suites)	W1	connect	SURVEY-EVIDENCE
45	ecb-policy-rates	rates	ECB key rates (DFR/MRO/MLF; key-rates page)	event-based	1999-01→	event	SDMX keyless	free-attribution	operational-facility changes (e.g. clear route since <1981) — tag (action-note: ingest)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	ecb-govcurves	rates-curve	ECB €-area gov bond curves (AAA + non-AAA spot; EMU convergence yields)	daily	2004?→ verify	D0	SDMX keyless	free-attribution	read ECB curve-construction docs once, pin version (action-note: ingest)	W0	connect	SURVEY-EVIDENCE
47	eurostat-sdmx	macro	Eurostat SDMX REST (HICP, LFS unemployment, trade)	monthly	1960s→ many series	monthly	SDMX keyless	free-attribution	country-specific HICP re-basing chains (action-note: ingest core families)	W0	connect	SURVEY-EVIDENCE
48	bis-credit-gaps	macro	BIS credit-to-GDP + credit-gap stats (bis.org/statistics/credigy)	quarterly	credit series: verify 1948 claim; gap computation 1999→	quarterly	SDMX + bulk-download	BIS permitted use	brief's "since 1948?" → validate at build; not all countries start same (action-note: ingest + span audit)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
49	bis-long-run-hist	macro·archive	BIS long-run historical monetary/credit statistics (bulk-download)	monthly/annual	claimed 1948→ (verify)	static	bulkdownload TXT/CSV	BIS permitted use	pre-modern-convention mix risk; verify spans before merging into modern frame (action-note: fetch bulk; validate spans)	W2	connect	UNVERIFIED-CANDIDATE
50	imf-ifs	macro-archive	IMF IFS monthly database (data.imf.org)	monthly	1957→ (print began Jan-1957)	monthly	SDMX (data.imf.org)	IMF free attribution	country coverage/breaks historically uneven (action-note: ingest IFS core)	W1	connect	SURVEY-EVIDENCE
51	imf-ifs-vintages	macro·archive	IFS vintage archive (eLibrary print scans + archived editions)	monthly editions	1957→ scans	static	eLibrary PDF → OCR	IMF use-terms	OCR pipeline cost; PIT = printed vintage (action-note: targeted OCR probes)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
52	imf-weo-archives	macro·archive	IMF WEO vintage archives (April + October edition datasets)	semiannual	prior editions back-separated (earliest edition verify)	biannual	data.imf.org edition files (xLS/SDMX)	IMF free	edition date-stamp = forecast PIT; use not only latest (action-note: archive all editions ≤ latest)	W1	connect	SURVEY-EVIDENCE
53	imf-cofer	macro-archive	IMF COFER (official FX reserve composi-tion; SDR rates)	quarterly	1999Q1→ pool currency split window	quarterly	SDMX/CSV	IMF free-attribution	currency-assignment revisions possible; shares may ex-ceed/under tend due to unallo-cated (action-note: ingest quar-terly)	W0	connect	SURVEY-EVIDENCE
54	imf-fsi	macro	IMF Financial Sound-ness Indica-tors	quarterly	2000s→ (verify per-country)	quarterly	SDMX	IMF free-attribution	NPL defi-nitions diverge; curation grid needed (action-note: ingest FSI core)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	wb-wdi	macro	World Bank WDI API (`api.worldbank.org/v2`; verified URL/arg queries, XML/JSON; bulk ZIPs + BigQuery mirror)	annual (some sub-annual)	1960→	annual	REST v2 + bulk ZIP	CC BY 4.0	no vintages — append revision LOG of published correc-tions (action-note: ingest (+ BigQuery mirror))	W0	connect	VERIFIED-UPSTREAM-DOC
56	wb-gpi	macro	WDI gender-parity (GPI) indicator family (SE.* / SG.*)	annual	per indicator	annual	same WDI API codes	CC BY 4.0	pin exact indicator codes + defini-tions at build (action-note: map GPI codes)	W2	connect	UNVERIFIED-CANDIDATE
57	wb-long-run	macro-archive	World Bank legacy historical long-run (older World Tables volumes + assisted series)	annual	mid-20th-C → 1960	static	bulk CSV + archive contain-ers	attribution	pre-1960 per-country coverage patchy; treat as supple-ment to WDI spine (action-note: deep-history fills)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
58	oecd-mei	macro	OECD SDMX (MEI/CPI monthly; composite leading indicators; quarterly national accounts)	monthly	1960s→	monthly	SDMX REST (JSON/XML)	CC BY 4.0	CLI composition revisions = factor breaks (action-note: ingest MEI + CLI)	W0	connect	SURVEY-EVIDENCE
59	boe-rates-curve	rates-curve	BoE rates + gilt curve (Bank Rate 1694→ documented; sterling 5/10/20y daily curve)	daily	Bank Rate 1694→	T+1	boeapps CSV/XLSX	UK OGL	read curve-construction docs once, pin version (HM Treasury linkage) (action-note: ingest Bank Rate + curve)	W0	connect	SURVEY-EVIDENCE
60	boj-jgb	rates-curve	BoJ JGB daily yield list (TTSS route)	daily	1970s?→ verify	D0	TTSS/BoJ CSV	BoJ terms	maturity panel depth varies by era (action-note: JGB curve ingest)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	pboc-chinamoney	rates	PBoC / Chinamoney (CFETS) — Shibor (2006→), LPR (2019→)	daily	per fixings	D+1	Chinamoney EN web CSV	CN public info — non-commercial	no API for org redistribution; scraping resilience needed (action-note: parse daily fixings)	W1	connect	SURVEY-EVIDENCE
62	rbi-dbie	rates	RBI DBie (SDMX; repo, G-sec curve, CPI/WPI)	daily/monthly	1991→ mostly	daily	DBie SDMX/CSV	public gov-data attribution	some legacy series PDF-only (action-note: SDMX ingest)	W0	connect	SURVEY-EVIDENCE
63	safe-cn	macro	SAFE reserves/settlement data (PDF/HTML releases)	monthly	2010→ verify	monthly	web scrape	public CN info (no API)	manual QA; CN capital-flow proxy (action-note: monthly scrape)	W2	connect	SURVEY-EVIDENCE
64	cnb-arad	rates	Czech National Bank ARAD time-series system (confirmed URL `cnb.cz/arad`)	daily/monthly	deep CZK curve + macro (1990s?→ verify)	D+1	ARAD app/API	CNB terms	confirm REST/CSV API spec at build (action-note: probe + ingest)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	boi-rates	rates	Bank of Israel data portal (rates/FX/macro)	daily/monthly	verify	daily	BoI web CSV?	BoI terms	bilingual (HE/EN) metadata QA; API existence uncon-firmed (action-note: probe)	W1	connect	UNVERIFIED-CANDIDATE
66	cbs-israel	macro	Central Bureau of Statistics — Israel (CPI)	monthly	deep (verify)	monthly	CBS web CSV	public gov	CPI rebase method-ology notes per year (action-note: CPI leg)	W2	connect	UNVERIFIED-CANDIDATE
67	sarb-stats	rates	South African Reserve Bank statdata (repo curve, S.A. macro)	daily/monthly	verify	daily	SARB statdata JSON?/web	SARB terms	API shape undocu-mented — probe first (action-note: probe)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
68	bcb-sgs	rates	BCB Brazil Série-Históricas API (`api.bcb.gov.br/dados/serie/bcdata.sgs.*`)	daily/monthly	Selic 1986?→; CPI 1980s?→ verify	D+1	REST JSON, keyless	open BCB data (CC-BY? confirm)	fetch 2026-10-24 got HTTP 406 (UA-block) — retry w/ browser UA; map SGS codes (e.g. 432 Selic) (action-note: probe w/ UA; code map)	W0	connect	UNVERIFIED-CANDIDATE
69	bcch-sdmx	rates	Banco Central de Chile SDMX REST (si3.bcentral.cl; TPM, UF, CPI)	daily/monthly	verify depth	daily	SDMX REST	BCCh terms	coverage nuances; personal data regs (action-note: probe)	W1	connect	UNVERIFIED-CANDIDATE
70	bsp-ph	rates	Bangko Sentral ng Pilipinas statistics (RRP, CPI, PHP fx)	daily/monthly	verify	daily	BSP web XLS	BSP terms	XLS/PDF mix; extraction fragility (action-note: probe)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
71	bls-cpi	macro-archive	BLS CPI (CU/CW/SU) — series database + Public Data API v2 + down-load.bls.gov/pub/time.series/cu text files	monthly	1913→ national	monthly	API v2 (free key) + text files	US-gov public	annual baseline revisions; chained-CPI (SU) separate lineage — never mix SA/NSA (action-note: ingest + seasonal-ity flags)	W0	connect	VERIFIED-UPSTREAM-DOC
72	bls-ppi	macro	BLS PPI (pc / wp commod-ity + FD-ID)	monthly	1980s?→ deep (verify)	monthly	API + text files	US-gov public	2011 FD-ID reform breaks (action-note: ingest)	W0	connect	VERIFIED-UPSTREAM-DOC
73	bls-jolts	macro	BLS JOLTS (job open-ings/hires/quits)	monthly	2000-12→	~5-6 wk lag	API + /jlt/ files	US-gov public	3-release revision ladder; flag final only (action-note: ingest)	W0	connect	VERIFIED-UPSTREAM-DOC
74	bls-eci-ecec	macro	BLS ECI + ECEC (compen-sation)	quarterly	ECI 1975?→ verify	quarterly	API	US-gov public	seasonally-adjusted subset only (action-note: ingest)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
75	census-adv-trade	macro	Census advance interna-tional trade (FT900 monthly)	monthly	1992?→ verify	~6-weeks lag	Census EP API (key free) + bulk	public	advance→final revision chain; joint Cen-sus/BP ITA con-solidation (action-note: ingest)	W0	connect	UNVERIFIED-CANDIDATE
76	census-construction	macro	Census construc-tion spending (VIP)	monthly	1964?→ verify	~1mo	Census API	public	seasonal + 1-month revision ladder (action-note: ingest)	W0	connect	UNVERIFIED-CANDIDATE
77	census-retail	macro	Census retail (MRTS / advance retail)	monthly	1992?→ verify	~2wk	Census API	public	NAICS re-basing cycles (action-note: ingest)	W0	connect	UNVERIFIED-CANDIDATE
78	bea-nipa	macro	BEA NIPA (GDP/investment/personal income; annual + quarterly tables)	quarterly + annual	annual 1929→; quarterly 1947→	monthly esti-mates/revision calendar	BEA API (free key)	US-gov public	summer annual-revision resets; PIT only via ALFRED [36] (action-note: ingest + PIT walk)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
79	conference-board	macro	Conference Board LEI/CLI + Consumer Confidence	monthly	LEI 1959?→ verify	monthly	subscription/API	licensed quote	proprietary index construction; component weights change (action-note: RFQ)	W3	connect	UNVERIFIED-CANDIDATE
80	spf-philly	macro-archive	Philadelphia Fed Survey of Professional Forecasters	quarterly	1968Q4→	quarterly	free XLS/CSV data files	US public (Fed bank)	free vintage archive of forecasts (consensus) — rare PIT grader (action-note: ingest + forecaster breakdowns)	W0	connect	SURVEY-EVIDENCE
81	michigan-sca	macro-archive	University of Michigan Surveys of Consumers (ICS/MCS)	monthly	1978 monthly (earlier reconstruction verify)	monthly	free site CSV/XLS archive	UMich terms (research)	keep prelim vs final distinction; revisions history (action-note: archive + register if needed)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
82	nfib-sbet	macro-archive	NFIB Small Business Economic Trends	monthly	1974?/1986?→ verify	monthly	NFIB web PDF/CSV	NFIB terms (some restricted)	archive availability + cost unconfirmed (action-note: archive index + subcomponents)	W1	connect	UNVERIFIED-CANDIDATE
83	ism	macro	ISM Manufacturing/Services Indices	monthly	mfg 1948?→ (verify)	monthly	ISM web (paywalled core)	licensed quote	free townhall PDFs lack historical depth (action-note: assess ISM web data routes)	W2	connect	UNVERIFIED-CANDIDATE
84	spg-pmi	macro	S&P Global (ex-Markit) PMIs (manufacturing/services/composite, 30+ countries)	monthly	1998→ majors (some 1980s-era legacy)	monthly	licensed (data/API)	licensed quote	PMI revision ladder — pin first print for PIT (action-note: RFQ with historical-archive scope)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
85	caixin-pmi	macro	Caixin China PMIs (manufacturing + services, CN)	monthly	2011→	monthly	licensed	licensed quote	CN business-cycle factor alternative (pairs w/ [61]) (action-note: RFQ)	W3	connect	UNVERIFIED-CANDIDATE
86	fred-credit-spreads	credit	FRED credit spreads (Moody’s Baa-Aaa 1983→; ICE BofA HY/IG OAS 1996→)	daily	per-row	D0/D+1	FRED API	public (underlying index values mirror ICE/Moody’s)	index-owner licence references documented; HY OAS = option-adjusted, not raw spread (action-note: ingest + licence notes)	W0	connect	SURVEY-EVIDENCE
87	moody-yield-arch	credit·archive	Moody’s long corporate yield averages (Aaa/Baa via published archives; FRED daily/pre-1983 monthly)	monthly→daily	1919→	historic	FRED (1983→) + archive PDFs	archive use (verify)	splice 1919→1983 monthly with 1983→ daily — break-tag the join (action-note: assemble archive panel)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
88	finra-trace	credit-trades	FINRA TRACE (corporate bond trade reports + aggregates)	intraday trades / EOD	1990s?→ verify era bounds	realtime/EOD	FINRA data subscriptions	licensed quote	fair-use rules; no redistribution; enhanced tier = extra (action-note: RFQ breadth + fair-use review)	W3	connect	UNVERIFIED-CANDIDATE
89	bbg-barclays-indices	credit-index	Bloomberg Barclays Aggregates (US/EU/Global + sub-indices)	daily	US Agg 1986?→ (verify per index)	D0	Bloomberg library	licensed quote	redistribution prohibited; TR vs price-index gates (action-note: RFQ only parent+travel series needed)	W3	connect	UNVERIFIED-CANDIDATE
90	jpm-embi	credit-index	JPMorgan EMBI (EM sovereign spread index family)	daily	1993?→ verify	D0	JPM subscription	licensed quote	index membership changes; hard vs local-currency split (action-note: RFQ)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
91	cma-cds	credit	CMA (S&P Global MI) sovereign CDS quotes	daily	2010s?→ verify	D0	enterprise API	licensed quote (row pre-verified in [S1])	daily close markers only (not intraday); currency/tenor panel spreads (action-note: RFQ)	W3	connect	SURVEY-EVIDENCE
92	sp-default-studies	credit-archive	S&P Ratings annual default/upgrade studies (public PDFs / XLS tables)	annual	1981→ cohort studies	annual	spglobal research publishes PDFs	public publication (data tables vary)	cohort methodology changes by year — pin per-edition definitions (action-note: scrape + parse yearly studies)	W2	connect	UNVERIFIED-CANDIDATE
93	cfed-nfci	credit	Chicago Fed National Financial Conditions Index	weekly	1971→	weekly	FRED/cfed CSV	public	weekly updates; sub-indices available (action-note: ingest)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
94	stlfsi4	credit	St. Louis Fed Financial Stress Index (STLFSI4)	weekly	1994→	weekly	FRED	public	v4 construction differs from prior versions — break-tag (action-note: ingest)	W1	connect	UNVERIFIED-CANDIDATE
95	msci-esg	esg-ratings	MSCI ESG ratings (AAA–CCC; feed/API)	monthly refresh (verify)	rating-change history needed	monthly	MSCI data licensed	licensed quote	PIT = rating-change events (need change log, not snapshot); methodology revisions frequent (action-note: RFQ — demand event history)	W3	connect	SURVEY-EVIDENCE
96	sustainalytics-esg	esg	Sustainalytics (Morningstar) ESG Risk Ratings	company snapshot	coverage list + rating history	monthly	data/API	licensed quote	risk-score scale ≠ MSCI ordinal; standardization layer required (action-note: RFQ)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
97	iss-esg	esg	ISS ESG ratings	monthly	verify	monthly	licensed	licensed quote	methodology version-ing check (action-note: RFQ)	W3	connect	UNVERIFIED-CANDIDATE
98	lseg-esg	esg	LSEG/Refinitiv ESG (pillar and com-ponent scores)	monthly	2002?→ verify	D+1	workspace/API	licensed quote	heavy backfill available but licence-limited (row pre-verified in [S1]) (action-note: RFQ)	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
99	eex-eua-auctions	esg-carbon	EEX EUA primary-market auction results (free CSV down-loads)	per-auction	Germany 2013?→; UK 2021→; per-platform	post-auction	EEX CSV down-loads	free (EU auction data open)	distribution pattern differs per member state (EEX/other platform) — my fetch of EEX path 404'd 2026-10-24; re-discover path at build (action-note: discover path + parse EUA results)	W0	connect	UNVERIFIED-CANDIDATE
100	icap-status	esg-carbon	ICAP Status Report + ETS Map data (annual Excel)	annual	2005?→ verify	annual	site Excel	free (verify licence)	coverage/inclusion rules differ per ETS — map schema first (action-note: annual Excel ingest)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
101	cdp	esg	CDP corporate climate disclo-sures	annually	2002?→ verify	annually	CDP data/API	licensed quote (some open)	disclosure vs score linkage; coverage gaps (action-note: RFQ)	W3	connect	UNVERIFIED-CANDIDATE
102	wb-esg	esg	World Bank sovereign ESG dataset (`data.worldbank.org` ESG col-lection)	annual	aligns to WDI rail 1960→ subset	annual	WB API/CSV (same rail [55])	CC BY 4.0 (assumed same as WDI — verify)	curated ESG-framework subset ≠ WDI superset — keep separate ID names-pace (action-note: piggy-back on WDI ingest)	W0	connect	UNVERIFIED-CANDIDATE
103	sdg-index	esg	UN/SDSN Global SDG Index (SDR Ex-cel/CSV)	annual	2015→ (vs baseline 2000/15) verify	annually	free Ex-cel/CSV	CC BY-NC? verify licence	methodology changes between editions — break-tag (action-note: annual ingest)	W1	connect	UNVERIFIED-CANDIDATE

08 — Commodities · energy · power · shipping (86 rows)

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	wb-pink-sheet	archive	World Bank CMO / Pink Sheet monthly + annual price XLSX (the-docs.worldbank.org fixed doc URLs)	monthly + annual	annual 1960→; monthly long-run 1990s-era	monthly (fixed ~1st of month)	file-drop XLSX → parquet; per-release vintage dir	free (WB dataset terms, CC BY catalog [^1])	rebasing changes historicals ⇒ snapshot per release date (action-note: ingest monthly+annual each release; keep vintage store)	W0	connect	VERIFIED-UPSTREAM-DOC
2	imf-pcp	archive	IMF Primary Commodity Price System (monthly XLSX + pre-rebase supplement)	monthly	current base: 1992→; supplement covers pre-1992 decades	monthly	XLSX file-drop	free w/ attribution	base-year rebases rewrite history ⇒ vintage; fetch blocked 429×3 today (action-note: resolve w/ retry/wget from another IP; then bulk)	W0	backfill	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	lme-official-prices	benchmark	LME Data licensing portal — Official daily metal prices	daily	1990s→	daily (official close)	licence portal / SFTP drop	exchange licence; **next-day distribution one-off ~$4k** vs $25k/yr realtime [^3]	delayed-only ⇒ no PIT perfect; cross-check CME/SHFE same metal (action-note: purchase next-day tier; map Cu/Al/Ni/Zn/Pb/Sn to CME twins)	W1	connect	VERIFIED-UPSTREAM-DOC
4	lme-stocks	benchmark	LME ware-house stocks (per metal, per region)	daily	2000s→	daily	same licence portal [^3]	same licence	warehouse-cornering months distort; join with CME de-stock proxy (action-note: buy with prices; build stocks-change signal)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	cme-comex-metalshist	benchmark	CME DataMine / COMEX per-metal per-contract (GC, SI, HG, PL) settle-ments+volume	daily (tick-class W2)	multi-decade	settlements EOD; ticks W2	DataMine API/request (paid)	enterprise licence; no redis-tribution	contract-roll math manda-tory; expired-contract historics (action-note: pilot GC+HG sample; wire to roll calendar (#18))	W2	connect	UNVERIFIED-CANDIDATE
6	databento-glbx	benchmark	Databento GLBX.MDP3 historical (CME Globex futures: metals, WTI, NG, agri)	1-min→tick	~15y Rt-era; per-GB	<1s live	REST batch → DBN files	paid per-GB ~$0.40–0.50, credits $125→$199 plan [^4]	roll math; schema stable (owned files) (action-note: free-credit prototype CL/GC/ZC bars)	W2	connect	VERIFIED-UPSTREAM-DOC
7	ice-eu-futures	benchmark	ICE Futures Europe Data Services (Brent, Gasoil, Cocoa, Coffee, Sugar, WTI-Europe?)	daily/tick	decades	realtime/delayed tiers	ICE Data Portal (paid line-items)	exchange licence	roll math; Euronext-ICE split pre-2012 caveats (action-note: first; only if energy-contango sim needed)	W2	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
8	platts-assessments	benchmark	S&P Global Commodity Insights (Platts) — crude/products/LNG assessments	daily	1980s→	realtime	enterprise feed	no redistribution	MOG window conventions; subscriber-only basis (action-note: first procurement; cross-check EIA spot free)	W3	quote	SURVEY-EVIDENCE
9	argus-media	benchmark	Argus Media (Argus-DC) non-US energy/chemicals assessments	daily	1990s→	daily	enterprise feed	no redistribution	same class as Platts (action-note: ; pick ONE of Platts/Argus first)	W3	quote	SURVEY-EVIDENCE
10	icis-assessments	benchmark	ICIS petrochemical / energy / LNG assessments	daily	1990s→	daily	enterprise feed	no redistribution	overlapping w/ Platts/Argus (action-note: only if chemical leg opens)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	fastmarkets-metallic’s	benchmark	Fastmarkets (battery-metals, Li/Co/Ni) assess-ments	daily	2010s→	daily	enterprise feed	no redis-tribution	new-index basis flux (action-note: only if battery-metals leg)	W3	quote	UNVERIFIED-CANDIDATE
12	sp-gsci	benchmark	S&P GSCI (S&P DJI) index levels + method-ol-ogy/constituents	daily	1970→	daily	S&P DJI portal; fact-sheet free; con-stituents paid	levels free-ish; method-ology licence	roll-window defini-tions in method-ology PDF (action-note: free index levels now; con-stituents via)	W1	quote	UNVERIFIED-CANDIDATE
13	bloomberg-bcom	benchmark	Bloomberg Commod-ity Index levels + annual con-stituents files	daily	1991→	daily	BCOM portal download / own terminal	Bloomberg licence (con-stituents)	index-vs-constituent Lump bindings (action-note: levels scrape; con-stituents one-off contract)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
14	yf-futures-proxy	benchmark	yfinance commodity front-months (CL=F, GC=F, BZ=F, NG=F, ZC=F…)	intraday/daily	full history of each prompt contract	delayed ~15 min	yfinance REST (no key)	delayed TOS; non-commercial	roll contamination; venue mix (CME/ICE); use proxy only (action-note: cross-check Pink/EIA vs futures spread; compute fu-adj)	W0	connect	SURVEY-EVIDENCE
15	shfe-ine-daily	benchmark	SHFE + INE (Shanghai Intl Energy crude) settlement/quotes via official site CSV/HTML	daily	SHFE 2000s→; INE crude 2018→	EOD +1d	web CSV/HTML (semi-official)	gov-published; TOS grey	unofficial endpoint drift (action-note: scrape with throttle; CN commodity anchor)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
16	cftc-disagg-cot	positioning	CFTC Commitments of Traders — Disaggregated (physical metals/energy) via Socrata API	weekly	disagg 2006→; legacy 1995→	weekly Fri EOD	REST publicreporting.cftc.gov (Socrata, keyless)	free US-gov	open-interest basis; contract codes rename (action-note: pull disagg + legacy; build positioning z-scores)	W0	connect	UNVERIFIED-CANDIDATE
17	eia-spot-crudes	benchmark	EIA PMM/APIv2 Cushing WTI spot + landed/refiner acquisition costs	daily product spot (monthly series)	Cushing WTI 1986→	monthly+weekly releases	APIv2	free US-gov	monthly revisions; PMM presets (action-note: fold into EIA spine row family (#19))	W0	connect	VERIFIED-UPSTREAM-DOC
18	roll-calendar-cme	methodology	contract-roll mapping for WTI/Brent/metal/agri front months	daily	spans full history	n/a	internal derivation from #5/#6 conventions	—	wrong roll ⇒ phantom PnL (action-note: build once, test vs Databento continuous)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
19	eia-api-v2-spine	energy-balance	EIA APIv2 — full family flagship (petroleum/petroleum, nat-gas/NG, electric-ity/EPB, coal, STEO)	→ annual	60+ years (per family varies)	15–60 min (rto)	REST free key per family	free US-gov	revisions disclosed per report (action-note: central spine already selected in stream-05 □)	W0	connect	VERIFIED-UPSTREAM-DOC
20	eia-grid-hourly	power-grid	EIA Hourly Electric Grid Monitor (rto family, BA-level demand/net-gen/interchange)	hourly	2019→ current; pre-2019 bulk EBA zip [^2]	≤1h	REST + EBA.zip bulk backfill [^2]	free [^2]	BA-code mapping; timezone normal-ization (action-note: bulk both zips + daily delta)	W0	backfill	VERIFIED-UPSTREAM-DOC
21	eia-wpsr-weekly	energy-balance	EIA Weekly Petroleum Status Report (crude/product stocks, refinery utiliza-tion, imports)	weekly	1980s→	weekly Thu 10:30 ET (market-moving)	APIv2 + PDF/CSV	free US-gov	TBD revisions; monthly bench-mark reset vs weekly (action-note: one-shot ETL + revision tracker)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
22	eia-ng-storage-weekly	energy-balance	EIA Weekly Natural Gas Storage Report (working gas, L48/regional)	weekly	1994→	weekly Thu 10:30 ET	APIv2 (nat-gas/stor) [^2]	free US-gov	Sept SGC audit dis-locations (action-note: deliver W0; volatility trigger row)	W0	connect	VERIFIED-UPSTREAM-DOC
23	eia-steo-current	energy-balance	STEO — monthly projec-tions (US sup-ply/demand/prices + global liquids)	monthly (1–25-month horizon)	forecasts ~2y	monthly Tue	APIv2 steo browser + tables [^2]	free US-gov	vintage store needed (STEO re-estimates) (action-note: pull each edition)	W0	connect	VERIFIED-UPSTREAM-DOC
24	eia-steo-archive	archive	STEO editions archive (vintage stack)	monthly	1990s→ vintage	monthly	ZIP/PDF archive	free US-gov	THE PIT spine for oil/US energy forecasts (action-note: snapshot every edition on release)	W0	connect	VERIFIED-UPSTREAM-DOC
25	eia-international	energy-balance	EIA Inter-national Energy Statistics (produc-tion, consump-tion, prices, trade)	monthly/annual	1980→ish	monthly	APIv2 interna-tional browser [^2]	free US-gov	country-renames; Iran-style gaps (action-note: ingest selected series)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
26	jodi-oil	energy-balance	JODI-Oil World DB (production, demand, refinery runs, stocks, trade)	monthly	Jan 2002→ (CSV annual files 2002-2026 confirmed) [^6]	monthly (+1-month lag)	annual CSV bulk + REST API v1.2 free account [^6]	free	participation quality colour-codes 1/2/3 in-file [^6]; country coverage gaps (action-note: bulk CSVs + REST deltas; keep colour flag)	W0	connect	VERIFIED-UPSTREAM-DOC
27	jodi-gas	energy-balance	JODI-Gas World DB (production, demand, LNG trade, storage)	monthly	2009→	monthly	CSV bulk down-loads [^6]	free	same quality codes class (action-note: bulk)	W0	backfill	VERIFIED-UPSTREAM-DOC
28	opec-momr-archive	archive	OPEC MOMR historical editions (global supply/demand/shipping tables)	monthly	2000s→ editions	monthly	PDF (slot 2/wk)	free w/ attribution	PDF parse brittleness; vintage store (action-note: parse library + release snapshotter)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
29	eurostat-nrg	archive	Eurostat nrg_* SDMX datasets (supply, stocks, gas/electricity prices by con-sumer band)	monthly/annual	1990s→ (prices 2007→)	monthly/annual	SDMX CSV REST	free w/ at-tribution	shortcode churn (nrg changes names) (action-note: select nrg_cb_gal/nrg_pc core series)	W0	connect	UNVERIFIED-CANDIDATE
30	entsoe-gen	power-grid	ENTSO-E Trans-parency — actual genera-tion per unit / per produc-tion type	15–60 min	2015→	near-RT (release ~1h)	REST API (token, per-dataset params)	free w/ ToS token	per-country gaps; unit-code churn; probe reach □ [^7] (action-note: bulk + daily)	W0	backfill	VERIFIED-UPSTREAM-DOC
31	entsoe-load	power-grid	ENTSO-E actual total load + day/week/season-ahead forecasts	15–60 min	2015→	near-RT	REST API [^7]	free w/ token	forecast-vintage retention (action-note: ingest + store forecast vintages)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
32	entsoe-transfers	power-grid	ENTSO-E cross-border physical flows + scheduled commercial exchanges	15–60 min	2015→	near-RT	REST API [^7]	free	border direction sign conventions (action-note: ingest)	W0	connect	VERIFIED-UPSTREAM-DOC
33	entsoe-capacity	power-grid	ENTSO-E installed generation capacity per unit/type	annual	2015→	annual	REST API [^7]	free	decommission date lag (action-note: annual refresh)	W0	connect	VERIFIED-UPSTREAM-DOC
34	entsoe-outages	power-grid	ENTSO-E unavailability of generation units + transmission (outage RDF)	event	2015→	up to RT	REST API	free	flag mergers/withdrawals (planned vs unplanned) (action-note: ETL + event store)	W0	connect	VERIFIED-UPSTREAM-DOC
35	entsoe-daa-prices	power-grid	ENTSO-E day-ahead auction (EUPHEMIA) + intraday/continuous prices (legacy free-download platform)	hourly	2015→	next-day	REST API + legacy CSV.gov	free	re-run/corrections of published prices (action-note: ingest for EU zonal price spine)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
36	ember-electricity	power-grid	Ember electricity dataset (generation mix per country, monthly/yearly CSVs on S3)	monthly/annual	2000→	monthly	CSV bulk (S3-published)	**CC BY 4.0**	methodology changes between editions (action-note: full download + hash pin)	W0	connect	VERIFIED-UPSTREAM-DOC
37	ember-wholesale-prices	power-grid	Ember wholesale day-ahead price dataset (EU zonal, daily)	daily	2015→	monthly	CSV	CC BY 4.0	market quirks per BZ (action-note: download)	W0	connect	VERIFIED-UPSTREAM-DOC
38	ei-statistical-review	archive	Energy Institute **Statistical Review of World Energy** incl. pre-fusion BP archive (75th edition, since 1952; XLSX tables 1965→) [^9]	annual	1965→ (tables); 1952 inception	annual June	email-gated XLSX download	free (“full, first, free, fair”)	2026 methodology updates; Ember partnership columns (action-note: download all-archive XLSX; harmonize ex-BP splice)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
39	iea-wreo-datasets	energy-balance	IEA WEO + IEA Data Browser free tiers (world energy balances)	annual	decades	annual	IEA portal down-loads	**WEO data = CC BY-NC-SA ⚠ NOT commercial/derivable** [^10]	year-base changes; NC-SA gating (action-note: only for dis-play/consensus; negotiate IEA reuse if core)	W1	connect	VERIFIED-UPSTREAM-DOC
40	iea-tech-datasets	energy-balance	IEA per-technology free datasets (solar PV, wind, EV fleet, heat pumps; CC BY subset)	annual	2000s→	annual	IEA CSV/portal	CC BY 4.0 for tech data folder	WEO subset ≠ tech subset — per-file licence check (action-note: tech panels)	W0	backfill	UNVERIFIED-CANDIDATE
41	owid-energy	archive	Our World in Data energy (legacy repo + new catalog CSV)	annual	1900→ country panel	annual	GitHub + cata-log.ourworldindata.org CSV/XLSX + codebook **[legacy frozen; switched 2026 method-ology]** [^11]	CC BY	2026 primary-energy re-measurement ⇒ old≠new (action-note: pin col-umn_mapping.csv; use new catalog)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
42	caiso-oasis-art-prices	power-grid	CAISO **OASIS** — PRICES publication group (Locational Marginal Prices DAM/RTM/interval, HASP, FMM, congestion shadow prices) [^12]	5-min / hourly / daily	DAM LMP 1998→	daily post-market + interval RT	OASIS `oasisapi` REST (keyless, reportName+querystring, ZIP download)	free ISO TOS	constant price-corrections log; ACR flags [^12] (action-note: by SPP reportName month-by-month)	W0	backfill	VERIFIED-UPSTREAM-DOC
43	caiso-demand-netload	power-grid	CAISO OASIS demand forecast / net-demand / fuel mix dashboards	5–60 min	2010s→	realtime-after	OASIS + public dashboards CSV	free ISO	jettisoned? no (action-note: attach to OASIS loop)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	ercot-rt-lmp	power-grid	ERCOT EMS Real-Time Market LMPs (15-min) + DAM hourly points / price caps	15-min / hourly	2010s→	real-time-after (2-min)	MIS public REST + CSV filedrop	free TOS	**doc blocked 403** today; endpoint naming unverified (action-note: curl-probe /mis paths; then bulk)	W0	connect	UNVERIFIED-CANDIDATE
45	ercot-fuelmix-outages	power-grid	ERCOT fuel mix (gen by source incl. weather-sensitive), outages (SCED telemetry summaries)	5/15-min	2010s→	real-time-after	MIS CSV + dashboards [^13]	free	mirror EMS naming (action-note: probe + ingest)	W0	connect	UNVERIFIED-CANDIDATE
46	pjm-dataminer2-lmp	power-grid	PJM **DataMiner 2** — DA/RT LMP per pnode, hourly/5-min	hourly / 5-min	2000s→	real-time-after	DataMiner2 REST (keyless; feed-class, month windows)	free ISO TOS	node mapping churn; price-corrections file (action-note: bulk monthly via page-fence)	W0	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
47	pjm-gen-fuel	power-grid	PJM generation by fuel (eDART/eDART Archive CSV) + interchange, load	hourly	2011→	daily	DataMiner2 / eDART CSV	free ISO	unit-type roll-ups (action-note: ingest alongside LMP)	W0	connect	UNVERIFIED-CANDIDATE
48	miso-openapi	power-grid	MISO Market Data APIs (DA/RT LMPs, gen by fuel, load, outages; REST + CSVs)	5–60 min	2005-ish→	realtime-after	MISO REST	free TOS	MISO south/CBC swap (action-note: probe; ingest)	W0	connect	UNVERIFIED-CANDIDATE
49	iso-ne-ais-express	power-grid	ISO-NE Express API (hourly gen by fuel, LMP by zone, demand+forecast JSON)	hourly	2000s→	1–5 min	Express REST JSON	free TOS	fuzzy time-epoch (posix vs local) (action-note: probe + ingest)	W0	connect	UNVERIFIED-CANDIDATE
50	nyiso-portal	power-grid	NYISO portal (DAM LMP, RT 5-min, gen by fuel, capacity, forecast-vs-actual)	5–60 min	2000s→	realtime-after	portal CSV/ZIP + custom MIS	free TOS	11-zone → generator-level translation (action-note: monthly CSV bulk loop)	W0	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
51	spp-mis	power-grid	SPP Market Data Portal (DA/RT LMP MP valuation slicing, gen by fuel, operating reserve)	5–60 min	2007-ish→	realtime-after	MDP CSV/report download	free TOS	MDP UI JS-app; API path fuzziness (action-note: probe; schedule)	W0	connect	UNVERIFIED-CANDIDATE
52	aemo-mms	power-grid	AEMO NEM MMS data model (dispatch 5-min, price tables, PLAP archives)	5-min	2000s→	realtime-after (settle-ment 30-min, **5-min pricing since Oct-2021**)	MMS CSV/TRI archives (month-fold)	free (AEMO TOS)	DST/Adelaide offset traps; Apr 2021 5-min rule transition (action-note: monthly MMS bulk + daily delta)	W0	connect	UNVERIFIED-CANDIDATE
53	aemo-aggregate-demand	power-grid	AEMO ag-gregator price & demand datasets (regional 30-min/now 5-min)	30→5-min	1998→ (price/demand run back)	daily/hourly	aggregate CSV per region	free	30-min→5-min stitch step (action-note: 1998→)	W0	backfill	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
54	elexon-bmrs	power-grid	Elexon **BMRS per-dataset** (System Prices, Market Index, DA/WD prices, FPN)	30–60 min + events	2000s→	realtime-after	BMRS site + Insights REST API [^14]	free (GFD attribu-tion)	app JS-only — probe Swagger via curl (action-note: probe Insights end-points)	W0	connect	UNVERIFIED-CANDIDATE
55	elexon-bods	power-grid	Elexon BODS — physical notifica-tions (FPN) / BM unit solar-wind forecast	30-min	2000s→	realtime	BODS REST API	free	FPN update fre-quency noise (action-note: probe + ingest for GB commodity-renewables z-score)	W0	connect	UNVERIFIED-CANDIDATE
56	gie-agsi	power-grid	GIE **AGSI+** — gas storage (working gas, injec-tion/withdrawal, per coun-try/operator; REMIT-aligned) [^15]	daily (prior day)	2010s→	daily	AGSI REST API (account token) [^15]	free w/ at-tribution	operator double-counting guards (action-note: token + REST pull)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
57	gie-alsi	power-grid	GIE ALSI — LNG terminal inventories, send-out, cargoes arrived/expected [^15]	daily	2010s→	daily	ALSI REST API	free w/ attribution	terminal granularity (action-note: pull with AGSI)	W0	connect	VERIFIED-UPSTREAM-DOC
58	gie-umm-iip	power-grid	GIE UMM / IIP Inside Information platform (TSO/operator urgent notices) [^15]	event	2010s→	realtime	iip.gie.eu	free w/ attribution	non-structured text (action-note: event feed → alert bus)	W1	connect	VERIFIED-UPSTREAM-DOC
59	jao-tcf	power-grid	JAO/ENTSO-E **TCF** (Transparency Capacity) cross-border capacity [^16]	daily	2010s→	daily	JAO CSV/REST feed	free	TCF versioning (action-note: probe; attach to ENTSO-E ingest)	W0	connect	UNVERIFIED-CANDIDATE
60	energinet-datashop	power-grid	DK Energienet Datashop API (elpriser, gen 15-min, gaslib)	5–60 min	2000s→	realtime-after	REST + CSV bulk	free	DK DK1/DK2 split (action-note: probe; Danish quality anchor)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	tennet-transparency	power-grid	NL TenneT Trans-parency portal (15-min gen, trade, system reserve, balance)	15-min	2010s→	realtime-after	portal REST/XML/CSV	free	just-sourced naming align-ment (action-note: probe; ingest)	W0	connect	UNVERIFIED-CANDIDATE
62	elia-opendata	power-grid	BE Elia Open Data portal (gen/load/balancing, per-CSV)	15-min	2014-ish→	realtime-after	REST open-data API	free	BE-A midjör split (action-note: probe; ingest)	W0	connect	UNVERIFIED-CANDIDATE
63	de-gen-by-source	power-grid	DE SMARD (Bun-desnetza-gentur) gen per source (incl. now-cast), EUR prices	15-min	2015→	up to 15-min	SMARD REST + CSV	free	national→TSO reconcili-ation (action-note: probe; ingest)	W0	connect	UNVERIFIED-CANDIDATE
64	baltic-exchange-indices	shipping	Baltic Ex-change BDI/BCI/BDTI/BDSI per-index historical	daily	1985→ (BDI base 1000)	daily	Baltic Exchange data licence	strictly licensed	composition re-bases (action-note: first procure-ment)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	bdi-open-mirror	shipping	BDI via public secondary mirrors/sites (non-licensed)	daily	1990s→	daily	web scrape (mirror risk)	**unofficial** = no basis to re-quote	lag/accuracy — proxy only, never quote-generating (action-note: only as compute-less proxy until W3)	W2	license-review	UNVERIFIED-CANDIDATE
66	xeneta-xsi	shipping	Xeneta XSI ocean/air benchmark indices	weekly	2010s→	weekly	enterprise feed	licensed	tender-pool basis	W3	quote	UNVERIFIED-CANDIDATE
67	freightos-fbx	shipping	Freightos **FBX** (Freightos Baltic Index ocean) + **FAX** (air); Terminal API portal (api-dos.freightos.com) [^17]	weekly (daily intraweek variants)	2010s-era→ weekly tracks	weekly	Terminal API / product pages [^17]	vendor TOS; header-registered data downloads	route-average basis (22 lanes) (action-note: register + API probe)	W1	connect	VERIFIED-UPSTREAM-DOC
68	drewry-wci	shipping	Drewry World Container Index weekly spot assessment + long-run CSVs	weekly	2011→	weekly	CSV download (email/register)	free-ish research-with-attribution	route-set changes (action-note: register downloader)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
69	tac-air-cargo	shipping	TAC Index (TAC Shanghai/TAC HK air cargo rates)	weekly	2010s→	weekly	enterprise feed	licensed	US-ban Asia-basis flux (action-note: only if air leg)	W3	quote	UNVERIFIED-CANDIDATE
70	aisstream-live-rail	shipping	aisstream.io free WebSocket AIS rail (PositionReport, ShipProperty, bounding-box filters) [^18]	realtime event (<5s)	live only (no history)	<5s	WS JSON (APIKey+v1 stream) [^18]	free live; **3 conn/IP**; browser-banned [^18]	event-driven drop risk — fast consumer needed [^18] (action-note: bounding-box subscribe + persist)	W0	connect	VERIFIED-UPSTREAM-DOC
71	kpler-ais-history	shipping	Kpler (incl. MarineTraffic+FleetMon since 2023) AIS events + history, vessel class flows	event / aggregates	2008→	realtime	enterprise	licensed — **one contract covers sector** [^19]	history pricing tiers (action-note: ; single contract)	W3	quote	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
72	kpler-lng-cargoes	shipping	Kpler LNG cargo-level flows (load/discharge) + GIE ALSI cross-check	daily	2008→	realtime	enterprise	licensed	AIS-inference basis (action-note: ; pre-post gas shocks)	W3	quote	UNVERIFIED-CANDIDATE
73	emodnet-port-calls	shipping	EMODnet Human Activities — port calls, vessel density rasters	monthly/annual	2017→	monthly/annual	CSV download + WMS/WFS	CC BY 4.0	raster pixel approxi-mations (action-note: download density + port-call tables)	W1	connect	UNVERIFIED-CANDIDATE
74	unctad-maritime	shipping	UNCTAD maritime statistics (freight-rate index, port calls, tonnage, canal transits)	annual/quarterly	2000s→	annual	web CSV/API	free w/ at-tribution	low fre-quency — context only (action-note: bulk static down-load)	W0	connect	UNVERIFIED-CANDIDATE
75	eia-psm-movements	shipping	EIA Petroleum Supply Monthly (regional product move-ments, pipelines, PADD transfers)	monthly	1981→	monthly	APIv2 + PDF	free US-gov	inter-PADD move-ments revisions (action-note: fold into APIv2 ETL)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
76	usda-nass-quickstats	agri	USDA NASS QuickStats (surveys + census: crops area/yield, livestock, prices; county/national)	annual/weekly	**1850/1860→** (year selector) [^20]	daily/weekly (NASS schedule)	QuickStats REST API (free key)	free US-gov	census vs survey gap (action-note: keyset-iterate commodities×years)	W0	connect	VERIFIED-UPSTREAM-DOC
77	usda-fas-psd	archive	USDA FAS PSD Online — world crop supply/use long-run (grains, oilseeds, cotton, sugar)	monthly	1960-ish→	monthly (WASDE cadence)	PSD Online query/API + CSV	free US-gov	mid-month vintage ⇒ snapshot each PSD update (action-note: bulk tables + vintage store)	W0	connect	UNVERIFIED-CANDIDATE
78	wasde-archive	archive	USDA WASDE monthly report archive	monthly	1970s→	monthly 12:00 ET	usda.gov archive PDF+XLSX	free US-gov	**12:00 release shock**; fetch blocked 403 today (action-note: schedule pull at 12:05 ET)	W0	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
79	faostat-fppi	agri	FAO FAOSTAT (production, trade, prices) + FAO Food Price Index (FPPI nom/real)	monthly/annual	FPPI 1990→; FAOSTAT 1961→	monthly	CSV bulk + FAO REST	CC BY 4.0	base-period resets (action-note: bulk series)	W0	connect	UNVERIFIED-CANDIDATE
80	cme-agri-hist	agri	CME Group agri contracts (CBOT ZC/ZS/ZW/ZM/ZO, ICE-11 sugar) via CME DataMine / Databento	daily/tick	decades	EOD→tick	API paid	licence	roll math (#18) (action-note: share pipeline with #5/#6)	W2	connect	UNVERIFIED-CANDIDATE
81	icco-cocoa-prices	archive & agri	ICCO daily cocoa prices (London £, New York $, ICCO avg) + monthly averages to 1960 in Statistics portal [^21]	daily (2-lag on site) + monthly	monthly 1960→ [^21]	daily	site table scrape + Statistics section	free w/ attribution	NY/LDN fix cadence; holiday gaps (action-note: pull daily + monthly series)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
82	igc-grains	agri	International Grains Council — grains/rice/oilseeds market report + supply-demand	monthly	2000s-era→	monthly	web/PDF + paid data portal	partial licence (archives paid)	annual projec-tion vintages (action-note: ; free monthly summary parse)	W1	quote	UNVERIFIED-CANDIDATE
83	mpob-palm	archive/agri	MPOB BEPI — Malaysian palm stats (produc-tion, stocks, exports, FFB; monthly release 10th 12:30 + annual Overview 2016→ PDFs) [^22]	monthly	deep (portal stats); Overviews 2016→ [^22]	monthly 12:30	portal login + monthly tables	free regis-tration (login-gated) [^22]	login-gate; MY-only official, world palm via PSD (action-note: register; sched-uled pull + snapshot [#22])	W0	connect	VERIFIED-UPSTREAM-DOC
84	iso-sugar-daily	agri	International Sugar Organiza-tion daily world sugar price	daily	1985-ish→	daily	website web/PDF + data subscrip-tion	free headline / archives paid	NY11-White premium context (action-note: pull daily series)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
85	eia-seds-longrun	archive	EIA State Energy Data System (SEDS) — 1960→ state energy by source/sector, annual [^2]	annual	1960→	annual	APIv2 + bulk [^2]	free US-gov	benchmark updates each fall (action-note: bulk ingest for US-energy panel)	W0	connect	VERIFIED-UPSTREAM-DOC
86	copper-warehouse-verify	benchmark	copper LME-COMEx-SHFE tri-market spread validation panel (per #3/#5/#15)	daily	full overlap window	daily	derived from 3/4/5/15	mixed	inventory-correction impossibility — must reconcile (action-note: build arbitrage panel test)	W1	connect	SURVEY-EVIDENCE

09 — Alternative data (75 rows)

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	Similarweb	web/app	Similarweb Digital Intelligence (site traffic, web segments)	site/per-month visits & channels	panel-derived backfill (publisher claim; years)	weekly-ish refresh	licensed REST API + file-export entitlement	commercial-vendor ⚠no-redistribute	estimates are panel-modelled, not census — record panel mix + revision policy per release (action-note: request + API T&Cs; pilot one competitor set before committing)	W2	quote	UNVERIFIED-CANDIDATE
2	Semrush/Ahrefs-class web intel	web/app	Semrush Traffic Analytics (second opinion vs Similarweb)	site/per-month	years backfill (vendor claim)	near-realtime to weekly	licensed API (per-plan limits)	commercial-vendor ⚠no-redistribute	clickstream/panel estimates with thin methodology disclosure — use as triangulation only, never spine (action-note: only; optional second-source for web-coverage)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	Sensor Tower (absorbed data.ai 2024)	web/app	Sensor Tower Digital Market Intelligence (app downloads/DAU/MAU, IAP estimates)	app/cohort/global weekly-monthly	global 5-year backfill (publisher claim)	weekly-ish refresh	licensed Query API / BI export	commercial-vendor ⚠no-redistribute	panel extrapolation; store-policy shocks create breakpoints — record panel version; **one invoice: data.ai/Sensor Tower consolidated** (action-note: request ; scope = 100–500 app universe only)	W2	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
4	App-store charts (App Store / Google Play)	web/app	public top-charts pages	app/daily rank by category	rank snapshot only (no history) → self-archive from day 0	realtime (page)	self-capture (⚠ both stores restrict auto-mated access — ToS-gray)	self-captured ⚠ToS-risk	charts re-rank intraday — pin fetch time; keep raw HTML/PNG evidence (action-note: build daily rank harvester (top ~200, core geos, price/type flags))	W2	connect	UNVERIFIED-CANDIDATE
5	Amazon market-place monitor-ing	web/app	Amazon best-seller ranks, offer counts, buybox price per ASIN	ASIN/daily (hourly pilots)	self-archive only	near-realtime	self-capture (⚠ Amazon actively bans scraping; consider partner vendors)	self-captured ⚠ToS-risk	rank is category-local and re-scores hourly; snapshot full PDP; dedupe offer set (action-note: pilot 1k-ASIN tracker; evaluate d alterna-tive before scale-up)	W2	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
6	Keepa (Amazon price/rank history)	web/app	Keepa API tokens (pre-collected price & rank series by ASIN)	ASIN/daily-hourly	multi-year price history (popular ASINs back to ~2015)	daily token quota (~monthly token packs; pricing UNVERI-FIED)	licensed API	commercial-vendor ⚠no-redistribute	rank units differ per category — normal-ize; gaps for cold ASINs; price-reads vs rounded market prices mixed (action-note: price 500-ASIN pilot coverage before commit-ting)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
7	Glassdoor (employer sentiment)	web/app	Glassdoor data partnerships (reviews, ratings, salary bands)	employer/month	vendor-claim years	event-level on page	licensed partnership feed	commercial-vendor ⚠PII (self-selected reviewer panel)	extreme selection bias (disgruntled/leavers); moderation edits republish — store first-seen timestamps (action-note: via partnerships desk; low priority vs LinkUp/Lightcast)	W3	quote	UNVERIFIED-CANDIDATE
8	G2 (software peer sentiment)	web/app	G2 research program (grid scores, review trends)	vendor/quarter/ scorecards	product-launch forward only	quarterly	licensed research-data program	commercial-vendor ⚠no-redistribute	tiny review volume per vendor pre-2022 — mid-quality confirmation only (action-note: ; bundle with Glassdoor decision)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
9	Google Trends	web/app	Google Trends UI + unofficial `pytrends` library	term/geo daily (hourly last 90d)	back to 2004	hours	library self-capture (⚠ unofficial endpoint, no SLA; rate-limit hard)	free (⚠ ToS-gray on bulk)	sampling noise — re-runs differ; snapshot per run and never join across reruns (action-note: `pytrends` pipeline for ~200-term watchlist; zero-vendor depen-dency)	W1	connect	UNVERIFIED-CANDIDATE
10	Wikimedia pageviews	web/app	`wikimedia.org/api/rest_v1/metrics/pageviews` + dumps	per-project-article daily-hourly	2015-08 →	near-realtime	REST API batch; per-month HTML dump files	free CC0-attribution (WMF)	page re-names/redirect chains break series — resolve canonical titles first (action-note: company wiki pages + ticker redirects as interest proxy)	W0	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	Indeed Hiring Lab posting aggregates	employment	hiringlab.org sector/state/geo postings trackers (vs Feb-2020 base)	country/state/sector weekly	2020→ (indices, rebased — not raw counts)	weekly	free PDF/CSV attachments (data page WAF-403 to bots — browser headers)	free-attribution (publisher credit terms)	index rebasing + methodology footnotes per release — snapshot every release; revisions not versioned (action-note: ingest weekly US/EU trackers; store attachments immutably)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
12	LinkUp (raw postings)	employment	LinkUp STS postings API (posting counts, skills)	posting/geo/pub daily	publisher claims 2008+ archive	daily	licensed API / flat-file entitle-ment	commercial-vendor ⚠no-redistribute	posting dedupe & spam-post inflation are known issues — demand method-ology changelog (action-note: ; bench-mark 1 country vs Indeed free + JOLTS)	W2	quote	UNVERIFIED-CANDIDATE
13	Lightcast (ex-Emsi Burning Glass)	employment	Lightcast post-ings/skills subscrip-tion	posting/firm/skill daily	~2010+ (claim)	daily	licensed API + flat files	commercial-vendor ⚠no-redistribute	multi-source scrape consoli-dation — request post-dedupe count method-ology doc (action-note: (firms + skills bundle))	W2	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
14	layoffs.fyi	employment	layoffs.fyi crowd-tracked tracker (firm/region/sector, staged per date)	firm/event daily	2020-03 →	near-realtime (community-edited)	self-capture weekly CSV/sheet snapshot	free-attribution (site terms)	crowd-edit races; sever-ance numbers arrive delayed — store last-modified times-tamps; cross-check WARN notices + 8-Ks (action-note: CSV ingest; free panel; add cross-check gate in pipeline)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
15	Sentinel-1 SAR	satellite/geo	Copernicus Data Space S1 GRD+SLC	swath/12d revisit	2014-10 →	hours	STAC product catalogue / openEO / S3 + OData; `sentinel-sat/openeo` SDKs ([^1])	free (Copernicus open-data licence)	processing-baseline versions change back-compat — archive by baseline ID (action-note: STAC + openEO extract for watch-list ports/fields)	W1	connect	VERIFIED-UPSTREAM-DOC
16	Sentinel-2 L1C/L2A	satellite/geo	Copernicus Data Space S2 MSI L1C/L2A (cloud-masked)	tile/10m	L2A from 2016-06 (L1C 2015-06)	hours	STAC / openEO / S3 ([^1])	free (Copernicus)	Sen2Cor version shifts L2A values — pin baseline; cloud-mask QA layer mandatory (action-note: same pipeline as row 15; NDVI/urban-sprawl indices)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
17	Landsat 5-9 archive	satellite/geo	CDSE Complementary Data (Landsat-5/7/8/9 + Bi-monthly Mosaics) + USGS EarthExplorer / Machine-to-Machine API	scene/pass ≈16d	**1972 →**	hours	USGS M2M API (bulk) — head of site is bot-challenged (202); CDSE STAC mirrors Landsat products ([^1])	free (USGS public-domain; Copernicus licence for CDSE path)	Collection 2 standard — pin collection version; CDSE coverage caveats — re-confirm geos (action-note: primary = USGS M2M bulk; CDSE as convenient mirror)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
18	Planet Insights Platform (incl. NICFI)	satellite/geo	Planet ~3 m daily + monthly mosaics; NICFI tropics basemaps	tile/daily	NICFI from 2015/2016 (tropics); commercial scope varies	daily	licensed APIs (tasking/orders)	commercial-vendor ⚠no-redistribute	cloud artifacts; NICFI program phases change mosaic availability — record mosaic release ID (action-note: for 3 m mosaics over watch-atlas zones (pilot))	W3	quote	UNVERIFIED-CANDIDATE
19	ICEYE	satellite/geo	ICEYE X-band SAR constellation + analytics	tile/task daily	archive on request	day (tasking)	licensed download & API	commercial-vendor ⚠no-redistribute	tasking vs archive pricing split; SLC/L1 access tiers; flood/freeze analytics separate SKUs (action-note: (flood/train-disruption case study))	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
20	Umbra	satellite/geo	Umbra SAR tasking + open-data distribu-tion	spot-task tiles	archive on request; some free samples	day	licensed API + free dis-tributor tracks	commercial-vendor (limited free tier)	task price varies by area/angle — record task plan (action-note: ; low priority vs IC-EYE/Planet)	W3	quote	UNVERIFIED-CANDIDATE
21	NASA FIRMS active fire	satellite/geo	FIRMS area API (MODIS/VIIRS/Landsat fire de-tections)	point/daily	2000+ MODIS; 2012+ VIIRS; 2024+ Landsat (US/CA NRT)	≤3h NRT; URT <1 min US	REST `/api/area/csv/…` with free MAP_KEY; 5000 tx/10-min cap ([^2])	free (US gov)	VIIRS vs MODIS pixel-size differ; confidence/low-confidence filtering manda-tory (action-note: ingest for crop/forest watch-geos; register MAP_KEY)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
22	VIIRS night lights (S-NPP/NOAA-20/21)	satellite/geo	NOAA/Colorado Mines EOG data center (daily + monthly composites)	grid/daily-monthly	2012-01 →	hours–days	file-drop (free w/ account) — **/products/viirs/ path returned 404 this census; domain live but portal restructured** ([^6])	free-attribution (EOG)	stray-light correction versions; radiance units nW/cm²/sr (action-note: re-locate current EOG catalogue page + bulk repo before ingest)	W2	connect	SURVEY-EVIDENCE
23	DMSP-OLS 1992–2013 night lights	satellite/geo	EOG/NOAA DMSP-OLS intercalibrated annual composites	grid/annual	1992 → 2013 (frozen)	n/a (never updated)	one-time bulk download	free-attribution	inter-calibration across F10–F18 satellites debated — use distributed intercalibrated set, pin version (action-note: pull full archive once; historical light index)	W0	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
24	GPM IMERG rainfall	satellite/geo	NASA GPM IMERG V07 early/final	grid/0.1°/30-min	2000-06 →	4h (early) / weeks (final)	GES DISC file-drop + Giovanni extracts	free (NASA)	Early vs Final differ — always FINAL for back-tests; griddle bias vs station rain (action-note: scripted GES DISC pull for pilot basins)	W1	connect	SURVEY-EVIDENCE
25	ERA5	weather	Copernicus Climate Data Store re-analysis	grid/0.25°/hourly	**1940-01** →	~5-day tail	`cdsapi` file-drop (NetCDF/GRIB), monthly release runs	free-attribution (licence per-dataset; re-check CDS licence banner)	ERA5T prelim-inary values revised next cycle — snapshot release month (action-note: CDS bulk pulls for key watch-grids)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
26	WorldPop population grids	satellite/geo	WorldPop hub (counts, density, age-sex, migration, settlement)	grid/100m country-year	2000 → 2030 projections	annual	file-drop + WOPR REST API ([^3])	free CC-BY-4.0 (confirmed on hub footer)	UN-adjusted vs unadjusted vintages per country; census covariate lag by country (action-note: ingest v2020/2021 rasters for GFI country set)	W1	connect	VERIFIED-UPSTREAM-DOC
27	Google Open Buildings v3 (+Temporal)	satellite/geo	GCS `gs://open-buildings-data/v3/` polygons (S2-sharded CSV/WKT); v3-Temporal = per-epoch presence	building footprint	v3 inference May-2023 on 58M km² (Africa/S/S-E Asia/LatAm/Caribbean); Temporal = presence series	batch (rare updates)	gsutil download / per-country Colab; Earth Engine asset; HDX packs ([^4])	free dual-licence CC-BY 4.0 / ODbL 1.0	publisher recommends ≥90%-precision confidence filter; Global-South coverage only (no G7) (action-note: download pilot footprints + Temporal presence)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
28	OSM-derived tables	satellite/geo	OpenStreetMap planet dump + full-history diffs + Overpass	Maple/daily	continuous 2007 → (history via diffs)	minutes–daily	weekly `.osm.pbf` file-drop + `pyosmium` diff application; Overpass self-capture	free ODbL 1.0 (attribution + share-alike on derived DB)	bulk-import waves create fake growth jumps — count real additions from full-history `.osh`; tag-usage drift (action-note: nightly POI-count time-series DB for watch city-set)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
29	Copernicus Marine (SST/chlorophyll)	satellite/geo	Copernicus Marine Service (model + satellite products)	grid/daily	SST 1982+; reanalysis 1992+	hours–days	`copernicusmarine` python file-drop	marine (Copernicus)	product version migrations break series — archive toolchain logs per version (action-note: optional for crop/fish-adjacent commodity overlay)	W2	connect	SURVEY-EVIDENCE
30	PortWatch (IMF + World Bank)	shipping/log	portwatch.imf.org — AIS-derived port calls, congestion, chokepoints	port/daily; chokepoint/daily	historical archive from ~2019 baseline (publisher claim — verify on first load)	daily (1–2d source lag)	public dashboard + curated Python package (package name/access to re-verify on portal) ([^7])	free-attribution (IMF open data — confirm licence on portal)	congestion-window definitions disclosed in pipeline notes; AIS panel composition not disclosed (action-note: ingest daily flow for listed ports; historical depth immediately)	W0	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
31	UNCTAD ports & liner connectivity	shipping/log	UNCTADstat datacentre (TEU throughput, PLSCI)	port/country monthly-yearly	throughput from ~1970s; PLSCI 2004+	yearly-quarterly	datacentre CSV/SDMX/Excel exports (JS-rendered UI — direct file URLs needed) ([^8])	free-attribution (UN)	port renames / operator change break continuity — use UNCTAD continuity notes (action-note: locate direct CSV/SDMX endpoints; ingest long TEU series)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
32	Kpler (post-2023 hub, absorbed MarineTraffic)	shipping/log	Kpler One (tankers/dry/LNG AIS flows, port calls, flow analytics)	vessel/port daily	vendor-claim multi-year backfill varies by product	hours	licensed platform/API entitlement	commercial-vendor ⚠no-redistribute	dark-fleet AIS gaps compress into estimates — flag; MarineTraffic integrations one-vendor since 2023 — record vendor version (action-note: (commodity-flow tier only))	W3	quote	UNVERIFIED-CANDIDATE
33	MarineTraffic (Kpler-owned)	shipping/log	MarineTraffic vessel/port AIS	vessel/port realtime-daily	platform holds historic AIS per plan	near-realtime	licensed API / platform	commercial-vendor ⚠no-redistribute	same AIS pool as Kpler — treat as one vendor from 2023; avoid double-purchase (action-note: fold into Kpler quotation (one invoice))	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	FleetMon	shipping/log	FleetMon vessel tracking & port ETD/ETA API (independent, DE)	vessel/port realtime-daily	long AIS history (claim)	near-realtime	licensed API/Explorer	commercial-vendor ⚠no-redistribute	independent vendor = hedge vs Kpler consolidation; benchmark AIS match rate on common sample (action-note: ; keep as vendor-risk hedge)	W3	quote	UNVERIFIED-CANDIDATE
35	aisstream.io free AIS feed	shipping/log	aisstream.io WebSocket relay (free key)	vessel/realtime	stream only (we self-archive)	realtime	WebSocket + key; spool MMSI-level into DuckDB	free (per site ToS)	community receiver coverage uneven (good Americas/EU, thin elsewhere) (action-note: 30-day pilot spooler to quantify realistic coverage before paying a vendor)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
36	Global Fishing Watch open data	shipping/log	GFW API + open datasets (fishing presence, transshipment, vessel identity)	vessel/grid daily	2017 →	days	open REST API + weekly GeoJSON tiles	free CC-BY-4.0	fishing-gear class probabilities — keep gear-class caveats (action-note: vessel-flow overlay on commodity corridors)	W2	connect	SURVEY-EVIDENCE
37	EMODnet Human Activities	shipping/log	EMODnet Human Activities WMS/WPS/CSV (ports, shipping density, EU maritime zones)	port/grid monthly-yearly	~2005+ by layer	monthly batch	open datasets + WPS services	free-attribution (EU)	EU-scope only — incomplete outside EU waters (action-note: ingest only if EU corridors on GFI list)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
38	Container freight-rate indices	shipping/log	Drewry WCI / Freightos FBX / Xeneta	lane/weekly index	WCI 2011+; FBX 2017+	weekly	licensed dash-board export + free weekly press-release attribu-tion share	commercial-vendor (chart at-tribution lenient)	spot vs contract mix per index — stay index-consistent; publica-tion gaps docu-mented (action-note: subscribe weekly CSV for ~8 long-haul lanes)	W2	connect	UNVERIFIED-CANDIDATE
39	BofA Institute card ag-gregates	consumer/credit	institute.bankofamerica.com	household aggregate weekly-monthly by cohort; US	2019 →	weekly release cadence	free PDF/CSV attach-ments ([^9])	free-attribution (issuer's own ag-gregate publica-tion)	issuer-sample (BofA cardhold-ers skew NE US/wealthy); index rebases at period start — track release footnotes (action-note: weekly scrape → CSV normal-ization)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
40	Facteus (de-identified card panel)	consumer/credit	Facteus Mimic panel (US card spend trends)	spend/category/merchant daily-weekly	vendor-claim from ~2014	near-realtime	licensed feed/API	commercial-vendor ⚠no-redistribute	merchant-category allocation (NAICS cross-walk) is the main quality risk — verify before use; panel composition doc required (action-note: ; only if BofA/Placer gap confirmed)	W2	quote	UNVERIFIED-CANDIDATE
41	Consumer Edge	consumer/credit	Consumer Edge card-spend data (US/global)	spend/brand daily-weekly	vendor-claim multi-year	daily	licensed feed/API	commercial-vendor ⚠no-redistribute	bank-sample panel churn — request panel-cohort doc; spend-share ≠ transaction-count (action-note: ; big-cap consumer coverage pitch)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
42	Placer.ai (Placer Labs)	consumer/credit	Placer.ai retail foot traffic	POI/daily visits	vendor-claim years	daily	licensed portal reports only (no bulk API on standard tier)	commercial-vendor ⚠no-redistribute	GPS-device panel mix churn; store chain-level defini-tions; POI re-geocoding mid-series (action-note: ; scope to ~50 tickers' POI list)	W2	quote	UNVERIFIED-CANDIDATE
43	Unacast	consumer/credit	Unacast site analytics deliveries (visits, cross-shop, com-mute)	POI/daily	vendor-claim years	daily	licensed S3 deliveries	commercial-vendor ⚠no-redistribute	device-panel sampling shift; record panel-vintage per snapshot (action-note: sec-ondary as Placer fallback)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	Google COVID-19 Commu-nity Mobility Reports	consumer/credit	Google health-research frozen archive (categori-cal mobility %)	country-region-category daily	2020-02 → 2022 (frozen)	n/a (frozen)	per-region PDFs/CSV via health.google research page (guessed help URL 404’d this census — locate via site nav)	free-attribution	frozen product — historical only; Jan/Feb-2020 baseline fixed (action-note: one-time pull of all target-region files)	W0	connect	SURVEY-EVIDENCE
45	Apple Mobility archive	consumer/credit	covid19.apple.com/mobility frozen CSVs per coun-try/region (driv-ing/walking/transit)	country-region daily	2020-01-13 → end-2022 (frozen; verify end date on pull)	n/a (frozen)	direct CSV/JSON download ([^12])	free-attribution	relative-vs-Jan2020 index only; transit series weak outside big cities (action-note: grab full historical JSON once)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	GTFS-RT transit feeds	consumer/credit	Mobility Database catalog + per-agency GTFS real-time feeds	agency/realtime	self-archive from ingest	realtime	protobuf + JSON per feed; daily schedule roll-up	free (per-agency licences — most open, some restrict auto-mated use)	feed down-time churn; per-agency licence variance (action-note: collect feeds for ~10 major GFI metros)	W2	connect	SURVEY-EVIDENCE
47	Panjiva (S&P Global)	trade/customs	Panjiva supply-chain intelli-gence (cus-toms/shipment records)	shipment/entity daily-monthly	US BoL baseline ~2007+; other countries frag-mented	daily–weekly	licensed plat-form/API	commercial-vendor ⚠no-redistribute	raw customs data quality issues (typos, delays) — S&P cleaning rules; CN data restricted (action-note: ; start with supplier-node graph for 1–2 in-dustries)	W2	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
48	ImportGenius	trade/customs	ImportGenius customs shipment data	shipment/daily	US BoL from ~2006 (claim)	daily–weekly	licensed platform/API	commercial-vendor ⚠no-redistribute	cleansing similar to Panjiva — buy only one of the two customs shops (action-note: ; prefer Census raw + Panjiva if budget limited)	W2	quote	SURVEY-EVIDENCE
49	US Census international trade (USA Trade Online + bulk API)	trade/customs	api.census.gov/data/timeseries/intltrade — imports/exports HS-2/4/10 by country + port + district	monthly	1989/1992 (varies by series; decades)	monthly (~T+6 wks after period close)	REST bulk API + USA Trade Online UI ([^15])	free (US gov public-domain)	HS/global classification revisions — store first-release vs latest-revision; cross-walks rebuilt yearly (action-note: full monthly imports/exports bulk pull to SQLite/S3; revision snapshotting)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
50	UN Comtrade Plus	trade/customs	comtradeplus.un.org + v2 API (v1 deprecated — migrate)	country × HS flow monthly-annual	**free back to 1962** (annual aggregates); monthly depth varies by reporter	T+4 weeks to months	REST API v2 + bulk files + BigQuery mirror ([^13])	free-attribution (UN)	reporter methodology changes (FOB/CIF, revisions); store reporter metadata; v1→v2 migration guard (action-note: monthly key flows (US/CN/EU27/JP/KR/IN) + historical annual full pull)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
51	Eurostat Comext	trade/customs	Eurostat international trade-in-goods (Comext) UI + bulk/API routes	reporter-partner-HS monthly	**1988 →**	monthly T+6 wks	REST bulk (`/api/comext`) + TSV bulkfiles ([^14])	free-attribution (EU)	intra-EU flows re-bound on enlargement (EU-27) — store break-down to avoid false series-shifts (action-note: bulk API for extra-EU27 focus flows)	W0	connect	VERIFIED-UPSTREAM-DOC
52	India DGFT / MoSPI trade statistics	trade/customs	DGFT tradestatistics + MoSPI open data (HS export-import)	HS/country monthly-yearly	~2010+ online; earlier via publications	monthly batch	web CSV/PDF self-capture + data.gov.in APIs	free-attribution (India gov)	formatting not versioned; HS alignment crosswalk maintained (action-note: bulk monthly pull; secondary to Comtrade for IN)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
53	China GACC customs HS statistics	trade/customs	GACC online statistics query (HS-8 monthly)	commodity/country-monthly	~2014+ reliably online; earlier in year-books	monthly (mid-month lag)	self-capture query UI (⚠ anti-bot) + yearbook PDFs	free-attribution (China gov; site discour-ages bulk scrape)	primary CN trade source; no public bulk API — record query cadence (mid-month) (action-note: build query-scraper capturing monthly CN im-port/export by com-modity)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
54	FEC API v2 (campaign finance)	politics/policy	api.open.fec.gov v2 — filings, contributions, eFiling	committee-contribution transaction	filings archives from 1980s; efilings stream 2017+	efilings near-realtime; rest T+0–7	REST API (free key; ~1000 calls/hr) + bulk flat files per cycle ([^17])	free (US gov)	form-type mapping (F3X/F3P…); committee re-registrations — dedupe `filer_ids`; itemized vs unitemized amounts (action-note: bulk 20-cycle historical pull + live efilings stream)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	OpenSecrets (contributors + lobbying)	politics/policy	opensecrets.org API + bulk data files (CRP); lobbying LDA remap	politician/committee/company-cycle	1989/1990-present (claim)	yearly/half-yearly	free key-by-request API (redistribution restricted) — **docs page WAF-403 to scripts this census** ([^18])	free-attribution ⚠no-redistribute	LDA→standard industry mapping re-versioned — snapshot mapping; prefer FEC raw as spine, OpenSecrets for aggregates only (action-note: confirm API access/terms hand-on before use)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
56	GovInfo bulk (Congressional Record / bills)	politics/policy	govinfo.gov bulk XML/PDF + api.govinfo.gov	document/daily	CR electronic 1994+; pre-1994 print	next-day	bulk per-year packages + free-key API	free (US gov)	CR daily vs bound revisions — store with revision flag; bill text versions pinned by version ID (action-note: new-docs API feed + CR 1994→ bulk archive)	W0	connect	SURVEY-EVIDENCE
57	Federal Register API v1	politics/policy	federalregister.gov/api/v1 (rules/notices/PRs/comment windows)	agency/doc/day	1994→	next-day	keyless REST JSON (fields param) + bulk XML mirrors	free (US gov)	docket re-corrections — snapshot docs, don’t re-pull (action-note: new-doc feed as event calendar)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
58	Regulations.gov API v4	politics/policy	regulations.gov API — dockets, documents, comments	docket/comment daily	electronic comments ~2003+; pre-2003 scans	hours–day	REST v4 (free key; rate-limited) + bulk dataset files	free (US gov)	v4 schema reshaped — pin `/datasets` schema; form-letter campaign dedupe (action-note: dockets for financials/energy/crypto/AI watch sectors)	W1	connect	SURVEY-EVIDENCE
59	Congress.gov API v3	politics/policy	congress.gov API v3 (bills, amendments, members, votes, committees)	bill/member daily	full release history as published	next-day	REST (free key; chunked JSON)	free (US gov)	use `latestAction.date` not index; roll-call depth varies by era (action-note: key + incremental bill tracking + committee rosters)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
60	Open States	politics/policy	v3.openstates.org API (50-state bills, legislators, committees)	state-bill daily	backfill varies by state (up to ~2007+)	near-realtime bill actions	free API key + bulk data downloads ([^19])	free-attribution (per-state source licences vary)	per-state cadence/quality differs; snapshot rather than re-pull; API key confirmed in docs ([^19]) (action-note: ingest bills feed for ~5 focus legislatures)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	US LDA lobbying disclo-sure (primary)	politics/policy	official LDA search / quarterly report feeds (House/Senate disclo-sure portals)	registrant/filing quarterly	electronic LDA era 2008+; paper filings remain	quarterly	downloadable XML/bulk + portal queries	free (US gov)	issue-code standard-ization; client-name entity res-olution needed; portal URL churn — locate bulk end-points (action-note: locate bulk XML export end-points — TBD this census)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
62	FiveThirtyEight data archive	politics/policy	github.com/fivethirtyeight/data (polls, pollster ratings, generic-ballot)	poll-level + aggregate daily	1948→ historical-polling archives + modern scrape archives	frozen (repo static)	GitHub clone ([^20])	free CC-BY-4.0 (per-folder licences — read READMEs)	repo frozen post-2023-ish — historical only; pollster-weight defini-tions per project; new ag-gregator (ABC538) separate (action-note: clone archive; generic-ballot 1948→ series)	W0	connect	VERIFIED-UPSTREAM-DOC
63	RealClearPolitics aggre-gates	politics/policy	realclearpolitics.com daily average tables (embed-ded, no API)	race/day avg	race-level rolling — self-archive	daily	self-capture HTML (⚠ ToS check before scale)	self-captured ⚠ToS-risk	methodology opaque; race links break across cycles — snapshot per race (action-note: harvest final-weeks averages per election cycle)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
64	White House visitor logs archive	politics/policy	NARA / white-house.gov disclo-sure portals per adminis-tration	visitor/day-batch	Obama-era bulk 2009–2017; later admins par-tial/policy swings	n/a (archive)	file-drop PDF/XLSX per release	free (US gov)	redaction tiers vary by admin-istration; record release policy per admin (action-note: one-time archive pull per admin; low priority)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	SEC EDGAR full-text search + n-gram build	corp-own-signal	`efts.sec.gov/LATEST/search-index` (200 OK this census) + FTS UI; filings corpus cross-ref sheet 00	filing/CIK/date; full-text n-gram level	**2001 →** (FTS coverage)	realtime-ish (index within ~min)	REST JSON; respect 10 req/s + declared User-Agent ([^21])	free (US gov)	FTS covers published text incl. exhibits (graphics excluded); filings resubmissions — dedupe versions in n-gram panel (action-note: corpus queries + n-gram extract for key themes (supply chain, layoffs, “generative AI”))	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
66	Earnings-call tran-scripts	corp-own-signal	→ cross-ref only to tran-scripts stream (sheet 05 overlap): Motley Fool free tier, Al-phaSense/CapIQ licensed, SA paywall	transcript/event	per source	hours after call	cross-ref only — licensing & scraper-risk live in sheet 05	mixed: free tier ⚠ToS; licensed ⚠no-redistribute	never splice n-grams across transcrip-tion providers — vendor-token differ-ences (action-note: depen-dency note only; owner-ship lives in sheet 05)	W1	connect	SURVEY-EVIDENCE
67	Annual-report text mining	corp-own-signal	→ cross-ref only to filings sheet (00): 10-K/20-F/annual reports + proxy state-ments PDFs	filing/doc-level	per filing	n/a	cross-ref only (text-extraction pipeline belongs to filings sheet)	free (US gov; foreign filings per juris-diction)	— (action-note: depen-dency note only)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
68	TSA passenger volumes	travel	tsa.gov/travel/passenger-volumes — daily checkpoint throughput CSV (with same-day prior-year compare)	checkpoint-daily	2019 → (older years via archived releases)	daily	CSV file-drop per year ([^22])	free (US gov)	holiday-ops peaks; counts = travelers, not flights — proxy only (action-note: full CSV archive pull + monthly updater)	W0	connect	VERIFIED-UPSTREAM-DOC
69	BTS Airline On-Time Performance	travel	TranStats BTS “Reporting Carrier On-Time Performance”	flight/route-daily	**1987 →**	next-month batch	TranStats monthly CSV/bz2 downloads (site WAF-hard to scripts — browser headers)	free (US gov)	carrier code changes across years; cancelled/diverted flags stable; route-truncation pre-2003 (action-note: monthly historical pull to 1987)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
70	UNWTO (UN Tourism) statistics	travel	UN Tourism data dash-board + World Tourism Barome-ter	country monthly-yearly arrivals, receipts	decades (reporter-based)	monthly-quarterly	dashboard + licensed analytics channel for depth	partially free (head-line), depth-tier licensed	arrivals defini-tions differ (tourists vs day visitors) per country — store defining metadata (action-note: barome-ter subscrip-tion + free headline tables)	W2	connect	UNVERIFIED-CANDIDATE
71	STR / CoStar hotel data	travel	STR via CoStar (occu-pancy, ADR, RevPAR)	country/CITY monthly-weekly	multi-decade (claim)	weekly/monthly	licensed plat-form/API	commercial-vendor ⚠no-redistribute	market-code sets; pre-1996 chain-census differ-ences — pin method-ology year (action-note: ; only if hospital-ity sectors in GFI list)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
72	Open Ownership (BODS)	ownership	openownership.org beneficial-ownership dataset (BODS standard)	entity-beneficial-owner snapshots	per implementation 2015+; country coverage uneven	periodic refresh	bulk downloads + API	free-attribution (BODS; per-source national licences vary)	per-jurisdiction quality highly uneven — build scorecard; US CTA BOI status volatile (policy) (action-note: bulk ingest + country-coverage scorecard)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
74	Bureau van Dijk Orbis	ownership	Moody’s BvD Orbis (global firm ownership structure + financials)	company/annual (monthly embed)	deep, varies by country (claim)	monthly embed	licensed platform / flat-file delivery	commercial-vendor ⚠no-redistribute	large-firm bias in some EMs; ownership linkages update as filings change — record snapshot date; cross-check open registries (action-note: quote; scope to watch-list firms)	W3	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
75	PACER (US federal court filings)	ownership/legal	pacer.uscourts.gov (documents, dockets; ~$0.10/page, waived under $30/quarter)	docket/document daily	online records ~1990s+ (varies by court)	realtime-ish	pay-per-page portal/API (judgement/fraud)	fee-for-service (nominal; not free redistribution license — check)	redaction quality varies by court; fees prevent bulk pulls — strategic query lists only (action-note: scope: targeted GFI-litigation watchlist; check current fee waiver rules)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
76	CourtListener / RECAP	ownership/legal	courtlistener.com (free PACER mirror + REST API + bulk archives)	docket/document-daily	RECAP archive coverage grows via user pulls	near-realtime updates	REST API + bulk tarballs	free-attribution (RECAP; per-term judicial data subset)	coverage is subset of PACER (grows via contri-butions) — never treat as full federal corpus (action-note: ingest API for watchlist dockets; check bulk archive terms)	W2	connect	SURVEY-EVIDENCE

10 — Registries · legal · IP · clinical · procurement (72 rows)

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	GLEIF LEI base pool (Level 1 LEI-CDF)	corporate-registry	GLEIF Golden Copy + LEI2 concatenated files (lei-data.gleif.org/api/v1); daily full files	record-per-LEI (~3.3M; 2026-05 validated ~3,304,415)	full since 2012 LEI launch; **previous Golden Copies retained for download**; daily deltas (intra-day/day/week/month)	daily	file-download (ZIP XML CDF) + GLEIF JSON API	open/attribution (CC-class free; re-confirm term page)	ContentDate vintage per file; registration.status alive-set for PIT (action-note: pin schema 2.1; ingest snapshot+delta cadence)	W0	connect	VERIFIED-UPSTREAM-DOC
2	GLEIF relationships (Level 2 RR-CDF)	corporate-registry-graph	RR-CDF Golden Copy/concatenated files; XML schema 2.1 (fund-relationship types since 2021)	edge-per-parent (consolidated/delta/direct/ultimate consolidating parent, fund-managed-by, sub-fund-of, feeder-to; ~647k records)	archived Golden Copies back to RR publication (Nov 2016 format v1.0; May 2017 v1.1; May 2021 v2.1); validFrom/validTo on edges	daily (8h Golden Copy cycle)	file-download + JSON API (`direct-parent-relationship`)	same as LEI base	relationship.periods[] + validity dates = PIT spine for ownership graphs (action-note: separate ingest stream from base pool; version history ≠ edge history (dedupe delta semantics))	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	GLEIF Reporting Exceptions (RepEx)	corporate-registry	RepEx CDF concatenated files (~5.9M records)	record-per-non-reported-parent (state-owned, natural person, legal restriction)	full archive via previous Golden Copies	daily	file-download	same	use as confidence labels, not edges (action-note: attach as edge-quality attribute)	W1	connect	VERIFIED-UPSTREAM-DOC
4	OpenCorporates	corporate-registry-aggregator	REST API JSON/XML (documented v0.4.8); free for open-data projects w/ attribution+share-alike; paid accounts remove share-alike restrictions	company record + officers + PSC/beneficial data + incorporation events (200d jurisdic-tions)	long-tail: some records back to 1600s-1800s; per-source depth varies	near-realtime API; snapshot via paid	API (key required)	attribution+share-alike (free) / paid-API	share-source provenance field; watch share-alike handling of indices (action-note: quote for commercial use BEFORE ingest)	W1	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	UK Companies House (registry)	corporate-registry	REST API (free, key) + bulk product downloads (free; company profile, filing history, persons-of-significant-control)	company/record; directors; PSC	bulk; filing history back to 1991 for electronic records; paper pre-1991 scanned; PSC since 2016	API near-realtime; bulk weekly	API + file-download	OGL-UK govt	cross-ref 05 skip duplicates: 05 handles accounts/filings consumed for fundamentals — this row = identity + directors + PSC only (action-note: dedupe vs class 05; PSC graph join to GLEIF)	W0	connect	VERIFIED-UPSTREAM-DOC
6	EU BRIS (business register interconnection)	corporate-registry	e-Justice BR portal (ejustice.just.fgov) — search UI per national BR; **no official bulk/API**	company stub per member state	current only	web	scraping only (ToS risk)	portal-only	patchwork formats (action-note: low priority; prefer national registries (INPI/BRIS members map))	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
7	US state corporate filings	corporate-registry	per-state SOS: DE SOS search portal (no API), NY Open-Data CSV, WA SOS bulk DL, CA bizsearch…	filing-level per state	mixed; some states keep 100y+ (DE pre-1900), most online since 1990s	daily-ish per portal	scraping (no unified API)	state govt public records	entity-number keys; name-change events unreli-able (action-note: jurisdiction-by-jurisdiction; start DE+NY+WA)	W2	connect	UNVERIFIED-CANDIDATE
8	Cayman Islands registry (CIMA/General Registry)	corporate-registry	cayman registry site — pay-per-search; no public API; occa-sional blocks on offshore-scraper clouds	company record	current; filing index history on request	pay-per-query	manual/pay portal	paid-portal	— (action-note: flag: unreach-able for automa-tion; use OpenCor-porates as proxy)	W3	connect	UNVERIFIED-CANDIDATE
9	BVI FSRA (VIR-RGIN)	corporate-registry	bviassa fsra viewer — pay-per-search, no bulk	company record	current	pay-per-query	manual/pay portal	paid-portal	— (action-note: flag: unreach-able; OpenCor-porates proxy)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
10	India MCA Master Data	corporate-registry	mca.gov.in view CIN/DIN master (free); API LOGIN-gated?; private proxies (SignCare ₹5/lookup, Deepvue) exist → direct API channel unclear 2026	CIN record: company status, directors, DINs	filing history depth ~2006+ (CIN); master = current snapshot	near-self (gov'ed) day up	web scraping or private API (paid)	restricted (gov) — scraping not clearly permit-ted	director identity PIT via DIN; no master-PIT (action-note: locate official API/registration; else out-source via proxy)	W2	connect	UNVERIFIED-CANDIDATE
11	India GSTN (GSTIN registry)	corporate-registry/tax-id	gst.gov.in GSTIN search (free portal); bulk API requires PAN-registration + ToS; periodic "auto dealers list"	GSTIN→PAN entity map	current only	near-realtime portal	portal + restricted API	portal-only (ToS-check needed)	PAN=XOR for Holly Henry dedupe (action-note: read GSTN API ToS; test search-page rate limits)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
12	Singapore ACRA (bizfile)	corporate-registry	ACRA data products + Free Corporate Search; API via ACRA API Gateway (subscription tiers; free “shell” master-data no confirmed)	company full particulars / UBO extracts (paid per doc HK$5.50-SGD free-tier? unclear 2026)	filings since 2014 digitized? — e-docs; pre-2014 microfilm	extract per-order	API gateway (paid) / manual extracts	paid-API	UBO/registered-status via BizFile extracts (action-note: current gateway pricing incl. free tier)	W2	verify	UNVERIFIED-CANDIDATE
13	Hong Kong Companies Registry (CR e-Search)	corporate-registry	e-services.cr.gov.hk — free: name search, doc index, charges index, disqualification index; paid: particulars HK$22/co, director index HK$11, images HK$9-35/doc	doc-image + field-report level	docs index back to incorporation year 1845eso? — images from 2014+; pre-2014 docs archived (order)	T+0 to T+3 images	pay-per-doc / account HK$500/yr principal + HK$100 subsequent	paid-portal	**shareholder data withheld** (NID/PI) — form route only (action-note: build doc-index free; budget §8(h)-(j) disclosure HK$10 for FI access)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
14	Japan □□□□ (Houjin Bangou, National Tax Agency CNPS)	corporate-registry	houjin-bangou.nta.go.jp — free CSV per prefecture (regular rolling)	legal-entity number record: corporate number, name, address, HOCR w/o historical field	published monthly per prefecture (full replacement); no history at source — archive-of-vintages needed	monthly (free ML)	file-download	public-domain (Japanese gov open data)	corporate-number PKeys; no prior names (action-note: store vintage-stamped archives (free → trivial))	W0	connect	VERIFIED-UPSTREAM-DOC
15	Korea DART (corp registry subset)	corporate-registry	dart.fss.or.kr — free w/ login-gated corp code list download; API via open-dart.fss.or.kr free (registered corp corp-publishers only)	corp code + entity metadata	since 1998 electronic filings (05-assets); corp-code file monthly full	API near-realtime; corp-code list monthly	OpenDART API (free key, gov open-data-portal license)	gov open data (KR open-API license)	cross-ref: mercados Stocks (03) skip duplicates — keep identity only (action-note: OpenDART API key; align corp-code vintage with filings)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
16	KVK (Nether-lands)	corporate-registry	kvk.nl — extracts (paid per doc), Cor-porate History product, “KVK APIs” bundle; **KVK Open Data** free since 2025? (open data file incl. ad-dresses/registration)	company + office + history product	extracts historical since 1983?; digital open data 2025	extracts on-order	manual extracts + data-API (licensed)	paid-portal + some open-data	SBI codes; multiples per KvK ID (action-note: Open-Data API scope + extract pricing)	W2	verify	SURVEY-EVIDENCE
17	Australia ASIC + ABR	corporate-registry	ASIC Connect extracts (search-free-info product, paid per doc); ABR (ABN Lookup) — free web services (abr.business.gov.au json/xml SOAP; data.gov.au free ABN bulk extracts)	ABN→entity; ASIC per-company	ABN since 2000; ASIC extracts current	extracts T+0	ABN web services free; ASIC paid extracts	ABN: free/attribution; ASIC: paid	ABN current-only; entity-status snap-shots (action-note: ABR first (free); ASIC extracts only for col issues)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
18	Germany Handel-sregister	corporate-registry	handelsregister.de (free web search + pdf extracts since 2022 free?); no bulk	company record, pdf extract	since 2007 elec-tronic register (pre-2007 paper)	extracts per-order	manual/scrape (base-rate offered by 3rd parties legally?)	portal-only	unique court+HRB key (action-note: no-bulk flag; use OpenCor-porates for multi-entity lift)	W2	connect	UNVERIFIED-CANDIDATE
19	France INPI Open Data	corporate-registry	data.inpi.fr free down-loads (com-pany identities, filings PSC-like since 2018, no bulk restric-tions), API-modifiable	SIREN record + filing index	DB (2001+ filings) 15y+ SIRENE registry	daily-ish	bulk CSV/API	free/attribution (CC-BY-class); declaration-based RSS	SIREN↔SIRET keys; Etalab identity join (action-note: bulk CSV archive loop + SIRENE cross-join)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
20	Brazil Receita Federal CNPJ	corporate-registry	arquivos.receitafederal.gov.br/cnpj/dados/ (free, monthly full CSV rotation) + gov.br page	establishment-per-CNPJ (~35M)	**monthly full-file archive avail-able** (2015+ online; older in file-set); historical rollovers	monthly	file-download (multi-table PB-classic fixed/CSV)	public-domain (BR gov open data)	CNPJ root+branch PIT via monthly vintages (action-note: download full historical stack; build CNPJ vintage DB)	W0	connect	SURVEY-EVIDENCE
21	IRS TEOS / EO BMF (chari-ties)	charitable-registry	irs.gov EO BMF extract (CSV per US state/region, updated monthly) + TEOS bulk (BMF + e-file + pub78 PDFs)	org record (EIN, sub-section code, ruling year)	**incl. “His-torical Con-tent” switch on TEOS page**; BMF monthly vintages; ruling-year on records	monthly	file-download (CSV)	public-domain (US Fed)	EIN unique-ness; subsec-tion codded path-fix for SOX501-type changes (action-note: pull historical BMF stack + pub78 letter versions)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
22	SEC EDGAR filer identities	corporate-registry-filing	sec.gov company tickers JSON + EDGAR full-text/company-search	filer identity (CIK, tickers, SIC)	CIK identities since 1993? filings; identities mostly historical-stable	EOD	API/file	public-domain	cross-ref 05 skip duplicates: 05 owns filings themselves (action-note: join CIK↔LEI↔CNPJ via GLEIF registeredAs)	W0	connect	SURVEY-EVIDENCE
23	UK Charity Commission register	charitable-registry	register-of-charities extracted CSV/documents (free; charity data incl. income, contact); historical register snapshots on archive	org record	**2007+ full register historic on archive**; annual returns per charity	weekly	file-download	OGL-UK	charity-number PIT (action-note: pair with IRS TEOS global charity coverage)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
24	OFAC SDN	sanctions	treasury sanction-slist + down-loads sdn.csv/sdn.xml/sdn_pstands + **OFAC SDN changes (delta) files**	entry-level; SDN changes delta (2003+)	SDN changes file incl. his-torical addi-tions/removals since 2016?; full list snap-shots — external archive	updated T+0-ish (list updates not sched-uled)	file-download, checker	public-domain	alias res-olution; update-date stamp; building-materials alias-match (action-note: download snapshot + changes-delta ingest)	W0	connect	VERIFIED-UPSTREAM-DOC
25	OFAC non-SDN lists (NS-MBS, NS-IC / CAPTA, NS-CMIC)	sanctions	same OFAC download tree (separate files)	entry-level	CMIC list since 2020 (EO 13959/14032 routes); separate delta	event-driven	file	public-domain	program-code routing (action-note: pair with SDN; program-tag)	W1	connect	VERIFIED-UPSTREAM-DOC
26	EU con-solidated sanctions list	sanctions	finance.ec.europa.eu consoli-dated list — CSV/XML/PDF per regime + entity-level unified feed	entry (per-son/organisation)	regime-by-regime archives; consoli-dated XML updated rolling (history: archived by month on fi-nance.ec.europa.eu)	event-driven	file-download	free (EU)	subject-type+regime-code mapping; OJ amend-ments (action-note: unified feed + regime codemap)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
27	UN 1267/Security Council consolidated list	sanctions	un.org security-council/sanctions consolidated list (free downloads; XML/HTML) + UNODC?	entry-level (individuals, entities)	since 1999 targeted sanctions; UN free	event-driven	file-download	free (UN)	narrative-report cross-check (action-note: ingest into canonical sanction DB)	W1	connect	SURVEY-EVIDENCE
28	UK OFSI consolidated list	sanctions	gov.uk OFSI consolidated list (CSV/XML) + OFSI case links	entry-level	since OFSI created 2016; consolidated includes 2002+ acts?	event-driven	file-download	OGL-UK? (gov.uk open)	sanctions-court-appeals track (action-note: ingest)	W1	connect	SURVEY-EVIDENCE
29	OpenSanctions	sanctions-PEP-aggregate	opensanctions.org — **data under AGPL/attribution license structure; hosted API paid**; datasets: Default (sanctions+PEP+others (maritime, securities, cross-KYus?)	entity + alias/network edges	own PEPs data since 2022; sources mirrored (OFSI/EU/UN/US regulatory),	daily	bulk pt.tar archives + API (licence-based)	**AGPL-3 + CC-BY?** verify licensing page (path moved)	entity dedupe/statement-level (datasetschema) — best PIT spine if attributed (action-note: free tier ingest + pricing for screening API)	W1	quote	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
30	World-Check-class (Refinitiv/LSEG World-Check One)	sanctions-PEP-commercial	LSEG — API + SFTP feeds; institutional subscription	entity + risk-level + adverse-media	curated global coverage years-deep	realtime-ish API; EOD file	SFTP/API	license quote RE-QUIRED	proprietary mapping (no source trans-parency) (action-note: vs Open-Sanctions hosted; PIT vs Open-Sanctions value debated)	W2	quote	UNVERIFIED-CANDIDATE
31	EveryPolitician / PEP-registry classes	sanctions-PEP	everypolitician.org (free CSV; per-country politi-cians data; back-ground via github repo)	person-level (role, dates)	archived per country (static snapshot model; repo-based)	repo-based	file-download CSV	CC-BY-SA? (mySoci-ety open)	merge into OpenSanctions-candidate PEPs (action-note: supple-ment to Open-Sanctions PEPs; not primary)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
32	Interpol notices	sanctions/notices	interpol.int notices (free public search for Red/UN-UNRF notices); API-эп? limited	notice-level (person/entity)	search UI only; no bulk archive at source — archive via own scrape	web	scraping-safe? (rate-limits unknown)	portal-only (rights restrictions on reuse — check per-notice)	notice status withdrawal lag (action-note: scrape + respect robots; use OpenSanctions mirror)	W1	connect	VERIFIED-UPSTREAM-DOC
33	CourtListener / RECAP Archive	litigation	courtlistener.com — bulk data service (daily dumps; case-law, dockets, opinions, oral args); **free incl. full historical dockets archive**; RECAP extension pulls PACER free copy	docket/entry; document; opinion	dockets from RECAP (2009+); opinions back to 1750s (via court packs); documents backfill continuing	daily dumps	bulk file / API	public-domain (SC/US court docs; some courts?) — mostly CC/research	best docket PST; document judge-coverage gaps (action-note: mirror daily bulk; build PACER-coverage gauge)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	PACER (via RECAP exten-sion)	litigation	pacer.uscourts.gov — pay-per-page $(0.10/page per $3 cap/doc, oval) + RECAP free-mirror encour-ages sharing	docket docu-ment	1988+ ECF era elec-tronic; legacy court PDFs	T+0 (pay)	extension + per-case payment	paid-per-doc	dedupe vs CourtLis-tener mirror before paying (action-note: RECAP-first; pay only for uncov-ered districts)	W1	connect	SURVEY-EVIDENCE
35	SEC litigation releases + admin proceed-ings	litigation/financial enforcement	sec.gov/litigation (free since 1995 archives; older via sec.gov/rules?) + admin-istrative proceed-ings archive	release-level	releases back to 1995 (2000s com-plete); earlier manual	EOD	scraping/RSS	public-domain	release-number ordering (action-note: mirror releases → entity-to-publisher case match)	W1	connect	SURVEY-EVIDENCE
36	DOJ press + action pages	litigation	justice.gov/news (press releases), case pages	release-level	~2010+ online search-able; older via DOJ archive	event	web	public-domain	entity-tag extrac-tion needed (action-note: flow into sanc-tions/ enforce-ment graph)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
37	FINRA AWC + BrokerCheck/CRD disciplinary	litigation/regulatory	finra.org BrokerCheck (free API/detail pages; AWC/CRD disciplinary via BrokerCheck PDF + finra arbitration awards)	rep/firm + action	CRD disciplinary history deep (decades); AWC/BR disclosure	event	API/scraper	free (non-commercial ToS)	CRD is firm-of-record for broker-identity PIT (action-note: BrokerCheck endpoint probe → CRD cross-join)	W1	connect	SURVEY-EVIDENCE
38	State AG enforcement actions	litigation	naag? per-office press pages (no unified dataset)	action-level	patchwork 2010+	web	scraping	varies	dedupe vs federal (action-note: low priority until CFPB/state-regulatorParsing unified)	W2	connect	UNVERIFIED-CANDIDATE
39	ICSID case database	arbitration	icsid.worldbank.org/cases-and-proceedings — free case records + awards PDF	case-level	since ICSID founding 1966; **full awards archive**	event	web/db download links	public + free (World Bank)	case status; award date ordering (action-note: archive PDFs + case metadata JSON)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
40	WTO dispute-settlement archive	arbitration	wto.org disputes (free case pages + docs since in-stitution 1995)	case-level	since 1995 full	event	web	free	panel/AB ruling dates (action-note: mirror)	W1	connect	SURVEY-EVIDENCE
41	PCA case database	arbitration	pca-cpa.org/cases — free	case-level	PCA cases since 1900s archives	event	web	free	— (action-note: mirror)	W1	connect	SURVEY-EVIDENCE
42	EU CELEX / EUR-Lex + Curia	litigation/regs	eur-lex.europa.eu (OJ, regu-lations) + cu-ria.europa.eu (judg-ments) — free, full historical	document-level	EU legal corpus complete since 1952; judg-ments all	EOD	basket scraping + cell (CELEX IDs)	re-use allowed (EU decision 2011/833)	procedure-type classifica-tion (action-note: CELEX ID graph join to sanctions regs)	W1	connect	SURVEY-EVIDENCE
43	UK BAILII (courts of UK/IE)	litigation	bailii.org — free full judg-ments; also “Find Case Law” (caselaw.nationalarchives.gov.uk, OGL)	judgment	deep (case law 1800s+, UKSC 2009+, courts archives)	updates irregular	scraping	free (BAILII condi-tions); TNA OGL for Find-Case-Law	— (action-note: prefer TNA UK Find-Case-Law (struc-tured))	W1	connect	SURVEY-EVIDENCE
44	Global Ar-bitration Review / Kluwer Arb	litigation-commercial	globalarbitrationreview.com; kluwerar-bitra-tionblog → license quote only	article/case commen-tary	subscription archive	paywall	license	paid license	— (action-note: if arbitrator-network analysis justifies)	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
45	USPTO PatentsView bulk	IP/patents	patentsview.org → **migrated to data.uspto.gov (transition guide live)**; bulk downloads (grants 1976+ — full text), applications (2001+)	patent/assignee/inventor/year	**grants 1976+ full text historical; entity disambiguation versions**	weekly-ish (grant lag increased 2023+)	file-download (TSV/parquet per entity tables)	public-domain (US Fed)	assignee disambig versioning = PIT trap (action-note: adopt disambiguation version pinning + grant-lag watch)	W0	connect	VERIFIED-UPSTREAM-DOC
46	USPTO Patent Examination Data (PDES)	IP/patents	developer.uspto.gov PDES API/public — prosecution-history events	application-transaction	since 1981 applications?	monthly	API/bulk free	public-domain	office-action ↔ claim-text timing (action-note: pair with PatentsView grants)	W1	connect	SURVEY-EVIDENCE
47	EPO OPS	IP/patents	developers.epo.org — REST XML bibliographic (worldwide), free limited tier w/ key	doc-level worldwide	EPO worldwide DB deep (1900s worldwide claims?)	near-realtime-ish (biblio)	RESTful XML; weekly bulk (pay?)	free limited tier; bulk on fee?	family/INPADOC linking is the hard part (action-note: OPS free-tier coverage mapping)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
48	WIPO PATENTSCOPE	IP/patents	patentscope.wipo.int — free search; bulk via subscription/CCC	2w-lag (PCT national phase)	PCT since 1978 full	live-ish search; scheduled bulk paid	search free; bulk license	free search / paid bulk	national-phase entry tracking (action-note: only if non-US/EPO filing weight justified)	W2	connect	SURVEY-EVIDENCE
49	Lens.org	IP/patents	lens.org — free API tier (attribution); Scholar+Patent linked graphs	doc/person/org graphs	deep patent collections (100y?); scholar layer growing	daily	REST API (key)	free tier attribution; commercial tiers	open-linkage graph (well-documented quality) (action-note: free-tier scoped pilot (citation graphs))	W1	connect	SURVEY-EVIDENCE
50	USPTO TSDR (trademark status)	IP/trademark	tsdr.uspto.gov — free per-mark docs/status; TSDR API	mark-level	marks since 1884? (records via TSDR since 1984 digital); pre-1984 paper	per-mark	free API, no bulk	public-domain	mark-status event log (action-note: status-list bulk alternate: USPTO trademark bulk XML)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
51	USPTO trade-mark bulk XML	IP/trademark	developer.uspto.gov trade-mark bulk XML (current applica-tion + registra-tion XML monthly)	nat'l	tradmark registry monthly snap-shots; historic dumps	monthly	file-download	public-domain	method: pair TSDR with bulk dumps (action-note: alternate to TSDR scraping)	W1	connect	SURVEY-EVIDENCE
52	EUIPO TMview / Design-View	IP/trademark	tmview.euipo.europa.eu/bulkdesign — free; EUIPO Open Data bulk (free CSV)	nat'l + EU per OHIM	EU-wide harmo-nize (member-state feeds per national office)	daily-ish	search + open-data bulk	free (EU)	status date fields; national-office lag asymme-try (action-note: gener-ated CSV archive loop)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
53	ClinicalTrials.gov API v2	clinical-trials	clinicaltrials.gov/api/api — REST JSON; full-field incl. results + full study record; NLM **archive snapshots exist** (compressed monthly public data files)	study/data-record (NCT)	since registry 2000 (FDAAA); protocol revisions via record-history (API field)	daily (record revisions differential)	REST API; monthly full-file alternative	public-domain / CC0-class (US gov)	record revision timelines as PIT (action-note: API-v2 full snapshot crawl + archive file mirror)	W0	connect	VERIFIED-UPSTREAM-DOC
54	EU CTIS (EMA clinical data portal)	clinical-trials	ema.europa.eu clinical data portal (CTIS) — public data; API unclear	study/event	since CTIS launch Jan 2022 (EU Reg 536/2014); pre-2022 in EudraCT	event	web (API unconfirmed)	free (EU)	transparency-rule rollouts phased; commercial-info redactions (action-note: probe CTIS API/bulk availability)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	openFDA (drug events/devices/recalls/enforcement)	clinical/regulatory	api.fda.gov — free bulk JSON; end-points drug/event, drug/label, de-vice/event, de-vice/recall, food/enforcement, etc.; **reports back to 2004 (FAERS), enforce-ment histori-cal archives**	record/data-level downloads	drug event 2004+; device adve 1990s? enforce-ment historical	monthly-ish updates	bulk JSON per endpoint + free API	public-domain (US Fed)	FAERS duplicate/lack-of-causality gates (action-note: full-archive mirror + endpoint row-limits plan)	W0	connect	VERIFIED-UPSTREAM-DOC
56	EMA company registry / XEVMPD	clinical/regulatory	ema.europa.eu medicines/company data (free web); “compa-nies” list	company/product-level	1995+ central-ized; MRP/DCP 2000s	EOD-ish	web/scraping + some data files	free (EU)	MA-holder↔manufacturer mapping quality (action-note: pair with openFDA global)	W1	connect	SURVEY-EVIDENCE
57	EudraGMDP	clinical/regulatory	eudragmdp.ema.europa.eu — EU GMP cer-tificates + GDP; public web	certificate-level	certificates EMA pub-lished; outages known	web	scraping	free (EU)	certificate with-drawal dates (action-note: low-priority scrape)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
58	WHO Prequalification (med products)	clinical/regulatory	WHO PQ database — free lists + PQ status docs (long-archived)	product/manufacturer	since 2001 WHO PQ; docs public	event	web + downloads	free (WHO)	product status “prequalified/suspended” (action-note: mirror lists + status PDFs)	W1	connect	SURVEY-EVIDENCE
59	SAM.gov (entity + procurement notices)	procurement/firmo	sam.gov/data-services — free API; bulk data downloads (`api.sam.gov` entity API, exclusions, HMDA? no) + forecast + notices	entity/notice/award record	entity-registrations historical?; **notices historical since ~2010 bulk**	near-realtime / daily extracts	REST API (key) + bulk download	public-domain (US Fed)	UEI replaces DUNS (2022) — ID migration gate (action-note: entity extract + notices bulk archive)	W0	connect	VERIFIED-UPSTREAM-DOC
60	EU TED (Tenders Electronic Daily)	procurement	ted.europa.eu — free API v3 + data extracts (**historical CSV dumps since ~2006/2007**; extended to 1980s-90s via national tegular sources?)	notice/cotract-award	historical extracts since 2006-ish online; older raw? (e-notices pre-2006 partial)	daily	API + bulk extracts	free (EU re-use)	award-decision date vs publication lag (action-note: extracts archive loop + API delta)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	UK Contracts Finder	procurement	contracts-finder.service.gov.uk — free API + CSV; OGL	notice/contract	since 2015 platform	daily	API/CSV	OGL-UK	award-value nulls common (action-note: CSV archive)	W1	connect	SURVEY-EVIDENCE
62	Australia AusTender	procurement	tenders.gov.au — free CSV exports; (403-bot-blocked this session)	contract notices	since 2007-ish consolidated	daily	portal CSV export	free (AU)	contract-value missing rates (action-note: retry with browser UA / schedule off-peak)	W1	connect	SURVEY-EVIDENCE
63	India GeM (Government e-Marketplace)	procurement	gem.gov.in — portal; public API limited; reports	bid/contract	since 2016	portal	scraping (limited)	portal-only	— (action-note: probe report downloads)	W2	connect	UNVERIFIED-CANDIDATE
64	Canada tender notices (Buyandsell.gc.ca; “Oppy?” token unresolved)	procurement	buyandsell.gc.ca (TBO/OJEC-class) — free; OPPA? token unparseable	notice	~2005+	daily-ish	web/CSV	free (CA)	token ambiguity to resolve with owner (action-note: owner-confirm “Oppy?” referent first)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	Korea Nara Market (□□□□) — PPS procurement	procurement	nara.go? (PPS portal) — free portal; API unclear	notice/bid	since ~2000	portal	unknown API	portal-only	— (action-note: probe emarket? availability)	W2	connect	UNVERIFIED-CANDIDATE
66	Japan government procurement (□□□□□□)	procurement	geppo? (□□□□□□/API?) — free portal; no unified API known	tender notices	~2010s	portal	scraping	portal-only	— (action-note: probe available CSV/feeds)	W2	connect	UNVERIFIED-CANDIDATE
67	World Bank procurement notices	procurement	/dev — **DNS NXDOMAIN on 2026-10-24 (host moved?)**; data.worldbank?	worldbank.org	vintage portal ~2001+	—	unknown post-move	free?	migrated to “World Bank Group procurement portal”? (action-note: re-locate current host; skip until)	W3	connect	UNVERIFIED-CANDIDATE
68	OpenOpps	procurement-commercial	openopps.com — commercial platform + paid API/data extracts	tender graph (cross-source EU+)	multi-year archive compiled	daily	API (paid)	paid license	aggregation quality vs TED (action-note: only if TED-extract gaps empirically bite)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
69	USAspending (archive)	procurement_spend	/spending_transaction/ — free bulk CSV; **monthly archives back to FY 2001**; API	transaction-level	FY2001+ (mandatory); award-level	monthly archive + daily API	file-download + API	public-domain	agency/org PK migration; CFDA→assistance-listing re-key (action-note: PhD ingestation of full archive — highest-value free fiscal history)	W0	connect	VERIFIED-UPSTREAM-DOC
70	FPDS-NG (federal procurementdata system)	procurement_spend	/api.gov / army equires? free contract-reports export (search→CSV); daily reports published	contract-transaction	linked into USAspending (same transactions pre-2007 via legacy FPDS)	daily	search-export free	public-domain	NAICS/PSC code codemapping (action-note: feed USAspending overlap only; not standalone primary)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
71	FCC ULS database downloads	telecom-filings	fcc.gov/wireless/data/publications-access-files + data.fcc.gov/download/pubs/ — free ZIPs (per radio service; SQL schema docs published)	application/license records per /download/pubs/{complete,daily,weekly}/	**full complete DB snapshots + daily/weekly transaction files; historical retention: rotating (own archive needed)**	daily	file-download ZIP (data + SQL defs)	public-domain (US Fed)	call-sign keyed; schema versioned (defs updated 2026-03-19) (action-note: standing snapshot+daily ingest w/ own archive)	W0	connect	VERIFIED-UPSTREAM-DOC
72	FCC spectrum auction results	telecom/spectrum	fcc.gov auction reports/results (public auction results since 1994; bidding-system docs)	auction-round/license	since Auction 1 (1994) PDF/reports; modern auctions CSV	per-auction	file-download	public-domain	bidder↔ULS entity cross-join (action-note: auction-history power users (charta PR foam))	W1	connect	SURVEY-EVIDENCE

11 — Crypto full-stack (90 rows)

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	binance-vision-spot-klines	CEX-archive	data.binance.vision → data/spot/{daily,monthly}/klines (URL-pattern walk)	OHLCV 1s,1m…1mo (12)	2017-08→ full (C2)	daily T+1; monthly 1st-Monday	S3-archive walk → Bronze	free (site ToS)	spot timestamps become µs from 2025-01-01 (C2) — schema gate; delist = files stop, kept (action-note: full-history walker; union daily+monthly dedup)	W1	connect	VERIFIED-UPSTREAM-DOC
2	binance-vision-spot-trades	CEX-archive	same tree /trades (historicalTrades fields per README)	tick	2017-08→ (C2)	daily T+1	S3-archive walk	free	tradeId gap check (first/last id); µs regime 2025+ (action-note: bulk-walk all d pairs → Parquet Silver)	W1	quote	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	binance-vision-spot-aggtrades	CEX-archive	/aggTrades files (/api/v3/aggTrades fields)	aggregated tick	2017-08→ (□C2)	daily T+1	S3-archive walk	free	aggId continuity; µs regime (action-note: cross-check agg vs raw trade counts)	W1	connect	VERIFIED-UPSTREAM-DOC
4	binance-vision-umfut-tape	CEX-archive (derivs)	data/futures/um/{daily,monthly}/{klines+tradesCHaGVTrades} (/fapi/v1 fields)	tick + 3H agg	2019-09→ per listing (□C2)	daily/monthly	S3-archive walk	free	funding-hour PIT (00:00/08:00 UTC); ms timestamps retained spot≠fut (action-note: um tape load; funding alignment)	W1	connect	VERIFIED-UPSTREAM-DOC
5	binance-vision-dapi (COIN-M)	CEX-archive (derivs)	data/futures/cm/{daily,monthly}/{klines+tradesCHaGVTrades} (/dapi/v1 fields)	tick + 3H agg	since dapi listing (older than um → verify start)	daily/monthly	S3-archive walk	free	base-asset volume semantics differ vs um (action-note: second-tier after um)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
6	binance-vision-fut-bookticker-diffdepth	CEX-archive L2	data/futures tree bookTicker / diffDepth daily files	book-ticker tick; depth-diff	later than klines family (verify start)	daily	S3-archive walk	free	partial-depth only; L2 rebuild via diff+snapshot replay, keep incident log (action-note: L2 reconstruction stage for futures book)	W2	connect	UNVERIFIED-CANDIDATE
7	binance-vision-fundingrate	CEX-archive (derivs)	data/futures/um fundingRate daily/monthly files	8h prints	since listing (verify start)	daily	S3-archive walk	free	funding print PIT vs premiumIndexKlines; mag-rate-change flags (action-note: cross-venue funding panel input)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
8	binance-vision-liquidationSnapshot	CEX-archive (deprecated)	data/futures/um liquidationSnapshot dumps (legacy product)	event	sweep window-limited (verify start)	daily	S3-archive walk	free	legacy dump vs live forceOrder stream semantics changed → PIT flag (action-note: archive-only; live liq via WS fapi)	W2	connect	UNVERIFIED-CANDIDATE
9	binance-spot-REST	CEX-API	api.binance.com /api/v3 klines, aggTrades, historicalTrades, depth	realtime	full via paging	realtime	REST keyless	free (weight caps)	geo-fencing caveat; weight-limit backoff; µs 2025+ (action-note: live tail fills W1 dump gaps)	W0	connect	SURVEY-EVIDENCE
10	binance-spot-WS	CEX-API	wss @trade @kline_1s @bookTicker @depth20 @diffDepth	tick/event	live tail	realtime	WS keyless	free	diff-depth lastUpdateId sync PIT; geo-fencing flag (row 12) (action-note: canonical realtime Bronze tail)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	binance-fapi-derivs-REST	CEX-API (derivs)	fapi/v1 fundingRate (full history), premiumIndexKlines; futures/data/openInterestHist	8h/5m	funding full since listing; OI history rolling-cap (verify window)	realtime	REST keyless	free	OI-history cap → use dumps for deep OI; funding PIT (action-note: full funding loader + OI cap workaround)	W1	connect	UNVERIFIED-CANDIDATE
12	cex-geo-fencing-metadata	compliance	Binance prohibited-jurisdictions page (bot-walled per stream-04); OKX DOJ DPA + 2025 plea ($505M) + US relaunch under ICE JV (□C11□C12); venue entity charts	ToS events	ToS history	batch	webfetch+Ujo care → GFI compliance table	public ToS text	ban-list drift detector; per-jurisdiction flag on every CEX row; Binance page bot-wall = curated manual copy (action-note: curate per-jurisdiction matrix; auto-diff releases)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
13	okx-v5-rest-ws	CEX-API	OKX V5 public REST + WS (candles w/ hist param since listing, funding-rate-history, OI history; WS books ≤400 levels)	tick + OHLCV 60s+	per-instrument since listing	realtime	REST+WS key-less/key	free tier	entity/jurisdiction pin (US policy line DOJ-verified □C11; re-launched US 2025 □C12); KYC-walled users; docs page timed out this session (re-verify params) (action-note: realtime tail; funding & OI loaders)	W0	connect	SURVEY-EVIDENCE
14	bybit-v5-rest-ws	CEX-API	Bybit V5 (spot, linear/inverse perps, funding, mark, insurance)	tick + 8h funding	candles since listing	realtime	REST+WS keyless	free	region-entity split; funding PIT; server-time skew (action-note: per-venue funding/liq live leg)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
15	bybit-derivatives-data-dumps	CEX-archive	bybit.com/derivatives-data downloadable exports	event/8h	Tardis anchors: inverse 2019-11-07; linear 2020-05-28; liquidations 2020-12-18 (□C5)	daily	batch-CSV walk	free	portal JS-rendered (listing □C21) — enumerate file index; Tardis mirror as cross-check (action-note: verify file list; ingest bybit)	W1	backfill	VERIFIED-UPSTREAM-DOC
16	coinbase-exchange-candles	CEX-API	api.exchange.coinbase.com /products/{id}/candles granularities 60/3600/86400…	1m–1d	~2014-12→ GDAX era (stream-04)	realtime	REST keyless	free	candle-gap detector; US-compliant reference venue; product rename continuity (BTC-USD etc.) (action-note: live leg; full candle W1)	W0	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
17	kraken-rest-ohlc	CEX-API	Kraken public REST /0/public/{OHLC,Trades,Depth}	1m+ OHLC, tick, L2 top	OHLC REST capped 720 candles back (stream-04)	realtime	REST keyless	free	720-candle cap → dumps (rows 18–19) for depth; Trades since market start via API (action-note: live tail only)	W0	connect	SURVEY-EVIDENCE
18	kraken-ohlcvt-dumps	CEX-archive	support.kraken.com/articles/360047124832 downloadable OHLCVT zips (per-market CSV ×8 intervals)	1/5/15/30/60/240/720/1440 min	from inception of each market (→2013+ era); quarterly incremental ZIPs	batch	batch-CSV + sha256 manifest verify	free	gaps NOT zero-filled (no-trade intervals absent); MANIFEST.json + checksum gate (action-note: full archive ingest + quarterly incrementals)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
19	kraken-trades-dumps	CEX-archive	support.kraken.com “time and sales” CSVs (one per pair)	tick	from beginning of each market	batch	batch-CSV	free	header schema (ts,price,vol,b,s,trade_id); manifest+sha; reassemble part-files in order (action-note: tick-grade free (US venue))	W1	backfill	VERIFIED-UPSTREAM-DOC
20	kraken-futures-paid-tick	CEX-derivs-paid	Kraken Futures tick+full-depth via CryptoStruct self-service	tick + L2	per-day instrument-days (vendor coverage July 2023→, verify)	batch purchase	CSV/Parquet per-day	paid (€1/day-class vendor)	vendor capture claims; low liquidity venue flag (action-note: slot only if Kraken perp features needed)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
21	bitstamp-dumps	CEX-archive	bitstamp.net/market/ data down-loadable trade files (page ×2 timeout this session) + CDD mirror (C20)	t/k/CHLC CSV	Amberdata claim: spot from 2011-08-18 (C8); Tardis mirror only 2019-03-30→ (C4)	batch	batch-CSV direct-dump walk	free	start date is vendor-claim (fetch to confirm); oldest continu-ous BTC tape (action-note: walk dump tree; fill 2011–2019 via dumps)	W1	connect	SURVEY-EVIDENCE
22	bitfinex-v2-rest-ws	CEX-API	Bitfinex v2 REST /candles (hist since listing), /trades; WS v2	1m+	2015→ (stream-04, verify)	realtime	REST+WS keyless	free	maintenance candle oddities (stream-04) → sec-ondary venue cap (action-note: sec-ondary his-tory/book)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
23	midtier-ccxt-venues-kucoin-gate-mexc	CEX-API	KuCoin v4 + Gate.io v4 + MEXC v3 REST/WS (via CCXT harness)	tick + 1m+	since listing each	realtime	REST+WS (CCXT)	free	≥1 filler venue read-only; new-listing signal from MEXC/Gate; volume-credibility discount (action-note: single adapter, three venue ids)	W2	connect	UNVERIFIED-CANDIDATE
24	upbit-bithumb-korea	CEX-API	Upbit V1 REST (KRW candles/trades; KCE/US companions) + Bithumb candle JSON	1m/1d	since listing (Upbit daily 2017-10→, verify)	realtime	REST keyless	free	KR fiat quote; regional walling; archive-depth verify (action-note: KRW parity strip)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
25	india-cex-wazirx-coindcx-zebpay	CEX-API	WazirX + CoinDCX (+ZebPay check-points) public OHLC/depth REST	1m/1d	since listing (verify)	realtime	REST	free	WazirX suspended-flow incidents → PIT gate; INR reference only (action-note: optional INR strip)	W3	connect	UNVERIFIED-CANDIDATE
26	coingecko-chart-history	aggregator-REST	CoinGecko /coins/{id}/market_chart/global (has /coins/{id}/history for exact ts)	5m/daily tiers	~2013→ global series com-puted via coingecko history + mar-ket_chart range (paid depth)	near-realtime	REST + demo key	free tier tight keyless limits; paid tiers	free depth cap (~365d keyless; verify); volume credibility flags; delist/market PIT at global level (action-note: daily global snapshot job + 2013 history loader)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
27	coingecko-onchain-geckoterminal	DEX-agg	CoinGecko onchain DEX-candle endpoint + GeckoTerminal pools API	pool OHLCV	per-pool listing	realtime-ish	REST	free tier	pool rebasing; router identity; new-pool spam (action-note: DEX parity vs sub-graphs)	W3	connect	UNVERIFIED-CANDIDATE
28	coinmarketcap-freemium	aggregator-REST	CMC Pro API open/labels/global + v2 historical	hourly/daily	freemium history windows limited	near-realtime	REST keyed	free tier	CMC method-ology; rate caps (action-note: only for parity strip)	W2	connect	SURVEY-EVIDENCE
29	coindesk-data-cryptocompare	aggregator-REST	CoinDesk Data ex-CryptoCompare legacy min-api + CSV historical archives	1m+	2011-era reference windows	near-realtime	REST keyed	freemium + paid	support-life-cycle risk (stream-04 □D3): migrate or budget replace-ment; endpoint deprecа-tion watch (action-note: 2011-era fallback history only)	W2	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
30	coinapi-flat-files	archive-flatfiles	CoinAPI Flat Files (S3/Snowflake/CSV.gz, gzip partitions)	tick + agg	400+ exchanges / 599K symbols / 632TB (vendor-claim; verify partition map)	batch	S3 bulk + license	paid (negotiated)	per-REQUEST vs bulk budgeting; venue-symbol map drift (action-note: cost-quantify vs Tardis/Massive)	W2	connect	VERIFIED-UPSTREAM-DOC
31	coinapi-unified-api	aggregator-REST	CoinAPI REST/WS/FIX (multi-venue trade+quote unified)	tick	since 2013+ per plan	realtime	REST+WS keyed	paid (free ~100 req/d order-of-magnitude; re-verify)	per-request cost discipline; normalized schema drift (action-note: multi-venue fallback when CCXT too costly)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
32	tardis-dev-archive	archive-flatfiles+replay	docs.tardis.dev Market Data API + downloadable CSVs (50+ venues)	tick L2/trades/liq	per-venue anchors: Bitstamp 2019-03-30 (□C4); Bybit inverse 2019-11-07, linear 2020-05-28, liquidations 2020-12-18 (□C5)	batch	HTTP tick-list API + CSV; Python/Node libs	free recent-N-days (vendor-claim); full history paid	per-venue channel availability vary; incident pages per venue; no redistribution (action-note: full-history buy eval for L2)	W2	backfill	VERIFIED-UPSTREAM-DOC
33	coinpaprika-api	aggregator-REST	api.coinpaprika.com/v1 tickers, OHLCV, markets, contracts	daily OHLCV (+hourly Business-up)	BTC first-data 2010-07-17; full OHLCV→2010 on Ultimate tier (vendor docs; verify tier price)	realtime	REST keyless 20k calls/mo	free/personal; enterprise paid	SEC-hosted A-grade data-quality study Oct-2025 (□C14); trust-scores per market (action-note: reference/market cross-check source)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
34	massive-crypto-flat-files	archive-flatfiles	massive.com (ex-Polygon) flat files: Coin-base/Bitstamp/Kraken/Bitfinex trades	hourly-partitioned trades/aggs	per-vendor coverage; verify starts per ticker	batch	S3 CSV bulk	paid	UTC-standardized; hourly zstd; first-party doc □C19 lists subset venues (action-note: evaluate as Coin-base/Kraken tick mirror)	W2	connect	VERIFIED-UPSTREAM-DOC
35	cryptodatadownload-csv	archive-flatfiles	cryptodatadownload.com CSV archive (Bit-stamp, Binance, Gemini, FTX-pre-collapse, OKX, Bybit…)	1m/hr/d	2017→ refreshed daily	batch	direct CSV download	free (no login, no paywall)	cleaned/derived ware-house — NOT raw ticks; dead-venue archive lane (Bit-trex/FTX for back-testing) (action-note: dead-venue archive lane only)	W3	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
36	dormant-aggregators-nomics-coinlore	dormant-flag	Nomics (defunct-risk) + CoinLore free	—	era-varying	—	—	—	confirm operational status before integration; Nomics API key docs long stale (action-note: kill-confirm then exclude)	W3	connect	UNVERIFIED-CANDIDATE
37	ccxt-library-route	library	ccxt unified connectors (multi-venue REST/WS adapters)	n/a	n/a	realtime	library integration	OSS; venue ToS inherit	adapter quirks patched unevenly → pin CCXT version per venue; arms races on REST shapes (action-note: W0 tail adapter harness)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
38	deribit-options-chains-realtime	derivs-API	Deribit v2 WS/REST (book.{instrument}, ticker, get_volatility_index)	tick	2016→ (venue start)	realtime	WS keyless	free	BTC/ETH options ground truth (stream-04 □D1); testnet mirrors; quote-quality legs (action-note: L2 + IV joiner (non-colo path))	W0	connect	VERIFIED-UPSTREAM-DOC
39	deribit-dvol-history	derivs-index-history	DVOL index history via Deribit get_volatility_index_data + monthly bulk	1m/1d series	DVOL since 2016 launch	realtime	REST	free	DVOL method-ology updates; cross-ref vendor-diligence census (06) (action-note: index-series loader + index refresh)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
40	okx-funding-oi-history	derivs-archive	OKX /funding-rate-history, /open-interest-history	8h + intervals	since listing (verify)	realtime	REST	free	OKX entity split context (rows 12–13); funding PIT (action-note: third funding leg after Bi-nance/Bybit)	W1	connect	SURVEY-EVIDENCE
41	bybit-funding-liq-archive	derivs-archive	Bybit funding records + liquida-tions via API + Tardis coverage anchors (□C5)	event/8h	funding since listing; liq mir-rorable from 2020-12-18	realtime	REST+WS	free	liq vs forced-order seman-tics; PIT on instru-ment rename (action-note: build fund-ing+liq panels)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
42	derivs-aggregators-coinglass-coinalyze	derivs-agg	Coinglass v4 (paid liq-heatmap/history; MCP beta exposes skill) + Coinalyze free OI/funding/liq cross-section	1s–1d	multi-month by tier	near-realtime	REST keyed	Coinglass paid; Coinalyze free-attribution	free-tier history caps; vendor-claim pricing; Coinalyze attribution req (action-note: free Coinalyze benchmark first; Coinglass only for heatmap feature)	W2	connect	VERIFIED-UPSTREAM-DOC
43	bitmex-dumps	derivs-archive	bitmex.com/app/tradeDownloads traded/quoted/liquidation zips	tick-level	~2014-11→ (BitMEX perp era; verify)	batch	batch-CSV	free	legacy dump staleness check; earliest perp liquidation tape (action-note: pre-Binance perp-era coverage)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	btc-fullnode-archive	onchain-node	bitcoind (full archive with txindex, or pruned tail node)	block/tx + mempool	since 2009-01-03 genesis	realtime (node RPC/ZMQ)	own-node RPC+ZMQ	chain data public; OSS client	reorg-window gate; disk budget; ZMQ rawtx vs hash-block subscrip-tion choice (action-note: node deploy plan; genesis→ present archive)	W1	connect	UNVERIFIED-CANDIDATE
45	mempool-space-rest	onchain-API	mempool.space REST explorer API (blocks, tx, addr, mempool, fees, mining-pool series incl. block audit scores, /api/v1/historical-price)	block/addr-level history	blocks from genesis queryable; mining series spans history	near-realtime	REST keyless	free	shared public instance rate-limit → throttle + self-host mempool for prod; index rebuilds PIT (action-note: BTC blocks/fees/ via API)	W0	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	mempool-space-ws	onchain-API	mempool.space Web-Socket API (blocks, mempool additions, stats)	tick	live	realtime	WS keyless	free	reconnect/backoff; sequence resync (action-note: BTC realtime tail (blocks+mem))	W0	connect	VERIFIED-UPSTREAM-DOC
47	eth-execution-archive-node	onchain-node	Geth/Reth/Erigon archive (eth_getLogs, ws subs; Reth snap-shots shorten sync)	block/log/tx	since 2015-07-30 genesis (archive mode)	realtime	own-node RPC/WS	chain data public; OSS clients	reorg handling; getLogs range caps vs Erigon full-index; disk ~2TB+ order-of-mag (stream-04) (action-note: ERC-20 transfer account-ing Gold layer)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
48	eth-mempool-subscribe	onchain-stream	self-host Geth/Reth ws eth_subscribe newPend-ingTrans-actions	tick	live	realtime	own-node WS	self	sampling NON-exhaustive (relay/private-orderflow blind); privacy handling for EMD (action-note: only for DEX execution-slippage research)	W2	connect	UNVERIFIED-CANDIDATE
49	beaconchain-api-eth	onchain-API	beaconcha.in API v2 (valida-tors, balances, effective balances, attesta-tion, Beacon-Score; endpoint v1→v2 migra-tion)	epoch/validator	beacon genesis 2020-12-01→	epoch cadence (near-realtime)	REST keyed	free w/ required attribu-tion (docs C15)	attribution require-ments; premium selectors pay-walled; rate-limit tiers (action-note: ETH staking yield panels)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
50	beacon-self-attestation	onchain-node	own consensus client (Lighthouse/Prysm) + Beacon API (duties, attestation history)	slot/epoch	since 2020-12-01	12s slots	own-node Beacon API	chain data public; OSS	attestation duties are not persisted by spec for old epochs → rebuild window; huge state sync cost (action-note: only if validator-level features beyond beacon-cha.in)	W2	connect	UNVERIFIED-CANDIDATE
51	base-node	onchain-node	op-stack (op-geth + op-node) self-host	block/log	since 2023 (target archive)	realtime	own-node	chain data public; OSS	rollup stage/fault-proofs (L2Beat); reorg semantics; L3 extras post-scope (action-note: L2 Gold feeds before buying indexers)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
52	arbitrum-nitro-node	onchain-node	Arbitrum nitro full/archive node	block/log	since 2021 (target archive)	realtime	own-node	chain data public; OSS	L1-posted data dependency for full replay; snapshot availability varies (action-note: L2 strip)	W1	connect	UNVERIFIED-CANDIDATE
53	optimism-node	onchain-node	op-stack (op-geth) self-host	block/log	since 2021 superchain era migration (verify pre-2023 depth)	realtime	own-node	chain data public; OSS	bedrock migration block semantics → PIT on pre/post while replaying (action-note: L2 strip)	W1	connect	UNVERIFIED-CANDIDATE
54	solana-rpc-geyser	onchain-node	validator-lite RPC + Yellowstone gRPC (geyser plugin streams)	slot/txn	RPC lookback bounded by snapshot window; archival beyond scope	realtime (sub-second slots)	own-node gRPC	chain data public; OSS	public RPC rate-caps; memory-heavy node; snapshot resurrection offsets (action-note: only if Solana-DEX venue play)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	avalanche-node	onchain-node	avalanchego node (C-Chain EVM, X/P chains)	block/log	since 2020 (C-Chain 2020-09)	realtime	own-node	chain data public; OSS	multi-chain index burden; subnet filtering (action-note: alt-L1 strip)	W3	connect	UNVERIFIED-CANDIDATE
56	polygon-pos-node	onchain-node	heimdall + bor (or erigon/LaGeth-class forks)	block/log	since 2020-05 mainnet PoS chain	realtime	own-node	chain data public; OSS	checkpoint vs block cadence split; snap-shot/pruning trade-offs (action-note: Polygon strip)	W2	connect	UNVERIFIED-CANDIDATE
57	bsc-node	onchain-node	geth-fork client (erigon-style for BSC)	block/log	since 2020-09 mainnet	realtime	own-node	chain data public; OSS	TB-class snapshot sync; parallel-exec reorg schema (2024+ eras) (action-note: BSC strip)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
58	tron-trongrid	onchain-API+node	TronGrid REST/HTTP (compat Java-tron RPC) or self-host java-tron + event indexer	block/contract-event	since 2018-05 mainnet	near-realtime	REST/node	TronGrid free tier keyed	energy/resource model events; indexer filter per contract (action-note: TRON strip via API first)	W3	connect	UNVERIFIED-CANDIDATE
59	ton-fullnode	onchain-node	ton-node (validator-lite) + tx explorer ingestion	block/tx mes-sages	since 2020-08 mainnet	realtime	own-node	chain data public; OSS	shard message splits; indexer immatu-rity — verify stability (action-note: TON strip)	W3	connect	UNVERIFIED-CANDIDATE
60	cardano-dbsync	onchain-DB	cardano-node + cardano-db-sync → Postgres SQL	block/UTxO/ledger	since 2017-09 mainnet	near-realtime (sync lag grows without snap-shot)	own-node SQL	chain data public; OSS	multi-day replay cycle; era-schema changes (Shel-ley/Vasil…) (action-note: Cardano strip (SQL-native))	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	aptos-sui-move-chains	onchain-node	Aptos fullnode + Indexer gRPC/GraphQL; Sui fullnode + GraphQL	block/tx events	Aptos 2022-10 / Sui 2023-05 →	realtime	own-node API	chain data public; OSS	Move event-model differences; indexer lag + version-manager maturity flags (action-note: Move-chain strip (both L1s, one worker))	W3	connect	UNVERIFIED-CANDIDATE
62	dune-sql-layer	onchain-warehouse	Dune SQL (Spellbook dbt, Trino connector, API credits, SDK-execute)	per-request	multi-chain from genesis	batch	SQL/rest + credits	free credits; paid tiers	query-credit cost management; Spellbook pinned model versions; output attribution policy (stream-04 □D5) (action-note: Gold-layer prototyping + s)	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
63	mev-eigenphi-api	mev-DB	EigenPhi API (arb/sandwich/liq MEV stats)	event + aggregated	~2021→ (verify)	near-realtime	REST keyed	free community tier / paid	docs endpoints unreachable ×2 this session (transport error) — verify before paid commit (action-note: DEX exec-cost slippage features)	W2	connect	UNVERIFIED-CANDIDATE
64	flashbots-mevboost-relay-datasets	mev-DB	Flashbots MEV-Boost relay public datasets + ZeroMEV / mev-boost.pics Dune lineage	slot-level	since Merge 2022-09	batch (block-lag)	file + SQL (Dune-backed)	open dataset lineage	relay participation share shifts; SSZ→decoded schema versioning (action-note: ETH execution-tax features; cross-check EigenPhi)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	uniswap-v2-v3-subgraphs	dex-subgraph	The Graph hosted sub-graphs (v2 factory 2018; v3 pools w/ sqrt-PriceX96 + tick liquidity)	per-block/event	since deploy block	near-realtime (minutes)	GraphQL	subgraph data open; Graph 100k free queries/mo per subgraph (stream-04 □D4)	index lag = minutes — NOT tick-exact; pin de-ployment ids (depreca-tions) (action-note: pool-day flows + liquidity-tick snap-shots)	W1	connect	VERIFIED-UPSTREAM-DOC
66	uniswap-v4-subgraph	dex-subgraph	V4 hooks/state indexing (per-chain subgraph deploy-ments)	per-block	2024→	near-realtime	GraphQL	data open	hooks make pools heteroge-neous → schema-drift vigilance; verify per-chain deploy-ment coverage (action-note: v3 indexer fork with hook-state columns)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
67	curve-balancer-sushi-subgraphs	dex-subgraph	Curve pools/rates + Balancer weighted pools + SushiSwap pairs (hosted sub-graphs)	per-block	since 2020-era deploys	near-realtime	GraphQL	subgraph data open	veCRV gauge ac-counting; Balancer weight drift; Sushi fork-of-v2 seman-tics (action-note: stable-swap + weighted-pool strips)	W1	connect	UNVERIFIED-CANDIDATE
68	aave-compound-lending-subgraphs	lending-subgraph	Aave v2/v3 + Com-pound v2/v3 (sup-ply/borrow/utilization/liquidations)	per-block	Aave 2020→; Com-pound 2020/2021→	near-realtime	GraphQL	data open	utilization PIT from index snap-shots; reserve-factor config history (action-note: DeFi lend/borrow rate panels)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
69	lido-eigenlayer-staking-subgraphs	staking-subgraph	Lido stETH subgraph + EigenLayer operator/AVS datasets	per-block	Lido 2020-12→; EigenLayer 2023→	near-realtime	GraphQL	data open	yield attribution split vs beacon-chain validators (row 49); restaking risk (action-note: staking-yield joins for ETH yield models)	W2	connect	UNVERIFIED-CANDIDATE
70	defillama-tvl	defi-aggregate	DefiLlama API /protocols + /charts (endpoint families per D7)	chain/protocol/token	2019→ per protocol	realtime-approx (m-refresh)	REST keyless	free + attribution	per-adapter TVL methodology quirks; double-count flags; stale-adapter audit (action-note: core DeFi aggregate feed)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
71	defillama-stablecoins	defi-aggregate	DefiLlama stable-coins mcap + pegSta-bility end-points	chain/token	issuers tracked since 2018-era	realtime	REST	free	mint/redemption double-count; mcap lag hours; issuance-bridge splits (action-note: stable-coin module pair w/ attesta-tions (77–79))	W1	connect	VERIFIED-UPSTREAM-DOC
72	defillama-yields	defi-aggregate	DefiLlama yields/pools history	pool-level	2021→ estab-lished pools	realtime-approx	REST	free	APY com-position (base vs reward) seman-tics; TVL-shift noise (action-note: yield bench-mark set (Stak-ingRe-wards cross-check))	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
73	defillama-bridges	defi-aggregate	DefiLlama bridges vol-ume/day	chain	2020→	near-realtime	REST	free	bridge-vendor volume overlap; bridge incident PIT (action-note: cross-chain flow features)	W2	connect	VERIFIED-UPSTREAM-DOC
74	defillama-volume-fees-raises	defi-aggregate	DefiLlama DEX spot volume + protocol fees/revenue + raises DB	chain/protocol	volume 2020+ era; raises listed	realtime	REST	free	fees vs revenue seman-tics differ per protocol; raises PIT (action-note: money-flow + cap-formation panels)	W1	connect	VERIFIED-UPSTREAM-DOC
75	l2beat-tvs-risk	L2-aggregate	L2Beat API (TVS, Stage 0/1/2, risk dic-tionary)	chain	since rollup launches	hours	REST unauth/json	free (at-tribution)	risk-dictionary method-ology changes → version pin; TVS ≠ TVL (action-note: L2 trust-scoring enrich-ment)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
76	dexscreener-pools	DEX-agg	DexScreener API (new-token pool pairs, liquid-ity/vol)	per-pool	since listing	realtime-ish	REST	free tier limited	rebase/exploit filter; canonical subgraph fallback (action-note: alt-DEX parity/scam-filter lane)	W3	connect	UNVERIFIED-CANDIDATE
77	chainlist-metadata	chain-metadata	Chainlist (De-fiLlama EVM registry: chain-ids, RPC/WSS end-points)	static registry	n/a	batch	REST/static	OSS (MIT)	RPC hygiene (public endpoint quality varies); registry drift (action-note: feed GFI or config)	W1	connect	UNVERIFIED-CANDIDATE
78	circle-usdc-attestations	stablecoin-attestation	Circle trans-parency (attesta-tion service + reserve disclo-sures)	monthly attesta-tion	from 2018	batch	PDF + public API fetch	public disclo-sure	issuer-commissioned attesta-tion; malformed-archive paths → check-sum gating (action-note: USDC collateral-risk feature)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
79	tether-attestations	stablecoin-attestation	Tether quarterly (BDO Italy assurance + CRR disclosure)	quarterly	from ~2014	batch	PDF	public disclosure	assurance breadth selectively scoped → directional only (action-note: USDT counterpart feature)	W1	connect	UNVERIFIED-CANDIDATE
80	paxos-pyusd-attestations	stablecoin-attestation	Paxos attestation service (USDP, PYUSD, USDG monthly)	monthly	from 2018	batch	PDF, URL-param per issuer	public disclosures	issuer coverage timeline; PYUSD conversion-PIT (verify) (action-note: lesser-issuer stable lane)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
81	pyth-price-feeds	oracle	Pyth Pro (ex-Lazer) WS/REST + history endpoint; classic on-chain pull feeds (stream-04 □D8)	tick	history via Pro REST endpoint	ms-class (vendor claim)	keyed WS/REST	Pro paid; on-chain pull free	confidence-interval attached per tick; per-asset feed ids drift; enti-tlement scope (action-note: multi-asset realtime overlay (incl. non-crypto))	W0	connect	VERIFIED-UPSTREAM-DOC
82	chainlink-feed-history	oracle	on-chain aggrega-tor reads (Round-Data/EVENT logs of classic feeds; Data Streams out of initial scope, stream-04 □D9)	round-based	since pair-feed deploy-ment (~2020+)	heartbeat/deviation-bound (NOT tick)	event-node eth_getLogs	on-chain output public	round PIT = update-block times-tamp (not heart-beat); devia-tion+heartbeat config per pair; L2/L1 chain splits (action-note: consensus-fallback leg of GFI price ag-gregator)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
83	redstone-api3-oracles	oracle	Redstone (WS/REST per feed model) + API3 OEV/dAPI claims	tick/round	since deploy per chain (verify)	near-realtime	REST/WS	free read	per-chain feed coverage verify first; coverage decay after Pyth/CL domi-nance (action-note: sec-ondary oracle strips only)	W3	connect	UNVERIFIED-CANDIDATE
84	tally-snapshot-governance	governance-votes	Tally API (EVM on-chain propos-als/votes historical) + Snapshot Hub API (propos-als/votes via IPFS archive)	proposal-level	Snapshot 2020→; Tally per DAO deploy	batch	REST	free (rate-limited)	Snapshot IPFS per-sistence gap risk; Tally = EVM-only; quorum-params PIT (action-note: governance-vote panel cross-vendor check)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
85	certik-immunefi-security	security-DB	CertiK Skynet scores (paid API) + Immunefi boun-ties/loss data (free REST)	asset-level	since audit listings	near-realtime	REST	CertiK paid; Immunefi free	score-methodology churn; bounty-TVL mapping staleness (action-note: risk-score enrich-ment features)	W3	connect	UNVERIFIED-CANDIDATE
86	rekt-chainabuse-incident-db	security-DB	rekt.news (histori-cal hacks w/ loss series) + Chainabuse reports + De-fiLlama raises (□D7)	incident-level	rekt archive 2020→	batch	web/REST scrape	attribution-license	incident attribu-tion error risk; loss-figure spread across sources; scam-report quality noise (action-note: loss-episode feature set for risk models)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
87	nft-dune-blur-os	nft-market	Dune-lineage NFT trade datasets (OpenSea/Blur/X2Y2; royalty-free era break-points)	per-tx	2021→ core marketplaces	batch	SQL credits (Dune □D5 lineage)	Dune output attribution	floor/trait quality; indexer lags; royalty-era PIT break-points (action-note: optional NFT feature lane)	W3	connect	UNVERIFIED-CANDIDATE
88	crypto-social-stocktwits-twitter	social-sentiment	StockTwits crypto streams (free API) + paid crypto-Twitter/social vendors (RavenPack/Meltwater-class)	msg-level	varies by vendor	near-realtime	REST	StockTwits free; paid-class for X/Telegram coverage	free Twitter API limits (2023-era pricing regime) — paid wedge for X; spam-filter PIT (action-note: optional sentiment lane)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
89	fear-greed-index-archive	sentiment-index	alternative.me F&G historical API (daily)	daily	from 2018-02-01	daily T+	REST keyless	free	index methodology opaque/vol-weighted; census anchor claims vol-11 verification — archive URL NOT re-fetched this session → re-verify endpoint live-ness (action-note: daily sentiment series (cheap W1))	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
90	paid-onchain-metrics-cryptoquant-glassnode-santiment	paid-metrics	CryptoQuant (exchange-flow), Glassnode Studio (limited free series), Santiment (API)	multi-resolution	multi-year per metric	daily/RT-class	REST keyed	paid tiers; limited free tiers	licenses forbid redistribution into vaults; cross-vendor triangulation needed (no single source of truth) (action-note: paid wedge only when exchange-flow deltas required)	W2	connect	UNVERIFIED-CANDIDATE

12 — Cross-domain historical archives (76 rows)

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	Jordà–Schularick–Taylor Macrohistory Database (JST R6)	market-history/macro	macrohistory.net direct file download (Stata/XLSX) [1]	Annual; 48 vars; 18 advanced economies	1870–present (~155y)	batch (release-based)	archive-HTTP bulk file; pin R.6 snapshot	free, CC BY-NC-SA 4.0 (non-commercial; no vendor redistribution)	fixed release tag; crisis-dummy alignment vs JST crisis chronology PDF; document splices per 100pp documentation (action-note: ingest R6 XLSX → country×variable long table; freeze source snapshot hash)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
2	JST crisis chronol-ogy + RORE returns docs	financial-stress/history	macrohistory.net — doc PDFs (same release) [1]	Annual event/series	1870–2015	batch	archive-HTTP PDF/XLSX parse	free CC BY-NC-SA	cross-check crisis flags in main table vs chronol-ogy PDF; PIT: cite release year (action-note: parse crisis chronol-ogy into event table (cross-ref row 1))	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
3	Shiller US equity/CAPE/US home prices (ie_data.xls, Fig3-1)	market-history/equity/	shillerdata.com direct XLSX [2]	Monthly; US	1871+ stock/CAPE; 1890+ home prices	batch (monthly)	archive-HTTP XLSX; scripted re-pull + diff	free (attribution; no warranty)	spliced Cowles/1926 sources documented; CPI splice at 1913; recompute CAPE from constituents to validate (action-note: ingest ie_data.xls all columns; version-by-download-date snapshot)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
4	BoE “A Millennium of Macroeconomic Data” (UK v3.1)	market-history/macro	bankofengland.co.uk/research/datasets page, 28MB XLSX direct [3]	Annual (some pre-1830 benchmarks); UK	C13th–2016 (1086 Domesday benchmarks)	batch (rare updates)	archive-HTTP XLSX	free (Crown/attributed via BoE)	v3.1 published; email-contact annotation; pre-1830 treated as benchmark-point, not continuous (action-note: ingest UK mega-sheet into long format; flag pre-1700 series provenance)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	MeasuringWorth US/UK (GDP, CPI, wages, $/£ER, interest rates, S&P/DJIA)	macro-history/deflator	measuringworth.com, datasets pages [12]	Annual, US+UK (+AU, ES)	US GDP 1790+; CPI 1774+; ER 1791+	batch	scrape/download per-dataset pages (no bulk API)	free for non-profit educational use w/ credit; permission needed otherwise	method-essay awareness (7 measures of worth); cite staff/Officer papers (action-note: ingest US CPI + $/£ ER as deflator backbone; keep per-dataset source notes)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
6	Global Financial Data (GFD)	market-history (longest runs)	globalfinancialdata.com (licensed subscrip-tion; custom data-file delivery)	Annual/monthly, 200+ countries	1265+ (header series), equities 1600s+	batch (subscrip-tion dumps)	licensed bulk file (SFTP/email zip)	licensed (paid, per-user/redistribution limits)	license quote req: seats, redistri-bution, annual update fee; cross-validate overlap vs Shiller/JST (action-note: request quote; scope: JST gap-fill + pre-1870 bench-marks only)	W2	license-review	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
7	Ibbotson SBBI (Morningstar)	market-history/equity	Morningstar Direct/IDX; licensed	Annual/monthly; US asset classes	1926–present	batch	licensed feed; CSV export	licensed (paid; chart-licensing separate)	quote cost; replicate-access only via WRDS unless licensed; PIT: annual ed. vintage (action-note: ; only if GFD fails for 1926–1970 returns)	W2	quote	UNVERIFIED-CANDIDATE
8	CRSP US equities (cross-ref 05)	securities/history	WRDS/Crsp; see vol 05	Daily; US listed universe	1925–present	batch	WRDS (sas xport ftplib)	licensed (academic quote)	coverage-gaps for delisted shares — survivorship-free requirement (action-note: cross-ref 05; here only for pre-1962 of 05’s route)	W2	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
9	Compustat North America (cross-ref 05)	securities/fund-history	WRDS; see vol 05	Annual/quarterly	1950s–	batch	WRDS	licensed	restatement vintages via comp_history tables (action-note: cross-ref 05 only)	W2	connect	SURVEY-EVIDENCE
10	WFE (World Federation of Exchanges) annual market stats archive	exchange-history	focus.world-exchanges.org; down-loaded WFE workbook archive	Annual by exchange	1970s+ (common members backfill)	batch	archive-HTTP XLSX per year	licensed-ish (terms: no redistribution, non-commercial survey use)	member-change breaks continuity; document coverage by exchange-year (action-note: pull archive year sheets; build exchange-years matrix)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
11	FRED bulk CSV (full history, by series id)	macro-archive bulk	fred.stlouisfed.org /fred/series/{id}/ + fred-graph + fredapi/FRED API + full-file dumps via external mirrors	Varies: daily-annual); global	Series-specific (many 1900s+)	batch	REST + bulk-CSV	free	cite St. Louis Fed as redistrib-utor of underly-ing owners; per-series owner/license check (action-note: nightly delta via FRED API; store vin-tage_id=term)	W0	connect	SURVEY-EVIDENCE
12	ALFRED (archival FRED vintage)	PIT/vintage	alfred.stlouisfed.org [10]; API base fred.stlouisfed.org/docs/api/fred/	Varies	Vintages since 2006 (data itself older)	batch (already archived)	REST API (`vintage_dates`) + CSV	free	the desig-nated PIT spine for US macro: ingest real-time-vintage series first (action-note: fetch `observa-tion_start`+vintage paths for top-200 US macro series)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
13	Maddison Project Database (GDP per capita, popula-tion)	academic/macro-history	ggdc.net/maddison (Project 2020 release, direct XLSX)	Admin1; 169+ coun-tries/regions	AD 1–2018 core; 1500+ denser	batch	archive-HTTP XLSX	free (CC BY 4.0-type; cite Bolt & van Zanden)	year-0 nan pattern; region-vs-country bound-aries; use 2012-boundary CLIO harmo-nization if mixing (action-note: ingest maddi-son2020; derive growth/level panels)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
14	CLIO-INFRA (86 datasets: GDP/capita 1500+, bond yields 1727+, FX 1500+, central debt 1692+, metals production 1681+, demography)	academic/multi-domain	clio-infra.eu direct Indicator XLSX downloads [4]	Annual-decadal; 224 countries	1500–2013 (varies 1543–2018)	batch (archived)	archive-HTTP XLSX per indicator	free (attribution / varied per dataset — check per-file terms)	boundaries=V2012-reference; mixed annual/decadal already harmonized; download per-indicator as-of-date provenance (action-note: bulk-pull finance block first (yields, FX, debt) then production)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
15	Our World in Data (owlid bulk + per-topic grapher CSVs)	academic/topic-archive	ourworldindata.org/grapher/{slug}.csv per chart + owid dataset repo (quiltify)	Annual; global topic-mapped	varies; energy 1800s+, poverty 1820+	batch	bulk-CSV (grapher endpoint) + GitHub repo mirroring	free (CC BY 4.0)	chart-slug volatility → pin slugs; underlying-source attribution chain per column (action-note: enumerate target slugs (energy, emissions, growth); subscribe repo diffs)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
16	Barro–Lee educational attainment 1950–2010	academic/human-capital	barrolee.com — direct XLSX	5-year interval; 146 countries	1950–2010	batch (static)	archive-HTTP XLSX	free (cite Barro & Lee 2013)	2010 home-age cutoff; no forward updates — pair with post-2010 Lutz witt (CLIO/OWED) (action-note: one-time ingest; register as frozen reference)	W1	connect	SURVEY-EVIDENCE
17	WID.world (DINA top wealth/income shares)	inequality-history	wid.world data portal full-dataset CSV export + `wid` Stata package [7]	Annual; 100+ countries	1820–present (col. vary)	batch (infrequent quarterly)	portal bulk CSV + REST-ish WID API	free (CC BY-NC? verify per-dataset; attribution required)	DINA-ratio method editions; collection-specific selection (match JST era) (action-note: bulk download WID-DINA; build top-1%/top-10% panels)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
18	World Bank WDI (bundled long-run archive + API archive)	macro/long-run	databank.worldbank.org/source/world-development-indicators + api.worldbank.org (M119.MWI/M…); plus legacy “WDI archives” page	Annual; 217 countries	1960–current (archival pdf back to 1980s)	batch	REST (JSON/CSV) `source=2`	free (CC BY 4.0)	country-code-history (e.g., remnant fran-chises), missing-x pattern; fallback to World Bank CSV dumps (action-note: API script by indicator-list; keep wrangler yearly in-stance_id)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
19	IMF IFS historical (classic IFS lines incl. exchange-era FX)	macro-archive	IMF Data (data.imf.org) "International Financial Statistics"; legacy line-set cut off series-wise (classic IFS retired, successor tables absorb)	Monthly; ~200 countries	1948–2022 (some pre-1948 via IFS PDFs-cans)	batch (monthly, archived)	SDMX bulk CSV (IFS dataset id: IFS_6.0-style key family)	free (IMF CC BY-NC condition; IFS carry copyright of underlying countries)	IFS line-number stability; skip IFS data-reference codes with patch notes (action-note: bulk SDMX (IFS) monthly; keep pre-1980 PDF scan backlog as W3 manual)	W0	connect	UNVERIFIED-CANDIDATE
20	UN Statistics Monthly Bulletin of Statistics (MBS) archive	macro-archive	unstats.un.org MBS archive (PDF/CD-era)	Monthly; ~200 countries	1950s–2010s (online archive)	batch (static)	archive-HTTP PDF	free (public UN)	table-code churn; only as gap-fill vs IFS/WDI (action-note: select ~20 tables (minting/M2/FX) if IMF gaps remain)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
21	OECD Historical Statistics (book-era series + OECD.Stat archive)	macro-archive	OECD Data Explorer “Historical Statistics 1960–1996” bound dataset + scanned Historical Statistics compilations (1947–2010 era) as OCR fallback	Annual	1960–1996 (book compilations)	batch (static)	SDMX bulk or manual scan	license: OECD reproduction terms (non-commercial ok w/ attribution for stats)	scanned-table OCR QC; series-merge notes vs current OECD.Stat (action-note: only where pre-1960 OECD gaps)	W2	connect	UNVERIFIED-CANDIDATE
22	Historical Statistics of the United States (Bicentennial Edition)	.US-history	1975 Bicentennial-edition scans via FRASER (fraser.stlouisfed.org); pay-walled on-line ed. at hsus.cambridge.org	Mixed (annual proxy-specific)	1610–1975 (Bicentennial edition)	batch (static)	archive-HTTP PDF scan + manual table extraction	public-domain scans (FRASER); hsus.cambridge.org licensed	OCR QC gates for scanned tables; series-code dropout doc (action-note: FRASER-selected tables (finance, prices, FX): manual extract)	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
23	PWT (Penn World Table v10.x)	academic/macro	pwt.ruhby? — rug.nl/ggdc/productivity/pwt; University of Groningen direct download	annual; 183 countries	1950–2022 (v10.0)	batch (release-based)	archive-HTTP XLSX	free (CC BY 4.0)	version pinning (use v10.x); rgdpna vs rgdpe marker (action-note: ingest v10.x — one-time + release watch)	W0	connect	SURVEY-EVIDENCE
24	GGDC productivity/structural-change databases (incl. 10-sector historical backfills pre-1947 for ~10 countries)	academic/industry	rug.nl/ggdc/productivity + ggdc historical database	annual; NACE-grouped	pre-1947 backfills via GGDC historical	batch	archive-HTTP XLSX	free (attribution)	pre-1947 output-concept v1 inconsistencies; cite-by-database-specific papers (action-note: ingest 10-sector historical block only (economies coverage thin))	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
25	UNIDO INDSTAT 2 (2022+ public interna-tional standard)	industry/history	stat.unido.org (public data down-loads since 2022)	Annual; 4-digit ISIC	1963–current (cover-age varies)	batch	portal bulk CSV	free w/ citation (UNIDO public)	classification revisions (ISIC rev 2→3→4 bridges) (action-note: download country-wide manufac-turing value-added & output)	W2	connect	UNVERIFIED-CANDIDATE
26	Freeman–Oostendorp wages database (IISG)	academic/wages	socialhistory.org project archive (CLIO per-petuation via CLIO-INFRA real-wages)	Annual; many countries	1200–2000 (frag-mented)	batch (static)	archive	free w/ citation	coverage sparse — prefer CLIO real-wage (row 14) for conti-nuity (action-note: skip unless target country wage-history needed)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
27	BoE daily GBP spot rates (exchange-rate database)	FX archive	bankofengland.co.uk/boeapps/database (Series “A” spot FX)	Daily; ~50 currencies vs GBP	1919–present (some w/ gaps)	batch	HTTP CSV export per series ID	free (attribution)	spliced pre-1971 string with dollar regimes; BoE-only series (not market-close) — note source; splice-flags required (action-note: export A-series FX block; document splice points)	W0	connect	VERIFIED-UPSTREAM-DOC
28	BIS Nominal/Real Effective Exchange Rate series (BIS/NBHR)	FX archive	bis.org/statistics/eer; historical long-run EERs	Monthly; 60–100 economies	Jan 1964–present (nominal; broader since 1994)	batch (monthly)	SDMX bulk CSV	free (BIS)	1964–93 thin trading caveat; rebase-pair vs 2010/2015 weights revisions (action-note: ingest NOMER/REALER; join vs row 27 splices)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
29	IMF IFS exchange-rate era series (FX cross of row 19)	FX archive	data.imf.org IFS RF-lines (end-of-period FX)	Monthly	1948 onward (cover-age historic via IFS)	batch	SDMX as row 19	free w/ condition	era breaks: gold/, SDR unifi-cation periods (action-note: extracted from row 19 dataset (RF% line subset))	W0	connect	UNVERIFIED-CANDIDATE
30	FRB FRASER H.10 archive (US $ daily/weekly FX 1971–)	FX archive	fraser.stlouisfed.org H.10 collection (scanned+API-era) + FRED series EX…	Daily/weekly	1971–present	batch	FRED API + FRASER PDF for pre-API classics	free	1971 start (Bretton-Woods collapse); pre-1971 via row 27 (action-note: FRED DEXUSUK… id-block pull)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
31	ICE/USDXY (USDX) index history	FX archive	ICE DataServices (licensed); free partial via market-media mirrors	Daily	1973–present	batch	licensed feed or WSJ-Yahoo free snapshot	licensed if ICE official; media-mirror licensed-free (non-commercial)	PIT: index-methodology changes (1999 euro); rebase weights history (action-note: methods note first; media-keyed snapshot acceptable for daily-close research)	W2	connect	UNVERIFIED-CANDIDATE
32	HISTDATA (free historical FX / MT5 tick)	FX tick archive	histdata.com per-pair per-month free zip	tick (2000s era) & MT5	1990s?–present	manual-batch	archive-HTTP zip per pair	free (non-commercial terms on site)	session-basis server-side; checksum; volume dn flags (action-note: only for pairs missing from Dukas/Yahoo)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
33	Dukascopy tick reposi-tory (cross-ref 07)	FX tick	datafeed.dukascopy.com historical tick HTTP (free)	Tick, 30+ pairs	2003–present	batch	free-release HTTP (cross-ref vol 07 tech)	free (terms: no redis-tribution of raw tick)	gap-patching + tick-clean; overlap vs row 32 (action-note: cross-ref 07; here for 2003–2010 span)	W2	backfill	SURVEY-EVIDENCE
34	RMB official/NDF-history (CNY fixed, managed float eras)	FX archive	PBoC + SAFE published daily USD/CNY from 1994-01; NDF-era via news/research captures	Daily	1994–2005 fixed; 2005–2015 crawling +NDF era	batch	portal CSV/web page parse (PBoC has limited API)	free (govt data)	regime-change labels manda-tory (peg/crawl/floating periods); use row 30 DEX-CHUS as US$-side cross-check (action-note: build CNY era-tagged series; NDF s from FRB/press archives only if gap)	W2	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
35	IMF Primary Commodity Prices (PCPS/PCP amenable long-run)	commodity archive	imf.org PCPS monthly bulk (index-level + USD/unit)	Monthly; 50+ commodities, 9 aggregates	1962–present (varies by commodity)	batch (monthly)	SDMX/API bulk CSV	free (IMF, attribution)	base-era rebases; unit-USD history; check long-spin cadence variance for softs (action-note: bulk pull PCPS; index s vs Pink Sheet (row 36))	W0	backfill	SURVEY-EVIDENCE
36	World Bank Pink Sheet Commodity Price Data archive	commodity archive	worldbank.org commodity-markets monthly XLSX archive (back decades)	Monthly; 45+ commodities	1960–present (archive extent varies)	batch (monthly)	direct XLSX archive-HTTP	free (attribution: World Bank Global Price Monitoring System)	pink-sheet monthly tiny-format churn; annual “commodity price data archive” join to herded archive (action-note: ingest full archive incl. annual-history sheets)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
37	EIA WTI/Brent/NYM spot & inventory long history	commodity archive	api.eia.gov / eia.gov series browser (WTI from Jan 1946, Brent from Jan 1987)	Monthly (daily post-1986)	1946–present	batch (monthly)	REST API v2 (api key)	free (US govt public domain)	use Cushing WTI era-notes; join vs FRED DCOILWTICO (action-note: API script for oil-gas-coal-nuclear spot set)	W0	connect	SURVEY-EVIDENCE
38	USDA PSD (Production, Supply & Distribution) historical archive	agriculture archive	apps.fas.usda.gov/psdonline/ historical bulk downloads (CSV/XLSX)	Annual marketing-year; commodities by country	1960+ (fixed backfill) to current	batch (monthly)	bulk zipped CSV + dataset-API (individual commodities CSV)	free (USDA public domain)	marketing-year (not cal-yr) alignment; rolling revisions — store snapshot code/year (action-note: bulk download PSD all-commodities; year-grained panel)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
39	FAOSTAT production/trade/supply archives	agriculture archive	fao.org/faostat bulk downloads (All-Data zip + FAOSTAT API)	Annual yields; 1961+ ; land use 1961+	1961–present	batch (annual)	bulk-HTTP zip + ETL coming FAOSTAT-API	free (CC BY 4.0)	production-country boundary corrections (post-USSR split handled natively) (action-note: All-Data pull for production + trade matrix)	W0	connect	SURVEY-EVIDENCE
40	Energy Institute (ex-BP) Statistical Review of World Energy	energy archive	energyinst.org/statistical-review (2023+); BP archive 1950s+ PDFs per decade	Annual; country×fuel	global 1950s+; per-country 1965+ with v2.2 OWID reframe	batch (annual)	XLSX download per year-reviewed; PDF archives pre-2001	free (attribution)	decade-boundary backfills; fuel-category reclass (nuclear appear later; wind cap IAWS) (action-note: pieced: XLSX 2001–now + PDF table-scan for pre-2001)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
41	USGS Historical Mineral Commodity Summaries + commodities consumption statistics	metals archive	usgs.gov MCA (annual PDFs/XLSX) + minerals consumption historical	Annual; ~90 minerals; countries	commodity consumption back to 19th century for many (HSUS-linked); summaries 1995+	batch (annual Jan/Feb)	XLSX/PDF bulk per year	free (public domain, US govt)	year-of-printed MCA version pinning; expect country-of-record break patterns (action-note: pull MCA last 10y + historical consumption stats heavy-mineral set)	W0	connect	SURVEY-EVIDENCE
42	NOAA ISD (Integrated Surface Data) global station archive	weather-archive	ncei.noaa.gov ISD: GHCN-D? separate — ISD via NCEI bulk (ftp-era + WebAPI now)	Hourly/sub-hourly stations	1901–present (Mexico? US 1901+; global thin pre-1940)	batch (annual)	bulk HTTPS zip per year	free (public domain)	station-id churn & relocations; pre-1940 QC flags quality-code (action-note: bulk pipeline for 15 pilot exchange-region stations 1901–2026)	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
43	NASA GISTEMP v4 global surface temp 1880+	climate archive	data.giss.nasa.gov/gistemp/ GLB tsp tables + netCDF [9]	Monthly & annual	1880–present	batch (monthly ~10th)	plain-text table + netCDF archive-HTTP	free (public-domain-ish; cite GISTEMP team)	1951–1980 base-period anchor-ing; re-base to 1991-2020 when combin-ing (action-note: pull GLB TSV monthly; join vs HadCRUT row 44)	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
44	HadCRUT5 (CRU/Met Office) 1850+	climate archive	crudata.uea.ac.uk/cru/data/temperature/ [8]	Monthly & annual; 5°×5° grid	1850–present	batch (every 1–2 mos)	CRU text format (global/nh/sh) + netCDF grid	free (CRU licence; attribution + scientific citation)	non-infilled vs Analysis choice: pick Analysis for wider coverage; late-arrival revisions near-real-time (action-note: CRU text GL/NH/SH series; netCDF only if grid modeling later)	W1	connect	VERIFIED-UPSTREAM-DOC
45	ICE softs (cof-fee/cocoa/sugar) official historical	commodity-market (gar-ner)	ICE DataSer-vices (RTH + daily historical per-mar.ket)	Daily	market birth (NYBOT-era continua-tions back to 1890s via stocks&commodity splices)	batch	licensed data feed (ICE Con-nect/API)	licensed (paid; per-market onboard-ing)	continuous contracts splice policy; volume caps by license tier (action-note: only if cof-fee/cocoa MVP ships (pilot 2 roster))	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	MPOB palm-oil long-run (ITC-pioneered histori-cal)	agriculture archive	MPOB (Malaysian Palm Oil Board) apsis mpob.gov.my an-nual/monthly statistical tables	Monthly/annual	1970s+ (statisti-cal bulletins)	batch	portal down-loads (XLSX/PDF)	free w/ at-tribution (MPOB Malaysian govt)	revise-panels; freight & FFA hedging refs only as index (action-note: monthly OPR/CPO price history manual collect)	W2	connect	UNVERIFIED-CANDIDATE
47	FRED Moody’s Aaa/Baa yields (long-run US credit)	rates archive	FRED AAA/BAA (origi-nally Moody’s; now ICE BofA carried), monthly series from Dec 1919	Monthly	Dec 1919–present	batch (monthly)	FRED REST/CSV (row 11 route)	free (require Moody’s→Moody’s/ICE 2016 source at-tribution)	series-switch (Moody’s exiting; FED’s-other-publisher churn); verify with release notes (action-note: FRED AAA/BAA ingest; yield-spread derived)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
48	H.15 vintages (Treasury/motional rates release archive)	rates archive	FRB H.15 via FRASER (scanned pre-API) + FRED (DGS-alle series)	Daily/weekly/monthly	H.15 earliest scanned 1950s/1960s; DGS backfill: 1962+, DFF 1954+	batch	FRED API + FRASER PDF scans	free (public domain)	vintage-aware for PIT: dailyTN concept changes (Wednesday-cycle pre-1997) (action-note: FRED daily-treasury block + FRASER scan backlog only if pre-1962 debt-model needs)	W1	connect	SURVEY-EVIDENCE
49	CRSP US Treasury Database (bond database incl. pre-1941 roots)	rates securities	WRDS CRSP fixed-income legacy + CRSP TreasuryF-BOND	Monthly	1926–present (bond db 1926+ via CRSP legacy)	batch	WRDS	licensed	missing-calls on pre-1926; PIT uses CRSP access dates (action-note: as complement to FRED H.15 (row 48))	W3	quote	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
50	JGB long-run yields (Japan MoF)	rates archive	mof.go.jp interest-rate historical (long-term/short-term rates, monthly since 1974, 10yJGB matrix)	Daily/monthly, JPY curve tenors	1974–present (long 10y rate from 1874 via aca-demic?” long-run DAIWAR)	batch (monthly/daily)	HTTP CSV/XLSX per page	free (govt data)	market-era discon-tinuity pre-1971 (fixed-rate issues); partici-pating vs settle-ment conven-tions (action-note: pull MoF rate history pages; mark era labels)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
51	ECB long-run sovereign yields (EMU aggrega-tion)	rates archive	ecb.europa.eu SDW series (e.g., monthly long-term yields for EMU-continuation & per-country)	Monthly	1960s–present via con-structed “continua-tion” member series, 1999+ direct	batch (monthly)	ECB SDW API/CSV	free (ECB data terms)	COMBI-of-country series construc-tion (rotate members joins); pre-1999 synthetic labeling (action-note: SDW API: long rates (LT) for EA countries; tag pre-euro coyness)	W0	connect	SURVEY-EVIDENCE
52	St. Louis Fed STLFSI2/STLFSI4 (financial-stress index)	stress archive	research.stlouisfed.org FRED STLFSI/STLFSI2/STLFSI4	Weekly/daily	1993+ (v1), 1994+ (v2/3/4)	batch (weekly)	FRED API	free	component-set key differ-ences across versions — keep all 3 con-tiguous (action-note: pull all three stlfsi series)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
53	Chicago Fed CFNAI (+CFNAI-MA3)	stress/archive	chicagofed.org + FRED CFNAI	Monthly	1967–present	batch	FRED/HTTP	free	month-of-release publication lag (release t+1) — PIT: use release-vintage mode of ALFRED if available (action-note: FRED pull CFNAI/CFNAIMA3)	W0	connect	SURVEY-EVIDENCE
54	OECD CLI (composite leading indicator) long run	archive indicator	oecd.org CLI SDMX	Monthly	1960s+ (varies country)	batch (monthly)	SDMX CSV (as row 21 route family)	free (OECD terms)	CLI historical construction revisions; per-station base-month labeling (action-note: OECD API pull per-country CLI)	W0	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
55	Conference Board LEI/US + world LEI historical	archive indicator	conferenboard.org (official, licensed); free partial series via FRED (USLEI)	Monthly	1959+ (US); later interna-tional	batch	FRED-part + licensed direct	mixed: FRED US-LEI free; full-international licensed	FRED partial cross-check only; PIT from CB release dates (action-note: FRED USLEI + USSLIND; CB if non-US LEI needed)	W1	quote	UNVERIFIED-CANDIDATE
56	NBER US business-cycle dates (peak/trough chronol-ogy)	archive reference	nber.org bcdc-page JSON/xlsx [6]	Monthly reference points	Dec 1854–present (32 cycles)	batch (committee-driven)	data.nber.org/cycles/ JSON API	free (as-is, attribution)	announcement-date (not turning-point-date) as PIT for “when was it known” vs “when it was” (action-note: JSON pull; freeze an-nounce-ment dates)	W0	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
57	Reinhart–Rogoff crisis chronology (historical currency/banking/debt crises)	stress archive	Harvard/PIIE hosted dataset page (Reinhart's crisis dataset page; canonical "external-banking-debt-crises" XLSX + IMF WP annex files)	Annual event table; 70 countries	1800–2016 (varies)	batch (static)	XLSX download from creator-affiliated post	free (academic; cite Reinhart & Rogoff)	cannot claim current-governance; update path = Laeven-Valencia (row 58) (action-note: locate canonical XLSX; parse events 1800–2020)	W1	connect	SURVEY-EVIDENCE
58	Laeven–Valencia Systemic Banking Crises Database	stress archive	IMF WP/20/112 annex dataset (4th ed 2020)	Annual events; 132 countries	1970–2018 (4th ed)	batch (static)	XLSX via imf.org article page	free (IMF)	event-code discipline; 5th-update watch (action-note: parse annex XLSX into crisis table)	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
59	IMF Financial Soundness Indicators (FSI)	stress archive	data.imf.org FSI dataset	Quarterly/annual	1990s+ (country varies), guidance-2000-era	batch	SDMX (row 19 route)	free w/ attribution	coverage-depth varies; compliance with IMF FSI guide quality flags (action-note: bulk FSI core set (NPL ratio, capital adequacy))	W2	connect	UNVERIFIED-CANDIDATE
60	GDELT (events, GKG, translingual, 1979+ full archive)	news/text archive	gdeltproject.org; BigQuery public dataset + raw file HTTP (data.gdeltproject.org) [5]	Event/article, 15-min update	Jan 1, 1979–present	near-realtime (this volume treats as batch historical sweeps)	BigQuery SQL + raw HTTP firehose	free (open data; attribution requested)	dedupe vs vol 08 real-time consumer; CAMEO coverage density pre-1990s thinner (action-note: BigQuery historical sweeps for W1 sample (cross-ref 08))	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
61	Common Crawl (quarterly web snapshots)	news/web text archive	commoncrawl.org; s3://commoncrawl/ + data.commoncrawl.org per-crawl indexes [11]	WebGraph, crawl	2008 Q3–present (~50+ crawls)	batch (quarterly)	S3/HTTP + CDX index; parallel compute via AWS Open Data Sponsorship	free (site-content rights vary — crawl-level free)	per-domain rights only apply downstream; pin crawl ID per research corpus (action-note: choose sample crawl for pilot text-measure (narrative-history proxy))	W2	connect	VERIFIED-UPSTREAM-DOC
62	Chronicling America (LoC historical US newspapers)	news/text archive	chroniclingamerica.loc.gov JSON/text API (bot-403 on root: use api path with UA)	issue/page OCR	1776–1963 (per digitization waves)	batch (static)	HTTP JSON (search API) + plain-text OCR	free (public domain)	OCR quality varies by era; accept per-page with OCR-confidence flag (action-note: targeted fetch for event-anchored narrative windows (~12 events))	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
63	Wayback Machine (CDX API snap-shots)	web/archive	web.archive.org CDX API (`/cdx?url=…&output=json`)	Snapshot times-tamps	1996–present	batch	REST CDX + WARC download	free (Internet Archive terms)	snapshot-density skew post-2015; capture-status (4xx/redirect) flags (action-note: CDX for pilot-firm domains (e.g., annual-report URLs))	W1	backfill	SURVEY-EVIDENCE
64	Internet Archive books/texts scholar collec-tions	text archive	archive.org ad-vanced-search API + item-download	Page-text (books/journals)	Pre-20th-century–present	batch	REST search API + HTTPS files	free (most items public-domain or IA lending)	inbound license bound-aries per item; OCR con-fidence noted (action-note: only for GFI-narrative feature search (pre-internet events))	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	Nikkei 225 long-run index archive	dormant/dead venue	indexes.nikkei.co.jp historical “long-term data” XLSX (base 1986 adjusted for returns to 1949)	Daily (pre-1960s recon-structed monthly in official down-load)	1949–present	batch	portal XLSX download	free for research w/ attri-bution (Nikkei Inc. dis-claimer)	splice: index-methodology changes 1970 (50→225) & 2010s changeovers; distribute via download-snapshots (action-note: pull official long-history XLSX; docu-ment method-change dates)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
66	Berlin Stock Exchange historical (dead venue)	dormant	pre-1945 Berlin Börse indices/turnover via academic digitization (FU Berlin / digitized course-of-the-day bulletins) — no official archive	Annual/monthly	1870s–1944	batch	digitized journal/PDF scans	free (scan archives) w/ citation	high manual-friction (bulletin scans where no digital index); defer if GFD (row 6) covered (action-note: scoped test: 1870s–1913 aggregate share-price index only)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
67	DNB/De Neder-landsche Bank historical money & debt stats	dormant/central-bank-archive	dnb.nl historical statistical publica-tions (Dutch guilder era scans)	Monthly/annual	1860s–2014 guilder-era	batch	portal XLSX/PDF	free (govt)	OCR + Dutch-language headers; prefer JST/CLIO where overlap-ping (action-note: pull guilder-era money-credit set if Nether-lands pilot)	W2	connect	UNVERIFIED-CANDIDATE
68	RBA 150-year statistics archive? Bank of Australia? RBA 2000-series historical	central-bank-archive	rba.gov.au “Measur-ing the size of the econ-omy” legacy — concrete: RBA statistical historical-150-year series XLSX (2021 release)	Annual/monthly	1901–2021	batch	XLSX direct	free (govt)	splice notes native in publica-tion; pre-1940 thin series flagged (action-note: ingest 150-year historical XLSX (Aus-tralia))	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
69	Danmarks Nationalbank historical statistics (pre-1950 stat-archive/“data on central interest 1824”)	central-bank-archive	nationalbanken.dk historical statistics page (long-run rates & credit series incl. Abildgren backfills to 1818)	Monthly/annual	1824–present (key rates series)	batch	XLSX download + published research-doc series	free (govt)	Abildgren-construction notes binding; document overlaps vs JST Denmark rows (action-note: pull Danmarks historical long-rate file (Abildgren 2020 update))	W2	connect	UNVERIFIED-CANDIDATE
70	Norges Bank historical monetary statistics	central-bank-archive	norges-bank.no/en/about/statistics/historical-monetary-statistics	Annual/monthly	1819–present (nominal series); FX & capital flows back to 1938	batch	XLSX direct	free (govt)	series-rebasing listed in historical-statistics com-pendium (action-note: pull kronie money+FX block)	W2	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
71	BoJ long-run historical statistics (BOJ “100-year” historical stats)	central-bank-archive	boj.or.jp/en/statistics/stat_history/classics archive (Historical Statistics, hundred-year series PDF/CGHTML)	Annual/step	1885/1886–2000 (hundred-year vol), post-2000 in main DB	batch	PDF/HTML transcribe + BOJ time-series lookup (BOJ main stats DB continues)	free (govt)	typo-reviewed transcription; weigh vs JST Japan rows (action-note: extract M2/long-rates columns from hundred-year table)	W2	connect	UNVERIFIED-CANDIDATE
72	Dead/delisted asset archive: CoinMetrics Community (free CSV incl. delisted assets incl. Mt.-Gox-era data)	dead-asset record	github.com/coinmetrics/data free CSV incl. delisted/dead markets	Daily	2009/2010–present (per asset)	batch (daily)	GitHub-HTTPS raw CSV	free (CC BY 4.0)	delisted assets freeze at last-print date — must not forward-fill; exchange-coverage flags per asset (action-note: pull CSV; split live vs delisted universe (cross-ref 07 for dex-live-live))	W0	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
73	Dead-coin/delisted marker: CoinGecko dead status + delisted-event history	dead-asset record	api.coingecko.com (public end-points; includes delisted coins history where rate-limited)	Daily + status flag	2013–now (per asset varies)	near-realtime (treat batch curate weekly)	REST (public, no key)	free (at-tribution, rate-limited)	delist-date cannot guaran-tee last-price quality (thin tape) — pair with row 72 oracle (action-note: weekly delist-flag delta; archive removed assets)	W1	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
74	CoinMarketCap historical archive dumps (early CSV snapshots incl. dead coins)	dead-asset record	community-crafted dumps (CMC 2013–2015 era official snapshot CSVs; CMC discontinued free full backfill long ago)	Daily	2013–2021 (community-file dependent)	batch (static archive)	download archive CSVs	license-risk (CMC data not redistributable freely) → use for PIT research only	snapshot completeness verified by row-count-per-date sanity; do not publish externally (action-note: acquire community dumps; internal-only use; no publication)	W2	connect	UNVERIFIED-CANDIDATE
75	DeadCoins.com / delisted-project registry	dead-asset record	deadcoins.com catalog (scrapeable) + cross-ref modest GitHub “deadcoin-dataset” mirrors	Event/record	2013–2022 (site state varies)	batch (static scrape)	HTTP scrape → CSV	attribution-needed (site unclear OTRL) — treat reference-only	site abandoned risk = archive.org mirror WARC as fallback (row 63 route) (action-note: archive.org WARC scrape into registry table)	W3	connect	UNVERIFIED-CANDIDATE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
76	Delisted ex-change/failure record archive (exchange-collapse dataset: Mt.Gox 2010–2014, Cryptsy 2014–16, Fcoin 2019)	dead-venue record	open-source recon: Mt.Gox recovered-trades commu-nity releases (github MtGox-recovered datasets) + published post-mortem/FMA-recovery reports	Tick/daily	2010–2019 era-specific	batch (static)	GitHub/archive release	free (forensic-community releases)	recovered-trade complete-ness labels; volume excluded from live-market aggre-gates (action-note: pull Mt.Gox recovery block; docu-ment coverage fraction)	W2	connect	UNVERIFIED-CANDIDATE

13 — Prediction markets (7 rows)

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	Polymarket market metadata + prices	prediction	Gamma API `gamma-api.polymarket.com/markets` + `/events` (S43 fetch 200 OK; alphalens spike-proven)	market/outcome snapshot	launch 2020→	minutes (poll)	REST polling	free (per site ToS; display-derived only)	outcome-token splits re-denominate markets — key on condi-tionId not slug; mar-ket→event 1:N	W1	connect	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
2	Polymarket CLOB order-book + trades	prediction	CLOB API `clob.polymarket.com` book + trades (alphalens spike-proven: 200)	book/tick	2020→ prices/markets/trades	poll (no free WS guarantee)	REST polling + own-spool	free (ToS)	thin long-tail books — store best-N levels only; resolution disputes move prices post-hoc	W1	connect	VERIFIED-UPSTREAM-DOC
3	Kalshi event-contract market data	prediction	Kalshi Trade API v2 `api.elections.kalshi.com/trade-api/v2` markets/trades/tickers (S43 rate-limited this census; alphalens probed)	contract/tick	2021→	poll	REST (key optional for reads)	free-read (trading needs account)	CFTC-regulated settlement = clean resolution truth; our IP got 429 — polite-poll cadence mandatory	W1	verify	SURVEY-EVIDENCE
4	Manifold Markets play-money forecasts	prediction	Open API `api.manifold.markets` + bets + positions	market/user forecast	2021→	near-realtime	REST	free (open API)	play-money ≠ real odds — calibration signal only; Brier scoring needs resolution labels	W2	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	Metaculus quantitative predictions	prediction	API2 `api2.metaculus.com` questions + community forecasts	question/forecast time-series	2015?→now (verify earliest)	daily-ish	REST (API key pending)	free w/ key	community vs author probabilities differ — store both; closed questions = resolution set	W2	verify	UNVERIFIED-CANDIDATE
6	PredictIt archive (retired)	prediction (archive)	predictit.org market files + web.archive.org CDX of market pages	market/day	2014?→retired (verify exact end)	n/a (frozen)	S3-archive walk	free (archive content)	retired venue — historical-only; fork (Manifold/Polymarket) continuity is NOT stitchable	W3	backfill	UNVERIFIED-CANDIDATE
7	India-relevant prediction-market odds	prediction	cross-venue roll-up from rows 1–5 filtered to India events (elections, RBI/rate, cricket World Cup, monsoon)	event/day	venue-limited	poll	derived view over rows 1–5	free (derived)	venue event-ID mapping is manual — keep venue-native IDs; odds ≠ realized probability (calibrate vs outcomes)	W2	build-parser	SURVEY-EVIDENCE

14 — India depth (alphalens gap-closure) (89 rows)

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	NSE 52-week highs/lows + price bands	market structure	nseindia.com all-reports band files (alphalens in_004)	daily per scrip	regime-dated; own-capture forward	EOD	file-drop	free	band-change events PIT per session; T2T vs normal series tags	W1	build-parser	SURVEY-EVIDENCE
2	NSE SLBS securities-lending volumes	flow	NSE SLBS daily reports (alphalens in_012)	daily aggregate + per scrip	depth to sample	EOD	file-drop	free	borrower/lender category buckets; haircut changes break series	W1	verify	SURVEY-EVIDENCE
3	NSE SME (Emerge) + BSE SME platform stats	universe	exchange SME report pages (alphalens in_014)	daily/monthly	SME-regime 2012→ approx	EOD	file-drop	free	SME ≠ main-board liquidity — never mix in main-board breadth stats	W2	verify	SURVEY-EVIDENCE
4	NSE index reconstitution press releases	corporate event	nseindia.com index press page (alphalens in_016)	event rows	index-history-long	event-driven	file-drop	free	announcement→effective lag = tradeable window; store both dates	W1	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	InvIT/REIT unit disclosures + metrics	corporate event	NSE/BSE InvIT-REIT sections (alphalens in_017)	per-filing + daily unit data	regime 2014→ approx (InvIT) / 2007→ (REIT)	EOD/batch	file-drop	free	distribution-yield definitions differ per trust — store trust methodology	W1	verify	SURVEY-EVIDENCE
6	NSE currency-derivative (CD) bhavcopy	derivative price	nseindia.com CD reports (alphalens in_018)	daily bar per contract	CD-regime 2008?→ to verify	EOD	file-drop	free	USD/INR dominance; expiry-calendar changes across regimes	W1	backfill	SURVEY-EVIDENCE
7	NSE commodity segment (NSE-COM) reports	derivative price	nseindia.com commodity reports (alphalens in_019)	daily bar	segment launch→	EOD	file-drop	free	thin segment — zero-volume days are real, not gaps	W2	verify	SURVEY-EVIDENCE
8	SGB secondary-market premium/discount	flow/valuation	exchange SGB series + RBI issue data (alphalens in_021)	daily per series	SGB era 2015→	EOD	file-drop	free	premium decays to gold-spot near maturity — model per series; series-ID mapping required	W2	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
9	Sovereign green-bond auctions	rates/policy	RBI auction results green tranche (alphalens in_022)	auction	green-bond era 2023→	auction-day	file-drop	free	twin-auction pricing vs conventional G-sec = spread signal	W2	verify	SURVEY-EVIDENCE
10	Combined-exchange OI report (NSE+BSE)	flow	exchange combined OI disclosure (alphalens in_031)	daily	recent regime	EOD	file-drop	free	combined vs single-exchange OI must not be stitched without regime flag	W1	verify	SURVEY-EVIDENCE
11	NSDL fortnightly FPI sector-wise holdings	flow	nsdl.com FPI reports (alphalens in_032)	fortnight × sector	FPI-regime 2019?→ (post-FEMA revamp)	fortnightly	file-drop	free	FPI→FPI-category renumbering 2019 — regime tag mandatory; sector taxonomy vintages	W1	backfill	SURVEY-EVIDENCE
12	Per-stock FII headroom disclosure	risk	exchange FII-headroom files (alphalens in_033)	daily per scrip	regime-dated	EOD	file-drop	free	headroom hits 0 = binding constraint event; MWPL cross-check	W1	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
13	Demat account additions (NSDL+CDSL)	flow/retail	nsdl + cdsl monthly stats (alphalens in_034)	monthly per depository	demat-boom 2015?→	monthly	file-drop	free	monthly net adds ≠ active accounts; KYC-rule changes shift series (Jan-2023 SHCIL halt documented)	W0	backfill	SURVEY-EVIDENCE
14	Broker-wise active clients (monthly mandate)	flow/retail	SEBI monthly broker-client disclosure (alphalens in_035)	monthly × broker	mandate-era 2018?→	monthly	file-drop	free	per-broker dedupe across NSE+BSE registrations; round-trip caps distort	W1	backfill	SURVEY-EVIDENCE
15	MF folio additions (equity vs debt split)	flow	AMFI monthly fact-sheet+statistical report (alphalens in_036)	monthly × category	depth to sample	monthly	file-drop	free	folio ≠ AUM; category reclassifications mid-series	W1	backfill	SURVEY-EVIDENCE
16	ETF AUM/flows + iNAV premium	flow	exchange ETF pages + AMFI (alphalens in_039)	daily iNAV; monthly AUM	ETF-era 2008?→	daily/monthly	file-drop	free	iNAV = indicative, not executable; premium spikes at close are artifacts	W2	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
17	PFRDA/NPS pension AUM + sub-scribers	flow	prda.gov.in + nsdl cra reports (al-phalens in_041)	monthly	NPS era 2004?→	monthly	file-drop	free	subscriber counts include exited accounts; scheme-tier splits	W1	backfill	SURVEY-EVIDENCE
18	IRDAI insurer premium + persis-tence	flow	irdai.gov.in annual + monthly state-ments (al-phalens in_042)	monthly/yearly × insurer	irdai digital era	monthly	file-drop	free	premium ≠ profit; renewal-persistence defini-tions per regulator vintage	W1	backfill	SURVEY-EVIDENCE
19	SEBI PIT insider-trading disclo-sures (SAST)	ownership	sebi.gov.in PIT filings (al-phalens in_046)	per-filing	SAST-regime 2002?→	T+2 publish	file-drop	free	trading-window rule changes; demat-pledge encum-bered sales misread as exits	W1	backfill	SURVEY-EVIDENCE
20	Credit-rating actions + rationale PDFs	corporate event	CARE/CRISIL/ICRA/SBIL + BRWR pages (al-phalens in_047)	per-action	rating-agency digital era	T+0-ish	file-drop + PDF	free	rating floor changes (SEBI scale revamps 2010s) — never stitch across scales; outlook vs action	W1	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
21	Auditor changes / resignations / qualified opinions	governance	MCA + exchange filings (alphalens in_048)	per-event	depth to sample	T+1	file-drop	free	qualified-opinion flag lives in financial-assertion PDFs — requires text parse; churn = distress signal	W1	build-parser	SURVEY-EVIDENCE
22	KMP exits (CEO/CFO/CS changes)	governance	exchange announcements + MCA DIN filings (alphalens in_049)	per-event	digital era	T+0	file-drop	free	name→DIN entity resolution; interim vs permanent appointments	W1	build-parser	SURVEY-EVIDENCE
23	ESOP grant/vesting filings	ownership	exchange filings + annual-report annexures (alphalens in_050)	per-event	ESOP-regime	T+1	file-drop	free	grant ≠ vest ≠ exercise — model 3 event types; dilution math per series	W2	build-parser	SURVEY-EVIDENCE
24	Buyback tenders + results	corporate event	exchange buyback pages (alphalens in_051)	per-buyback + tender window	buyback-regime 2000s→	event	file-drop	free	tender-vs-market route; acceptance-ratio math; buyback-entitlement date joins	W1	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
25	Reg-23 related-party transaction disclosures	governance	annual reports + quarterly RPT filings (alphalens in_052)	per-filing	LODR-regime 2015?→	quarterly	file-drop + PDF	free	RPT materiality thresholds revised 2021+ — threshold regime tags; arm's-length flag text	W2	build-parser	SURVEY-EVIDENCE
26	SEBI enforcement orders database	regulatory	sebi.gov.in orders page (alphalens in_056)	per-order	SEBI orders digital era 2000s→	weekly-ish	file-drop + PDF	free	order→entity resolution needed; appeal outcomes post-order	W1	build-parser	SURVEY-EVIDENCE
27	NCLT/CIRP insolvency process states	legal/corporate	nclt.gov.in orders + ibbi.org.in (alphalens in_058)	per-case state	IBC era 2016→	weekly	file-drop + PDF	free	CIRP stage machine (admission→CoC→plan→liquidation) — store state transitions as events; order PDFs are truth	W1	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
28	eCourts judgments entity-matched	legal	ecourts.gov.in + High Court portals (alphalens in_059)	per-case	eCourts digital era	weekly-ish	portal self-capture	free (portal ToS check)	case-number reuse across courts; party-name entity resolution is the hard part; scrape-politeness	W2	build-parser	SURVEY-EVIDENCE
29	RBI weekly credit/deposit growth	macro	rbi.org.in weekly statistical supplement (alphalens in_065)	weekly	deep weekly archive	weekly	file-drop	free	fortnightly-vs-weekly rhythm; festival-shifted reporting weeks	W1	backfill	SURVEY-EVIDENCE
30	CCIL yields/volumes + FBIL benchmarks	rates	ccil.in + fbil.org.in (alphalens in_068)	daily benchmarks	CCIL digital era	EOD	file-drop	free	FBIL fix methodology revisions (0900/1430/1600 cuts); GCA vs settlement volumes	W1	connect	SURVEY-EVIDENCE
31	G-sec auction results + curve	rates	rbi.org.in auctions (alphalens in_069)	per-auction	auction archive deep	auction-day	file-drop	free	cut-off vs notified amount; non-competitive bids; auction calendar shifts	W1	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
32	CGA central accounts (capex/receipts)	macro/fiscal	cgaindia.nic.in monthly state-ments (al-phalens in_070)	monthly	FY-digital era	monthly	file-drop + PDF	free	FY vs calendar tagging; revised-vs-provisional flags per release	W1	build-parser	SURVEY-EVIDENCE
33	CBDT direct-tax collec-tions	macro/fiscal	incometax.gov.in + CBDT releases (al-phalens in_071)	monthly × tax head	digital era	monthly	file-drop	free	gross vs net; refunds-netted series; FY pacing noise	W1	backfill	SURVEY-EVIDENCE
34	GST col-lections (state × month)	macro/fiscal	gstcouncil.gov.in + cbic releases (al-phalens in_072)	monthly × state	GST era 2017-07→	monthly	file-drop	free	gross-domestic vs IGST-import splits; rate-rationalization break-points (Sep-2019)	W1	backfill	SURVEY-EVIDENCE
35	Economic Survey corpus (annual PDF)	macro/doc	indiabudget.gov.in (al-phalens in_073)	annual docu-ment	1950s→ (digital copies ~2010→)	annual	file-drop	free (Govt of India)	one-shot per year; chapter text revisions between in-terim/final prints	W1	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
36	Union Budget document corpus	macro/doc	indiabudget.gov.in full budget + annexures (alphalens in_074)	annual document set	digital era 2010s→ (older via archive)	annual	file-drop	free (Govt of India)	interim vs full budget per FY; amendment finance bills; speech-vs-text deltas	W1	backfill	SURVEY-EVIDENCE
37	State budget documents	macro/doc	state finance portals (alphalens in_075)	annual × state	2010s→ varies per state	annual	portal self-capture	free (state gov)	FRBM classification changes per state; state fiscal-year tagging	W2	build-parser	SURVEY-EVIDENCE
38	RBI FSR + Trend & Progress of Banking	macro/doc	rbi.org.in publications (alphalens in_076)	half-yearly / annual	FSR digital era	half-yearly	file-drop	free	financial-stability narrative = regime label source; bank-sample changes (NPA cycles)	W2	backfill	SURVEY-EVIDENCE
39	CAG audit reports (union + state)	governance	cag.gov.in (alphalens in_077)	per-report	CAG digital era	periodic	file-drop + PDF	free (CAG)	paragraph-level extraction needed; audit→allegation lag	W2	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
40	NITI Aayog indices (SDG, MPI)	macro	niti.gov.in (al-phalens in_078)	annual × state/district	index-vintage only	annual	file-drop	free	index method-ology rebasing across editions — store per-edition, never stitch	W2	verify	SURVEY-EVIDENCE
41	PRS Leg-islative Research bill tracker	policy	prsindia.org + sansad.in (al-phalens in_079)	per-bill stage	PRS digital era	near-realtime	REST-ish + file-drop	free (PRS attribu-tion)	bill num-bering resets per session; standing-committee stage dates	W1	connect	SURVEY-EVIDENCE
42	SEBI con-sultation papers + final circulars	regulatory	sebi.gov.in consult pages (al-phalens in_080)	per-document	digital era	weekly	file-drop	free	paper→circular pairing = policy-reaction event pair; comment-window close date	W2	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
43	DPIIT startup recogni-tions + FDI inflows (sector × country)	macro/policy	dpiit.gov.in + rbi FDI reports (al-phalens in_081, in_082)	monthly FDI × sector-country; recogni-tion stock	FDI digital era	monthly	file-drop + PDF	free	press-note policy changes (2020 CASO carve-outs) = regime tags; route vs guess counts	W1	build-parser	SURVEY-EVIDENCE
44	EPFO payroll additions	macro/labor	epfindia.gov.in monthly press releases (al-phalens in_083)	monthly	EPFO digital era 2018?→	monthly	file-drop	free	establishment-count growth ≠ new jobs; re-subscription vs first-time split (2022+ split pub-lished)	W1	backfill	SURVEY-EVIDENCE
45	Naukri JobSpeak hiring index	macro/labor	naukri.com JobSpeak monthly PDF (al-phalens in_084)	monthly × sec-tor/city	JobSpeak era 2008?→	monthly	file-drop	free (pub-lisher terms)	panel-based index from one job-board — non-census; rebasing events; cross-check EPFO/PLFS	W1	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
46	PLFS employment-unemployment survey	macro/labor	mospi.gov.in PLFS annual + usual-status CWS (alphalens in_085)	annual × state/sector	PLFS 2017-18→ (NSS 68th superseded)	annual	file-drop	free (MoSPI)	PLFS ≠ NSS-68th basis — never splice; CWS vs UPS definitions	W1	backfill	SURVEY-EVIDENCE
47	NPCI UPI volumes + bank-wise share	macro/payments	npci.org.in transactions page (alphalens in_086)	monthly (+bank share)	UPI era 2016→	monthly (live counter volatile)	portal self-capture	free (NPCI terms)	counter values revised; bank-share restatements; value vs volume	W1	build-parser	UNVERIFIED-CANDIDATE
48	Grid-India 15-min all-India demand CSV	power/nowcast	grid-india.in reports (alphalens in_087; V1-verified 15-min CSV exists)	15-min × region	digital era, depth to sample	15-min	file-drop	free	regional-balance vs all-India; holiday dips real; despatch vs consumption	W0	backfill	SURVEY-EVIDENCE
49	PFC discom finances annual dataset	power	pfc.org.in DISCOM reports (alphalens in_090)	annual × discom	DISCOM dataset era	annual	file-drop	free (PFC)	AT&C loss definitions revised; discom-ID churn; state-vs-utility mapping	W2	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
50	IMD gridded rainfall 1901+ + AWS stations	weather	mausam.imd.gov.in + metgrid (al-phalens in_093)	grid/daily; sta-tion/hourly	gridded 1901→	days	file-drop	free (IMD)	gridded-product version shifts; AWS station-network churn; QA-flag columns	W1	backfill	SURVEY-EVIDENCE
51	CPCB AQI per city + GRAP stages	environment	cpcb.nic.in AQI bulletins (al-phalens in_095, in_096)	city/hourly + stage events	AQI era 2014→	hours	portal self-capture + file-drop	free	station additions mid-series; GRAP stage = event calendar for Delhi pollution trades	W1	build-parser	SURVEY-EVIDENCE
52	CWC reservoir storage (150+ reser-voirs)	water/power	cwc.gov.in weekly reservoir bulletins (al-phalens in_097)	reservoir/weekly	bulletin era	weekly	file-drop	free	live vs avg storage; monsoon-season tagging; reservoir set churn	W1	build-parser	SURVEY-EVIDENCE
53	Flood forecasts + cyclone warnings	weather/risk	CWC flood forecasts + IMD cyclone bulletins (al-phalens in_098, in_099)	event/alert	bulletin era	event-driven	portal self-capture	free	alert-grade escala-tion events; forecast-track revisions store first-seen	W2	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
54	Soil-moisture satellites + CGWB groundwater	water	SMAP/ESA CCI + cgwb.gov.in (alphalens in_101, in_102, in_160)	grid/3-day (sat); district/year (GW)	SMAP 2015→; GW rounds	days–annual	file-drop	free (NASA/ESA/GOVT)	SMOS vs SMAP product IDs; groundwater round cadence irregular; irrigation-season tagging	W2	build-parser	SURVEY-EVIDENCE
55	AGMARKNET mandi prices + arrivals	agri/nowcast	agmarknet.gov.in daily 3000+ mandis (alphalens in_103; parser-sprint target #1)	commodity × mandi/day	digital era, depth to sample	daily	file-drop + scrape	free (Agmarknet)	arrival = volume not price; quality-variant grades; mandi-ID churn; duplicate rows across reports	W1	build-parser	SURVEY-EVIDENCE
56	e-NAM integrated market trades	agri	enam.gov.in (alphalens in_104)	trade × mandi/day	e-NAM era 2016→	daily	portal self-capture	free	e-NAM subset of mandis — never treat as national; traded-vs-offered	W2	verify	SURVEY-EVIDENCE
57	Weekly sowing progress (state × crop)	agri/nowcast	agri.nic.in + ministry dashboards (alphalens in_105)	state × crop/weekly	sowing season era	weekly (kharif/rabi)	file-drop	free	sowing ≠ harvest; area-revision mid-season; kharif/rabi calendars	W1	backfill	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
58	MSP notifications (kharif/rabi)	agri/policy	agri.gov.in MSP pages (alphalens in_106)	crop × season	deep archive	annual (event)	file-drop	free	announcement-date PIT; effective-vs-announced pricing; A2+50% formula shifts	W1	backfill	SURVEY-EVIDENCE
59	Fertilizer sales (DoF monthly)	agri	fertilizer.gov.in (alphalens in_107)	month × state × product	digital era	monthly	file-drop	free	pant-season year (Oct–Sep) ≠ calendar; subsidy-rate regime changes	W2	build-parser	SURVEY-EVIDENCE
60	FCI food-grain stocks (central pool)	agri	fci.gov.in stock reports (alphalens in_108)	weekly/monthly	digital era	weekly	file-drop	free	buffer norms revised 2013+; season-tagged (Oct onset); state-wise split	W1	backfill	SURVEY-EVIDENCE
61	Daily retail prices of 22 essentials □	agri/nowcast	consumer affairs price monitoring (alphalens in_114; parser-proof committed)	commodity × city/day	digital era	daily	file-drop	free	parser already proven in alphalens sprint — port their extractor; city-set churn	W1	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
62	CEA daily coal stock per thermal plant □	energy/nowcast	cea.nic.in daily coal-stock reports (alphalens in_115; PDF spike proven)	plant/day	spike-era 2010s→	daily	file-drop (PDF/xlsx)	free (CEA)	norm-days math (15/20/25 by plant class); PDF table extractor needed; plant-ID mapping	W1	build-parser	SURVEY-EVIDENCE
63	PPAC fuel consumption + crude imports by source	energy/trade	ppac.gov.in monthly (alphalens in_118, in_131)	month × product × source-country	PPAC digital era	monthly	file-drop	free	source-country switch events (Russia flows 2022+) are the signal; SKU definitions	W1	build-parser	SURVEY-EVIDENCE
64	DGH crude/gas field production	energy	dgh.gov.in monthly (alphalens in_120)	field/month	digital era	monthly	file-drop	free	field-ID mapping; ODL vs gross; block transfers break continuity	W2	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
65	PNGRB city-gas network + CNG/PNG connec-tions	energy	pngrb.gov.in (al-phalens in_121)	network/month	PNGRB digital era	monthly	file-drop	free	network-extension phase tagging (2018–2030 phases); connec-tion vs consump-tion	W2	verify	SURVEY-EVIDENCE
66	Toll revenue plaza × conces-sionaire □	logistics/nowcast	ihmcl.gov.in NHAI toll disclo-sures (al-phalens in_122; format-proven)	plaza × month	concession-era 2010s?→	monthly	file-drop (PDF/xlsx)	free	HAM vs BOT con-cession mix; COVID-2020 toll waiver = docu-mented gap; plaza additions mid-series	W1	build-parser	SURVEY-EVIDENCE
67	Major-port cargo traffic (IPA daily)	logistics/nowcast	indianports.gov.in IPA daily (al-phalens in_123; table-scrape proven)	port/day	digital era	daily	scrape + file-drop	free	cargo ≠ container TEU — keep separate series; monsoon seasonal-ity	W1	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
68	DGCA airline traffic by carrier	logistics/nowcast	dgca.gov.in monthly PDFs (alphalens in_125)	carrier × month	digital era	monthly	file-drop (PDF)	free (DGCA)	PDF extractor templates; carrier-code churn (Jet 2019 exit = structural break); ASM vs RPK vs passengers	W1	build-parser	SURVEY-EVIDENCE
69	AAI airport passengers + cargo	logistics/nowcast	aai.aero monthly (alphalens in_126)	airport × month	digital era	monthly	file-drop	free	airport reclassification (international status changes); COVID gap real	W2	build-parser	SURVEY-EVIDENCE
70	Railway freight loading	logistics/nowcast	indianrailways monthly stats (alphalens in_127)	commodity/zone × month	digital era	monthly	file-drop	free	freight-rate revamp 2015+; zone reorganizations; container-vs-break-bulk splits	W1	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
71	VAHAN vehicle registrations + EV dashboard	logistics/now-cast	parivahan.gov.in VAHAN (alphalens in_128)	category × state × month/week	VAHAN digital era	weekly/monthly	portal self-capture	free (VAHAN ToS check)	registration ≠ sales (dealer vs RTO lag); BS-IV→BS-VI switch Apr-2020 = break; EV split era 2019→	W1	build-parser	SURVEY-EVIDENCE
72	Metro ridership (7 metros)	logistics/now-cast	metro rail corporation stats (alphalens in_129)	metro × month	line-openings era	monthly	portal self-capture	free	new-lines create fake growth — per-line tagging; COVID gap real	W2	build-parser	SURVEY-EVIDENCE
73	US FDA import refusals + 483s (Indian plants)	pharma-regulatory	FDA DUINS/RII + import-alert queries (alphalens in_132; V1 query-builder proven)	per-event	FDA digital era	daily-ish	REST (openFDA-class) + file-drop	free (US gov)	firm-name entity resolution (Indian plant names in romanization/transliteration); 483 ≠ refusal — separate series	W1	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
74	DGTR anti-dumping / safe-guard cases	trade-regulatory	dgtr.gov.in case search (al-phalens in_133)	per-case	DGTR digital era	weekly-ish	portal self-capture	free	case→product HS mapping; duty-levy vs case-filing events; sunset reviews	W2	build-parser	UNVERIFIED-CANDIDATE
75	RoDTEP/RoSCTL incentive rates	trade/policy	dgft.gov.in RoDTEP sched-ules (al-phalens in_134)	HS × rate schedule	RoDTEP era 2021→	periodic	file-drop	free	rate sched-ules revised per review cycle — store schedule-vintage; scrip vs cash modes	W2	verify	SURVEY-EVIDENCE
76	BIS QCO import-restriction tracker	trade/regulatory	bis.gov.in + dgft QCO noti-fications (al-phalens in_135)	per-QCO × HS	QCO wave 2020s→	event	file-drop	free	notification→effective-date lag; scope creep per sector	W2	build-parser	SURVEY-EVIDENCE
77	Export councils monthly (GJEPC/Pharmexcil/EEPC/APEDA…)	trade/nowcast	council monthly releases (al-phalens in_136)	council × month	digital era	monthly	portal self-capture	free	per-council HTML/PDF hetero-geneity — 6 parser families; gem/jewellery volatility	W2	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
78	RBI services exports (quarterly)	macro/trade	rbi.org.in BoP services (alphalens in_137)	quarterly	BoP digital era	quarterly	file-drop	free	BoP revisions (2013 India-usage vs US-usage basis note); software vs other-services	W2	verify	SURVEY-EVIDENCE
79	India earnings-call transcripts (LODR-mandated)	news/corpus	exchange audio + LODR transcripts (alphalens in_141)	per-call	LODR era 2018?→	hours	file-drop + self-transcribe	free / self-derived	LODR mandates calls — audio archived by exchanges; ASR quality per language; Q&A segment splits	W1	build-parser	SURVEY-EVIDENCE
80	Investor presentations + annual-report library	corpus	exchange corp-interview pages + company IR (alphalens in_142, in_143)	per-doc	digital era	T+1	file-drop + PDF	free (company filings)	presentation decks are unaudited — flag vs financials; version churn post-IR edits	W1	build-parser	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
81	Board-meeting / results calendar	corporate event	exchange board-meeting pages (alphalens in_145)	event rows	digital era	near-realtime	file-drop	free	outcome (approved/postponed) arrives after date — store both; extended-date events	W0	connect	SURVEY-EVIDENCE
82	Reddit investing-sub senti-ment	attention	PRAW pull r/IndianStreetBets + r/IndiaInvestments (alphalens in_149)	post/thread hourly	sub-era 2015?→	hourly	REST (PRAW; API terms)	free (Reddit API terms; rate-capped)	bot/spam share; API pricing shock 2023 — terms can change; PII retention policy needed	W2	connect	SURVEY-EVIDENCE
83	YouTube earnings-call video attention	attention	YouTube Data API v3 (alphalens in_154)	video/day	API-era	daily	REST (key; quota-capped)	free (API terms; quota)	view counts re-vised/removed; comment API terms; quota 10k units/key/day	W3	connect	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
84	TROPOMI NO2 / methane plumes (refineries, ports)	satellite/energy	Sentinel-5P via CDSE/Tropomi (alphalens in_158)	grid/3-day	2018→ (S5P launch)	days	STAC / file-drop	free (Copernicus)	overpass-filter (cloud); reprocessing-version pins; refinery-activity proxy only	W2	build-parser	SURVEY-EVIDENCE
85	Reservoir / waterbody extent change	satellite/water	JRC Global Surface Water + Sentinel-1 (alphalens in_161)	grid/monthly	JRC 1984→	monthly	file-drop	free (EC JRC)	JRC extent vs CWC gauge = two different truths — keep separate; monsoon-seasonal	W2	build-parser	SURVEY-EVIDENCE
86	Seismic events, India region	macro/risk	USGS + NCS (alphalens in_164)	event	deep archive	near-realtime	REST	free (USGS public-domain)	magnitude/depth quality flags; aftershock clustering; NCS-vs-USGS dedupe	W2	connect	SURVEY-EVIDENCE
87	NOTAMs + GNSS-jamming advisories (aviation risk)	risk/aviation	notams.aim.gov.in + NAV CANADA-class mirrors (alphalens in_166)	advisory	digital era	hours	portal self-capture	free (ICAO terms)	NOTAM decode formats; jamming-event dedupe across issuers	W3	verify	SURVEY-EVIDENCE

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
88	India news corpus + ticker senti-ment (6-year HF set) □	news/corpus	HuggingFace `mveen3/Sixty_Year_Indian_Stock_Market_Dataset-News_and_Ticker` — 4.2 GB, MIT ([HF-s43])	article × ticker	~6y window (verify exact span on pull)	daily-batch	HF datasets clone (parquet)	MIT (dataset card; source-article ToS sits up-stream; dedupe wire copies; ticker-tagging accuracy unknown — sample-audit before training; PIT = article publish times-tamp (first-seen)	source-article copyright still third-party — internal use, no redistri-bution)	W0	backfill	VERIFIED-UPSTREAM-DOC
89	GDELT news tone 15-min (cross-check rail)	news	GDELT DOC API (al-phalens in_152; global rows exist in TODO)	15-min tone	GDELT 2.0 2015→	15-min	REST	free (GDELT terms)	tone is model-derived; language-filter errors; cross-ref to existing GDELT row in sheet 09 — dedupe on ingest	W1	verify	SURVEY-EVIDENCE

15 — External resources (S43 intake) (18 rows)

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
1	Jhanvi NSE/BSE 1-minute bars	price/trades	HuggingFace `Jhanvi-29112000/indian-stock-market-minute-data` — 10.5 GB snappy parquet (125 GB raw), MIT ([HF-jhanvi])	1-min bar × symbol	2022 → 2026-10 (rolling; verify exact start on pull)	daily-batch (mirror)	HF datasets clone → DuckDB/Parquet store	MIT (dataset card; underlying exchange data ToS still applies to redistribution — internal use)	1-min aggregation convention (bar close vs tick) unknown — sample vs True-Data/Kite on 1 overlapping day; symbol-set churn; delisted coverage unknown	W0	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
2	Jhanvi NSE/BSE daily OHLCV	price/trades	HuggingFace same-repo daily partition — 2000 → 2026 ([HF-jhanvi])	daily bar × symbol	~2000 → 2026	daily-batch	HF clone → store	MIT	survivorship unknown (delisted names?); adjust-ment conven-tion unknown (raw vs split-adjusted) — verify vs NSE bhavcopy row 01 on 5 sample days before use as spine	W0	backfill	VERIFIED-UPSTREAM-DOC
3	Vishnu Indian market daily OHLCV	price/trades	HuggingFace `vishnun0027/indian-market-historical-ohlcv` — 290.1 MB, MIT ([HF-vishnu])	daily bar × symbol	multi-year (verify exact span on pull)	daily-batch	HF clone → store	MIT	smaller daily universe — cross-check coverage vs row 2 (union both, prefer official bhavcopy where overlap-ping)	W0	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
4	India index-options 1-minute	derivative price	HuggingFace `thetrademarkk/india-index-options-1m` — 4 GB, CC-BY-NC 4.0 ([HF-options])	1-min bar × contract	rolling window (verify start on pull)	daily-batch	HF clone → store	**CC-BY-NC — internal-use only, no redistribution/commercial derivative display**	NC license conflicts with any future commercial product — quarantine rows by license-tier from day 1; IV/OI presence unknown; contract-ID mapping to exchange tokens needed	W0	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
5	Hyperliquid perpetuals full history □	crypto derivative price	HuggingFace `Chainticks/hyperliquid-data` — 89.5 GB, CC-BY 4.0 ([HF-hyperliquid])	tick/1-minute × perp	Hyperliquid era 2023→ (verify exact start)	daily-batch	HF clone → object store	CC-BY 4.0 (attribution; redistribution OK with credit)	funding/OI columns presence to verify; cross-check vs in-house Hyperliquid node (latency UNVERIFIED per S38) — dataset is the backfill, node is the live rail	W1	backfill	VERIFIED-UPSTREAM-DOC
6	XAUUSD tick data	fx/commodity ticks	HuggingFace `CarlosSilvaJr/xauusd-ticks` — 2.65 GB, CC-BY 4.0 ([HF-xauusd])	tick × pair	multi-year (verify span)	daily-batch	HF clone → store	CC-BY 4.0	broker-feed provenance (not exchange tape) — spread/bid-ask conventions per feed; DST/timezone alignment; cross-check vs Dukascopy on 1 day	W1	backfill	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
7	Shadowbroker OSINT feed spine	reference/pattern donor	GitHub `BigBody-Cobain/Shadowbroker` — AGPL-3.0, 11,399★, pushed 2026-10-27 ([GH-shadowbroker])	repo (FastAPI + Next.js, 60+ live OSINT feeds cat-alogued)	n/a (code + feed registry)	n/a	**pattern donor**: polling spine (60s fast / 5min slow cycles, gzip/ETag/dedup, feed registry = candi-date source list for sheet 09/10/14	**AGPL-3.0 — internal-use code only; any network deploy-ment of derived code must stay internal (no SaaS)**	feed registry freshness must be re-probed per feed (catalog ≠ live); AGPL derivative-use boundary = O11 checklist item	W2	license-review	VERIFIED-UPSTREAM-DOC
8	worldmonitor country-metrics + freshness tracker	reference/pattern donor	GitHub `koala73/worldmonitor` — AGPL-3.0, 89.7k★ ([GH-worldmonitor])	repo (feed-freshness tracker, Welford anomaly detector, MCP server)	n/a	n/a	**pattern donor**: source-health dash-board design (fresh/stale/error), anomaly detection for ingest QA, macro-regime context factor	**AGPL-3.0 — internal-use only**	same AGPL boundary as row 7; metric defini-tions per country need own valida-tion	W2	license-review	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
9	OpenBB integration patterns	reference/pattern	GitHub `openbq-org/OpenBB` — NOASSERTION, 74,821★, org renamed, pushed 2026-10-24 ([GH-openbb])	repo (Open Data Platform; provider-adapter architecture)	n/a	n/a	**pattern donor**: adapter-interface design (Vol-1 §3 one-adapter philosophy), provider fallback chains	**NOASSERTION license — treat as proprietary until legal review; no code copy**	License unresolved = no code reuse; design-pattern study only	W2	license-review	VERIFIED-UPSTREAM-DOC
10	FinanceDatabase asset-class taxonomy	reference/data	GitHub `JerBouma/FinanceDatabase` — MIT, 9.5k★ ([GH-financedb])	repo (structured ticker/asset-class/sector metadata, JSON)	continuous (repo current)	weekly-ish	clone → reference table (asset classification spine)	MIT (attribution)	taxonomy is editorial (not exchange-official) — use for classification joins, never as exchange universe truth	W1	backfill	VERIFIED-UPSTREAM-DOC
11	gods-eye-view county geoJSON	reference/data	GitHub `bilawalsidhu/gods-eye-view` — 60.8k★ ([GH-godseye])	repo (US county boundary geoJSON set)	static snapshot	n/a	clone → geo reference table	per-repo (verify license file on clone)	US-only geo; vintage must be pinned; overlaps sheet 09 World-Pop/OSM geo stack — keep OSM as spine	W2	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
12	OpenTerminal	reference/pattern	GitHub `ErTas-selli/OpenTerminal` — MIT, 2.1k★ ([GH-openterminal])	repo (terminal-style data UI)	n/a	n/a	**pattern donor**: terminal UX + data-fetch patterns	MIT	UI reference only — no data claims	W3	verify	VERIFIED-UPSTREAM-DOC
13	OracleX	reference/pattern	GitHub `Yigtwxx/OracleX` — MIT, 2★ ([GH-oraclex])	repo	n/a	n/a	low-star repo — evaluate for ideas only before adopting anything	MIT	tiny repo — credibility check before any reliance	W3	verify	VERIFIED-UPSTREAM-DOC
14	evans-0/finance	reference/pattern	GitHub `evans-0/finance` — MIT, 2★ ([GH-evans])	repo	n/a	n/a	low-star repo — same low-priority screen as row 13	MIT	tiny repo — credibility check	W3	verify	VERIFIED-UPSTREAM-DOC
15	osint-worldview	reference/pattern	GitHub `amanimran786/osint-worldview` — 69★ ([GH-osintwv])	repo (geocoded places + OSINT map layers)	n/a	n/a	candidate geo-OSINT layer for sheet 09/14 — probe layer sources before adopting	per-repo (verify on clone)	low-star; layer prove-nance unknown — audit each layer	W3	verify	VERIFIED-UPSTREAM-DOC

#	dataset	class	source & route	granularity	history depth	latency	integration pattern	license class	quality/PIT gates	wave	TODO action	verification
16	OpenSpaceData (Copernicus EO access)	reference/pattern	GitHub org `OpenSpaceData` — Hippocratic-2.1, dormant since 2021 ([GH-openspace])	repo (ESA Copernicus EO access tooling)	n/a	n/a	**dormant 2021** — read-only reference; sheet 09 CDSE/sentinel rails already cover this need	Hippocratic-2.1 (ethical-use license — restrictive; internal reference only)	dormant = unmaintained against CDSE API changes; do not build on it	W3	verify	VERIFIED-UPSTREAM-DOC
17	QGIS Dashboard plugin	reference/tool	GitHub `isaacenage/QGIS-Plugins` — GPL-3.0, v0.2 ([GH-qgis])	repo (QGIS visualization plugin)	n/a	n/a	**visualizer tool, NOT a data source** — optional local map-QA tool for AOI sanity checks	GPL-3.0 (desktop tool use fine; no redistribution of modified plugin)	v0.2 = immature; tool-only row — zero data claims	W3	verify	VERIFIED-UPSTREAM-DOC
18	FinceptTerminal	reference/pattern	GitHub `Fincept-Corporation/FinceptTerminal` — NOASSERTION, 33.2k★ ([GH-fincept])	repo (multi-asset terminal, broad data rails)	n/a	n/a	**pattern donor**: multi-asset rail inventory for coverage-gap cross-check	**NOASSERTION — proprietary-risk; design reference only, no code copy**	License unresolved; audit its rail list vs our catalog for missed sources (not for direct reuse)	W2	license-review	VERIFIED-UPSTREAM-DOC

Build-log problems (not in table)

- duplicate dropped: `FINRA daily short-sale volume` (in 03, first occurrence kept; also in 05-fundamentals-filings.md)